TFPT — Changelog

All changes and new results, by date

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This file is the canonical, dated record of every change to the TFPT theory development: new verification modules (`vN`), status changes, editorial passes and infrastructure. It is maintained *in the same change* as the work it records, and it is also imported at the end of the `introduction` document. Module numbers `vN` refer to `verification/vN_*.py`; claim IDs refer to rows of `verification/status_ledger.csv`.

2026-10-18: Project separation and retained physics history

The combined publication history was separated by research ownership. Complete original dated entries remain byte-preserved in the independent archive, including mixed entries that also recorded physical work. The following existing TFPT modules retain their original identifiers and their currently scoped physical, compiler or verification roles. This entry records ownership and preserves history navigation; it does not re-date the original results, report a new execution, alter assumptions or promote any status. The current registry and claim ledger govern each module's precise scope.

- `v241_dhr_sectors.py` — retained dhr sectors.
- `v242_spin_statistics_modular.py` — retained spin statistics modular.
- `v561_cp_channel_identities.py` — retained cp channel identities.
- `v564_cp_fiber_pin.py` — retained cp fiber pin.
- `v578_native_jets.py` — retained native jets.
- `v616_su2_internal_kinematic.py` — retained su2 internal kinematic.
- `v621_interacting_semigroup.py` — retained interacting semigroup.
- `v626_e8_code.py` — retained e8 code.
- `v628_orbifold_casimir.py` — retained orbifold casimir.
- `v629_root_incidence.py` — retained root incidence.
- `v632_ftransfer_pgl2.py` — retained ftransfer pgl2.
- `v633_orbit60_quotient.py` — retained orbit60 quotient.
- `v634_st31_structure.py` — retained st31 structure.
- `v638_code_semantics.py` — retained code semantics.
- `v639_twist_ope.py` — retained twist ope.
- `v645_klein_rp.py` — retained klein rp.
- `v646_rm13_reverse.py` — retained rm13 reverse.
- `v647_st31_degree24.py` — retained st31 degree24.
- `v650_orbifold_modular.py` — retained orbifold modular.
- `v651_orbifold_assembly.py` — retained orbifold assembly.

- `v653_bond_defect.py` — retained bond defect.
- `v654_st31_degree8.py` — retained st31 degree8.
- `v664_look_elsewhere.py` — retained look elsewhere.
- `v679_orbifold_continuum_os.py` — retained orbifold continuum os.
- `v689_gaussian_code_bridge.py` — retained gaussian code bridge.
- `v690_quartic_half.py` — retained quartic half.
- `v700_orbit60_census.py` — retained orbit60 census.
- `v705_chain_deck_sector.py` — retained chain deck sector.
- `v715_qgeo_car_continuum.py` — retained qgeo car continuum.
- `v722_phys_ramified_ns_r.py` — retained phys ramified ns r.
- `v726_phys_car_pp_index.py` — retained phys car pp index.
- `v736_orbit_packet.py` — retained orbit packet.
- `v737_seam48_intertwiner.py` — retained seam48 intertwiner.
- `v745_qgeo_car_l1_sector_lemma.py` — retained qgeo car l1 sector lemma.
- `v746_phys_gnet_local_functor.py` — retained phys gnet local functor.
- `v752_projective_hamming_incidence.py` — retained projective hamming incidence.
- `v757_fano_disc7_bridge.py` — retained fano disc7 bridge.
- `v774_arf_spinor_compiler.py` — retained arf spinor compiler.
- `v775_gaussian_class_d5_purity.py` — retained gaussian class d5 purity.
- `v776_boundary_hamming_uniqueness.py` — retained boundary hamming uniqueness.
- `v778_qgeo_gridsup_logsum.py` — retained qgeo gridsup logsum.
- `v779_gnet_gns_arf.py` — retained gnet gns arf.
- `v781_one_object_clock_census.py` — retained one object clock census.
- `v782_e8_transition_bus.py` — retained e8 transition bus.
- `v783_two_qubit_clifford.py` — retained two qubit clifford.
- `v784_curve_code_2torsion.py` — retained curve code 2torsion.
- `v792_ftransfer_kms_coset.py` — retained ftransfer kms coset.
- `v796_duad_syntheme_bridge.py` — retained duad syntheme bridge.
- `v798_seam_clifford_modular_s.py` — retained seam clifford modular s.
- `v799_seam_code_typeii.py` — retained seam code typeii.
- `v800_e8_torsor_fourier.py` — retained e8 torsor fourier.
- `v802_gnet_martingale.py` — retained gnet martingale.
- `v803_e8_jetcode_affine_nsr.py` — retained e8 jetcode affine nsr.
- `v805_e8_syndrome_algebra.py` — retained e8 syndrome algebra.
- `v807_lorentz_nullselector.py` — retained lorentz nullselector.
- `v808_petersen_sixthroot.py` — retained petersen sixthroot.
- `v809_curve_code_doily.py` — retained curve code doily.
- `v810_cutcode_moment.py` — retained cutcode moment.

- `v812_flavor_graph_filtration.py` — retained flavor graph filtration.
- `v814_k5_sixstep_transport.py` — retained k5 sixstep transport.
- `v817_positive_descent_master.py` — retained positive descent master.
- `v832_anchor_flavor_checksum.py` — retained anchor flavor checksum.
- `v833_gaussian_ramification_ladder.py` — retained gaussian ramification ladder.
- `v834_pg32_flag_completion.py` — retained pg32 flag completion.
- `v844_message_doily_rank.py` — retained message doily rank.
- `v845_cfin_normal_form.py` — retained cfin normal form.
- `v849_cfin_unique_cofinal_lean.py` — retained cfin unique cofinal lean.
- `v852_arf_vacuum_ovoid.py` — retained arf vacuum ovoid.
- `v853_bent_css_mangoldt.py` — retained bent css mangoldt.
- `v857_simplex_fourier_winding.py` — retained simplex fourier winding.
- `v858_g31_clock_alphabet.py` — retained g31 clock alphabet.
- `v874_feedback_decoder_lean.py` — retained feedback decoder lean.
- `v875_shell_global_lean.py` — retained shell global lean.
- `v878_ledger_integrity.py` — retained ledger integrity.
- `v880_finite_anchor_theorems.py` — retained finite anchor theorems.
- `v888_finite_pfaffian_signature.py` — retained finite pfaffian signature.
- `v896_wick_block_functor.py` — retained wick block functor.
- `v898_kms_schur_mixing.py` — retained kms schur mixing.
- `v903_seam_rp_exclusion.py` — retained seam rp exclusion.
- `v908_seam_equilibrium_wiring.py` — retained seam equilibrium wiring.
- `v911_wiring_freedom.py` — retained wiring freedom.
- `v973_seam_route_narrowing.py` — retained seam route narrowing.
- `v975_dimension_selector_4d.py` — retained dimension selector 4d.
- `v987_os_dilation_package.py` — retained os dilation package.
- `v988_psi_lambda_reduction.py` — retained psi lambda reduction.
- `v989_lattice_gate_battery.py` — retained lattice gate battery.
- `v990_wj_transduction_mechanism.py` — retained wj transduction mechanism.
- `v991_detline_bulk_edge.py` — retained detline bulk edge.
- `v1018_e8_directed_readout.py` — retained e8 directed readout.
- `v1019_coxeter_euler_completion.py` — retained coxeter euler completion.

2026-09-29 (CLXIII): Executable TFPT explorer and consolidated session synthesis

The new `tfpt_explorer` package connects the origin principles, algebraic compiler, geometric and topological structures, physical readouts, and their evidence in an executable process with a guided graphical tour. It includes source browsing, verification and experiment discovery, and a Lean bridge that preserves the distinction between formal statements and executable numerical

checks. Six recent source documents and the session synthesis, calculations and visual snapshots are versioned alongside it.

The latest research path connects oriented flavor transport and neutral source response to an internal spin lift and graded Majorana pair channels. The last focused regression run passed 39 tests; the Explorer evaluated 343 internal checks across 24 stages. These are scoped implementation and mathematical consistency results, not a new full verification-suite or Lean build result. Native source selection and the complete four-dimensional physical identification remain explicit open questions. Historical snapshots retain their original counts. No canonical ledger verdict, empirical claim, or TOE/RH status is upgraded by this archival integration.

2026-09-09 (CLXII): Source fields, full lattice dynamics and Clock/E8 bridge consolidated

The post-Round 27 research archive, including Rounds 28–50 and subsequent named branches, is integrated without rewriting source-pinned historical records. The contracts paper gives a consolidated derivation of local charged CAR fields from the original filled QWZ sea: both adjoints, integer charge, finite-energy generator convergence and the top-edge charge minimum $q^2/2 - q/4$. Neutral four-point collision control and the improved auxiliary-window estimate are included with their exact scope. Neither result supplies the microscopic half-charge E_8 spin field.

For the separate, declared compact-$U(1)$ rotor/CAR parent, the new text records all-electric resummation, full all-time quasi-local automorphisms, the failure of point-norm continuity on the full bounded algebra, and neutral physical ground-state representations. Stronger coercivity and Wilson-loop spectral moments yield nonzero positive-energy weight, not a mass or a many-body gap theorem. The actual Clock phase classification, interaction provenance, eight-bit lattice cocycle refinement and scalar-symbol Weyl obstruction identify the still missing common-parent bridge. The sharp half-twist implementability obstruction is retained; renormalized inter-sector fields are the next construction target.

The related double-cover, inter-sheet and prime-event investigations are archived with their negative controls and separate arithmetic scope. The current charged-field regression chain contains 131 tests per mode (ordinary and optimized Python); these are execution controls, not independent mathematical review or physical completion. Frontier, introduction via this changelog, research overview, open-problem map, website metadata, PDF mirrors and reproducibility instructions are updated together. All T1–T8, TOE, empirical and RH markers are unchanged.

2026-09-07 (CLXI): Non-RH research through Round 27 integrated into repository, papers and website

The previously local theory-contract archive is shipped with its proofs, source-pinned checkers and immutable per-round validation snapshots. The intervening geometry, positive-parent, constraint-domain and charge transport branches retain their distinct hypotheses and additional off-shell inputs; they are not combined into a claimed microscopic TOE. The contracts paper now includes the full-reference argument for the original finite-regulator $g = 0$ scalar/charge model: $e^{-48\beta NJ} Z_D \le Z \le Z_{\mathrm{abs}} \le Z_D$, retaining every charge, the E8 cocycle and the charge-dependent scalar determinants. This removes the artificial extra charge heat-trace factor from the sign, memory and hop bounds, but not the extensive exponential sign budget.

For $N = 27, T = 3, \beta = 1, J = 0.1$, the certified 1% omission planner improves from $K4096/P962$ to $K2670/P354$; refining the old inequalities to the same integer search gives $K4035/P956$. A declared weak-hopping example at $L = 9, T = 33, \delta = 3/4, m = a = 1, J = 1/866052$ certifies $R = 12$: 25 of 33 quadratic time offsets, memory error at most 0.2045%, and combined omission error below 1% with $K12235/P5$. The full eliminated determinant remains a full-history function. The partition itself is not evaluated, and the naive history counts remain astronomical. Round 27 has two checkers with 23/25 exact groups and 18 input pins each, 37 isolated fixtures passing

normally and under optimization, and 29 prerequisite reruns. No T1–T8, physical TOE, empirical or RH status is promoted.

The existing correction to `v472_quillen_detline_moduli.py`, its ledger description and the corresponding contract is included unchanged: the finite result is an occupied Berry line and integrated Chern response, not a globally nonvanishing Dai–Freed section or a continuum Quillen connection. Its seven checks pass. Frontier prose, public research text, document metadata, PDFs and generated mirrors are updated together.

2026-09-06 (CLX): QWZ zero-mode portability correction and a free 3D matter contract

Linux reproduction exposed an eigensolver-dependent choice of a pure zero-mode vacuum in `v1033_charged_disorder.py`. The numerical diagnostic now uses the basis-independent fixed-particle ground-space mixture and computes variational energy bounds over every pure zero-space filling. At $N = 64$ the smooth N-scaled mixture excess is 6.904290, with full ground-space interval $[5.333652, 8.474929]$; the former 8.475 value was filling-dependent, not a unique vacuum coefficient. The charged-disorder proof is amended explicitly. The same 35/35 typed checks pass without weakening an exact identity; the integration guards now report subprocess output on failure and include a degenerate-basis regression. All 20/20 integration tests pass in isolated Linux Python 3.11 and 3.12 environments. The registered suite remains 1028 modules, with 233 Round-7 checks.

A separate, unpromoted theory contract in `experiments/theory-contracts/free-scalar-3d/` supplies a finite-range symmetric free-scalar stress on the original staggered sites, links and plaquettes without changing the energy density or current. Its complete real-space derivation and 68/68 exact checks include all three momentum Ward identities, genuine negative controls, bilinear Bloch transitions, periodic conserved charges and the quadratic Weyl operator lift. This is not yet a new registered verification module or a TFPT parent construction. Interacting matter, nonlinear gravity feedback and the missing homogeneous gravity receiver remain open. No T1–T8, TOE or RH marker moves.

2026-09-05 (CLIX): non-RH round 7 — free quantum curvature, charged twists and interaction boundaries

Five modules `v1031`–`v1035` add 233/233 typed checks (183 exact, 13 source-contract, 37 floating) to the registered suite (1023 → 1028 modules), including the round-6 quantum prerequisite. `v1031_quantum_tensor_curvature.py` (69/69) constructs the zero-mean Weyl/Fock target and its distribution limit; `v1032_covariant_curvature.py` (34/34) supplies exact identities for the positive local Lorentz-covariant free curvature field with helicities ± 2. Neither derives a common microscopic TFPT parent. `v1033_charged_disorder.py` (35/35) constructs auxiliary $\mathbb{Z}_4$ flux-register corners and proves the quarter-twist mismatch $h = 1/4 \neq h_\lambda = 1$ and the integer-dressing parity obstruction. `v1034_parent_mirror.py` (36/36) proves the factorized onsite lower gap $1 + \sqrt{3}/2$ and refutes a proposed bare-hopping vacuum-relative form bound, not every interacting gap argument. The newly chosen bond 3/5 is distinguished from the source coefficient 3/10. `v1035_matter_coupling.py` (59/59) keeps the physical Fourier factors in the prescribed-source Ward identities and exposes the dynamical-source, nonlinear scalar and zero-mode limitations. Full written proofs, five narrowly typed ledger subclaims, paper-body citations and the public status mirrors are integrated. Source-reading v1033/v1034 require the complete repository and are not advertised as browser-only certificates. No T1–T8 or TOE marker moves; no RH claim. Fourteen representative exact Wolfram checks are authored and included in the master runner; local execution is deferred because the installed kernel requires license activation. The last verified 116/116 count is unchanged; no successful independent-engine run is claimed.

2026-09-05 (CLVIII): non-RH round 4 — uniform TEL-B norm and explicit T1–T8 boundaries

Five modules graduate as `v1026`–`v1030`; the suite grows from 1018 to 1023 modules. Their standalone checks pass 353/353. `v1026_telb_round4_closure.py` (199/199, native python-flint 0.9.0) executes CF, DG, one-sided CROSS, 512 validated Acb eigensystems, sixteen Fourier modes, all infinite aliases and 154 numerical regressions. With v1022/v1025 inputs it proves $\|R_N\|_{\mathrm{HS}} < 2.995906 < 3$ at fixed $M = 1$, $N_y = 8$ for all even $N \geq 16$. The former finite 1.7833 smooth bound is withdrawn; the valid finite bound is 1.800594. The historical 2.11 and sharp Bound-B 0.62 targets are not proved. `v1027_signed_det_car_wall.py` (47/47) constructs canonical DET excitations and a signed $z = 1$ wall for fixed classical gauge backgrounds, changing the flat local parent to an additive one; there is no hard bare-mirror gap or chiral continuum claim. `v1028_gauss_d4_cap.py` (36/36) counts full 192-mode site Gauss characters modulo five (one site 3; cube 0 on the specified nine-irrep link truncation), constructs a symmetric internal gap-one cap, and retains the symmetry and Majorana selection-rule obstructions. `v1029_local_tensor_constraints.py` (44/44) gives a local two-mode radiative target only after explicit global zero-mode removal; the unmodified periodic target is unstable. `v1030_alg2_sector_frame.py` (27/27) proves the conditional sector-connectivity/actual-word frame bridge, with joint-adjoint and high-output tails still required. The complete proof supplement is included in the contracts paper, with intro/frontier/MMST, ledger, website and reproducibility surfaces synchronized. An independent-engine representative Wolfram mirror is supplied but its execution is unavailable locally (no kernel found); no verified Wolfram count is increased. The static discipline census (v649) now recognizes genuine renamed imports of the shared check function; seven regression tests reject comment/string-only and unrelated-import false positives. Its 18-check audit and twelve deterministic replays pass without exemptions. The 68-check v1006 battery now requires the native norm certificate together with the explicitly open FE-GEN/ALG-EXH/MMST boundaries in the memorandum; three document-guard regression tests reject a missing premise or the old external-box text alone. The 31-check v1016 battery likewise replaces its historical scalar-increment document guard by the current norm certificate and open-boundary guard, with three negative regression tests; its numerical thresholds are unchanged. All three regression sets run in CI. The atlas generator removes trailing whitespace introduced by one-line truncation. All T1–T8 and the TOE contract remain open. No RH claim.

2026-09-05 (CLVII) (non-RH T1–T8 Round 3 graduates as `v1022`–`v1025`; suite 1014 $\to$ 1018**, ledger rows** 1178 $\to$ 1184 **through six narrow subcontracts;** 156/156 **module checks; NO marker move;** `TFPT.TOE.COMPLETE.01` **and every T1–T8 gate remain [O]).** `v1022_telb_round3_certificates.py` **(**83/83**; native** `python-flint==0.9.0`**, 256-bit Arb/Acb) replaces the float/Richardson TEL-B input by 5812 rational cells and the exact cut jump, proving** $\mathrm{TV}_p(F'') \leq 15.817695962988924\ldots < 15.928674$ **and** $K_3^{\mathrm{TV}} < 2.517464500835$**. Together with the stated analytic digamma/midpoint lemmas, its executable Arb arithmetic gives** $c_\varepsilon = 0.288547359238391\ldots$**,** $c_\zeta = 0.110677064900715\ldots$ **and** $c_{\mathrm{potential}} = 0.399224424139106\ldots < 0.400$**; full C2b remains open exactly at CF** $\leq 0.139/\sqrt{N}$ **and DG** $\leq 0.060/\sqrt{N}$**. Its exact one-mode** M_2 **countermodel refutes the former ALG2 Vandermonde-to-generated-algebra implication: FE-GEN or an exact gauge/code-projected microscopic-algebra theorem is still required, and ALG1 remains conditional on the generated projection.** `v1025_telb_c1_c2a_certificates.py` **(**8/8**; native 100-bit Arb/Acb) executes the existing analytic C1/C2a proofs:** $C_\star \leq 0.1520453048\ldots < 0.1525$ **and** $N \sup_k |\phi_N(k)| \leq 0.508228779140631\ldots < 0.509$ **for every even** $N \geq 16$**.** `v1023_derivative_filtered_wall.py` **(**16/16**) proves only the finite quasi-free singular-value block: the high continuation branch is separated, but the bare** b **subspace**

is not invariant, the low branch has $z = 2$ rather than a Weyl cone, and no DET16/DET32, interacting-parent, measure/anomaly/index or 3+1D theorem follows. `v1024_parent_internal_boundaries.py` (49/49) supplies finite parent-internal T6–T8 witnesses and no-gos: a positive seam/neutrino-pair intertwiner with projective full-Fock D_4 lift but no mass texture; the uniform-gap and direct-TT locality obstructions but no graviton; and a gapped internal Z_5 cap that is chosen, D_4-breaking and complement-trivial, hence no global ρ_0. T1 retains only finite axiom-core/W-bridge support while unconditional provenance and the dimension selector remain open; T3 remains at Hamiltonian-class level; T5 retains finite IR witnesses and the v1020 spectator obstruction while prime completion, OS reconstruction, confinement and scattering remain open. The common missing object is still one shared 3+1D parent satisfying T3–T8. Native-only handling is explicit in the website reproducer; v1022/v1025 are not represented as Pyodide certificates. NO RH CLAIM. Surfaces: `tfpt_research_contracts`, `tfpt_4_frontier`, `introduction`, `tfpt_2_standard_model`, the MMST article/PDF, ledger, generated verification maps/index, website paper/DAG/gap/FAQ mirrors and changelog.

2026-09-04 (CLIV) (TEL-B reduction absorbed into the `SEAM.MMST.TYPEIII.CHARGED.01` contract text and ledger row; NO marker move, status stays [O]). The standalone note `articles/2026-09-04/telb_bound_b_sketch.tex` is superseded and removed: the contract is now the single home. Cover identity + sawtooth split $R_N = R_N^{\mathrm{sm}} + M_N$; bound (A) certified numerically for all N ($\|R_N^{\mathrm{sm}}\| \le 1.7833$, all-mode BV, one float64 TV constant); bound (B) reduced to $\sum|\mathrm{Res}_N|^2 \le 0.2769$ (measured 0.1550), open items C1/C2a/C2b. Residual restated as TEL-B-EXTERNAL (2026-09-04 reduction); the $A_R < 43$ dyadic-increment route is superseded (measured $A_R = 0.1238$ at r645, no longer the route). Numerically $2.57 < 3$, conditional analytic $2.9495 < 3$; analytically open. Evidence (experiments, not `veri`-citable): `mmst_telb_trace_decomposition_probe.py`, `mmst_telb_cover_split_probe.py`, `mmst_telb_bound_a_probe.py` (9/9), `mmst_telb_bound_b_probe.py` (17/20, G18–G20 open by design). Surfaces: `tfpt_research_contracts` (new TEL-B reduction subsection); ledger row claim text only; website `changelog.ts` regenerated.

2026-08-31 (CXLIX) (Monday-morning harvest — axiom-core closure + state/gap batteries graduate as `v1015`–`v1016`; suite $1007 \to 1009$, ledger 1173 no new rows, existing contract rows updated, NO status-marker upgrades; Python-only / Wolfram mirror deferred as for v974–v1014, engine still `DEFERRED_NO_ENGINE`). (1) `v1015_axiom_core_closure.py` (11/11): character-blind P-dem response DERIVED — $r = (1, 1, 1)$ exactly at collar sizes 12 and 16, democracy residual 0; mutants split $(1, 0, 0)$ and $(5/4, 1, 5/4)$. Together with P-anch (v1014) BOTH W-bridge premises are derived at finite level; the axiom-core remainder is ZERO modulo the externalized MMST identification. `AX.P2.01` stays an axiom (finite shadows; continuum identification is the same external leg). T1 note on `TFPT.TOE.COMPLETE.01`: the structure-postulate route is now fully premise-supported at finite level. (2) `v1016_state_gap_batteries.py` (31/31): R_4 seam-modular cap is the unique finite-level non-entropic selector ($D_{\mathrm{tr}} = 0.1434$, SK RMS 3.038×10^{-3}, $3\pi/5$ saddle preserved; finite-proxy caveat; `FTRANSFER.SK.RHO0.01` stays [O]). TEL-B certified-tail BLOCKED ($N^2 D_N$ $530 \to 19032$, UV-supported remainder); strictly smaller external target $A_R < 43$ with factor-~ 350 measured headroom; N-uniform CAR nuclearity PROVED ($C_\beta = (67.97, 29.73, 12.15)$) as ALG specialist input; ALG-EXH unchanged (`SEAM.MMST.TYPEIII.CHARGED.01` stays [O]). DFP volume-uniform gap cited-verified for $\mathbb{Z}_2/\mathbb{Z}_4$ open chains ($a = 0.0233$, $a/16 = 0.00146$); dynamical-link

leg closed at that level; remaining rotor/wall/3+1D (`CHIRAL4D.MIRROR.DET16.01` stays [C]). Surfaces: `tfpt_research_contracts` executed-update paragraphs (five items + L1 nuclearity note); `tfpt_4_frontier` + introduction Monday-morning note; MMST handoff HTML update-2026-08-31 block; website `papers.ts` / VerificationDag.

2026-08-30 (CXLVIII) (late-evening harvest — thermodynamic-dynamics theorem + detline/W-bridge refinements graduate as `v1013`–`v1014`; suite $1005 \to 1007$, ledger 1173 no new rows, existing contract rows updated, NO status-marker upgrades; Python-only / Wolfram mirror deferred as for v974–v1012, engine still `DEFERRED_NO_ENGINE`). (1) `v1013_thermodynamic_dynamics.py` ($12/12$): the mandatory-dynamics leg of `QFT4D.LATTICE.FUNDAMENTAL.01` is CLOSED at Hamiltonian-class level. Cited `articles/2026-08-30/tfpt_thermodynamic_dynamics_en.tex/.pdf`: uniform Lieb–Robinson bound $J = 12/5$, $R = 2$, $z = 4$, $\kappa = 16043/450$, $v_{\mathrm{LR}} = 32086e/225 \sim 387.64$; norm-convergent τ_t; gauge-invariant quasilocal $\mathcal{A}^G$ (Gauss caveat resolved on the invariant subalgebra); existence of ground and β-KMS states. Numeric twin: measured $v = 2.0 \ll$ proved bound; analytic finite-volume bound dominates nested differences at buffers $8/16/32$; $1/r$ mutant ratio 4.84; Gauss-positive control commutes, violating embedding has commutator norm 1. Remaining: phase uniqueness, limit gap, IR universality. Decision typing UNCHANGED. (2) `v1014_bridge_refinements.py` ($17/17$): finite detline restriction isomorphism VERIFIED (c_{phase} constancy 3.7×10^{-16}, conjugation covariance exact, $C = W_{\mathrm{bulk}} = W_{\mathrm{seam}} = +1$ / mirror -1); orientation fixes only the conjugate branch ($2 \to 1$); the residual constant $U(1)$ is the A_0 reference normalization — recorded on `ALPHA.QUILLEN.EXACT.01`. W-bridge: P-anch DERIVED ($e_1 = R^{-1}(1,1,2)^T$ via the winding lock); P-dem PARTIAL (the $\mathbb{Z}_4$ average annihilates the nontrivial character space); missing object = the character-blind determinant-response map — the same MMST/Quillen externalized leg as TEL-B/ALG-EXH/P1. `AX.P2.01` stays an axiom. Convergence: every remaining axiom-core and continuum gap now runs through ONE externalized theorem. Surfaces: `tfpt_research_contracts` executed-update paragraphs; `tfpt_4_frontier` + introduction late-evening note; website `papers.ts` / VerificationDag.

2026-08-30 (CXLVII) (evening TOE-gate wave — four executed witnesses graduate as `v1012`; suite $1004 \to 1005$, ledger 1173 no new rows, existing contract rows updated, NO status-marker upgrades; Python-only / Wolfram mirror deferred as for v974–v1011). `v1012_toe_gate_witnesses.py` ($40/40$): T5 IR universality on the physical QWZ edge — $c \to 1$ (limit 0.99993), clustering dichotomy $\xi\cdot\mathrm{gap} \approx 1.089$ vs power-law exponent 2.034, Lieb–Robinson cone $v_{\mathrm{LR}} = 5/3 \geq c$ with leakage 0.015, cubic curvature $\to 1/6$ with volume; TYPED SPLIT: gauge holonomy + seam clock are not propagating modes, so the common-c test needs dynamical gauge excitations. T8 ρ_0 eight-candidate table ALL unique; hard sectors kill the SK response ($\sim 10^{-31}$); every response-alive set is exactly ρ_{KMS} even without μ_4 — entropic-proximity selection is VACUOUS at finite level; verdict `NO_SET_WORKS` with the double diagnosis; binding requirement = a non-entropic principle (Euclidean-cap / orientation-branch). T6 κ six-fold NULL: six pre-declared D4-odd candidates, all parity-checked, all NULL (five at 10^{-14}; $K_4 = 3.55\times10^{-3}$ is $18.8\times$ too large); the 1.885×10^{-4} cancellation requires the true 4D functional — third and final scaffold-level nu-mechanism bound this week. T7 interacting TT: conserved interaction-complete lattice stress $\sim 10^{-15}$, TT spectral positivity exact, sum rules $\sim 10^{-15}$, dominant positive-Z

pole persists under ϕ^4 (gapped); Z trend k_1 suppressed 0.958 / k_2 enhanced 1.033; free analytic match. Missing: TFPT content, $k \to 0$ volume scaling, Ward identities. Ledger notes only on `TFPT.TOE.COMPLETE.01`, `FTRANSFER.SK.RHO0.01`, `FLAV.NU.TEXTURE.MECHANISM.01`, `GRAV.SPIN2.EMERGENCE.01` (all stay [O]). Surfaces: `tfpt_research_contracts` executed-update paragraphs; `tfpt_4_frontier` + introduction evening-wave note; website `papers.ts` / VerificationDag.

2026-08-30 (CXLVI) (afternoon harvest — simplicity campaign + 3+1D ladder graduate as `v1010`–`v1011`; suite $1002 \to 1004$, ledger 1173 no new rows, existing contract rows updated, NO status-marker upgrades; Python-only / Wolfram mirror deferred as for v974–v1009). (1) `v1010_simplicity_bridge_census.py` ($42/42$): derivation matrix over four frozen seeds, unique remainder min $S_4 = 2$; anchor $(1, 1, 2)$ is not among the three declared Milnor maps; φ_0 four novel candidates NULL. Bridge Q CANONICAL — the bit-selector ladder forces $c = 2$ and family $n = 3$ forces Q column sums $(9, 5, 1)$; C's carrier-looking 5 is inherited from R column 3, not inserted (the v1005 circularity flag is corrected as forced provenance). Bridge W CHOICE-but-conditionally-unique: spectrum $\{5/2, 1, 1\}$ alone has 1078 solutions; democratic-image + carrier-anchor kernel leave exactly the corpus shift (count 1); remaining obligation = seam-geometric derivation of those two directional premises. Milnor strong all-structure bridge NONEXISTENT at $\mathbb{F}_2$ (clock 2 vs 4, CP rank 2 vs 4) and at every upstairs home ($\mathbb{Z}/4$, Gaussian $k = 2, 3$, affine); only the coarse D-module+pairing bridge exists (witness constructed; $|\mathrm{Aut}(X)| = 36864$, $|\mathrm{Aut}(Y)| = 8$). Ring-internal identities of v1004 (rank clock, socle CP, self-clock) unaffected. `SEAM.MILNOR.LOCALRING.01` stays [O] (canonical status SHARPENED); `AX.P2.01` stays an axiom (pencil rehabilitated, W premises remain). (2) `v1011_3p1d_ladder_battery.py` ($32/32$): B6–B9 compressed — minimal 3+1D viable (exact Gauss 32768, confinement $W(A{=}1, g{=}2) \approx 0.063$ vs $W(A{=}1, g{=}0.8) \approx 0.995$, exact in-face isotropy, $g_\star \approx 1.915$); exact tree-fixing $524288 \to 4096$ with tree-independence $\sim 10^{-14}$; DET survives coupling; back-reaction; periodicity restores holonomy K; charged channels $\sim 2.014\times$ at weak g; charge-0 control matches B8 to 10^{-10}; K collapses above $g \sim 1.6$. BINDING: the matching point of the coupling functional must sit in the deconfined/weak regime. `QFT4D.LATTICE.FUNDAMENTAL.01` Decision typing UNCHANGED (Kronecker limitation removed); `GAUGE.DETLINE.FIXPOINT.01` stays [O] with the matching-scale constraint recorded. Surfaces: `tfpt_research_contracts` executed-update paragraphs (MILNOR sharpening, P2 rehabilitation, QFT4D ladder, DETLINE matching-scale); `tfpt_4_frontier` + introduction afternoon note; website `papers.ts` / VerificationDag.

2026-08-30 (CXLV) (review wave 7 — four exploration strands graduate as `v1006`–`v1009`: the MMST lemma battery, the DET16 stability battery, the 2+1D master-object assembly scaffold, and the ρ_0 constrained minimizer; suite $998 \to 1002$, ledger 1173 no new rows, existing contract rows updated, NO status-marker upgrades; Python-only / Wolfram mirror deferred as for v974–v1005). (1) `v1006_mmst_lemma_battery.py` ($68/68$): five lemmas in-house on `SEAM.MMST.TYPEIII.CHARGED.01` [O] — L3 crossed-product; UGF $K_G = \pi^2/4$ (measured plateau ~ 1.023, max measured K 1.029399 at $N = 16$); TEL-a N^{-2} with exact isometries ($A_{\mathrm{TEL}} = 5120(1 + \pi/2)$, measured envelope $3.420384 < 8$); integer D5+A3 pairings; $\mathbb{Z}_4$ outerness in the limit + finite-inner correction. Residual boxed: TEL-B-EXTERNAL (Hankel C_R; HS remainder sourced from the paper max 2.105079358 / fit 2.107565497 / v998 bound 2.11, not recomputed) + ALG-EXH (two Buchholz–Verch estimates). Dis-

play stays [O]. Cited `articles/2026-08-30/mmst_charged_scaling_limit_en.tex` (21 pp) + `externalization_mmst_handoff_v2_en.html`. (2) `v1007_det16_stability_battery.py` (67/67): T1/T2 proved (v1002); T3 Michalakis–Zwolak cited-verified (LTQO, $t_* > 0$, gap $\geq 1/2$). Counting line $n = 2, 3$; $n = 2$ closing scan through $t = 4$; dynamical $\mathbb{Z}_2/\mathbb{Z}_4$ uniquely gapped (full-probe min 0.728854416 pinned; compressed suite). SW $1 - \alpha t^2 + O(t^4)$; c_{geo}/L bounded. $1/192$ boxed (ordinary hopping not block-diagonal); BDL boxed (no volume-uniform gap remainder). `CHIRAL4D.MIRROR.DET16.01` stays [C]; `CHIRAL4D.NOMIRROR.01` unmoved. Cited `articles/2026-08-30/det16_mirror_gap_theorem_en.tex`. (3) `v1008_master_assembly_scaffold.py` (22/22): 2+1D master-object scaffold-coherent — combinatorial $K = 0$; frozen-link $q^2 \sim 4$, $K(q = 0) = 0$; unique-contractive $|R'| < 1$; QWZ wall slope ≈ -0.9989, DET gap ≥ 1.5 through $t = 0.6$; assembly $\mathbb{Z}_6$ $(1, 2, 3, 0)$, clock $[0, 1, 1, 2]$. $L = 3$ spaces 262144 / 129024 out of suite. Decision `QFT4D.LATTICE.FUNDAMENTAL.01` and `GAUGE.DETLINE.FIXPOINT.01` [O] unmoved. (4) `v1009_rho0_minimizer.py` (13/13): finite uniqueness of the KMS compression (Klein); Hessian strictly positive; centre dim 1; identity $\sim 8{\times}10^{-17}$. TYPED: `V1000_CONSEQUENCE_MISMATCH` (compression kills the SK response vs frozen contrast $3.550{\times}10^{-3}$) + `CENTER_FLATNESS_PREMISE_FALSE`. Canonical formula needs a weaker admissible set. `FTRANSFER.SK.RHO0.01` stays [O]. Surfaces: `tfpt_research_contracts` executed-update paragraphs; `tfpt_4_frontier` wave-7 paragraph; introduction status card; website `papers.ts` / VerificationDag.

2026-08-30 (CXLIV) (review wave 6 — two exploration probes graduate as `v1004`–`v1005` plus two registrations and display corrections: the Milnor–Frobenius–Gray finite algebra, the circularity-flagged relative pencil, the local-ring identification contract, the Weinberg–Witten spin-2 firewall, and honest rest/SM/gravity wording; suite $996 \to 998$, ledger $1171 \to 1173$: two new rows, existing contract rows updated, NO status-marker upgrades; Python-only / Wolfram mirror deferred as for v974–v1003). (1) `v1004_milnor_frobenius_gray.py` (40/40): CLASSICAL (cite, no claim) — Milnor algebra of $x^2 + y^3 + z^5 \cong \mathbb{Q}[y, z]/(y^2, z^4)$ dimension 8, spectral numbers $= E_8$ exponents $\{1, 7, 11, 13, 17, 19, 23, 29\}$. NEW-IN-CORPUS EXACT [E]-finite: rank clock $(2 - a)(4 - b)$ table-sum $30 = h(E_8)$; $S = pqr \cdot \mu/8$; self-clock iff $\mu = 8$; $(2, 3, 5)$ unique among ordered spherical triples; Galois–Gray $11/7/-1$ (binary Gray order $0 \to 1 \to 3 \to 2$); CP = socle/Frobenius; integral dual-number bridge $\mathbb{Z}[y, z]/(y^2, z^4) \otimes \mathbb{F}_2$ free rank 4 over $D = \mathbb{F}_2[\varepsilon]/(\varepsilon^2)$, basis-dependent. INTERPRETIVE [C]: $PG(3, 2) = AG(3, 2) \sqcup PG(2, 2)$. Mutants: $(2, 3, 7)$ $S = 63 \neq 42$; A_8 self-clock survives but exponents fail; wrong socle rank 4. NEW ledger row `SEAM.MILNOR.LOCALRING.01` [O]: required canonical geometric/mechanical identification of the raw seam collar / E_8 transition bus with $\mathbb{F}_2[y, z]/(y^2, z^4)$ preserving dual-number action, Frobenius pairing, Gray torsor, deck action and CP (rank clock and socle-CP as outputs); kill = second inequivalent identification or μ_4-equivariance failure; currently [O] because Milnor algebra + exponents are classical and the corpus map is unproven. (2) `v1005_relative_pencil_bridges.py` (43/43): EXACT [E]-finite Hankel $p_n p_{n+2} - p_{n+1}^2 = 2^{n+1}$ + Jacobi spec $\{1, 2\}$ diag $(4/3, 5/3)$ (v106/v119) and winding spec $\{5/2, 1, 1\}$ + fixed plane (v4). EXACT BUT CIRCULARITY-FLAGGED [C]: unique $t = 2$, $\lambda = 7/3$, factorisation $(3\lambda - 7)(\lambda^2 - 2\lambda + 2)$, CM $J_0^2 = -I$, dual number mod 2, index $25 = g_{\text{car}}^2$ SNF $(1, 1, 25)$ — BUT carrier 5 already sits in the column sums of Q_2 and C: NOT a P2 selector until Q, C, ε are derived from seam geometry independently; `AX.P2.01` typing UNCHANGED. INTERPRETIVE [C]: Klein-four SNF $(2, 2)$ only after $D = 2d$ (side-blind nulls v521, not a theorem). Mutants: $t = 0$ fails;

perturbed C breaks factorisation; $2+3^n$ breaks Hankel. (3) registrations (no marker moves). NEW `GRAV.WEINBERG.WITTEN.01` [O]: binding firewall BEFORE any spin-2 marker move — the microscopic Hamiltonian family must NOT possess a fundamental Lorentz-covariant local conserved $T_{\mu\nu}$; Lorentz, diffeomorphism Ward identities and the spin-2 mode must emerge in the SAME IR limit; acceptance = positive TT density, two helicities, massless pole, emergent diff Ward, universal soft coupling, named evaded Weinberg–Witten premise; kill = microscopic covariant local $T_{\mu\nu}$ together with a massless composite spin-2 claim; child of T7, cross-link `GRAV.SPIN2.EMERGENCE.01` (not back). (4) display corrections (honest wording, no claim changes). Status-at-a-glance: compiler rest $v_{\mathrm{geo}} \oplus G_{\mathrm{net}} \oplus F_{\mathrm{transfer}}$ now sits NEXT TO the strict-physical-TOE rest (ten-component $\mathrm{Rest}_{\mathrm{TOE}}$ / eight `TFPT.TOE.COMPLETE.01` gates, one line). SM completeness retyped: the discrete Standard-Model ***structure*** (charges, families, flavor operators) is derived under the axioms; 4D existence, chiral measure, mirror gap, couplings and neutrino dynamics are open. Gravity display: "parameter-free" headlines replaced by "the classical covariant field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C]". Surfaces: `tfpt_research_contracts` (new sections MILNOR.LOCALRING, WEINBERG.WITTEN; SPIN2 firewall note; T7 child; v358/v359 keybox titles); `tfpt_4_frontier` wave-6 paragraph; introduction status card + SM/gravity lines; `tfpt_2` gravity heading + conclusion; website StatusMatrix / WhatTfptClaims / ClaimStack / GravityEmergence / OpenGates / VerificationDag.

2026-08-29 (CXLIII) (review wave 5 — two exploration probes graduate as `v1002`–`v1003` plus three registrations: the DET16 finite-cluster hit, the two neutrino-texture structural nulls, the master scaling-limit theorem `SEAM.MMST.TYPEIII.CHARGED.01`, the gauge-stiffness step-scaling divergence, and the holomorphic-KMS document; suite $994 \to 996$, ledger $1168 \to 1171$: three new rows, existing contract rows updated, NO status-marker upgrades; Python-only / Wolfram mirror deferred as for v974–v1001). (1) `v1002_spin10_mirror_projector.py` (39/39): DET16 rank-one projector on the full 2^{16} cluster — $\Omega^\dagger\Omega = P_{16}$, $\Omega\Omega^\dagger = P_0$, defect 0; P_ϕ hermitian idempotent 2.8×10^{-18}, rank 1; h_{mir} spectrum $\{0^1, 1^{65535}\}$, gap exactly 1; all 45 $\mathfrak{so}(10)$ Fock commutators $= 0$ ($\det \rho_{16} = 1$); parity commutator 0; $|0\rangle$ and $|F\rangle$ singlets. Hopping stability on the 4-mode analogue: gap 0.90 (2 clusters) / 0.86 (3 clusters) at $t = 0.2$; spectators untouched. NEW ledger row `CHIRAL4D.MIRROR.DET16.01` Candidate [C] (exact finite-cluster gapping; open: dynamical gauge fields, domain-wall geometry, 4D volume theorem). (2) the Casimir kill, permanent. Number-preserving Casimir projector uniqueness fails at every cluster size: one $16 = \{N{=}0{:}1,\ N{=}16{:}1\}$ (none at $N{=}8$); $k{=}2$ half-filling multiplicity 27; $k{=}3{:}1463$ (exact Weyl alternation, $|W(D_5)|{=}1920$). Ledger note on `CHIRAL4D.NOMIRROR.01`: number-preserving Casimir projector route excluded; display stays [O]. (3) `v1003_nu_structural_nulls.py` (24/24): charged-sector data $h = (1,1,1)$, $d = (64,60,64)$ [E]; Schur-texture NULL — all 8 preregistered (K,B) fail, D4 reflection forces $\lambda_1 = \lambda_3$, hierarchy $(\varepsilon, 2\varepsilon, 3)$ structurally unreachable from D4-even seam data, required $K \sim [3536.998, 1768.499, 1/3]$ matches none of 9 comparators; orientation-doubling structure theorem — $\mathrm{sgn}\, J$ odd sector generically lifts the degeneracy (exact rational witness); scale NULL — 12/12 fail, required cancellation $1 - |c/a| = 1.885\times10^{-4}$ not supplied, η-branches isospectral. NEW ledger row `FLAV.NU.TEXTURE.MECHANISM.01` [O]: both nulls are binding constraints (mechanism must be D4-odd AND supply a $\sim 2\times10^{-4}$ even-odd cancellation; pure seam data excluded [X]-typed).

FLAV.NUSCALE.05/.06 notes only; candidates unmoved. (4) registrations (no marker moves). NEW SEAM.MMST.TYPEIII.CHARGED.01 [O]: ONE scaling-limit theorem $\Psi_{\lambda,N} \to \Psi_\lambda$ strongly on the finite-energy core + crossed-product continuity, from which $B \cong (E_8)_1$, $\mu = 1$, type III_1, unique rotational KMS, unique CE, deck equidistribution and modular invariance follow as corollaries; children SEAM.SIMPLECURRENT.GENERATOR.01, SEAM.STATE.RPMIXING.01 continuum, ALPHA.QUILLEN.EXACT.01 rigidity, AX.P1.01 reading (listed on the parent, not back). Cited document articles/2026-08-29/holomorphic_kms_extension_en.tex (13 pp): unconditional rotational-KMS uniqueness via KLM + Longo–Tanimoto GIVEN the identification; P1 = theorem-conditional on seam identification + $\beta_{\mathrm{angle}} = 2\pi$; axiom typing UNCHANGED. GAUGE.DETLINE.FIXPOINT.01: mechanism executed at fixed volume (unique fixed point, contraction -0.632, exact q^2 and $SU(2)$ Dynkin ratio 4.0) BUT the preregistered thermodynamic step-scaling $L = 2..8$ is DIVERGENT ($g_\star$ oscillates $0.65..1.69$, $|R'| > 1$ at $L = 4, 5, 8$); typed hurdle: the 1+1D toy has no plaquette self-interaction, the real contract needs $\geq$ 2+1D; display stays [O]. v_{geo} boundary note: unit choice closed; dimensionless hierarchies ($M_{\mathrm{Pl}}/v_{\mathrm{EW}}$, M_R/M_{scal}, $\Lambda_{\mathrm{QCD}}/v_{\mathrm{EW}}$, $T_{\mathrm{reh}}/M_{\mathrm{scal}}$) remain predictive obligations. Surfaces: tfpt_research_contracts (new sections MIRROR.DET16, TYPEIII.CHARGED, NU.TEXTURE.MECHANISM; DETLINE stiffness update; NOMIRROR Casimir kill); tfpt_4_frontier wave-5 paragraph; tfpt_2 (texture-mechanism note); introduction status card v_{geo} hierarchy sentence.

2026-08-29 (CXLII) (review wave 4 — four exploration probes graduate as v998–v1001 plus the lattice-fundamental quasilocal-family amendment, the v_{geo} closed-metrology retype, the ALPHA.QUILLEN relative-determinant note, and the master contract TFPT.TOE.COMPLETE.01; suite $990 \to 994$, ledger $1166 \to 1168$: two new rows, existing contract rows updated, NO status-marker upgrades; Python-only / Wolfram mirror deferred as for v974–v997). (1) v998_seam_modular_closure.py (14/14): two-edge HS remainder bounded by 2.11 on the frozen plateau $N_x \in (16, 24, 32, 48, 64, 96)$; lattice character θ_{E_8}/η^8 EXACT four orders $[1, 0, 248, 0, 4124, 0, 34752]$; $\Theta_{E_8}(\tau) = E_4(\tau)$ as Eisenstein series, q-expansion match to ≥ 8 orders exact (Jacobi identity; the reviewer's C7 sharpening); $j^{1/3}$ numeric transform at 60 digits with S-weight bookkeeping. Ledger notes on SEAM.EQUIV.MMST.01 and SEAM.SIMPLECURRENT.GENERATOR.01: C7 arithmetic shadow now exact; net-level C7 stays cited/open. (2) v999_weak_collision_hp.py (14/14): the continuous-time dilation leg — generator extraction Richardson 1.098×10^{-10}; $\mathrm{diag}\, L_{\mathrm{GKSL}} = Q = \log B$ symbolic; $e^Q = B$ exact; GNS splitting defect 0; locality 1.355×10^{-20}; strong-collision singularity contrast rank 9/81. DYN.UNITARY.DILATION.01: the finite ladder is now COMPLETE (discrete/kernel/field-free/interacting-classical/quantum-coupled/continuous-time); open = the thermodynamic field limit only; contract stays [O]. (3) v1000_spin2_sk_witnesses.py (26/26): lattice graviton — positivity exact, 2 TT helicities at all 5852 momenta ($N = 6..12$), $\omega = k - k^3/24$ exact; SK — KMS/FDT $\sim 10^{-17}$, ρ_0-dependence contrast 3.550×10^{-3} vs static 10^{-16}, saddle 5-fold degenerate with $3\pi/5$ unique in the μ_4 lift [C]. GRAV.SPIN2.EMERGENCE.01 quadratic Hamiltonian half executed; FTRANSFER.SK.RHO0.01 mechanism executed; both contracts stay [O]. (4) v1001_nu_pentagon_chain.py (14/14): $U_e = I$ inventory theorem; misalignment $U = U_{v9} R_{13}(\theta, \phi)$; pentagon double hit $\phi = 288^\circ = 4(2\pi/5)$ frozen, all three measured angles $\leq 0.56\sigma$ (honest max pull 0.557) at $\theta = 2\pi/35$, unique LEE survivor 2.7%; v3 chain SHA-16 a4c28732fa687620 with $\Sigma = 0.0599$ eV, $m_\beta = 9.0$ meV, $m_{\beta\beta} \in [1.5, 3.8]$ meV, $\delta_{\mathrm{CP}} = 287.66^\circ$; v270–$\theta_{23}$ tension 1.85σ

TYPED; census null $0/1607$. NEW ledger row `FLAV.NUSCALE.06` Candidate [C] (pentagon-class misalignment; mechanism = the Q_+-to-flavor operator [O]; kills DESI/DUNE 287.7 vs 240/JUNO). NO seesaw closure; `FLAV.NUSCALE.05` unmoved. (5) Amendments (typed; no marker moves). `QFT4D.LATTICE.FUNDAMENTAL.01`: a single finite box is not a world — the fundamental object is the quasilocal consistent family $\{H_\Lambda\}$ with $\tau_t = \lim e^{itH_\Lambda} A e^{-itH_\Lambda}$; continuum ($a \to 0$) optional, thermodynamic limit and controlled IR universality mandatory (dated 2026-08-29). `ANCHOR.VGEO.01/.02`: display retyped from open-physics/primitive phrasing to CLOSED METROLOGY STRUCTURE ($\mathbb{R}_+$ unit torsor; one dimensionful calibration necessary by dimensional analysis — No-Unit `v153/v725`); display stays [O], no physical [E]. `ALPHA.QUILLEN.EXACT.01`: the correct object is the relative determinant $Q(A) = \log[\det' \Delta_A / \det' \Delta_{A_0}] + 8\, b_1\, c_3^6 \log \varphi(A)$; BFK 2^{-4} is $A/\tau/\alpha$-independent at fixed cut number so $\delta \log 2^{-4} = 0$ — no physical compensation (consistent with `COMPENSATION_INTERNAL`). NEW master contract `TFPT.TOE.COMPLETE.01` [O]: eight gates T1–T8 with the binding separation $\mathrm{TOE}_{\mathrm{Complete}} := \mathrm{AND}(\mathrm{T1..T8})$ vs $\mathrm{TOE}_{\mathrm{Validated}} := \mathrm{Complete}$ AND independent holdouts; every existing 4D/dilation/detline/SK/flavor/gravity contract is a child of one gate. Surfaces: `tfpt_research_contracts` (executed-update paragraphs at MMST/SIMPLECURRENT, DILATION, GRAV.SPIN2, SK.RHO0; new sections TOE.COMPLETE and NUSCALE.06; lattice-fundamental amendment; ALPHA relative-det note); `tfpt_4_frontier` + introduction status card / v_{geo} row; `tfpt_2` (NUSCALE.06 candidate paragraph).

2026-08-28 (CXLI) (the completeness wave — four exploration probes graduate as `v994`–`v997` (suite $986 \to 990$, ledger $1166 \to 1166$: eight contract rows updated, NO status-marker upgrades; Python-only / Wolfram mirror deferred as for v974–v993). (1) `v994_mmst_criteria_battery.py` ($11/11$): 5 of 7 MMST identification criteria executed [E]-measured — measured $c = 8.005$ (16 Majorana copies), conformal edge spectrum $[0.506, 1.516, 2.523, 3.524]$, current normalisation $k = 1.000000000000$, sector collapse $\det D_8 = 4 \to \det E_8 = 1$, and the flagship: μ_4/GSO-projected lattice character sums reproduce the $(E_8)_1$ vacuum character $[1, 0, 248, 0, 4124, 0, 34752]$ three orders deep; C6 is now *written out* in `articles/2026-08-28/psi_lambda_convergence_theorem_en.tex` (analytic log coefficient $2/\pi^2 = 0.20264$ vs measured 0.203, 0.18%); C7 modular invariance stays cited/open. Ledger notes on `SEAM.EQUIV.MMST.01` (stays [C]) and `SEAM.SIMPLECURRENT.GENERATOR.01` (stays [O]; open rest = C7 + net G2/G3/G4). (2) `v995_kms_subnet_rigidity.py` ($10/10$): all 33 v972 mixing directions killed by KMS alone on finite $\mathrm{CAR}_8 = M_{256}$; covariance provably redundant; unique E-invariant crossed-product extension $196608 \to 0$. `SEAM.STATE.RPMIXING.01` finite KMS rigidity [E]; type-III/continuum stays [O]. (3) `v996_gauge_grammar_kill.py` ($17/17$): H0 structural kill — negative b_2/b_3 flip the stationary-root signs ($SU(2)$ inverse -151.2, $SU(3)$ -115.9); 64 conventions 0 hits; $U(1)$ control 137.0359992168 survives all mutants; detline curvature $\to 2\pi$ exponentially with rate $0.6936 \sim \ln 2$ (numerical Bismut–Freed shadow). `GAUGE.DETLINE.FIXPOINT.01` naive grammar excluded [E]; genuinely nonabelian structure required — typed; $\alpha_s(M_Z)$ remains an external input; contract stays [O]. (4) `v997_spin2_smg_witnesses.py` ($24/24$): Barnes–Rivers quadratic decomposition — $R{+}R^2$ clean (one massless TT pole, two helicities, scalaron at M_{scal} exact); the local a_4 Weyl^2 truncation necessarily carries an opposite-residue spin-2 ghost — typed; FK SMG toy: $N = 8$ unique $\mathrm{Spin}(7)$-singlet ground state, Schmidt rank 1, volume-independent gap 14 at $L = 1, 2, 3$; $N = 4$ contrast 2^L-degenerate; physical chiral edge keeps π/L exactly.

GRAV.SPIN2.EMERGENCE.01 quadratic witnesses [E], content pinned on the untruncated form factor; CHIRAL4D.NOMIRROR.01 SMG toy-proved [E], 4D $\mathrm{Spin}(10)$ stays [O]; both contracts stay [O]. Canonical note (not a module) on FLAV.NUSCALE.05: NU_TEXTURE_CENSUS_NULL $0/1607 + 0/200$ — frozen 3×3 inventory provably insufficient; mechanism [O] sharpened. Surfaces: tfpt_research_contracts (executed-update paragraphs at MMST/SIMPLECURRENT, RPMIXING, GAUGE.DETLINE.FIXPOINT.01, GRAV.SPIN2.EMERGENCE.01, CHIRAL4D.NOMIRROR.01, ALPHA.QUILLEN rigidity/ln-2 notes); tfpt_2 (census-null note); tfpt_4_frontier + introduction card addendum. Exploration record: articles/2026-08-28/big_picture_simplicity_first_en.md.

2026-08-28 (CXL) (review wave 3 — the third external bird's-eye review adjudicated and integrated: ONE new module v993 plus THREE new [O] contracts plus a v986 motivation retype plus the dual-rest formula; suite $985 \to 986$, ledger $1163 \to 1166$; NO status-marker upgrades; Python-only / Wolfram mirror deferred as for v974–v992). (1) v993_minimal_defect_selector.py ($29/29$): the FULL rank-8 ADE census (SNF of Cartan) with cyclic $\mathbb{Z}_4$ glue on BOTH factors selects (D_5, A_3) uniquely ($D_3 \cong A_3$ aliases collapsed) — the $D{+}A$ architecture of v624 is an OUTPUT of (rank 8 + cyclic $\mathbb{Z}_4$), not an INPUT; kill tests $D_4 \oplus D_4$ / $A_7 \oplus A_1$ (product 16, fail cyclicity) and the $\mathbb{Z}_3$ control $E_6 \oplus A_2$; $[\lambda]^2 = [v]$ exact (lattice shadow of $U^2 = (-1)^F$); the μ_4 average is the UNIQUE trace-preserving finite conditional expectation (uniqueness needs the module property; kernel dim $36/720$ without it). Ledger notes on AX.P1.01/AX.P2.01: axiom typing UNCHANGED; the equidistribution identification stays the open [C] of v813; cited-not-proved remain $c_- = 8$ and holomorphic uniqueness of $(E_8)_1$. SEAM.SIMPLECURRENT.GENERATOR.01 canonical status grows the $[\lambda]^2 = [v]$ parity-shadow line. (2) Three new research contracts, all [O], no scripts: GAUGE.DETLINE.FIXPOINT.01 (all three gauge couplings as unique stationary points of one nonabelian zeta-det/inflow functional; generalizes ALPHA.QUILLEN.EXACT.01; honest: $\alpha_s(M_Z)$ is currently an external input); GRAV.SPIN2.EMERGENCE.01 (massless transversal spin-2 pole from the same spectral determinant; Einstein equation stays downstream/conditional; GRAV.NONCIRCULAR.01 binding); FTRANSFER.SK.RHO0.01 (cosmological transfers need (S, ρ_0), not $W[J]$ alone; typed candidate $\theta_i = 3\pi/5$, μ_4 orientation picks $k = 0$; FTRANSFER.GENERATING.01 covers equilibrium only). (3) v986 docstring retype: the $\mathrm{Tr}\, I = 3$ motivation is loose ($\mathrm{Spec}(I) = \{1, 1, 1\}$); the operator reading is the Q_+ eigenvalue route $\mathrm{Spec}\{1, 2, 3\}$; texture data-constrains-texture; check logic and pass/fail UNCHANGED. (4) Dual rest: introduction status card (and the research-contracts keybox, with a brief frontier mirror) now carries $\mathrm{Rest} = v_{\mathrm{geo}} \oplus G_{\mathrm{net}} \oplus F_{\mathrm{transfer}}$ (compiler) beside $\mathrm{Rest}_{\mathrm{TOE}}$ (strict-physical-TOE accounting of the 2026-08-27/28 contract wave); nothing in the second formula is claimed closed. Exploration not promoted: the wave-3 Quillen zero-mode Gram identity is a reformulation ($|\det \mathrm{DFT}_4| = 16$), not a Quillen-metric object.

2026-08-28 (CXXXVIII) (the master-object wave, part 2 — the second external bird's-eye review ("one 4D lattice + one seam simple current + one unitary dilation + one determinant line + one $W[J]$") adjudicated: its four machine-checkable candidate results verified and integrated as FOUR new modules (suite $975 \to 979$, ledger $1161 \to 1162$: one new row FLAV.NUSCALE.05; three contract rows updated, NO status-marker moves). (1) v983_simple_current_generator.py: the SEAM.SIMPLECURRENT.GENERATOR.01 generator is ***identified*** — the glue vector $\lambda = (\omega_s, \omega_f)$ with $\|\lambda\|^2 = 2$, $h_\lambda = 1$, order 4 in the discriminant group $\mathbb{Z}_4 \times \mathbb{Z}_4$ of $D_5 \oplus A_3$; isotropic glue $\Rightarrow$ even unimodular rank-8 overlattice (E_8); coset root

census $[52, 64, 60, 64]$ — the odd fusion powers of the ONE class carry the full 128 spinor sector; kill tests (vector-class / mismatched glue) fire; holomorphy shadow $\det 1$ vs $\det D_8 = 4$ at the lattice level; the analytic half (G1–G4) stays [O]. (2) `v984_markov_qca_dilation.py`: the DISCRETE leg of `DYN.UNITARY.DILATION.01` realized constructively — the collision step $V = C(U_B \otimes I)$ is exactly unitary, the Stinespring identity $\Phi(\rho) = \sum_j P_j U_B \rho U_B^\dagger P_j$ holds symbolically, $\operatorname{diag}\Phi(\operatorname{diag} p) = Bp$ and six fresh ancillas give $B^6 = T$ exactly (radius-1 collision band); must-fail fires ($|U_B^2|^2 \neq B^2$); typed remark AGAINST the review text: a reused mod-3 adder ancilla still reproduces the populations exactly; the continuous Hamiltonian/OS legs stay [O]. (3) `v985_quillen_channel_swap.py`: the v974 CHANNEL-SWAP face of `ALPHA.QUILLEN.EXACT.01` resolved at the finite level by μ_4-character grading — the deck swap $r \to r{+}2$ maps the jump spectrum $[0, 2, 4, 2]$ onto the mark pattern $|[4, -2, 0, -2]|$ exactly (odd swap fails), the graded $\det'$ difference is c_3-free $= \log(\ln 2/4)$, the channelwise unit ratio is uniform $4/\ln 2$; continuum Bismut–Freed, KMS rigidity and the 2^{-4} compensation stay [O]. (4) `v986_nu_scalaron_rung.py`: the SCALARON-TRACE RUNG $M_R = N_{\mathrm{fam}} M_{\mathrm{scal}} = 3 c_3^{7/2} \bar{M}_{\mathrm{Pl}} = 9.18 \times 10^{13}$ GeV — a structured point INSIDE the v481 window, -1.8% from the 1-loop requirement (the review claimed $\sim$1%; the honest value is 1.8), $\sim 20\times$ sharper than the declined integer rung; exact ladder position $3\sqrt{c_3} = 0.598$, 0.26% from the closed $3/5$ Clebsch door — adjacency TYPED, not consumed (v488 killed the Clebsch mechanism; the generation-blind scalaron trace is a different class, itself [O]); kill test $\Sigma m_\nu = 0.059$ eV; new row `FLAV.NUSCALE.05` [C]/[N], no seesaw closure. The remaining review content is typed, not consumed: the axion line ($f_a = M_{\mathrm{scal}}/128$, $\theta_i = 3\pi/5$) and the Riemann–Roch carrier ARE the existing v185/v211/v189; the 4D action, chiral measure, OS package, bulk dictionary, determinant-line unification, $W[J]$, transduction matrix and joint likelihood stay open contracts exactly as registered on 2026-08-27. Paper surfaces: `tfpt_research_contracts` (three executed-update paragraphs), `tfpt_2_standard_model` (scalaron-trace rung paragraph).)

2026-08-27 (CXXXVI) (the transport-lift wave — the two external-review rounds on the quantum structure of the seam transport graduate from the experiments tree into the load-bearing suite as `v976_seam_lift_birkhoff.py` and `v977_transfer_unistochastic_wilson.py` (suite $968 \to 970$, ledger $1154 \to 1156$: new rows `TRANSFER.HIDDEN.CIRCULATION.01` and `TRANSFER.COHERENT.WILSON.01`, both "executed half [E] / selection open [O]"; the exploration probes `transfer_birkhoff_circulation_probe.py` 42/42 and `transfer_coherent_wilson_probe.py` 28/28 stay in `experiments/tfpt-discovery/` as the discovery record). (1) v976 — the Birkhoff fibre and the entanglement-breaking no-go. The Birkhoff (random-permutation) fibre of the seam single step B (exact sixth root of the v221 kernel, $B = \frac{1}{2}I + \frac{1}{18}P_{12} + \frac{2}{9}P_{13} + \frac{2}{9}P_{23}$ bit-exact) is exactly a one-parameter line $t \in [0, 1/18]$; the hidden coordinate is the sign-character Fourier coefficient $\widehat{w}_t(\mathrm{sgn}) = 6t$, populations are exactly t-free, and no fibre member is chiral ($w(C_{123}) = w(C_{132})$ forced). The one-step coherence spectrum is $\{1, 6t, (2/3)^{\times 2}, (1/3)^{\times 2}\}$ and the triangle loop current is an exact eigen-observable $\Psi_t(A) = 6t\,A$ — one step plus one current readout measures t without an ancilla. On the contrast qubit the Choi partial transpose has spectrum $\{-3t/2,\ 1/6 + 3t/2,\ 1/3 + 3t/2,\ 1/2 - 3t/2\}$: the deployed sixth-root spectrum sits exactly on the entanglement-breaking boundary because $2/3 + 1/3 = 1$ (mutated spectra miss by $\mp 1/12$); at the full three-level the family stays NPT at every tested iteration with $\lambda_{\min} = -(1/3)(2/3)^n$, t-independent. The no-go that sharpens `DYN.UNITARY.DILATION.01`: the deployed Kraus dilation

$K_{ij} = \sqrt{T_{ij}}\,|i\rangle\langle j|$ is measure-and-prepare, hence entanglement breaking with quantum capacity 0 and $C_0(T) = 0$ exactly — any dilation closing that contract must be a genuinely different object. Shannon KKT certificates: $p^* = (1/2, 1/2, 0)$ is exactly capacity-achieving for both B and T ($C(T) = 0.0084$ bit, $C(B) = 0.4890$ bit, ratio $1.71\,\%$) — the surviving classical message is binary on every scale. The defect–Birkhoff value coincidence (weights $(9, 1, 4, 4)/18$ vs the v652 ladder $g^2/18$) is recorded typed [C] with its honest multiplicity audit; the forcing functor is the named open contract `TRANSFER.DEFECT.BIRKHOFF.01`. (2) v977 — the unistochastic $SU(3)$ lift and the 5-12-13 Wilson code. B is unistochastic: an exact $U \in SU(3)$ with $|U_{ij}|^2 = B_{ij}$ entrywise, angles forced by B ($\sin^2\theta_{13} = 2/9$, $\sin^2\theta_{12} = 1/14$, $\sin^2\theta_{23} = 2/7$), $\cos\delta = -4/\sqrt{65}$ unique, lift set $= \{U, U^*\}$ up to rephasing; Jarlskog $J = \pm 1/27$ exactly, and the real part of every Wilson plaquette is fixed by B through unitarity — the full rephasing-invariant Wilson data is classical up to the $\mathbb{Z}_2$ orientation bit $\operatorname{sgn} J$. The main plaquette is $(-5 + 12i)/324$, normalized $(-5 + 12i)/13$ with $5^2 + 12^2 = 13^2$ (atoms: $5 = g_{\mathrm{car}}$, $12 = \dim \mathfrak{g}_{\mathrm{SM}}$, $13 = \Delta_Q = N(3{+}2i)$); all nine plaquettes close on four Gaussian phase types with norms exclusively $\{10, 13, 130, 169\}$. The center bridge: $\operatorname{disc}\chi_A = 65 = \det(A{-}I)^2 + (\det A/2)^2 = 4^2{+}7^2$ for the v530 atom matrix, $e^{i\delta} = (-4 \pm 7i)/\sqrt{65} = (1 \pm 2i)(2 \pm 3i)/\sqrt{65}$, and the v530 sibling quotients fail the identity — the bridge selects the atom quotient. Rationality selection: J_n is rational only at $n = 1$ (the deployed step hits $3(1 + 2b - 3a^2) = 1$ exactly, so $J = 1/27$); census over 238 valid family points: exactly one full-code point = the deployed $(2/3, 1/3)$ ($0.42\,\% \ll$ the preregistered $10\,\%$ kill). No fixed microstep: $B^2 \neq |U^2|^2$ and a Dirichlet recurrence witness ($n^* = 967909$) shows no single closed unitary generates the relaxing sequence — repeated steps require dephasing between steps, the six-step macro-gate $U_6 \neq U_1^6$ (the coherent sharpening of clock-vs-walk). The PMNS near-miss stays a negative control (240.2551° vs 240°; $-1/27 \neq -0.02965$; no PMNS derivation). Surfaces. Research-contracts document: the `DYN.UNITARY.DILATION.01` section gains the executed lift-structure sharpening, the CCC reduced-channel prose and the two former "exploration-only" citations of rounds DCXCVI/DCXCVII are upgraded to `\veri` citations; `origin_theory` gains the honest "what the classical transport does not determine" paragraph next to the v221 recoverability code; `tfpt_2` gains the $65 = 4^2 + 7^2$ bridge corollary in the v530 center-quotient block. No status marker moves; both contracts stay open; no PMNS/CP/flavor claim.)

2026-08-27 (CXXXI) (the external deep-review round — dimension / dynamics / gravity firewalls and the 4D construction ladder: the review diagnosis "chiral 1+1D seam net $\not\Rightarrow$ local, interacting 3+1D quantum world" is implemented as eight new contract IDs in `tfpt_research_contracts` plus one new load-bearing module. (1) `DIMENSION.UPLIFT.FIREWALL.01` (standing rule, in force): no 1+1D theorem closes a 3+1D claim; every contract closure states the spacetime dimension of hypotheses AND conclusion; audit census recorded (SEAM.EQUIV.01 is 1+1D in and out — even fully proved it is a boundary building block, not the 4D theory; the v957 kinematic uplift does not close it; the anomaly-inflow bulk is invertible/topological, never to be silently equated with physical spacetime); the realistic route ordering (seam as boundary module → genuine 4D Euclidean lattice → OS package → chirality/no-mirror → only then BW/CHM + entanglement gravity → bulk–seam equivalence) is registered as the programme skeleton. (2) `SEAM.BULK4D.RECON.01` [O]: the previously implicit seam→bulk theorem made explicit (4D local algebras/Schwinger functions; locality/microlocality, Poincaré covariance + spec-

trum condition, spin-statistics + CPT, both helicities, clustering, nontrivial S-matrix, unique operator dictionary). (3) the dynamics split `DYN.MARKOV.EMBED.01` / `DYN.UNITARY.DILATION.01` ("three meanings of dynamics": relaxation T^n / dissipative semigroup $e^{t \log T}$ / unitary e^{-itH}): the finite kernel half of MARKOV.EMBED is executed [E] as `v971_markov_embedding_generator.py` (15/15, sympy-exact) — $Q = \log T$ of the v221 seam kernel is an EXACT classical Markov generator: $e^Q = T$ by spectral calculus, row sums symbolically zero, off-diagonals exactly $\{\ln(9/8),\, 2\ln 3,\, 2\ln 3\} = \{0.117783,\, 2.197225,\, 2.197225\} > 0$ (the reviewer's independently verified numbers), uniformization certificate $Q + (4\ln 3)I \geq 0$ entrywise $\Rightarrow$ e^{tQ} positivity-preserving doubly stochastic for all $t \geq 0$, $\mathrm{spec}(Q) = -\mathrm{spec}(H_{\mathrm{mod}})$ (gap $= \Delta = 6\ln(3/2)$), negative control: second eigenvalue $-1/10$ gives $\det = -1/200 < 0$ — provably NOT embeddable; the field level and the local size-consistent unitary dilation (Stinespring + Lieb–Robinson + OS continuation) stay [O]; binding precision: the shared F_transfer relaxation shape is a universal CONTRACTION CLASS, not one physical clock (v723/v724/v777). (4) `QFT4D.OS.RECON.01` [O] (the genuine 4D Euclidean lattice OS package) with the binding phrasing rule: the v381 all-orders leg is a ***local perturbative construction under the stated EG/BRST hypotheses*** (Duch weak adiabatic limit cited, hypotheses unverified for the concrete model; no confinement / mass gap / asymptotic charged states / nonperturbative Higgs vacuum / full S-matrix follows) — `tfpt_2` and `tfpt_5` phrasing restricted accordingly. (5) `CHIRAL4D.NOMIRROR.01` [O] (explicit GW/overlap chiral lattice operator; local gauge-invariant measure, perturbative + global anomaly freedom, no Witten $SU(2)$ anomaly, no mirrors, correct index/charges, stable continuum limit; Mastropietro `arXiv:2303.02790` as calibration). (6) `GRAV.NONCIRCULAR.01` (standing rule, in force): BW is a theorem WITHIN a local Poincaré-covariant QFT (Mund/Morinelli counterexample landscape cited), never a generator of Lorentz spacetime from a modular flow; every BW/CHM use declares assumption-vs-result; audit census: the entanglement-equilibrium Einstein derivation (v323/v358/v359) is typed ASSUMPTION (conditional on the 4D package, never citable as evidence for it), QG.AMB.01 correctly declared, the seam KMS 2π chain dimension-typed, the intrinsic-BW programme the honest result-direction attempt. (7) `OBS.TRANSDUCTION.01`: the transduction principle (internal eigenvalue + nonvanishing projection $BP_r \neq 0$ + physical coupling = observable signature; without the middle arrow neither non-observation falsifies nor observation confirms; minimal model machine-checked at exploration level in `fo01`, uniform functional blind to 10^{-14}, character channel recovers $64/729$ to 10^{-10}; the ACT+Planck birefringence proximity typed coincidence-level pending the shared-action proof). (8) `PRED.JOINTLIKELIHOOD.01` [O]: evidence reorganized from hit-counting into overdetermined relation bundles (flavor sum rule; $r = 3(1-n_s)^2$, $\alpha_s = -r/6$; one normalized axion action; character-resolved transfer) with the statistical demand (joint likelihood, atom-induced covariance, effective dof, prespecified null + formula grammar, real holdout); the v100 "13/13, $< 10^{-30}$" statement retyped as a look-elsewhere quantifier under its declared null — the 2026-06-09 freeze protects FUTURE measurements, it does not retroactively make historically known data blind. Docs corrected in the same change: `docs/OPEN_PROBLEMS.md` (the Einstein-derivation overstatement rewritten to the conditional Jacobson-route typing with the explicit does-NOT-follow list; "zero open internal mechanisms" fixed to match the contract scope fence — ledger wins; the all-orders phrasing restricted) and `docs/FALSIFICATION.md` (the null-model section retitled and rescoped, freeze and kill tests unchanged). Ledger $+9$ rows; suite 963 modules; `bash build.sh audit` and the manifest regeneration

are the joint exit gate)

2026-08-24 (CXVIII) (the CCC-crossover promotion — the kinematics dictionary of the conformal-cyclic crossover graduates from the experiments tree into the load-bearing suite as `v957_ccc_crossover_kinematics.py` (suite $949 \to 950$; six discovery probes of rounds DL–DLV re-run GREEN AS TYPED at promotion and embedded BYTE-EXACT with pinned SHA-256 hashes: `ccc_crossover_gates_probe.py` $28/28$, `ccc_uplift_4d_probe.py` $29/29$, `ccc_fork_kms_holonomy_probe.py` $18/18$, `ccc_kernel_gamma_kms_probe.py` $21/21$, `ccc_transfer_selector_probe.py` $14/14$, `ccc_kappa_freeze_probe.py` $7/7$): (1) Penrose's reciprocal gauge $\hat{\Omega}\check{\Omega} = -1$ is the element $\tau = \rho^2\sigma : z \mapsto -1/z$ of the PROVEN μ_4 Möbius stabiliser D_4 — τ reverses the clock (= the $\mathbb{Z}_2$ sheet parity), swaps the aeon poles $\{0, \infty\}$, and the crossover locus $|\hat{\Omega}| = |\check{\Omega}|$ is the unit circle carrying ALL FOUR marks; (2) conformal-factor rigidity: on the 2D seam the clock alone leaves exactly ONE constant-curvature modulus (Liouville-exact dilation family) and the reciprocal $\mathbb{Z}_2$ kills it (unique zero at $\lambda = 1$); on the 3D crossover the deck ALONE kills all conformal moduli (the $so(4,1)$ deck-commutant is 4-dimensional $su(2)+u(1)$, ALL Killing, boost content 3.3×10^{-16}; global uniqueness on the explicit Obata boost family, 200-axis scan bounded below by 0.53) — TFPT's candidate answer to CCC's open unique-conformal-factor problem, v_{geo} staying pure unit calibration; (3) the entropy reset of the cyclic [C] reading is a REDUCED CHANNEL: the v221 transport is CPTP not unitary ($\|T^tT - 1\|_2 = 0.9999$), its Stinespring isometry explicit ($V^\dagger V = 1_3$ exact), $\|T^n - P_\star\|_2 = (2/3)^{6n}$ EXACTLY, the global overlap invariant and the environment retaining the full distinguishability — "unitarity + rank-1 attractor" is isometry + partial trace; (4) the kinematic $2\text{D} \to 3\text{D} \to 4\text{D}$ uplift from corpus objects only: stereographic(Hopf(v)) $= z_1/z_2$ EXACTLY (the seam sphere IS the Hopf base), the v492 deck $\mathrm{diag}(i, i^{-1})$ acts freely so the crossover $X = S^3/\mathbb{Z}_4$ is the smooth lens boundary of the corpus A_3 ALE $XY = Z^4$, clock $\mathrm{diag}(i, 1)$ and the reciprocal lift J normalise the deck and descend (J to τ, inverting the clock), the four marks lift to four Hopf fibre circles paired by the deck into the TWO $\mathbb{Z}_2$ sheet orbits with Gauss linking number 1.0000, the 4D bridge is the cone $\mathbb{C}^2/\mathbb{Z}_4$ with the inversion $x \mapsto x/|x|^2$ a conformal involution (pullback $|x|^{-4}\,\delta$ symbolic) fixing the unit lens POINTWISE and swapping the aeon ends, and the Lorentzian shadow $(1/\cos^2 T)(-dT^2 + g_{S^3})$ is Einstein $R_{ab} = 3g_{ab}$ (symbolic Ricci) with a SPACELIKE round-S^3 crossover and $\check{\Omega} = -\cos T$ vanishing to first order; (5) the KMS/modular closure of discrete $\to$ dynamics at the free level: detailed balance EXACT, $\mathrm{spec}(-\ln T) = \{0, 6\ln\frac{3}{2}, 6\ln 3\}$ EXACTLY (the seam gap IS a modular energy), $T = e^{-H_{\text{mod}}}$ (one clock tick = one unit of KMS time at $\beta = 1$), OS Gram PSD, and the v526 normalisation $\beta_{\text{angle}} = 2\pi = 1/(4c_3)$ replicated exactly = the Gibbons–Hawking normalisation on the crossover fibre; (6) the holonomy dictionary: the two clock-pole circles descend to the lens core geodesics with the full μ_4 monodromy $\chi(D) = i$ while EVERY seam string carries exactly the 2-torsion $\chi(D^2) = -1$, and $D^2 = -1$ is trivial on the base while flipping the spinor coordinates — the v522 statement "gaugeable datum = the 2-torsion $\{1, (-1)^F\}$; full wrap = fermion parity" in geometric form (typed [C] correspondence); (7) the selector necessity (the γ-facing theorem): the admissible OS reflections of the v512 mark family $M(\delta)$ form the pair-exchanging V_4 for every $\delta \neq \pi/2$ and close to D_4 EXACTLY at $\delta = \pi/2$ (the order-4 clock factorises into admissible reflections, $\rho = \tau \circ s_{\text{mid}}$, iff μ_4), and at Majorana operator level on the $N = 16$ seam ring with fermionic reordering signs kept the free parent is Θ_a-real on ALL 16 axes

(v521 side-blindness replicated) while the interacting family is Θ-real on exactly $\{k \neq 4 : 2,\ k = 4 : 4\}$ axes and the clock-axis cuts ONLY at $k = 4$, asymmetric couplings breaking reality — the UNIQUENESS half of the v534 straddle selector ($\delta = \pi/2$, symmetric couplings) is thereby DERIVED as a Θ-reality necessity (the survival half and the coupling sign stay v534's nonperturbative toy facts, cited; reality is sign-blind, checked); (8) the frozen Gate-6 kernel: exactly two decay modes $\ln(729/64)$ and $\ln 729$ (ratio $\ln 3/\ln\frac{3}{2} = 2.709511$), the frozen defect-class overlap table $(\pm 1/\sqrt{2}, +1/\sqrt{6})$ / $(0, -2/\sqrt{6})$ with the sheet-parity sign rule, disc radius $\theta_{\max} = \eta_{\rm rec}/(\eta_0 - \eta_{\rm rec}) \approx 1.16°$, and κ DERIVED from the v526 KMS normalisation up to a declared convention band — $u_{\rm rec} \leq 8.3 \times 10^{-5}$ ticks across the whole band, relic contrast $\leq 3.3 \times 10^{-4}$: the template degenerates ROBUSTLY to the sharp causal top-hat disc; three freeze hashes pinned and recomputed in-run. NEW LEDGER ROWS: `CCC.SEAM.KINEMATICS.01` [E] (the certified dictionary) and `CCC.SEAM.CROSSOVER.01` [O] (the dynamical-crossover research contract: D1 interacting algebra on the crossover = WOIT γ in the crossover geometry, D2 field-level Stinespring, D3 the cosmology fork decision — Starobinsky $r = 12/N_\star^2 \in [0.0033, 0.0048]$ priced decidable by $\sigma(r) \sim 10^{-3}$ experiments —, D4 the executed disc search; kill criteria frozen incl. the forbidden mark/Hawking-point count identification); new contract section in `tfpt_research_contracts`; the origin-theory Penrose-CCC bullet SHARPENED (asymptotic de Sitter does NOT imply conformal flatness — $C^2 = 48M^2/r^6$ for Schwarzschild–de Sitter, machine-checked; the CCC smoothness/Weyl conditions are additional crossover data) and pointed at the certified kinematics; PREREGISTERED SEARCH scaffolded at `experiments/ccc-crossover-disc/` (frozen hypotheses `ccc_disc_v1.yaml` with kill criteria K1–K6, null battery and BH-$q < 0.01$ + replication decision rule; freeze-check CLI 3/3 hashes intact; 5/5 guard tests; catalog entry §7b.9 of the experiments README) — verdict slot `data_limited (preregistered)`; the Planck PR3 SMICA data pass executed separately at experiments level (first data contact logged in the project README). FENCES CARRIED EVERYWHERE: the cyclic/CCC reading stays [C] and is NOT upgraded by the promotion; `WOIT.OS.TWISTOR.01` stays [O] with all kill tests live (the module certifies kinematics + free level + the necessity direction ONLY); the deck fork is decided only CONDITIONALLY on the v522 wrap identification; a null disc search damages nothing in the compiler core. Research notes DL–DLV appended to `experiments/next.txt`; FULL `run_all.py` ALL CHECKS PASSED at 950 modules; `bash build.sh release` green; `bash build.sh audit` → AUDIT OK; manifests regenerated LAST. No omega closed beyond the two new rows; NO claim that CCC follows from TFPT)

2026-08-20 (CXI) (the externalization contracts — the two open non-RH kind-(A) external-math targets `SEAM.EQUIV.01` and `ALPHA.QUILLEN.EXACT.01` packaged as short, self-contained EXTERNALIZATION CONTRACTS for outside specialist mathematicians (conformal nets/VOA/AQFT for the seam; determinant lines/η-invariants/index theory for alpha), added as two new sections at the end of `tfpt_research_contracts` (before "Certifiability and recommended order"), plus a one-sentence pointer in the head keybox; DOCUMENTATION ONLY — no module touched, no number changed, no status marker moved, no gate closed or narrowed, NO RH claim: (1) the `SEAM.EQUIV.01` contract states the minimal theorem (the scaling limit of the gapped μ_4-equivariant CAR collar exists in local-net topology; the index-4 simple-current extension functor commutes with the limit; the OS reconstruction is the holomorphic $(E_8)_1$ lattice net) broken

into the four review-named lemmata L1 (uniform energy/nuclearity bounds — honestly typed: NO uniform nuclearity estimate exists in the corpus), L2 (convergence of local algebras, the cited Osborne–Stottmeister scaling-limit theorem with the μ_4-equivariant structure retained), L3 (crossed product commutes with the limit; the finite index-4 Q-system is explicit, `v125/v154/v469`) and L4 (uniqueness of the holomorphic $c = 8$ reconstruction; $\det K = 1$ vs the same-c rival $SO(16)_1$), with the complete typed hypothesis list (12 PROVEN-FINITE rows citing `v302/v113/v155/v160/v356/v366/v367/v368/v461/v376/v377/v378/v379/v125/v154/v216/v286`, 5 CITED-CLASSICAL rows, 1 OPEN target), the runnable finite models, the HONEST Lean axiom surface (`SeamScalingLimit.lean` kernel-proves only the arithmetic hypotheses by `decide`; the analytic content is named cited axioms over opaque `Props`; no `sorry`; explicitly flagged: L1–L4 have ZERO Lean content beyond the composition audit), five must-fail controls (MF-S1: $c = 8$ alone must NOT suffice — the red-team point; MF-S2: index-2 glue must yield $SO(16)_1$; MF-S3: the trivial $M = 3$ collar must give no chiral limit; MF-S4: non-order-4 clock breaks the marks; MF-S5: the `v286` import firewall) and four acceptance tests; (2) the `ALPHA.QUILLEN.EXACT.01` contract states the two-step theorem T1 (gap-stable continuum limit $c_1(\det D_{\mathrm{finite}}) \to c_1(\det_\zeta D_{\mathrm{seam}})$ under norm-resolvent convergence with the Fermi gap uniformly open over the twist torus — the Dai–Freed section has no zero; finite side computed, `v472`) and T2 ($\delta_\tau(\log\det_\zeta \Delta_{U(1)} + 8b_1c_3^6 \log\varphi_{\mathrm{seam}}) = 0 =$ Bismut–Freed curvature = inflow response with the exact corpus coefficients $k_0 = |C| = 1$, $k_Y = 5/3$, $b_1 = 41/10$, $c_3 = 1/(8\pi)$, exponent $-5/4 = -q(D_5)$), the two named open faces (the analytic diagonal-ζ/multiplicity face `v484/v485` and the topological `EM1` lemma `v48/v435`), the typed hypothesis list (10 PROVEN-FINITE rows citing `v3/v48/v341/v382/v434/v470/v435/v342/v367/v472/v484/v485/v391/v433`, 5 CITED-CLASSICAL rows, 1 OPEN target), the runnable finite models, the HONEST Lean statement (there is NO Lean formalisation of this target — stated plainly), four must-fail controls (MF-A1: no experimental α anywhere in the construction; MF-A2: deck/carrier/Chern-level ablations must destroy the CODATA match; MF-A3: $M = 3 \Rightarrow C = 0$, $M = -1 \Rightarrow C = -1$; MF-A4: the $\{0, 3, 6\}$ π-power/metric-independence partition) and the review's three acceptance tests verbatim (no experimental α input; unique physical fixed point 137.0359992168; ablations destroy the hit) plus one structural test; (3) the website paper mirror of `tfpt_research_contracts` (`papers.ts`) gains one section entry covering both contracts; `tfpt_research_contracts.pdf` and `changelog.pdf` recompiled and mirrored via `bash build.sh website`; maps and `changelog.ts` regenerated via `bash build.sh gen`; `bash build.sh audit` → AUDIT OK; manifests regenerated LAST. Both targets stay [O]; NO RH claim anywhere)

2026-08-04 (LVIII) (the news round: a pure data-refresh against the August-2026 literature — no new analysis of our own, honest in both directions. CONFIRMING: JUNO's Neutrino-2026 update (207 days) lands at $\sin^2\theta_{12} = 0.3036 \pm 0.0064$ — the sharpest single measurement of the solar angle, $+0.49\sigma$ from the frozen prediction 0.306747, central value moving ***toward*** it from the 59.1-day run; ICARUS's first standalone result (no ν_μ disappearance, new $3{+}1$ exclusions at 90% CL) adds another sterile null exactly as $N_{\mathrm{fam}} = 3$ requires; the sequentially-valid DESI reanalysis localises the evolving-dark-energy signal almost entirely in the LRG2 bin (without it the evidence collapses, mildly pro-ΛCDM — the dissolution trend the $w = -1$ watchdog predicts); two new independent birefringence analyses ($\beta = 0.30^\circ \pm 0.05^\circ$ constant-model-preferred; PR4 field-level $0.32^\circ \pm 0.12^\circ$)

stay consistent with the $0.2424°$ seed; CMB-SPA $H_0 = 67.19 \pm 0.38$ lands $+0.11\sigma$ from the parameter-free budget 67.15 (correlated legs, never a new independent hit). WEAKENING, equally prominent: CMB-SPA (SPT-3G D1+ACT+Planck) gives $n_s = 0.9679 \pm 0.0033$ — the central value sits *above* the TFPT band edge 0.967 for the first time (edge at -0.27σ, preferred branch 0.9611 at -2.06σ; not yet the dated kill "robustly ≥ 0.967", typed an honest mild weakening, scorecard row tension + watch-flag); the DESI DR2 effective-mass posterior keeps the $\Sigma m_\nu = 0.0588\,\text{eV}$ floor under $+3.1\sigma$ model-dependent pressure (unchanged, on record). ONE BRIDGE DIES: the muon $g-2$ seam-vertex bridge's dated kill condition (2026-07-02) has FIRED — WP25 (Phys. Rep. 2025) adopts the lattice HVP as the SM baseline and the BMW hybrid evaluation (Nature 2026) confirms it, experiment−SM $= 0.5\sigma$, so the converged Δa_μ lies outside $2.879 \times 10^{-9} \pm 0.5 \times 10^{-9}$: the vertex identification is excluded in its present form, the scorecard row is typed kill_channel, ledger `FR.MUONG2.01` carries the dated kill note, and `tfpt_4_frontier` gets the dated update paragraph + summary-table move [C]→[X] on the bridge — the exact compiler number $\delta_2/(2\pi)$ and the compiler core are untouched (no core marker moves): the watchdog delivered exactly what it was built for. OPERATIONAL: the Vela GO-programme preregistration is FROZEN as `experiments/p9-forecast/hypotheses/p9_go_programmes_v1.yaml` ahead of the expected 2026/27 giant-glitch window (no glitch since 2024-04-29; long-component decision rule and the 8-TOAs/day cadence requirement inside; draft SHA recorded as provenance); CHIME/FRB Catalog-2 + RepCat4 data-availability notes recorded in `repeater-cascade` and `frb-ontology`. Scorecard $120 \to 121$ typed rows (consistent $49 \to 47$, null 27, parked 4, kill_channel 1); `predictions.ts` mirrors the JUNO/n_s/$g-2$ status; no verification module touched, no core marker moves)

- **Scorecard (regenerated).** θ_{12} row: JUNO 207-day value 0.3036 ± 0.0064 ($+0.49\sigma$; supersedes the 59.1-day 0.3092). n_s row: CMB-SPA 0.9679 ± 0.0033 added, status CONSISTENT→TENSION + watch-flag (central value above the 0.967 band edge). Flat-budget row: CMB-SPA H_0 leg $+0.11\sigma$ (correlated). Sterile watchdog: ICARUS null leg. w watchdog: LRG2-localisation note. Birefringence row: two new supporting legs. $g-2$: WP2025-lattice row TENSION→KILL_CHANNEL (kill condition fired), WP2020-dispersive row PARKED (baseline withdrawn by WP25).
- **Ledger.** `FR.MUONG2.01`: dated kill note "(6) KILL FIRED 2026-08-04" — bridge dissolved, exact value stays **[E]**, core untouched. No other row touched.
- **Papers.** `tfpt_4_frontier`: dated update paragraph in the muon section + summary-table bridge move **[C]→[X]** (core untouched); no other paper touched.
- **Experiments.** Vela/Auger GO-prereg frozen (`p9_go_programmes_v1.yaml`); CHIME Cat2/RepCat4 data notes; news-check note `experiments/news_check_2026-08-04.md` on record.
- **Website.** `predictions.ts`: JUNO 207-day confrontation, n_s watch-flag (+ CMB-SPA numbers), the dated news-round entry (verdict MIXED), audit counters 121 rows; `papers.ts`: muon section mirror updated (bridge dissolved).

2026-08-01 (XXXVI) (the covered seam lands (`v623` promoted) — the lattice skeleton of the covering-level identification: the seam lifted to the μ_3-cover is ONE 48-site NS circle for the uniform weights (the alternating pair DISCONNECTS it — a new independent kill), the lifted clock has EXACT ORDER 12 with $L^4 =$ deck ($r^4 = \omega$ as lattice combinatorics), the v622 kernel identity generalizes verbatim to $N = 48$, and the deck-character mode grid is the 12-grid ($\{1/6, 1/2, 5/6\}$) with

the untwisted sector equal to the base seam VERBATIM and the twisted pair carrying ω on bilinears (the $t = \omega/\bar{\omega}$ sheets); the fermionic deck lift satisfies $\mathrm{deck}^3 = -1$ **(the spin double))**

- `v623_covered_seam.py` **(`QGEO.COVERSEAM.01`,** 9 **checks; probe 9/9, COVERED-SEAM-SKELETON).** Connectivity census ($j = 1, 2$: one circle; trivial and alternating: 3 components); the explicit 48-walk (three base turns, 12 crossings); $L^4 =$ deck and $L^{12} =$ id on all 48 points with proper order 12; the $N = 48$ mode-sum identity; three 16-mode deck classes with the offset-1/2 class = the base NS frequencies exactly; $\mathrm{deck}^3 = -1$ on modes, bilinear eigenvalue $= \omega$; the 12 lifted marks in a single 12-cycle. Residue named: the interacting/CFT orbifold statement (twist-field OPE).
- **Sync.** Ledger row + note on `QGEO.SEAMID.01`; `tfpt_3` (audit block); diary; suite 616 → 617 scripts; website/README/zenodo counters; mirrors regenerated.

2026-08-01 (XXXV) (the seam identification lands (`v622` promoted) — THE PHYSICAL SEAM IS THE CONFORMAL SEAM: the v519 chiral seam kernel is EXACTLY the discrete NS (antiperiodic) chiral mode sum on the 16-site circle, including the even-distance zeros (the flat band IS the NS spectrum; the Ramond sum fails as must-fail control); the four mark-bond midpoints sit EXACTLY at μ_4 under the site map $\theta_a = (2a+1)\pi/16$ (no site touches a mark — the v519 "marks at bond midpoints" becomes literal geometry); the group dictionary is exact (clock $= z \mapsto iz$, RP reflections = diameter reflections); and the v534 STRADDLE LAW becomes geometry — straddled cuts = exactly the through-mark axes; the second v617 bedrock residue closes at the kernel + dictionary level, leaving the covering-level identification)

- `v622_seam_identification.py` **(`QGEO.SEAMID.01`,** 10 **checks; probe 10/10, SEAM-IDENTIFIED).** Mode-sum identity for all $d = 1..15$ with the closed form $\sum_{j<8} \sin((2j+1)x) = \sin^2(8x)/\sin(x)$; Ramond must-fail; mark midpoints $= \mu_4$ exact; reflection axes $(k+1)\pi/16$ with cuts $\mapsto \{\pi/4, \pi/2, 3\pi/4, \pi\}$; through-mark axes = the straddled cuts $\{7, 15\}$; v617's bond reflection = cut $k = 11$ with the v599 $(k, 5-k)$ pattern.
- **Sync.** Ledger row + notes on `QGEO.SEAMFORCE.01` and `GATE.QGEO.01` (bedrock state after v617/v620/v622; no status move); `tfpt_3` (audit block); diary; suite 615 → 616 scripts; website/README/zenodo counters; mirrors regenerated.

2026-08-01 (XXXII) (the interacting gamma slice lands at toy level (`v615` promoted): the gamma battery executed ON the RP-surviving interaction class (the alignment bit $\delta = \pi/2$, positive coupling, v534) — THE INTERACTING MIRROR FLIP: at the forced twist $\eta = +i$ the mirror state's odd sector is STRICTLY NEGATIVE DEFINITE on every cut and every coupling $g \in \{1/32..8\}$ (bounded away from zero: kill test (3) cannot fire on the surviving class — v608 upgraded from $g = 0$ to the full grid); the mark transport exists interacting (kill test (7) does not fire); the odd transfer spectrum stays multiplicity-free; and the dead member has no RP home — the dynamical bit selection and the chirality data CO-LOCATE on the same member)

- `v615_gamma_toy_interacting.py` **(`WOIT.GAMMA.TOY.01`,** 11 **checks; probe 11/11, GAMMA-TOY-LANDED).** Chiral/mirror parents ($I \pm iC$, gap 1); v534 survivor anchor reproduced; sector-exact mirror flip (even $(29, 0, 0)$ PD, odd $(0, 8, 0)$ ND, max odd eigenvalue $\leq -1.5 \cdot 10^{-3}$); $\eta = -i$ convention control; mark transport α_4 exact 16/16 with Gram and transport form Hermitian PD at every g; odd transfer spectrum simple at every g (honest finding: strong-coupling near-doubling at $g = 8$ without closing; $g = 0$ top 1.641438

= numerically the v524 clock top); dead-member control (m=2, s=+) indefinite. Kill tests (3)/(6)/(7) stay formally live on $\mathcal{A}_{\mathrm{hol}}$; kill test (6) untouched.

- **Sync.** Ledger row + note on `WOIT.GAMMA.FREE.01`; `tfpt_research_contracts` (γ-stage note); `tfpt_3` (audit block); diary; suite 608 → 609 scripts; website/README/zenodo counters; mirrors regenerated.

2026-08-01 (XXXI) (the Gauss–Manin transport round lands (`v614` promoted) — THE DECK STEP IS THE BOUNDARY MONODROMY: transporting the segment cycles along the rigid quarter-rotation loop, the discriminant $\lambda(\tau) = e^{2\pi i\tau}$ winds ONCE around 0 and the transported periods factorize EXACTLY, $I_m(\tau) = e^{i\pi\tau(m+1)/2}e^{-2\pi i\tau j/3}I_m(0)$; at $\tau = 1$ the deck factor is $\omega = t^4$ on EVERY sheet (the exponent $4 = |\mu_4|$ forced by the puncture count); the canonical transport matrix in the segment basis is EXACTLY ωM — the v613 dictionary $r \sim \omega M$ is CANONICAL, no conjugator freedom left, and $(\omega M)^4 = \omega$ re-derives $r^4 = \omega$; v613's residue F1 closed at the transport level)

- **`v614_transport_deck.py` (`QGEO.TRANSPORT.01`, 12 checks; probe 12/12, TRANSPORT-CANONICAL).** Exact rotation substitution identity; uniform deck ω across all three omega-sheet rows (must-fail: $t^3 \neq \omega$, $t^5 \neq \omega$); char(transport) = char(r) exact; 25-digit branch-tracked certificates at $\tau = 1/3, 1/2, 2/3, 1$ (dps 80, max $5{\cdot}10^{-28}$; $\tau = 1$ at $2.7{\cdot}10^{-81}$). The reading: the braid fixes the boundary, the rigid rotation does not — their homological difference is one boundary loop of local-system monodromy.

- **Sync.** Ledger row + note on `QGEO.PERNORM.01`; `tfpt_3` (audit block); diary; suite 607 → 608 scripts; website/README/zenodo counters; mirrors regenerated.

2026-08-01 (XXX) (the canonical period normalization lands (`v613` promoted) — THE ANALYTIC BRIDGE CLOSES: the quarter rotation acts DECK-FREE on the four branch segments (all 12 wrap factors exactly 1), and the exact dictionary is r = deck ∘ rotation — char(r) = char(ωM) EXACTLY, explaining $r^4 = \omega$ structurally, with the deck power unique; the period rows (hol dx/y^2, hol $x\,dx/y^2$, antihol $\overline{dx/y}$) realize the rotation eigenframe; the Hodge weights are explicit Gamma monomials ($h_1 = (3\sqrt{3}/16\pi)\Gamma(1/3)^2\Gamma(1/6)^2$, complex Beta formula); and $J = -h$ PER CHARACTER: the compiler polarization is MINUS the analytic Hodge form — Riemann–Hodge positivity realized, v612's named step B5 executed)

- **`v613_canonical_periods.py` (`QGEO.PERNORM.01`, 19 checks; probe 19/19, PERIOD-NORMALIZED).** Deck-free rotation on segments (40-digit certificates) with the must-fail control (raw period rows are NOT r-eigenvectors, residual 0.42 — the naive ansatz falsified before the honest construction); char(r) = char(ωM) exact with explicit conjugator G; character match antihol ↔ ζ_{12}; omega-sheet signature (2, 1) with the unique negative on the antiholomorphic line = J's unique positive line (v612): all three character-matched sign products negative. Residue named: the canonical fork-basis identification (basepoint bookkeeping for G).

- **Sync.** Ledger row + notes on `QGEO.POLFRAME.01` and `QGEO.PERIODS.01`; `tfpt_3` (audit block); diary; suite 606 → 607 scripts; website/README/zenodo counters; mirrors regenerated.

2026-08-01 (XXIX) (the polarization frame lands (`v612` promoted): the braid rotation has three simple eigenvalues $\zeta_{12}{\cdot}\{1, -1, -i\}$, the polarization J diagonalizes EXACTLY in the rotation eigenbasis (values $(1+\sqrt{3})/2$, $(1-\sqrt{3})/2$, -1), and THE UNIQUE POSITIVE (HODGE) DIRECTION IS THE PRIMITIVE ζ_{12} EIGENSPACE — the analytic Hodge structure and the finite rotation agree on

where positivity lives; the canonical period normalization (Beta weights) is the named next step)

- `v612_polarization_frame.py` (`QGEO.POLFRAME.01`, 9 **checks; probe 9/9, POLARIZATION-FRAMED).** Simple spectrum, $\det r = -1$; exact diagonalization (honest typing: diagonality follows from J-unitarity, the content is the values and the signature distribution); product of diagonal values $= \det J / |\det W|^2 = 1/2$ exactly.
- **Sync.** Ledger row + note on `QGEO.PERIODS.01`; `tfpt_3` (audit block); diary; suite 605 → 606 scripts; website/README/zenodo counters; mirrors regenerated.

2026-08-01 (XXVIII) (the periods round lands (`v611` promoted): the analytic layer of the cover is CLASSICAL — all five basic period integrals are exact Beta values with Gamma denominators on the ζ_{12} grid ($\Gamma(11/12), \Gamma(7/12), \Gamma(5/12), \Gamma(1/12)$); the μ_4 rotation acts on periods with the exact characters i^{k+1}; $\Omega_E = B(1/3, 1/6)/2$ exactly; and the cross-factor period ratio admits no low-degree integer relation (PSLQ null) — the two CM factors are period-independent, as the v610 splitting demands)

- `v611_periods.py` (`QGEO.PERIODS.01`, 6 **checks; probe 6/6, PERIODS-CLASSICAL).** Five symbolic Beta identities; three 40-digit rotation-character certificates; the classical ω-CM period; the PSLQ-null cross-factor consistency; the period dictionary named as the Hodge/RP resource.
- **Sync.** Ledger row + note on `QGEO.LANDSCAPE.01`; `tfpt_3` (audit block); diary; suite 604 → 605 scripts; website/README/zenodo counters; mirrors regenerated.

2026-08-01 (XXVII) (the curve landscape lands (`v610` promoted): the Jacobian SPLITS into a $\mathbb{Q}(\zeta_{12})$-CM abelian surface — the field of deck AND marks together — times the $\mathbb{Q}(\omega)$-CM elliptic quotient $E : y^3 = u^2 - 1$, point-count-confirmed ($a_p(C) = a_p(E)$ exactly for all ten tested $p \not\equiv 1 \bmod 12$); the integer-spectrum census honestly relativizes single-char hits (7894 integer entries — single spectra are cheap; the anchor class is the most frequent nontrivial family at 286; the compiler's distinction is the joint structure); two new canonical period resources named)

- `v610_curve_landscape.py` (`QGEO.LANDSCAPE.01`, 9 **checks; probe 9/9, LANDSCAPE-MAPPED).** Differential characters $(12, 12, 6)$; splitting confirmed at 14 primes (ten exact zero-differences; surface traces $(-8, 4, -20, -32)$ at $p \equiv 1 \bmod 12$); $a_p = 0$ for $p \equiv 2 \bmod 3$; census: anchor class 286 = max nontrivial, $\{1, 1, 3\}$ at 254; v598/v599 char hits retroactively confirmed as correctly typed candidates.
- **Sync.** Ledger row + notes on `QGEO.COVER.01`/`QGEO.DICT.01`; `tfpt_3` (audit block); diary; suite 603 → 604 scripts; website/README/zenodo counters; mirrors regenerated.

2026-08-01 (XXVI) (the translator factorization lands (`v609` promoted): the geometric identity of c is CLOSED — the mirror intertwiner d ($dA = \Gamma(A)d$) is unique, unimodular ($\det d = 1$), $\mathbb{Z}[\omega]$-integral and Γ-involutive ($\Gamma(d)d = 1$ exactly), and the v605 element factorizes exactly as $c = \lambda K_1 d$ with $K_1 = J^{-1}\overline{F_2}^{T} M^{-1}$: $\theta_{\text{corpus}} = J\text{adj} \circ \text{inner}(K_1) \circ \text{inner}(d)$ — all three v605 residuals are now resolved (seam reversal v603, zero-mode rule v607, c-identity v609))

- `v609_translator_factorization.py` (`QGEO.CFACTOR.01`, 8 **checks; probe 8/8, C-FACTORIZATION-LANDED).** Intertwiner space 1-dim; $\det d = 1$; $d \in M_3(\mathbb{Z}[\omega])$; $\Gamma(d)d = 1$; $c = \lambda K_1 d$ exact ($\lambda = 1/7 - 3\sqrt{3}i/7$); inner$(K_1 d)$ reproduces φ on the pair; honest

scope: F_2 carries the corpus datum by definition; the v606 arithmetic negative explains why the cyclotomic route had to fail.

- **Sync.** Ledger row + notes on `QGEO.TRANSLATOR.01`/`QGEO.MODESEP.01`; `tfpt_3` (audit block); diary; suite 602 → 603 scripts; website/README/zenodo counters; mirrors regenerated.

2026-08-01 (XXV) (the free gamma slice lands (`v608` promoted) — the first executed step of the WOIT γ milestone: mirror modes are STRICTLY OS-negative sector-exactly (every mirror fermion vector has negative OS norm — no mirror survives reconstruction, kill test (3) cannot fire at the free level), the chiral generation is multiplicity-free (clock spectrum exactly $\{1, \sqrt{2}-1\}$), and the mark transport exists (τ_2 is the mark-A→B transport, PD — kill test (7) does not fire free); γ itself stays open on $\mathcal{A}_{\mathrm{hol}}$; side finding: $14c$ is unipotent mod $(1-\omega)$)

- **`v608_gamma_free_slice.py` (`WOIT.GAMMA.FREE.01`, 9 checks; probe 9/9, GAMMA-FREE-SLICE-LANDED; the v524 machinery imported read-only).** Chiral $(16,0,0)$; anti-chiral $(8,8,0)$ with even $(8,0,0)$ PD and odd $(0,8,0)$ strictly ND; clock spectrum $\{1, \sqrt{2}-1\}$ multiplicity-free on the fermion domain; mark algebras PD $(4,0,0)$ with $\dim 4 = |\mu_4|$, products generate the full 16-dim basis, α_2 transports mark A to mark B.
- **Sync.** Ledger row + notes on `WOIT.OS.TWISTOR.01`/`WOIT.BETA2.OS.01`; `tfpt_research_contracts` (γ-stage free-slice note); `tfpt_3` (audit block); diary (incl. the c-unipotency side finding); suite 601 → 602 scripts; website/README/zenodo counters; mirrors regenerated.

2026-08-01 (XXIV) (the selection rule lands (`v607` promoted): the infinity torsion is a genuine $\mathbb{F}_3$ space, the deck rotation acts with char $(x+1)(x^2+1)$ mod 3 — EXACTLY ONE projective eigenline, and it IS the top mode; the three modes carry mutually exclusive canonical labels (seam = R-real, top = deck-eigen, zero = the only unlabeled) and S_0 flips exactly the labeled pair: BOTH v590 demands expressed through cover structure (typed reading; the gate residual is the continuum derivation))

- **`v607_selection_rule.py` (`QGEO.SELRULE.01`, 10 checks; probe 10/10, SELECTION-RULE-LANDED).** ω-multiplication acts as the identity on the torsion; the induced r-action is $[[0,1,0],[1,0,2],[2,2,2]]$ mod 3 with only fixed class 0; x^2+1 irreducible over $\mathbb{F}_3$ gives structural uniqueness of the eigenline ($\lambda=-1$), census-confirmed; the top-mode class is that eigenline, the zero mode and the seam are not projectively fixed (nor under r^2).
- **Sync.** Ledger row + notes on `QGEO.MODESEP.01`/`QGEO.INVOL.01`; `tfpt_3` (audit block); diary; suite 600 → 601 scripts; website/README/zenodo counters; mirrors regenerated.

2026-08-01 (XXI) (the translator round lands (`v605` promoted): the corpus RP anti-automorphism IS the J-adjoint twisted by an explicit inner element c ($\varphi = J\mathrm{adj} \circ \theta_{\mathrm{pull}}$ is inner, c unique up to scale, $14c$ integral over $\mathbb{Z}[\omega]$ — the single-sheet negatives resolved, the $\bar{\omega}$ detour unnecessary); the seam is a NONZERO pure 3-torsion class at infinity (torsion coordinates $(2,2,0)$); the 81 survives frame-joining as the saturation defect (chain indices 3^5, 3^4); the J_{fix} duality certified with a Lean addendum)

- **`v605_translator.py` (`QGEO.TRANSLATOR.01`, 11 checks; probe 11/11, TRANSLATOR-LANDED).** $\varphi(A) = cAc^{-1}$ exact (1-dim intertwiner space, c invertible); $14c$ $\mathbb{Z}[\omega]$-integral (denominator $14 = 2 \cdot 7$); seam lift zero in the free part, $(2,2,0)$ in the torsion of $\mathrm{coker}(1-T_\infty)$; order chain $O_1 \subset O_{\mathrm{joint}} \cap \mathrm{span}_1 \subset P_1$ with indices 243 and 81; $C_V^T J_{\mathrm{fix}} = J_{\mathrm{fix}} C_{V\,\mathrm{dual}}$ integer; c's independent geometric identity typed open.

- **Lean.** `CoverEmbedding.lean` addendum: `Jfix_symmetric`, `Jfix_det` = 72, `CV_dual_certificate`, `CVdual_cubic`, `dual_joint_trace`; lake build green (3375 jobs).
- **Sync.** Ledger row + notes on `QGEO.EQORDER.01`/`QGEO.SEAMMARKS.01`/`QGEO.DUALFORMS.01`; `tfpt_3` (audit block); diary; suite 598 → 599 scripts; website/README/zenodo counters; mirrors regenerated.

2026-08-01 (XX) (the equivariant-order round lands (`v604` promoted): R IS an anti-isometry of the polarization ($M^\dagger J M = J^T$ exact — correcting a diary side remark), J restricts to the integer form J_{fix} with $\det = 72 = 8 \cdot 9$; the four mark frames generate a FULL order of index $27 = 3^3$ in $M_3(\mathbb{Z}[\omega])$; and the 3-torsion localizes at infinity ($\text{coker}(1 - T_\infty) = \mathbb{Z}^2 \oplus (\mathbb{Z}/3)^3$, finite cusps torsion-free): 81 at the marks, 27 at infinity — the integral infinity/order-torsion bridge; translator negatives extended (the $\bar{\omega}$-sheet named))

- **`v604_equivariant_order.py` (`QGEO.EQORDER.01`, 12 checks; probe 12/12, EQUIVARIANT-ORDER-LANDED).** Anti-isometry exact; $J_{\text{fix}} = [[16, 2, 4], [2, -2, 2], [4, 2, -2]]$; all 28 rotated words $\mathbb{Z}[\omega]$-integral, joint lattice full rank 18, Smith divisors $(1^{15}, 3, 3, 3)$; $\text{coker}(1 - T_\infty)$ divisors $(1, 1, 1, 3, 3, 3, 0, 0)$; each finite cusp twist torsion-free; three objects share $(\mathbb{Z}/3)^3 = 27$ (infinity monodromy, equivariant order, $\det(1 - \text{deck})$); F_2 vs J_{fix}-menu and algebra-twist negatives quantified.
- **Sync.** Ledger row + notes on `QGEO.SEAMMARKS.01`/`QGEO.COVER.01`; `tfpt_3` (audit block); diary (incl. the exploration round XX: ∞-embedding, R-orbit classes, translator hunt); suite 597 → 598 scripts; website/README/zenodo counters; mirrors regenerated.

2026-08-01 (XIX) (the seam-and-marks round lands (`v603` promoted): the seam line is a REAL line of the cover (unique common eigenline, R-fixed) and S_0 reverses it — the v590 seam-reversal demand holds geometrically; the infinity cusp separates in the unreduced Burau module; the first per-mark $\mathbb{Z}/3$ witness lands (E_1, class order 3 in the word-order cokernel); the corpus-transpose translator negatives are quantified)

- **`v603_seam_marks.py` (`QGEO.SEAMMARKS.01`, 15 checks; probe 15/15, SEAM-REAL-LANDED).** Seam: $Uv = 0$, $Vv = v$ unique, $v = (0, (\sqrt{3} - i)/(\sqrt{3} + i), 1)$, $Rv \sim v$, $S_0 v = -v$; infinity: unreduced Burau braid relations, invariant column $(1, 1, 1, 1)$, full twist eigenvalues $\{1, \omega^{\times 3}\}$; marks: E_1 integer restriction, in the word algebra with denominator lcm 3 (order-3 cokernel class); E_4 trivial, E_2/E_3 outside (naive global map fails, equivariant framing typed); translator: $F_2 = g^T \Sigma g$ matches no canonical menu form (all typed open).
- **Sync.** Ledger row + notes on `QGEO.DUALFORMS.01`/`QGEO.INVOL.01`; `tfpt_3` (audit block); diary; suite 596 → 597 scripts; website/README/zenodo counters; mirrors regenerated.

2026-08-01 (XVIII) (the duality-and-forms round lands (`v602` promoted): the J-adjoint exchanges the two parabolic types exactly ($(U_1, V_1^J) \sim$ compiler pair); the dictionary completes in the J-projector language (the corpus anchor formula $H = 1 + V(V-1)/2$ holds verbatim on the curve); Weil positivity holds exactly (the Riemann bilinear relations on the ω-sheet); the canonical bilinear form $B = M^\dagger J$ is symmetric with $\det B = 8$; the v600 integer core is Lean-certified (`FORM.QGEO.04`); the contracts paper gains the μ_3-cover model paragraph)

- **`v602_duality_forms.py` (`QGEO.DUALFORMS.01`, 17 checks; probe 17/17, DUALITY-FORMS-LANDED).** U_1 J-self-adjoint, V_1^J the exact product reversal, (U_1, V_1^J) conjugate to the compiler pair; H, Q, Q_+, ladder all exact in the J-projector language; Weil-twisted form

positive definite ($\{2\sqrt{2}-2, 2, 2+2\sqrt{2}\}$); honest negatives: $g^T g \neq \lambda B$, naive mark-local map fails, seam normal not a deck eigenvector (all typed open).

- **Lean (`FORM.QGEO.04`).** `TfptCarrier/CoverEmbedding.lean`: the v600 integer core via determinant-free Smith certificates ($W = TS$, $PT = \mathrm{diag}(1,1,1,3,3,3,3)$ with unimodular P, S-primitivity via unimodular minor); kernel `decide`; lake build green (3375 jobs).
- **Sync.** Ledger row + notes on `QGEO.CANON.01`/`QGEO.EMBED.01`; `tfpt_research_contracts` (μ_3-cover model paragraph after the bedrock chain), `tfpt_3` (audit block); diary; suite 595 → 596 scripts; website/README/zenodo counters; mirrors regenerated.

2026-08-01 (XVII) (the canonicity round lands (`v601` promoted): the J-adapted cusp projectors give a manifestly D_4-equivariant presentation of the pair (mark choice = winding choice), which realizes the DUAL parabolic exactly (conjugate to the transposed compiler pair; the direct intertwiner is singular); the v590 Klein four-group lives integrally on the curve; the Weil-twisted RP comparison and the seam identification stay open)

- **`v601_equivariant_dual.py` (`QGEO.CANON.01`, 14 checks; probe 14/14, EQUIVARIANT-DUAL-LANDED).** $GJ_k = (1-T_k)J^{-1}(1-T_k)^\dagger J$ idempotent rank-1 J-projectors rotating exactly under D_4; pair family $U_k = 3\,GJ_k$, $V_k = 1 + GJ_{k-1} - GJ_k GJ_{k+1}$: $\mathbb{Z}[\omega]$-integral, compiler-exact, equivariant by construction; duality: conjugate to (U_c^T, V_c^T) exactly (must-fail to (U_c, V_c) documented); Klein group: all eight spectral-sign involutions integral, S_0 explicit, $AB = S_0$; honest: naive J-word-Gram inertia $(3,2,2) \neq (4,0,3)$ (Weil twist named open).
- **Sync.** Ledger row + note on `QGEO.EMBED.01`; `tfpt_1`/`tfpt_2` (canonicity clauses), `tfpt_3` (audit block); diary; suite 594 → 595 scripts; website/README/zenodo counters; mirrors regenerated.

2026-08-01 (XVI) (the joint embedding lands (`v600` promoted): the compiler pair (U, V) is REALIZED on the μ_3-cover — a Γ-even degree-2 pair, $\mathbb{Z}[\omega]$-integral, seven-word rank 7, simultaneous conjugation to the compiler pair solved exactly, integer restrictions on the saturated $\mathrm{Fix}(R)$ lattice, and saturation index $81 = 3^4$ with the exact v566 divisor pattern; canonicity typed open, the gate does not move)

- **`v600_joint_embedding.py` (`QGEO.EMBED.01`, 15 checks; probe 15/15, JOINT-EMBEDDING-LANDED).** The pair $U^* = -2E_1 + (G_1\Gamma(G_1) + \Gamma(G_1)G_1) + 3$, $V^* = -E_3 + \frac{1}{2}(G_2\Gamma(G_2) + \Gamma(G_2)G_2) + 1$: chars and joint traces compiler-exact ($\mathrm{tr}(UV) = 3$, $\mathrm{tr}(UV^2) = 6$, $\mathrm{char}(U{+}V) = \mathrm{char}\,Q$); seven words independent; explicit invertible g with $gU^*g^{-1} = U_c$, $gV^*g^{-1} = V_c$; the real structure integer on $\mathbb{Z}^6$ with explicit saturated fixed-lattice basis (SNF certificate); integer restrictions; Smith divisors $(1,1,1,3,3,3,3)$, index 81 — the v566 self-code order structure on the curve. Must-break control passes; canonicity open (existence claim only).
- **Sync.** Ledger row + notes on `QGEO.REAL.01`, `QGEO.DICT.01`, `QGEO.COVER.01` and `DIAMOND.PARA.SELFCODE.01`; `tfpt_1`/`tfpt_2` (realization sentences), `tfpt_3` (audit block); diary; suite 593 → 594 scripts; website/README/zenodo counters; mirrors regenerated.

2026-08-01 (XV) (the real structure lands (`v599` promoted): the anti-holomorphic D_4 reflection on the μ_3-cover exists and is unique up to scale with $R^2 = +1$; its grading functor closes all three dictionary gaps at char level (V, H with the anchor spectrum $(1,1,2)$, Q_+); the Γ-decomposition carries 5, 9, 27; the joint algebra embedding stays open (quantified negative); and the invariant form has

$\det J = 8$ — **a sheet-diamond determinant-list member — with Lorentz signature** $(1,2)$**)**

- **`v599_real_structure.py` (`QGEO.REAL.01`, 18 checks; probe 18/18, REAL-STRUCTURE-LANDED).** Existence and uniqueness of $R(v) = M\,\overline{v}$ (1-dimensional intertwiner space, $R^2 = +1$ exact, four twist intertwinings $RT_kR^{-1} = T_{5-k}^{-1}$); the grading functor $\Gamma(A) = M\overline{A}M^{-1}$ locates $V \sim G_2+G_3-\Gamma(G_3)$, $H \sim 1+G_2-\Gamma(G_3)$ (the anchor spectrum) and $Q_+ \sim 1+G_2+G_3-\Gamma(G_3)$ exactly at char level; the Γ-even/odd cusp elements carry $\{0,1,5\}$, $\{0,3,9\}$, $\pm\sqrt{5}$, $\pm\sqrt{27}$ and one nilpotent; the joint negative quantified (4×6 candidates, zero pairs pass $\mathrm{tr}(UV) = 3$ and $\mathrm{char}(U+V) = \mathrm{char}\,Q$; kill not triggered); the invariant hermitian form is unique with $\det J = 8$ and signature $(1,2)$ — Strategy II's Hodge prediction at form level, pinned.
- **Sync.** Ledger row + notes on `QGEO.DICT.01`/`QGEO.COVER.01`; `tfpt_3` (audit block); diary; suite 592 → 593 scripts; website/README/zenodo counters; mirrors regenerated.

2026-08-01 (XIII) (the cover dictionary reaches THREE exact entries (`v598` promoted): $U \sim G_2$, $Q_- \sim G_1 - G_3$ — the μ_4 sheet coupling is the diamond-diagonal Gram difference — and the binary AnchorLadder $(1,2,4) \sim 1 + G_1 + G_3$; the grading half is absent from 7296 canonical combinations, with the real structure as the named suspect)

- **`v598_cover_dictionary.py` (`QGEO.DICT.01`, 4 checks; probe 4/4, DICT-THREE-ENTRIES).** All three entries exact (numeric prefilter, symbolic confirmation); supporting structure: opposite-twist products nilpotent, $Q_-^2 \sim x(x-3)^2$; negatives quantified (7296 combinations, 0 matches for $Q_+/V/H/Q$); the Strategy-I kill criterion not triggered; GATE.QGEO does not move.
- **Sync.** Ledger row + note on `QGEO.COVER.01`; `tfpt_1`/`tfpt_2` (dictionary sentences updated), `tfpt_3` (audit block); diary; suite 591 → 592 scripts; website/README/zenodo counters; mirrors regenerated.

2026-08-01 (XI) (the dictionary's first entry: U IS LOCATED — the winding operator appears as a cusp Gram on the μ_3-cover; exploration round, documented as a note on `QGEO.COVER.01`)

- **The hit.** The cusp Gram $(1-T_2)(1-T_2)^\dagger$ has characteristic polynomial $x^2(x-3) = \mathrm{char}(U)$ exactly — the rank-one winding operator (trace $3 = N_{\mathrm{fam}}$) as a Gram of a D_4-conjugate cusp twist: the fifth fingerprint. Additional exact structure: opposite-twist products $(1-T_1)(1-T_3)$ are nilpotent (x^3), G_1+G_3 has spectrum $\{0,1,3\}$. Honest negative: $V/H/Q$ not located in the scanned canonical ranges (products/sums/ω-twists of length ≤ 2) — the V-dictionary likely requires the parabolic flag (the seam-cusp filtration); named next step; kill criterion unchanged.
- **Sync.** Ledger note on `QGEO.COVER.01`; diary; mirrors regenerated.

2026-08-01 (X) (Strategy I/II executed: the μ_3-cover $y^3 = x^4 - 1$ hits FOUR compiler fingerprints (`v597` promoted) — rank-3 Eisenstein H_1, projective D_4 via the deck, cusp-class group $(\mathbb{Z}/3)^4$ of order 81, and the unique invariant hermitian form of LORENTZ signature $(1,2)$ matching Paper II's polarization: the first geometric candidate that could close GATE.QGEO and the RP slot with one construction)

- **`v597_mu4_cover_model.py` (`QGEO.COVER.01`, 6 checks; probe 6/6, FINGERPRINTS-HIT).** Genus 3, five cusps ($\mu_4 + \infty$), Chevalley–Weil $H^{1,0} = (1,2)$; Burau at $t = \omega$ with verified braid relations, $r^4 = \omega \cdot 1$, $r^{12} = 1$; cusp classes $(\mathbb{Z}/3)^5$/one relation $= (\mathbb{Z}/3)^4$, order $81 = 3^4$ with the per-mark $\mathbb{Z}/3$ mechanism ($\det_{\mathbb{Z}}(1 - \mathrm{deck}) = 27$ complementarily); invariant form unique up

to scale, eigenvalues $\{-2, 2-2\sqrt{2}, 2+2\sqrt{2}\}$, signature $(1,2)$; single-cusp Gram spectrum $\{0,0,2\}$ all-integer. OPEN and named: the Q/V operator dictionary, the real-structure/RP map to the v572 Gram data; kill criteria live, not triggered; GATE.QGEO does not move.

- **Sync.** Ledger row + notes on `QGEO.INVOL.01` and `DIAMOND.PARA.SELFCODE.01`; `tfpt_1`/`tfpt_2` (candidate paragraphs), `tfpt_3` (audit block); diary; suite 590 → 591 scripts; website/README/zenodo counters; mirrors regenerated.

2026-08-01 (IV) (the sheet involution exists unconditionally (`v590` promoted): the integral V-commuting involutions are exactly $(\mathbb{Z}/2)^3$ (rigidity, both twins); the v574 type theorem + seam reversal + zero-mode fixing single out the explicit S_0 with exact Σ-signature uniquely — GATE.QGEO.01's involution-existence input relocates to DERIVED)

- **`v590_involution_existence.py` (`QGEO.INVOL.01`, 7 checks; probe 7/7, EXISTENCE-UNCONDITIONAL).** All eight spectral sign choices on V's eigenspaces are integer, involutive and order-stabilizing on both twins; narrowing $6 \to 3 \to 2 \to 1$: the trace $+1$ half dies by the v574 rank theorem, $Se_3 = -e_3$ and $+1$ on $\ker V$ single out $S_0 = [[1,-1,0],[0,-1,0],[0,0,-1]]$; honest scope: the corpus RP reflection does not commute with V (different structure), the alt twin has the same classification (not a bit selector); the two demands' continuum derivation $(\mu_4/D_4/H^1)$ stays the named open.
- **Sync.** Ledger row + note on `QGEO.WINDING.SELF.01`; `tfpt_1` (Stage-B passage), `tfpt_2` (new paragraph), `tfpt_3` (audit block); diary; suite 583 → 584 scripts; website/README/zenodo counters; mirrors regenerated.

2026-07-31 (XVII) (the interacting RP route of the alignment bit, executed: EXT-SOURCE COUPLING SEPARATES THE 81-TWINS DIRECTIONALLY IN THE UNIT BOX (`v584` promoted) — exact rational flip thresholds with the v572 pencil factors as numerators; positive coupling $J \le 15/16 < 1$ heals exactly the compiler twin, the alt twin is rigid; response pole on the golden polynomial)

- **`v584_ext_source_response.py` (`DIAMOND.BIT.EXTSOURCE.01`, 8 checks; probe 8/8, UNIT-BOX-SEPARATED).** The interacting kernel $K(s,J) = K_0 + sK_\Pi + J\,vv^T$ (v from the shared Ext arrows E_{31}/E_{32}) has exact thresholds $J_c(s) = -1/(v^T K(s)^{-1} v)$: compiler $(15s^2-16s-16)/(16(s^2-s-1))$ and $(15s^2-16s-16)/16$, alt $(s^2+16s+16)/(16(s+1))$ and $-(s^2+16s+16)/16$ — numerators exactly the v572 pencil factors; verified numerically (16 flip events). In the crossed phase the compiler is healed by positive $J \le 15/16 < 1$ (either arrow); the alt twin cannot be flipped by any positive coupling in $|J| < 1$ (E31 threshold ≥ 1, equality exactly at $s = 0$; E32 negative). Negative coupling never heals (direction consistent with v534's toy RP selection). Exact anchors $15/16$, 1, -1; $J_c(s^*) = 0$ via Ext-visibility of the critical mode; response pole on s^2-s-1 ($-1/\varphi$, next to s^*) — observed. Honest: response-level, no PSD anywhere, the alignment-bit contract stays open.
- **Sync.** Ledger row + note on `DIAMOND.BIT.RPFORM.01`; `tfpt_1` (follow-up passage extended), `tfpt_2` (new interacting-route paragraph), `tfpt_3` (audit block); diary; suite 577 → 578 scripts; website/README/zenodo counters; mirrors regenerated.

2026-07-31 (XI) (the continuous half of the cross-ratio check executed and closed: UNIFLOW-DEGENERATE (`v575` promoted) — the native exports are forced degenerate by the single-flow theorem, so the continuous half closes onto `v425`; both halves of the preregistered check are now executed)

- **`v575_cr_continuous_uniflow.py` (`FTR.CRCONT.01`, 7 checks; probe 7/7, UNIFLOW-DEGENERATE).** The frozen protocol exports, recomputed from primitives (F_pole $= e^{-\Delta} =$

$(2/3)^6$ via the v99 generator; F_Boltzmann the SAME contraction by the v425 single-flow theorem; F_QCD $b_3 = -7$; F_relic adiabatic 1), form the quadruple $\{(2/3)^6, (2/3)^6, 1, -7\}$ with a forced repetition — and a repeated quadruple gives CR $\in \{0, 1, \infty\}$ in every corner assignment (theorem + all 24 permutations): the continuous half can never pass natively. It closes onto the single-flow reduction; the anchors (v_{geo}, M_R, C_p, θ_i) stay **[O]** walls; the meaningful cross-ratio content remains the discrete half (`v571`). Contract markers unchanged.

- **Sync.** New ledger row `FTR.CRCONT.01`; update notes on `FTR.CRDISC.01` and `CONTRACT.F.01`; contracts-paper protocol block extended with the execution result; suite 568 → 569 scripts; diary; README/zenodo/website counters; mirrors regenerated.

2026-07-31 (X) (closing the round: the winding form's content located exactly (`v574` → 11 checks — without it a four-column family survives, exactly one uniform: the compiler), and the half-sided RP restrictions shown blind (`v572` → 11 — the interacting construction is the one remaining RP route))

- **`v574` extended (probe 12/12).** Dropping the winding demand entirely (arbitrary first column), the self-code package leaves the four-column family $\{(0,1,1), (0,3,3), (3,3,3), (5,1,5)\}$ — exactly one uniform: the compiler. The winding demand *is* the uniformity choice (1 of 4). Final declared list of the reconstruction: involution existence, column uniformity, $N_{\text{fam}} = 3$ (E8-glue).
- **`v572` extended (probe 11/11).** The Gibbs kernels restricted to the Levi and radical sublattices have identical inertia on both twins across the sampled range: the RP-positivity selection resists every kinematic and half-sided design tried — the genuinely interacting construction (the v534 FK-quartet analogue) is the one remaining route, typed open.
- **Sync.** Ledger/registry/run_all counts; `tfpt_1`/`tfpt_2` passages; diary; mirrors regenerated.

2026-07-31 (VIII) (two decisive deepenings: the anchor-Hamiltonian Gibbs pencil delivers the first DYNAMICAL distinction of the reconstruction bit (`v572` → 9 checks: the compiler twin is Gibbs-critical at $s^* = (8 - 4\sqrt{19})/15$, the alternative never), and the redundancy ablation shrinks the reconstruction inputs to three (`v574` → 9 checks: (M) and (L) fall relative to the self-code demands))

- **`v572` extended (probe 9/9).** Along the Gibbs family $\omega_g \propto \mathrm{tr}(e^{-gH}\cdot)$ of the v566 anchor operator $H = I + \Pi$, the pencil determinant factors exactly differently on the twins — compiler $\frac{6561}{4096}(s^2{+}16s{+}16)^2(15s^2{-}16s{-}16)$, alternative $-\frac{6561}{4096}(s^2{+}16s{+}16)^3$ ($6561 = 81^2$) — and only the compiler's extra factor has a root inside the physical Gibbs range: the compiler twin's RP kernel degenerates at a finite critical temperature, the alternative twin's never does. Honest typing: a rank-drop marker, not a positivity selection; the interacting RP question stays open.
- **`v574` extended (probe 10/10).** Redundancy ablation: relative to the self-code demands (parabolic + ladder + saturation 81) the monodromy budget (M) and the seam-line demand (L) are redundant at $B = 6$ — dropping both widens the class to 1274 members and the self-code still picks exactly the compiler Q: the declared geometric input list shrinks to (S)-existence, (W)-form and (E); `GATE.QGEO.01` narrowed by four items, not closed.
- **Exploratory (diary only).** The pair-mechanism hunt on the reference window: no dominant pair exists (max |contribution| $\sim 0.006 \times \det S$ — consistent with the dead two-atom witness), and the fine structure by smaller atom shows the smallest primes $(2, 3)$ dragging negative while medium primes (11..41) carry positive net.
- **Sync.** Ledger/registry/run_all counts; `tfpt_1`/`tfpt_2` passages; diary; mirrors regenerated.

2026-07-31 (III) (the residual reconstruction bit located and given a canonical witness: the ladder self-consistency selects the compiler Q uniquely (SELECTOR-FOUND); the split geometry excluded; Priority 2 sharpened + `v568` promoted + ledger/paper/diary/website sync)

- **Probe `bit_selector_probe.py` (10/10, verdict SELECTOR-FOUND).** Priorities 2+3 of the self-code review, fused with the `v567` residual bit. *The bit is integrally visible:* the two 81-twins ($q_{32} = 2 = |\mathbb{Z}_2|$ vs 0) generate *different* integral orders (equal index, equal Smith divisors, different lattices). *The selector:* the dominant mode of the sheet direction is $v(c) = (1, 2, 2c)$ identically, and the *ladder self-consistency* — the dominant mode reproduces its own eigenvalue as the geometric ladder $(2^0, 2^1, 2^2) = (1, 2, 4)$ — selects $c = 2$, the compiler Q, uniquely on all 28 members of the carrier-free class (no 81-pin, no column budget); the witness is prior structure (binary AnchorLadder steps 2^n, $2 = |\mathbb{Z}_2|$, the `v566` trace code). *The split geometry located and excluded:* a module complement exists iff $q_{32} = 1$, and that member's algebra collapses to dimension 5 — the A_2 Ext arrow is nonzero on the whole reconstruction class; neither twin splits: the bit toggles *which* nonsplit integral order is realised, not split-vs-nonsplit. *Priority-2 candidates, honest both ways:* the μ_4-grading fails (the order is not graded; the 81 stays a fingerprint, the kill condition does not fire), while left V-multiplication induces a canonical $\mathbb{F}_3$-action on the cokernel $(\mathbb{Z}/3)^4$ — *computed*: its characteristic polynomial is $x(x-1)^2(x+1)$ mod 3 (eigenvalues $\{0, 1, 1, -1\}$, NOT the regular μ_4 pattern $x^4 - 1$; U acts as zero; both twins identical) — after two candidates the 81 stays a fingerprint.
- **Promotion `v568_bit_selector_ladder.py` (`DIAMOND.BIT.SELECT.01`, 11 checks, ∼1 s).** New ledger row; update notes on `QGEO.QRECON.01` and `DIAMOND.PARA.SELFCODE.01`; suite 561 → 562 scripts. **The physical selection is not derived:** why the dominant mode should satisfy the ladder self-consistency is the relocated open question — the same structural slot as the RP-selection of the alignment bit (`v528`/`v534`); no marker moves, no P2 reduction.
- **Sync.** `tfpt_1` (selector appended to the Stage-A passage), `tfpt_2` (a dedicated bit paragraph after the audit paragraph); diary entry in `experiments/next.txt`; README/zenodo/website counters 562; mirrors regenerated.

2026-07-31 (II) (the No-Carrier-Reconstruction audit, Stage A: the anchor self-code is carrier-free, and Q reconstructs up to ONE binary sheet-coupling choice (RECONSTRUCTS-TO-FAMILY) + `v567` promoted + ledger/paper/diary/website sync)

- **Audit probe `carrier_free_q_reconstruction_probe.py` (10/10, verdict RECONSTRUCTS-TO-FAMILY).** Priority 1 of the self-code review executed. *The circularity located:* `v11`'s uniqueness certificate for Q uses row sums $(4, 5, 6)$, column sums $(9, 5, 1)$ and $\chi_Q = (t-1)(t^2-5t+3)$ — the $5 = g_{\mathrm{car}}$ in the budgets and in χ_Q; all of it *forbidden* in the audit. Under the corpus' own carrier-free data (sheet involution, `v10` winding normalisation, A_3 exponent spectrum, monodromy budget, integrality, seam-line demand): exactly $52 \to 28$ (parabolic) admissible reconstructions — and **all** 28 **code the same anchor** ($\chi_{H'} = x^3 - 4x^2 + 5x - 2$, spectrum $(1, 1, 2) = a$): the anchor self-code sits *upstream* of the carrier budgets. The single family invariant $81 = N_{\mathrm{fam}}^4$ (saturation index) pins Q to a *two-element* residual family ($q_{32} \in \{2 = |\mathbb{Z}_2|, 0\}$, difference $2E_{32}$) on which every classical carrier-free certificate coincides (anticommutator spectrum $\{6, -4, -2\}$, commutator charpoly $t(t^2+12)$, determinant, saturation) — the one separating certificate is the carrier column budget $(9, 5, 1)$: exactly the forbidden input. Ablations: dropping the winding input widens the bound-3 set $28 \to 1376$ (real load); bound-stable at $B = 9$.
- **Promotion `v567_carrier_free_q_reconstruction.py` (`QGEO.QRECON.01`, 9 checks, ∼1 s; integer prefilters take the probe's 555 s to 1.2 s).** New ledger row; update notes on `FLAV.QGEO.01` and `DIAMOND.PARA.SELFCODE.01`; suite 560 → 561 scripts. **Stage A only, no**

marker moves: whether the geometric inputs themselves derive from $\mu_4/D_4/H^1$ parabolic geometry is exactly `GATE.QGEO.01` (the audit is relocated, not closed); the carrier-free selector of the residual bit is open, its resonance with the split/nonsplit alignment bit a typed observation; no P2 reduction, `AX.P2.01` untouched.

- **Sync.** `tfpt_1` (Stage-A result appended to the self-code paragraph), `tfpt_2` (a dedicated audit paragraph after the self-code block); diary entry in `experiments/next.txt`; README/zenodo/website counters 561; mirrors regenerated.

2026-07-30 (II) (WOIT β_3 executed: the $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$ duality typed against σ_{std} — the OS cut and the $(2,2)$ signature forced, the kill branch empty by algebra, the induced member = the Θ_{phys} carrier (SUCCESS) + `v565` promoted + contract/paper/diary/website sync)

- **Discovery probe `woit_os_beta3_pt_dual_probe.py` (25/25, 2.4 s, verdict SUCCESS per the frozen preregistration; contract wording verbatim: success = Woit's Minkowski conjugation identified with the family-D conjugation on the quotient, kill = the duality forces a clock-centralising (family A) structure).** *The class (block theorem):* a $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$ duality is a nondegenerate Hermitian form Φ on $\mathbb{C}^4$; clock compatibility has entrywise residual $(\bar{\lambda}_j\lambda_k - 1)\Phi_{jk}$ and forces the four cross-eigenspace entries to zero: two 2×2 Hermitian blocks, $16 \to 8$ real parameters (the deck adds nothing; the wrong clock lift breaks the split). *The dichotomy and the forced* $(2,2)$*:* the ρ_{tw} pairing $M^\dagger \Phi M = \varepsilon\bar{\Phi}$ splits the class exactly in two — the Euclidean branch (σ_{std}'s home) has seam-line norm $a(|z|^2+1)$, no null point, no OS cut ever; the Minkowski branch has norm $a(|z|^2-1)$: *the null locus meets the seam line exactly in the seam circle* for every member (the OS cut is forced, not imposed), and $\Phi_1 = -\bar{\Phi}_i$ exactly (charpoly mirrored), so the total signature is $(2,2)$ for every nondegenerate member — Woit's Minkowski signature is a theorem of clock + ρ_{tw} pairing, not an input. *The kill is empty by algebra:* every compatible duality inverts clock and deck (family D); family A would need $\rho_4^2 =$ scalar, but $\rho_4^2 = \text{diag}(-1,1,-1,1)$ — no solution, decided not scanned; $s_\Phi^2 > 0$: the Kramers-free class. *The identification:* every seam-preserving Minkowski member induces the same conjugation $z \to -\bar{z}$ ($\mu = -1$, fixed points $\pm i$), perpendicular to σ_{std}'s $z \to +\bar{z}$; under the declared RP-side placement (exact rational-π arithmetic, offsets $\mp\pi/2$) it *is* $\theta_\perp$ — the mirror whose descent `v524` certified: Θ_{phys} is the quotient shadow of the duality (regressions re-verified: Gram PD $(16,0,0)$, anti-unitary with $\eta_\perp = +i$, $\Theta^2 = +1$ per sector, time reversal, torsor closure). Role separation complete: $\sigma_{\text{std}} \sim$ the OS cut (the metric), the Φ-twisted duality $\sim$ the descending mirror. Free/quotient level only; $\mathcal{A}_{\text{hol}}$ untouched; all seven kill tests stay live; no marker moves.
- **Promotion `v565_woit_beta3_pt_duality.py` (`WOIT.BETA3.DUALITY.01`, 22 checks, ∼2 s, sympy-exact + two 40-digit inertia certificates).** New ledger row; `WOIT.OS.TWISTOR.01` updated (β_3 executed, γ next); suite 558 → 559 scripts.
- **Sync.** `tfpt_research_contracts`: title line ($\alpha/\beta_1/\beta_2/\beta_3$ executed, γ next), the β_3 executed block, the roadmap item, the status paragraph, the straddle-hurdle wording (γ and beyond); `tfpt_3`: new Step 32, story counters thirty-one → thirty-two (title, intro, preamble), the Woit-bridge paragraph gains the β_3-stage status; diary entry in `experiments/next.txt`; website: `papers.ts` WOIT milestone list + `v565`, `suite.ts` 558 → 559, `ScriptIndex`/changelog mirrors regenerated; README/zenodo counters 559.

2026-07-24 (I) (`v534` — the straddle cone executed: THE FIRST DYNAMICAL SELECTION OF THE ALIGNMENT BIT — reflection positivity keeps exactly ONE member of the interacting mark family alive, $\delta = \pi/2$ with positive coupling;

the literal leading-order cone formalization dies; NO marker moves)

- **New module `v534` (`SEAM.STRADDLE.CONE.01`).** The 2026-07-23 review-round contract (recorded "not promoted" in entry (X)) is *executed*, double-verdicted. **V2 = ERFOLG — the selection law (the central datum):** with independent couplings λ_q on the four mark quartets of $M(\delta = m\pi/8)$ and the equivariant (uniform) direction, reflection positivity holds on *every* admissible cut over the whole grid $g \in \{1/32..8\}$ for *exactly one* member $\times$ sign: $\delta = \pi/2$ ($m = 4$) with *positive* coupling — all four Grams PD $(37, 0, 0)$ *including the previously untested straddled clock axes* 7/15, minimum eigenvalue *lifted* off the free RP boundary ($1.78 \cdot 10^{-6} \to 5.5 \cdot 10^{-5}$ near $g = 1/4$: the $\pi/2$ interaction pushes *away* from the boundary), Gibbs $\beta = 2$ identical **[E]**. The asymmetric members die: $m = 2/6$ with $s = +$ already at $g = 1/32$, with $s = -$ from $g = 1/8$. The `v529` straddle law *refines*: asymmetric straddling kills RP, the *symmetric* straddling of the self-mirror member protects it.
- **V1 = KILL (the literal cone formalization).** The leading-order soft-block cone $K_{\mathrm{RP}} = \mathrm{Fix}(G_\delta) \cap \bigcap_c \{\lambda : G_c^{\mathrm{lead}} \succeq 0\}$ is *empty* for every member and both signs (first-order PT via the exact reduced resolvent, Hermitian to 10^{-15}, central-difference regression $2 \cdot 10^{-8}$) — the linear pencil cannot carry the selection; the protection at $\pi/2$ is *nonperturbative*. Recoverable leading-order content: the straddled-cut drift signs are exact ($\pi/2$: $+1.4 \cdot 10^{-4}$ for $s = +$, $-3.7 \cdot 10^{-2}$ for $s = -$; asymmetric members reversed) — necessary, not sufficient.
- **The exact geometry behind it [E].** $\delta = \pi/2$ is the *unique symmetric-straddle member*: stabilizer = the full clock+mirror group (rotations $\{0, 4, 8, 12\}$, reflection axes $\{3, 7, 11, 15\}$ = four admissible bond cuts, straddled $\{7, 15\}$), the four quartets *partition* the 16-circle, and every straddling quartet is reflection-*closed* across its cut with exact sign $+1$; $m = 2/6$ straddle asymmetrically; odd m carry no bond axis (the typed `v521`/`v525` placement artifact). $\mathrm{Fix}(G_\delta)$ is 1-dimensional throughout: uniform on $m = 2..6$, NS-wrap-*twisted* $(1, -1, 1, 1)/(-1, -1, -1, 1)$ on the tight members $m = 1/7$. Typed cross-link: the RP-selected member $m = 4$ is exactly the carrier of the `v528` twist-class order parameter $O = 1/2$ — the dynamical selector and the kinematic definition point at the *same* member.
- **Honest fences [C].** One toy, one interaction class (FK quartets), flat-band parent, deg ≤ 2 OS bases, finite coupling grid — toy-level evidence *for* the bit's dynamical origin, *not* a derivation; nothing about the continuum $\mathcal{A}_{\mathrm{hol}}$ is claimed; `WOIT.OS.TWISTOR.01` stays Open, NO marker moves. Transparency: the run-1 preregistered verdict logic conflated formalization and substance (UNENTSCHIEDEN, mismatch typed); run 2 split V1/V2 with no tolerance change and no number changed — note kept verbatim in the probe. Discovery provenance: `experiments/tfpt-discovery/seam_straddle_cone_probe.py` (15/15, 2026-07-24).
- **Wolfram mirror.** Four exact mirrors (stabilizer census, $\pi/2$ partition + straddle incidence, reflection-closure signs, equivariant Fix rays incl. the wrap-twisted $m = 1/7$ rays) in `wolfram/tfpt_readouts_extension.wl`, verified engine run 2026-07-24, 584 $\to$ 588 (the 256-dim Fock spectra, PT corrections and finite-g scans stay Python-only, flagged); suite 527 $\to$ 528 modules.

2026-07-23 (X) (`v530`–`v533` — the external review round promoted: center quotient compiler, Coxeter Stage A tensor clock, dual-degree modular checksum 744, sheet-transfer disc -7 norm line; every identity machine-confirmed exactly before promotion (probe 76/76); NO marker moves)

- **New module `v530` (`DIAMOND.CENTER.QUOTIENT.01`).** The Sheet-Diamond center $C = M(1, 0)$ carries the fixed primitive line $Cv = v$ ($v = (-1, 1, 0)$), and the induced operator on $\mathbb{Z}^3/\mathbb{Z}v$ is *exactly* the atom matrix $A = \left(\begin{smallmatrix} 8 & 2 \\ 5 & 3 \end{smallmatrix}\right) = \left(\begin{smallmatrix} \mathrm{rank}\, E_8 & |\mathbb{Z}_2| \\ g_{\mathrm{car}} & N_{\mathrm{fam}} \end{smallmatrix}\right)$ **[E]** with $\mathrm{tr}\, A = 11$, $\det A = 14$, $\mathrm{disc} = 65 = 5 \cdot 13$, $\mathrm{perm}\, A = 34 = p_5(a)$, $\prod A_{ij} = 240$, $\det(A - I) = 4$. The center *self-codes*: row sums $C\mathbf{1} = (7, 11, 13) = (\det A/|\mathbb{Z}_2|, \mathrm{tr}\, A, \mathrm{disc}\,/g_{\mathrm{car}})$; with the fixed eigenvalue

1 this is $(1, 7, 11, 13)$ = the CP-halved E_8 Coxeter exponents. The `v414` resolvent ladder factorises: $\det(C{+}kI) = (k{+}1)\det(A{+}kI)$ identically, quotient ladder $(14, 26, 40, 56)$, differences $(12, 14, 16)$, endpoint $224 = 248 - 24$. *Honest correction of the review:* eigenvalue 1 is NOT unique to C (K, Q share it) — uniqueness holds for the *conjunction* fixed line + atom quotient + CP self-code; the full selector-stratum ablation stays **[O]**. New paragraph in `tfpt_2` (Sheet Diamond section).

- **New module `v531` (`COX.STAGEA.TENSOR.01`).** The `tfpt_1` two-stage Coxeter audit (Stage A: a natural order-30 operator with $\chi = \Phi_{30}$ on the even-flip atlas *or its* 8*-dim primitive-character space*) is *closed on the character-space alternative*: $T_{30} = \mathrm{Comp}(\Phi_5) \otimes \mathrm{Comp}(\Phi_6)$ has exact order 30 and $\chi_{T_{30}} = \Phi_{30}$ **[E]** (CRT, $\gcd(5, 6) = 1$: carrier clock × family clock = Coxeter clock as *one integer operator*); Ramanujan trace tomography $\mathrm{tr}(T_{30}^k) = c_{30}(k) = c_5(k)c_6(k)$ for all k, signed table $(-1, 1, 2, 4, -2, -4, -8, 8)$, atom readout $(2, 4, 8) = (|\mathbb{Z}_2|, |\mu_4|, \mathrm{rank}\, E_8)$; honest kill scope ($\Phi_{15}$ passes the unsigned triple — the *signed* table discriminates; $\Phi_{16}/\Phi_{20}/\Phi_{24}$ die on the triple). The flip-atlas intertwiner (`COX.FLIP.INTERTWINER`) and Stage B stay **[O]**; `tfpt_1` Stage A prose updated.

- **New module `v532` (`E8.DEGREE.MODULAR.01`, audit-only).** The dual products of the E_8 invariant degrees, $(60, 192, 240, 252)$, satisfy simultaneously $\sum = 744 = 3 \cdot 248$ (= the j-function constant, recomputed from the q-series E_4^3/Δ), $\prod = |W(E_8)|$, gcd = 12, $\frac{1}{12}(\cdot) = (g_{\mathrm{car}}, \dim S^+, \det L, 7N_{\mathrm{fam}})$ **[E]**. *Mandatory fence:* D_{16} passes the same checksum ($1488 = 3 \cdot 496 = 2 \cdot 744$ — heterotic, NOT E_8-exclusive); six non-heterotic controls fail. *Prime-totative fingerprint (audit):* $\{2, 3, 5\} \cup (\mathrm{Exp}(E_8) \setminus \{1\})$ = all primes below 30, and 30 is the *largest* integer whose totatives are all 1 or prime (scan to 10^4) — h_{E_8} as the maximal prime-totative Coxeter number; per the no-free-pattern rule nothing here is load-bearing. New keybox in `tfpt_3`, cross-note in the `tfpt_1` 496 box.

- **New module `v533` (`FTR.DISC7.NORM.01`).** The branch-preserving transfer path $J, K, C, F = M(1, -2..1)$ is a *norm line*: $\det M(1, t) = 4t^2{+}14t{+}14 = ((4t{+}7)^2{+}7)/4 = N(\alpha_t)$ with $\alpha_t = (4t{+}7{+}\sqrt{-7})/2$ integral in $\mathbb{Z}[(1{+}\sqrt{-7})/2]$, the path one translation $\alpha_{t+1} = \alpha_t + 2$, norms $(2, 4, 14, 32)$, constant discriminant -7, vertex $7/4 > 0$ **[E]**; 2 splits in $\mathbb{Q}(\sqrt{-7})$ ($-7 \equiv 1 \bmod 8$, class number 1); controls: winding axis linear, anchor block disc +105. **[C]**/**[O]** preregistered next test: do the four external transfer solvers share *one* fractional-linear action on α_t (kill: four incompatible actions)? `GATE.FTRANSFER` unchanged. New paragraph in `tfpt_2` + frame note in `tfpt_research_contracts`.

- **Wolfram mirror.** All four modules mirrored exactly in `wolfram/tfpt_readouts_extension.wl` (20 new checks, verified engine run 2026-07-23, 564 → 584); suite 523 → 527 modules.

- **Not promoted (honest fences).** The review's remaining proposals stay research contracts, not verifications: the straddle-cone selector on the interacting FK quartets (the highest-leverage attack on the alignment bit after `v529`) and the metrology-torsor formalisation of v_{geo} are recorded as next steps in `experiments/next.txt`; the primes-below-30 identity stays an audit fingerprint exactly as the no-free-pattern rule demands.

2026-07-23 (IX) (website entry redesign: narrative home, honesty band, proof strip, guided-tour route — editorial only, NO new modules, NO marker moves)

- **Website home.** Slim one-viewport hero (proof strip: 523 machine-checked modules · Wolfram mirror · Lean carrier · 0 external replications yet; copyable `python run_all.py → ALL CHECKS PASSED`; CTA trio film / browser check / kill board). Narrative order: Hero → IntroVideo (`#intro-video` kept) → In one breath → WhyThisMatters (plain-English column) → Trust-Contract → ClaimStack → HonestyBand (honest negatives: RXTE/NICER nulls, 2DCS/v529, ten bit tests) → Safeguards (four core stats) → Overview (TheoryUnpacking/ThreeDecoderMap behind progressive disclosure) → PathChooser → Downloads teaser.

- **Archive routes.** Full PapersSection → `/papers`; PredictionsSection → `/predictions`; ReconstructionChain + HorizonStory + Gravity → `/architecture`; RosettaLexicon → `/orientation`; OpenGates → `/verification`; EcosystemSection off home (footer adjacent link). Legacy hash redirects on home for old anchors.
- **Nav / design.** Primary nav: Start · Tour · Verification · Kill board · Replication · Papers · FAQ (More dropdown for secondary). SuiteBadge (static per release from `suite.ts/version.ts`). Design pass: drop purple-gradient pills/glow in hero and accent tokens; lab-notebook proof strip as above-the-fold visual.

2026-07-23 (VIII) (prominence pass — the v519–v529 breakthroughs lifted into the FRAME texts of the papers and the website data layer; purely editorial, NO new modules, NO marker moves — (1) `tfpt_5_redteam` gains the seam round Target G: the ten-test side-blind scoreboard on the alignment bit (`v512/v521/v525/v529`) and the first *firing* kill test of the programme — the interacting straddle law (`v529`), documented as a named threat AND the first hard selection principle for $\mathcal{A}_{\mathrm{hol}}$, fenced at toy level; (2) the bit-reduction statements in `tfpt_1`, `tfpt_2` and `origin_theory` extended from the v512 state to the current state (ten side-blind tests, the twist-class definition with gauge-robust order parameter, "in principle interferometrically readable", `v528` — the bit stays formal input); (3) the thermal seam lifted into the `origin_theory` horizon chapter (new keybox: $T_{\mathrm{seam}} = 4c_3$ measured, temperature the third leg of c_3, `v526`) and into the `tfpt_horizon_readouts` head keybox; (4) `tfpt_3` title/keybox now carry the thirty-step story and "the measure chain is derived" (`v520/v523`) prominently; (5) `tfpt_research_contracts` title/intro keybox now show the executed stages of `CELEST.SEAM.01` (M1–M3, δ-chains decided, w_m derived) and `WOIT.OS.TWISTOR.01` ($\alpha/\beta_1/\beta_2$ executed, β_3 next under the straddle constraint); both contracts stay Open)

- **Papers.** `tfpt_5_redteam` (Target G + method/outcome/title honesty pass); `tfpt_1`, `tfpt_2` and `origin_theory` (bit-reduction statements → ten tests + twist-class definition); `origin_theory` (thermal-seam keybox, horizon chapter); `tfpt_horizon_readouts` (head keybox third leg); `tfpt_3` (title + keybox: thirty-step story, measure chain derived); `tfpt_research_contracts` (title + intro keybox: executed contract stages).
- **Website.** `papers.ts` (redteam Target G, horizon third leg, audit/contracts abstracts, architecture reduction state); `orientation/StatusMatrix.tsx` (v512 → current state); `glossary.ts` (new terms: twist-class choice, thermal seam, straddle law, side-blind scoreboard); new visual components `ThermalSeamLegs` (three legs of c_3: geometry `v58` + anomaly/modular `v208` + temperature `v526`) and `WoitBridgeProgress` ($\alpha/\beta_1/\beta_2$ done, β_3 open, straddle warning) on the home and verification pages.
- **Counters.** Predictions counter made consistent with `predictions.ts` (27 falsifiable test surfaces): `README.md` badge/hero/prose and the intro-film description (23 → 27; the film narration already says twenty-seven).
- **Honesty.** No marker moves anywhere; `SEAM.EQUIV.01` and `WOIT.OS.TWISTOR.01` stay Open; the bit stays input; all fences ("derived at probe level", "at the free level", "toy model") preserved verbatim.

2026-07-23 (VII) (`v529` — the interacting Fidkowski–Kitaev seam toy promoted as the ledger row `SEAM.INT.FKTOY.01`: the FIRST firing Kill-Test-2 shadow — triple verdict F1 = KILL / F2 = KILL / F3 = ERFOLG; the celestial/twistor story in `tfpt_3` gains Step 30 and the header goes twenty-six → thirty steps — (1) F1 =

KILL [E]: Θ exists exactly on the interacting model ($U_r^2 = 2^8$, $\Theta^2 = +1$, clock-tower inversion, $[\Theta, H_g] = 0$) so Kill-Test 1 does *not* fire; but Kill-Test 2 *fires at toy level*: RP breaks for every $g > 0$ — inertia ladder $(37, 0, 0) \to (33, 4, 0) \to (31, 6, 0) \to (29, 8, 0)$; Gibbs $\beta = 2$ same ladder; mechanism = interference; STRADDLE LAW $(24/24)$; [C] fence one toy / one interaction class / flat-band parent — narrows the Woit route, does *not* close it; `WOIT.OS.TWISTOR.01` stays Open with the toy shadow documented as a named Kill-Test-2 threat; (2) F2 = KILL [E]: the TENTH side-blind test $(9 \to 10)$ — Θ_{15} maps $H(m) \to H(8-m)$ EXACTLY despite δ-seeing spectra; (3) F3 = ERFOLG [E]: grading + $\mathbb{Z}_8$ tower + FK anchor + constructive OS quotient at $\pi/2$; Wolfram extension $552 \to 564$ (verified engine run); suite $519 \to 523$; *NO marker moves*)

- **Suite.** New module `v529_seam_interacting_toy_fk.py` (24 checks, ~ 97 s); registered in `run_all.py` + `script_registry.csv`; suite 523 modules green.
- **Ledger.** New row `SEAM.INT.FKTOY.01`; `WOIT.OS.TWISTOR.01` text-precised (Kill-Test-2 toy shadow + straddle law); scoreboard rows text-precised $(9 \to 10)$; no status moves.
- **Papers.** `tfpt_3` Step 30; `tfpt_research_contracts` (Kill-Test-2 toy threat + bit scoreboard 10); `introduction` (tenth side-blind test).
- **Wolfram.** Three exact mirrors (Z8 identities; straddle combinatorics; FK binomial ladder).
- **Website.** `papers.ts` (story 30 steps + scoreboard 10), generated indices via build; `README.md`, `zenodo_description.html` (523 modules, highest ID `v529`, Wolfram 564/564), `next.txt` notes.

2026-07-23 (VI) (`v528` — the alignment bit as an exact measurable TWIST-CLASS choice promoted as the ledger row `SEAM.BIT.TWISTCLASS.01`: the physical definition — verdict SUCCESS; the celestial/twistor story in `tfpt_3` gains Step 29 — (1) Facets #14 (η-flip $\Leftrightarrow \delta = \pi/2$) and #15 (twist-frustration $\Leftrightarrow \delta = \pi/2$) both directions exact [E](cardinality lemma $|A_+| = |A_-| \Rightarrow m = 4$; g-class census 224 axes: flip class exactly at $\{15, 31\}$); (2) order parameter $O = 1/2$ at $m = 4$ else 0 [E], gauge-robust; (3) unification #14=#15=flag-transitivity = ONE $\mathbb{Z}_2$ invariant [E](`v512` web $13 \to 15$); Kramers = POSITION half (derived), twist class = MODULUS half (the input); (4) [C] measurement sketch (interferometric in principle); the bit remains formal input; *NO marker moves*)

- **Suite.** New module `v528_seam_bit_twist_class_definition.py` (18 checks, ~ 46 s).
- **Ledger.** New row `SEAM.BIT.TWISTCLASS.01`; `SEAM.TAU.FLAG.01` text-precised (web $13 \to 15$); no status moves.
- **Papers.** `tfpt_3` Step 29; contracts + introduction ("symmetry-lift bit" now also "the twist-class choice, in principle interferometrically readable").
- **Wolfram.** Three exact mirrors (cardinality lemma; 224-axis census; O-values).

2026-07-23 (V) (`v527` — the silver clock eigenvalue EXPLAINED and DEMYSTIFIED, promoted as the ledger row `SEAM.CLOCK.SILVER.01`: double verdict STRUCTURE = SUCCESS / CONNECTION = KILL (anti-numerology style); the celestial/twistor story in `tfpt_3` gains Step 28 — (1) STRUCTURE SUCCESS [E]: $\{1, \sqrt{2}-1\}$ closed via diagonal Hankel pencil at $N = 8$, $\tan(\pi/8) = C(3)/C(1)$, doubling = exact second quantisation, carrier = silver-midpoint axes, gap = $\operatorname{arcsinh}(1) = \mathrm{gd}^{-1}(\pi/4)$; N-scaling = pure $N = 8/16$ lattice datum (not continuum); positive clock only for $N \equiv 0 \pmod 8$; (2) CONNECTION KILL [E]: preregistered 11466-candidate census over 16 frozen numbers: 0 hits above null;

placebos comparable; (3) cross-link `v526`: $\sqrt{2}-1$ = thermal kernel ratio; *NO marker moves*)

- **Suite.** New module `v527_seam_clock_silver_spectrum.py` (22 checks, ~ 10 s).
- **Ledger.** New row `SEAM.CLOCK.SILVER.01`; no status moves.
- **Papers.** `tfpt_3` Step 28; contracts thermal/silver follow-on.
- **Wolfram.** Three exact mirrors ($\tan(\pi/8)$ pencil; N-scaling witness; gap = $\operatorname{arcsinh}(1)$).

2026-07-23 (IV) (`v526` — the thermal seam promoted as the ledger row `SEAM.THERMAL.KMS.01`: KMS of the reconstructed free OS quotient CLOSED onto the Nariai/BH temperature chain — the THIRD LEG of c_3; verdict SUCCESS; the celestial/twistor story in `tfpt_3` gains Step 27 — (1) β MEASURED $= N$ at both levels [E]($N = 8$: $\omega = \operatorname{arcsinh}(2^{-3/4})$, $\kappa = q^{-8}$; $N = 16$: pairing exponent $(N/2)+1 = 9$; wrong β excluded); (2) $\beta_{\text{angle}} = 2\pi$ EXACT $= 1/(4c_3)$, $T_{\text{seam}} = 4c_3$ = Bisognano–Wichmann/Hawking normalisation (`v208`) [E]; (3) modular Hamiltonian closed [E]($\nu_1 = \cos(\pi/24)$, $\nu_2 = \cos(7\pi/24)$; Tomita KMS-$\beta = 1$ on all 256 pairs); (4) c_3 third-leg class-1 selection [E]/[C](Hawking/Nariai chain; `v104` cross-link); (5) entropy $S(8/16/32) = 0.52187/0.63510/0.75006$, $\Delta S/\ln 2 \to 1/6$, horizon fractions do NOT transfer [E]; (6) [C]-5 silver $= C(3)/C(1)$ thermal kernel ratio; non-compact $\Rightarrow T = 0$; *NO marker moves*)

- **Suite.** New module `v526_seam_thermal_kms_nariai_bridge.py` (22 checks, ~ 8.4 s).
- **Ledger.** New row `SEAM.THERMAL.KMS.01`; no status moves.
- **Papers.** `tfpt_3` Step 27; `tfpt_horizon_readouts` (third leg of c_3); `tfpt_research_contracts` (thermal follow-on); `introduction`.
- **Wolfram.** Three exact mirrors (Q-polynomial/β measurement; $2\pi = 1/(4c_3)$ chain; ν charpoly).

2026-07-23 (III) (`v525` — the twist-state attack on the alignment bit DECIDED and promoted as the ledger row `SEAM.BIT.TWISTBLIND.01`: "mark-decorated states do not see δ" — an honest, decided NEGATIVE result (`v508` style), the NINTH side-blind test on the `v512` scoreboard ($8 \to 9$); the free-plus-twist class (everything Wick-computable) is EXHAUSTED; the celestial/twistor story in `tfpt_3` gains Step 26 and the header goes twenty-three $\to$ twenty-six steps — (1) ALL LIFTINGS WELL-DEFINED AND BLIND [E]: (a) the σ-gauge arc decoration ($\mathbb{Z}_2$) satisfies $\sigma \circ r = \sigma$ on the swap axes $\Rightarrow$ the twisted Gram is a diagonal unitary congruence DMD, spectra IDENTICAL to untwisted (dev 0.0); (b) the Kadanoff–Ceva 4-twist insertion $\omega(D\,\cdot)/\omega(D)$ is well-defined on the $N = 32$ ladder ($\omega(D)$ real nonzero; the $N = 16$ odd-m degeneration is the $C(\text{even}) = 0$ checkerboard artifact) and its RP SPECTRA are the FIRST free-class data of the programme that depend on δ (pairwise up to 1.05) — yet the INERTIA is $(16, 0, 0)$ PD with the same $\eta = +i$ for EVERY $m = 1..7$: the decoration sees δ, positivity does not; (c) the μ_4 twist at $\beta = \pi/2$ is pure plane gauge; (c′) the genuine Bogoliubov defect at $\beta = \pi/4$: Θ-compatibility selects exactly 2 of 16 sign patterns (NS-wrap-forced), the winner genuinely non-monomial — still PD, spectra identical; STRUCTURE THEOREM (mechanism exact): the defect planes never straddle a cut bond, so every Θ-compatible quasi-free mark decoration is RP-spectrally INVISIBLE; (2) SUB-RESULTS [E]: the defect energy $E_{\text{def}} = 1.2752872$ for EVERY m (spread 4×10^{-40}) — δ-blind; the 4-twist Casimir $\ln|\omega(D)|$ measures δ ($-0.890.. - 1.348$) but

exactly mirror-symmetric under $m \leftrightarrow 8-m$ — a family gauge, never a side selector; Kramers $\langle\Gamma\rangle_{\rm tw} = +1$ for every m; (3) the HARVEST [E]: two bit-presupposing $\pi/2$ structures (the η flip $+i \to -i$ on the mark-fixing axes via $\sigma \circ r = -\sigma$; the twist frustration of the mark-fixing mirror, $(4,4,0)$ indefinite for both η — the lattice shadow of the Ising twist field's mirror-oddness) — both formulable only on the fixing axes existing iff $\delta = \pi/2$ (the `v512` facet class); (4) NEGATIVE CONTROLS [E]: the twist-free limit = v519/v521 exactly; the site cut keeps failing; the `v512` counterwitness is arc-equivalent at $N = 16$ to the blind member $\pi/4$ — no state selector can exclude it; a single twist has $\omega(D) = 0$ identically; [O]: the door narrows to a genuinely INTERACTING $\mathcal{A}_{\rm hol}$ whose OS data are NOT Pfaffian-reducible to the chiral vacuum; Wolfram extension (with v523/v524) $540 \to 552$ (verified engine run 2026-07-23, $552/552$ + base $116/116$); suite $516 \to 519$ modules; *NO marker moves — the alignment bit remains genuine discrete input*)

- **Suite.** New module `v525_seam_bit_twist_blind.py` (20 checks, ~ 80 s; Pfaffian-exact insertion machinery cross-validated three ways, 40-digit certificates); registered in `run_all.py` + `script_registry.csv`; suite 519 modules green.
- **Ledger.** New row `SEAM.BIT.TWISTBLIND.01` (KILL, honest negative, preregistered); `SEAM.BIT.RPBLIND.01` and `SEAM.TAU.FLAG.01` text-precised (scoreboard $8 \to 9$, "the free-plus-twist class is exhausted"); no status moves.
- **Papers.** `tfpt_3` Step 26 + the story header (twenty-six steps) + the Step-22 gap sentence; `tfpt_research_contracts` (the `QGEO.REALIZE.01` reduction paragraph $8 \to 9$ and the WP2 relocation note extended by `v525`); `introduction` (the ninth side-blind test added to the reduction-state sentence).
- **Wolfram.** Three exact mirrors (the $N = 32$ integer frame + the $\sigma \circ r = \pm\sigma$ congruence mechanism; the $\pi/2$ existence solveset + the $(3+4i)/5$ root-of-unity census; the $\delta \leftrightarrow \pi - \delta$ mirror equivariance); the Pfaffian spectra, the 2-of-16 pattern census and $E_{\rm def}$ stay Python-only.
- **Website.** `papers.ts` (tfpt_3 story mirror 26 steps + scoreboard 9), generated indices (`ScriptIndex.tsx`, `changelog.ts`, maps) via build; `README.md`, `zenodo_description.html` (519 modules, highest ID `v525`, Wolfram $552/552$), `next.txt` notes.

2026-07-23 (II) (`v524` — the WOIT-β_2 milestone of the OS twistor bridge `WOIT.OS.TWISTOR.01` executed and promoted as the ledger row `WOIT.BETA2.OS.01`: "the OS quotient of the free system made explicit" — verdict SUCCESS per the frozen preregistration, [C]-typed per contract precision (iii); the celestial/twistor story in `tfpt_3` gains Step 25 — (1) $\mathcal{H}_{\rm phys}$ EXPLICIT [E]: $N = 16$ ($\deg \le 2$): the 37×37 bond-cut OS Gram exactly Hermitian, parity-block-diagonal, inertia $(37,0,0)$ PD (min eigenvalue 1.7801×10^{-6} at 40 digits), null space $\{0\}$, $\dim 37 = 29 \oplus 8$; $N = 8$ (complete half algebra): $(16,0,0)$ PD (min 3.3471×10^{-3}), $\dim 16 = 8 \oplus 8 = 4^2$ — compact euclidean time reconstructs a THERMAL (KMS) representation; exact certificate $\sin^2(3\pi/8) - \sin(\pi/8)\sin(5\pi/8) = 1/2$; (2) the TRANSFER SEMIGROUP (Klein–Landau) [E]: τ_k exactly Hermitian on all domains ($k = 1..7$, dims $29, 22, 16, 11, 7, 4, 2$), chain identities exact, vacuum fixed; the positivity pattern = the site/bond dichotomy: even k PSD via the exact square identity $T(2j) = A_j^* A_j$ ($(22,0,0)/(11,0,0)/(4,0,0)$), odd k indefinite ($(10,12,7)/(5,6,5)/(2,2,3)$) — the one-step transfer is NOT positive (the chirality datum, kill-3 shadow sharpened), its even powers are; (3) THE CLOCK $= T^{N/4}$ [E]: $N = 16$: τ_4 PD $(11,0,0)$ (min 6.6821×10^{-5}), compressed spectrum $[0.0194517, 1.64144]$, vacuum eigenvalue exactly 1, spectral projections $\sim 10^{-40}$; $N = 8$: spectrum EXACTLY

$\{1, \sqrt{2}-1\} = \{1, 1/\delta_{\text{Silber}}\}$ **in both parity sectors — the silver axes of the** `v519` **torsor return as the clock eigenvalue; the reconstructed rotation group** $U(s) = e^{isH}$ **unitary with the group law; (4) the** `v522` **RESOLUTION [E]: the census** $(745, 96, 0)$ **reproduced; EVERY anti-matching entry is a wrap overlap, every overlap-free entry matches — the pre-quotient non-Hermiticity is EXACTLY the domain/wrap artifact; on the quotient the clock has spectral calculus; pre-registered KMS deviation: no contraction on the compact circle** ($C(1)/C(3) = 1 + \sqrt{2} = \delta_S$ **exactly,** $\det(G - \tau_4) < 0$**); (5) GSO/**Θ **[E]: the grading survives; the perpendicular torsor mirror descends anti-unitarily** ($\eta_\perp \in \{\pm i\}$)**,** $\Theta^2_{\text{phys}} = +1$ **on every sector (Kramers-free),** $\theta_{\text{cut}} \circ \theta_\perp = \alpha_{N/2}$ **exactly; (6) CONTROLS [E]: site cut** $(3, 3, 1)/(2, 2, 4)$ **— the kill branch fires there; family A** $(4, 4, 0)$**; anti-chiral** $(8, 8, 0)$ **no quotient; wrong direction non-Hermitian (all mismatches wrap-overlap); KILL TESTS: (1)/(2) strengthened, (3) shadow sharpened, none fires, all seven stay live on** $\mathcal{A}_{\text{hol}}$**; TRANSPARENCY: run 1** $= 18/20$ **(a sympy simplify failure on a TRUE** $\pi/16$ **product formula, fixed by a Laurent-polynomial certificate; no criterion changed);** ***NO marker moves — `WOIT.OS.TWISTOR.01` stays Open,*** β_3 ***is next*****)**

- **Suite.** New module `v524_woit_beta2_os_quotient.py` (20 checks, $\sim$ 5 s; exact sympy + Pfaffian-Wick + 40-digit inertia certificates); registered in `run_all.py` + `script_registry.csv`.
- **Ledger.** New row `WOIT.BETA2.OS.01` (SUCCESS, [**C**]-typed per precision (iii)); `WOIT.OS.TWISTOR.01` claim/canonical status extended (β_2 executed, β_3 next); no status moves.
- **Papers.** `tfpt_3`: Step 25 plus the Woit-bridge paragraph with the β_2-stage status. `tfpt_research_contracts`: a full "β_2 stage — executed" block with `\veri`; the β/γ roadmap entry marked executed; the scope-fence status sentence extended.
- **Wolfram.** Four exact mirrors (the $\alpha_1^N = (-1)^F$ wrap arithmetic + the certificate identity; the Klein–Landau one-particle transfer forms with odd-k zero diagonal and the even-k square identity; the exact $N = 8$ clock spectrum $\{1, \sqrt{2}-1\}$ + the transparency $\pi/16$ product formula certified symbolically; the KMS silver witnesses); the $\mathcal{H}_{\text{phys}}$ Grams, inertia spectra and the rotation group stay Python-only.
- **Website.** `papers.ts` (tfpt_3 + contracts mirrors), generated indices via build.

2026-07-23 (I) (`v523` — the constructive w_m **derivation executed and promoted as the ledger row `CELEST.WP5E.WM.01`: "**$1/\det_j$ **is the Atiyah–Bott / zeta-determinant fixed-point factor, COMPUTED from three independent sources" — verdict ERFOLG per the frozen preregistration; the residual [O] of `v516`/`v520` is closed at the fibre-zero-mode level and narrows to the GLOBAL BCOV integral on** $\mathbb{PT}/\mathbb{Z}_4$ **beyond the fibre zero-mode factor; the celestial/twistor story in `tfpt_3` gains Step 24 — (1) ATIYAH–BOTT (route i) [E]: the equivariant mode ledger of the fibre** $\mathbb{C}^2$ **equals the closed form** $1/((1 - q\,i^j)(1 - q\,i^{-j}))$ **exactly in** $\mathbb{Q}(i)$ **for all** $n \le 120$**; REGULAR at** $q = 1$ **with Abel value** $(1/2, 1/4, 1/2) = 1/\det_j$**,** $\det_j = \det(1 - g^j) = (2, 4, 2)$**;** $(C, 2)$ **Cesàro convergence in exact arithmetic; fixed-point splitting at EVERY truncation level** $d \in \{7, 12, 25\}$ **for the phased (skeleton) AND the `v520`-forced unphased (completion) trace — the Abel-limit** $\text{contact}_j = (Q^{(0)} - Q^{(j)})/\det_j$ **is the `v516` contact vector COMPUTED, not declared; (2) ZETA/QUILLEN (routes ii/iii) [E]:** $\det_\zeta$ **of the** g^j**-twisted circle Laplacian** $= 4\sin^2(\pi j/4) = \det_j$ **EXACTLY (Lerch + reflection formula symbolically, Hurwitz certificates** $\sim 10^{-41}$**); Quillen split** $\det \Delta = \det_j^2$**,** $|\det \bar\partial| = \det_j$**, the real positive holomorphic section UNIQUE via the** $SU(2)$ **conjugation pairing; the** $\delta_{1f}\ f_1 f_3$ **block at the** $(0, b)$ **nodes carries the exact constant term** $1/\det_b$ **(deviations** $\le 5.2 \times 10^{-28}$**); (3) the CONSISTENCY**

CHAIN [E]: $w_m = (0, \frac{3}{2}, 2, \frac{3}{2}) = 4h_m = |\mu_4|h_m = -4\,\mathrm{ch}_2(T_m)$**;** $T_5 = 0$ **at every scale;** $h_2 : h_1 = 4 : 3$**; the** T_3 **budget** $32 - 8c$ **forces** $c = 4 = |\mu_4|$ **uniquely (leftovers** $24/16/8/0/-8$**); per-sector Okubo squares** $(18, 9, 18)\langle x, x\rangle^2$**; total** $45\langle x, x\rangle^2$**;** $\Phi_{T_3} : 32 \to 0$**,** $\Phi_P : 72 \to 0$**;** $\psi(A_{\mathrm{fix}}) = +64$**,** $\psi(\mathrm{contact}) = -64$**; (4) NEGATIVE CONTROLS [E]:** $1/\det^2$ **gives** $w' = (0, 5/8, 1, 5/8)$**, leftovers** $(2, 12)$**, Quillen contradiction;** $1/|1-i^j|$ **gives** $w_1'' = 1+\sqrt{2} \notin \mathbb{Q}$**; the** $\mathbb{Z}_2$**/Eguchi–Hanson anchor PASSES with the same derivation (**$\det = 4$**,** $w = 1/2 = |\mathbb{Z}_2|h^{A_1}$**,** $9\langle x, x\rangle^2 = \frac{1}{4} \times 36$**,** $c = 2 = |\mathbb{Z}_2|$**); the** $SO(16)/D_8$ **KILL fires;** $\mathrm{diag}(i, i)$ **breaks the zeta = AB identity itself;** $k = 3, 5$**:** $w_m = m(k-m)/2 = k\,h_m$ **— the weights wander correctly, only** $k = 4$ **carries the** E_8 **chain; TYPED PREMISES [C] (the mandatory fence): TP-REG (Abel/Cesàro/zeta regularisation = the definition of the equivariant character), TP-Q (Quillen structure of BCOV** F_1 **=** `v520` **TP-4), TP-NUM (unphased numerator =** `v520` **ERFOLG-A), TP-CH (insertion channel** $(0, j)$**); [O]: the global BCOV integral beyond the fibre zero-mode factor, the resolved-ALE route not built,** `SEAM.EQUIV.TWISTOR.01` **stays Open;** ***NO marker moves*****)**

- **Suite.** New module `v523_celestial_wp5e_wm_derived.py` (26 checks, $\sim$ 1 s; exact $\mathbb{Q}(i)/\mathbb{Q}(\sqrt{2})$/cyclotomic arithmetic + 40-digit Hurwitz/block certificates); registered in `run_all.py` + `script_registry.csv`.
- **Ledger.** New row `CELEST.WP5E.WM.01` (ERFOLG); the `v516` row `CELEST.WP5E.M2.01` and the `v520` row `CELEST.WP5E.MEASURE.01` text-precised ("the w_m normalisation is now DERIVED at probe level; the residual **[O]** narrows to the global BCOV integral beyond the fibre zero-mode factor"); no status moves.
- **Papers.** `tfpt_3` Step 24 + the "What remains, honestly" narrowing; `tfpt_research_contracts` (the WP5e measure-question block extended by the executed derivation with `\veri`, the M2 fence, the celestial summary blocks and the section status list); `SEAM.EQUIV.TWISTOR.01` stays Open.
- **Wolfram.** Five exact mirrors (the mode ledger = closed form with Abel value; the zeta/Quillen route with the symbolic reflection formula and the $\mathrm{diag}(i, i)$ control; the derived-weight contact vectors + success table + $\mathbb{Z}_2$/EH anchor on the 240 glue roots; the consistency chain + wrong-weight controls; the $k = 3, 5$ cyclotomic wandering controls); the Cesàro/Hurwitz/stripped-block numerics stay Python-only.
- **Website.** `papers.ts` (tfpt_3 + contracts mirrors), generated indices via build.

2026-07-22 (XIII) (HFQPO scorecard extension — the NICER/MAXI J1820+070 third-tooth null and the AstroSat/LAXPC block integrated as a new evidence-scorecard row (search target, never a claim): the preregistered single-QPO extension of the H3 ladder discriminator (frozen `hypotheses/hfqpo_ext_v1.yaml` **BEFORE any photon download) ran on** 6 **NICER ObsIDs (**13.22 **GB,** 2013×16 **s =** 32.2 **ks stack at** 19.05 **kc/s) — sanity gate PASS (the published** 55.12 **Hz anchor reproduced at** 55.03 **Hz,** 0.94% **rms,** 3.8σ**), injection calibration PASS (**$\geq$ 90% **recovery at** 0.93% **rms), the geometric tooth at** 82.68 **Hz NOT detected (**0.69σ **single-trial;** 3σ **upper limit** 0.75% **rms BELOW the detected anchor strength** $\Rightarrow$ `null_with_sensitivity`**), the integer line at** $\sim$ 110.6 **Hz at** 3.82σ **after trials — BELOW the preregistered** 4σ **threshold, documented as a sub-threshold excess consistent with the weak second harmonic of ATel #11951, NOT a hit; AstroSat/LAXPC (GRS 1915+105, scientifically the strongest channel)** `infrastructure_blocked` **(ISSDC login-gated); the RXTE + NICER nulls now cover five sources on two instruments — the tooth channel is recommended DORMANT; anti-double-counting: the parent RXTE row and the new extension row**

share `independence_group hfqpo_ladder_tooth` (one hypothesis, two instruments, never two independent nulls); scorecard regenerated (121 rows), `predictions.txt` HFQPO row extended ("Kanal ruhend"); experiment-only integration — no verification claim, no marker moves)

- **Scorecard.** New row "BH HFQPO third-tooth extension on non-RXTE archives" (NICER/MAXI J1820+070 plus AstroSat/LAXPC; `hfqpo-ladder` extension) in the generator `experiments/build_evidence_scorecard.py` (status `null`, stage `search_target`, signature S1e); `OVERRIDES` give both HFQPO rows the shared `independence_group hfqpo_ladder_tooth`; generator re-run: `evidence_scorecard.json` 121 rows, README stats block refreshed.
- **Experiment surfaces.** `experiments/hfqpo-ladder/extension-nicer-laxpc/` (preregistration, frozen manifest, download log, per-ObsID caches, stack/sanity/injection/scan reports, `results_ext.json`, README) — executed 2026-07-22; `predictions.txt` HFQPO row updated with the NICER finding and the dormant-channel recommendation; `next.txt` round notes.

2026-07-22 (XII) (`v522` — the WOIT-β_1 milestone of the OS twistor bridge `WOIT.OS.TWISTOR.01` executed and promoted as the ledger row `WOIT.BETA1.GSO.01`: "the clock is time-like, GSO is the gauge datum" — a TYPED result (verdict UNDECIDED per the frozen preregistration, the mechanism identified exactly, gauge-fixed RP established under the corrected typing; the celestial/twistor story in `tfpt_3` gains Step 23) — (1) the HERMITICITY WITNESS [E]: the μ_4 clock average violates Hermiticity EXACTLY (the one-step insertion $T_1 = \langle \theta(e_a), \alpha_S(e_b) \rangle$ has entry $-i/(8\sin(5\pi/16))$ where the conjugate transpose demands $+i/(8\sin(5\pi/16))$; entrywise census 745 Hermitian-matching, 96 ANTI-matching, 0 violations — the pairing is complex-SYMMETRIC, strictly weaker than Hermitian: "positivity after gauge fixing" is not well-posed for the clock reading); (2) TIME-LIKE TYPING [E]: ALL 16 dihedral reflection axes of the NS seam circle invert the clock lift ($RSR^{-1} = S^{-1}$), NONE commutes — the μ_4 clock IS Woit's euclidean rotation; the gaugeable (θ-compatible) part of the $\mathbb{Z}_8$ Fock tower is EXACTLY its 2-torsion $\{1, (-1)^F\}$ = the GSO/fermion-parity $\mathbb{Z}_2$ (Hermiticity census over all powers; α^4 = the full wrap = the 2π rotation of the seam = fermion parity); (3) the POSITIVE PART [E]: GSO-fixed RP holds EXACTLY — $N = 16$ even $\deg \leq 2$ parity-projected bond-cut Gram $(29, 0, 0)$ PD (min eigenvalue 1.78×10^{-6} at 40 digits), $N = 8$ complete half algebra: all odd rows/columns killed exactly, even block $(8, 0, 0)$ PD (3.35×10^{-3}), no degree truncation; the site-cut defect SURVIVES gauge fixing ($(7, 9, 6)$ — the bond placement is gauge-invariant information); family A keeps failing ($(17, 12, 0)$ indefinite vs family D's $(29, 0, 0)$); Kramers: $\Theta_t^2 = \mathrm{id}$ on $\mathcal{A}^G$, Θ_{Fock} restricts with $\Theta^2 = +1$; Ramond control: the wrap $+1$ "true $\mathbb{Z}_4$" lift fails state invariance — the NS/$\mathbb{Z}_8$ tower is forced; (4) KILL-TEST BOOKKEEPING: kill test (2)'s free shadow does NOT fire (no legitimate Hermitian gauge-fixed form has a negative direction), kill test (1) stays discharged also gauge-invariantly; both stay formally live on $\mathcal{A}_{\mathrm{hol}}$; (5) TRANSPARENCY: the first frozen run scored 6/13 (the clock-reading checks failed exactly as the mechanism predicts) and the preregistered deg-2 invariant dimension 30 was a combinatorial error (correct census $28 + 4 = 32$) — documented in full, the verdict by the frozen [C-3] logic (Hermiticity failure $\Rightarrow$ UNDECIDED); CONTRACT PRECISION (iii) (prose, no marker): "gauge-invariant subalgebra" at the free level means the GSO/fermion-parity $\mathbb{Z}_2$ — the only θ-compatible part of the μ_4 tower; the clock is the euclidean rotation itself

(timelike), its equivariant statement moves to β_2 (OS quotient first, then the clock as reconstructed transfer operator); Wolfram extension (with v520/v521) $522 \to 540$ (verified engine run); *NO marker moves — `WOIT.OS.TWISTOR.01` stays Open, β_1 proper re-routed via β_2*)

- **`v522_woit_beta1_gso_gauge.py` (`WOIT.BETA1.GSO.01`, 16 checks, ∼5 s; exact sympy Pfaffian–Wick + exact Cl(16) Fractions; definiteness via 40-digit spectra, tolerance 10^{-25}; deterministic).** G1 the gauge datum (Z8/NS tower, Ramond control, invariant dimensions $120 > 64 > 32$); G2 Θ normalises the tower and restricts as an involution; G3 the $(-1)^F$ class does not penetrate $\mathcal{A}^G$; R1 the tower average is OS-symmetric, the odd sector dies; M1 the time-like census; M2 the exact Hermiticity witness; M3 the GSO 2-torsion census; R2 GSO-fixed RP ($N = 16/N = 8$); C1–C3 site-cut/wrong-projection/family-A controls; V1 the frozen verdict. Registered in `run_all.py` (suite 516 modules, green), `script_registry.csv` (cluster *frontier*), ledger row `WOIT.BETA1.GSO.01` (new).
- **Contract updated (markers unchanged).** `WOIT.OS.TWISTOR.01`: the β_1 stage moved to *done* (typed UNDECIDED) with its own block, precision (iii) added, the β_1 roadmap bullet annotated *executed*, the closing status line carries the β_1 state; ledger `WOIT.OS.TWISTOR.01` text extended (beta-1 executed via v522).
- **Paper integration.** `tfpt_3` (new Step 23 "the clock is time-like, GSO is the gauge datum"; the Woit-bridge paragraph extended by the β_1 status); `veri` citations in `tfpt_research_contracts` and `tfpt_3`.
- **Wolfram.** Six exact mirrors (the Z8/NS tower $S^4 = -1$ + the Ramond witness $1/(4\sin(5\pi/16))$; the invariant-dimension chain by exact ranks; the time-like reflection census; the exact Hermiticity witness with the full 745/96/0 census via exact Pfaffian–Wick; the GSO 2-torsion census; the family-A tower-normalisation failure) added to `tfpt_readouts_extension.wl`; the GSO-fixed RP inertia certificates stay Python-only, flagged in the `.wl` comment.
- **Provenance.** Discovery probe `woit_os_beta1_gauge_invariant_theta_probe.py` (16/16, ∼3 s) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated; `next.txt` round notes.

2026-07-22 (XI) (`v521` — the RP/Θ attack on the alignment bit DECIDED and promoted as the ledger row `SEAM.BIT.RPBLIND.01`: "free OS positivity does NOT see δ" — an honest, decided NEGATIVE result in the `v508` style (the celestial/twistor story in `tfpt_3` gains Step 22): does free reflection positivity (`v519`) or anti-linear Θ existence break for $\delta \neq \pi/2$ on the deformed mark family $M(\delta) = \{\pm 1, \pm e^{i\delta}\}$ — which would DERIVE the `v512` alignment bit from OS positivity? NO — (1) the CENSUS [E] (exact solvesets, symbolic δ): for EVERY δ exactly the two mark-SWAPPING reflections $\Theta_{\pm e^{i\delta}}$ exist (axes $\delta/2$, $\delta/2 + \pi/2$; all 20 cut/mark incidence solvesets EMPTY); mark-FIXING reflections exist exactly at $\delta = \pi/2$ (solveset $\cos\delta = 0$, lost immediately at $\pi/2 - \varepsilon$); the `v510`/`v512` counterwitness $\{\pm 1, \pm(3+4i)/5\}$ carries the swap pair exactly; (2) RP IS δ-BLIND [E]: bond-cut Gram inertias $(8, 0, 0)$ PD for all tested δ with the full sorted spectra IDENTICAL to 40 digits (deviation 0.0) — the ENTIRE free-RP spectral datum carries ZERO δ information; the $N = 16$ odd-m failure (site axes, $(3, 3, 1)$, $\det = 0$) is a lattice-PLACEMENT artifact: at $N = 32$ ALL $m = 1..7$ have mark-compatible BOND axes with PD inertia $(16, 0, 0)$; continuum: the OS kernel in axis-adapted coordinates is EXACTLY $1/\sin((s+t)/2)$ — the axis position drops out identically (Cauchy–Stieltjes PD); the counterwitness PASSES free RP (6-point Gram $(6, 0, 0)$); (3) Θ EXISTENCE IS δ-BLIND [E]: $\Theta^2 = +1$, deck normalisation and Fock implementability ($U^2 = +2^7/2^8 > 0$, exact Cl(16)) for every

δ; the ONLY δ-sensitive `v519` clause ($\Theta\rho\Theta = \rho^{-1}$) is for $\delta \neq \pi/2$ not violated but NOT FORMULABLE (a mark-compatible order-4 rotation exists iff $\delta = \pi/2$) — it PRESUPPOSES the bit; same for the clock-tower clause; (4) the battery HAS TEETH at every δ [E]: family A/antipode fails RP structurally (zero diagonal, $(4, 4, 0)$), the chirality–η pinning persists ($(0, 8, 0)$ ND) — RP separates the two real-structure FAMILIES, never the SIDES; (5) precondition controls: hexagonal (marks not concyclic — no seam circle), silver (deck not free), lifting audit (B = index relabelling, δ-blind by construction; B' fails the $\pi/2$ negative control — discarded); [C] FACE #13 of the `v512` equivalence web: the group generated by the admissible OS reflections is V_4 for every $\delta < \pi/2$ and closes to D_4 exactly at $\delta = \pi/2$ ($\Theta_i \circ \Theta_1$ = the clock) — 12 → 13 faces, a typed REFORMULATION of the bit, not a derivation; [O] the honest gap: marks decorating the STATE (defect/twist insertions, interacting $\mathcal{A}_{\mathrm{hol}}$) = exactly `v519`'s [O] scope; VERDICT: KILL exactly as preregistered — the free RP/Θ battery is the EIGHTH side-blind test on the `v512` scoreboard (7 → 8); the alignment bit is NOT derivable from OS positivity and remains genuine discrete input; *NO marker moves — the bit stays the one [C] carrier input*)

- **`v521_seam_bit_rp_blind.py` (`SEAM.BIT.RPBLIND.01`, 25 checks, ∼25 s; exact sympy solvesets + exact Gram entries + exact Cl(16) Fractions; inertia via 40-digit spectra, tolerance 10^{-25}; deterministic).** S the anti-linear census (family D/A, counterwitness, face #13 closure, cut incidence, side conditions); L the lattice δ ladder (frame, $\pi/2$ reproduction, $\pi/4$ PD, odd-m artifact, $N = 32$ refinement, site-axis control, spectra blindness, family-A rejection, chirality pinning); C the continuum axis elimination + counterwitness; LB the lifting audit; T Fock implementability + the clock-tower clause; W scoreboard + hex/silver controls + the preregistered KILL verdict. Registered in `run_all.py`, `script_registry.csv` (cluster *core*), ledger row `SEAM.BIT.RPBLIND.01` (new).
- **Ledger/paper text precisions (markers unchanged).** `SEAM.TAU.FLAG.01`: scoreboard text extended (7 → 8 side-blind tests, face #13, web 12 → 13); `tfpt_research_contracts` (the `v512` reduction paragraph now carries the `v521` kill and the honest [**O**] gap; the WP2 residual sentence extended), `introduction` and `tfpt_3` Step 2 (12-fold → 13-fold, the eighth side-blind test); `veri` citations in `tfpt_research_contracts`, `introduction` and `tfpt_3`.
- **Wolfram.** Six exact mirrors (the census solvesets incl. the exact-rational counterwitness; the $V_4 \to D_4$ closure / face #13 as symbolic composition identities; the 20 incidence solvesets + the eps jump + the integer lattice frame; the continuum axis elimination + Cauchy–Stieltjes minors + the counterwitness cut ordering; the clock-tower clause + hex/silver controls; the δ-blind side conditions) added to `tfpt_readouts_extension.wl`; the 40-digit lattice inertia spectra stay Python-only.
- **Provenance.** Discovery probe `seam_alignment_bit_rp_selector_probe.py` (25/25, ∼24 s) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated; `next.txt` round notes.

2026-07-22 (X) (`v520` — the WP5e BCOV measure question DECIDED at probe level and promoted as the ledger row `CELEST.WP5E.MEASURE.01`: "single-valuedness is derived, the completion reading wins" — the $\delta_{1e} + \delta_{1f}$ consolidation in the `v518` style, ERFOLG-A on the preregistered decision layer (the celestial/twistor story in `tfpt_3` gains Step 21): which of the two exact, contradictory measures on $\mathbb{PT}/\mathbb{Z}_4$ is the true BCOV measure — (A) the DECLARED completion reading (`v516`) or (B) the DERIVED chiral measure (`v518`)? — (1) the INVARIANT SUBSPACE [E]: $\dim_{\mathbb{Q}}\{v : \rho(T)v = v,\ \rho(S)v = v\} = 8 = 4 \times 2$ in the dressed 16-component Weil system (exact rational nullspace); EVERY kernel vector lies

in the $\mathbb{Q}(\zeta_8)$-span of $\{e_H, e_{H'}\}$ — the two Lagrangian gluings, both theta $= E_4$, level-1 quartic $= Q^{(0)}$ = the UNPHASED trace; strict trivial-character solutions at physical weight $(m,k) = (0,0)$: ZERO for every dressing (11 combinations); (2) SOURCE 1 — the F-LEMMA [E]: the only strict chiral families are χ_4 (dims $(3,3)$) and χ_{10} (dims $(1,1)$), both with nontrivial T-character EXACTLY; the canonical column-matched member obeys $G(\gamma\tau) = \chi_4(\gamma)G(\tau)$ pointwise (certificate 3.9×10^{-16}, $|G| \geq 59.6$); EXACT change of variables ⇒ fundamental-domain independence forces $\int = 0$ for EVERY chiral integrand: the `v518` route was never a well-defined nonvanishing moduli integral (the kill is SHARPENED — its unfolded values are convention bookkeeping); (3) SOURCE 2 — the QUILLEN PAIRING [E]: $|t\bar{t} - 1| < 8.3 \times 10^{-40}$ on all 15 pairs and both generators; $|\chi_4|^2 = 1$ exactly vs one-sided $\chi_4^2 = e(2/3) \neq 1$ (T-fixed: $640 \neq 0 \bmod 960$, 15/15 defective); the doubled 256-dim hol×antihol system closes STRICTLY at trivial character and physical weight (nullspaces $(28,17) \geq 11$), contains the unphased diagonal ($\sim 2\times10^{-15}$) AND the physical columns $w_b \otimes \bar{w}_b$ (σ $4.9/8.3\times10^{-16}$); negative control hol⊗hol: EMPTY nullspace $(0,0)$; (4) the CONSISTENCY CLOSURE [E]: the forced unphased numerator in the `v518` scaffold reproduces $\sum_m V \propto (1,2,1,0,0) = \langle x,x\rangle^2$ EXACTLY on every level $n \leq 8$; J-weighted spreads $\sim 5\times10^{-41}$, $T_5/T_3 \sim 7\times10^{-42}$, zero negative cells; $\psi(\text{skeleton}) = +0.2302 = -\psi(\text{contact})$ — the J-weighted mirror image of $\psi(A_{\text{fix}}) = +64/\psi(\text{contact}) = -64$; linearity 4.6×10^{-41}: the circle to `v516` is CLOSED; (5) D3 — the $\mathbb{Z}_2$/EH ANCHOR [E]: only the declared reading hits ($9\langle x,x\rangle^2 = \frac{1}{4}\times36$, $\psi = -8$, scale tooth $c = 1/2 = |\mathbb{Z}_2|h^{A_1}$); all three derived instantiations fail (spreads $1.12/0.895/0.516$, ψ sign wrong); (6) D1 — COLUMN CANONICALISATION [E]: the physical AB column $\sum_m i^{bm} e_{(m,m)}$ is contained EXACTLY in the χ_4 family (σ $1.8/1.7\times10^{-16}$, one scalar per orbit — canonicalising (N_1,N_2)); the canonical member STILL fails every tester; NEGATIVE CONTROLS: $SO(16)/D_8$ ($T_5 = 20/T_3 = -40$ exact, μ_2-only), wrong form ($|\text{Gauss}|^2 = 64$), wrong weight (dev 0.78), shuffle ($T_5 = -14/T_3 = 36$), planted family detected (1.5×10^{-16}), hol⊗hol empty; [C] TYPED PREMISES (the mandatory fence): TP-1 (F-independent moduli integral), TP-2 (nonvanishing — must source $\psi = 64$), TP-3 (`v518` kernel convention), TP-4 (Quillen structure of BCOV F_1); [O] the constructive BCOV derivation of the w_m normalisation itself; WIRKUNG (text only, no marker moves): the `v516` fence and the `v518` tension now read "decided at probe level for the declared reading under TP-1..TP-4 (v520)"; `SEAM.EQUIV.TWISTOR.01` stays Open; *VERDICT ERFOLG-A per the preregistered criteria — both constructive sources fire independently, the consistency closure carries*)

- **`v520_celestial_wp5e_measure_decision.py` (`CELEST.WP5E.MEASURE.01`, 39 checks, ∼20 s; exact $\mathbb{Q}(\zeta_8)$/Fraction/sympy rational nullspaces + 1/960-grid phase arithmetic + mpmath 30-digit kernels + numpy double SVD, deterministic).** S0 exact foundations (roots, ledgers, A-core + $\mathbb{Z}_2$-anchor targets, Weil relations, characters); S1 transports + derived families + the $k = 0$ slice; S2 D1 column canonicalisation + the sharpened kill; S3 D2 the exact invariant subspace; S4 D3 the $\mathbb{Z}_2$/EH anchor at the integral level; S5 source-1 (the F-lemma); S6 source-2 (the Quillen pairing, doubled 256-dim system); S7 the point-4 consistency closure; S8 negative controls; S9 the ERFOLG-A verdict + typed premises. Registered in `run_all.py`, `script_registry.csv` (cluster *frontier*), ledger row `CELEST.WP5E.MEASURE.01` (new).
- **Ledger/paper text precisions (markers unchanged).** `CELEST.WP5E.M2.01` (the `v516` fence: the completion reading is now DERIVED at probe level from two independent constructive sources under TP-1..TP-4; residual [O] = the constructive w_m derivation), `CELEST.WP5E.DELTA1.01` (the `v518` kill sharpened, the tension decided); `tfpt_research_contracts` (the WP5e measure-question prose updated throughout: object

diagram, WP4 arc, ε_1/M2 fences, the measure-question target block now carries the executed decision, the closing status lines); `tfpt_3` (Step 17/19 fences updated, new Step 21, the "what remains" block narrowed to the constructive w_m derivation); `tfpt_4_frontier` (celestial keybox updated). `veri` citations in `tfpt_research_contracts` and `tfpt_3`.

- **Wolfram.** Six exact mirrors (the 16-component Weil data with both Lagrangian gluings; the invariant-subspace forcing $\dim = 2 = \mathrm{span}\{e_H, e_{H'}\}$ by exact NullSpace + rank; the character arithmetic of the two sources $|\chi_4|^2 = 1$ vs $\chi_4^2 = e(2/3)$ with the exact grid defects; the A-core + $\mathbb{Z}_2$/EH-anchor targets with the scale teeth; the level-1 Okubo identity of BOTH gluings (240 norm-2 vectors each, summed quartic $36\langle x, x\rangle^2$); the so16/shuffle/diag(i, i) controls) added to `tfpt_readouts_extension.wl`; the Harvey–Moore integrals, SVD memberships and J-weighted closures stay Python-only, flagged in the `.wl` comment; combined engine run with the v521/v522 mirrors: extension $522 \to 540$ (verified, Wolfram Engine 14.3; `GATE.WOLFRAM.02`, Wolfram README, `tfpt_safeguards`, website counters and `zenodo_description.html` synced $116/116 + 540/540$).
- **Provenance.** Discovery probes in `experiments/tfpt-discovery/` (2026-07-22): `celestial_seam_wp5e_delta1e_bcov_measure_decision_probe.py` (28/28, ∼15 s) and `celestial_seam_wp5e_delta1f_single_valuedness_probe.py` (23/23, ∼17 s); suite $513 \to 516$ modules (green); website mirrors regenerated (`ScriptIndex`, DAG *qft/P2* nodes, `papers.ts` story counters, glossary, changelog mirror); `next.txt` round notes.

2026-07-22 (IX) (`v519` — the WOIT-α milestone of the OS twistor bridge `WOIT.OS.TWISTOR.01` executed and promoted as the ledger row `WOIT.THETA.FREE.01`: "the real structure exists — and free reflection positivity picks the same family"; the celestial/twistor story in `tfpt_3` is now TWENTY steps — (1) THETA CLASSIFICATION COMPLETE [E]: exactly TWO families of anti-linear structures on $\mathbb{C}^2$ normalise the clock $\rho = \mathrm{diag}(i, 1)$ — family D ($z \mapsto \mu\bar{z}$, $|\mu| = 1$): $\Theta\rho\Theta = \rho^{-1}$ EXACTLY at matrix level, $\Theta^2 = +1$ (-1 IMPOSSIBLE: $M\bar{M} = \mathrm{diag}(|\mu|^2, 1)$ — Kramers-free), inverts the deck, fixed set = the great circle through the deck poles; family A ($z \mapsto \mu/\bar{z}$): centralises the clock projectively (NEVER inverts it), $\Theta^2 = \pm 1$ per μ, fixed set $|z| = 1$ resp. empty; ROLE SEPARATION: family A DEFINES the euclidean section (Woit's ρ_{tw} side: $\rho_{\mathrm{tw}}^2 = -1$, no real points — replicated exactly on $\mathbb{C}^4$), family D REFLECTS it (the OS conjugation, σ_{std} = conjugation with real points $\mathbb{RP}^3$); (2) MARK PINNING [E]: $z \mapsto \mu\bar{z}$ preserves the μ_4 marks iff $\mu \in \mu_4$ — a 4-element μ_4-TORSOR (2 mark axes, 2 silver-midpoint axes), clock conjugation $\rho\,\Theta_\mu\,\rho^{-1} = \Theta_{-\mu} \Rightarrow 2$ clock orbits; (3) SPIN/FOCK [E]: the $\mathbb{Z}_8$ plane has no phase leaks ($\overline{\zeta_8^k} = \zeta_8^{-k}$); exact Cl(16): $\Theta_{\mathrm{Fock}} = U_r \circ K$ with $\Theta_{\mathrm{Fock}}^2 = 2^7 > 0$ (normalised $+1$), inverts the clock tower ($V \mapsto 4096\,V^{-1}$) and normalises the deck; the KRAMERS DICHOTOMY: the deck-induced candidate $\Theta_t = \tilde{U} \circ K$ has $\Theta_t^2 = (-1)^F$ — ***the `v510` split/nonsplit dichotomy IS the* $\Theta^2 = +1$ ***vs*** $(-1)^F$ ***dichotomy***; the deck does NOT furnish Θ, the seam-circle REFLECTION does; (4) KILL TEST 1 (free level) [E]: does NOT fire — Θ exists with $\Theta^2 = +1$ AND $\Theta\rho\Theta = \rho^{-1}$ and satisfies all three side conditions simultaneously (marks, deck normalisation, spin/Fock); (5) FREE RP [E]: the NS vacuum $C(d) = (2/N)/\sin(\pi d/N)$ has $C^2 = -1$ exactly; preconditions exact ($RCR^T = -C$, $SCS^T = C$, $RSR^{-1} = S^{-1}$; the deck FAILS anti-invariance); BOND CUT: one-particle Gram $(8, 0, 0)$ PD (min eigenvalue 1.888×10^{-3}, 40 digits), even $\deg \le 2$ sector $(29, 0, 0)$ PD, $N = 8$ complete half-sided algebra PD (even + odd each $(8, 0, 0)$) — RP with NO degree truncation; twist $\eta = +i$ FORCED ($\eta = 1$ non-Hermitian); CONTROLS: the site cut fails exactly ($\det = 0$, inertia $(3, 3, 1)$; continuum control: Cauchy–Stieltjes strictly positive — a lattice-PLACEMENT artifact, RP selects the site placement**

relative to the marks); the clock-CENTRALISING family-A structure fails RP structurally ($(4,4,0)$) — *RP and* $\Theta\rho\Theta = \rho^{-1}$ *select the SAME family*; bonus: the anti-chiral state flips the odd sector to $(0,8,0)$ ND (the free shadow of kill test 3); [C]/[O] NOT shown: the interacting algebra $\mathcal{A}_{\mathrm{hol}}$ (kill test 1 stays live there), gauge-fixed RP (kill test 2), OS reconstruction, chirality without mirrors (kill tests 3/6), μ_4 incidence (kill test 7) — the WOIT-β/γ roadmap named (β_1 Θ on the gauge-invariant subalgebra + gauge-fixed RP on the equivariant SDYM(E_8) sector; β_2 the OS quotient of the free system explicit; β_3 the $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$ duality vs σ_{std} typed; γ the chirality theorem + the mark incidence); TWO CONTRACT PRECISIONS (prose, no marker): (i) "Θ induced by the seam reflection" means the seam-circle REFLECTION, not the deck — the deck furnishes the $(-1)^F$ Kramers class (the `v510` bridge); (ii) the marks sit RP-side on the BOND MIDPOINTS of the 16-Majorana circle; Wolfram extension $516 \to 522$ (verified engine run); *NO marker moves — `WOIT.OS.TWISTOR.01` stays Open: kill test 1 is discharged on the FREE level only and stays formally live on the interacting algebra*)

- **`v519_woit_theta_rp_free.py` (`WOIT.THETA.FREE.01`, 34 checks, ∼20 s; exact sympy + exact Cl(16) Fractions; RP definiteness via 40-digit spectra of exact Hermitian matrices, tolerance 10^{-25}; deterministic).** Part W: W-S1 the two-family classification (symbolic solve); W-S2 family D (relations, Kramers-free, mark pinning, seam action, torsor); W-S3 family A (equatorial/ρ_{tw}); W-S4 the $\mathbb{C}^4$ twistor level (ρ_{tw} vs σ_{std}); W-S5 the $\mathbb{Z}_8$ spin level; W-S6 the Cl(16) Fock level ($\Theta^2_{\mathrm{Fock}} = 2^7$, $V \mapsto 4096\, V^{-1}$ via the implementer/centre route — a consolidation of the 105-s discovery probe to ∼20 s with identical claims and tolerances — deck normalisation, Kramers dichotomy, $\tilde{U}V = V\tilde{U}$ control); W-S7 verdict. Part R: R1 the exact preconditions; R2 the site cut fails; R3 the bond cut passes (incl. the complete $N = 8$ half-algebra); R4 the continuum control; R5 controls (twist forced, antipode fails, chirality flip); R6 verdict. Registered in `run_all.py` (suite 513 modules, green), `script_registry.csv` (cluster *frontier*), ledger row `WOIT.THETA.FREE.01` (new).
- **Contract updated (markers unchanged).** `WOIT.OS.TWISTOR.01`: the α stage moved to *done* with the core numbers and `veri{v519}`; the two precisions added as their own sentences (the input definition of Θ now names the seam-circle reflection, with the deck carrying the $(-1)^F$ Kramers class; the marks at the bond midpoints of the 16-Majorana circle); the β/γ roadmap bullets added with success and kill criteria; the object-diagram OS row and the closing status line carry the α-stage state. Ledger: `WOIT.THETA.FREE.01` new; `WOIT.OS.TWISTOR.01` text extended ("alpha stage executed via v519 … kill test 1 does NOT fire at the free level and stays live on the interacting algebra; beta/gamma milestones named"). NO marker moves anywhere.
- **Paper integration.** `tfpt_3` (new Step 20 "the real structure exists — and free reflection positivity picks the same family" — the celestial/twistor story is now twenty steps; the Woit-bridge paragraph extended by the α status: "the missing global object now exists at the free level; the interacting statement stays [**O**]"). `v519_woit_theta_rp_free.py` is cited via `veri` in `tfpt_research_contracts` and `tfpt_3`.
- **Wolfram.** Six exact mirrors (the two-family classification; mark pinning + the μ_4 torsor with the clock-orbit map and the cut-point census; the $\mathbb{C}^4$ level ρ_{tw} vs σ_{std}; the $\mathbb{Z}_8$ tower conjugation; the Cl(16) dichotomy 2^7 vs $(-1)^F$ on explicit Jordan–Wigner gammas incl. $U_r V U_r V = 2^{19}$ and $\tilde{U}V = V\tilde{U}$; the free-RP preconditions + site-cut degeneration + the Cauchy–Stieltjes continuum control) added to `tfpt_readouts_extension.wl` and verified with the active engine, $516 \to 522$; the 40-digit PD inertia certificates of the bond-cut Grams stay Python-only, flagged in the `.wl` comment; `GATE.WOLFRAM.02`, Wolfram README and the website counters synced ($116/116 + 522/522$).

- **Provenance.** Discovery probes `woit_os_theta_realstructure_probe.py` (21/21, ∼105 s) and `woit_os_theta_rp_freelevel_probe.py` (13/13, ∼5 s) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (`ScriptIndex`, DAG *qft* node + `v519`, `papers.ts` "20 steps" + the WOIT-contract mirror, glossary Θ/Woit-bridge entries, changelog mirror); `next.txt` round notes.

2026-07-22 (VIII) (`v518` — the WP5e δ_1 chain ($\delta_{1b}+\delta_{1c}+\delta_{1d}$) DECIDED and promoted as the ledger row `CELEST.WP5E.DELTA1.01`: "the derived chiral measure kills the δ_1 route" — an honest, decided NEGATIVE result in the `v508` style, with the `v516` tension stated: does the Harvey–Moore/BCOV fundamental-domain integral on $\mathbb{PT}/\mathbb{Z}_4$, evaluated under a measure DERIVED from blockwise $SL(2,\mathbb{Z})$ covariance of the dressed 16-component Weil system (solved for — not declared, not scanned), deliver the twisted contact vector $-A_{\rm fix}=-(9,-30,-15,0,32)$ or the $\psi=64$ slice? NO — (1) the 16-COMPONENT WEIL COMPLETION (δ_{1c}): the discriminant module of $D_5\oplus A_3$ is $\mathbb{Z}_4\times\mathbb{Z}_4$ with $q=(5x^2+3y^2)/8$, Gauss sums $2\zeta_8^5\times 2\zeta_8^3=4=\sqrt{16}$ (signature $0 \bmod 8=$ rank E_8), the Weil relations hold EXACTLY in $\mathbb{Q}(\zeta_8)$ ($S^2=C$, $(ST)^3=S^2$, $S^4=1$, $T^8=1$, tensor split $S=S_{D_5}\otimes S_{A_3}$), the diagonal H and anti-diagonal H' are the two invariant Lagrangians (both counts $=E_4$: two E_8 gluings), $P_HSP_H=\frac14 J$ with 3/4 of every S-image weight on the 12 shifted classes, and the 16-vector theta S-covariance CLOSES: the δ_{1b} E1.5 residual $2.91=O(1)$ (naive 4-character rule) drops to $\sim 10^{-39}$; (2) the OBSTRUCTION IS A FINITE μ_4 CHARACTER, IDENTIFIED (δ_{1d}): the dressed gauge blocks $M=G[a,b]\,\eta^{-8}$ carry constant transport (t_p,s_p) (28+ digits, recognised as exact 1/960-grid phases at 10^{-25}), the T-fixed pairs $(0,b)$ carry $t=e(-1/2)=-1$ EXACTLY (the naked level-matching defect), the transport-groupoid loops land in $\Gamma_1(4)$ resp. $\Gamma_0(2)^{\pm}$ with joint holonomy eigenspaces dim 8/16 resp. 6/16 and every eigen-phase of order ≤ 4; the KOBOUNDARY TEST: the $SL(2,\mathbb{Z})$ relation defects c_{S^4}, $c_{(ST)^3S^{-2}}$, $c_{[S^2,T]}$ are $(1,1,1)$ EXACTLY on all 15 pairs — the multiplier system is a CHARACTER of the orbit stabilisers, NOT a genuine 2-cocycle, and on $\Gamma_1(4)$ it is $\lambda(\gamma)=i^{2B+C/4}$; (3) the CANCELLATION EXISTS AND IS THE TWISTED FIBRE BLOCK f_1f_3, NOT THE THREE SPHERE AXIONS: $G[a,b]=f_1f_3$ is an exact identity (30+ digits; the axion T-phases are `v502` vacuum-energy data), the T-fix mechanism is one line of exact phase arithmetic — $(-1)\times e(-1/6)=e(1/3)=\chi_4(T)$ at every T-fixed node — and the exact CANCELLATION TABLE reads: none $\to$ residual order 4, $f_1f_2f_3$ (the three sphere axions) $\to 4$, $f_2\to 6$, $f_1f_3\to 1$ (hol AND anti), wrong weights $(1,1,2)/(2,2,2)\to 4$; the strict solutions are exactly χ_4 (dims $(3,3)$) and χ_{10} (dims $(1,1)$) at η^0, certified on the dressed functions at a fresh τ (10^{-15}, NO residual multiplier); (4) the DECISION — ALL THREE PREREGISTERED TESTERS FAIL: under both derived solutions the integral (30-digit kernels, negative-exponent cell audit, cut-6 vs cut-8 scan) fails $T_5(V)=0$ (fractions 0.5/0.83), fails $W_2{:}W_{13}=4{:}3$ ($W_{13}=0$), fails $V=-A_{\rm fix}N$ (spreads 1.05/1.47) and the $\psi=-64N$ slice, and the honest (N_1,N_2) orbit rebalance rescues nothing in the positive cone (χ_4: $T_5=0$ only at $\psi>0$; χ_{10}: only at $N_2/N_1=-4<0$) — a GENUINE KILL on the derived surface; (5) NEGATIVE CONTROLS: $\mathbb{Z}_2$/Eguchi–Hanson (gauge-only residual order 2 — the μ_4 defect is $\mathbb{Z}_4$-specific — and the SAME fibre dressing cancels it to 1), $SO(16)/D_8$ (only even sectors, all axion b-characters in μ_2: the order-4 supply structurally absent), wrong form $(x^2+y^2)/4$ ($|G|^2=64\neq 16$, its S-rule fails at $O(1)$), wrong signature ($S^2=e(-1/2)C$ exactly); [C] the $f_1f_3=$ KS-weight identification (deck weights $(1,3)=$ the twistor fibre weights forced by the `v514` incidence ledger — exact match, NOT a complete

BCOV derivation) + the kernel-family convention + the discrete χ_k choice (both evaluated, both fail); the MANDATORY TENSION FENCE: the DECLARED completion reading (`v516`/M2) delivers the $\psi = 64$ slice and cancels $32\,T_3$; the DERIVED chiral measure (this module) fails all three testers — both are exact; the sharp open question is which of the two is the true BCOV measure — neither result is hidden behind the other; [O] the BCOV-integral derivation on $\mathbb{PT}/\mathbb{Z}_4$ that decides between the two measures; Wolfram extension $510 \to 516$ (verified engine run); *NO marker moves — the δ_1 route is decided negative under the derived measure, the cubic-GS question stays dissolved, `SEAM.EQUIV.TWISTOR.01` stays Open, `SEAM.EQUIV.01` untouched*)

- **`v518_celestial_wp5e_delta1_derived_kill.py` (`CELEST.WP5E.DELTA1.01`, 30 checks, ∼20 s; exact $\mathbb{Q}(\zeta_8)$/Fraction/1-960-grid phase arithmetic + mpmath 30-digit kernels + numpy double SVD, deterministic).** S0 conventions + 16-class ledgers + targets; S1 the 16-component Weil system (exact) + the E1.5 closure; S2 gauge transport + the μ_4 multiplier system (koboundary test, character identified); S3 the cancellation ($G = f_1 f_3$, T-fix mechanism, exact table); S4 the strict solutions (constructive scan, $m \in \{0, 8, 16\}$); S5 the decision (function certificates, integrals, three testers, (N_1, N_2) scan); S6 negative controls; S7 verdict + the tension fence. Registered in `run_all.py` (suite 512 modules, green), `script_registry.csv` (cluster *frontier*), ledger row `CELEST.WP5E.DELTA1.01` (new — "DECIDED: kill under the derived measure, tension with v516 stated").
- **Contract updated (markers unchanged).** `CELEST.SEAM.01`: the δ_1 strand moves from "UNDECIDED at exploration level" to DECIDED by `v518`; the named remaining target is now the BCOV MEASURE QUESTION (the declared `v516` completion reading vs the derived `v518` chiral measure — both exact, which is the true BCOV measure is the named **[O]**); the cubic-GS question stays dissolved (`v518` does not revive the d-channel). `SEAM.EQUIV.TWISTOR.01` dependencies extended by `v518` (preparation range now `v492`–`v518`; the route stays Open). `CELEST.WP5E.M2.01` cross-referenced (its $[O]$ pointer "$\delta_{1b/1c/1d}$ territory" is executed and decided). NO marker moves anywhere.
- **Paper integration.** `tfpt_research_contracts` (the δ_1 stage moved to *done, DECIDED* with the four exact results and the mandatory tension fence as its own paragraph; "WP5e proper" renarrowed from the δ_{1d} derivation to the declared-vs-derived measure question; the object-diagram row, the honest status line and the per-ID status list extended by `CELEST.WP5E.DELTA1.01`); `tfpt_3` (new Step 19 "the derived measure decides — and disagrees with the declared one" — the celestial story is now nineteen steps; the Step-16/17 fences and "what remains" updated to the measure question); `tfpt_4_frontier` (keybox synced). `v518_celestial_wp5e_delta1_derived_kill.py` is cited via `veri` in `tfpt_research_contracts` and `tfpt_3`.
- **Wolfram.** Six exact mirrors (the 16-component Weil system with Gauss sums and tensor split; the two invariant Lagrangians + $P_H S P_H = \frac{1}{4} J$ + the 3/4 shifted-class weight; the koboundary test $(1, 1, 1)$ on all 15 pairs from the recorded exact transport phases; the $\Gamma_1(4)$ character formula $\lambda(\gamma) = i^{2B+C/4}$ on the recorded loops; the T-fix mechanism $(-1)\,e(-1/6) = e(1/3) = \chi_4(T)$ uniform for $f_1 f_3$ vs inconsistent $\{\chi_3, \chi_4\}/\{\chi_5, \chi_6\}$ for $f_1 f_2 f_3/f_2$; the wrong-form/wrong-signature controls) added to `tfpt_readouts_extension.wl` and verified with the active engine, $510 \to 516$; the Harvey–Moore integrals, SVD contraction solves, theta covariance certificates and block transport measurements are numerical and stay Python-only, flagged in the `.wl` comment; `GATE.WOLFRAM.02`, Wolfram README and the website counters synced (116/116 + 516/516).
- **Provenance.** Discovery probes `celestial_seam_wp5e_delta1b/1c/1d_*.py` — the Harvey–Moore unfolding (30/30), the Weil-16 completion (31/31) and the order-4 multiplier cancellation (36/36) — in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated

(`ScriptIndex`, DAG *qft* node + `v518`, `papers.ts` "19 steps", changelog mirror); `next.txt` round notes.

2026-07-22 (VII) (`v517` — WP5e back-reaction milestone M3 "the a_0 uplift" executed (SUCCESS on the preregistered `v514` S8.3 criterion), promoted as the ledger row `CELEST.WP5E.M3.01`: the $(4, \pm 4)$ GH multipole (= the a_0 direction) is uplifted to the twistor-potential kernel $\chi = \log P_4$ — the LOG of the `v515` family polynomial — and the induced Kähler-potential correction is a log-type correction COUPLED TO THE CENTRE COUNT ON FOUR SCALES — (1) the KERNEL BRIDGE: the $O(2)$ section is a NULL coordinate ($\Delta_{3d} G(\eta_p) = 0$ identically for an ARBITRARY kernel G: the uplift is well-typed), the RESIDUE IDENTITY $\partial_x[\text{transform}(\log \eta_p)] = 1/r_p$ holds EXACTLY (residue $1/(2r)$), so the GH potential $V = \sum_p 1/r_p$ IS the x-derivative of the contour transform of χ; flux -4π per centre, source charge $4 = |\mu_4|$ (the V-ledger matched); (2) the ASYMPTOTIC KERNEL LOG: $\chi = 4 \log \eta + a_2/\eta^2 + (a_0 - a_2^2/2)/\eta^4 + \ldots$ with LOG COEFFICIENT $4 = |\mu_4|$ = centre count (the CPS "$N \log$" analogue at kernel level); at the clock-fixed seam fibre $\zeta = 0$ the a_2 slot closes and the first correction is EXACTLY a_0/η^4 = the $(4, \pm 4)$ multipole in kernel clothes; exact m-grading ($m = \pm 4$ pieces PURE a_0, $m = 0$ piece $-6t_0^4$ φ_0-free: the $U(1) \to \mathbb{Z}_4$ breaking sits exclusively in the a_0 pieces); NO $\log \times$power terms (the `v514` honest zero preserved — the a_0 correction is power-law at infinity); the once-integrated GLT tower $\sum_p \eta_p \log \eta_p = 4\eta \log \eta + p_4/(12\eta^3) + p_8/(56\eta^7) + \ldots$ with $p_n = 0$ unless $n \equiv 0 \bmod 4$ and $p_{4k} = 4(-a_0)^k$: EVERY correction carries the prefactor 4 = centre count; (3) the EXCEPTIONAL-LOCUS LOG: $\chi(\eta = 0) = \log a_0 = 4 \log t_0 + i(4\varphi_0 + \pi)$ (t_0-coefficient $4 = |\mu_4|$, clock phase $4i$); the modulus response $a_0 \partial_{a_0} \chi = a_0/(\eta^4 + a_0)$ equals 1 AT the exceptional locus and falls off as a_0/η^4 at infinity (no constant, no log): the a_0-log lives at the exceptional locus, exactly as `v514` S6.3 demanded; (4) the PERIOD RESPONSE: $\Pi_j(\lambda)$ is LINEAR in t_0 (uniform first-order shift: lockstep + phase ratios $(1, i, i^2)$ preserved exactly), $d \log \Pi_j / d \log a_0 = 1/4 = 1/|\mu_4|$ uniformly, and the integrated shift around the a_0 circle is the monodromy $e^{2\pi i/4} = i$ = ONE Coxeter clock step (the `v493` monodromy REPRODUCED from perturbation theory; the finite monodromy is the cyclic centre permutation $q_p \to q_{p+1}$); the node λ-support is t_0-FREE (the $\mathbb{Z}_8$ seam bridge is a_0-rigid), all 6 pair discriminants $\sim t_0^2 \neq 0$ (K_4 connectivity: the forced uniform flux vector persists), the BM kernel is modulus-free with $\int_{S^3} K = (2\pi i)^2$ (the quantised fluxes cannot move); (5) NEGATIVE CONTROLS: the $(4, 0)$ multipole is φ_0-independent in V AND in the kernel (clock-invariant, breaks nothing); the $\mathbb{Z}_2$/Eguchi–Hanson analogue reads $2 = |\mathbb{Z}_2|$ on EVERY dial ($(2, \pm 2)$ amplitude $\frac{3}{2} \sin^2 \theta$, kernel log 2, exceptional log $2 \log t_0$, response $1/2$, monodromy -1); wrong centre counts $k = 3, 5$: kernel log coefficient k, modulus slot $O(6)/O(10) \neq O(8)$ — the observable MOVES with the centre count; forbidden family $Z^4 - Z$: $p_3 = 3 \neq 0$ (selection rule broken), $e_4 = 0$ (no a_0-log) — fails BOTH uplift dials; [C] the GLT dictionary ($\eta_p \log \eta_p$ = the GH Kähler-potential kernel, Lindström–Roček / Ivanov–Roček); [O] the full nonlinear Kähler potential of the resolved A_3 ALE; with `v516` the `v514` fence M1–M3 is FULLY WORKED OFF; Wolfram extension $504 \to 510$ (verified engine run); *NO marker moves — the KILL (decoupling from the centre count) does NOT fire, `SEAM.EQUIV.TWISTOR.01` stays Open (preparation, not construction), `SEAM.EQUIV.01` untouched*)

- **`v517_celestial_wp5e_m3_a0_uplift.py` (`CELEST.WP5E.M3.01`, 23 checks, ∼7 s; exact sympy, no floats; deterministic).** S0 replication (GH multipole ledger + the twistor family

at general orbit phase, `v515` slice); S1 the kernel bridge (null coordinate, residue V-ledger identity, source flux); S2 the uplift (asymptotic log 4, seam-fibre correction a_0/η^4, exact m-grading, GLT tower + selection rule); S3 the exceptional-locus log; S4 Ω_N consistency (period response $1/4$, monodromy i, a_0-rigid nodes and fluxes); S5 negative controls; S6 verdict. Registered in `run_all.py` (suite 511 modules, green), `script_registry.csv` (cluster *frontier*), ledger row `CELEST.WP5E.M3.01` (new, SUCCESS).

- **Contract updated (markers unchanged).** `CELEST.SEAM.01` now reads: M2+M3 executed via `v516`/`v517` — the `v514` fence M1–M3 is fully worked off: the twisted KS measure (declared completion reading, ch_2-weighted) cancels the $32\,T_3$ residual and delivers the $\psi = 64$ slice without the cubic d-channel, and the a_0 uplift couples to the centre count on four scales; what stays **[O]**: the BCOV-integral derivation of the completion reading (δ_{1d}) and `SEAM.EQUIV.TWISTOR.01`. `SEAM.EQUIV.TWISTOR.01` dependencies extended by `v516`/`v517` (preparation; the route stays Open).
- **Paper integration.** `tfpt_research_contracts` (the M3 bullet moved to *done* with the four coupled scales and the GLT fence; "what remains" narrowed to δ_{1d} + the BCOV derivation); `tfpt_3` (new Step 18 "the a_0 uplift: four coupled centre-count scales" — the celestial story is now eighteen steps; "what remains" updated). `v517_celestial_wp5e_m3_a0_uplift.py` is cited via `veri` in both.
- **Wolfram.** Six exact mirrors (multipole ledger + family at general phase; kernel bridge (null coordinate, residue $1/(2r)$, flux -4π); asymptotic log + m-grading + GLT tower; exceptional-locus log; period response $1/4$ + monodromy i + a_0-rigidity; negative controls) added to `tfpt_readouts_extension.wl` and verified with the active engine, $504 \to 510$; `GATE.WOLFRAM.02`, Wolfram README and the website counters synced ($116/116 + 510/510$).
- **Provenance.** Discovery probe `celestial_seam_wp5e_m2m3_a0_uplift_probe.py` (23/23) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (`ScriptIndex`, DAG *qft* node + `v517`, `papers.ts` "18 steps", changelog mirror); `next.txt` round notes.

2026-07-22 (VI) (`v516` — WP5e back-reaction milestone M2 "the twisted Kodaira–Spencer measure" executed (SUCCESS on the preregistered `v514` S8.2 criterion, ON THE DECLARED COMPLETION MEASURE — verdict B), promoted as the ledger row `CELEST.WP5E.M2.01`: the one-loop box coefficient per sector on $\mathbb{PT}/\mathbb{Z}_4$ under the completion contact term $\mathrm{contact}_j = (Q^{(0)} - Q^{(j)})/\det_j$ — the twisted-sector LOOP contributes the UNPHASED sector trace with the g^j zero-mode normalisation $1/\det_j$, of which the Atiyah–Bott skeleton kept only the phase-weighted insertion part — (1) the COMPLETION-WEIGHT IDENTITY (the centrepiece): per glue class $w_m = \sum_j (1 - i^{jm})/\det_j = (0, \frac{3}{2}, 2, \frac{3}{2}) = 4h_m = |\mu_4| h_m = -4\,\mathrm{ch}_2(T_m)$ EXACTLY ($h_m = m(4-m)/8$ the `v502` sector Casimir energies, $\mathrm{ch}_2(T_m)$ the McKay/Kronheimer charges, `v505` S4.3 index bridge): THE THREE SPHERE AXIONS PAIR THROUGH THEIR OWN ch_2 CHARGES — an exact identity, NO free scale, no fit; (2) PARAMETER-FREE LOCKS: $T_5 = 0$ for ANY scale c (the ch_2 pattern alone kills T_5); ratio $h_2 : h_1 = 4 : 3 =$ the δ_1-forced leading ratio (REPRODUCED, not fitted); the T_3 budget fixes $c = 4 = |\mu_4|$ uniquely (leading system det 256, unique weights $(\frac{3}{2}, 2) = (4h_1, 4h_2)$); (3) the SUCCESS TABLE: $\mathrm{skeleton}_j + \mathrm{contact}_j = Q^{(0)}/\det_j = 36\langle x, x\rangle^2/\det_j$ — a PERFECT Okubo square ($T_5 = T_3 = 0$, $\tilde{\lambda}^2 = 36$ in EVERY twisted channel: $(18, 9, 18) \times \langle x, x\rangle^2$); total $A_{\mathrm{fix}} + \mathrm{contact} = 45\langle x, x\rangle^2 = \frac{5}{4} \times 36\langle x, x\rangle^2 =$ Dedekind × Okubo = the UNIQUE quartic-free weighting of `v505` S3.8 (the rigidity theorem is exactly what the measure lands on); both `v508` certificates KILLED ($\Phi_{T_3} : 32 \to 0$, $\Phi_P : 72 \to 0$); the `v511` slice $\psi = 64$ SUPPLIED EXACTLY ($\psi(\mathrm{contact}) = -64$) — NO cubic d-channel needed (c_d free $= 0$); remainder inside the exchange span ($45\langle x, x\rangle^2 = \frac{45}{16} E_{22} = \frac{1}{80} E_{00}$, $\Phi_P = 0$): the BULK axion cancels every channel with the SAME $\tilde{\lambda}^2 = 36$, the twisted propagator carrying the AB zero-mode factor $1/\det_j$; the `v514` measure chain stays μ-blind; (4) NEGATIVE CONTROLS: wrong scale $c \in \{1, 2, 3, 5\}$ leaves $T_3 = (24, 16, 8, -8)$ — only $c = 4 = |\mu_4|$ clears the budget; the $h_1 \leftrightarrow h_2$ shuffle breaks $T_5 = -14$ AND $T_3 = 36$; $SO(16)$ glue keeps $T_5 = 20$ / $T_3 = -40$ — the KILL branch FIRES there (Okubo failure inherited: E_8 doubly special); $\mathrm{diag}(i, i)$ degenerates to the $\mathbb{Z}_2$ target and breaks the conjugation pairing + the index bridge; the $\mathbb{Z}_2$/Eguchi–Hanson anchor PASSES with the same mechanism ($A_2 + \mathrm{contact} = 9\langle x, x\rangle^2 = \frac{1}{4} \times 36\langle x, x\rangle^2$, coupling scale $2 = |\mathbb{Z}_2| =$ centre count); (5) the MULTIPLIER NOTE (cross-reference to the parallel δ_{1d} strand): the contact weights $4h$ are REAL RATIONALS with no multiplier system of their own; the δ_{1c} residual phases are the SQUARES of the A_3 discriminant T-phases $e(3a^2/8)$ — orders $(1, 8, 2, 8)$ squaring to $(1, 4, 1, 4)$, a finite μ_4 system; Gauss sums $2e(3/8)$, $2e(5/8)$, full Weil sum 4; [C] the MANDATORY FENCE: the completion reading (the loop of twisted states carries the UNPHASED sector trace with the same zero-mode normalisation) is DECLARED, supported by the δ_1 modular-completion finding, NOT derived from the BCOV integral; [O] the BCOV-integral derivation ($\delta_{1b}/\delta_{1c}/\delta_{1d}$ territory); Wolfram extension $498 \to 504$ (verified engine run); *NO marker moves — the KILL (leftover independent quartic in ANY sector) does NOT fire, the completion reading stays [C]-declared, `SEAM.EQUIV.TWISTOR.01` stays Open, `SEAM.EQUIV.01` untouched*)

- **`v516_celestial_wp5e_m2_twisted_ks_measure.py` (`CELEST.WP5E.M2.01`, 23 checks, ∼1 s; exact sympy/Fraction, no floats; deterministic).** S0 replication (roots, class quartics, A_{fix} two routes, bulk Okubo); S1 the index bridge + the completion weights (no free scale, parameter-free locks); S2 the contact term (two forms, genuine-contact certificate); S3 the success table (per-sector Okubo squares, Dedekind × Okubo, certificate kills, exchange span, μ-blind chain); S4 negative controls; S5 the multiplier note (cross-reference to δ_{1d}); S6 verdict. Registered

in `run_all.py` (suite 511 modules together with `v517`, green), `script_registry.csv` (cluster *frontier*), ledger row `CELEST.WP5E.M2.01` (new, SUCCESS on the declared measure).

- **Contract updated (markers unchanged).** `CELEST.SEAM.01`: the M2 bullet moves to *done*; the `v511` burden paragraph is closed out — the $\psi = 64$ slice is DELIVERED by the declared KS measure (`v516`); the cubic d-channel is not needed — its physical-justification question dissolves unless δ_{1d} revives it. The completion reading itself stays **[C]**-declared; its BCOV-integral derivation stays **[O]** (δ_{1d}).
- **Paper integration.** `tfpt_research_contracts` (the M2 bullet moved to *done* with the weight identity and the success table; the `v511` fence paragraph extended); `tfpt_3` (new Step 17 "the twisted KS measure: every sector the same Okubo square" — on the way to the eighteen-step story). `v516_celestial_wp5e_m2_twisted_ks_measure.py` is cited via `veri` in both.
- **Wolfram.** Six exact mirrors (completion-weight identity + index bridge; root/quartic replication; parameter-free locks; contact term + success table + certificates + μ-blind chain; negative controls; multiplier note) added to `tfpt_readouts_extension.wl` and verified with the active engine, 498 → 504; `GATE.WOLFRAM.02` and the Wolfram README synced.
- **Provenance.** Discovery probe `celestial_seam_wp5e_m2m3_twisted_ks_measure_probe.py` (23/23) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (`ScriptIndex`, DAG *qft* node + `v516`, changelog mirror); `next.txt` round notes.

2026-07-22 (V) (`v515` — WP5e back-reaction milestone M1 "the A_3 Ω_N" executed (SUCCESS on the preregistered `v514` S8.1 criterion), promoted as the ledger row `CELEST.WP5E.M1.01`: the back-reacted holomorphic 3-form Ω_N is CLOSED FORM on the A_3 twistor family, every $S^3/\mathbb{Z}_4$ period is $(2\pi i)^2$-integral with the lockstep flux vector, and the lens geometry FORCES the source charge $4 = |\mu_4|$ — (1) the RESIDUE FORM DERIVED: on $F = XY - P(Z)$ all three ambient representatives satisfy $dF \wedge \omega_2 = -dX{\wedge}dY{\wedge}dZ$ exactly and pull back to $\omega_2 = dX{\wedge}dZ/X$ (Poincaré residue, derived not assumed); orbifold-cover pullback at $a_0 = 0$: $\omega_2 = 4\,dz_1{\wedge}dz_2 = |\mu_4|\times$(flat form) — the residue normalisation CARRIES the source charge 4; clock $\omega_2 \to i\,\omega_2$ (`v492` S5 replicated); weight ledger $4{+}4{+}2{-}8 = 2$, $\Omega \in O(4) = -\deg K_{\mathbb{PT}}$; (2) the TWISTOR FAMILY CLOSED: from the four $O(2)$ centre sections $q_p(\lambda) = i^p t_0 - i^{-p} t_0 \lambda^2$ exactly $XY = Z^4 + 4t_0^2\lambda^2 Z^2 - t_0^4(1-\lambda^4)^2$ ($e_1 = e_3 = 0$ identically; $\lambda = 0$ gives exactly the `v493` family; $a_2 \in O(4)$ opens only off-seam), CY-compatible clock lift $\gamma : (Z, \lambda) \mapsto (iZ, -i\lambda)$, $\gamma^4 = 1$, $\Omega \to +\Omega$; (3) Ω_N CLOSED + PERIOD TABLE: $\Omega_N = \Omega_0 + \sum_p N_p K_p$ with CPS/Bochner–Martinelli kernels on the centre twistor lines, $dK = 0$ off-source; boundary $S^3/\mathbb{Z}_4$ at the orbifold point $(2\pi i)^2 \times 1$ (minimal invariant charge $4 = |\mu_4|$ — the lens fundamental domain FORCES $N \in 4\mathbb{Z}$), deformed phase $(2\pi i)^2 \times 4$, linking S^3 uniformly $(2\pi i)^2 \times N$, per-sphere flux vector $(2\pi i)^2 \times N(1,1,1)$ (forced uniform: clock orbit + K_4 connectivity), seam-fibre phase vector $2\pi i\, t_0(i-1)(1, i, i^2)$; Ω_0 carries ZERO 3-flux (all flux sourced); $(2\pi i)^2$ = fibre residue × transversal residue (density $-\sin 2\theta$); (4) CLOCK COVARIANCE: $\Pi_{j+1}(-i\lambda) = i\,\Pi_j(\lambda)$ for ALL λ; the 12 collision nodes sit exactly on the 8 eighth roots of unity ($8 = 2|\mu_4|$, the $\mathbb{Z}_8$ bridge), clock orbits $4{+}4{+}4$, honest conifold points (Hessian det $512t_0^6$, quadric rank 4); (5) the HONEST KILL PROBE: forbidden family $Z^4 - Z$: 0/24 lockstep (vs 8/24), λ-support 12 twelfth roots $\neq \mathbb{Z}_8$, no antipodal pairing — BUT the $(2\pi i)^2$ quantisation holds there TOO: integrality alone is NOT the discriminator, the discriminator is the lockstep phase structure + the clock forcing; the `v509` dial survives; (6) ANCHORS + CONTROLS: EH/$\mathbb{Z}_2$ (charge $2 = |\mathbb{Z}_2|$ forced), CPS original (every integer N allowed), $\mathrm{diag}(i, i)$ ($\Omega_0 \to -\Omega_0$, collapse at step zero), fractional charges $N = 1, 2, 3$ excluded; [C] global kernel patching / Hitchin small resolution / CPS brane dictionary; [O] M2 (twisted KS measure), M3 (a_0 uplift), the quantised BCOV coefficient; Wolfram extension $490 \to 498$ (verified engine run); *NO marker moves — M2/M3 and BCOV stay [O], `SEAM.EQUIV.TWISTOR.01` stays Open (`v515` is preparation, not construction), `SEAM.EQUIV.01` untouched*)

- **`v515_celestial_wp5e_m1_omega_n.py` (`CELEST.WP5E.M1.01`, 30 checks, ∼1 s; exact sympy, no floats; deterministic).** S1 the residue form (three routes, cover factor 4, weight ledger, EH anchor, wrong-weight control); S2 the twistor family (closed form, $O(2)$ reality, clock lift, section-level dichotomy); S3 periods (generic reduction, lockstep, covariance, collision ledger on $\mathbb{Z}_8$); S4 Ω_N (BM anchor, lens descent, source curve, conifold nodes, M1 criterion); S5 the KILL probe (honest); S6 negative controls; S7 fences + verdict. Registered in `run_all.py` (suite 509 modules, green), `script_registry.csv` (cluster *frontier*), ledger row `CELEST.WP5E.M1.01` (new, SUCCESS).
- **Contract updated (markers unchanged).** `CELEST.SEAM.01` now reads: M1 executed via `v515` — the back-reacted Ω_N is closed-form on the A_3 twistor family, all $S^3/\mathbb{Z}_4$ periods are $(2\pi i)^2$-integral with the lockstep flux vector and clock covariance, and the lens geometry FORCES the source charge $4 = |\mu_4|$; honest fence: integrality alone does not discriminate — the discriminator is the lockstep phase structure (0/24 on the forbidden family); M2/M3 stay **[O]**. `SEAM.EQUIV.TWISTOR.01` dependencies extended by `v515` (preparation; the route stays Open).
- **Paper integration.** `tfpt_research_contracts` (the M1 bullet moved to *done* with the core

numbers; the object-diagram row $\mathbb{C}^2/\mathbb{Z}_4 \to \mathbb{PT}/\Gamma$ extended by `v515` with the M1 kill retired to δ_1/M2–M3; WP4 handover, WP5 narrative, honest status line and status inventory synced); `tfpt_3` (new Step 16 "the back-reacted Ω_N: closed form, integral periods, forced charge 4" — the celestial story is now sixteen steps; "what remains" updated: M2/M3 are the remaining preregistered milestones); `tfpt_4_frontier` stage list synced. `v515_celestial_wp5e_m1_omega_n.py` is cited via `veri` in both.

- **Wolfram.** Eight exact mirrors (residue-form identities; twistor family closed; period reduction + lockstep + covariance; collision ledger on $\mathbb{Z}_8$; BM anchor + lens forcing $N \in 4\mathbb{Z}$; source curve K_4; conifold nodes; KILL probe + EH/diag(i,i) controls) added to `tfpt_readouts_extension.wl` and verified with the active engine, 490 → 498; `GATE.WOLFRAM.02`, Wolfram README and the website counters synced (116/116 + 498/498).
- **Provenance.** Discovery probe `celestial_seam_wp5e_m1_a3_omega_n_backreaction_probe.py` (30/30) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (`ScriptIndex`, DAG *qft* node + `v515`, `papers.ts` "16 steps", changelog mirror); `next.txt` round notes.

2026-07-22 (IV) (external review integrated: route renamed, object diagram, group-extension definition, SEAM.EQUIV route split (MMST/TWISTOR ids, no status change), A_3 role table, `WOIT.OS.TWISTOR.01` contract — a pure structure/text integration, NO new physics claims, NO marker moves, suite unchanged at 508)

An external expert review of the celestial/twistor branch confirmed the substance and asked for *semantic* precision — role separation, status clarity, and the exact naming of what is missing. Six changes plus one new central research contract, all editorial/structural:

- **(Ä1) Route renamed + the Woit bridge.** The `tfpt_3` celestial section and the contract heading are now **"The celestial and twistor continuum route"** (the old "from the μ_4 clock to the $(E_8)_1$ boundary shadow" survives as the narrative-arc subtitle). A new paragraph block in `tfpt_3`, *"The Woit bridge: chiral Wick rotation and the real structure"* **[O]**, introduces Woit's Euclidean Twistor Unification (`arXiv:2104.05099`) as a *named external programme* (chiral Wick rotation problematic for purely chiral theories; Euclidean-holomorphic theory on $\mathbb{PT}$, Lorentzian theory via conjugation structure/boundary values), lists TFPT's existing ingredients (RP `v379`, OS reconstruction, ρ = clock `v492`, σ = deck/seam reflection `v510`), and names exactly what is missing: a global anti-linear $\Theta : \mathcal{A}(\mathbb{PT}/\Gamma) \to \mathcal{A}(\mathbb{PT}/\Gamma)$ with $\Theta^2 = 1$, $\Theta\rho\Theta = \rho^{-1}$, plus RP of the interacting BCOV+SDYM functional — "until that exists, the Woit connection is a strong analogy, not a mathematical integration". No namedropping as confirmation.
- **(Ä2) The object diagram.** `tfpt_research_contracts` (CELEST section, up front) now opens with the chain $(\Sigma, \mu_4, \rho, \Theta) \to \mathbb{P}^1\backslash\mu_4 \to \mathbb{C}^2/\mathbb{Z}_4 \to \mathbb{PT}/\Gamma \to \mathcal{A}_{\text{hol}} \to_{\text{OS}} \mathcal{A}_{\text{Mink}}$ as a table with one row per arrow (claim id, status, load-bearing modules `v492`/`v493`/`v497`–`v514`, kill test); the last two arrows are honestly **[O]** with `WOIT.OS.TWISTOR.01` as the target — where mathematics ends and physics begins is visible at a glance.
- **(Ä3) The group extension, exactly.** A definition block: $\Gamma_{\text{ALE}} = \langle \text{Deck} \rangle \cong \mathbb{Z}_4 \subset SU(2)$ (triholomorphic, preserves Ω; the quotiented $\mathbb{Z}_4$); $\rho = \text{Clock} = \text{diag}(i, 1) \in U(2)$ (Kähler, *not* triholomorphic, det $= i$ rotates Ω by i — `v492` S5; normalises, is not quotiented); the global object is cyclic $\langle\rho\rangle$ with $\rho^2 = \text{Deck}$ (projectively $\mathbb{Z}_4$, spin lift $\mathbb{Z}_8$; `v492`/`v506`/`v507`), plus the action table on $K_{\mathbb{PT}}$, Ω, BCOV fields, open-string fields and boundary states, each entry **[E]** (arithmetic) or **[O]** (quantisation question); "$\mathbb{PT}/\mathbb{Z}_4$" is defined once for the whole contract (quotient by Γ_{ALE}, normalised by $\langle\rho\rangle$) with a cross-reference sentence in `tfpt_3`.
- **(Ä4) SEAM.EQUIV route split (ledger restructuring, no status change in substance).** New ledger rows `SEAM.EQUIV.MMST.01` (the conditional scaling-limit route — closed

modulo cited theorems, typing taken verbatim from the pre-split rows/prose; residual [**O**] = the cited continuum existence, `v336`) and `SEAM.EQUIV.TWISTOR.01` (the open Costello–Li construction, [**O**]; dependencies on `CELEST.SEAM.01`/WP modules and `WOIT.OS.TWISTOR.01`). `SEAM.EQUIV.01` *remains* as the parent target with the OR logic as claim text: *closed if MMST.01 OR TWISTOR.01 closes; currently: MMST conditional-closed modulo cited theorems, TWISTOR open ⇒ parent stays **[O]** as an unconditional claim.* Every surface that said "`SEAM.EQUIV.01` closed modulo cited theorem" (status cards in `introduction`, `tfpt_1/3/4`, the central status card, README, `origin_theory`, `tfpt_2`, `tfpt_5`, website StatusPyramid/OpenGates/StatusMatrix/`papers.ts`/glossary/VerificationDag) now attaches the statement to the MMST route id — the assertions are unchanged, only the ids are clean.

- **(Ä5) The A_3 role table.** In the contract: A_3^{family} (lattice family factor, $\mathfrak{su}(4)$ flavour), A_3^{ALE} (ADE type of $\mathbb{C}^2/\mathbb{Z}_4$), the three exceptional spheres (Coxeter/PL structure), D_5^{carrier}, and — as external reference rows — Woit's $SU(3)_{\text{color}}$ and $SU(2)_{\text{weak}}$: per pair either the connecting functor with its `vN` witness (McKay type-level; Kronheimer–Nakajima quiver `v479`) or an explicit *"not identical — no identification claimed"*; in particular $A_3^{\text{family}} \not\cong SU(3)_{\text{color}}$, and the Woit rows are pure programme references.
- **(Ä6) Anti-number-hunting programme note.** `experiments/next.txt` (dated 2026-07-22 (IV)): the next real gain lies at positivity, incidence, locality and the Θ real structure — *not* at further factorisations; recorded as working discipline per the review.
- **New central contract `WOIT.OS.TWISTOR.01` (Open, research contract; new ledger row).** In `tfpt_research_contracts` after `CELEST.SEAM.01`: input $X_+ = (\mathbb{PT}/\Gamma)_+$, $\mathcal{A}_{\text{hol}}$ (interacting open+closed twistorial algebra), ρ (order-4 clock), Θ (seam-reflection-induced anti-linear real structure), $\mu_{\text{BCOV+SDYM}}$; target theorem $\Theta^2 = 1$, $\Theta\rho\Theta = \rho^{-1}$, RP of the gauge-invariant algebra, OS quotient $\Rightarrow (\mathcal{H}, \Omega, U(\mathcal{P}_+^{\uparrow}), \mathcal{A}_{\text{Mink}})$ with positive Hilbert metric, positive energy, local causality, *one chiral generation without mirror doubling*, the TFPT charge lattice, internal $SU(3)\times SU(2)\times U(1)$ action, and the $(E_8)_1$ seam net as an *actual* boundary net; *seven* preregistered kill tests (Θ–clock incompatibility; RP failure after gauge fixing; vector-like mirror generation; Penrose transform only free/self-dual; right character but wrong OPE/net equivalence; $SU(2)$ stays a spacetime factor; the four μ_4 marks not incidence-compatibly extendable); typing "this is the actual bridge from compiler to physics — everything in `CELEST.SEAM.01` is preparation for it"; honest scope fence (self-dual ≠ full 4D physics: both helicities, generic amplitudes, local matter, full Einstein dynamics, EWSB, confinement are explicit non-claims of the celestial branch). Mirrored on the website (`papers.ts` section, glossary entries "Woit bridge" and "Real structure Θ").
- **No marker moves; no new `vN` modules; suite unchanged (**508 **modules, green).**

2026-07-22 (III) (`v514` — WP5e-ε_1 "the $O(-2)$ bulk-axion slot" executed (verdict B), promoted as the ledger row `CELEST.WP5E.EPS1.01`: the $s = 0$ tower field of the celestial route is now a CONSTRUCTION, $\tilde{\lambda} = 6$ is pinned three ways, and the first honest back-reaction step is taken in Gibbons–Hawking form — (1) the EQUIVARIANT $O(-2)$ SLOT: the pushforward ledger $\pi_* O(-2)$ on $\mathbb{PT}' = \mathrm{Tot}(O(1) \oplus O(1) \to \mathbb{P}^1)$ gives per degree d exactly $(d+1)^2$ classes = the exact wave-operator nullspaces ($1, 4, 9, 16, 25, 36$ for $d = 0..5$), the `v492` action on the fibre coordinates $(\mu_1, \mu_2) \mapsto (i\mu_1, i^{-1}\mu_2)$ with incidence-forced column weights $(+1, +1, -1, -1)$ closes the equivariant bookkeeping BLOCK BY BLOCK (all $d \leq 6$, all 4 characters); character series $P_0 = 1 + 3t^2 + 15t^4 + 21t^6 + 45t^8 + \ldots$, $P_1 = P_3 = 2t + 8t^3 + 18t^5 + \ldots$, $P_2 = 6t^2 + 10t^4 + 28t^6 + \ldots$; the $d = 0$ slot has multiplicity $(1, 0, 0, 0)$ — THE BULK AXION SURVIVES THE PROJECTION; Molien invariant ring $= (1 - t^8)/((1 - t^4)^2(1 - t^2))$ — $Z \in O(2)$, $X, Y \in O(4)$, relation degree 8 = the a_0 weight (`v492`/`v493` bridge); E3 bijection sharpened (twisted minimal content $(2t, 6t^2, 2t)$ = the Coxeter eigenvalues, $\det(A - 1) = -4$); graviton control: $O(+2)$ invariant only from fibre degree 4 with multiplicity $3 = \{X, Z^2, Y\}$; (2) $\tilde{\lambda} = 6$ TRIPLY PINNED: Okubo $\mathrm{Tr}_{\mathrm{adj}}\, X^4 = 36\langle x, x\rangle^2 = (6\langle x, x\rangle)^2$ on the 240 glue roots ($36 = h^\vee + 6$, $\tilde{\lambda} = 6 = |\mathbb{Z}_2| N_{\mathrm{fam}}$); the measure chain cancels $\mu = 1/4$ EXACTLY (vertex^2 μ^2, propagator $1/\mu$, anomaly μ; $\tilde{\lambda}^2_{\mathrm{eff}} = 36$ for arbitrary μ; wrong bookings EXCLUDED: anomaly-only $\Rightarrow 3$, missing propagator renormalisation $\Rightarrow 12$); flux (minimal CPS quantum $N = 1$, single channel iff $k = 1$, $(\kappa/c_3)^2 = 12$ exact); (3) the GH/A_3 BACK-REACTION STEP: invariant 4-centre configurations = exactly 2 branches, the free orbit re-derives the `v493` family ($\prod_p (Z - i^p z_0) = Z^4 - z_0^4$, $a_0 = -z_0^4$, monodromy = one clock step), the charge-k GH point is exactly $\mathbb{R}^4/\mathbb{Z}_k$ ($k = 4$: seam boundary $S^3/\mathbb{Z}_4$), periods $\Pi_j = 4\pi t_0 (i - 1)(1, i, i^2)$ LOCKSTEP (`v509` counter-check), Coxeter clock from geometry ($\Pi \cdot A = i\Pi$), source charge $4 = |\mu_4|$, EH asymptotic log coefficient EXACTLY 0 (the CPS log is an exceptional-locus statement; Burns contrast $\det g \neq 1$), multipole rule $m \equiv 0 \bmod 4$ with $(4, \pm 4)$ carrying $-a_0$; controls $\mathfrak{so}_8$ ($\tilde{\lambda}^2 = 12$ irrational, squares only $\{9, 36\}$), $\mathrm{diag}(i, i)$ (Veronese), $k = 2$ (three channels); [C] conditional on Costello's flat-$\mathbb{PT}$ matching; [O] the quantised BCOV coefficient and the twisted channels ($32\, T_3$) — M1–M3 preregistered (the A_3 Ω_N / the twisted KS measure / the a_0 uplift, each with success + kill); Wolfram extension $482 \to 490$ (verified engine run); ***NO marker moves — WP5e stays [O], `SEAM.EQUIV.01` untouched, `CELEST.SEAM.01` stays Open*)**

- **`v514_celestial_wp5e_eps1_axion_slot.py` (`CELEST.WP5E.EPS1.01`, 34 checks, ∼4 s; exact sympy/Fraction, no floats; deterministic).** S1 Penrose accounting; S2 equivariant ledger + character series + Molien + E3 sharpening + graviton control; S3 the $\tilde{\lambda} = 6$ residue (Okubo / measure / flux) with honest gaps; S4–S5 GH centre dichotomy, periods, Coxeter clock; S6 the CPS "N log" ledger (kernel charge, multipoles, EH log); S7 negative controls; S8 the M1–M3 roadmap; S9 verdict. Registered in `run_all.py` (suite 508 modules, green), `script_registry.csv` (cluster *frontier*), ledger row `CELEST.WP5E.EPS1.01` (new, verdict B).
- **Contract updated (markers unchanged).** `CELEST.SEAM.01` now reads: ε_1 executed via `v514` — the slot is a construction, $\tilde{\lambda} = 6$ pinned by three exact ledgers, the GH/A_3 step re-derives the `v493` family and the Coxeter clock from geometry; the quantised BCOV coefficient and the twisted channels stay **[O]** (M1–M3 preregistered).
- **Paper integration.** `tfpt_research_contracts` (the ε_1 stage moved to *done* with its own stage block; the named remaining targets now carry the M1–M3 milestones verbatim; WP4 handover and honest status line synced); `tfpt_3` (new Step 15 "the bulk axion is a construction; $\tilde{\lambda} = 6$ is pinned three ways" — the celestial story is now fifteen steps; "what remains" updated).

`v514_celestial_wp5e_eps1_axion_slot.py` is cited via `veri` in both.

- **Wolfram.** Eight exact mirrors (equivariant Penrose ledger; character series + bulk-axion survival; Molien hypersurface + Veronese control; Okubo perfect square + $\mathfrak{so}_8$ control; measure cancellation excluding 3 and 12 + flux dial; GH centre dichotomy; period lockstep + Coxeter clock; multipole rule + EH log ledger) added to `tfpt_readouts_extension.wl` and verified with the active engine, 482 → 490; `GATE.WOLFRAM.02`, Wolfram README and the website counters synced (116/116 + 490/490).
- **Provenance.** Discovery probes `celestial_seam_wp5e_eps1_o2_bulk_axion_probe.py` (20/20) and `celestial_seam_wp5e_eps1_gh_a3_backreaction_probe.py` (14/14) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (`ScriptIndex`, DAG *qft* node, `papers.ts` "15 steps", changelog mirror); `next.txt` round notes.

2026-07-22 (II) (`v513` — the c_d negative certificate, promoted as the ledger row `CELEST.DTERM.NONDERIV.01` (anti-numerology certificate in the `v508` style, honest rigid NEGATIVE result): can geometry derive $c_d = 1920$? NO — the `v511` [C] fence "$1920 = |W(D_5)|$" does NOT survive the honest four-route battery with its PRE-DECLARED look-elsewhere ledger: (1) the CONVENTION TABLE gives $c_d \in \{1920, 120, 256, 32, 69120\}$ across five normalisations (probe tr-fund/e_8-Killing; Gell-Mann; $\mathfrak{su}(4)$-Killing; plain trace; ch_3-normalised $d/6$) — only 1 of 5 hits $|W(D_5)|$, while the convention-STABLE content is the factorisation $c_d = \Phi_{T_3}(A_{\mathrm{fix}}) \times 2h^\vee(E_8) = 32 \times 60$, both factors [E] (Killing propagator 60 from $\sum_\alpha \langle \alpha, e\rangle^2 = 120$ over the actual roots; T_3-slot fixing via $\det(\text{P-block}) = -64 \neq 0$); (2) K1 DEAD: the charge-legal flux pairings $(1,3)/(2,2)$ give 2048/1800 — they BRACKET 1920 and both MISS; the only flux hit $64 \cdot 60/2$ is the FORBIDDEN $(1,2)$ pairing (charge 3 mod 4, `v511` innerness); look-elsewhere: 924 expressions from 21 [E] numbers, 11 hits on 1920, controls $1800 \to 8$, $2048 \to 5$, $2016 \to 0$; (3) K2: quantisation delivers only lattices ($60\,\mathbb{Z}$ with 1920 the 32nd multiple, no minimality; strict ch_3 units $2160\,\mathbb{Z}$ EXCLUDE 1920); the cubic quantum is lattice-dependent (6 on coroots, 3/8 on weights), the integer 32 has 12 factorisations; (4) K3 PROVENANCE CLASH (the death blow): cubic indices $A_m = (0, 16, 0, -16)$ with $\sum_m i^m A_m = 32i$ numerically $= \Phi_{T_3}(A_{\mathrm{fix}})$ — but the quartic class decomposition $(10, -4, 30, -4)$ comes from classes 0/2 vs the cubic 16+16 from classes 1/3: same number, disjoint mechanism = coincidence; the Weyl side is WRONG (d-channel is A_3, $|W(A_3)| = 24$; D_5 has no cubic; $|W(E_8)|/|W(A_8)| = 1920$ a second accidental 1920); (5) K4 controls ($SO(16)$, $k = 2$ doubling 7680, false $\mathfrak{g}_0$ with six cubics, $\mathfrak{so}(10)$) separate consistently; VERDICT: NUMEROLOGY as a derivation claim — the `v511` [C] fence stays a fence, now typed look-elsewhere-loaded; the honest [E] core is $c_d = 32 \times 60$; Wolfram extension 476 → 482 (verified engine run); *NO marker moves — the fence stays [C], the physical generation of the* 32 *stays [O], `CELEST.SEAM.01` stays Open*)

- **`v513_celestial_dterm_nonderivation.py` (`CELEST.DTERM.NONDERIV.01`, 24 checks, ∼2 s; exact sympy/Fraction, no floats; deterministic).** N0 baseline (240 roots $(52, 64, 60, 64)$, $A_{\mathrm{fix}} = (9, -30, -15, 0, 32)$, Q_{dd} two routes, c_d pinned by the T_3 slot alone); N1 convention table + the stable factorisation 32×60; K1 flux products + look-elsewhere ledger; K2 quantisation lattices; K3 cubic/quartic provenance clash + Weyl data; K4 negative controls. Registered in `run_all.py` (suite 507 modules, green), `script_registry.csv` (cluster *frontier*), ledger row `CELEST.DTERM.NONDERIV.01` (new).
- **Fence sharpened (text only, marker unchanged).** `CELEST.WP5E.DELTA2.01` and the `CELEST.SEAM.01` contract prose now carry: "the $1920 = |W(D_5)|$ reading is look-elsewhere-loaded (11/924 vs. 8/924 for the control target 1800) and convention-contingent — `v513`"; the

honest remaining target of the cubic-GS item is the physical generation of the factor 32, not the Weyl fingerprint.

- **Paper integration.** `tfpt_research_contracts` (δ_2 fence environment, named remaining targets, status list — new `CELEST.DTERM.NONDERIV.01` status line); `tfpt_3` (Step-14 paragraph + the "what remains" sentence); `tfpt_4_frontier` (celestial-route clause). `v513_celestial_dterm_nonderivation.py` is cited via `veri` in the δ_2/fence environments.
- **Wolfram.** Six exact mirrors (propagator anchor + T_3-slot pinning; five-way convention table; flux enumeration + look-elsewhere counts; K2 lattices; K3 provenance clash + Weyl data; K4 controls) added to `tfpt_readouts_extension.wl` and verified with the active engine, $476 \to 482$; `GATE.WOLFRAM.02`, Wolfram README and the website counters synced ($116/116 + 482/482$).
- **Provenance.** Discovery probe `celestial_seam_wp5e_dterm_coefficient_derivation_probe.py` in `experiments/tfpt-discovery/` (2026-07-22, 24/24); website mirrors regenerated (`ScriptIndex`, DAG *qft* node, `papers.ts`, changelog mirror); `next.txt` round notes.

2026-07-22 (I) (`v512` — the flag-transitivity equivalence web promoted as `SEAM.TAU.FLAG.01`: the $\tau = i$ attack executed on the free RP seam circle — BARE MARK-TRANSITIVITY IS FALSIFIED as a selector (on the test family $M(\delta) = \{\pm 1, \pm e^{i\delta}\}$, $\delta \in (0, \pi/2]$, the symmetry group is the pair-exchanging $V_4 = \{z, -z, b/z, -b/z\}$ for EVERY δ, simply transitive on the four marks; complete census: deck-commuting Möbius maps are diagonal/antidiagonal; hardening: ALL undirected per-mark data are identical for every δ — no local jet sees the bit), and THE SHARP SELECTOR is FLAG transitivity (marks AND their two sides indistinguishable) $\iff \delta = \pi/2 \iff \tau = i \iff$ the clock — stabiliser jump $1 \to 2$, flag/arc orbits $2 \to 1$, order-4 rotation ($b^2 = -1$ solveset $\{\pi/2\}$), mark mirror $z \mapsto 1/z \iff \cos\delta = 0$; welded into a 12-FOLD EQUIVALENCE WEB (every link exact): clock rotation, D_4, mark mirror, flag transitivity, three spectral degeneracies (odd-doublet split $= (2/\pi)\ln\cot(\delta/2)$, zero only at $\pi/2$ — counterwitness exactly $(2/\pi)\ln 2$; arc-Laplacian degeneracy $4/\pi$; the `v215`-K3 indicator in CLOSED FORM $(2\sqrt{2}/\pi)|\ln\tan(\delta/2)|$ — the `v503` lattice curve analytically), ρ-twist existence ($(m^2 - n^2)(t - 1/t) = 0 \iff t = 1$; Atiyah–Bott rationality δ-BLIND — $\mathrm{Tr}(\sigma e^{t\Delta}) = 1$ for all δ, $j_{\mathrm{cw}} = 148176/25$ rational non-CM: rationality selects NOTHING), cusp-4-cycle implementability (H4 rigidity: V_4 with orders $\{1, 2, 2, 2\}$ cannot induce it; centraliser scalar, U^2 nonscalar $\Rightarrow$ no projective involution — projective order exactly 4, $\mathbb{Z}_8$-shadow consistent `v510`), harmonic, $j = 1728$; NEGATIVE CONTROLS: hexagonal (not concyclic, fails preconditions), silver (harmonic but topologically dead), 3 marks (impossible on the free circle: 0 vs 28 invariant subsets; bare $n = 3$ transitivity already forces equal spacing — "4 counts"); [C] THE REST: the last input reduces from a continuous modulus to ONE discrete symmetry-lift bit ($V_4 \to D_4$: side-indistinguishability of the four equally-decorated marks), `v216` local equality does NOT deliver it, its negation is now concretely measurable; scoreboard: the counterwitness passes all 7 established unsided tests, fails all 9 clock-equivalent ones; Wolfram extension $470 \to 476$ (verified engine run); ***NO marker moves — the bit stays [C], `QGEO.REALIZE.01` and `P2.TYPING.01/AX.P2.01` text-precised only, `SEAM.BIT.FREEDOM.01` cross-referenced*)**

- **`v512_seam_tau_flag.py` (`SEAM.TAU.FLAG.01`, 29 checks, ∼6 s; sympy-exact + mpmath 30 digits + one numpy lattice commutator; deterministic).** S0 family frame (counterwitness inside the family: $\lambda = 5/4$, $j = 148176/25$, cross-ratio $-1/4$); H1 complete symmetry census (census basis, rotation solvesets, stabiliser jump, flag/arc orbits, `v216` separation, five-member

table); H2 spectral criteria in closed form; H3 heat-trace rationality (δ-blind) + twist existence; H4 monodromy rigidity (Schur); NC negative controls; SYN the 12-fold web + counterwitness scoreboard + the typed sentence. Registered in `run_all.py`, `script_registry.csv` (cluster *core*), ledger row `SEAM.TAU.FLAG.01` (new).

- **Reduction state re-stated (text only, markers unchanged).** `AX.P2.01` and `P2.TYPING.01` synthesis notes now read: "four marks + one discrete symmetry-lift bit (flag transitivity $\iff \tau = i$) + π"; `QGEO.REALIZE.01` precision: "the remaining input can equivalently be stated as the flag-transitivity postulate — `v512`; the search space for a future [**E**] closure is narrowed to the RP positivity of the odd sector"; `SEAM.BIT.FREEDOM.01` cross-referenced (the modulus half gains the discrete form).
- **Paper integration.** `tfpt_1` (the P2 reduction keybox re-dated 2026-07-22 + Theorem-7 skeleton keybox), `introduction` (narrative chain, consolidated reduction state, status-card addendum), `tfpt_3` (celestial Step 2 + the honest-status reduction bullet), `tfpt_research_contracts` (QGEO kill-test and marks-derived blocks, WP2 note); secondary mirrors `tfpt_2`/`origin_theory` aligned. `v512_seam_tau_flag.py` is cited via `veri` in all four primary documents.
- **Wolfram.** Six exact mirrors (V_4/D_4 census table incl. flag/arc orbit counts; solveset equivalences + family identities; spectral faces in closed form; twist mode identity + AB δ-blindness + j-rationality control; H4 Schur rigidity; negative controls) added to `tfpt_readouts_extension.wl` and verified with the active engine, 470 $\to$ 476; the mpmath heat-trace numerics and the numpy lattice commutator stay Python-only (flagged).
- **Provenance.** Discovery probe `seam_tau_i_established_selector_probe.py` in `experiments/tfpt-discovery/` (2026-07-22, 29/29); website mirrors regenerated (`ScriptIndex`, DAG *P2/masses* nodes, `papers.ts` reduction-state mirrors, `StatusMatrix`); `next.txt` round notes.

2026-07-21 (XXIII) (`v511` — WP5e-δ_2 of `CELEST.SEAM.01` "the full-tensor ledger" promoted as `CELEST.WP5E.DELTA2.01` (intermediate verdict, exact): the named escape route of the `v508` exchange no-go executed on the FULL adjoint tensor structure of $\mathfrak{e}_8$ — the COLLAPSE IS CONFIRMED full-tensorially AND arity-crossing (the INNERNESS THEOREM: $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_3$ semisimple with NO $\mathfrak{u}(1)$ (class-0 roots 40+12, root span rank 8), h in the Cartan OF $\mathfrak{g}_0 \Rightarrow$ every invariant tensor carries total charge $0 \bmod 4$: the bilinear Hom table is nonzero ONLY at $j' = -j$ (dims 2/1/1/1) and ALL 15 non-neutral trilinear sector triples have $\mathrm{Hom} = 0$), BUT ONE CUBIC DOOR OPENS: of the neutral trilinears ($\{0,0,0\}: 3 = 1$ sym $+2$ antisym, $\{0,1,3\}: 2$, $\{0,2,2\}: 2$, $\{1,1,2\}: 1$, $\{2,3,3\}: 1$; all cross-sector ones pure brackets) the UNIQUE totally symmetric survivor is the $\mathfrak{su}(4)$ d-symbol ($\mathfrak{so}(10)$ has no cubic Casimir, $\mathrm{Inv}(S^3\,45) = 0$ protects T_5), whose exchange quartic $Q_{dd} = (0, 0, -\frac{1}{240}, 0, \frac{1}{60}) = \frac{1}{60}(T_3 - P_3/4)$ (TWO independent routes, Killing propagator $2h^\vee = 60$ from the roots) has $\Phi_{T_3}(Q_{dd}) = \frac{1}{60} \neq 0$ — the `v508` master kill (product theorem) does NOT extend to cubic vertices; the pairing stays OBSTRUCTED in the charge reading with the genuinely WEAKER certificate $\{\Phi_{T_5}, \psi = \Phi_P - \Phi_{T_3}/4\}$, $\psi(A_{\mathrm{fix}}) = 72 - 8 = 64$ ($M = [E_{13}, E_{22}, E_{00}, Q_{dd}]$ rank 3, augmented 4), and becomes EXACTLY solvable relaxed: $A_{\mathrm{fix}} = -u + 8v + 2w + 1920\,Q_{dd}$ (residual 0); [C] fences $64 = \dim \mathfrak{g}_1 = 2^6$, $1920 = |W(D_5)| = 8 \cdot 240$; controls: $SO(16)/D_8$ has NO symmetric cubic and no A_3 block (E_8/μ_4 doubly special), false $\mathfrak{g}_0$/sector assignments separate; the δ_1 burden SHARPENED to the $\psi = 64$ slice or the selection-rule relaxation — and the δ_1 exploration run itself stands UNDECIDED at exploration level (strict holomorphic q^0 reading REFUTED at the $\mathbb{Z}_2$/Eguchi–Hanson anchor = method boundary, not a kill; contact term = the MODULAR COMPLETION of the AB data, a Harvey–Moore τ-integral with forced leading (T_5, T_3) ratio 4:3; massive coset levels $n \geq 2$ mandatory); Wolfram extension $463 \to 470$ (verified engine run); *NO marker moves — WP5e proper stays [O], `SEAM.EQUIV.01` stays [O]*)

- **`v511_celestial_wp5e_delta2_tensor_ledger.py` (`CELEST.WP5E.DELTA2.01`, 41 checks, ∼4 s; integer-scaled weight multisets + Kostant/Weyl alternating sums + sympy/Fraction, no floats; deterministic).** D0/D1 structure: 240 roots, class split $(52, 64, 60, 64)$; $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_3$ SEMISIMPLE without $\mathfrak{u}(1)$ (40+12 class-0 roots, no mixed root, root span rank 8); sectors $\mathfrak{g}_1 = (16_s, 4)$, $\mathfrak{g}_2 = (10, 6)$, $\mathfrak{g}_3 = (\mathfrak{g}_1)^*$ as multiplicity-free minuscule Weyl-orbit products, each irreducible (64/60/64); the grading is INNER ($h = (2, 2, 2, 2, 2; 0^4)$ in the $\mathfrak{g}_0$ Cartan). D3/D4 Hom tables: bilinear nonzero ONLY at $j' = -j$ (dims 2/1/1/1; all 12 charged pairs zero — the inner-torus theorem at arity 2); trilinear: ALL 15 non-neutral triples Hom = 0 (arity 3), neutral dims $(3, 2, 2, 1, 1)$ with all cross-sector trilinears ANTISYMMETRIC in their repeated legs — the unique symmetric survivor is the $\mathfrak{su}(4)$ d-symbol ($\mathrm{Inv}(S^3\,45) = 0$, $\mathrm{Inv}(S^3\mathfrak{a}_3) = 1$). D5 the new channel: $Q_{dd} = \frac{1}{60}(T_3 - P_3/4)$ by Killing-propagator contraction AND the hand route $|x^2 - (\mathrm{tr}\,x^2/4)\mathbf{1}|^2/60$; $\Phi_{T_3}(Q_{dd}) = \frac{1}{60} \neq 0$; brackets/tadpoles/$d\times f$ die exactly; no T_5 analogue. D6 decision: strict reading unreachable (rank 2 vs 3); generous reading STILL unsolvable (rank 3 vs 4) with the weaker certificate $\psi(A_{\mathrm{fix}}) = 64$; relaxed reading EXACTLY solvable with $c_d = 1920 = 32 \times 60$. D7 controls: Killing control = `v508` replication (A_{fix}, E_{13}, E_{22}, $E_{00} = 225E_{22}$); $SO(16)/D_8$ bilinears $(1, 0, 1)$, trilinears $(1, 0, 1, 0)$, $\mathrm{Inv}(S^3\mathrm{adj}_{D_8}) = 0$; false $\mathfrak{g}_0 = \mathfrak{d}_5 + \mathfrak{a}_2 + \mathfrak{u}(1)$: $(5, 2, 2) \neq (2, 1, 1)$, $\mathrm{Inv}(S^3\mathfrak{g}_0) = 6 \neq 1$; false sector content $(16_s, \bar{4})$ kills every pairing. Discovery provenance: `experiments/tfpt-discovery/celestial_seam_wp5e_delta2_full_tensor_ledger_probe.py` (41/41), same day.
- **The δ_1 fence (exploration-level, NOT promoted — verdict UNDECIDED).** The twisted BCOV contact-term probe (`celestial_seam_wp5e_delta1_bcov_contact_probe.py`, 27/27, stays sandbox; v-numbered on demand) contributes its four exact sharpenings as fence

text: (i) the strict holomorphic q^0 reading is REFUTED at the $\mathbb{Z}_2$/Eguchi–Hanson anchor (method boundary, not a kill); (ii) the forced leading (T_5, T_3) ratio is 4:3 (sector 2 : sectors 1+3); (iii) the $SL(2,\mathbb{Z})$ orbit decomposition $15 = 12 + 3$ identifies the contact term as the MODULAR COMPLETION of the AB data (a Harvey–Moore-type τ-integral), not a free number; (iv) massive coset levels $n \geq 2$ are mandatory. The remaining δ_1 target is the τ-integral, evaluated against the δ_2-sharpened criterion ($\psi = 64$ slice or selection-rule relaxation).

- **Ledger + papers + Wolfram + website.** Ledger: new row `CELEST.WP5E.DELTA2.01` ([**E**] Hom tables/innerness/Q_{dd}/certificate; [**C**] the number fences; [**O**] arity ≥ 4, cubic GS term, δ_1); `CELEST.SEAM.01` updated (delta-2 executed via v511 — the full-tensor collapse is confirmed (innerness), but the cubic d-symbol channel opens T_3 with the weaker certificate $\psi = 64$; delta-1 stands UNDECIDED at exploration level). Papers: `tfpt_research_contracts` WP5e bullet gains the executed δ_2 stage + the δ_1 exploration fence, WP5e-proper targets renamed (τ-integral, cubic GS term, ε_1, back-reaction); `tfpt_3` celestial section extended to the FOURTEEN-step story (new Step 14 "the full-tensor ledger: the collapse is real, and one cubic door opens") with the "What remains" block updated. Wolfram: seven mirrors (innerness count + $\mathfrak{g}_0$ structure; the bilinear and trilinear Hom tables by exact Kostant alternating sums; the unique symmetric survivor; the $SO(16)$/false-$\mathfrak{g}_0$/ false-sector controls; Q_{dd} by two routes; the ψ certificate + the relaxed solution $(-1, 8, 2; 1920)$) verified with the active engine the same day ($463 \to 470$, `ALL WOLFRAM EXTENSION CHECKS PASSED`). Registry + `run_all` (505 modules, suite green: `ALL CHECKS PASSED`). No gate closes, no marker moves — WP5e proper stays [**O**].

2026-07-21 (XXII) (P2 consolidation — the axiomatics reduction state written as ONE coherent statement across the papers (editorial, NO new claims, NO marker moves): the compiler's dimensionless inputs reduce to *four marks* (derived from P1-side topology, `v216`), *ONE square-modulus datum* ($\tau = i$ — equivalently: the clock / the CM-fixed half-period / the non-split extension class on the free seam circle; `v491/v499/v506/v507/v510`) and π, plus the [C] identifications R1/R2 of `v499`; `AX.P2.01` stays a DECLARED axiom and `QGEO.REALIZE.01` stays [C]/[O] — the consolidation describes the *reduction state*, not an abolition)

- **What is consolidated (and why now).** The P2 state of affairs was spread over five module rounds: the partition corollary (`v491`: $4 = 2{+}1{+}1$ unique $\Rightarrow g_{\text{car}} = e_2 = 5$), the weight typing (`v499`: Deligne + Birkhoff–Grothendieck + Mehta–Seshadri force $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$, residuals R1/R2 [**C**]), the residual convergence (`v503`: QGEO-R1 ≡ P2-R1), the clock rigidity and bit origin (`v506/v507`: one alignment bit, order fermionically forced, deck class derived) and the freedom theorems (`v510`: the position half is topology, the edge class excluded). The honest consolidated statement, now in one place: *the compiler's dimensionless inputs reduce to four marks (derived from P1-side topology), one square-modulus datum ($\tau = i$, equivalently: the clock / the CM-fixed half-period / the non-split extension class on the free seam circle), and π* — plus the [**C**] identifications R1/R2 of `v499`. No new claims; conservative wording; no "axiom abolished" language anywhere.
- **Papers + ledger + website.** `tfpt_1_architecture_e8`: new keybox "The current reduction state of P2" next to the foundational-postulates keybox, with `v491/v499/v506/v507/v510` citations. `tfpt_3`: mirror bullet "The current reduction state" in the honest-status section. `introduction`: the anchor paragraph gains the consolidated one-sentence state and the status card row $a = (1, 1, 2)$ now names the reduction state. Ledger: SYNTHESIS NOTE appended to `AX.P2.01` ("reduction state 2026-07-21: four marks + one square-modulus datum ($\tau = i$) + π"), marker *Axiom unchanged.* Website: `papers.ts` mirrors of the changed sections. No gate closes, no marker moves.

2026-07-21 (XXI) (`v510` — the freedom attack on the `v506`/`v507` alignment bit promoted as `SEAM.BIT.FREEDOM.01`: the POSITION half of the bit is TOPOLOGY — the collar deck is a covering deck (the $\mathbb{Z}_2$ seam of the Möbius double), and covering decks act FREELY; the DIHEDRAL THEOREM (on the seam circle the free involution is UNIQUE = the antipode: 17 involutions in D_{16}, exactly 1 free; $\mathrm{Aut}(C_{16}) = D_{16}$ complete census, brute force $8!$ at N=8; quotient topology circle/antipode = CIRCLE χ=0 vs circle/reflection = INTERVAL χ=1 — the 6π Gauss–Bonnet budget needs closed circles, so freeness IS the established P1 datum); NONSPLIT $\iff$ FREE over the complete 17-involution census in exact Cl(16) (free $\Rightarrow \tilde{U}^2 = 256\,\gamma_1 \cdots \gamma_{16} = (-1)^F$ nonsplit with 2 roots = the $\mathbb{Z}_8$ tower; ALL 16 reflections $\Rightarrow U^2$ = scalar $\{2^7, 2^8\}$, split, 0 roots); the REAL-STRUCTURE classification (deck-commuting families = exactly two; $\mu_4 \to$ equatorial, deck FREE; silver $\to$ real axis THROUGH the deck poles, NOT free — the edge class is the restriction of the BRANCHED pillowcase involution; hexagonal $\to$ NO mark-fixing real structure, RP-incompatible); HARMONIC + FREE $\Rightarrow \mu_4$ exactly ($b = \pm i$); honest counterwitnesses ("nonsplit $\Rightarrow$ clock" is FALSE: $\{0, 1, 8, 9\}$ and $\{\pm 1, \pm(3{+}4i)/5\}$ with $j = 148176/25 \neq 1728$); the alignment bit REDUCES from "harmonic AND central" to the square-modulus datum $\tau = i$ alone; Wolfram extension 457 $\to$ 463 (verified engine run); *NO marker moves — `QGEO.REALIZE.01` stays [C]/[O], `AX.P2.01` stays an axiom, P2-R1 halved*)

- **`v510_seam_bit_freedom.py` (`SEAM.BIT.FREEDOM.01`, 30 checks, < 5 s; sympy + exact Cl(16) Fractions, no floats; deterministic).** Part A (dihedral/fermionic): the involution census in D_N on the refined $2N$-point seam circle ($N = 16$: 17 involutions, EXACTLY ONE free = the antipodal shift by 8; all 16 reflections have exactly 2 circle fixed points — sites for even axis, edge midpoints for odd; robustness $N = 8, 12$; odd control $N = 9$: ZERO free — an even carrier count is needed, consistent with $16 = 2^{g_{\mathrm{car}}-1}$); the census is COMPLETE because $\mathrm{Aut}(C_{16}) = D_{16}$ exactly (brute force over all $8! = 40320$ permutations at $N = 8$, propagation at $N = 16$); quotient topology: circle/antipode = CIRCLE ($\chi = 0$, closed), circle/reflection = INTERVAL ($\chi = 1$, two boundary corners) — a fixed-point deck would break the $6\pi = 3 \times 2\pi$ closed-circle count, so "deck free on the seam circle" is exactly the established "the seam is a closed circle", not a new premise; the complete Fock census: ALL 17 involutions implemented in exact Cl(16) — NONSPLIT $\iff$ FREE with no exception, root dichotomy 2 vs 0; the chain: free $\Rightarrow$ shift 8 $\Rightarrow$ crossing (all 28 invariant 4-sets) $\Rightarrow U^2 = (-1)^F \Rightarrow$ central class $\Rightarrow$ the edge/silver arrangement is EXCLUDED; Klein control (covering decks are free regardless of the gluing); branch control (the pillowcase involution has exactly 4 fixed points = the marks and restricts to a REFLECTION: the silver arrangement mistakes the mark-creating branch $\mathbb{Z}_2$ for the seam deck). Part B (Möbius/RP): the cylinder-double-cover deck is free on all 120 model points and restricts to the antipode on the $\mathbb{Z}_2$ seam fibre; one-sidedness TYPES the $\mathbb{Z}_2$ as the free antipode (quotient $\mathbb{RP}^2$ = the $|\mathbb{Z}_2|$ of $c_3 = 1/(|\mathbb{Z}_2|2\pi\chi)$); the deck-commuting real structures are EXACTLY two families ($\mu\bar{z}$: fix line through the poles, NOT free; $\mu/\bar{z}$, $\mu > 0$: fix circle $|z| = \sqrt{\mu}$, FREE; $\mu < 0$: empty); the per-configuration table (μ_4 equatorial/free, silver real-axis/not free, generic-unit free without harmonicity, hexagonal RP-incompatible); harmonic + free solves EXACTLY to $b = \pm i$ (on free circles the silver family "harmonic AND edge" is EMPTY — harmonicity becomes sufficient); honest counterwitnesses: the discrete $\{0, 1, 8, 9\}$ and the continuous $\{\pm 1, \pm(3{+}4i)/5\}$ (free, central, $j = 148176/25$, no clock) — the Fock dichotomy measures the CLASS, not the modulus. Premise typing: (P-i) deck = Möbius covering deck $\Rightarrow$ free [standard topology; the one-sidedness strand `origin_theory`/`v456`]; (P-ii) marks on the Θ-fix seam circle = existing REALIZE **[C]** content (`QGEO.REALIZE.02` hypotheses, `v264`) — NO new input. Audit finding documented: "circle-freeness" was nowhere typed before (only mark-freeness, `v216`); `v507` F1.1 explicitly admitted the reflection reading of the edge class — `v510` closes that door (a

reflection is not a covering deck). The [**C**] rest, verbatim: *the alignment bit reduces from "harmonic AND central" to the square-modulus datum $\tau = i$ alone; the position half is topology (edge class excluded), the modulus half stays the one carrier input.* Discovery provenance: `experiments/tfpt-discovery/qgeo_realize_deck_freedom_dihedral_probe.py` (14/14) and `qgeo_realize_rp_fixcircle_moebius_probe.py` (16/16), same day.

- **Ledger + papers + Wolfram + website.** Ledger: new row `SEAM.BIT.FREEDOM.01` ([**E**] dihedral theorem + complete Fock census + real-structure classification + harmonic-free collapse + counterwitnesses; [**C**] the square-modulus datum); TEXT precisions (markers unchanged) in `SEAM.CLOCK.RIGIDITY.01`, `SEAM.BIT.ORIGIN.01`, `QGEO.REALIZE.01` (bit → square-modulus datum; edge class topologically excluded) and `P2.TYPING.01` (R1 halved). Papers: `tfpt_3` celestial/QGEO passage extended ("the position half is topology"), `tfpt_research_contracts` QGEO kill-test + QGEO.MARKS.03 + celestial-contract passages extended, one-sentence precisions in `introduction`, `tfpt_1`, `tfpt_2`, `origin_theory` where `v506`/`v507` are cited. Wolfram: six mirrors (dihedral census + quotient topology; Aut completeness incl. the 8! brute force; the Cl(16) dichotomy census over all 17 involutions on 256×256 Jordan–Wigner gammas; the real-structure table; harmonic + free $\Rightarrow \pm i$; the counterwitnesses) verified with the active engine the same day ($457 \to 463$, `ALL WOLFRAM EXTENSION CHECKS PASSED`; base file re-verified 116/116). Registry + `run_all` (504 modules, suite green: `ALL CHECKS PASSED`). No gate closes, no marker moves.

2026-07-21 (XX) (`v509` — WP5e-ε_2 of `CELEST.SEAM.01` "the CPS level-from-flux dial" promoted as `CELEST.WP5E.EPS2.01`, verdict B: the Costello–Paquette–Sharma Burns-holography mechanism ("level = flux quantum × Dynkin index") executed on the lockstep spheres — the CPS SKELETON exact (S^3 period $(2\pi i)^2 N$ with pullback density $-\sin 2\theta$, exceptional-sphere Kähler flux $2\pi N$, boundary level magnitude $2N$ by the symplectic-boson contraction, $T(\text{vector}) = 1$); the PAIRING MATRIX pinned, not postulated (ch_2 ledger + integrality + unimodularity + effectivity $\Rightarrow P_{mi} = \delta_{mi}$ by complete enumeration: rows 8/6/8, 48 unimodular, exactly 2 effective = identity + diagram flip), fluxes $F_i = (64, 60, 64) = \dim \mathfrak{g}_i$ with uniformly ONE quantum per charged current; "level = total flux" KILLED by the lockstep test itself (F_i unequal = flavour multiplicity), the per-current reading $(1, 1, 1)$ anchored by the lattice current count $(240, 0, 0, 0)$ and embedding index 1; ord-4-vs-level-1 RESOLVED (the clock order counts fractional flux sectors: $\mu = 16 = \mathrm{ord}^2$, Lagrangian μ_4 glue, KLM $16 \to 1$) with the NEW sector-counter dial $\#\mathrm{prim}((E_8)_k) = (1, 3, 5, 10, 15, 27)$ for $k = 1..6$ — exactly ONE sector iff $k = 1$; the LOCKSTEP THEOREM lifts the `v505` "one scale ⇒ one level" to a THEOREM of clock invariance ($|\Pi_j| = \sqrt{2}\,t$ equal, monodromy = pure phase i, $\det(A-1) = -4$: no invariant flux vector; NEW falsifier $Z^4 - Z$: 0/24 orderings lockstep); honest limit: uniform doubling keeps the lockstep (equality, not the value); controls k=2/$SO(16)$/A_2 separate honestly; Wolfram extension $450 \to 457$ (verified engine run); *NO marker moves — WP5e stays [O] (the CPS dictionary on $\mathbb{PT}/\mathbb{Z}_4$ typed [C], the type-I B-model back-reaction [O]), `SEAM.EQUIV.01` stays [O]*)

- **`v509_celestial_wp5e_eps2_level_from_flux.py` (`CELEST.WP5E.EPS2.01`, 28 checks, < 1 s; sympy/Fraction exact, no floats; deterministic).** Target: derive the boundary level $k = 1$ from flux quantisation on the lockstep spheres, following the Costello–Paquette–Sharma Burns-holography mechanism (PRL **130**, 061602; `arXiv:2306.00940`). *CPS skeleton replicated exactly*: the S^3 period of the brane back-reaction 3-form is $(2\pi i)^2 N$ (their eq. 3.33; large-gauge invariance quantises N), the exceptional-sphere Kähler flux is $2\pi N$ exactly ($K = |u|^2 + N \log |u|^2$, the flat part dies at $t \to 0$), and the boundary level magnitude is $2N$ via the symplectic-boson double contraction ($N = 1, 2, 3$; their eq. 9.4), Dynkin normalisation $T(\mathrm{so}_8 \text{ vector}) = 1$, $T(\mathrm{adj}) = 6$. *Pairing matrix pinned*: the `v505` ch_2 ledger ($f(m) = -(C^{-1})_{mm}/2 = (-\frac{3}{8}, -\frac{1}{2}, -\frac{3}{8})$)

+ integrality + unimodularity + effectivity force $P_{mi} = c_1(T_m).[\Sigma_i] = \delta_{mi}$ by COMPLETE enumeration (row candidates 8/6/8; 48 unimodular solutions of $PC^{-1}P^T = C^{-1}$; with $P \geq 0$ exactly 2: identity + A_3 diagram flip); fluxes $F_i = (64, 60, 64) = \dim \mathfrak{g}_i$, total $188 = 248 - 60$, flip-invariant; flux per charged root current uniformly ONE quantum through exactly one sphere. *The level formula, honest*: "level = total open-string flux" is KILLED by the lockstep test itself ($k_i = (64, 60, 64)$ unequal — F_i is the FLAVOUR multiplicity, the brane matter content); what works: k = flux per adjoint current = $F_i / \dim \mathfrak{g}_i = (1, 1, 1)$, anchored non-circularly by the lattice current count $(240, 0, 0, 0)$ for $k = 1..4$ (`v505` S4.8) and the embedding index $j(D_5 \subset E_8) = j(A_3 \subset E_8) = 1$ ($60 = 2h^\vee$). *Ord-4-vs-level-1 resolved*: the clock order counts FRACTIONAL flux sectors, not the level ($\mu = \det(D_5)\det(A_3) = 16 = \text{ord}^2$; the μ_4 glue diagonal isotropic + Lagrangian; KLM $16 \to 16/4^2 = 1$, zero residual fractional flux); NEW DIAL: #primaries$((E_8)_k) = (1, 3, 5, 10, 15, 27)$ for $k = 1..6$ from the affine marks — exactly ONE sector iff $k = 1$ (independent of the CPS transplantation). *The lockstep theorem* (the lifting of `v505` S4.10): clock invariance $P(iZ) = P(Z)$ forces $a_3 = a_2 = a_1 = 0$ sharply, the four branch points are ONE μ_4 orbit, $|\Pi_j| = \sqrt{2}\,t$ EQUAL on all three spheres, the monodromy is the pure phase i and $\det(A - 1) = -4 \neq 0$ (no invariant flux vector) — $k_1 = k_2 = k_3$ IS A THEOREM of clock invariance; NEGATIVE PROBE (new falsifier): the clock-forbidden family $Z^4 - Z$ (smooth, disc -27, branch points $\{0, 1, \omega, \omega^2\}$ not a μ_4 orbit): over ALL 24 orderings ZERO chains with three equal period moduli (1 vs $\sqrt{3}$); HONEST LIMIT: uniform flux doubling keeps the lockstep — the theorem proves EQUALITY, not the value (1 from the counters and the CPS minimal quantum $N = 1$). *Negative controls*: $k = 2$ (current count 0, residual 360, prefactor $-31/48$, #prim 3 — while the lockstep dial honestly does NOT fire); $SO(16)$ (fluxes $(0, 60, 0)$: spheres 1/3 FLUX-DARK, defect $-30 \neq -78$, condensation stuck at $4 = \det \text{Cartan}(D_8)$); A_2 ($\det 3 \neq 4$, glue diagonal $23/24 \notin \mathbb{Z}$, $\mu = 12$ no perfect square: NO Lagrangian subgroup, Coxeter $3 \neq 4$). Typed honestly: the CPS dictionary itself on $\mathbb{PT}/\mathbb{Z}_4$ stays **[C]** (their geometry is the $\mathbb{C}^2$ blow-up with one (-1)-sphere and $\mathfrak{so}(8)$ at negative level — not the A_3 ALE); the type-I B-model back-reaction stays **[O]** (the same fence as `v502`/`v505`).

- **Ledger + papers + Wolfram + website.** Ledger: new row `CELEST.WP5E.EPS2.01` ([**E**] CPS skeleton/pairing/lockstep/sector arithmetic + [**C**] CPS dictionary + [**O**] back-reaction); `CELEST.SEAM.01` updated ("ε_2 executed via v509: the one-level statement is now a THEOREM (lockstep from clock invariance, falsifier $Z^4 - Z$), level = 1 carries the new sector-counter dial #prim$((E_8)_k) = 1 \Leftrightarrow k = 1$; the $\mathbb{PT}/\mathbb{Z}_4$ back-reaction stays **[O]**"); the `v505` "one scale ⇒ one level" sentence LIFTED to theorem status in ledger and prose (text move, no marker move). Papers: `tfpt_research_contracts` WP5e section extended by the ε_2 block (done, verdict B; roadmap now $\delta_1/\delta_2/\varepsilon_1$ + back-reaction); `tfpt_3` celestial section becomes a THIRTEEN-step story with Step 13 "one level is a theorem; the sector counter pins $k = 1$" and the "What remains" paragraph updated. Wolfram: seven mirrors (the CPS period integrals; the $2N$ contraction + Dynkin normalisation; the pairing enumeration $48 \to 2$; the fluxes $(64, 60, 64)$; the #prim table $(1, 3, 5, 10, 15, 27)$; the lockstep periods + the 0/24 negative probe; the $SO(16)/A_2$ controls) verified with the active engine the same day ($450 \to 457$, `ALL WOLFRAM EXTENSION CHECKS PASSED`). `VerificationDag`: `v509` on the QFT/celestial node; registry + `run_all` (503 modules, suite green). No gate closes, no marker moves — WP5e proper stays **[O]**.

2026-07-21 (XIX) (`e8-ladder-bed` v2 — the 2DCS extension executed with an honest preregistered `kill_2d` verdict at the DETECTOR level: the known-answer map recovers $8/8$ rungs incl. m_7/m_8 via F3 difference coordinates (MC $p = 2\times10^{-4}$), but the 2D control battery breaks (scramble 27%, jitter 18% at the destructive 30% level) — the v1 1D results (BCVO $8/8$, $CoNb_2O_6$ $6/8$) are untouched; scorecard row updated, status/stage unchanged; *experiments level only — no verification module, no gate closes, no marker moves*)

- **`experiments/e8-ladder-bed` v2 (2DCS): amendment preregistered, run, and killed by its own controls.** Trigger: arXiv:2512.16829 (Watanabe, Trebst, Hickey) — a THEORY-only paper (four-kink/ED localized-limit spectra at $B_y = 0$–3 T, not the E8 point; no measurements); as of 2026-07-21 *no experimental 2DCS measurement of $CoNb_2O_6$ exists anywhere* (web-checked). `amendment_v2` was frozen in `hypotheses/e8_ladder_bed_v1.yaml` BEFORE any 2D recovery run (detector, catalog F1/F2/F3, gates, null model, verdict criteria); `amendment_v2_1` discloses a null-region correction (add `n_matched` to the false-alarm region) found on synthetic diagnostics, timing disclosed. Runs (`results/results2d.json`, deterministic): known-answer KA2D $8/8$ rungs with m_7/m_8 recovered ONLY via F3 difference coordinates ($\chi^2/\mathrm{dof} = 1.30$, MC $p = 2.0 \times 10^{-4}$, $0/5000$) — the mechanism that pulls m_7/m_8 inside a low-pass window works; the NEG2D localized-limit map (the paper's own non-E8 closed forms) is correctly rejected ($p = 0.029$); offset and wrong-ladder controls rejected. BUT `scramble2d` fires at $54/200 = 27\%$ and `jitter2d` shows no dose-response decay ($29/21/17/18\%$; 18% at the destructive 30% level) $\Rightarrow$ preregistered verdict `kill_2d`: permuting the detection axis preserves the pump-axis marginal (which carries the full 1D ladder) and the 120-entry catalog is dense enough that rung coverage follows from the marginal alone — the v2 detector does not test the 2D *pairing*; failure kills the detector, not TFPT. The $CoNb_2O_6$ leg stays $6/8$ ($p = 0.0038$) per the preregistered `m7_m8_upgrade_rule`; a future v3 would need coincidence-based coverage and a sparser catalog, preregistered BEFORE real data exists.
- **Surfaces synced (scorecard row updated; counters unchanged, 120 rows).** Scorecard row "E8 mass-ladder blind recovery bed": `data_value` extended ("CoNb2O6 6/8 unchanged; 2DCS extension killed at detector level (scramble 27%, jitter 18% @30%); KA 8/8 incl. m7/m8 via F3, p=2e-4; no experimental 2DCS data exists (arXiv:2512.16829 theory-only)") and `status_note` extended (amendment v2/v2.1 preregistered+disclosed; real-data leg blocked; v3 criteria) via `build_evidence_scorecard.py`; status/stage UNCHANGED (`consistent/not_applicable`, `analog_positive_control` basket — the v1 detector validation stands). Experiment README v2 section, hypotheses YAML amendments, `experiments/next.txt` note; data-request plan on record (Armitage lab/JHU, Wang lab/Ames; 2D peak table at $B \parallel b$, 4.7–5.5 T, sub-Kelvin). No claim; firewall intact.

2026-07-21 (XVIII) (`v508` — WP5e-γ of `CELEST.SEAM.01` "the sphere-axion pairing check" promoted as `CELEST.WP5E.GAMMA.01`, an honest rigid NEGATIVE result: the VERTEX-SPACE THEOREM — the $W(D_5)\times W(A_3)$-invariant quadratics on the 8-dim glue Cartan are EXACTLY $\mathrm{span}\{s_5, s_3\}$ (dim 2) and the quartics EXACTLY $\mathrm{span}\{P_1, P_2, P_3, T_5, T_3\}$ (dim 5), with the PRODUCT THEOREM as the master kill (any two invariant quadratics multiply into $\mathrm{span}\{P_1, P_2, P_3\}$ — zero T_5/T_3 content, so Φ_{T_3} annihilates EVERY exchange image while $\Phi_{T_3}(A_{\mathrm{fix}}) = 32 \neq 0$); the strict two-index $\mathbb{Z}_4$ rule COLLAPSES (the sphere axions have no invariant bilinear Cartan vertex at all), the twist-insertion channels give a rank-2 exchange matrix with $\mathrm{rank}([M \mid A_{\mathrm{fix}}]) = 3$ and the annihilator certificate $(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\mathrm{fix}}) = (0, 32, 72)$; side discovery $K^{(0)} = -15\,K^{(2)}$; naturalness DISSOLVES (ch_2-natural and AB-weight couplings both certify $(0, 0)$ vs required $(32, 72)$ — scale-independent); controls: $SO(16)$ has no sphere partners AND uncancelled T_5, D_8 no T_3 structure; the O-fence is SHARPENED: the exchange sub-branch is dead, the $32\,T_3$ residual MUST be carried by twisted BCOV contact terms (δ_1 binary: coefficient exactly $(9, -30, -15, 0, 32)$, computed not fitted; escape δ_2: the full-tensor ledger); Wolfram extension $444 \to 450$ (verified engine run); *NO marker moves — `CELEST.SEAM.01` stays Open, the slot bijection untouched, the preregistered level-kill still does not fire, `SEAM.EQUIV.01` stays [O]*)

- **`v508_celestial_wp5e_gamma_pairing.py` (`CELEST.WP5E.GAMMA.01`, 27 checks, ∼4 s; sympy/Fraction exact, no floats; deterministic).** Question: can the three twisted sphere axions η_m (Coxeter eigencharacters $\{i, -1, -i\}$ on H^2(ALE), `v505` S5.8) cancel the rigid $32\,T_3$ residual of the Atiyah–Bott ledger ($A_{\mathrm{fix}} = Q^{(1)}/2 + Q^{(2)}/4 + Q^{(3)}/2 = 9P_1 - 30P_2 - 15P_3 + 32\,T_3$, `v505` S3.7) by Green–Schwarz-type EXCHANGE with sector-compatible quadratic vertices? *No, with a certificate.* *Vertex-space theorem*: exact Weyl nullspace arithmetic over all bidegrees gives quadratics dim 2 (bidegrees $(1, 0, 1)$) and quartics dim 5 (bidegrees $(2, 0, 1, 0, 2)$) — the vertex basis is COMPLETE; *product theorem* (the master kill): $(as_5 + bs_3)(cs_5 + ds_3) = ac\,P_1 + (ad + bc)P_2 + bd\,P_3$ identically — every exchange image (any axion count, any rule, any couplings) is annihilated by Φ_{T_3}, and $\Phi_{T_3}(A_{\mathrm{fix}}) = 32 \neq 0$. *Selection enumeration*: the strict two-index reading collapses ($C_{j,j'} \neq 0$ only for $j' = -j$, forcing $m = 0$: 16 raw triples, 4 nonzero, all $m = 0$); twist insertion: $\eta_1 \leftrightarrow K^{(3)}$, $\eta_2 \leftrightarrow K^{(2)}$, $\eta_3 \leftrightarrow K^{(1)}$, $\eta_0 \leftrightarrow K^{(0)}$; channels $E_{13} = (16, -96, 144, 0, 0)$, $E_{22} = (16, 32, 16, 0, 0)$, $E_{00} = 225\,E_{22}$; SIDE DISCOVERY: $K^{(0)} = -15\,K^{(2)}$ (the even-sector quadratics are PARALLEL — this collapse drives the Φ_P obstruction). *Unsolvability with certificate (the core)*: M is 5×2, rank 2, kernel 0; $\mathrm{rank}([M \mid A_{\mathrm{fix}}]) = 3$; annihilator certificate $(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\mathrm{fix}}) = (0, 32, 72)$ — two independent obstructions ($32\,T_3$ structural, rule-independent; $\Phi_P = 72$ selection-rule-driven); $\ker M = 0$ forces the unique non-disturbing coupling $c = 0$ (the sphere axions are RIGIDLY SILENT); $\Phi_P(Q^{(0)}) = 0$ (the sector-0 Okubo mechanism undisturbed); relaxed control: dropping charge conservation reaches the P-block ($\det = -64$) but $T_5 = T_3 = 0$ persists. *Naturalness resolution*: ch_2-natural couplings $(9/64, 1/4)$ give $(25/4, -11/2, 97/4, 0, 0)$ and AB-weight couplings $(1/2, 1/4)$ give $(12, -40, 76, 0, 0)$ — BOTH with certificate $(0, 0)$ vs required $(32, 72)$: the failure is scale- and coupling-INDEPENDENT (no "unnatural factor"). *Negative controls*: invariant-only = `v505` S3.9 replicated + sharpened; flipped rule $m + j - j'$: Φ_{T_3} kill flip-invariant, Φ_P rule-dependent; $SO(16)$: odd classes EMPTY and AB sum $(15, -18, -15, -4, 40)$ — T_5 does not even cancel (E_8 doubly special); D_8: $16\,T_8 + 12\,s_8^2$, no T_3 structure — the question cannot even be posed. Number fences **[C]** only: $32 = (h, h) + (h', h') = 20 + 12$, $72 = 2\tilde{\lambda}^2_{e_8}$.
- **Ledger + papers + Wolfram + website.** Ledger: new row `CELEST.WP5E.GAMMA.01` (**[E]** vertex space/selection/certificate + **[C]** fences + **[O]** contact terms); `CELEST.SEAM.01` stays Open with the WP5e targets UPDATED ("the exchange sub-branch is closed by v508; the named remaining targets are the twisted BCOV contact terms (δ_1, binary test) and the full-tensor ledger (δ_2)"). Papers: `tfpt_research_contracts` WP5e section extended by the γ block (γ

done — negative result with certificate; the $\delta_1/\delta_2/\varepsilon_1/\varepsilon_2$ roadmap with success/kill criteria); `tfpt_3` celestial section becomes a TWELVE-step story with Step 12 "the exchange no-go and the contact-term criterion" and the "What remains" paragraph updated. Wolfram: six mirrors (the vertex-space dims 2/5 by generator-wise nullspace arithmetic; the twisted quadratics with $K^{(0)} = -15\,K^{(2)}$; the product theorem, symbolic; the rank-3 certificate $(0, 32, 72)$ with channels and naturalness images; the $SO(16)$ AB sum $(15, -18, -15, -4, 40)$; the D_8 quartic $16\,T_8 + 12\,s_8^2$) verified with the active engine the same day ($444 \to 450$, `ALL WOLFRAM EXTENSION CHECKS PASSED`). `VerificationDag`: `v508` on the QFT/celestial node; registry + `run_all` (502 modules, suite green). No gate closes, no marker moves — the O-fence is only sharpened.

2026-07-21 (XVII) (`v507` — the tautology attack on the `v506` alignment bit, executed and refuted, promoted as `SEAM.BIT.ORIGIN.01`: the ORIGIN THEOREM — marks = fixed points of the hyperelliptic involution = the 4 half-periods $\Lambda/2\Lambda$ of the seam double, and EVERY mark-free collar deck IS a half-period translation (in all three stabiliser types $(4,3,3)/(8,5,3)/(12,3,3)$ the free involutions are exactly the three σ_c: the CLASS of the deck is forced, its POSITION is not); the CORE SOLVE (symbolic λ): a marks-preserving root of σ_c exists $\iff$ $\lambda \in \{-1, 2, \frac{1}{2}\}$ (one per c; intrinsic: the deck pairs separate harmonically), union = the harmonic orbit $\iff$ $j = 1728$ $\iff$ $\tau = i$; the EQUIVALENCE TETRAD "Uhr existiert $\iff$ ($\tau{=}i$ AND deck = CM-fixed half-period) $\iff$ (Stab = D_4 AND deck central) $\iff$ deck pairs harmonic $\iff$ NS Fock lift NONSPLIT"; the FERMION DICHOTOMY (the new physical face): central $\tilde{U}^2 = 256\,\gamma_1 \cdots \gamma_{16} \Rightarrow U^2 = (-1)^F$ nonsplit vs edge $\tilde{U}_e^2 = +2^7 \Rightarrow U^2 = +1$ SPLIT (2 vs 0 roots — the bit IS a Fidkowski–Kitaev-type extension class, distinguished but not derived); the silver set = the in-family misalignment witness; Wolfram extension $438 \to 444$ (verified engine run); *NO marker moves — the bit stays [C], `SEAM.CLOCK.RIGIDITY.01`/`QGEO.REALIZE.01`/`P2.TYPING.01` text precised only*)

- **`v507_seam_bit_origin.py` (`SEAM.BIT.ORIGIN.01`, 26 checks, ∼2.6 s; Part A sympy-exact torus/Möbius arithmetic + Part B exact Cl(16) Fractions; deterministic).** Conjecture under attack: marks and deck both descend from the SAME torus structure (the seam double $T^2_\tau \to S^2$, `v214`/`v260`), so the common origin would force the central alignment automatically and the `v506` bit would be a tautology. *Origin theorem (die Herkunft)*: the 4 marks are the fixed points of $w \mapsto -w$ on $\mathbb{C}/(\mathbb{Z}{+}\tau\mathbb{Z})$ = the half-periods $\Lambda/2\Lambda$ (a $\frac{1}{3}$-coefficient never qualifies); t_c descends iff $2c \in \Lambda$ — exactly the three half-period translations (control $c = \frac{1}{3}$ fails); the descended maps are the EXPLICIT rational involutions $\sigma_i(x) = e_i + (e_i{-}e_j)(e_i{-}e_k)/(x{-}e_i)$ with the exact curve-lift identity $\prod(\sigma_i{-}e_l) = \psi_i^2 \prod(x{-}e_l)$, ψ_i RATIONAL (the $\wp$-addition formula as pure algebra), closing to Klein V_4; CONVERSELY the freely-acting involutions in ALL three stabiliser types (generic V_4 $(4,3,3)$, harmonic D_4 $(8,5,3)$, hexagonal A_4 $(12,3,3)$) are EXACTLY the three $\sigma_c \Rightarrow$ every mark-free collar deck (`v216`) IS a half-period translation. *Core solve (symbolic λ)*: σ_c admits a marks-preserving PGL_2 root $\iff$ $\lambda = -1$ (σ_1), 2 (σ_2), $\frac{1}{2}$ (σ_3) — per half-period exactly ONE solution, intrinsically "the deck pairs separate harmonically" (pair cross-ratio -1); generic modulus: V_4 abelian, ALL σ_c central yet clockless ("central alone is empty" — the bit is the CONJUNCTION $D_4 \wedge$ central); at $\tau = i$ the stabiliser jumps to D_4 with σ_1 central (2 roots = the clock pair), σ_2/σ_3 edge-type (0 roots); the `v506` seam deck lands on the central $\sigma_1 = -1/x$ under all 8 mark bijections; CM dictionary: mult-by-i fixes exactly $c^* = (1{+}\tau)/2$ (mult-by-ρ 3-cycles all three), the clock lift needs the CM unit i ($\psi = 2i/(x{+}1)^2$, $t^2{+}1$ irreducible over $\mathbb{Q}$) while half-period decks lift rationally. *Equivalence tetrad + circularity audit*: clock exists $\iff$ ($\tau{=}i$ AND deck = the CM-fixed half-period) $\iff$ (Stab = D_4 AND deck central) $\iff$ deck pairs harmonic $\iff$ NS Fock lift nonsplit; the `v493` converse independently re-verified — two independently computed converses of ONE equivalence, no circle. *Fermion dichotomy (Part B)*: central arrangement (shift by

8): $\tilde{U}^2 = 256\,\gamma_1\cdots\gamma_{16} \Rightarrow U^2 = (-1)^F$, nonsplit $\mathbb{Z}_4$, 2 roots in the 32-element dihedral lift (the $\mathbb{Z}_8$ tower); edge arrangement (reflection $j \mapsto 4-j$, fixed sites = non-marks = the silver $\{0,\infty\}$ between neighbour marks): the implementer EXISTS but $\tilde{U}_e^2 = +2^7\cdot\mathbf{1} \Rightarrow U^2 = +1$, the extension SPLITS, 0 roots, no $\mathbb{Z}_8$; one-particle $(\lambda^2{+}1)^8$ vs $(\lambda^2{-}1)^8$; census: all 28 shift-invariant 4-sets cross (central type), all 21 mark-free reflection-invariant sets do not; robust at 8 sites; R-sector edge control splits too — the bit IS a physical extension class (Fidkowski–Kitaev-type): distinguished, but not derived (WHICH class nature realises stays the input). *Silver placement*: $\{\pm1, \pm(3{+}2\sqrt{2})\}$ = "$\tau = i$ with deck = a non-CM-fixed half-period translation" (all 8 bijections $\to \sigma_2/\sigma_3$, never σ_1) — the in-family witness: TAUTOLOGY REFUTED. *The one **[C]** rest (verbatim)*: the alignment bit remains a carrier datum; "no tautology — the origin forces the CLASS of the deck (half-period translation), not its POSITION". No overclaim.

- **Ledger + papers + Wolfram + website.** Ledger: new row `SEAM.BIT.ORIGIN.01` ([**E**] origin theorem/core solve/tetrad/fermion dichotomy/silver placement + [**C**] the bit); TEXT precisions at *unchanged* markers: `SEAM.CLOCK.RIGIDITY.01` ("the bit is now fully characterised as a tetrad … and carries a physical face: the non-split NS Fock extension class — v507"), `QGEO.REALIZE.01` ("the class of the deck IS derived (half-period translation, v507); only its position stays the input", stays [**C**]/[**O**]), `P2.TYPING.01` (origin cross-reference on the common R1 bit). Papers: `tfpt_3` celestial Step-1/2 environment + QGEO paragraph, contracts marks/kill-test paragraph, `introduction/tfpt_1/tfpt_2/origin_theory` one-sentence updates where `v506` is cited. Wolfram: six mirrors (half-period descent $2c \in \Lambda$ with the rational σ_i lift identity; the core solve $\lambda \in \{-1, 2, \frac{1}{2}\}$; the stabiliser counts $(4,3,3)/(8,5,3)/(12,3,3)$ via the complete 24-triple enumeration; the CM fixed point c^* + the lemniscatic $\psi = 2i/(x{+}1)^2$; the Cl(16) dichotomy $256\,\gamma$-monomial vs scalar 2^7 on explicit $256{\times}256$ Jordan–Wigner matrices + the 2-vs-0 root census in the 32-element dihedral lift; the silver bijection census) verified with the active engine the same day ($438 \to 444$, `ALL WOLFRAM EXTENSION CHECKS PASSED`); the `v493` audit direction and the 28/21 arrangement census stay Python-side (flagged). `VerificationDag`: `v507` on the P2/carrier and QGEO nodes; registry + `run_all` (501 modules, suite green). No gate closes, no marker moves.

2026-07-21 (XVI) (`v506` — seam clock rigidity, Route B of the `v503` classification, promoted as `SEAM.CLOCK.RIGIDITY.01`: TWO exact theorems reduce the order-4 clock input to ONE alignment bit — Theorem A (Möbius): the deck $z \mapsto -z$ has EXACTLY TWO square roots $z \mapsto \pm iz$ in $\mathrm{PGL}_2(\mathbb{C})$, both *automatically* order 4, and a marks-preserving root exists $\iff$ the deck is the CENTRAL involution of the mark-D_4 (counterexample $\{\pm1, \pm(3{+}2\sqrt{2})\}$: harmonic with full D_4 but deck non-central $\Rightarrow$ 0 roots — harmonicity alone insufficient); Theorem B (Fock, the core): the NS deck implementation is FORCED to order 4 — $\tilde{U}^2 = 256\,\gamma_1\cdots\gamma_{16}$, i.e. $U^2 = (-1)^F$ EXACTLY, a nonsplit $\mathbb{Z}_4$, with the $\mathbb{Z}_8$ tower $V^2 \propto U$, $V^4 \propto (-1)^F$ ($8 = 2|\mu_4|$ fermionically explained) and the R-sector control split ($U_R^2 = +1$); the one [C] rest: "the collar deck is the central involution of the mark-D_4" — the fermionic $\mathbb{Z}_4$ forces the ORDER, not the spatial alignment; Wolfram extension $434 \to 438$ (verified engine run); *NO marker moves — `QGEO.REALIZE.01` stays [C]/[O] (text precised), `P2.TYPING.01` R1 mapped onto the same bit*)

- **`v506_seam_clock_rigidity.py` (`SEAM.CLOCK.RIGIDITY.01`, 34 checks, ∼3.5 s; Part A sympy-exact Möbius/divisor arithmetic + Part B exact Cl(16) Fractions; deterministic).** `v503` localised the one remaining identification input ("why does the raw seam carry the order-4 clock", QGEO-R1 ≡ P2-R1); `v506` reduces it. *Theorem A (Möbius side, complete)*: Cayley–Hamilton + the complete case analysis give $g^2 = \text{deck} \Rightarrow \operatorname{tr} g \neq 0$ forces g diagonal with $(a/d)^2 = -1$, and $\operatorname{tr} g = 0$ forces $\det g = 0$ (impossible) — exactly two roots $z \mapsto \pm iz$, both projective order 4, mutually inverse ($\mathrm{Aut}(\mathbb{Z}_4)$), conjugation-covariant; marks filter: a deck-invariant deck-free 4-set $\{u, -u, w, -w\}$ admits a marks-preserving root $\iff$

$w = \pm iu \iff$ marks = one μ_4 orbit $\Rightarrow$ cross-ratio class $\{2, \frac{1}{2}, -1\}$, $j = 1728$, $\tau = i$, free 4-cycle, deck = clock^2, only the scale free; group form: $|\mathrm{Stab}(\mu_4)| = 8 = 2|\mu_4|$ with order profile $\{1{:}1, 2{:}5, 4{:}2\} = D_4$, center = $\{\mathrm{id}, \mathrm{deck}\}$, exactly 2 order-4 elements = the clocks — "deck central in the mark-D_4" $\iff$ "clock exists"; the counterexample $\{\pm 1, \pm(3{+}2\sqrt{2})\}$ is deck-free AND harmonic ($j = 1728$, cross-ratio $\frac{1}{2}$) with full D_4, but the deck sits *non*-centrally (the central involution crosses the $\pm$ pairs) $\Rightarrow$ 0 roots; the RP filter passes BOTH roots (the $\mathrm{Aut}(\mathbb{Z}_4)$ ambiguity is irreducible, `v492` NC3). *B3/B4 (n-arithmetic + controls)*: free orbit $\Rightarrow n \mid 4$; transitivity $\Rightarrow n = 4$; faithfulness kills $n \geq 5$ (kernel argument through S_4); $n = 3$ fixes a mark; $n = 6$ dies independently at the cusp degree 3 (`v117`); the deck is modulus-blind (`v503` B5 exact); $n = 8$ impossible on 4 marks; hexagonal stabiliser A_4 (0 order-4 elements, squares only orders $\{1, 3\}$); generic stabiliser V_4 (exponent 2); stabiliser trichotomy $V_4/D_4/A_4$ with order-4 counts $(0, 2, 0)$. *Theorem B (Fock side, the core result)*: S^2 has $b_1 = 0 \Rightarrow$ unique spin structure $\Rightarrow$ NS/bounding on every circle (established P1 datum); one-particle: $D^2 = -1$ (charpoly $(\lambda^2{+}1)^8$), quarter-shift $C^2 = D$, $C^8 = 1$ (charpoly $(\lambda^4{+}1)^4$, ζ_8 spectrum); Fock (exact Cl(16)): $\tilde{U} = \prod_j (1 - \gamma_j \gamma_{j+8})$ implements the deck on all 16 generators and $\tilde{U}^2 = 256\, \gamma_1 \cdots \gamma_{16}$ $\Rightarrow U^2 = (-1)^F$ EXACTLY; $(cU)^2 = c^2(-1)^F$ is *never* 1 (center of Cl(16) = scalars): the $\mathbb{Z}_4$ lift is FORCED, $1 \to \mathbb{Z}_2^{(-1)^F} \to \mathbb{Z}_4 \to \mathbb{Z}_2^{\mathrm{deck}} \to 1$ nonsplit; the quarter-shift lift V (unique even block implementer) has $V^2 \propto \tilde{U}$, $V^4 \propto (-1)^F$, projective order $8 = 2|\mu_4|$ — the `v492` $\mathbb{Z}_8$ spin bridge realised on the 16-Majorana Fock space; R control $U_R^2 = +1$ (split — the forcing rests on NS = established geometry); 16-fold way $\theta_v = 1$ at $\nu = 16$; robustness: the forcing already holds at 8 sites (16 = the carrier count $2^{g_{\mathrm{car}}-1}$, not a tuning). *The one **[C] rest (verbatim)***: "the collar deck is the central involution of the mark-D_4" ($\iff$ square-root existence on the mark data) — ONE alignment bit; the fermionic $\mathbb{Z}_4$ forces the ORDER, not the spatial alignment. No overclaim.

- **Ledger + papers + Wolfram + website.** Ledger: new row `SEAM.CLOCK.RIGIDITY.01` ([**E**] both theorems + [**C**] the alignment bit); TEXT precisions at *unchanged* markers: `QGEO.REALIZE.01` ("the rest is now ONE bit: the central position of the deck in the mark-D_4; order/uniqueness/orbit/$\tau{=}i$ follow — v506", stays [**C**]/[**O**]), `P2.TYPING.01` (R1 mapped onto the same bit, fermionic halving), `QGEO.EMERGE.LIGHT.01` context sharpened. Papers: `tfpt_3` celestial Step 1 (fermionic $U^2 = (-1)^F$ sentence) + Step 2 (the one-bit reduction), contracts QGEO kill-test paragraph + marks paragraph + CELEST hypothesis note, `introduction/tfpt_1/tfpt_2/origin_theory` one-sentence updates at the `v499/v503` citation sites. Wolfram: four mirrors (the complete square-root solve; D_4 center + order-4 counts + the silver counterexample; the Cl(16) identity $\tilde{U}^2 = 256\, \gamma_1 \cdots \gamma_{16}$ on explicit $256{\times}256$ Jordan–Wigner matrices + the split R control; the n-arithmetic + generic kill) verified with the active engine the same day (434 $\to$ 438, `ALL WOLFRAM EXTENSION CHECKS PASSED`); the 24-triple stabiliser enumeration and the A_4 profile stay Python-side (flagged). `VerificationDag`: `v506` on the P2/carrier node; registry + `run_all` (500 modules, suite green). No gate closes, no marker moves.

2026-07-21 (XV) (`v505` — WP5e-β of `CELEST.SEAM.01` "the equivariant anomaly ledger on twistor space" promoted as `CELEST.WP5E.BETA.01`: the Atiyah–Bott/Lefschetz fixed-point skeleton of the one-loop twistor inflow is EXACT — denominators $(2,4,2)$ with Dedekind sum $\frac{5}{4} = (|\mathbb{Z}_4|^2-1)/12$, equivariant characters $(248,0,-8,0)$ by two routes, invariant Okubo $36\langle x,x\rangle^2$ with the twisted sectors NOT reducing and the AB sum leaving the RIGID residual $32\,T_3$, the index bridge $f(m) = \mathrm{ch}_2(T_m)$ exact with glue defect -78 both routes, the level dials $k=1$ geometric (current count $240/0/0/0$), and the honest E3 REFUTATION: a_0 fills the GRAVITON slot $O(2)$, NOT the axion slot $O(-2)$ — the three sphere classes fill the three twisted axion slots instead; the preregistered kill ("inflow demands a level $\neq 1$") does NOT fire on the equivariant skeleton; WP5e: α (CFT side, `v502`) + β (equivariant twistor skeleton, `v505`) executed, the global BCOV quantisation stays [O]; Wolfram extension $429 \to 434$ (verified engine run); `SEAM.EQUIV.01` stays [O], no marker moves)

- **`v505_celestial_wp5e_beta_equivariant_ledger.py` (`CELEST.WP5E.BETA.01`, 47 checks, ∼1.3 s; sympy/Fraction exact, no floats; deterministic).** The twistor side of WP5e pushed as far as exact arithmetic reaches: the Atiyah–Bott/Lefschetz skeleton of the one-loop box anomaly (Costello; Bittleston–Sharma–Skinner) on $\mathbb{C}^2/\mathbb{Z}_4$ with the E_8 μ_4-glue monodromy. *AB ledger*: $\det(1-g_j^{-1}) = (2,4,2)$; $\sum_j 1/\det_j = \frac{5}{4} = (|\mathbb{Z}_4|^2-1)/12$; $\mathrm{tr}_{g_j}(\mathrm{Ad}\,E_8) = (248,0,-8,0)$ two routes; invariant average 60 = carrier; Frobenius 61568; $\mathrm{tr}_{g_2} = -8 = 120-128$. *Okubo per sector (honest sharpening)*: the invariant sector is exactly Okubo-quadratic ($Q^{(0)} = 36\langle x,x\rangle^2$, $36 = \tilde{\lambda}_{e_8}^2$ — `v495` re-derived through the glue grading); the twisted sectors do NOT reduce (T_5: $(-8,32,-8)$, T_3: $(48,-64,48)$); the AB sum cancels the D_5 quartic exactly and leaves the rigid $32\,T_3$ (the only quartic-free weighting is the $\mathbb{Z}_4$ projector; the graded GS exchange is rank-obstructed in sectors 1–3); heterotic identities (Tr_{120}, Tr_{128}, $\mathrm{Tr}_{E_8} = 9(\mathrm{tr}^2)^2$) exact. *Index bridge*: $f(m) = \frac{1}{4}\sum_j (i^{jm}-1)/\det_j = (0,-\frac{3}{8},-\frac{1}{2},-\frac{3}{8}) = \mathrm{ch}_2(T_m) = -(C^{-1})_{mm}/2$ EXACT (E1 = E2 arithmetic); glue ch_2 defect -78 both routes (classes $(-24,-30,-24)$); Killing sums $(320,320,240,320)$, total $1200 = 2h^\vee(h,h)$. $k=1$ *geometric*: lattice current count 240 at k=1, exactly 0 at $k=2,3,4$; embedding residual $47k^2+219k-266 = (0,360,814,1362)$; integrality alone fixes nothing (honest, as `v502`); one scale ⇒ one level (`v493` periods). *E3 refutation (prominent)*: $a_0 \in O(8)$ fills the GRAVITON slot $O(2)$, NOT the BSS axion slot $O(-2)$ ($X = -2 \neq +2 = Y$; mismatch $4 = |\mu_4|$ as a [C] coincidence) — "the theory brings its GS axion as a_0" is REFUTED; instead the three H^2(ALE) classes carry exactly the Coxeter eigencharacters $\{i,-1,-i\}$ = the three twisted slots (bijection); slot count $1+3 = 4 = |\mu_4|$ soft. *Negative controls*: $\mathrm{diag}(i,i)$ breaks the ledger four ways ($\sum 1/\det = \frac{1}{4}$, ω-weight -1, 5 generators); $SO(16)$ glue: characters $(120,0,+120,0)$, defect $-30 \neq -78$, bulk Okubo fails (independent 4th Casimir); $k=2$: current count 0, c-prefactor $-31/48 \neq -\frac{1}{3}$. [C] dictionaries ($\mathrm{CS}(\rho)$ mod 1 = discriminant weight; $32 = (h,h)+(h',h')$; $-30 = -h^\vee$; Ω-contraction). [O] the global BCOV/KS quantisation, non-fixed-point contributions, the $O(-2)$ bulk axion, CPS level-from-flux, the pairing $32\,T_3 \leftrightarrow$ sphere axions. VERDICT B; the kill test does not fire on the equivariant skeleton.
- **Ledger + papers + Wolfram + website.** Ledger: new row `CELEST.WP5E.BETA.01`; `CELEST.SEAM.01` updated to "WP5e: α (CFT side, `v502`) + β (equivariant twistor skeleton, `v505`) executed; the global BCOV quantisation stays [O] — with the rigid $32\,T_3$ twisted residual and the sphere-axion pairing as the named remaining targets" at *unchanged* Open status. Papers: contracts WP5e bullet → "α + β done" with the a_0 refutation as an honest finding (kill-test and honest-success paragraphs consistent); `tfpt_3` celestial section becomes the *eleven*-step story (new Step 11 "the equivariant anomaly ledger on twistor space", "what remains" = the global BCOV quantisation with named targets). Wolfram: five mirrors (AB denominators + characters two routes; the index bridge + -78 defect; the per-sector Okubo sums over the 240 glue roots with the $32\,T_3$ residual; the lattice current counts + embedding residuals; the

axion-slot weight arithmetic + twisted-slot bijection) verified with the active engine the same day (429 → 434, `ALL WOLFRAM EXTENSION CHECKS PASSED`). `VerificationDag` qft node gains `v505`; registry + `run_all` (499 modules at this step; 500 with `v506`, suite green). No gate closes, no marker moves — `SEAM.EQUIV.01` stays **[O]**.

2026-07-21 (XIV) (`v504` — WP5d-β of `CELEST.SEAM.01` "split + strong additivity from the lattice" promoted as `CELEST.WP5DB.01`: the two remaining KLM legs of complete rationality witnessed for the orbifold prescription — strong additivity ALGEBRAICALLY EXACT with the shared boundary Majorana ($\mathrm{Even}(A) \vee \mathrm{Even}(B) = \mathrm{Even}(A \cup B)$, GF2 spans full, rank $32/32$; disjoint exactly index $2 =$ the `v501` $\ln 2$ bit), the entropic touching defect BOUNDED $< \ln 2$ with the Ising $\frac{1}{4}$-exponent approach ($p = 0.2444$) while the preregistered $U(1)$/Dirac control DIVERGES (Klich–Levitov pinned, infinite index), the split property at the elliptic-nome rate $\pi K(1-x)/K(x)$ (1.3–2.0%) with EXACT orbifold inheritance ($C \to -C +$ the $\Lambda^2 C$ compound — Longo heredity), and Pimsner–Popa $E(a) - a/2 = PaP/2$ identically ($\lambda = \frac{1}{2} = 1/[F{:}F_{\mathrm{even}}]$, $\lambda_{E_4} = \frac{1}{4} = 1/\mu$, $e^{\Delta_\infty} = 2 = 1/\lambda_{\mathrm{PP}}$ — two independent index routes); with `v501` ALL THREE KLM ingredients carry lattice witnesses; the continuum uplift stays [O] — "finite-group orbifolds of completely rational nets are completely rational" is XU'S THEOREM, cited not claimed; Wolfram extension $424 \to 429$ (verified engine run); `SEAM.EQUIV.01` stays [O], `CELEST.SEAM.01` stays Open, no marker moves)

- **`v504_celestial_wp5d_klm_completeness.py` (`CELEST.WP5DB.01`, 37 checks, ∼12 s; numpy/scipy Gaussian machinery ED-validated + exact GF2/integer algebra; deterministic).** Complete rationality (Kawahigashi–Longo–Müger) = split + strong additivity + finite two-interval μ-index; `v501` (WP5d-α) witnessed the index, `v504` witnesses the other two legs on the same critical Majorana seam edge, with the $U(1)$/Dirac number-conserving subnet as the preregistered negative control (the current net, where strong additivity famously *fails* in the continuum — Buchholz–Schulz–Mirbach). *Strong additivity, exact algebra*: "touching = shared boundary Majorana" gives $\mathrm{Even}(A) \vee \mathrm{Even}(B) = \mathrm{Even}(A \cup B)$ EXACTLY (GF2 spans full 64/64, 256/256, 512/512, 1024/1024; literal matrix rank 32/32); disjoint intervals give exactly HALF (rank 16/32, index 2 — the missing sector odd $\otimes$ odd is the `v501` $\ln 2$ bit, localised at the split point); the neutral $U(1)$ algebras do NOT generate the union even with the shared site (gaps 2/10/52, growing). *Entropic witness*: the fermionic touching MI carries the pure UV point-gap law ($(c/3)\ln 2$ increments, 0.115525, dev 1.2×10^{-6} — present for every net, no discriminator); the honest discriminator is *bounded vs divergent*: the $\mathbb{Z}_2$/orbifold deficit rises $0.4045 \to 0.5899$ ($N = 64 \to 4096$), stays $< \ln 2$, and $(\ln 2 - \Delta_2) \sim N^{-p}$ with $p = 0.2444$ vs $2\Delta_\mu = \frac{1}{4}$ (Ising disorder exponent); split-point scan reflection-symmetric to 3.9×10^{-14}; ED vN same direction; gap continuity to the `v501` $x \to 1$ endpoint; honest note: "defect $\to 0$" would be *false* at the sharp lattice split — bounded $\Leftrightarrow$ finite index (Longo–Xu) is the correct statement; the $U(1)$ control bursts $\ln 2$ from $L = 128$ and grows as $\frac{1}{2}\ln \mathrm{Var}\, Q_A$ (Klich–Levitov slope 0.10134 vs $1/\pi^2 = 0.10132$). *Split/nuclearity*: coupling-block σ_k fall exponentially at the elliptic-nome rate $\pi K(1-x)/K(x)$ to 1.3–2.0% (rates 2.29/3.24/4.82/5.81 at $d = 16/64/256/512$); $\sigma_1 \sim x^{0.5044}$ (vs $\frac{1}{2}$); trace norm summable; dimerised control exponential. *Orbifold inheritance exact*: P_A flips $C \to -C$ (σ_k identical, 8.3×10^{-17}) and the even-bilinear coupling Gram is the second compound $\Lambda^2 C$ (singular values $\{\sigma_a \sigma_b\}$, 2.8×10^{-14}) — the same ladder at doubled rate (Longo heredity, **[C]**). *Pimsner–Popa, exact*: $E(a) - a/2 = PaP/2$ identically; attainment in integers (all 16384 (even, odd) monomial pairs at $N = 8$: 8192 extremal with $M^2 = 4$, $\mathrm{Tr}\, M = 0$, $PMP = -M$, 0 violations; 2048/2048 at $N = 12$; pencil $\min \lambda = 0.5000000000$); $\lambda_{E_4} = \frac{1}{4} = 1/\mu$; index consistency $e^{\Delta_\infty} = 2.000000000 = 1/\lambda_{\mathrm{PP}}$ and $e^{2\Delta_\infty} = 4 = \mu$; $U(1)$: $\lambda = 1/(m{+}1) \to 0$. *Controls + honesty*: dimerised phase trivial (drift 3.6×10^{-15}); wrong prescription $\{1, P_{AB}\}$ deficit exactly 0; $\mathbb{Z}_2$ finite-size honesty (14.9% below

$\ln 2$ at $N = 4096$ — witness, not proof). VERDICT: with `v501` all THREE KLM ingredients of complete rationality are witnessed on the lattice; the continuum theorem for finite-group orbifolds is *Xu's* (cited, not claimed), the concrete seam quotient net and the condensed $(E_8)_1$ net stay WP5e/Costello–Li.

- **Ledger + papers + Wolfram + website.** Ledger: new row `CELEST.WP5DB.01`; `CELEST.SEAM.01` updated to "WP5d: α (μ-index) + β (split + strong additivity) executed via `v501/v504` — all three KLM ingredients witnessed on the lattice; continuum uplift stays **[O]** (Xu's theorem cited)" at *unchanged* Open status. Papers: contracts WP5d bullet → "both stages done" with the β-stage numbers + Xu fence (kill-test and honest-success paragraphs consistent); `tfpt_3` celestial section becomes the *ten*-step story (new Step 10 "the KLM completion on the lattice", "what remains" = WP5e twistor inflow only); `tfpt_4` keystone note updated. Wolfram: five mirrors (GF2 span theorem; Pimsner–Popa identity + attainment on explicit gamma matrices; two-interval group $\lambda_{E_4} = \frac{1}{4}$ + index consistency; the Λ^2/Minors compound with Cauchy–Binet and Minors$[-C, 2]$ = Minors$[C, 2]$; the $U(1)$ dephasing $\lambda = 1/(m{+}1)$ family) verified with the active engine the same day (424 → 429, `ALL WOLFRAM EXTENSION CHECKS PASSED`); the lattice curves stay Python-only (flagged). `VerificationDag` qft node gains `v504`; registry + `run_all` (498 modules, suite green). No gate closes, no marker moves — `SEAM.EQUIV.01` stays **[O]**.

2026-07-21 (XIII) (`v503` — the `v215` deferred emergence test (lever 7) executed *light* and promoted as `QGEO.EMERGE.LIGHT.01`: NO dynamical selection of the square modulus — the free field's global $\det'$ winner is the *hexagonal* torus, $\tau{=}i$ is a saddle, criticality at the square is Palais symmetry; the square is pinned by *rigidity* alone (the raw DtN is mod-4 block-diagonal *iff* $\tau{=}i$, an isolated zero); and the residual *converges*: QGEO-R1 ≡ P2-R1 = one common rest, "why does the raw seam carry the order-4 clock"; Wolfram extension 420 → 424 (verified engine run); *NO marker moves — `QGEO.REALIZE.01` stays [C]/[O], `AX.P2.01` stays a declared axiom*)

- **`v503_qgeo_emergence_light.py` (`QGEO.EMERGE.LIGHT.01`, 17 checks; mpmath 40 digits + `v280`-style lattice DtN + sympy exact).** The `v215` lever 7 ("the decisive kill is deferred: compute the raw seam-collar DtN *without* inputting the marks and check whether four square marks *emerge*") executed in its first honest *light* version. *No dynamical selection*: $\log F = \log(\tau_2|\eta|^4)$ (Osgood–Phillips–Sarnak) is exactly S-invariant (dev $< 10^{-35}$), so $t{=}1$ is symmetry-forced critical on the rectangular family $\tau = it$; there $g''(1) = -1.298 < 0$ (strict maximum) but the τ_1-direction has $h''(0) = +0.298 > 0$: $\tau{=}i$ is a *saddle* of the full moduli space; the only critical points are the CM points ($E_2^*(i) = E_2^*(\rho) = 0$ at 40 digits, generic $0.163 \neq 0$); the *global* $\det'$ winner at fixed area is the *hexagonal* torus ($\log F(\rho) = -1.03352 > \log F(i) = -1.05469$, diff 0.021169); criticality at the square is Palais symmetric criticality (the independent heat trace is equally critical). *Sharp rigidity*: the DtN mod-4 off-character indicator (the `v215`-K3 lever as a *modulus function*) vanishes to 8×10^{-15} exactly at $\tau{=}i$ and rises strictly monotonically both ways (min 8.4×10^{-2} away) — an isolated zero; $R^2 = \mathbb{Z}_2$ exact as permutation matrices (the `v492` $\mathbb{Z}_8$ bridge on the lattice); $j(it) \geq 1728$ with equality only at $t{=}1$. *The $\mathbb{Z}_2$ chain is modulus-blind*: 4 marks + $[\mathbb{Z}_2, \Delta_t] = 0$ on *every* torus (`v216` confirmed without mark input); pillowcase $\det' = \frac{1}{2}$ torus $\det'$ exactly ($\pm$-pair structure) — only the order-4 clock distinguishes $\tau{=}i$; $\mathrm{Aut}(E_\rho) = \mathbb{Z}_6$ has *no* order-4 element (the dynamical winner cannot carry the clock); given the clock the modulus freezes automatically ($j = 1728$ identically, `v493` mini). *Residual convergence (the core finding)*: the order-4 element on H^1 is a Coxeter element of $W(A_3)$ (char poly $\lambda^3{+}\lambda^2{+}\lambda{+}1$, order $4 = |\mu_4| = N_{\mathrm{fam}}{+}1$) = the P2 cusp-class carrier datum of `v499` R1 ⇒ *QGEO-R1 ≡ P2-R1, one common residual*; chain: marks derived (`v216`), square derived *given* the clock (`v493`), clock = carrier input [[**C**]], rest = its dynamical realisation **[O]**; route classification (`v347` analogue): B-rigidity in the sharpened form "RP seam data ⇒ order-4 clock"

is the live route. v215 lever 7 thereby moves from "deferred, not executable" to "light version executed: rigidity, not dynamical selection" — documented in v503, v215 itself unchanged.

- **Ledger + papers + Wolfram + website.** Ledger: new row QGEO.EMERGE.LIGHT.01; QGEO.REALIZE.01 text *precised* (rest = the order-4 clock, identical with the P2 typing rest R1; square + mark count derived) at *unchanged* marker **[C]/[O]**; cross-reference added in P2.TYPING.01 (convergence theorem). Papers: contracts kill-test paragraph (lever-7 light result + the residual-convergence sentence, both with v503), tfpt_3 Step 2 (light-converse sentence), origin_theory necessity paragraph, introduction/tfpt_1/tfpt_2 one-residual cross-references. Wolfram: four mirrors (CM-point identities $E_2(i) = 3/\pi$, $E_2(\rho) = 2\sqrt{3}/\pi$ $\Rightarrow E_2^* = 0$ exact, j/DedekindEta special values; the no-dynamical-selection witnesses; the pillowcase halving as an exact algebraic identity; clock $\Rightarrow$ square + Coxeter) verified with the active engine the same day (420 → 424, ALL WOLFRAM EXTENSION CHECKS PASSED); the lattice DtN curves stay Python-only (flagged). VerificationDag masses node gains v503; registry + run_all (497 modules, suite green). No gate closes, no marker moves.

2026-07-21 (XII) (v502 — WP5e-α of CELEST.SEAM.01 "prefactor and level bookkeeping" promoted as CELEST.WP5E.ALPHA.01: the $q^{-1/3}$ prefactor of E_4/η^8 IS exact μ_4 vacuum-energy bookkeeping (the clock is inner $\Rightarrow$ shift orbifold $\theta = 0^8$, common $-c/24 = -\frac{1}{3}$; discriminant form = Casimir via spectral flow AND 16-Majorana free fermions), and $k = 1$ is forced THREE independent ways (current condition; conformal embedding; central charge = the prefactor itself; plus the WP5b singular-vector dial $31124 - 27000 = 4124$); honest sharpening: glue-h integrality holds for ALL k and fixes nothing; the BCOV/Costello–Li twistor inflow stays [O] — WP5e is subdivided (α done, WP5e proper open); Wolfram extension $416 \to 420$ (verified engine run); SEAM.EQUIV.01 stays [O], no marker moves)

- **v502_celestial_wp5e_prefactor_level.py (CELEST.WP5E.ALPHA.01, 33 checks, exact sympy/Fraction, no floats).** The two exactly computable consistency anchors of the WP5e twistor uplift established. *Part A — the prefactor*: Hurwitz-ζ exact ($\zeta(-1,\theta) = -B_2(\theta)/2$; $E_b(\theta) = -\frac{1}{24} + \theta(1-\theta)/4$; Ising shift $\frac{1}{16}$ per Majorana); the clock is *inner* ($h = (2,2,2,2,2;0^4)$ and $h' = (0^5;1,1,1,-3)$ both read the class mod 4 on all 240 roots; $(h,h) = 20$, $(h',h') = 12$, sum 32) $\Rightarrow$ twist $\theta = 0^8$ in all four sectors — a *shift* orbifold, NOT a rotation orbifold — so $E_{\rm osc} = -\frac{1}{3} = -c/24$ is sector-independent ($c = 8 =$ rank $=$ Sugawara 248/31; one common η^8); sector ground exponents $-\frac{1}{3} + (0,1,1,1)$; *discriminant form = Casimir energy* two ways (spectral flow $j^2(h,h)/32$ mod 1; free fermions exact: $D_5 = 10$ Majorana $\to \frac{5}{8}$, $A_3 = D_3 = 6 \to \frac{3}{8}$, glue diagonal $= 16 = 2^{g_{\rm car}-1} =$ the WP5d seam carrier $\to 1$); the rotation reading *fails* (deck $\frac{3}{16} \neq \frac{3}{8}$ — "clock ≠ deck" at the Casimir level); the four sector characters sum exactly to $E_4/\eta^8 = q^{-1/3}(1, 248, 4124, 34752)$. *Part B — the level*: $k = 1$ forced three independent ways — (i) the current condition $h(J) = k = 1$ (adjoint closure 60+64+60+64 = 248 at spin 1); (ii) the conformal embedding ($47k^2+219k-266 = 47(k-1)(k+266/47) = 0$; D_8: $128k(1-k) = 0$); (iii) $c = 8 \Leftrightarrow 240(k-1) = 0$ (= the prefactor); plus the v498 singular-vector dial ($\|(E_{-1}^{\theta})^m 0\|^2 = m!\,k(k-1)\cdots(k-m+1)$, singular vector at grade $k+1$; $31124 - 27000 = 4124$ at k=1 only, $\langle s|s\rangle = +4$ at k=2). *Honest sharpening (mandatory)*: glue-h integrality $h(J^j;k) = k\,(0,1,1,1)$ is integer for ALL $k = 1..8$ — the originally conjectured integrality route fixes *nothing*; the three other dials carry. *Negative controls*: $D_8/SO(16)_1$ (same $c = 8$, SAME $q^{-1/3}$ prefactor, but $h = (0, \frac{1}{2}, 1, 1)$: the discriminator is glue-h integrality, not the prefactor; $\Theta_{D_8} + \Theta_s = E_4$); $\mathbb{Z}_2$ rotation ($E = +\frac{1}{6}$, $h = \frac{1}{2}$); μ_4 rotation (sector-dependent $(-\frac{1}{3}, \frac{1}{24}, \frac{1}{6}, \frac{1}{24})$ — the common prefactor breaks); wrong $k \in \{2,3,4\}$ fails all five dials coherently (residuals 360/814/1362; $\langle s|s\rangle = 4/12/24$; $31124 \neq 4124$). [[**C**]] Kac generation for $k \geq 3$ ($k = 1,2$ machine-anchored by WP5b/WP5c). **[O]** the BCOV/Costello–Li twistor inflow itself (Costello's one-loop axion $\lambda_g/(8\pi\sqrt{3})$ on $\mathbb{PT}$, the Costello–Paquette–Sharma level-from-flux mechanism) — v502 anchors the *CFT side* of WP5e only; the KILL branch ("inflow demands a level ≠ 1") is not triggered

on the CFT side, its twistor side is untested.

- **Contract + ledger + Wolfram + website.** `tfpt_research_contracts`: the WP5e bullet *subdivided* — "WP5e-α (done) [**E**]/[**C**]: the CFT-side prefactor + level pinning" with the core numbers and `v502_celestial_wp5e_prefactor_level.py`; "WP5e proper [**O**]: the twistor inflow"; summit header, honest-success paragraph and status roster updated ("WP5a–WP5d plus WP5e-α landed; WP5e proper the remaining open piece"). `tfpt_3` celestial section: new Step 9 (prefactor + level, one paragraph); `tfpt_4` keystone note updated. Ledger: new row `CELEST.WP5E.ALPHA.01`; `CELEST.SEAM.01` status → "WP5e: alpha stage (CFT-side prefactor + level pinning) executed via v502; the BCOV/twistor inflow itself stays [**O**]". Wolfram: four exact mirrors (Hurwitz-ζ vacuum energies; discriminant = Casimir with the deck failure; the k-fixing equations incl. the "fixes nothing" integrality table; the character sums + D_8 control) verified with the active engine the same day (416 → 420, `ALL WOLFRAM EXTENSION CHECKS PASSED`). `wolfram/README.md`, `GATE.WOLFRAM.02`, `tfpt_safeguards` Layer-6 count and the website counters (116/116 + 424/424 after (XIII)) updated; `VerificationDag` qft node gains `v502`; registry + `run_all` (497 modules, suite green). `SEAM.EQUIV.01` untouched; no marker moves.

2026-07-21 (XI) (`v501` — WP5d-α of `CELEST.SEAM.01` "the two-interval index from the lattice" promoted as `CELEST.WP5D.01`: the KLM Haag-duality discriminator measured entropically on the seam edge — the 16-layer carrier saturates the $\ln 2$ offset at machine precision, $\mu_{\text{gauged}} = 4$ = the `v490` census, and the condensation arithmetic $16 \to 4 \to 1$ closes with $\mu = 1$; the preregistered KILL ("μ-offset $\neq 0$ after condensation") does NOT fire; contract now "WP1–WP5d executed; *WP5e the single remaining milestone*"; Wolfram extension $412 \to 416$ (verified engine run); `SEAM.EQUIV.01` stays [O], no marker moves)

- **`v501_celestial_wp5d_two_interval_index.py` (`CELEST.WP5D.01`, 39 checks, numpy/scipy Gaussian machinery + exact Fractions).** The WP5d roadmap discriminator — $(E_8)_1 \Rightarrow \mu$=1 (Haag-dual two-interval net, one sector) vs $SO(16)_1 \Rightarrow \mu$=4 — executed at the α (lattice-index) stage on the `v367` seam edge (16 critical Majorana layers, $c_- = 8$, `v450`-style ring; Peschel-exact, every Gaussian formula ED-validated to $\leq 1.5 \times 10^{-15}$; the superselection identity $[\rho, P_{AB}] = 0$ machine-exact). *The $\mu = 1$ reference*: the fermionic (fixed-spin-structure) two-interval mutual information is *extensive* — c fit 0.5000, $\max|F(x)| = 5.97 \times 10^{-5}$ at N=512, falling monotonically to 1.3×10^{-6} at N=1024 (Casini–Huerta extensivity as a lattice witness). *The μ-offset*: Rényi-2 exactly $\Delta_2 = -\ln[(1+R)/2]$; the 16-layer seam carrier saturates $\ln 2$ at machine precision ($|\Delta_2 - \ln 2| = 1.1 \times 10^{-15}$ at N=512; $R \sim N^{-0.500}$); index reading $[F : F_{\text{even}}] = 2.000000$, $\mu_{\text{gauged}} = 4.000000$ = the `v490` parity census (*two independent lattice witnesses*); per-layer gauging costs $16 \ln 2$ (the offset measures ln[index], not a blob). *Haag-duality indicator*: fermionic $S(E) = S(E')$ to 8.2×10^{-12}; the orbifold *breaks* complementarity ($D(E) - D(E') = 0.427$ at a=64); the complementary-pair sum is exchange-symmetric and $\leq \ln 4 = \ln \mu(SO(16)_1)$ with max 1.2637 at the self-dual $x = \frac{1}{2}$ — a *double-limit* statement (the first naive fixed-x check was wrong and was replaced — honestly typed in the module). *Condensation arithmetic anchored at both measured ends*: $\det \text{Cartan}(D_5) \cdot \det \text{Cartan}(A_3) = 16$; KLM/Longo–Rehren $16/4^2 = 1$ and $16/2^2 = 4 \to 4/2^2 = 1$ (both routes exact); $\Sigma d_i^2 = (4, 4, 1)$; $\theta_v = 1$ exactly at $\nu = 2c_- = 16$, rivals $\neq 1$ — so $\mu = 1$ after condensation ([[**C**]] the condensed net is interacting, the `v490` fence): KILL not triggered. *Negative controls*: ν=1 single Majorana (offset non-removable, $\theta_v = e^{i\pi/8} \neq 1$ — the discriminator has teeth); trivial phase (all $\leq 10^{-7}$); wrong sector sum $\{1, P_{AB}\}$ (offset exactly 0). [**O**] WP5d-β (continuum Haag duality / local normality / split of the quotient net; runs as exploration) and WP5e (Costello–Li) stay open.

- **Contract + ledger + Wolfram + website.** `tfpt_research_contracts`: the WP5d bullet → "α stage done" with the core numbers and `v501_celestial_wp5d_two_interval_index.py`; status line and honest-success paragraph now "four of five milestones landed; WP5e (twistor

uplift) is the remaining open piece". `tfpt_3` celestial section: new Step 8 (one paragraph, conservative); `tfpt_4` keystone note updated to WP1–WP5d. Ledger: new row `CELEST.WP5D.01` (fine types Formal + Lattice/Numerical + [[**C**]] dictionary); `CELEST.SEAM.01` status → "WP1–WP5d executed; WP5e **[O]** (single remaining milestone)". Wolfram: four exact mirrors (Cartan determinants; KLM/Longo–Rehren quotients; Σd^2 + index-reading arithmetic; 16-fold-way θ_v) verified with the active engine the same day (412 → 416, `ALL WOLFRAM EXTENSION CHECKS PASSED`); the lattice curves stay Python-only (flagged). `wolfram/README.md`, `GATE.WOLFRAM.02`, `tfpt_safeguards` Layer-6 count and the website counters (116/116 + 416/416) updated; `VerificationDag` qft node gains `v501`; registry + `run_all` (495 modules, suite green). `SEAM.EQUIV.01` untouched; no marker moves.

2026-07-21 (X) (`v500` — WP5c of `CELEST.SEAM.01` "the GNS limit state" promoted as `CELEST.WP5C.01`: the quasi-free family ω_w exists, is positive, stabilises exactly at the WP5a threshold, and its limit carries the null ideal in its GNS kernel — the full 9361-block exact level-2 Gram has rank 4124 exactly with kernel $27000 = V(2\theta)$ weight by weight; $|s\rangle$ IS the GNS zero vector (the WP5b S4.10 tension resolved at the state level); a CCR obstruction proves the family formulation *necessary*; preregistered SUCCESS criterion met, KILL not triggered; Wolfram extension 408 → 412 (verified engine run); `SEAM.EQUIV.01` stays [O], no marker moves)

- **`v500_celestial_wp5c_gns_limit_state.py` (`CELEST.WP5C.01`, 35 checks, exact integer/Fraction, no floats, deterministic).** WP5a (`v497`) made the boundary limit precise, WP5b (`v498`) built the deleting operator; WP5c constructs the *state*. ω_w is quasi-free: loop sector = the affine k=1 vacuum n-point functions via the *machine-determined* compact anti-involution $\theta(e_\alpha) = -e_{-\alpha}$ (the unique sign that is an anti-automorphism on all 61504 basis pairs; contravariance exact); radial sector = oscillator pairings x^{wr} (the exact Gibbs regulator, x-adic). *Convergence sharp*: $\omega_w = \omega_\infty \bmod x^{N+1}$ for all $w \geq N{+}1$ (N=8), not yet at $w = N$ — the WP5a threshold $w = n{+}1$ holds *identically at the state level*; positivity of every ω_w checked. *The kernel (core, fully machine)*: the complete exact level-2 Gram — 31124 monomials in 9361 weight blocks (profile 164:1, 37:240, 7:2160, 1:6960) — has level-1 rank 248 positive definite (Cartan block = the E_8 Cartan matrix, det 1), level-2 rank 4124 *exactly* (the preregistered target check), kernel 27000, every block PSD; rank per Weyl orbit $(0, 0, 1, 8, 44)$; kernel = $V(2\theta)$ *weight by weight* (Freudenthal cross-check on all 9361 weights); highest-weight norms $(3844, 49, 2, 0)$ at $(0, \omega, \theta, 2\theta)$; direct ideal checks $F_i|s\rangle$, $F_jF_i|s\rangle$ norm 0. *μ_4-equivariance*: $\omega_w \circ \text{clock} = \omega_w$ for all w; the clock descends to GNS; level-2 rank split by clock class $(1036, 1024, 1040, 1024) = \Theta_{C_j}/\eta^8$ at q^2 — the two-routes identity at the *state* level. *Resolution of WP5b S4.10*: $|s\rangle$ *is* the zero vector of GNS(ω_∞) — the twisted two-C_1-mode floor (6 quarters = $q^{3/2}$) concerns only pre-limit objects (GNS has no state there, 6 mod 4 ≠ 0). *Negative controls*: k=2 ($\langle s|s\rangle = +4$, dominant blocks full rank — no ideal in the kernel); k=0 (level-1 Gram ≡ 0 — no 248 layer without the central extension); D_8 (rank 2076 = 7380 − 5304, kernel = $V(2\theta_{D_8})$ PSD, but $h = (0, \frac{1}{2}, 1, 1)$: one block of four — one-block completeness stays E_8/μ_4-specific); wrong family $E'_w = mw{+}r$ (the 248 layer erased, 710955 ≠ 248); no damping (897266 ≠ 248); *CCR obstruction*: at fixed radial pairing $c > 0$ positivity forces $\omega(bb^*) \geq c$ — no w-uniform damping state exists, the family formulation is *necessary* and the family *exists*: the preregistered KILL is probed and not triggered. [**C**] quasi-free extension to the full $*$-algebra; Kac positivity beyond level 2. [**O**] local normality, net structure/Haag duality (WP5d), Costello–Li (WP5e).
- **Contract + ledger + Wolfram + website.** `tfpt_research_contracts`: the WP5c bullet → done with the core numbers (state-level threshold $w = N{+}1$, the full 9361-block Gram rank 4124/kernel 27000 weight by weight, the clock-class split, the CCR necessity of the family), citing `v500_celestial_wp5c_gns_limit_state.py`; the WP5b handover marked answered. `tfpt_3` celestial section: new Step 7 (one paragraph, conservative). Ledger: new row `CELEST.WP5C.01` (fine type Formal/Identity [**E**] + [**C**] identifications + [**O**] net questions). Wolfram: four exact

mirrors (level-2 block census from the root data; rank/kernel arithmetic with $\dim V(\omega) = 3875$ by Weyl; the glue-frame clock-class split $(1036, 1024, 1040, 1024)$; threshold + level-dial + D_8 census) verified with the active engine the same day (408 → 412); the exact Gram/PSD computation stays Python-side (flagged). `VerificationDag` qft node gains `v500`; registry + `run_all` (494 modules at this step, suite green). `SEAM.EQUIV.01` untouched; no marker moves.

2026-07-21 (IX) (`v499` — the P2 weight-typing postulate HARDENED: the anchor $a = (1, 1, 2)$ typed as the Birkhoff–Grothendieck splitting exponents of the Deligne canonical extension, so integrality/positivity/sum-4 become theorems given the [E] anchors; the `v491` check-9 residual sharpened, the tfpt_2 "balanced" selection typed as stability; Wolfram extension 403 → 408 (verified engine run); *no gate closes, no marker moves — `AX.P2.01` stays a declared axiom, `QGEO.SYM` stays [O]*)

- **`v499_p2_weights_deligne_bg.py` (`P2.TYPING.01`, 23 checks, sympy exact, no floats).** `v491` reduced P2 to "3 positive integers summing to 4"; its residual was the *typing* (why integers, why positive, why sum 4). Now typed: given the **[E]** anchors (4 marks, `v216`; rank $3 = \#\text{marks} - 1$, `v137`/`v228`; cusp-class local monodromy $\lambda^3 - 1$ *and* exact irreducibility $|\langle U, M_0\rangle| = 24 = |S_4|$, $\sum_g |\chi|^2 = 24$, `v117`; unitarity (U)), the Deligne canonical extension E of the flavor connection satisfies: $\deg E = -4$ *as a theorem* (residue exponents $\{0, \frac{1}{3}, \frac{2}{3}\}$ the unique lift to $[0, 1)$, trace 1 per mark; par-deg 0; ∞ apparent, monodromy product $= \mathbf{1}$ exactly); $h^0(E) = 0$ *as a theorem* (Mehta–Seshadri stability; all 324 section-saturation cases; BG dictionary $h^0{=}0 \Leftrightarrow a_i \geq 1$); integrality = Birkhoff–Grothendieck. Hence $\{a\} = \{1, 1, 2\}$ unique and $g_{\text{car}} = e_2 = 5$. *Selector typed*: the Schur–Horn permutohedron has integer points exactly $\{(2, 2, 0), (2, 1, 1)\}$ (the two options of the tfpt_2 "Global computation" theorem) and $h^0{=}0$ kills $\{0, 2, 2\}$ — the "balanced/minimal-variance" heuristic is now stability/positivity. *The h^1 discrepancy, real and resolved*: $h^1(\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2) = 1 \neq 3$, so "generations $= H^1$ of the rank-3 bundle" is FALSE (negative control: forcing it demands $\deg -6$ and loses uniqueness, $e_2 \in \{9, 11, 12\}$ never 5); correct two-object typing: $L = \mathcal{O}(-D)$ carries the generations ($h^1(L) = 3 = N_{\text{fam}}$, the `v137` basis $= H^0(\Omega^1(\log D))$), E carries them as its *fiber*. *Biswas explicit* (independent witness): the weight denominator $3 = N_{\text{fam}}$ forces the $\mathbb{Z}_3$ cusp cover $y^3 = (z{-}1)(z{-}i)(z{+}1)^2(z{+}i)^2$ (genus 2; NOT the pillowcase double, genus 1, $j = 1728$ — the deck side); $p_*\mathcal{O}_Y = \mathcal{O} \oplus \mathcal{O}(-2)^2$; the regular-representation model lands on $\{0, 2, 2\}$ = the *unstable* Schur–Horn companion ($h^0 = 1$): integrality and the sum rule come *out* of the correspondence, stability is the selector. *Negative controls*: no positivity → 4/31 solutions; $\deg -3/-5$ breaks; n-mark scan $e_2 = (n{-}2)(n{+}1)/2 = 5$ ONLY at $n = 4$ (the value, not the uniqueness, is the $n{=}4$ content); MS-rational $e_2 = \frac{16}{3}, \frac{44}{9}$. *Honest residuals* (P2 stays an axiom): R1 module identification **[C]** (`QGEO.SYM` territory), R2 (U) **[C]** (`v33`/`v40`), R3 analytic monodromy (Numerical, `v117` check 4); non-circular (neither `v23`'s splitting input nor `v115`'s $\operatorname{diag} A_0 = a/4$ input used); the `v491` direction-reversal note carries over. Ledger: new row `P2.TYPING.01`; the `v491`/`P2.PARTITION.01` row records "typing residual sharpened by `v499`" (`v491` stays active, no supersede). Registry + `run_all` (493 modules, suite green).
- **Wolfram second path + papers + website.** Five exact mirrors added to the extension file `tfpt_readouts_extension.wl` (partition enumeration under the full typing + naive-typing control + h^1 discrepancy; Deligne residue-trace degree with the $A_0^\star$ witness char poly; Schur–Horn lattice points; Biswas $\mathbb{Z}_3$-cover degree arithmetic incl. deck double $j = 1728$; the n-mark formula), verified with the active engine the same day (403 → 408, `ALL WOLFRAM EXTENSION CHECKS PASSED`); `wolfram/README.md` + `GATE.WOLFRAM.02` + `tfpt_safeguards` Layer-6 count + website counters (116/116+408/408). Papers (one sentence each, conservative): `introduction` anchor paragraph, `tfpt_1` $(5, 3)$ keybox, `tfpt_2` H2 winbox *and* the "Global computation" theorem (the balanced selection typed as stability), `origin_theory` $(5, 3)$ keybox — all citing `v499_p2_weights_deligne_bg.py`; no status card changes. Website: `VerificationDag` P2

node gains `v499`; generated indices via build; no marker moved anywhere.

2026-07-21 (VIII) (Editorial: dedicated celestial-route section in Paper 3 + website mirror — the scattered celestial sentences consolidated into one coherent six-step section; *no new scripts, no status changes, no marker moves, `SEAM.EQUIV.01` untouched*)

- **`tfpt_3_e8_audit_bootstrap`: new section "The celestial route: from the μ_4 clock to the $(E_8)_1$ boundary shadow".** The celestial-contract material that had accumulated as one long paragraph inside "The Möbius origin of the irreducible π" is consolidated into a dedicated section (placed after the bootstrap audit, before "Honest status"), telling the executed WP1–WP5b story in six typed steps: (1) setup — the E_8 μ_4-glue as a flat $\mathbb{Z}_4$ monodromy on the A_3 ALE space, clock^2 = deck (spin $\mathbb{Z}_8$, $8 = 2|\mu_4|$) **[E]** (`v492`); (2) deformation — $XY = Z^4 + a_0$ pure scale, $j = 1728$ frozen, clock = Picard–Lefschetz/Coxeter monodromy **[E]** (`v493`); (3) consistency honestly demoted — $(\kappa/c_3)^2 = 12$ exact but the alignment format passes 8/8: alignment survives, selectivity does not **[E]**/**[C]** (`v495`); (4) type mismatch + boundary limit — the $(E_8)_1$ character is not a jet-grading conformal block (growth $n^{2/3}$ vs $n^{1/2}$) but the μ_4 period sum 248 holds exactly: a boundary-limit shadow of `SEAM.EQUIV.01`/MMST shape **[E]** (`v496`); (5) the null ideal quantitative — stabilisation threshold $w = n{+}1$, $27000 = (h^\vee)^3$ derived, quotient = sector sum = 4124, $SO(16)_1$ contrast **[E]** (`v497`); (6) the deleting operator explicit — $|s\rangle = (E^\theta_{-1})^2|0\rangle$, $J^a_1|s\rangle = 0$ on all 248 (tally 190/57/1), level dial $2(1{-}k)$, one-block closure E_8/μ_4-specific **[E]** (`v498`); plus the WP5c–e roadmap and the honest status line (the route does *not* close `SEAM.EQUIV.01`; the GNS/limit question stays **[O]**). All pre-existing script citations (`v492`–`v498`) are carried into the new section; the Möbius section now points to it with one sentence. `tfpt_research_contracts`: a reader's note at the head of the `CELEST.SEAM.01` contract cross-references the new Paper-3 section (the contract stays the full technical reference). `tfpt_4_frontier`: one sentence in the metric-sector keybox records the celestial route as a third, quantitative approach to the keystone (not closing it).
- **Website mirror.** `papers.ts`: the Paper-3 section list gains the celestial-route section (six-step summary in the existing style) and a "Celestial route" highlight; the contracts mirror gains the same cross-reference sentence. `glossary.ts`: three new entries (celestial route, ALE space, null ideal) in the established technical style. Generated indices via build; no status component changes (no marker moved anywhere).

2026-07-21 (VII) (CELEST.SEAM.01 WP5b executed and promoted: `v498` — "the singular vector as an operator", the deleting object of the `v497` null ideal constructed explicitly and machine-verified singular; contract now WP1–WP4 + WP5a + WP5b done, WP5c–e [O] (roadmap); Wolfram extension 398 → 403 (verified engine run); *no gate closes, no marker moves, `SEAM.EQUIV.01` untouched*)

- **`v498_celestial_wp5b_singular_vector.py` (`CELEST.WP5B.01`, 53 checks, exact integer/Fraction, deterministic, no floats).** Success on the preregistered WP5b criterion: the deleting object exists and is explicit. *Construction*: $|s\rangle = (E^\theta_{-1})^2|0\rangle$ (weight 2θ, level $2 = 8$ quarter units $= q^2$, an *integer* level) in a machine-built Chevalley/Frenkel–Kac basis — bimultiplicative 2-cocycle with asymmetry verified on all 57600 ordered root pairs, the $[e_\alpha, e_{-\alpha}]$ sign *forced* by the Jacobi identity (SGN = -1, not chosen), the invariant form κ derived ($\kappa(\theta^\vee, \theta^\vee) = 2$), Jacobi exact on all 57600 $\mathfrak{sl}_2$-slice triples + 6000 seeded + all 496 θ-adjacent, all $N_{\alpha\beta} = \pm 1$. *Singularity (machine)*: $J^a_1|s\rangle = 0$ for all 248 basis elements (0 violations), with the case classification 190 (first bracket) / 57 (second) / 1 ($a = F^\theta$ via the central-term cancellation — the *only* case that sees the level k); plus $J^a_2|s\rangle = 0$, $E^a_0|s\rangle = 0$, exact weight, Shapovalov norm 0; the affine PBW engine unit-tested on all 61504 basis pairs. *Level dial*: the F^θ_1 coefficient

is $2(1-k)$ — $2/0/-2$ at $k = 0/1/2$: without the central extension the deletion operator does not exist. μ_4 *data*: glue class $j(\theta) = 1$ (machine-built h-adapted chamber, $\langle\theta, h\rangle = 5$, height $29 = h^\vee - 1$), clock phase $i^{2j} = -1$, class$(2\theta) = 2$ sheet-even; the deleting θ-$\mathfrak{sl}_2$ crosses the sheet-odd classes $(1, 3)$ while its square is sheet-even; cross-table θ-grading $\times$ glue class consistent (rows $(1, 56, 126, 56, 1)$, columns $(52, 64, 60, 64)$); 8 quarters $= q^2$ via the per-period dictionary. *Level-2 match*: weight 2θ has multiplicity 1 in the level-2 Fock, so $U(\mathfrak{g})|s\rangle$ is *the* $27000 = (h^\vee)^3$; the direct $\mathfrak{g}_0$-orbit BFS matches the Freudenthal multiplicities of $V(2\theta)$ exactly through depth 4 ([**C**] Weyl complete reducibility beyond); quotient $31124 - 27000 = 4124$; lattice reading $(2\theta, 2\theta)/2 = 4 > 2$: $|s\rangle$ *is* the kernel generator of Fock $\to E_4/\eta^8$. *Negative controls (honest separation)*: the level-1 current state $E^\theta_{-1}|0\rangle$ (one witness, $\to -k|0\rangle$ — the 248-current state that must remain) and 4 generic level-2 states (14–33 witnesses) are *not* singular; at k=2 the *cube* is (generic Kac $(E^\theta)^{k+1}$ mechanics, honestly typed); $SO(16)_1$ has the same singular vector but keeps three extra level-1 primaries ($h = \frac{1}{2}$ breaks one-block fusion; $5304 \neq 14^3$) vs E_8 (comarks all ≥ 2, glue weights $(0, 1, 1, 1)$ integer) — the *one-block closure* is the E_8/μ_4-specific part. *Honest* [**O**]: in the twisted quarter-slot moding two sector-C_1 modes never sum to 8 quarters (minimum $6 = q^{3/2}$) — the per-period dictionary (`v496`), not the per-slot identification, carries $|s\rangle$ to q^2: exactly the WP5c GNS/limit-state question; WP5c–e stay open; `SEAM.EQUIV.01` untouched.

- **Contract + surfaces synced.** `tfpt_research_contracts`: the WP5b bullet moved to *done* with the key numbers (case tally 190/57/1; level dial $2(1-k)$; $j(\theta) = 1$, clock phase -1; orbit BFS through depth 4; the D_8 contrast) and the twisted-slot tension as the explicit WP5c handover; honest status line and K4 paragraph updated ("WP1–WP4 + WP5a + WP5b executed; WP5c–e [**O**]"). `tfpt_3_e8_audit_bootstrap`: the celestial-contract paragraph extended by the WP5b sentence. Ledger: `CELEST.WP5B.01` (new) + `CELEST.SEAM.01` ("WP1–WP4 + WP5a + WP5b executed; WP5c–e [**O**] (roadmap)"). Wolfram: 5 exact `v498` mirrors added to `tfpt_readouts_extension.wl` and *verified* with the active engine the same day (398 → 403, `ALL WOLFRAM EXTENSION CHECKS PASSED`); `wolfram/README.md` + `GATE.WOLFRAM.02` counters advanced; `tfpt_safeguards` Layer-6 count + website mirrors 116/116 + 403/403. Website: `papers.ts` contract mirror ("WP1–WP4 + WP5a + WP5b done, WP5c–e roadmap"), DAG `qft` node + `v498`, generated indices via build. Suite now 492 scripts, all green; `SEAM.EQUIV.01` stays [**O**].

2026-07-21 (VI) (CELEST.SEAM.01 WP5a executed and promoted: `v497` — "the null ideal from the limit", the `v496` boundary-limit shadow made a precise coefficientwise limit and the level-2 null ideal derived from root data; contract WP5 subdivided, now WP1–WP4 + WP5a done, WP5b–e [O] (roadmap); Wolfram extension 394 → 398 (verified engine run); *no gate closes, no marker moves, `SEAM.EQUIV.01` untouched*)

- **`v497_celestial_wp5a_null_ideal.py` (`CELEST.WP5A.01`, 34 checks, exact integer/Fraction, no floats).** Three results make the WP4 limit reading precise. *Stabilisation theorem*: the one-parameter family χ_w of graded Fock characters on the chiral jet generators (loop energy m, radial number r; $E_w = m + wr$ in quarter units $u = q^{1/4}$) contains the chiral jet grading as its w=2 member (generator counts $64, 120, 128, 180, 192, 240, 256, 300$ per degree), and its u^n coefficient equals the quarter-moded loop Fock for *all* $w \geq n+1$, strictly larger for every $w \leq n$ (verified $n \leq 8$, $w \leq 10$) — the `v496` "boundary-limit shadow" is a precise *coefficientwise* limit $\varepsilon = 1/w \to 0$ with explicit threshold $w = n+1$, not a slice (convention clarified: 897266 = the quarter-moded loop Fock at u^4 = *integer* level 1). *The null ideal derived, not cited*: Freudenthal multiplicities + the Weyl dimension formula + generic character peeling give $\mathrm{Sym}^2(248) = V(2\theta) \oplus V(\omega) \oplus V(0) = 27000 + 3875 + 1$ with residual exactly zero (independent Weyl-orbit cross-check); $27000 = 30^3 = (h^\vee)^3$ exactly; $C_2 = (124, 96)$, Dynkin indices 6750/750; level-1 integrability ($\langle 2\theta, \theta^\vee\rangle = 4$, $\langle\omega, \theta^\vee\rangle = 2$) leaves

the vacuum as the only level-1 primary (Kac, used only through its counting consequence); level-2 quotient $31124 - 27000 = 4124 = 1 + 248 + 3875$ exactly. *Two routes, one number*: the μ_4 theta-split sector sum at $q^2 = (1036, 1024, 1040, 1024)$ gives the *same* 4124 — the untwisted null-ideal quotient and the boundary/twisted lattice presentation count one space (q^3: $34752 = \chi_3$). Negative controls: $SO(16)_1$ through the same pipeline gives *four* components ($5304 + 1820 + 135 + 1$), $5304 \neq 14^3$ (the $h^{\vee 3}$ identity is not generic), quotient $2076 = \Theta_{D_8}/\eta^8$ at q^2 (the recipe validated on a second algebra), block weights $(0, \frac{1}{2}, 1, 1)$ break one-block fusion (vs E_8's integer $(0, 1, 1, 1)$); only $P = 4 = |\mu_4|$ periodisation reproduces the 248 layer; swapped glue breaks the zero-mode anchor ($64 \neq 60$). *Honest* **[O]**: the limit does *not* generate the truncation — the singular vector as an *operator* in the limit net (WP5b), the GNS limit state (WP5c) and local normality/Haag duality (WP5d) stay open; the celestial route now has the same two-step shape as MMST (limit + maximal ideal), with the ideal quantitatively fixed; `SEAM.EQUIV.01` untouched.

- **Contract + surfaces synced.** `tfpt_research_contracts`: WP5 subdivided — WP5a moved to *done* with the key numbers (stabilisation threshold $w = n{+}1$; $27000 = (h^\vee)^3$ derived; quotient = sector sum = 4124; $SO(16)_1$ contrast), WP5b–e added as roadmap bullets with success/kill criteria (WP5b singular-vector operator in the quarter-moded current Fock; WP5c GNS limit state, kernel $\supseteq$ ideal; WP5d local net/Haag duality via the KLM route; WP5e BCOV/Costello–Li twistor uplift, partition function $= E_4/\eta^8$ incl. the $q^{-1/3}$ prefactor); K4 paragraph and the honest-success status line updated. Ledger: `CELEST.WP5A.01` (new) + `CELEST.SEAM.01` ("WP1–WP4 + WP5a executed; WP5b–e **[O]** (roadmap)"). Wolfram: 4 exact `v497` mirrors added to `tfpt_readouts_extension.wl` and *verified* with the active engine the same day (394 → 398, `ALL WOLFRAM EXTENSION CHECKS PASSED`); `wolfram/README.md` + `GATE.WOLFRAM.02` counters advanced; `tfpt_safeguards` Layer-6 count + website mirrors 116/116 + 398/398. Website: `papers.ts` contract mirror ("WP5a done, WP5b–e roadmap"), DAG `qft` node + `v497`, generated indices via build. Suite now 491 scripts, all green; `SEAM.EQUIV.01` stays **[O]**.

2026-07-21 (V) (`comb-meta-limit` re-run with the PG.06b FULL Vela recovery ingested — the stale single-obs PG.06b reference (noise-peak $F_0 = 11.19275$ Hz) corrected in the channel definitions; the new X-ray channel enters *censored* (no absolute ε), both aggregate limits numerically unchanged; *experiments level only — no verification module, no gate closes, no marker moves*)

- **`experiments/comb-meta-limit` channel update (methodology untouched).** The Vela X-ray channel (`channels.py`) now ingests the PG.06b FULL reduction (`pg06b_full_vela.json` in `pulsar-glitch-recovery/results`: 665 obs → 491 usable → 119 segment-coherent $\nu(t)$ points) instead of the superseded single-obs `pg06b_vela.json`; the stale code note quoting the noise-peak "detection" $F_0 = 11.19275$ Hz is corrected on record (true Vela $F_0 = 11.1861692$ Hz), and the channel key is fixed from the mislabel `PG.07` to `PG.06b`. Censoring handled exactly by the existing Tier-B rule (nothing invented): the comb observable is a *linear* $\delta\nu(\tau)$ residual (μHz, not a fractional ln-flux modulation), both legs sit below the 2.8-period gate (2019: 2.17, 2021: 1.38), and the surrogate-calibrated injection never reaches ε_{90} below $\varepsilon = 1$ ($\varepsilon_{50} = 0.55$ in 2019 only) ⇒ no absolute per-channel ε-UL exists for the DL/HKSJ combination; the channel stays enumerated-only ("no constraint"), carrying the normalised context amplitude $\sqrt{2g} = 0.40$ (leg gains 0.046/0.114, min $p = 0.014$ — the 2019 audit-level chance feature: Bonferroni $\times 10 = 0.14$, shuffle $p = 0.40$).
- **Full re-run (seed 0, deterministic): both limits numerically unchanged.** Boundary/horizon-scoped: `data_limited` (still no horizon channel with an absolute ε); all-channel universal-DSI 95% UL $\varepsilon < 0.11954$ ($\approx 12.0\%$; HKSJ $t_2 = 2.92$, $\rho = 0.3$-inflated 0.137; 6.91× the predicted $\varepsilon = 0.0173$), still driven by the surface log-flux channels A1/A4/A5 (A4 GRB the anchor, per-channel UL 0.108); now 10 channels enumerated / 3 usable. Injection self-consistency unchanged (inject 5% → recover 4.8%, UL 5.5%;

coverage 0.92/150 trials). Surfaces synced: experiment README (tier table, ingest list, PG.06b FULL bullet), `results/results.json`, scorecard row `data_value`/source via `build_evidence_scorecard.py` (status stays `data_limited`; counters unchanged, 120 rows), `experiments/README.md` catalog row. No claim; firewall intact.

2026-07-21 (IV) (Wolfram engine reactivated — the deferred second-path mirrors VERIFIED: base 116/116 + extension 394/394 (378 → 394, the 16 mirrors accumulated since 2026-07-14 now counted); `GATE.WOLFRAM.01/02` and all counter surfaces synced; *no gate closes, no marker moves*)

- **Verified engine run 2026-07-21 (Wolfram Engine 14.3, license reactivated).** `wolframscript -file verification/wolfram/tfpt_readouts.wl` → `116 passed, 0 failed, ALL WOLFRAM CHECKS PASSED`; `tfpt_readouts_extension.wl` → `394 passed, 0 failed, ALL WOLFRAM EXTENSION CHECKS PASSED`. The 16 exact mirrors added while the engine was unactivated all enter the counted total (378 → 394): `v479` Kronheimer bridge (2), `v491` partition corollary (3), `v493` celestial WP2 clock deformation (4), `v495` celestial WP3 Green–Schwarz alignment (3), `v496` celestial WP4 character shadow (4).
- **Counter surfaces synced.** `verification/wolfram/README.md` (current-state block + the five deferral notes resolved; `v484`/`v485`/`v486`/`v487`/`v488`/`v492` stay mirrorable-but-unmirrored, Python-exact, so they are *not* counted); ledger `GATE.WOLFRAM.01` (re-verified 2026-07-21) + `GATE.WOLFRAM.02` (394/394, UPDATE note); `tfpt_safeguards` Layer-6 count + website `papers.ts` mirrors (116/116 + 394/394); `v479` registry/`run_all` deferral note resolved. No numeric claim changes; the second computational path is simply verified-current again.

2026-07-21 (III) (CELEST.SEAM.01 WP3 + WP4 executed and promoted: `v495` Green–Schwarz alignment (verdict B, honestly non-selective) + `v496` character-vs-S-algebra type mismatch (verdict B(ii), boundary-limit shadow); contract now WP1–WP4 done, WP5 [O]; *no gate closes, no marker moves, `SEAM.EQUIV.01` untouched*)

- **`v495_celestial_wp3_gs_alignment.py` (`CELEST.WP3.01`, 25 checks, exact Fraction/sympy).** The pre-registered WP3 "alignment test only": the Okubo coefficient $\mathrm{Tr}_{\mathrm{adj}}X^4/(\mathrm{Tr}_{\mathrm{adj}}X^2)^2 = 5/(2(\dim\mathfrak{g}+2))$ is *derived* as an exact polynomial identity for all 8 algebras on Costello's list (sl_2, sl_3, g_2, so_8, f_4, e_6, e_7, e_8; $\mathrm{e}_6/\mathrm{e}_7$ carved out of the `v492` E_8 root set; negative control: sl_4 fails, 5/32 vs 3/32); the closed form $\tilde{\lambda}^2 = 10h^{\vee 2}/(\dim+2) = h^\vee+6$ holds exactly across the Deligne series; E_8: $\tilde{\lambda}=6$ (unit-trace) resp. $\lambda_{\mathrm{fund}}=1/10$ (adjoint-trace) exact, so $(\kappa/c_3)^2 = 12 = |\mu_4|N_{\mathrm{fam}}$ resp. $1/300$ — with κ/c_3 itself *irrational* ($2\sqrt{3}$). Byproduct: the printed $\lambda^2_{\mathrm{so}_8} = 3/2$ in Costello's appendix A is a factor-2 slip (his own weight content gives 3). **Look-elsewhere (the hard fence):** the alignment format passes 8/8 across the whole list (zero selectivity), $\tilde{\lambda}$-integrality 2/8 (shared with sl_3), only $g_{\mathrm{car}}{=}5 \mid h^\vee$ is e_8-selective (1/8), the isolating conjunction is post hoc. Kill test K3 does *not* fire but is scope-demoted: *alignment survives; selectivity does not* — compatibility, not e_8-selective evidence. Verdict B.
- **`v496_celestial_wp4_character_shadow.py` (`CELEST.WP4.01`, 25 checks, exact integer/Fraction/sympy).** WP4, the type mismatch head-on: the $(E_8)_1$ character $E_4/\eta^8 = (1, 248, 4124, 34752, 213126, \dots)$ is *not* a conformal block of the glue-equivariant $E_8[\mathbb{C}^2]$ S-algebra in its own jet grading — localised three ways: (a) spin $1 - d/2$ unbounded below and 248 never a tower dimension ($d \le 100$); (b) growth $n^{2/3}$ vs $n^{1/2}$ (quadratic cumulative count $31s^2{+}92s{+}60$, $31 = k{+}h^\vee$; f_n/χ_n strictly increasing $n = 1..12$); (c) the level-2 null ideal deletes exactly $27000 = 30^3 = (h^\vee)^3$ out of $\mathrm{Sym}^2(248) = 1{+}3875{+}27000$ — no jet analogue. *But* the boundary/period reading holds exactly at the current stratum (zero modes $= 60 = C_0$ currents, jet slice cycles $(60, 64, 60, 64)$, one μ_4 period sums to 248, glue weights $(0, 1, 1, 1)$ integer; free loop Fock $897266 \gg 248$, so the truncation must come from a limit) — the same MMST

scaling-limit shape as `SEAM.EQUIV.01` (`v336`/`v449`). K4 fires only against the exact-block reading; the constructive limit question passes to WP5. Verdict B(ii).

- **Contract + surfaces synced.** `tfpt_research_contracts` WP3/WP4 bullets moved to *done* with the look-elsewhere demotion in the text, K3/K4 marked evaluated, status line now "WP1–WP4 executed (verdicts B, B, B, B(ii)), WP5 [**O**]"; one conservative sentence added to the `tfpt_3` celestial paragraph; ledger rows `CELEST.WP3.01` + `CELEST.WP4.01` (new) and `CELEST.SEAM.01` updated; `v495`/`v496` Wolfram mirrors added to `tfpt_readouts_extension.wl` (3 + 4 exact checks; count with the next verified engine run — engine still unactivated); website `papers.ts` contract highlight "WP1–WP4 done" + DAG `qft` node extended. Suite now 490 scripts, all green; `SEAM.EQUIV.01` stays [**O**].

2026-07-21 (II) (PG.06b FULL: the complete NICER Vela archive reduced — the first X-ray $\nu(t)$ leg of the recovery-comb search, verdict `data_limited`; the old PG.06b "pipeline proven" fold corrected on record; *experiments level only — no verification module, no gate closes, no marker moves*)

- **`experiments/pulsar-glitch-recovery` PG.06b FULL executed (`tfpt-pulsar vela-full`).** The full NICER Vela-pulsar (PSR B0833-45) L2 archive is reduced, with the manifest frozen *before* the bulk download (`manifest_full_reduction.json`: all 665 ObsIDs, preregistered 2019/2021/2024 glitch windows, 20-GB abort gate): 665/665 obs downloaded (2.6 GB, raw gitignored) → 491 usable → 240 per-obs detections → **119 segment-coherent $\nu(t)$ points** (108 fully coherent), median $\sigma_\nu \approx 0.6\,\mu$Hz; barycentring validated against PINT to 3.4 μs — the *first X-ray $\nu(t)$ probe* of the search program. Sanity gate PASS: $\Delta\nu/\nu$ 2019 $= 2.39 \times 10^{-6}$ vs 2.5012×10^{-6} published, 2021 $= 1.272 \times 10^{-6}$ vs 1.26×10^{-6}; the 2024 glitch is not reducible (only ∼150-s snapshots, 9.7 ks — documented).
- **Frozen-kernel comb ($\omega = 2.5827$, $\varepsilon_{\rm pred} = 0.0173$): verdict `data_limited` (range- and amplitude-limited).** Reach 2019: 2.17 comb periods (< the 2.8-period gate; beats the PG.08 public Vela reach 1.83), 2021: 1.38; the 2019 raw $p = 0.014$ has the kernel as smallest-p battery member BUT Bonferroni $\times 10 = 0.14$ and shuffle $p = 0.395$ ⇒ chance-level by the preregistered escalation rule (audit flag, NOT a candidate); 2021 $p = 0.731$; $\ln\tau$ stack $p = 0.049$/shuffle 0.307. Surrogate-calibrated injection at the real sampling and noise: $\varepsilon_{50} \approx 0.55$ (2019 only), ε_{90} never reached below $\varepsilon = 1$, 0% power at the predicted $\varepsilon = 0.0173$ — a weaker amplitude probe than the PG.08 radio residuals, but it independently reproduces the PG.06 range-blindness on real data. Second off-kernel flag: $\lambda = e$ in the 2021 leg (raw $p = 0.002$, Bonferroni 0.018; the same non-kernel periodicity class as PG.08 J0742−2822 — flagged, not claimed). The sub-day early window that would clear the gate is absent from the archive (first usable obs 3.4 d / 7 d post-glitch); next decider: <3-d post-glitch X-ray observations of a future Vela glitch (eXTP class).
- **Correction on record.** The old PG.06b single-obs "pipeline proven" detection at $F_0 = 11.19275$ Hz ($H = 18.4$) was a noise peak; the true Vela frequency at that epoch is $F_0 = 11.1861692$ Hz ($H = 99.7$, consistent with the phase-connected PuMA ephemeris). The pipeline remains proven — by the corrected fold. Scorecard row updated to "recovery comb on the full NICER Vela X-ray nu(t) (PG.06b FULL)" (status stays `data_limited`; counters unchanged, 120 rows: 49 consistent / 9 tension / 26 null / 33 data-limited / 3 parked); `experiments/README.md` catalog + §1 prose, pulsar README, `experiments/next.txt` synced. No claim; firewall intact.

2026-07-21 (the two 2026-07-20 prediction-of-record rows promoted to the suite: `v494` — α_s running + μ-distortion branch discriminator machine-checked, ledger rows `COSMO.RUNNING.01`/`COSMO.MUDIST.01`, prediction surfaces synced; *no gate closes, no marker moves*)

- **`v494_cosmo_running_mu_record.py` (23 checks, all green).** The α_s-running and μ-distortion scorecard rows of 2026-07-20 (IV) get their load-bearing module so the new paper/website statements are machine-backed. **[E]** leading-order identities (sympy exact): $\alpha_s = \mathrm{d}n_s/\mathrm{d}\ln k = -2/N_\star^2 = -r/6$, $\beta_s = -4/N_\star^3$ (NOT $-8/N^3$); LO band $N_\star \in [50, 60]$: $[-8.0, -5.6] \times 10^{-4}$. **[E]** exact slow roll on the Einstein-frame Starobinsky potential (third-order Lyth–Riotto + finite-difference cross-check $< 1\%$): $\alpha_s(51.4) = -7.09 \times 10^{-4}$, $\beta_s(51.4) = -2.70 \times 10^{-5}$. **[C]** data: Planck 2018 (-0.0045 ± 0.0067) $+0.57\sigma$; record leg P-ACT-LB $(+0.0062 \pm 0.0052)$ -1.33σ (watch channel: positive central value; kill: robust $\alpha_s < -5 \times 10^{-3}$ or $> +3 \times 10^{-3}$ at $\geq 5\sigma$; any robust POSITIVE running kills the plateau branch). **[C]** μ-distortion (Chluba window, +21% calibrated on the ΛCDM anchor): sharp branch $N_\star$=51.4 $\mu = 1.6 \times 10^{-8}$, profiled record $N_\star$=56.1 $\mu = 1.94 \times 10^{-8}$, band 1.5–2.3 $\times 10^{-8}$; branch split $\Delta\mu = 3.0 \times 10^{-9} \Rightarrow 3\sigma$ $A_s/N_\star$ branch decision at $\sigma(\mu) = 10^{-9}$ (Voyage-2050 class; PIXIE baseline 9×10^{-8} does not reach; kill: robust $\mu < 0.9 \times 10^{-8}$ or $> 4 \times 10^{-8}$). **[E]** comb blindness (honest S15): the frozen log-comb ($\omega = 2.5827$, $\varepsilon = 0.0173$) is structurally invisible in the μ window ($|F(\omega)| = 0.033$, $\delta\mu = 7.4 \times 10^{-12} = 0.007\sigma$ even at Voyage-2050). Fences: $N_\star$ stays an external reheating input (`GATE.NSTAR.01`); the frozen band $[50, 60]$ stays the surface of record (v84/v86). Ledger rows `COSMO.RUNNING.01` and `COSMO.MUDIST.01` added.
- **Prediction surfaces synced (the missing mirror of 2026-07-20 (IV)).** `predictions.txt` gains the two rows (α_s **[C]** with the -1.33σ P-ACT-LB pull and the positive-running kill; μ-distortion **[C]** data-limited with the Voyage-2050 branch split); website `lib/predictions.ts` gains the matching cards `alpha-s-running` and `mu-distortion` (confrontation + experiment blocks pointing at `experiments/tfpt-discovery`); the cosmology DAG node lists `v494` and states the running/μ readouts.
- **Paper bodies.** tfpt_1 (Inflation interface) gains the running consistency-relation paragraph ($\alpha_s = -2/N_\star^2 = -r/6$, exact slow roll, pulls, positive-running kill) with `\veri{v494_cosmo_running_mu_record.py}`; tfpt_2 (inflation-pivot keybox) gains the μ-distortion branch-decider sentence (sharp 1.6×10^{-8} vs profiled 2.0×10^{-8}, 3σ at Voyage-2050) with the same `\veri`. No marker of any existing claim changes. (v494 is numerical/quadrature, Python-only by the suite convention, flagged in the docstring; `GATE.WOLFRAM.01/02` untouched.)

2026-07-20 (VIII) (E8 mass-ladder blind-recovery bed EXECUTED: verdict `validated` at the detector level — the Cartan–PF ladder pipeline blindly recovers the published Zamolodchikov-E8 spectra of $BaCo_2V_2O_8$ (8/8) and $CoNb_2O_6$ (6/8 in-window) and rejects every control class; *experiments level only — confirms Zamolodchikov E8, not TFPT; no verification module, no gate closes, no marker moves*)

- **`experiments/e8-ladder-bed/` executed (preregistered `hypotheses/e8_ladder_bed_v1.yaml`).** The blind ladder detector (assignment DP + scale fit + 5000-ladder Monte-Carlo null; the same detector class TFPT uses on search surfaces, e.g. `hfqpo-ladder`) on the only public *known-answer* mass-ladder dataset: $BaCo_2V_2O_8$ (Zou+ PRL **127**, 077201; INS at the 4.7 T 1D QCP, Fig. 3a extraction QA-validated, 17 peaks blind) recovers **8/8 rungs**, $\chi^2/\mathrm{dof} = 0.35$, MC $p = 0.0056$ (singles-only run $p = 0.0004$; σ-variant robust); $CoNb_2O_6$ (Amelin+ PRB **102**, 104431; verbatim THz values) recovers **6/8 in-window rungs**, $\chi^2/\mathrm{dof} = 0.52$, $p = 0.0038$ (m_7/m_8 lie outside the 0.6-THz low-pass window; the broad 0.43-THz (m_1+m_2)-continuum onset is correctly left unassigned). Cartan–PF self-test: the sorted PF eigenvector of $2I - A(E_8)$ equals the Zamolodchikov ladder to 1.5×10^{-15} ($m_2/m_1 = \varphi$ to 2.2×10^{-16}; lowest Cartan

eigenvalue $2-2\cos(\pi/30)$ to 4.5×10^{-14}).

- **Control battery — every class rejected.** Scramble 0.5%/1.5% false-positive rate (200 log-gap permutations, near-duplicate accounting); offset $\pm\{0.3, 0.5\}m_1$: 0/8 detections; jitter dose-response $\leq 5\%$ at the destructive 30% level; wrong ladders on the real data (harmonic 1..8, A_3-PF, D_8-PF, Airy confined-spinon) all beaten by E8; the synthetic Airy bed (the true zero-field physics of these compounds) yields **0** valid rungs. The one disclosed v1.1 criteria amendment (jitter as dose-response, scramble near-duplicates, χ^2-gated wrong-ladder comparator) is fully documented in the hypotheses YAML with all v1 numbers preserved; the recoveries themselves were never touched.
- **Integration (firewall intact).** Scorecard row "E8 mass-ladder blind recovery bed" moves `parked` → `consistent` (stage `not_applicable`, tier `analog_positive_control`, leakage `detector_control` — the executed-analog basket of `qc-recovery-kernel` and `qgeo-eit-soff`); counters 120 rows: 49 consistent / 3 parked; `experiments/README.md` catalog row + stats (generator); website `predictions.ts EXPERIMENTS_AUDIT` counters; `experiments/next.txt`. This validates the *ladder-detector pipeline* — it confirms Zamolodchikov E8, *not* the TFPT axioms: no [**E**], no `\veri`, no ledger row.

2026-07-20 (VII) (`v493` — WP2 of `CELEST.SEAM.01` executed: the clock-invariant deformation $XY{=}Z^4{+}a_0$ is a PURE SCALE with the $\tau{=}i$ pillowcase frozen, the μ_4 clock IS the Picard–Lefschetz/Coxeter monodromy of the seam deformation, and the BHS deformed-algebra pattern transfers $\mathbb{Z}_2\to\mathbb{Z}_4$ with no sector leak; *verdict B, contract updated to WP2 done, no gate closes, no marker moves, `SEAM.EQUIV.01` and the `v216` residual stay [O]*)

- **`v493_celestial_wp2_clock_deformation.py` (`CELEST.WP2.01`).** WP2 of the celestial contract, sympy exact (47 checks), promoted from the `experiments/tfpt-discovery/` probe `celestial_seam_wp2_clock_deformation_probe.py`. *Geometry:* $P(iZ){=}P(Z)$ forces $a_3{=}a_2{=}a_1{=}0$ sharply and two-sided ($\prod_k(Z-i^kw)=Z^4-w^4$); $XY{=}Z^4{+}a_0$ is smooth iff $a_0{\neq}0$ ($\mathrm{disc}=256\,a_0^3$); the branch points $Z^4{=}{-}a_0$ are one free μ_4 orbit in *every* fibre with cross-ratio 2; the binary-quartic invariants of $w^2{=}Z^4{+}a_0$ are $I{=}12a_0$, $J{=}0$ *identically*, so $j{=}1728$ and the $\tau{=}i$ pillowcase shape (`v168`/`v214`) is *frozen* for all a_0 — a_0 is a pure scale, no shape modulus survives clock-invariance. *Periods/monodromy:* $\Pi=t(i{-}1)(1,i,i^2)$, all three vanishing spheres carry the same volume $\sqrt{2}\,t$ (lockstep); the clock acts on H_2 as a *Coxeter element* of $W(A_3)$ (char $x^3{+}x^2{+}x{+}1$, eigenvalues $\{i,-1,-i\}$ = the A_3 exponents, $h(A_3){=}4{=}|\mu_4|{=}N_{\mathrm{fam}}{+}1$), fixes no cycle ($\det(A{-}1){=}{-}4$), and *is* the Picard–Lefschetz monodromy of the family ($a_0\to e^{2\pi i}a_0$ gives the 4-cycle); the surviving direction is the weight-1 χ_1-Fourier diagonal $(1,i,i^2)$, *not* the naive invariant diagonal; the versal weights $(a_3,a_2,a_1,a_0)=(1,2,3,0)$ are the regular μ_4 representation. *Algebra (BHS $\mathbb{Z}_2\to\mathbb{Z}_4$, `arXiv:2305.09451`):* mode dictionary $\{w[m,n]:m{\equiv}n \bmod 4\}$; the fibre bracket $-4\,\mathrm{Nambu}(XY{-}Z^4{-}a_0)$ anchored at the $a_0{=}0$ orbifold ($\{X,Y\}{=}16Z^3$, Jacobi + Casimir exact, uniform clock-degree -1); S-algebra corrections exactly *linear* in a_0 at the $\mathbb{Z}_4$ wrap, conserving the μ_4 grade — no sector leak, the `v492` equivariant sector deforms consistently; $a_0\leftrightarrow$ BHS's weight-0 c^2 slot; matrix factorisations deform along all orbit splittings, the clock fixing exactly the antipodal 2+2 sector (= the EH/$\mathbb{Z}_2$ middle node, $4{=}2{\times}2$). *Negative controls:* $a_1Z\Rightarrow j{=}0$ (hexagonal); pure a_2Z^2 stays singular (A_1); $a_2{\neq}0\Rightarrow j{=}1556068/81{\neq}1728$; explicit grading leak — doubly sharp. *Verdict B* via three named [**C**] identifications (-4Nambu = the $k{=}4$ CCA bracket; period = root difference; seam-scale reading via clock^2=deck); the `v216` residual is *typed*, not moved (given the order-4 clock the square is automatic — a relocation of the same input, no new derivation).
- **Contract updated to WP2 done.** `tfpt_research_contracts` `CELEST.SEAM.01` section: WP2 bullet "finite, next" → *done (verdict B)* with the two sharpenings (clock = Coxeter/Picard–Lefschetz monodromy; surviving direction = χ_1-diagonal, a_0 pure scale); kill-test paragraph K1

"survived by WP1 and WP2"; status line `CELEST.WP2.01` [**E**]/[**C**] executed, WP3–WP5 stay [**O**]. Ledger rows `CELEST.WP2.01` (new) + `CELEST.SEAM.01` (WP2 done) + `GATE.WOLFRAM.02` (four exact `v493` mirrors added to `tfpt_readouts_extension.wl`, counted with the next verified engine run — engine unactivated, same convention as `v491`); registry + `run_all` (487 modules, suite green); website mirrors (`papers.ts` contract section, generated script index). `SEAM.EQUIV.01` stays [**O**]; no marker moves anywhere. Exact (sympy).

2026-07-20 (VI) (HFQPO "third tooth": the preregistered archival RXTE PCA scan has RUN — a well-powered null with limits; scorecard row S1.e `data_limited`→`null`, no marker moves)

- **H3 stage 2 executed (`experiments/hfqpo-ladder/results/archival/`).** The preregistered archival scan (`hypotheses/hfqpo_v1.yaml`, `archival_design`) ran end-to-end on public HEASARC data: 77 ObsIDs / 4.42 GB across all four 3:2-pair sources (GRO J1655−40, XTE J1550−564, GRS 1915+105, H1743−322). Sanity gate PASS (11/12 published pair lines reproduced, e.g. 450.3 Hz at 5.37% rms 4.4σ, 280.1 Hz at 3.71% 9.9σ); injection calibration PASS in all four sources (90%-recovery sensitivities 5.11/2.68/0.89/1.65% rms at the tooth frequencies).
- **Blind-scan verdict: NULL with limits.** Neither the geometric tooth $\nu_3 = 1.5\,\nu_u$ (661.5/414/252/363 Hz) nor the integer control line $4\nu_0$ (588/368/224/322.7 Hz) is detected in any source ($\sim 0\sigma$ single-trial; trials correction $N{=}8$ as preregistered). 3σ rms upper limits (tooth, primary bands): 3.06/1.61/0.53/1.26% — in every source at or below the strength of the detected upper pair line. The ladder reading gains no support but is not killed (the `harmonic_hit` branch did not fire either; the anti-kernel record 92=184/2, 34/68 stands); GR parametric resonance stays the favorite. Caveats: GRS 1915+105 scanned in a proxy state (the 113/168 Hz epochs were never published at ObsID level); H1743's program is 80146 (YAML said 80138); GRO hard-band sensitivity 5.1%.
- **Integration.** Scorecard row S1.e (hfqpo-ladder) `data_limited`→`null` (120 rows: 26 null / 33 data_limited), next decider eXTP-class sensitivity below the 0.5–3.1% rms limits; HFQPO search-target paragraph updated in `tfpt_horizon_readouts` (+ the `papers.ts` mirror); website `predictions.ts` (`hfqpo-tooth` entry + `EXPERIMENTS_AUDIT` counters); `predictions.txt`; `experiments/README.md`; `experiments/next.txt`. No `verification/` module, no ledger row, no marker move (the search target stays [**O**]).

2026-07-20 (V) (`v492` + the new research contract `CELEST.SEAM.01` — the celestial-holographic route: the E_8 μ_4-glue is the flat $\mathbb{Z}_4$ monodromy of the equivariant celestial chiral algebra on the A_3 ALE space, with the exact correction clock^2=deck (spin bridge $\mathbb{Z}_8$, $8{=}2|\mu_4|$); *verdict B, no gate closes, no marker moves, `SEAM.EQUIV.01` stays [O]*)

- **`v492_celestial_z4_orbifold.py` (`CELEST.WP1.01`).** WP1 of the new contract, sympy exact (36 checks): the `v128` μ_4-glue grading of E_8 ($240 = 52{+}64{+}60{+}64$, additive on all 6720 root pairs) is *inner* — $h = (2,2,2,2,2;0^4)$ reads the glue class mod 4 on all 240 roots, so the glue is $\mathrm{Ad}(\exp(2\pi i h/4))$, an order-4 inner torus automorphism = a flat $\mathbb{Z}_4$ monodromy in the Kronheimer–Nakajima sense; the A_3-side detector $h' = (0^5;1,1,1,-3)$ reads the same class = the diagonal $(1,1)$ of the discriminant group $\mathbb{Z}_4{\times}\mathbb{Z}_4$ (the `v92`/`v125` Lagrangian glue). $\mathbb{C}^2/\mathbb{Z}_4$ is verified as the A_3 singularity $XY{=}Z^4$ (Hilbert series $(1{+}t^4)/((1{-}t^2)(1{-}t^4))$, Molien exact); the glue-equivariant SDYM(E_8) sector *closes* with graded dimensions $60(d{+}1)/64(d{+}1)$ — possible only because $\dim \mathfrak{g}_j = (60,64,60,64)$ — with zero modes = the carrier $D_5{\oplus}A_3{+}$Cartan = 60 and density $1/4 = 1/|\mathbb{Z}_4|$; the four glue-sector characters sum exactly to $\chi_{(E_8)_1} = 1{+}248q{+}4124q^2{+}34752q^3$ (`v377`), with sector weights = the discriminant form $(5x^2{+}3y^2)/8$ and integer glue-diagonal $h = (0,1,1,1)$ (= locality of the $(E_8)_1$ extension). *Critical correction*

(verdict B): the A_3 deck acts on the celestial sphere as $z \mapsto -z$ (order 2), *not* as the order-4 clock; the clock is the $U(2)$ phase $\mathrm{diag}(i, 1)$ (normalising the deck, $\det = i$ rotating the holomorphic symplectic form by the μ_4 generator), with the exact spin bridge (spin clock)^2 = deck ($\mathbb{Z}_8$, $8 = 2|\mu_4|$, the $c_3 = 1/(8\pi)$ winding integer). Clock-invariance selects the 1-parameter A_3 deformation $XY{=}Z^4{+}a_0$ whose four branch points are one μ_4 orbit with cross-ratio 2 (`v168`/`v214` pillowcase). Negative controls: the false action $\mathrm{diag}(i, i)$ killed three ways, the false glue breaks additivity on 3112/6720 pairs, rigidity = $\mathrm{Aut}(\mathbb{Z}_4)$.

- **New research contract `CELEST.SEAM.01`** in `tfpt_research_contracts` (own section, the fourth contract alongside $U_{\mathrm{wall}}/G_{\mathrm{metric}}$/`CONTRACT.F.01`): hypothesis (seam = glue-equivariant $\mathbb{Z}_4$-orbifold sector of the CCA on the A_3 ALE; twistorial bulk = the *self-dual* sector only), typing/non-circularity (the continuum $(E_8)_1$ existence — the `SEAM.EQUIV.01` target — is *not* an admissible input), work packages WP1 (done, verdict B) through WP5 (second continuum route, Costello–Li grade), pre-registered kill tests K1–K4 (K1 survived by WP1), and an honest success estimate (full success low, partial findings high). Ledger rows `CELEST.WP1.01` + `CELEST.SEAM.01`; website mirror (`papers.ts` section, script index). `SEAM.EQUIV.01` stays **[O]**; no marker moves. Exact (sympy).

2026-07-20 (IV) (evidence scorecard: five new typed rows — the α_s running prediction of record, the μ-distortion $A_s/N_\star$ branch discriminator, and three parked analog beds (bosonic-E8 thermal Hall, E8 mass-ladder recovery, Efimov DSI tuning); *experiments level only — no verification module, no gate closes, no marker moves*)

- **α_s running (prediction of record, consistent).** The Starobinsky branch fixes the running of the spectral index: $\alpha_s = -2/N_\star^2 = -7.6 \times 10^{-4}$ at $N_\star$=51.4 (exact slow roll -7.09×10^{-4}; band $N_\star \in [50, 60]$: $[-8.0, -5.6] \times 10^{-4}$) and $\beta_s = -4/N_\star^3$ (leading order; NOT $-8/N^3$) — the identity $\alpha_s = -r/6$ holds only at leading order ($\sim$7% off exact). Data: Planck 2018 -0.0045 ± 0.0067 ($+0.57\sigma$), P-ACT $+0.0060 \pm 0.0055$ (-1.22σ), record leg P-ACT-LB $+0.0062 \pm 0.0052$ (-1.33σ, Calabrese et al. 2025 JCAP); β_s (Planck run+runrun) 0.010 ± 0.013 (-0.77σ). Forecast: LiteBIRD+CMB-S4 $\sigma(\alpha_s) \approx 4.9 \times 10^{-3}$ (0.14σ); CMB-S4+DESI+Euclid statistical $\sim 3.3 \times 10^{-4}$ (2.2σ, systematics-limited). Same `N_star_reheating` independence group as the $n_s/r/A_s$ rows (anti double-counting: never an independent hit); kill: robust $\alpha_s < -5 \times 10^{-3}$ or $> +3 \times 10^{-3}$ at $\geq 5\sigma$; any robust POSITIVE running kills the plateau branch. Probe: `experiments/tfpt-discovery/alpha_s_running_probe.py`.
- **μ-distortion (prediction of record, data-limited).** Chluba window integral (calibrated +20% to the full computation): sharp branch $N_\star$=51.4 (A_s=1.76×10^{-9}) $\mu \approx 1.6 \times 10^{-8}$, profiled record $N_\star$=56.1 (A_s=2.10×10^{-9}) $\mu \approx 2.0 \times 10^{-8}$, band 1.5–2.3 $\times 10^{-8}$; the 23% branch split ($\Delta\mu = 3.0 \times 10^{-9}$) gives a 3σ $A_s/N_\star$ branch decision at $\sigma(\mu) = 10^{-9}$ (Voyage-2050 class; the PIXIE baseline $|\mu| < 9 \times 10^{-8}$ does NOT reach — factor 4–7 above the band; FIRAS $|\mu| < 9 \times 10^{-5}$). Honest S15 typing: the frozen log-comb ($\omega = 2.5827$, $\varepsilon = 0.0173$) is STRUCTURALLY BLIND in the μ window — window attenuation $|F(\omega)| = 0.033$, $\delta\mu = 7.4 \times 10^{-12} = 0.007\sigma$ even at Voyage 2050 — the theory itself predicts where it is invisible, so NO separate comb-search row is opened. Shares the `Nstar_branch` alternative group with the two A_s rows (one branch question, several readings). Kill: robust $\mu < 0.9 \times 10^{-8}$ or $> 4 \times 10^{-8}$. Probe: `experiments/tfpt-discovery/mu_distortion_probe.py`.
- **Three parked analog beds (`parked_analog`; honest typing: they validate E8 edge/spectrum physics or the comb DETECTOR, never the TFPT axioms).** (i) Bosonic E8 phase / thermal Hall: $\kappa_{xy}/T = 8 \times (\pi^2 k_B^2/3h)$ with no charge Hall (seam edge c_-=8, `v450`/`v451`/`v489`); no engineered material yet (α-$RuCl_3$ c_-=1/2 contested: ultraclean half-quantized plateau, npj QM 2025, vs phonon-dominated κ_{xy}, PRX 12, 021025). (ii) E8 mass-ladder validation bed ($CoNb_2O_6$/$BaCo_2V_2O_8$): all 8 one-particle masses published (Zou et

al. 2021 PRL 127, 077201, NMR+INS; Amelin et al. 2020 PRB 102, 104431, THz m_1–m_6; Coldea et al. 2010 Science $m_2/m_1=\varphi$) — blind Cartan-PF ladder recovery on the published spectra (8/8 masses + rejection of scrambled controls) as detector validation; confirms Zamolodchikov E8, not TFPT. (iii) Efimov DSI tuning: $s_0 = \pi/(6\ln(3/2)) = 1.2914 \Leftrightarrow \lambda = (3/2)^6 = 11.391$ at $M/m = 8.67$ (2 resonant pairs); best candidate $^{52}Cr_2$–^{6}Li (M/m=8.635, s_0=1.2890, λ=11.44, +0.44% vs $(3/2)^6$, gap deviation +0.18%); alternatives $^{39}K_2/^{41}K_2$–^{6}Li ($\lambda = 11.59/11.20$, needs double Feshbach control), $^{176}Yb_2$–^{23}Na (+5.9%); solver validated on the Naidon–Endo anchors (homonuclear 22.69, Cs_2Li 4.877); first end-to-end HARDWARE injection of the comb detector at tunable $\lambda \approx 11.39$; kill condition honestly "none on TFPT axioms — detector validation only". Probe: `experiments/tfpt-discovery/efimov_s0_massratio_probe.py`.

- Scorecard regenerated (`experiments/build_evidence_scorecard.py`): 120 rows — 48 consistent, 9 tension, 25 null, 34 data-limited, 4 parked; website mirror (`EXPERIMENTS_AUDIT` in `predictions.ts`) synchronised. Experiments level only; the verification suite, ledger and papers are untouched.

2026-07-20 (III) (`v491` — $g_{\rm car}$=5 as the partition COROLLARY of the four marks: 4 into exactly 3 positive parts is uniquely $\{1,1,2\}$, sharpening `v53`; the residual is the weight-typing postulate — `AX.P2.01` stays a declared axiom; *no gate closes, no marker moves*)

- **`v491_p2_partition_corollary.py` (`P2.PARTITION.01`).** Sympy-exact: a multiset of exactly 3 POSITIVE INTEGERS summing to 4 is UNIQUE — $\{1,1,2\}$ — so $e = (e_1, e_2, e_3) = (4, 5, 2)$ and hence $g_{\rm car} = e_2 = 5$, $|\mathbb{Z}_2| = e_3 = 2$, $N_{\rm fam} = e_2 - e_3 = 3$ are *corollaries*, not inputs (consistency fixed point $(2^4-1)/5 = 3 = \operatorname{rank} H^1$). [**E**] negative controls, each assumption load-bearing: drop positivity → 4 solutions, e_2 ambiguous $\{0,3,4,5\}$; sum 3 → e_2=3; sum 5 → not unique; 2/4 parts → $e_2 \in \{3,4\}/\{6\}$; Mehta–Seshadri RATIONAL weights → $e_2 = \frac{16}{3}$ resp. $\frac{44}{9} \neq 5$ (integrality = the Birkhoff–Grothendieck splitting-type typing, not MS weights). [**E**] the `v53` sharpening: `v53` pins $(1,1,2)$ with ALL THREE targets $(4,5,2)$; the partition needs only e_1=4. [**E**] sum rule: $4 \times \frac{N-1}{2} = 4$ iff N=3; the `v115` route $|a|_1 = |\mu_4| \operatorname{tr}(A_0) = 4$ agrees. Independence chain: #marks = 4 from `v216` (P1-side, no $g_{\rm car}$ input); $\operatorname{rank} H^1 = 3$ pure topology (`v137`/`v228`); $|a|_1 = 4$ $g_{\rm car}$-free (`v115`). [**C**] honest residual: P2 is NOT made axiom-free — the weight-TYPING postulate remains (integrality/positivity h^0=0/sum "given (U)"); `AX.P2.01` stays a declared axiom, and the $N_{\rm fam}$ direction reversal vs `tfpt_constants` is typed as an over-determination check (never double-counted). Integrated in `introduction` (anchor half-forced paragraph), `tfpt_1` ((5, 3) keybox), `tfpt_2` (H2 winbox) and `origin_theory` ((5, 3) keybox); ledger row `P2.PARTITION.01`; three exact Wolfram mirrors added to `tfpt_readouts_extension.wl` (counted with the next verified engine run, engine currently unactivated); website mirror (DAG node `P2`). Exact (sympy).

2026-07-20 (II) (`v490` — the T^2 spin-structure parity census: $\mu_{\rm pre}$=4 witnessed on the lattice, the 16-fold-way bridge θ_v=1 on measured numbers, and the honest TEE non-discriminator fence; *no gate closes, no marker moves*)

- **`v490_seam_parity_census.py` (`SEAM.MU.01`).** The T^2 ground-state fermion parity of the `v367` $p{+}ip$ collar, resolved per spin structure (AA, AP, PA, PP) and determined TWICE — Read–Green TRIM product AND real-space Majorana Pfaffian (Schur), relative to the trivial M=3 phase, with the twisted real-space spectra matching the Bloch bands to 9×10^{-15}. [**E**] per layer $(+,+,+,-)$ (the ν=1 Ising signature, TRIM ≡ Pfaffian in every sector); [**E**] the 16-layer carrier has parity^{16} = + in ALL FOUR sectors ⇒ gauged torus GSD = 4 = $\mu(SO(16)_1)$ = $|\det \text{Cartan}(D_8)|$ — the lattice ground-state witness of "$\mu_{\rm pre}$=4 before condensation" (step 1 of the `v378` KLM chain); [**E**] the 16-fold-way bridge on MEASURED numbers: $\theta_v = e^{2\pi i\nu/16} = 1$ exactly at $\nu = 2c_- = 16$ (`v489`), rivals $\nu = 1, 2, 8$ not holomorphically condensable; [**E**]

honest fence: the fermionic Kitaev–Preskill TEE is identically zero for BOTH M=1 and M=3 (γ_f=0.000001/0.000000) — it does NOT discriminate $(E_8)_1$ vs $SO(16)_1$. **[C]** the KLM step $\mu : 4 \to 4/2^2 = 1$ stays the cited theorem (the condensed model is interacting); the numerics pin only the premise package $\{c_-$=8, $\mu_{\rm pre}$=4, θ_v=1$\}$. **[O]** `SEAM.EQUIV.01` stays **[O]** (continuum existence, MMST `v336`). Integrated in `tfpt_research_contracts` (new S3-stack paragraph); ledger row `SEAM.MU.01`; website mirror (DAG node `qft`). Python (numpy + scipy Schur + sympy exact θ_v).

2026-07-20 (`v489` — the chiral central charge c_-=8 measured DIRECTLY from the bulk ground state via the Kim–Shehab–Kim modular commutator: the first ground-state witness for premise (i) of `SEAM.EQUIV.01`, independent of the Chern route (`v367`) and the KLM count (`v378`); *no gate closes, no marker moves*)

- **`v489_seam_modular_commutator.py` (`SEAM.CCC.01`).** The Kim–Shehab–Kim modular commutator $J(A,B,C) = i\,\mathrm{Tr}(\rho_{ABC}[K_{AB},K_{BC}]) = (\pi/3)\,c_-$ (PRL 128 (2022); `arXiv:2110.06932/2110.10400`) is evaluated on the EXACT BdG ground state of the `v367` $p{+}ip$ seam collar model, entirely from the Peschel Majorana covariance ($W_X = -2\,\mathrm{artanh}\,\Gamma_X$, $J = \frac{1}{4}\,\mathrm{Tr}([W_{AB},W_{BC}]\Gamma_{ABC})$), on a disk about the torus centre cut into three 120° wedges. **[E]** ground-state purity $\mathrm{eig}(i\Gamma) = \pm 1$ to 6×10^{-15}; **[E]** per-layer $c_- = 0.499998$ at L=32 (convergence 0.499793 at L=16 $\to$ 0.500000 at L=48, deviation -1.8×10^{-7}); **[E]** controls: trivial M=3 $\to$ 0.000000, orientation flip M=−1 $\to$ −0.499986; **[E]** layer additivity exact (1.4×10^{-14}); **[E]** the 16-layer carrier lands on c_-=8=$g_{\rm car}$+$N_{\rm fam}$ (direct 7.9967 at L=16, additivity route 8.0000); **[E]** clip invariance 5×10^{-8}. The FIRST ground-state witness for premise (i) of `SEAM.EQUIV.01` (c_-=8) that is INDEPENDENT of both the Chern-number route (`v367`) and the counting route (`v378`) — read from entanglement data alone. **[O]** honest fence: does NOT close `SEAM.EQUIV.01`; the continuum scaling limit (MMST, `v336`) and the cited KLM condensation step are untouched. Integrated in `tfpt_research_contracts` (new S3-stack paragraph) and `tfpt_1` (the `SEAM.EQUIV.01` package paragraph); ledger row `SEAM.CCC.01`; website mirrors (DAG node `qft`, papers/glossary/FAQ inventories). Numerical (numpy), Python-only by nature.

2026-07-19 (editorial — positioning section *TFPT in the light of algorithmic physics: Zuse–Schmidhuber–Wolfram* added to `introduction` + website mirror; *no result, no status change*)

- **`introduction.tex` — new section "TFPT in the light of algorithmic physics".** Positions the compiler against the Zuse–Schmidhuber–Wolfram algorithmic-physics lineage: three shared instincts (minimum description length as the selection principle, deterministic resource-bounded generation, compressibility as the aesthetic) and three deliberate divergences (one *forced* program vs. an ensemble of all universes; an algebraic Lie/lattice/VOA description language vs. Turing bit-strings; and falsifiability — dated, kill-tested predictions — as the load-bearing difference). Explicitly typed as *positioning*: it introduces no formula and moves no status marker; its only machine-checked hooks are already in the suite (the grammar-internal null model and the icosahedral/McKay bedrock).
- **Website mirror.** A matching curated section was added to the `introduction` entry in `website/lib/papers.ts`.

2026-07-15 (III) (`v488` — the $3/5$ Clebsch door ENUMERATED AND CLOSED: no minimal Pati–Salam/$SO(10)$ Majorana channel can produce $5/3$ — the `v482` escape is decided dead, the `v481` window candidate stands as the honest endpoint; *a clean negative result, no gate closes*)

- **`v488_majorana_clebsch_door.py` (`FLAV.NUSCALE.04`).** v482 named ONE escape for the integer seesaw rung $M_R = c_3^3 \bar{M}_{\rm Pl}$: a third-generation Majorana factor $r \approx 1.6696$, 0.18% from $g_{\rm car}/N_{\rm fam} = 5/3$ — declined for lack of mechanism. The mechanism search is now EXECUTED: **[E]** NO RENORMALISABLE CHANNEL — $\mathrm{sym}^2(\mathbf{16}) = 136 = \mathbf{10} + \mathbf{126}$ and the $\overline{\mathbf{126}}$ is NOT in the E_8 hull $\{1, 10, 16, 45\}$ (`v247`), so M_R only via the d=5 operator $(\mathbf{16}_F \overline{\mathbf{16}}_H)^2/\Lambda$; channels $\mathbf{16} \times \overline{\mathbf{16}} = \mathbf{1} + \mathbf{45} + \mathbf{210}$ with only singlet + $\mathbf{45}$ admissible. **[E]** CHARGE TABLE: $(B{-}L, T_{3R}, Y)(\nu^c) = (+1, -\frac{1}{2}, 0)$, $\mathrm{Tr}_{16}\, Y^2 = \frac{10}{3}$ (`v485`). **[E]** THE $\{2,3\}$-SMOOTH THEOREM (the door closes): every nonzero ν^c channel weight $\{1, \frac{1}{4}, 1, -\frac{1}{2}, \frac{3}{8}\}$ is $\{2,3\}$-smooth, and the multiplicative closure of $\{2,3\}$-smooth rationals never contains $\frac{5}{3}$ (no factor 5); the UNIQUE natural 5 of the embedding, $k_Y = \frac{5}{3}$ (Ginsparg, `v470`), is a full-multiplet trace whose direction has $Y(\nu^c) = 0$ EXACTLY — structurally decoupled from the Majorana operator. **[E]** GENERATION-BLINDNESS: Clebsches are family-space scalars; $\mathrm{diag}(1, 1, \frac{3}{5})$ cannot arise from group theory by construction. **[O]** compiler-flavor inventory typed and declined: exact $\frac{3}{5}$'s only in wrong slots (K lepton gen2/gen1; $R + L$ lepton gen3/gen2; never gen3 vs a degenerate base). NET: the rung$+5/3$ rescue is DEAD as a mechanism candidate; the `v481` one-parameter window candidate (M_R free inside the PS window, the $\Sigma m_\nu = 0.059$ eV floor as kill test) stands. Contract provenance: `experiments/theory-contracts/clebsch01_majorana_35_door.py`. Integrated in `tfpt_2_standard_model` (new paragraph closing the seesaw arc); ledger row `FLAV.NUSCALE.04` + update note on `.03`; Python-only (Fraction/sympy exact), mirrorable, deferred to the next verified engine run.

2026-07-15 (II) (`v487` — the lazy split FORCED by the resummed clock: the walk's one-step spectrum is the complete `v124` clock ladder below the wall, deck parity = rung index, and the two comb tones are one bend apart — the `v486` [O] "split selection" closes onto established structure; *no gate closes, no marker moves*)

- **`v487_transfer_clock_rungs.py` (`HYP.REWRITE.03`).** v486 left the split selection open ("$(\frac{1}{2}, \frac{1}{6})$ vs uniform $(\frac{1}{3}, \frac{1}{3})$ is selected by the physical spectrum, not yet derived"; named candidate: the vague "sheet-symmetric choice"). Closed: **[E]** RUNG READING — the lazy rule's one-step spectrum is EXACTLY the complete resummed-clock ladder below the wall (`v124`: $\lambda_n^{1/p_2} = 1 - n/N_{\rm fam}$, pole at n=$N_{\rm fam}$): $\mathrm{eig}(M) = \{1, \frac{2}{3}, \frac{1}{3}\} = \{1 - n/3 : n = 0, 1, 2\}$, while `v327`'s uniform rule puts its second pair mode ON the wall ($0 = 1 - 3/3$) — the v486 degeneracy re-read as "odd channel collapsed to the wall". **[E]** DECK PARITY = RUNG INDEX: $[B, \sigma] = 0$; the σ-even pair mode carries rung n=1 (survival $\frac{2}{3}$), the σ-odd mode rung n=2 (survival $\frac{1}{3}$) — each clock step flips the deck parity; this is the structural home of the v486 parity assignment (ω_1 even-channel, ω_2 odd-channel) that `experiments/frb-parity-comb` tests. **[E]** FORCING: demanding the faithful ladder (no rung missing, no physical mode on the wall — definitional for a full-spectrum generator) + $\mathbb{Z}_2$ equivariance + non-negative rates forces UNIQUELY both the split $(\frac{1}{2}, \frac{1}{6})$ AND the rung assignment even$\to$1, odd$\to$2 (the swapped assignment needs $h = -\frac{1}{6} < 0$); the "sheet-symmetric choice" candidate is superseded by this proof. **[E]** BEND COROLLARY: $\omega_1/\omega_2 = \mathrm{rate}(2)/\mathrm{rate}(1) = \log_{3/2} 3 = 2.70951\ldots$ — the parity-comb tone ratio IS the `v107`/`v124` bend of the pulsar-clock search. **[O]** what remains open is exactly what was open before v486: the arity $\{2,3\}$ (anchor input) and the clock's semiclassical derivation (R1); no independent choice remains at the walk level. Contract provenance: `experiments/theory-contracts/lazy01_clock_rung_forcing.py`. Integrated in `origin_theory` (new paragraph after the v486 block); ledger row `HYP.REWRITE.03` + update note on `.02`; Python-only (sympy exact), mirrorable, deferred to the next verified engine run.

2026-07-15 (`v486` — generator hunt: the FULL transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ from ONE forced local rule — the unique lazy $\mathbb{Z}_2$-pair walk $(\text{stay}, \text{hop}, \text{leak}) = (\frac{1}{2}, \frac{1}{6}, \frac{1}{3})$; the previously generator-less $(1/3)^6$ mode gets its origin; `v327`'s uniform rule superseded as full-spectrum generator — *no gate closes, no marker moves*)

- **`v486_transfer_full_rule.py` (`HYP.REWRITE.02`).** The verified seam transfer spectrum is $\{1, (2/3)^6, (1/3)^6\}$ (`v54`/`v221`), but the only local generator in the suite (`v327`) produces $\{1, \frac{2}{3}, 0\}$ — its zero mode persists under every power, so no iterate reaches the third mode. **[E]** UNIQUENESS: for the symmetric 3-channel rule $M(s,h) = [[1,0,0],[0,s,h],[0,h,s]]$ (one absorbing family channel + $\mathbb{Z}_2$ pair) the survival spectrum is $\{s+h, s-h\}$; demanding the PHYSICAL pair $\{\frac{2}{3}, \frac{1}{3}\}$ forces UNIQUELY $(\text{stay}, \text{hop}, \text{leak}) = (\frac{1}{2}, \frac{1}{6}, \frac{1}{3}) = (1/|\mathbb{Z}_2|, 1/(|\mathbb{Z}_2|N_{\text{fam}}), 1/N_{\text{fam}})$ — the lazy $\mathbb{Z}_2$-pair walk, every rate an atom expression (honest direction: spectrum $\to$ rates). **[E]** SIX-HAND GENERATION: over the order-6 $= 2N_{\text{fam}} = |R^+(A_3)|$ hand, $\text{eig}(B^6) = \{\frac{64}{729}, \frac{1}{729}\}$ exactly — both decay gaps now have one recursive generator: $6\ln\frac{3}{2}$ (recovery, `v76`) and $6\ln 3$ (subdominant, previously generator-less). **[E]** SUPERSESSION: the uniform rule's zero mode is power-persistent; its λ_2 arity story ($\frac{2}{3} = |\mathbb{Z}_2|/N_{\text{fam}}$) is unchanged. **[O]** the arity $\{2,3\}$ stays the anchor input; the lazy-vs-uniform split is SELECTED by the physical spectrum, not yet first-principles-derived (named candidate: stay $= 1/|\mathbb{Z}_2|$ as the sheet-symmetric choice). Discovery provenance: `experiments/tfpt-discovery/generator_hunt_full_transfer_rule.py` (plus a typed anchor-bilinear coverage inventory — known hits a^TKa=41, Koide 53, and the recorded-only $\mathbf{1}^TQa$=16 — and the lepton-exponent observation $(5,3,2) = K$-row = atom triple; no unforced claims, `v354`/`v355` discipline). Integrated in `origin_theory` (new paragraph after the `v327` rewrite block); ledger rows `HYP.REWRITE.02` + update note on `.01`; Python-only (sympy exact), mirrorable, deferred to the next verified engine run.

2026-07-14 (IV) (`v485` — the merged analytic target's computable side COMPLETED: the diagonal self-energy is renormalised to ZERO exactly at the KMS seam scale, the mark contact calculus resums in closed form (BFK route), and the 48/41 multiplicities are ONE state set under two response weights — the residual collapses onto the `SEAM.EQUIV.01` face alone; *no gate closes, no marker moves*)

- **`v485_contact_diagonal_closed.py` (`SEAM.CONTACT.UNIT.02`).** The v484 merge left one analytic step: the diagonal (self-energy) zeta-renormalisation at the marks + the multiplicity matching. Settled at the computable level: **[E]** DIAGONAL ZERO AT THE KMS SCALE — the renormalised coincident-point Green function of $|D|$ on circumference ℓ is $G_{\text{reg}}(0;\ell) = \frac{1}{\pi}\ln(\ell/2\pi)$ EXACTLY (symbolic limit + 30-digit checks); it vanishes iff $\ell = 2\pi = 1/(4c_3)$, the `v239` KMS seam circumference — the seam scale is the unique zero-self-energy scheme point, and the log is the pure SCALE channel (matching the structure: $F_{U(1)}$ carries exactly ONE log term, the transport $8b_1c_3^6\ln(1/\varphi_{\text{seam}})$, while the φ_0^{ret} split is log-free). **[E]** CLOSED-FORM RESUMMATION (the BFK/gluing route already in the suite, `v151`): with the diagonal zero, the finite-rank mark determinant resums exactly — $\det(I - uC) = (1-4u)(1+2u)^2$, $u = \varepsilon c_3\ln 2$, all orders, the linear term ABSENT precisely because the diagonal vanishes ($\text{Tr}\, C = 0$); the series matches `v484`'s $\text{Tr}(C^k) = 4^k + 2(-2)^k$ term by term. **[E]** WEIGHT DICHOTOMY: the SAME $48 = 3 \times 16$ admissible states carry BOTH multiplicities — counted flat they give the φ_0^{ret} puncture count $48 = \Omega_{\text{adm}}$; Y^2/loop/GUT-weighted they give the alpha budget $41 = 10b_1 = 40 + 1$ ($\text{Tr}_{16}\, Y^2 = \frac{10}{3}$ exact, $\nu^c = 0$; Ginsparg $(3/5)(41/6) = 41/10$, `v470`) — one state set, two response weights, no new number. **[C]**/**[O]** CONSEQUENCE: with `v484` EVERY finite/computable piece of the merged `ALPHA.QUILLEN.EXACT.01` + `HYP.PHI0.PUNCTURE.01` target is proven; the single remaining **[O]** is the identification of the abstract seam ζ-det with this glued mark determinant — a face of `SEAM.EQUIV.01`, exactly the `v382` typing, now substantiated computationally. Integrated in `tfpt_1_architecture_e8` (eighth step of the ALPHA.QUILLEN arc) and `origin_theory`; ledger rows `SEAM.CONTACT.UNIT.02` + update

notes on `.01` and both targets; Python-only by nature, flagged in the Wolfram README.

2026-07-14 (III) (`v484` — the two open analytic targets MERGE: the shared "c_3 per boundary insertion" rule of `ALPHA.QUILLEN.EXACT.01` and `HYP.PHI0.PUNCTURE.01` is the KMS seam unit $2\pi = 1/(4c_3)$ with $\frac{1}{4} = 1/|\mu_4|$, derived on the seam circle for the finite cycle sector — *no gate closes, no marker moves*)

- **`v484_seam_contact_unit.py` (`SEAM.CONTACT.UNIT.01`).** The v483 round left the two analytic targets sharing one rule ("weight c_3 per boundary contact insertion": the $F_{U(1)}$ ladder $\{0,3,6\}$, the $\varphi_0^{\rm ret}$ puncture $\{1,4\}$). This module shows the shared rule is NOT a new unknown: **[E]** the unit chain $\frac{1}{2\pi} = 4c_3 = |\mu_4|c_3 = |\mathbb{Z}_2|\chi(S^2)c_3$ — one *bare* boundary insertion divided by the orbit order $|\mu_4|$=4 is exactly c_3; this is the v239 KMS seam unit and the v58/v73 one-sided Gauss–Bonnet in a single identity. **[E]** the mechanism is DERIVED on the actual seam circle for the finite cycle sector: the bare Green function $G(d) = -\frac{1}{\pi}\ln|2\sin(d/2)|$ (polylog identity, 40 digits) takes INTEGER multiples of $c_3\ln 2$ at the μ_4 separations ($G(\pi/2) = -4c_3\ln 2$, $G(\pi) = -8c_3\ln 2$; mark matrix $-4c_3\ln 2\cdot\mathrm{circ}(0,1,2,1)$, spectrum $\{4,-2,0,-2\}$), so the μ_4-invariant (orbit-averaged) insertion carries $-(c_3\ln 2)C$ exactly and the log-det contact expansion is exactly graded in c_3 per insertion ($\mathrm{Tr}\,C^k = 4^k + 2(-2)^k$). **[E]** the grammar is suite-wide: the Λ prefactor $\frac{3}{4\pi^2} = (8\pi)^2 d_{\rm top} = 48c_3^2$ (`v60`) carries the SAME multiplicity $\Omega_{\rm adm}$=48 as the $\varphi_0^{\rm ret}$ puncture at TWO insertions; the $g-2$ seam loop divides by the same $2\pi = 1/(4c_3)$. **[C]/[O]** THE MERGE: the two tracked **[O]** targets collapse into ONE named analytic step — the diagonal (self-energy) zeta-renormalisation at the marks + the state-multiplicity matching ($48 = \Omega_{\rm adm}$, $41 = 10b_1$); the finite half is proven, no target coefficient is claimed (the integers are forced by the square configuration, `v354`/`v355`-clean). Integrated in `tfpt_1_architecture_e8` (seventh step of the ALPHA.QUILLEN arc) and `origin_theory` ($\varphi_0^{\rm ret}$ split paragraph); ledger rows `SEAM.CONTACT.UNIT.01` + update notes on both targets; Python-only by nature (mpmath polylog), exact identities flagged in the Wolfram README.

2026-07-14 (II) (`v482`–`v483` — the deep-dive round: the ν-scale decision computation EXECUTED (the integer c_3^3-rung excluded unstructured, the $5/3$-proximate structure escape recorded and declined) and the $\varphi_0^{\rm ret}$ puncture target HALVED (the geometric side proven exact: all twisted pillowcase heat traces are t-independent rationals, so the π^{-4} must sit in the per-mark coupling weight) — *no gate closes, no marker moves*)

- **`v482_seesaw_rung_decision.py` (`FLAV.NUSCALE.03`).** The decision computation NAMED by `v481` is EXECUTED in bracketed (assumption-robust) form: the integer rung $M_R = c_3^3\bar{M}_{\rm Pl}$ needs a rescue factor $\times 1.670$ in m_3 (0.0301 vs 0.0503 eV), while $m_t(\overline{\rm MS}) \pm 3$ GeV ($> 3\sigma$), $\alpha_3^{-1} \mp 0.10$ ($> 3\sigma$), a $\pm 10\%$ band on the ADKLR κ run-down and a PS-leg Yukawa-beta envelope $\times[0.5, 1.5]$ move it by at most $\times 1.10$ singly and $\times 1.165$ ALL-favourably combined — the *unstructured* rung is EXCLUDED conditional on the carrier premise $y_\nu = y_t$ (the `v481` 1-loop miss is not an RG artifact; no exact higher-loop coefficients trusted, pure envelope argument). **[C]** The only escape is a third-generation Majorana STRUCTURE factor $r \sim 1.67$; its post-hoc 0.18% proximity to $g_{\rm car}/N_{\rm fam} = 5/3$ is recorded and DECLINED per `v354`/`v355` (no forcing mechanism; the $t_0\sim 30$ discipline of `v478`). **[O]** the `v481` one-parameter window candidate (NO floor Σm_ν=0.059 eV) never used the integer rung and STANDS. Integrated in `tfpt_2_standard_model` (ν-scale paragraph) and `tfpt_5_redteam`; ledger rows `FLAV.NUSCALE.03` + `.02` update note; Python-only by nature.

- **`v483_pillowcase_exact_traces.py` (`HYP.PHI0.GEOM.01`).** The GEOMETRIC side of the $\varphi_0^{\rm ret}$ puncture target (`HYP.PHI0.PUNCTURE.01`/`v408`) is EXACT: every twisted/orbifold heat trace of the flat $\tau{=}i$ pillowcase is a t-INDEPENDENT RATIONAL — $\mathrm{Tr}(\gamma e^{t\Delta}) = 1$ for

$\gamma = \sigma, \rho, \rho^3$ at every tested t (position-space Gaussian image sums $< 10^{-9}$; mode-space proof: only n=0 is γ-diagonal), matching the Atiyah–Bott fixed-point counts $4\times\frac{1}{4}$ (det = 4=$|\mu_4|$) and $2\times\frac{1}{2}$ (det = 2=$|\mathbb{Z}_2|$); the pillowcase contact term is EXACTLY $\frac{1}{2} = 4 \times \frac{1}{8}$ per mark and the clock-equivariant trace exactly 1, for all t. THEREFORE the π^{-4} in $d_{\text{top}} = 48c_3^4 = \frac{3}{256\pi^4}$ cannot come from flat orbifold geometry at ANY order: it must sit in the per-insertion COUPLING normalisation (4=$|\mu_4|$ marks of weight c_3 give c_3^4, multiplicity Ω_{adm}=48; the bare $1/(2\pi)$-propagator 4-cycle differs by the exact rational $\frac{3}{16}$). The target NARROWS from "derive the term" to ONE rule — "per-mark insertion weight = c_3" (the same residual class as the EM-Ward c_3-ladder, `EM.WARD.02`) — and stays **[O]**. Complements the experiments-side Donnelly probe (honest negative). Integrated in `origin_theory` (φ_0^{ret} split paragraph); ledger rows `HYP.PHI0.GEOM.01` + `HYP.PHI0.PUNCTURE.01` update note; Python-only by nature (quadrature verification of exact statements).

2026-07-14 (website + figure refresh — the `/verification` journey views become *live* HTML (the stale timeline/residual-chain images replaced by interactive components wired to the in-browser reproducer), the prediction surface's empirical-audit wall of text restructured into dated, expandable rounds, and the generated timeline/residual-chain figures extended to the current suite — *presentation only, no content or marker moves*)

- **Website `/verification`: live journey views.** The static `script_timeline.png` and `residual_chain.png` images (stale at the `v303`–`v407` figure state, with alt text and captions still describing the old eleven-phase/`v237` state) are replaced by two interactive components, `SuiteTimeline.tsx` (17 phases, `v1`–`v481`) and `ResidualChain.tsx` (the reduction chain through the certification round to the bedrock): every phase/step lists representative `vN` scripts that run live in the browser via the existing Pyodide reproducer, exactly like the dependency graph. Stale counts fixed ("∼295-script journey", "260 scripts" → the generated suite total). The dependency graph itself was verified current (`v479`–`v481` present; all 288 referenced scripts exist in the registry and the website mirror).
- **Prediction surface: the empirical-audit narrative restructured.** The single ∼13 000-character audit paragraph in `lib/predictions.ts` is split into twelve dated, verdict-badged rounds (`AuditRound`: date, title, verdict, two-sentence summary, full unabridged record on expand) plus a domain-chip row and the anti-double-counting note — same content, readable surface. Long-token overflow fixed (`break-words`) in the dependency-graph detail panel and the universal-gap-lab residual cards; the over-long `ALPHA.QUILLEN` residual note tightened to its current state (`v470`/`v472`, residual unchanged).
- **Generated figures + captions synced.** `make_figures.py`: the script timeline gains the `v408`–`v481` certification-round phase (title now ∼475 scripts) and the residual chain gains the `v458`–`v480` certification step (Kronheimer `v479`, four-interval `v480`, crossed-product `v469`) with sub-text wrapping fixed; figures regenerated (PDF + website PNG). The stale figure captions updated: `introduction` (timeline caption now spans the full arc), `tfpt_research_contracts` (residual-chain caption now states the `SEAM.EQUIV.01` bedrock and certification round; QFT-skeleton caption now names the one **[O]** residual instead of the outdated "two open items"). No numbers, statuses or claims change anywhere.
- **KaTeX macro fix (broken formula rendering).** Formulas copied from the document set that use the shared TeX macros (`\gcar`, `\Nfam`, `\cthree`, …) rendered as red KaTeX errors on the site (e.g. the φ_0^{tree} icosahedral-combinatorics formula on the Origin-Theory page). A shared macro map (`lib/katex-macros.ts`, mirroring the `changelog.tex` preamble) is now passed to every KaTeX render; an automated scan of all 19 site routes finds zero `katex-error` nodes.
- **Readability pass (text walls clamped, content unchanged).** A reusable `Expandable` component clamps long mirror prose with a fade + "Show more" toggle — applied to the

paper-section bodies and front-box bullet lists (`PaperSection.tsx`, the home set pages and `/papers/[slug]`), the dependency-graph node summaries and the prediction-card descriptions. Nothing is deleted; the full text expands in place.

- **Hylaean ecosystem card corrected.** The `hylaean.ai` card spoke of "Field AI" with structure tags that did not describe the project. It now speaks consistently of *Hylaean* and is grounded in the project documentation (`_documentation/core-physics/big-picture.md`) and the site's own status log: one cognitive substrate behaving like a physical material, knowledge = attractor valleys, question = boundary condition, answers committed only on physical landing, capability claims gate-checked against a machine-checked claim registry (honest negatives published) — borrowing TFPT's structural discipline, not its physics predictions. The Agentic-Commerce card reference updated accordingly.
- **Live-site correctness pass (counts + reading order).** Sitewide audit against the deployed page found stale counts and fixed them: the document set is stated everywhere as *10 documents = 1 reading guide + 5 numbered papers + 4 companions* (the safeguards companion was missing from the papers/downloads headers, the JSON-LD collection and the `tfpt_docset.tex` box, which still said "three companions" next to "four"); the stale "23 falsifiable predictions" claims (site metadata, OG image, `llms.txt`) now state the live count of the prediction surface (25 test surfaces, sourced from `predictions.length` where possible), while the interactive compiler page is honestly labelled as its own *23-readout prediction board* (the standalone app's board genuinely has 23 entries; the intro film keeps its spoken "twenty-three"); and the `/orientation` reading-order lists are fixed to the canonical numbering (Red Team = Doc 5 added, Appendix H = 6, Origin = 7, contracts = 8, safeguards = 9 — the old list mislabelled 6/7 and omitted three documents). Presentation/consistency only; no physics content changes.

2026-07-14 (`v479`–`v481` — the bird's-eye "overlooked tools" round: R3's graph→geometry bridge discharged to Kronheimer 1989; the four μ_4 marks realised as the exactly solvable *four-interval* multilocal modular geometry with $\omega \circ \rho = \omega$ manifest; and the absolute ν-scale collapsed to a one-parameter, PS-window-bracketed candidate under the carrier normalisation y_ν=y_t — *no gate closes, no marker moves*)

- **`v479_kronheimer_quiver_bridge.py` (`SEAM.KRONHEIMER.01`).** The R3 residual of the $\det K$=1 keystone (`v344`: "the rewrite attractor graph IS the du Val singularity") is an ESTABLISHED THEOREM — Kronheimer's ALE hyper-Kähler quotient construction + Torelli classification (J. Diff. Geom. 29 (1989) 665 and 685) — with all finite input data already compiler outputs, machine-checked **[E]**: the Kac marks $\{1,2,2,3,3,4,4,5,6\}$ are the null vector of the affine-E_8 Cartan matrix AND the Perron vector $A\delta$=2δ of the rewrite attractor (`v312`); the hyper-Kähler quotient dimension count lands exactly on 4 (the $\mathbb{C}^2/\Gamma$ geometry; $\sum_{\text{edges}} d_i d_j = \sum d_i^2 = 120$ forced by the null identity); the gauge datum is $|2I| = 120 = |\mu_4| h(E_8)$, $h(E_8) = 30 = 2 \cdot 3 \cdot 5$; the resolution carries the unimodular $-E_8$ form ($\det = 1$, SNF = identity — the Poincaré link) with $3 \cdot 8 = 24$ deformation parameters. **[C]** the theorems are cited, not re-proved (import class of NPW26/MMST/AGT); **[O]** the residual premise *transforms* to "the raw seam supplies ALE hyper-Kähler data asymptotic to the $(2,3,5)$ cone" — the same single arrow as `SEAM.EQUIV.01` (`v346` L2). Integrated in `origin_theory` (R3 keybox + new Kronheimer paragraph) and `tfpt_research_contracts` (Kummer/du Val); ledger rows `SEAM.KRONHEIMER.01` + `SEAM.DETK.01` update note; Wolfram mirror added to the extension file (+2 exact checks, counted with the next verified engine run — the local engine is currently unactivated; the identical statements are sympy-exact in `v479`). LEAN: the finite input data are kernel-checked in `TfptCarrier/KronheimerMarks.lean` (marks = null/Perron vector over $\mathbb{Z}$, $\sum \delta_i^2 = 120 = 4 \cdot 30$, $30 = 2 \cdot 3 \cdot 5$, dimension count = 4; lake build clean, `#print axioms` = the three standard kernel axioms only).

- **v480_multilocal_four_interval.py (QGEO.MULTILOCAL.01).** The seam circle minus its four marks IS the n=4 multi-interval region of a chiral free fermion — the class whose multi-interval modular theory is exactly known (Casini–Huerta mixing term; Longo–Xu; Rehren–Tedesco multilocal fermions; KLM: holomorphic $\Leftrightarrow$ trivial multi-interval inclusion, μ=1 — the same one bit as $\det K$=1); until now the suite used only the single-ball CHM boost (v358). On the exactly solvable antiperiodic critical ring: [**E**] the quarter-turn clock is a signed permutation with ρ^4=-1 EXACTLY (the fermionic μ_4 clock is the order-8 double-cover lift — the same $\mathbb{Z}_2$-lift pattern as $2I$ over I); [**E**] its primitive-8th-root character projectors decouple the 4-interval covariance at 10^{-15}, each sector a single-interval quarter-ring kernel with boundary twist (10^{-14} spectra); [**E**] $\rho C\rho^{-1}$=C at 10^{-14}, so $\omega\circ\rho$=ω holds *manifestly* at the 4-interval state level, and a one-site displacement re-couples the sectors at 10^{-2} (negative control); [**E**] the cross-interval modular coupling peaks on the conjugate-point diagonal (the Casini–Huerta mixing shadow). [**C**] continuum theory cited; [**O**] the raw-seam premise (QGEO.SYM.01/SEAM.EQUIV.01) unchanged. Integrated in tfpt_research_contracts (marks keybox + rigidity contract + v478 leg A), origin_theory (BW paragraph), tfpt_4_frontier (v358 companion note); Python-only by nature.
- **v481_seesaw_carrier_ladder.py (FLAV.NUSCALE.02, CANDIDATE class, genre v467/v468).** FLAV.NUSCALE.01's y_ν=1 probe was not the carrier normalisation: one $SO(10)$ **16** per family + the minimal Yukawa sector ($\mathbf{10}_H$ / PS $(1,2,2)$) forces y_ν=y_t at the matching scale (named premise). With explicit 1-loop RG (gauge/y_t/λ up; ADKLR Weinberg-operator running down, hep-ph/0108005) the (y_ν, M_R) trade-off collapses to M_R alone: the observed m_3=0.0503 eV demands M_R=9.3×10^{13} GeV — INSIDE the PS window $[4.2\times10^{13}, 2.4\times10^{15}]$ (v249) at $\log_{c_3}(\bar{M}_{\rm Pl}/M_R)$=3.15 (the y_ν=1 inversion gives the structureless 2.58). [**X**] honesty gate: the integer rung M_R=$c_3^3\bar{M}_{\rm Pl}$ predicts m_3=0.030 eV — 40% low at 1-loop (κ run-down 0.794) — the ladder pin is DECLINED per v354/v355; the decision computation is named (2-loop SM + explicit PS thresholds). What survives: a *one*-parameter, window-bracketed [**O**] candidate with the falsifiable band $m_3\in[0.002, 0.115]$ eV and the NO floor Σm_ν=0.059 eV; frozen record untouched. Integrated in tfpt_2_standard_model (new paragraph after v272), introduction status card (third named candidate), tfpt_5_redteam; ledger rows FLAV.NUSCALE.02 + FLAV.NUSCALE.01 update note; Python-only by nature.
- *Honest negative (experiments-only, no promotion):* the Donnelly/Atiyah–Bott equivariant heat-kernel route for the ϕ_0 puncture target (HYP.PHI0.PUNCTURE.01) types the KIND of term (order-4 equivariant contact datum; the exact weights $\sum_{k=1}^{n-1}\frac{1}{4\sin^2(\pi k/n)}$=$\frac{n^2-1}{12}$ verified) but does NOT force the value $48c_3^4$ (pure rationals vs π^{-4}) — recorded in experiments/theory-contracts/phi0_donnelly_equivariant_probe.py; the target stays [**O**].

2026-07-13 (v478 — first computable steps on the two remaining legs of the entropic-action bridge: the state-side modular data flows to the CHM/Bisognano–Wichmann geometric form (Calabrese–Cardy $c_{\rm est}\to1$, CHM parabola, even bands exactly zero) — the bridge's modular side meets TFPT's Einstein-derivation input; and the measure condition reduced to one exact KMS time $t_0=\ln\frac{72}{17}+9\ln(8\pi)=30.461$ with the $h(E_8)$=30 near-miss explicitly declined; both legs stay [O], *no status moves*)

- **v478_entropic_continuum_legs.py (GRAV.ENTROPIC.LEGS.01).** *Leg A (AP2 continuum, exhibited):* on the exactly solvable critical chain (infinite-volume sinc kernel, 50-digit precision) the compressed interval state has $S(L)$=$(c/3)\ln L$+k with $c_{\rm est}$=1.00089/1.00038/1.00021 (monotone $\to$ the Dirac c=1=$2\times\frac{1}{2}$, the v450 reading; gapped control $\sim6\times10^{-8}$, area law); the Peschel modular Hamiltonian K=$\ln((1-C_A)/C_A)$ organises on the CHM weight $\beta(x)$=$x(L-x)/L$ (corr 0.87$\to$0.99, shrinking plateau drift; constants deliberately not asserted), the even-range bands

vanish identically (particle–hole, $\sim 10^{-36}$) and the odd tail is 1.8% with parabolic envelope — the state-side Δ_Σ (the reading forced by `v476`) flows to the CHM/BW geometric form, exactly the object TFPT's Einstein derivation consumes (`v323`/`v358`): the bridge's modular side and the gravity side meet in the continuum limit; a lattice exhibit, NOT a continuum proof — AP2 stays **[O]**. *Leg B (the measure):* **[E]** single-scale corollary — $d\mu{=}\delta(t{-}t_0)dt$ gives $\mu_2/\mu_1^2{=}e^{t_0}$ exactly, so the `v477` one-moment condition forces $t_0{=}\ln\frac{72}{17}{+}9\ln(8\pi){=}30.461$ (one dimensionless KMS time, closed form); the near-miss to $h(E_8){=}30$ (difference 0.461) is explicitly DECLINED per the `v354`/`v355` no-free-pattern rule — deriving the measure from the entropic structure stays OPEN. Keybox extended in `tfpt_research_contracts`; high-precision numerical module, Python-only by nature (Wolfram count unchanged, 378/378; flagged in the Wolfram README); ledger row `GRAV.ENTROPIC.LEGS.01`; both legs stay **[O]**, no marker moves.

2026-07-13 (`v477` — the scale-flow representation of the entropic action: the surviving R^2 route typed as *one moment condition* $\mu_2/\mu_1^2 = \frac{72}{17}(8\pi)^9$ with the exact closure $\frac{4608\pi^2/17}{(72/17)(8\pi)^9} = c_3^7$ — the kill-test gap is a scale-measure datum, satisfied by the same KMS moment TFPT already fixes (`v36`); zero new dials, *no status moves*)

- **`v477_entropic_scaleflow.py` (`GRAV.ENTROPIC.SCALEFLOW.01`).** **[E]** The Frullani identity upgrades to a scale-flow representation: $-\ln a = \int \frac{dt}{t}(e^{-ta}{-}e^{-t}) = 2\int \frac{d\chi}{\chi}(e^{-a/\chi^2}{-}e^{-1/\chi^2})$ (symbolic + 10^{-20} quadrature) — Bianconi's log action IS the *flat* scale-integral of relative heat-kernel actions, while TFPT's $S_{\mathrm{rel},\chi}$ evaluates *one* KMS scale with a weight: the bridge between the two action principles is a choice of SCALE MEASURE. **[E]** Moment dictionary: a weight $w(t)\,dt/t$ gives $3\beta R\,\mu_1 + \frac{17}{24}\beta^2R^2\mu_2$; the flat weight has $\mu_1{=}\mu_2{=}1$ and reproduces the `v475` raw coefficients identically. **[E]** The ONE moment condition: demanding $m^2{=}c_3^7\bar{M}_{\mathrm{Pl}}^2$ forces exactly $\mu_2/\mu_1^2{=}\frac{72}{17}(8\pi)^9$ — the `v475` kill-test number IS a moment ratio — and the closure identity $\frac{4608\pi^2/17}{(72/17)(8\pi)^9}{=}c_3^7$ holds identically: the "13 orders" are a scale-measure datum, not a bridge mismatch. **[E]** v36 consistency: $f_0{=}6(4\pi)^2/c_3^7 \Leftrightarrow M^2/\bar{M}_{\mathrm{Pl}}^2{=}c_3^7$ — one KMS moment, two parametrisations, zero new dials. **[C]** A consistency statement, not a derivation (the KMS weight is TFPT input; the entropic action supplies the Einstein + Λ level and the scale-flow form, TFPT supplies the measure). Negative side-record: an exact lattice-BW commutator attempt for AP2 (Peschel–Eisler simple forms) did NOT hold on the half-filled open chain — kept out of the suite, logged in the research notes. Keybox extended in `tfpt_research_contracts`; Wolfram mirrors both exact blocks (378/378); ledger row `GRAV.ENTROPIC.SCALEFLOW.01`; the R^2 gate and the `v359` typing stay **[O]**, no marker moves.

2026-07-13 (`v476` — the AP2 compression conjecture of the entropic-action bridge made *well-posed* on an exactly solvable model: the literal operator-side reading is ill-posed on a pure bulk, the state-side reading is forced, the mismatch is exactly quadratic in the cross-cut correlations and gap-suppressed; AP2 stays [O], *no status moves*)

- **`v476_entropic_compression_toy.py` (`GRAV.ENTROPIC.COMPRESS.01`).** The last open work package of the `v473` round, exercised on a gapped staggered-mass chain ($N{=}16$). The seam situation is reproduced (pure bulk: the ground-state covariance is a projection to 10^{-15}; thermal boundary: $\mathrm{spec}(PCP) \subset (0,1)$, Peschel $K{=}\ln((1{-}C_A)/C_A)$ finite, $S{>}0$ — `v155` in the bridge context). **The reading is decided:** on the pure bulk $f(x){=}x/(1{-}x)$ is singular on $\mathrm{spec}(C){=}\{0,1\}$, so the literal operator-side reading $Pf(C)P$ of the `v473` conjecture is *ill-posed* while the state-side $f(PCP)$ is finite — the compression identity can only mean "build Δ_Σ from the *compressed* relative metric", which matches Bianconi's own local-metric construction (AP2 retyped, not weakened). **[E]** Block algebra (symbolic): $[C^2]_{AA}{-}(C_{AA})^2{=}BB^t$, $[C^3]_{AA}{-}(C_{AA})^3{=}ABB^t{+}BB^tA{+}BDB^t$ — first order in the cross block vanishes identically,

so for any analytic f the two readings agree to *second* order in the cross-cut correlations: the modular (entanglement) content lives exactly in that quadratic correction. Gap control at the bounded (covariance) level: the Bianconi-shaped bulk identity $f(C_T)=e^{-h/T}$ holds to 10^{-11} relative; the mismatch obeys the quadratic law $d \le x^2$ ($d/x^2 \in [0.4, 1.0]$) and falls with the gap in the gapped regime, the decoupled cut agrees exactly — the two readings converge in the gap-dominated regime, where TFPT operates ($\Delta_{\text{eff}}>0$, `v76`/`v337`). Keybox extended in `tfpt_research_contracts`; numerical module, Python-only by nature (Wolfram count unchanged, 376/376; the block identity flagged in the Wolfram README); ledger row `GRAV.ENTROPIC.COMPRESS.01`; AP2 (the continuum seam statement) stays **[O]**, no marker moves.

2026-07-13 (`v475` — the R^2 kill test of the entropic-action bridge, *executed*: exact tensorial factors on the maximally symmetric background, raw entropic scalaron $m^2 = \frac{4608\pi^2}{17}\bar{M}_{\rm Pl}^2$ ($\approx 51.7\,\bar{M}_{\rm Pl}$, trans-Planckian) $\Rightarrow$ the light-trace-mode reading is dead, KMS renormalisation the only surviving R^2 route; plus the explicit Lorentzian-positivity witness; *no status moves*)

- **`v475_entropic_scalaron_gate.py` (`GRAV.ENTROPIC.SCALARON.01`).** Work package 5 of the `v473` bridge. **[E]** On a maximally symmetric background the relative curvature operator $\widetilde{\mathcal{R}}\widetilde{g}^{-1}$ has eigenvalues *exactly* $\{R;\ R/4{\times}4;\ R/6{\times}6\}$ (Bianconi's Appendix-B flattening verified symbolically, incl. the $6{\times}6$ two-form block), giving the exact vacuum action $\mathcal{L}_{\rm B}(R) = 3\beta R + \frac{17}{24}\beta^2 R^2 + O(R^3)$ — the 3 re-derives her Eq. (45) (= the `v473` pin), $\frac{17}{24}{=}\frac{1}{2}(1{+}\frac{1}{4}{+}\frac{1}{6})$ is the exact tensorial R^2 factor. **[E]** With the pinned $\beta'{=}c_3/6$ both mass routes (Starobinsky matching and $f'/3f''$) agree: the raw entropic scalaron is $m^2_{\rm raw}{=}\frac{4608\pi^2}{17}\bar{M}^2_{\rm Pl}$, i.e. $m_{\rm raw}{\approx}51.7\,\bar{M}_{\rm Pl}{\approx}1.3{\times}10^{20}$ GeV — TRANS-PLANCKIAN, not light; `v473`'s interpretation 2 (the TFPT scalaron as the light $\mathcal{G}$-trace mode) is *dead in naive form*; a viable mechanism must supply exactly $\frac{72}{17}(8\pi)^9{\approx}1.69{\times}10^{13}$ in mass^2 (the kill-test number now WITH tensorial factors). **[E]** Domain: first log pole at $R{=}1/\beta{=}384\pi^2\bar{M}^2_{\rm Pl}$ (trans-Planckian) — the obstruction is the mass, not the domain. **[E]** Lorentzian-positivity witness: a timelike gradient gives the Gg^{-1} eigenvalue $1-\alpha v^2 \le 0$ for $v^2 \ge 1/\alpha$ (ln undefined), spacelike/Euclidean stay positive — the `v473` **[C]** caveat exhibited, the OS route sturdier. **[C]** Verdict: interpretation 1 survives (Einstein + Λ level), 2 is dead naive, 3 (KMS-spectral renormalisation) is the ONLY surviving R^2 route. Keybox extended in `tfpt_research_contracts`; Wolfram mirrors all three blocks (376/376); ledger row `GRAV.ENTROPIC.SCALARON.01`; no gate closes, no marker moves.

2026-07-13 (`v474` — the operator level of the entropic-action bridge: the D_5 Clifford/spinor structure exhibited on the carrier Fock space, the Hodge fold identified as the $5{\to}\bar{5}$ conjugation, and the `v473` Q-target *decided* — supports exactly $\{|\mathbb{Z}_2|, \operatorname{rank} E_8, 2^{g_{\rm car}}\}$, the naive pair-block reading killed; *no status moves*)

- **`v474_entropic_hodge_carrier.py` (`GRAV.ENTROPIC.HODGE.01`).** Work packages 1 and 4-algebra of the `v473` bridge, machine-checked on the finite carrier Fock space $\Lambda^\bullet\mathbb{C}^5$ ($\dim 2^{g_{\rm car}}{=}32$). **[E]** CAR$\Rightarrow$Clifford: the ten Γ's from $(a_k^\dagger, a_k)$ satisfy $\{\Gamma_a,\Gamma_b\}{=}2\delta_{ab}$ exactly (all 100 pairs, integer matrices); the 45 bilinears $[\Gamma_a,\Gamma_b]/4$ commute with $\gamma{=}(-1)^N$, span exactly $45{=}\dim\mathfrak{so}(10)$ and leak nothing from the 16-dim even subspace — the carrier half-spinor *is* the even exterior algebra *with* its spinor structure (the operator level of `v2`/`v44`/`v197`). **[E]** the Hodge–Dirac symbol: $c(\xi){=}a^\dagger(\xi){-}a(\xi)$ satisfies $c(\xi)^2{=}-|\xi|^2\,\mathrm{Id}$ for symbolic ξ — Bianconi's $D{=}d{+}\delta$ is Clifford multiplication at symbol level. **[E]** the fold is the $5{\to}\bar{5}$ conjugation: the traceless $\mathfrak{u}(5)$ weights of Λ^4 are the negatives of those of Λ^1, so her fiber $1{+}5{+}10$ becomes the GUT half-spinor $16{=}1{+}\bar{5}{+}10$ (the SM carrier decomposition). **[E]** honest obstruction: $\Omega^{\le 2}$ is *not* Clifford-invariant ($\Lambda^2{\to}\Lambda^3$ leak) — the bridge must go through the fold, never through degree truncation (a recorded constraint on the AP2 compression). **[E]** the Q-target enumeration (the `v473` **[C]** conjecture decided at the algebraic level): $\mathrm{Tr}\,Q^2{=}d\,k^2c_3^4{=}32c_3^4$ with integer k has exactly the supports

$(d,k)=(2,4),(8,2),(32,1) = \{|\mathbb{Z}_2|,\ g_{\rm car}+N_{\rm fam}=\operatorname{rank} E_8,\ 2^{g_{\rm car}}=\dim\Lambda^\bullet\}$ — minimal uniform solution $q=c_3^2=1/(64\pi^2)$ per Fock mode, bonus identity $\operatorname{Tr} Q^2/\delta_{\rm top}=32/48=2/3=\dim\Lambda^\bullet/\Omega_{\rm adm}$; the two-form block 10 (the problem-note "pair sector" conjecture) and the fiber 16 need irrational multipliers $\sqrt{16/5}$, $\sqrt{2}$ — *killed* in uniform form. **[C]** the physical choice among the three admissible supports; **[O]** unchanged (AP2 compression + the R^2 mechanism). Keybox extended in `tfpt_research_contracts` (Research Contract 2, the `v473` keybox); Wolfram mirrors the enumeration and the weight-conjugation identity (373/373); ledger row `GRAV.ENTROPIC.HODGE.01`; no gate closes, no marker moves.

2026-07-13 (`v473` — the entropic-action bridge: Bianconi's *Gravity from entropy* (Phys. Rev. D 111, 066001 (2025); `arXiv:2408.14391`) quantified as an *external candidate* for the missing action level of the parameter-free Einstein equation; her free constant pinned $\beta'_{\rm B}=c_3/6=1/(48\pi)$, the R^2 gap $3(8\pi)^9$ pre-registered as kill test; *no status moves*, the `v359` equation-of-state typing stays [O])

- **`v473_entropic_action_bridge.py` (`GRAV.ENTROPIC.ACTION.01`).** TFPT's classical field equation $G_{ab}+\Lambda g_{ab}=c_3^{-1}T_{ab}$ is derived thermodynamically (`v358`/`v359`) and honestly typed "equation of state, not a from-action quantisation" **[O]**. Bianconi's entropic action $S_{\rm B}=-\operatorname{Tr}_F\ln(\widetilde{G}\widetilde{g}^{-1})$ (the quantum relative entropy between the spacetime metric and the matter-induced metric, Araki-related via $\Delta^{1/2}=\widetilde{G}\widetilde{g}^{-1}$) is a local covariant action whose variation gives Einstein + an emergent nonnegative Λ at low coupling — an explicit *external candidate* for exactly that missing variational layer. Machine-checked: **[E]** the carrier Hodge count (her $\Omega^{0,1,2}$ fiber on the five-slot carrier $\mathbb{C}^5$ counts $1+5+10=16=2^{g_{\rm car}-1}=\dim S^+_{D_5}$, while on 4d spacetime it is only 11 — the 16 *requires* the carrier); **[E]** the one quantitative pin — matching her weak-coupling $3\beta R$ to $1/(16\pi G)=c_3/(2G)$ fixes her free constant *exactly*, $\beta'_{\rm B}=c_3/6=1/(48\pi)$; **[E]** her $\Lambda_{\mathcal{G}}=\frac{1}{2\beta}\operatorname{Tr}(\mathcal{G}-I-\ln\mathcal{G})$ is nonnegative and quadratic at the fixed point, and with $\varepsilon_\star=e^{-\alpha^{-1}}Q$ it reproduces the closed `v60` branch $(3/4\pi^2)e^{-2\alpha^{-1}}$ with the exact target $\operatorname{Tr}Q^2=32c_3^4=1/(128\pi^4)=\frac{2}{3}\delta_{\rm top}$ (**[C]** until Q is constructed); **[E]** the R^2 kill test — the raw log expansion sits *exactly* $3c_3^{-9}=3(8\pi)^9\approx 1.2\times10^{13}$ below the TFPT Starobinsky coefficient $1/(12c_3^7)$, so a direct identification is *false* without a mechanism (pre-registered); **[C]** the compression conjecture $P_\Sigma(\widetilde{G}\widetilde{g}^{-1})P_\Sigma=\Delta_\Sigma^{1/2}$ (the bridge proper) stays open, with Lorentzian positivity flagged (TFPT's OS route is the sturdier construction). New keybox in `tfpt_research_contracts` (Research Contract 2, after the `v359` keybox); residual pointers updated in `introduction/tfpt_4_frontier`/status card. Wolfram mirrors the three exact blocks (371/371); ledger row `GRAV.ENTROPIC.ACTION.01`; *no* gate closes, *no* marker moves, zero TFPT knobs added.

2026-07-03 (surfacing round — three already-computed unique readouts promoted to the public prediction surfaces: the axion–photon coefficient $g_{a\gamma\gamma}=-4c_3$, the free-seam Higgs band $m_H\approx 129$–$134\,\mathrm{GeV}$, and the never-run BH HFQPO $\times1.5$ ladder-tooth search; no new `vN`, no status moves)

- **Uniqueness scan follow-up (no new results — presentation only).** A structure scan for "what does TFPT predict that nobody else pins?" found three concrete, falsifiable readouts that were fully computed and ledgered but missing from the public prediction surfaces; they are now surfaced, with their existing typing unchanged. (i) **Axion–photon coefficient**: the determinant-line anomaly coefficient $g_{a\gamma\gamma}=-4c_3=-1/(2\pi)$, $y^2=16c_3^2=1/(4\pi^2)\approx0.0253$ (`v207`, the **[E]** leg of `GRAV.ASYMP.01`; the **[C]** caveat — coefficient, not a parameter-free $g_{a\gamma\gamma}$ in GeV^{-1} — stated verbatim) joins the axion card in `website/lib/predictions.ts` and the frontier dark-matter mirror in `papers.ts`; the TeX source (`tfpt_4_frontier`) already carried it. (ii) **Higgs free-seam band**: the double-criticality prediction $\lambda(\bar{M}_{\rm Pl})=\beta_\lambda(\bar{M}_{\rm Pl})=0\Rightarrow m_H=133.5\,\mathrm{GeV}$ (band 129–134, m_t/α_s-sensitive; measured 125.25 sits a few GeV below — the known slight

metastability, Buttazzo NNLO $\lambda(M_{\rm Pl}) = -0.0143 \pm 0.0057$, 2.5σ) becomes a prediction card ([**C**]; `v166`, `HIGGS.FREESEAM.01`, kill: settled-(m_t, α_s) RGE pull off the surface at $> 5\sigma$). (iii) **BH HFQPO ladder tooth**: the one discriminating HFQPO test — a third tooth at $\nu_3 = (3/2)\nu_u$ (661.5/414/252/363 Hz), integer harmonics forbidden — that *no published search ever targeted* although the teeth lie in the searched RXTE band (`experiments/hfqpo-ladder`, preregistered): added as an [**O**] search-target bullet to the `tfpt_horizon_readouts` "Search targets (not claims)" gapbox (firewall language verbatim: non-canonical mapping, selection null 18.5%, J1859+226 $+9.2\sigma$, GR resonance favored; even a tooth hit would be [**C**]) and as an [**O**] prediction card. Website mirrors updated in the same change (`predictions.ts` 23 → 25 cards, `papers.ts` frontier + horizon sections incl. the new search-targets mirror section); scorecard, ledger and all markers untouched.

2026-07-03 (`v472` — the determinant line over the $U(1)$-twist moduli of the collar model carries curvature = the inflow level: the `v470` bridge lemma exhibited at the finite/model level; `ALPHA.QUILLEN.EXACT.01` stays [O]. Plus: bird's-eye rounds — electron-sector/Cabibbo whitespot probes, Calderón UV-universality, parameter-free flat-budget closure (sandbox); the flat-budget closure enters the evidence scorecard)

- **`v472_quillen_detline_moduli.py` (`ALPHA.QUILLEN.DETLINE.01`).** `ALPHA.QUILLEN.EXACT.01` names the α^3 level as the curvature of the *determinant line over the $U(1)$ seam moduli* (Quillen 1985 / Bismut–Freed CMP 106 (1986) curvature; Dai–Freed JMP 35 (1994) section); `v470` computed the bulk Chern invariant only over the Bloch Brillouin zone (the translation-invariant shortcut). `v472` computes the Quillen-shaped object itself: the same collar Hamiltonian ($h(k) = \sin k_x\, \sigma_x + \sin k_y\, \sigma_y + (M - \cos k_x - \cos k_y)\, \sigma_z$, `v367`/`v470`) in real space on an $L \times L$ torus with *twisted* boundary conditions — the twist torus (θ_x, θ_y) *is* the moduli space of flat $U(1)$ connections — and the Fukui–Hatsugai–Suzuki curvature of the *determinant line of the occupied frame* over that torus: $\oint = 2\pi \cdot 1$ exactly (integer to 10^{-9}) at M=1 for L=4 AND L=6 (size-independent); controls: trivial collar M=3 → 0, orientation flip M=−1 → −1; the twist-moduli integer equals the Bloch-BZ integer for all three M (Niu–Thouless–Wu PRB 31 (1985), exhibited on the S3 collar model); the Fermi gap (= 2.0) stays open over the whole twist torus, so the Dai–Freed section has no zero. **Reading [C]:** det-line holonomy over the $U(1)$ moduli = inflow level $k_0 = |C| = 1$ — the `v470` bridge lemma "$\delta \log \det_\zeta(\text{seam}) =$ the inflow response" holds *verbatim at the finite/model level*, where log det of the occupied frame replaces $\log \det_\zeta$. **Not closed [O]:** the *continuum* ζ-determinant identification on the abstract seam (= the `SEAM.EQUIV.01` face); `ALPHA.QUILLEN.EXACT.01` stays [**O**]; α^{-1} stays [**E**]. Structurally the same manoeuvre as `v471` for `SEAM.THEOREM.01`'s replica side. 7/7 checks; suite green through `v472` (466 modules); numerical (dense eigh + FHS twist grid), Python-only by nature. Papers updated: `tfpt_1` (sixth honest step in the EM closure), `tfpt_research_contracts` (residual certification), `tfpt_safeguards` (α-form lever), root `README`, `zenodo_description.html`. Exploration original: `experiments/tfpt-discovery/quillen_detline_twist_moduli.py` (same day).
- **Bird's-eye whitespot rounds (sandbox, `experiments/tfpt-discovery/`, nothing else promoted).** (i) `electron_sector_cabibbo_probe.py`: the α metrology triangle (TFPT $\alpha^{-1} = 137.0359992168$ vs Rb-2020 $+0.99\sigma$ / CODATA22 $+1.90\sigma$ / a_e-route $+3.39\sigma$ / Cs-2018 $+6.33\sigma$ — TFPT takes the Rb side of the 5.5σ metrology split; new honest pressure point recorded); the `v204` seam-vertex universality catch (a flavor-universal reading of $a_\mu^{\rm seam} = 2.879 \times 10^{-9}$ is excluded by the electron at $\sim 2 \times 10^4 \sigma$; linear scaling at $\sim 10^2 \sigma$; quadratic survives — the [**C**] bridge must scale at least quadratically in m_ℓ); the Cabibbo-angle anomaly as a third dissolution watchdog ($\lambda_C = 0.2243762$ exact sits $+0.08\sigma$ on the PDG-2026 rescaled kaon average, between $K_{\ell 3}$ and $K_{\mu 2}$; exact unitarity puts $V_{ud} = 0.97450$ — superallowed is the outlier at $+2.6\sigma$, all neutron/pion routes agree). (ii) `kernel_calderon_uv_universality.py`: the raw-collar

Calderón/DtN datum is UV-universal (mode-by-mode first-order convergence to the exact Bessel symbol $mI'_\nu(mR)/I_\nu(mR)$, Richardson residual 3×10^{-4}; N_θ-independent; $|k|$ principal symbol; μ_4 block structure at machine precision; both spec-T masses; 3-fold mesh-defect negative control) — the operator equality demanded by `QGEO.KERNEL.01` has a well-defined discretization-independent left-hand side (gate stays **[O]**). (iii) `flat_budget_closure.py` + scorecard row: see next bullet.

- **Flat-budget closure enters the evidence scorecard (experiments-only, pattern candidate).** The zero-dial closure $\{\Omega_b = \varphi_0^{\rm ret}(1-\frac{1}{4\pi})$ [frozen], $\Omega_c = \frac{2}{7}(1-\frac{1}{4\pi})$ [the 2026-07-02 numerology-flagged Nebenbefund], $\rho_\Lambda/\bar{M}_{\rm Pl}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$ [frozen], $\Sigma m_\nu = 0.0588\,$eV [`v468` route]$\}$ + spatial flatness determines the whole flat ΛCDM budget: $H_0 = 67.15\,$km/s/Mpc (-0.39σ Planck18, -5.67σ SH0ES22 — the parameter-free Planck side of the Hubble tension), $\Omega_m = 0.3133$ (-0.27σ), $t_0 = 13.86\,$Gyr; honest stresses recorded: ω_b -2.02σ (BBN-only -0.73σ), ω_c -1.19σ, DESI-DR2 H_0 -2.16σ (one correlated low-h direction). Scorecard row `cosmo / flat budget closure` (`consistent`, composite: shares the `phi0_seed` group (and the ρ_Λ leg the α engine), Ω_c leg in `alternative_group=axion_branch` vs the axion-spine DM reading — never double-counted); nothing promoted to `verification/`.

2026-07-02 (`v471` — the Seam–Horizon replica chain exercised numerically on the discretized collar: the kernel-identification premise of `SEAM.THEOREM.01` discharged at the finite/model level; residual retypes to the continuum MMST leg + the anchor; gate stays [O]. Plus: the GW ringdown-echo channel hardened — 10 loudest events incl. GW250114, redshift-corrected templates, stacked search and a 12-variant signature battery, all null)

- **`v471_seam_horizon_replica.py` (`SEAM.EHMODEL.04`).** The residual isolated by `v150`/`v151`/`v152` — "the RP seam kernel *is* this BFK Calderón datum" — is exercised numerically on a discretized seam collar with the seam's *own* kernel and *real* replica sheets: the matched conical deficit has slope $2C(\gamma)$ on deficit cones (0.01–0.05%) and replica sheets n=2 (0.8%), n=3 (1.5%); the EH coefficient is cutoff-independent (0.5% drift) while the ζ-scale $\mu_{\rm lat}$ drifts — the `v152` anchor exhibited on data, with *no* canonical $\ln(m/\mu)$=3/4 emerging (recorded honestly); the BFK/Schur split is exact ($\sim10^{-16}$) with the Calderón factor conically clean *measured on the kernel* (pure cut-edge law, tip term 0, power-checked) and the halves carrying $2C_{\rm cone}$ via Kac doubling; the `v302` transfer masses $6\ln(3/2)$, $6\ln 3$ sit on the calibrated EH line ($\leq 0.05\%$); and the Perron/attractor mode (m=0) is IR-divergent — **the recovery gap Δ=$6\ln(3/2)$ is what makes the induced-gravity coefficient finite** (the same gap that makes the attractor unique makes $1/G$ finite). `SEAM.THEOREM.01` stays **[O]**; its residual retypes to the *continuum* scaling limit (the MMST class — the same single residual as `SEAM.EQUIV.01`, `v336`) plus the one declared anchor. 16/16 checks; suite green through `v471` (465 modules); numerical (sparse lattice determinants), Python-only by nature. Papers updated: `origin_theory` (new collar paragraph in the Seam–Horizon gapbox + residual retyping), `tfpt_research_contracts` (R3 premise), `tfpt_4_frontier` (3/4 audit note), root `README`, `verification/README`, `zenodo_description.html`. Exploration original: `experiments/theory-contracts/seam_horizon_replica.py` (17/17, same day).

- **GW ringdown-echo channel hardened (`experiments/gw-ringdown-echo`, scorecard row `data_limited`→`consistent`).** (i) Strain coverage extended to the *10 loudest* ringdown events incl. GW250114 (network SNR 78.6, O4b; new `fetch_strain_4096.py` crops the 4096 s GWOSC archive segments). (ii) **Redshift correction**: the GWTC catalogue reports source-frame masses, the observed ringdown is at $f_0/(1+z)$ — all templates moved to detector-frame $M(1+z)$ (before the fix GW190521, z=0.56, was filtered $\sim$1.6× too high); all stages re-run. (iii) Stage 1b *stacked* search over 23 detector streams: stacked p=0.262, kernel-consistent streams 0/23 — STACKED NULL. (iv) Stage 1c *signature battery* (Bonferroni ×12): ratio semantics (amplitude $(2/3)^6$ vs *energy reading* $(2/3)^3$ vs step 2/3) × μ_4 per-bounce phases $\{0,\pi/2,\pi,3\pi/2\}$

(the boundary-birefringence analogue — path birefringence from parity violation c_-=8 cancels in the inter-echo ratio; only a per-reflection phase does not) × lags 0.5–350 ms (ECO through Planckian window), with joint $(2,2,0)$+$(2,2,1)$ subtraction and a template-agnosticism control: `NO_VARIANT_ECHO` on all 10 events — the null is robust against alternative signature readings. Search-target firewall unchanged: upper bound, no detection, no tension claim.

2026-07-02 (`v295` robustness patch — external-review follow-up: the numeric a_2-kernel tolerance must dominate its own finite-difference floor)

- **`v295_flataway_a2_exact.py` (`FLATAWAY.A2.01`).** Alessandro's Git-baseline audit (commit `8cb8ad0`) hit a platform-marginal failure in check 3: the exact divided-difference a_2 kernel is compared against an ε=10^{-3} second finite difference whose *own* truncation error is $O(\varepsilon^2) \sim 10^{-6}$, yet the tolerance was 10^{-6} — on his machine k=3 landed at 1.1×10^{-6} (ours: 2.9×10^{-7}). Fix: tolerance widened to 5×10^{-6} so it *dominates* the control's error floor, with the reasoning recorded in the check text; the load-bearing anchor remains the *closed form* at machine precision (`v296`), the numeric witness stays as the independent cross-check. A numerical-robustness patch, no claim or status change; ledger row updated; suite green (ALL CHECKS PASSED through `v470`).

2026-07-02 (`v469`/`v470` — the closure-route round on the two named targets: the `SEAM.EQUIV.01` extension leg re-founded on 1995–2001 peer-reviewed crossed-product theorems + the realisation axiom reduced to invariant level; the α^3 level = the *computed* bulk Chern invariant + the seam F-normalisation = the embedding index k_Y=5/3; both targets stay [O])

- **`v469_seam_crossedproduct_route.py` (`SEAM.EQUIV.CROSSEDPRODUCT.01`).** The 128-spinor extension leg of `SEAM.EQUIV.01` no longer rests on 2024/2025 preprints: [**E**] the $SO(16)_1$ spinor has h_s=N/16=16/16=1∈$\mathbb{Z}$, so its statistics phase is +1 — *exactly* the Longo–Rehren locality criterion — and the $\mathbb{Z}_2$ simple-current crossed product $SO(16)_1 \rtimes \{1, s\}$ is a *local* index-2 extension by the peer-reviewed subfactor package (Longo–Rehren, Rev. Math. Phys. 7 (1995); the level-1 $so(N)$ net with its DHR sectors from even-CAR: Böckenhauer, Rev. Math. Phys. 8 (1996); simple-current extensions of nets: Böckenhauer–Evans, Comm. Math. Phys. 197 (1998)); KLM (Comm. Math. Phys. 219 (2001)) gives $\mu(B)$=$4/2^2$=1 ⇒ holomorphic ⇒ $(E_8)_1$ by the `v463` pin. The AGT/AMT lattice-VOA route (`v459`) is *demoted to an independent second witness.* [**E**] the index-4 μ_4 glue runs on the same integer: $h(J^k)$=$\{1,1,1\}$ for k=1,2,3 (the `v125` isotropy $q(k(1,1))$=k^2 at net level), μ=$16/4^2$=1, same endpoint. [**E**] the realisation input is reduced from model fiat to *invariants* (R1′): quasi-free [**C**] (`v155`/`v160`) + gap [**E**] (`v302`) + class D + c_-=8 [**E**] (`v456`, from P1), with the invariants *computed* — per-copy FHS Chern $|C|$=1 vs 0 (control), ν=16×$|C|$=16, c_-=$\nu/2$=8 — and $\nu \equiv 0$ (mod 16) is exactly Kitaev's class-D phase whose edge admits a purely bosonic description = the $(E_8)_1$ level-1 state (Ann. Phys. 321 (2006), sixteen-fold way); the phase-invariance of c_- is cited (Kapustin–Spodyneiko, PRB 101 (2020); Kim et al., PRL 129 (2022)). [**O**] *not closed*: the cited theorems stay cited; `SEAM.EQUIV.01` stays [**O**].
- **Lean hardening.** `TfptCarrier/SeamResidualAxiom.lean` gains the *parallel* peer-reviewed derivation `seamResidualClosed′`: the invariant-level axiom `collar_invariants` (R1′) + *one* combined 1995–2001 package `crossedproduct_route_theorem`; the new joints are kernel facts with *no* axioms — `lr_locality_integer` (h_s=16/16∈$\mathbb{Z}$), `glue_locality_integers` (μ_4 $h(J^k)$∈$\mathbb{Z}$), `klm_holomorphic` ($4/2^2$=1, $16/4^2$=1), `sixteenfold_pin` (ν=16≡0 mod 16, c_-=8), bundled as `crossedproduct_arithmetic` (`[propext]` only); `#print axioms seamResidualClosed′` = exactly four; the original `seamResidualClosed` (AGT/AMT witness) is kept unchanged as the independent second route; `lake build` OK (3285 jobs).
- **`v470_alpha_inflow_level.py` (`ALPHA.QUILLEN.INFLOW.01`).** Both residuals left

on `ALPHA.QUILLEN.EXACT.01` after `v433`–`v435` are upgraded. [**E**] target re-verified (α^{-1}=137.0359992168; Quillen split $<10^{-35}$). [**C**] *the level is computed, not minimal*: the a^3 coefficient k_0=1 *equals* the FHS bulk Chern invariant $|C|$=1 of the *same* p+ip collar that realises S3 (`v367`/`v461`) — "why exactly 1" is answered by computation plus the cited inflow identification (TKNN PRL 49 (1982) / Avron–Seiler–Simon PRL 51 (1983) quantisation; Callan–Harvey, Nucl. Phys. B250 (1985) inflow; APS 1975 / Alvarez-Gaumé–Della Pietra–Moore 1985 / Witten, Rev. Mod. Phys. 88 (2016) η=CS part of $\delta \log \det$; Quillen 1985 / Bismut–Freed 1986 determinant-line integrality), *replacing* `v435`'s minimality assumption. [**E**] $k_Y = \mathrm{tr}(Y^2)/\,\mathrm{tr}(T_3^2)$=(5/6)/(1/2)=5/3 (Ginsparg, Phys. Lett. B197 (1987)): $(3/5)\times\frac{41}{6}=\frac{41}{10}=b_1$ — the "GUT 3/5" *is* the inverse embedding index; $\frac{41}{6}=\frac{20}{3}+\frac{1}{6}$ (carrier decomposition). [**C**] the seam F-normalisation (the one [**C**] of `v434`) is *retyped*: once the seam net is $(E_8)_1$ level 1, $\langle JJ\rangle = k/(z-w)^2$ is fixed by k_Y — level-1 current-algebra rigidity, zero independent content, a face of `SEAM.EQUIV.01` (per the `v382` typing). [**E**] *structural unification*: one invertible phase, two quantised responses — c_-=8 (gravitational, feeds `SEAM.EQUIV.01`/S3) and C=1 ($U(1)$, feeds EM1) — so closing R1′ feeds *both* named targets. [**O**] *not closed*: the bridge lemma "$\delta \log \det_\zeta$(seam)= the inflow response" stays the named cited step; `ALPHA.QUILLEN.EXACT.01` stays [**O**]; α^{-1} stays [**E**].

- **Integration.** Both registered in `run_all.py` + `script_registry.csv` (cluster `frontier`); ledger rows `SEAM.EQUIV.CROSSEDPRODUCT.01` + `ALPHA.QUILLEN.INFLOW.01`; `FORM.SEAM.RESIDUAL.01` updated (the `seamResidualClosed′` route); the exact joints Wolfram-mirrored (365 → 368: the LR/glue locality integers, the KLM μ-arithmetic, the sixteen-fold pin, the k_Y=5/3/b_1 identities); cited in `tfpt_1` (α fixed point + EM-Ward residual prose), `tfpt_research_contracts` (seam residual + Lean section), `introduction` (closure ledger), `tfpt_5_redteam` (Target A residual wording) and the README/website status surfaces; exploration provenance `experiments/tfpt-discovery/closure_route_*.py` (2026-07-02 note in `experiments/next.txt`). Suite ALL CHECKS PASSED through `v470`. Curated version bumped to **v5.4** (Zenodo deposit updated).

2026-07-02 (`v467`/`v468` — two named post-hoc [O] flavor candidates, record unchanged: the three $\sim 2\sigma$ mixing tensions as *one* third-generation $-\varphi_0$ pattern, and the splitting ratio $\Delta m^2_{21}/\Delta m^2_{31} = |J_{\mathrm{PMNS}}|$)

- **`v467_thirdgen_mixing_pattern.py` (`FLAV.THIRDGEN.PATTERN.01`).** The suite's three $\sim 2\sigma$ mixing tensions ($\sin^2\theta_{13}$ $+2.0\sigma$, $\sin^2\theta_{23}$ $+1.76\sigma$, $|V_{cb}|$ $+1.79\sigma$; `v307`) are exactly the three *third-generation* channels and share *one* relative deviation $-5.36\% \pm 1.67\%$ — non-zero at 3.2σ, -0.03σ from $-\varphi_0$. [**E**] vocabulary identities: $\lambda_C^2 = \varphi_0(1-\varphi_0)$ (the frozen Cabibbo); $\sin^2\theta_{23} = (1-\varphi_0)/2 \Leftrightarrow \cos 2\theta_{23} = \varphi_0$; the suppression size φ_0=5.3% sits inside the seam-gapped band $(2/3)^6$=8.8% (`v393`). [**N**] the ladder-base forms $\lambda_C^2 e^{-5/6}$=0.021880 / $(1-\varphi_0)/2$=0.473414 / $\lambda_C^2/(1+\lambda_C)$=0.041119 drop the joint four-observable χ^2 from 10.4 to 0.11 with *zero* new parameters; $|V_{ub}|$ (no bare φ_0) and the 1–2 channels stay untouched and correct (dressing λ_C would fail at -24σ — scope checked, not chosen); Daya Bay alone already prefers the pattern form ($\Delta\chi^2$=4.3). [**C**] **honest:** post-hoc (5/16 equally-simple dressings fix the two PMNS tensions; $(1-\varphi_0)/(1-c_3)$/resolvent degenerate $<1.4\%$). [**O**] no seam-transport derivation (the Bernoulli-variance reading $\lambda_Y^2 = \mathrm{Var}[\mathrm{Bernoulli}(\varphi_0)]$ is an exact identity but only an interpretation) — the frozen record (`v84`/`REG.FREEZE.01`) is *unchanged*. [**X**] pre-registered, symmetric: θ_{13} stable at 0.0231 kills the pattern, at $\sim$0.0219 (sub-1%) kills the record's bare-φ_0 reading, exclusive $|V_{cb}|\sim$0.0395 kills both — JUNO-era reactors + Belle II decide. Python-only.

- **`v468_dm2_ratio_jarlskog.py` (`FLAV.DM2RATIO.01`).** The splitting ratio $\Delta m^2_{21}/\Delta m^2_{31}$ was previously *unpredicted*; the frozen PMNS matrix (`v9`/`v268`/`v270`, $\delta = 4\pi/3$) has one canonical derived CP invariant, and [**N**] $|J_{\mathrm{PMNS}}|$=0.029653 meets the NuFIT 6.0 measured ratio 0.029805 ± 0.000792 at -0.19σ (equivalently $m_2/m_3 = \sqrt{|J|}$=0.17220 vs 0.17264(229)) — zero

dials. [**N**] the informal `tfpt_2` table heuristic $m_2/m_3 \sim \pi\varphi_0 = 0.16704$ (a "$\sim$" entry, never frozen) is now -2.44σ and is *superseded as the comparator*. [**C**] spectrum consistency: $\Sigma m_\nu = m_3(1+\sqrt{|J|}) = 0.0588$ eV reproduces the documented 0.0586 eV (`v272`); no absolute number is added. [**O**] no mechanism (the $\mu\tau$ texture gives $\theta_{13}=0$ at leading order). [**X**] JUNO shrinks the window $\sim 4\times$; adopting the `v467` pattern would shift $|J| \to 0.028849$ (-1.21σ today) — the two candidates *cross-discriminate* at JUNO precision. Python-only.

- **Integration.** Both registered in `run_all.py` and `script_registry.csv` (cluster `neutrinos`); ledger rows `FLAV.THIRDGEN.PATTERN.01` and `FLAV.DM2RATIO.01` (both [**O**] candidates); cited in `tfpt_2` (§ Neutrinos: pressure-point block + new candidate paragraphs, the m_2/m_3 table row retyped to the $|J|$ comparator, the octant prose kept honest — record does not select, the candidate would), `introduction` (closure ledger), `tfpt_4_frontier` (live kill-test board) and `tfpt_5_redteam` (seed-hyperplane caption); `v328`'s "single outlier" wording scoped to the seed observables; exploration provenance `experiments/tfpt-discovery/pattern_hunt_pmns_dressing.py` (2026-07-02 note in `experiments/next.txt`); Python-only (flagged in the Wolfram README).

2026-07-01 (`v466` — a sixth seed channel from a new measurement sector: the charged-lepton mass ratio $m_e/m_\mu = \frac{12}{7}\varphi_0^2$ back-solves the same axiom seed to -0.11%, extending the empirical over-determination to *five* sectors)

- **`v466_seed_leptonmass_channel.py` (`SEED.LEPTONMASS.01`).** The seed over-determination of `v306`/`v465` used five observables spanning three measurement sectors (neutrino mixing, CMB, quark mixing). This module adds the *charged-lepton mass* sector: the ratio $m_e/m_\mu = \frac{12}{7}\varphi_0^2$ is a sixth independent readout of the *one* axiom seed $\varphi_0 = 1/(6\pi) + 48c_3^4$. [**E**] plumbing: the axiom φ_0 reproduces the frozen $m_e/m_\mu = \frac{12}{7}\varphi_0^2 = 0.00484673$ to 10^{-9}. [**N**] **new sector (headline):** back-solving φ_0 from the measured *pole* ratio $m_e/m_\mu = 0.004836$ (PDG 2024) gives $\varphi_0 = 0.053113$, only -0.11% from the axiom — a fourth measurement sector beyond {neutrino mixing, CMB, quark mixing}. [**C**] **transport caveat:** the formula is the *source* ratio and the datum the *pole* ratio; the source→pole shift is the seam-gapped correction class (bound $(2/3)^6 = 8.78\%$, `v393`) but largely cancels for m_e/m_μ, so the 0.11% residual sits far inside the band — the channel is added as a *consistent new sector*, not a sharper pin. [**N**] **extends:** the six-channel cluster (`v306`'s five + this) has $\chi^2/\mathrm{dof} < 1$ and the error-weighted mean stays at the axiom to $<0.1\%$; with the corroborating heavy-quark m_t/m_b (-0.23%) it is a seven-channel, five-sector agreement. [**N**] **power:** a data↔formula shuffle explodes the cluster χ^2 by $>100\times$, and the scheme-ambiguous $m_c/m_s = (34/47)/\varphi_0$ (FLAG 11–14) is deliberately not counted. [**C**] **honest:** theoretically still the *one* seed (`v305` multiplicity 1) — an *empirical* strengthening, not theoretical independence; φ_0 is fixed by the axioms, so it upgrades *no* prediction and closes *no* gate. Python-only (floats).

- **Integration.** Registered in `run_all.py` and `script_registry.csv` (cluster `registry`); ledger row `SEED.LEPTONMASS.01`; cited in `tfpt_5_redteam` (the seed-cross-validation firewall winbox, alongside `v305`/`v306`/`v465`); Python-only (flagged in the Wolfram README).

2026-07-01 (`v465` — the cross-sector one-parameter seed test: a CMB birefringence angle and a quark Cabibbo angle point to the same seed within 0.01σ, the θ_{13}-independent externally-runnable slice of `v306`)

- **`v465_seed_crosssector_joint.py` (`SEED.CROSSSECTOR.01`).** The sharpest falsifiable signature of a *shared-origin* theory is a correlation no non-shared-origin model predicts. In the Standard Model the cosmic-birefringence rotation β (a CMB EB/TB polarisation observable) and the Cabibbo angle $\lambda_C = |V_{us}|$ (a quark-flavour observable) are unrelated; TFPT reads both as functions of the *one* axiom seed $\varphi_0 = 1/(6\pi) + 48c_3^4$. [**E**] plumbing: the axiom φ_0 reproduces the frozen λ_C, $\sin^2\theta_{12}$, β to 10^{-9}. [**N**] **cross-sector (headline):** back-solving φ_0

independently from β (CMB; ACT DR6 + Planck PR4 combined) and from λ_C (quark; PDG 2024) gives the same seed within $0.01\sigma_{\rm comb}$. **[N] one-parameter fit:** the error-weighted mean of the three back-solved φ_0 reproduces the axiom to 0.06%; the joint χ^2 at the axiom (zero free parameters, 3 dof) is 0.007, $\chi^2/\mathrm{dof}<1$, combined pull 0.08σ. **[N]** θ_{13} **exclusion declared:** adding $\sin^2\theta_{13}$ raises χ^2 by $\sim$4 (the documented $\sim 2\sigma$ pressure point, `v328`/`v338`), so this slice is the honest θ_{13}-independent statement, not a hidden dilution. **[N] power:** a data$\leftrightarrow$formula shuffle explodes the cluster χ^2 by $>5\times10^4$. A consistency / out-of-sample statistic (a subset of `v306`); φ_0 is fixed by the axioms, so it upgrades *no* prediction and closes *no* gate. Python-only (floats).

- **Integration.** Registered in `run_all.py` and `script_registry.csv` (cluster `registry`); ledger row `SEED.CROSSSECTOR.01`; cited in `tfpt_5_redteam` (the seed-cross-validation firewall winbox, alongside `v305`/`v306`) and in the `origin_theory` birefringence dependency chain; Python-only (flagged in the Wolfram README).

2026-06-30 (`v463`/`v464` + Lean + Wolfram — the c=8 holomorphic-uniqueness pin and the one-particle realisation rigidity: the `SEAM.EQUIV.01` residual reduced to *entirely certification* on both halves)

- **The bird's-eye reframing.** The recent G-block modules all sharpened the *target* of the scaling limit; this round attacks the two halves that actually remain. The *continuum existence* is already citable (`v458` MMST + `v459` AGT/AMT); what was implicit was (B) *why* a c=8 holomorphic limit must be $(E_8)_1$, and (A) the *realisation* input R_1 that the abstract seam IS the concrete quasi-free collar. `v463` closes B; `v464` reduces A to its one-particle data.

- **`v463_seam_c8_holomorphic_uniqueness.py` (`SEAM.EQUIV.UNIQ.01`).** The c=8 holomorphic-uniqueness pin. **[E]** c=8 is NOT unique at level 1: $c=\dim\mathfrak{g}/(1+h^\vee)$=8 has THREE simple solutions A_8 ($\dim 80$), $D_8=\mathfrak{so}(16)$ ($\dim 120$), E_8 ($\dim 248$), so the $\det K$=4 $SO(16)$ rival is a genuine same-c competitor. **[E]** HOLOMORPHY FORCES $\dim V_1$=248: the only weight-0 form with leading $q^{-1/3}$ and integer coefficients is $j^{1/3}=E_4/\eta^8$, q^1 coefficient 248 (the 1-dim candidate space, vacuum a_0=1). **[E]** $\dim 248$+ level-1 c=8 $\Rightarrow E_8$ uniquely (248/(1+30)=8); tower $\{1, 248, 4124, 34752, 213126\}=E_4/\eta^8=j^{1/3}$. **[C]** Dong–Mason/Schellekens: the holomorphic c=8 VOA is unique = V_{E_8} (even unimodular rank-8 lattice unique = E_8, Mordell–Witt) — so the identification half is classification-forced.

- **`v464_seam_oneparticle_rigidity.py` (`SEAM.EQUIV.ONEPARTICLE.01`).** The one-particle rigidity attacking R_1. The seam is quasi-free (`v113`/`v160`/`v161`), so by Araki self-dual CAR the net is the second quantisation of its one-particle symbol P. **[E]** RIGIDITY: the gapped ground state is the UNIQUE quasi-free state with $P=$ projection onto $K<0$ ($P^2=P$ to $<10^{-12}$), K fixed by gap + chirality + reflection (`v426`). **[E]** ONE-PARTICLE LIMIT EXISTS: the kernel is Cauchy on a fixed window ($\max|P_{2N}-P_N|\sim 1/N\to 0$). **[E]** the limit is c=1 Dirac: entanglement $S(L)=(c/3)\ln L$ gives $c\to1$, 16 copies $\Rightarrow c_-$=8 (ties `v450`). **[C]** Shale–Stinespring makes the Bogoliubov map implementable, so R_1 is upgraded from a bald axiom to "the unique quasi-free realisation, modulo the named CAR functor". Python-only (numpy).

- **Lean (`SeamResidualAxiom.lean`).** Adds the `v463` kernel facts `c8_three_candidates` ($A_8/D_8/E_8$ each $\dim=8(1+h^\vee)$) and `holomorphy_selects_e8` (forced $\dim V_1=248=E_8\neq 80, 120$) to `residual_arithmetic` (still `[propext]`-only); `v464` is numerical. `#print axioms seamResidualClosed` still lists exactly FOUR axioms.

- **Wolfram + integration.** `v463` mirrored (the three candidates, the forced 248, the E_4/η^8 tower); the extension count is now 365/365. Integrated across the contracts paper (attacks (xxx)/(xxxi)), README, zenodo, ledger and website. Net effect: across (v)–(xxxi) the `SEAM.EQUIV.01` residual is now *entirely certification* — a named, hypothesis-audited package

(MMST, AGT/AMT, Dong–Mason, Araki, Shale–Stinespring) with no open internal mechanism — yet `SEAM.EQUIV.01` stays **[O]** because it still rests on cited continuum-existence theorems we do not re-prove.

2026-06-30 (`v461`/`v462` + Lean axiom merge — the strict-locality obstruction made explicit, the 128-spinor extension exhibited at character level, and the `SEAM.EQUIV.01` residual reduced to ONE realisation axiom + ONE combined cited theorem)

- **`v461_seam_strict_locality.py` (`SEAM.S3.LOCALITY.01`).** Sharpens `v460`'s verdict from a *cited* statement to an *exhibited* topological obstruction on the same `v367` $p{+}ip$ collar. **[E]** CHERN INTEGER: the FHS Chern is $C{=}{+}1$ (chiral $M{=}1$), 0 (trivial $M{=}3$), -1 (opposite mass). **[E]** WILSON-LOOP / WANNIER-CENTRE WINDING = CHERN: the hybrid Wannier centre $\theta(k_y)$ winds by $|C|{=}1$ (chiral) and 0 (trivial) over the BZ; exponentially-localised Wannier exist IFF the winding is 0 (Brouder–Panati et al., PRL 98 046402; Thouless), so the chiral collar has NONE. **[E]** OBSTRUCTION INTEGERS TIED: winding $= |C|{=}1{\neq}0$ and $c_-{=}8{=}g_{\rm car}{+}N_{\rm fam}{=}\operatorname{rank} E_8{\neq}0$ (the same 8 as c_3's one-sided count, `v456`). **[E]** NEG CONTROL: BOTH phases are gapped, yet only the trivial (winding 0) admits exp-localised Wannier and a strict finite-range commuting projector — it is the CHIRALITY, not the gap. **[C][O]** Kapustin–Fidkowski (`arXiv:1810.07756`): a strictly finite-range commuting projector would force winding 0, $C{=}0$; since $C{=}1{\neq}0$ NONE exists, so `v424` sub-step (i)'s realisation is PROVABLY the quasi-local NPW26 LTO net (strict locality topologically forbidden, not a missing premise). `SEAM.EQUIV.01` stays **[O]**. Python-only (numpy); the winding/Chern integers Wolfram- and Lean-mirrored.
- **`v462_seam_spinor_continuum.py` (`SEAM.EQUIV.SPINOR.01`).** The constructive companion of the `v458` (MMST audit) / `v459` (lattice-VOA) residual: the 128-spinor extension $SO(16)_1{\to}(E_8)_1$ exhibited three ways. **[E]** $(E_8)_1$ TOWER FROM $SO(16)_1$ SECTORS: the exact Jacobi/E_8 identity $\theta_2^8{+}\theta_3^8{+}\theta_4^8{=}2E_4$ gives $\chi_{(E_8)_1}{=}E_4/\eta^8{=}\chi_o^{SO16}{+}\chi_s^{SO16}$ — the holomorphic $(E_8)_1$ character IS the $SO(16)_1$ vacuum + spinor sector sum. **[E]** THE 128 SPINOR WEIGHT-1 STATES FILL 120$\to$248: χ_o gives 120, χ_s gives 128, summing to the exact tower $\{1, 248, 4124, 34752\}$ (248=120+128). **[E]** FINITE 16-MAJORANA FOCK: $C(16,2){=}120$ NS bilinear currents and $2^{16/2}{=}256$ Ramond ground states = 128 spinor + 128 cospinor. **[E]** FINITE-L $\to$ CONTINUUM CONVERGENCE: the critical free-fermion ring Casimir gives $c_-{\to}8$ and the lowest scaled NS gap $x_1{\to}1$ with $|x_1{-}1|{\sim}1/L^2$ ($L{=}64..1024$). **[C][O]** AGT/AMT (`v459`) supply the rigorous extension; `SEAM.EQUIV.01` existence (`v336`) stays **[O]**. Mixed exact (θ identity + tower + 120/128/256, Wolfram-mirrored) + numerical (finite-L, Python-only).
- **Lean (`TfptCarrier/SeamResidualAxiom.lean`, `FORM.SEAM.RESIDUAL.01`).** The two former cited axioms `mmst_existence` ∘ `agt_lattice_extension` are MERGED into ONE combined external axiom `seam_realisation_theorem` (dropping the intermediate `ChiralCFT_c8`), so `#print axioms seamResidualClosed` now lists exactly FOUR axioms (down from six). Two more kernel facts are added axiom-free: `strict_locality_forbidden` (`v461`: winding $= |C|{=}1{\neq}0$, $c_-{\neq}0$) and `fock_sectors` (`v462`: $C(16,2){=}120$, $2^8{=}256$, spinor 128, 120+128=248); `residual_arithmetic` (now including both) depends only on `[propext]`. `lake build` OK. So the residual is reduced to ONE realisation axiom + ONE combined cited theorem.
- **Wolfram.** `tfpt_readouts_extension.wl` adds 3 exact mirrors (total 363/363): the `v461` obstruction integer-logic and the `v462` θ-identity + 120+128=248 sector / Fock counts.

2026-06-30 (`v460` — the explicit flat-band commuting-projector (LTO) realisation of the S3 input: advancing the `SEAM.EQUIV.01` sub-step (i) from abstract existence to an explicit gap-protected quasi-local projector net carrying the $\mathbb{Z}_2$ reflection and the μ_4 clock)

- **`v460_seam_s3_lto.py` (`SEAM.S3.LTO.01`).** NPW26 (`arXiv:2605.10693`) prove the intrinsic Bisognano–Wichmann lemma for *commuting-projector* (local-topological-order) models with a $\mathbb{Z}_2$ reflection; `v424` reduced the keystone BW residual to that theorem modulo two open sub-steps, of which (ii) was settled by `v426` and the continuum lift of BW by the uniform-in-N tower `v455`. This module attacks the *commuting-projector* side of sub-step (i) on the same explicit `v367` $p{+}ip$ collar. **[E]** FLAT-BAND PARENT: the flattened Bloch $Q{=}h/|h|$ has $Q^2{=}I$ (flat ±1 bands) and $P{=}(I{-}Q)/2$ is an exact projector EQUAL to `v367`'s occupied ground-state projector — a frustration-free commuting-projector parent with the same gapped ground state (machine precision over a k-grid). **[E]** GAP-PROTECTED QUASI-LOCALITY: the gapped projector kernel $|P(n,0)|$ decays rapidly ($\xi{\approx}1.14$ at $M{=}1$, exp $R^2{=}0.995$, $|P(20,0)|{<}10^{-6}$) and ξ grows as the gap closes (2.7 at gap 0.02); at the gapless point it decays as a clean power law $|R|^{-2.1}$ ($R^2{=}0.999$) — locality needs the gap. **[E]** $\mathbb{Z}_2$ REFLECTION + μ_4 INHERITED: the flattened modular Hamiltonian $K_{\rm flat}{=}\,\mathrm{sign}(K)$ inherits $\Theta K_{\rm flat}\Theta{=}{-}K_{\rm flat}$ ($u_\Theta{=}J$) and $[\rho, K_{\rm flat}]{=}0$ EXACTLY (sign is odd; ρ commutes with any $f(K)$), with side-even and off-μ_4 neg controls firing. **[C][O]** the chiral $c_-{=}8{\neq}0$ (FHS Chern $|C|{=}1$) obstructs STRICT finite-range commuting-projector locality (Kapustin–Fidkowski, `arXiv:1810.07756`; $C{\neq}0$ forbids exponentially-localised Wannier), so the realisation is the quasi-local LTO net NPW26 use — the form sub-step (i) asks for. Advances `v424` sub-step (i); `SEAM.EQUIV.01` stays **[O]** (the continuum theorem MMST `v336` is the backbone). Python-only (numpy); registered in `run_all.py`, the registry, the ledger and cited in `tfpt_research_contracts` (route (xxvii) of the convergent attacks).

2026-06-30 (`v454`–`v459` + Lean + Wolfram — the post-F "G-block": the lattice Sugawara current algebra, a uniform-in-N Tomita–Takesaki tower, chirality forced by P_1, an $(E_8)_1$ character kill test, the exact MMST citation audit and the complementary lattice-VOA route)

- **`v454_seam_edge_virasoro.py` (`SEAM.EQUIV.EDGE.VIRASORO.01`).** Tests the cited MMST "Koo–Saleur lattice generators → continuum Virasoro with the correct c" fact on our own free-fermion edge (G2). **[E]** the Affleck–Cardy Casimir energy fits $c_{\rm Dirac}{=}1.000$ ($\frac{1}{2}$ per Majorana, $L{=}64..1024$); **[E]** the lattice $U(1)$ current closes its affine level exactly ($\langle 0|J_nJ_{-n}|0\rangle{=}n$, $n{=}1..8$; a gapped control vanishes); **[E]** the level-1 Sugawara $c{=}\dim G/(1{+}h^\vee){=}8$ for *both* $SO(16)_1$ (120/15) and $(E_8)_1$ (248/31). **[C]** $16{\cdot}\frac{1}{2}{=}8{=}g_{\rm car}{+}N_{\rm fam}$ is the level-1 current-algebra c_-; `SEAM.EQUIV.01` stays **[O]**. Sugawara Wolfram-mirrored.

- **`v455_seam_bw_uniform.py` (`SEAM.EQUIV.BW.UNIFORM.01`).** A uniform-in-N Tomita–Takesaki tower for the intrinsic Bisognano–Wichmann condition $u_\Theta{=}J$ — the inductive-limit companion to `v426`, advancing `v424` sub-step (i) (G1). **[E]** $\|\Theta K\Theta{+}K\|{=}0$ exactly at every N ($N{=}8..128$); **[E]** an N-independent gap $6\log\frac{3}{2}$; **[E]** $[\rho, K]{=}0$ exactly at every N; **[E]** an N-independent neg-control margin $2\|K\|$. **[C][O]** the inductive limit inherits $u_\Theta{=}J$; `SEAM.EQUIV.01` stays **[O]**. Python-only.

- **`v456_seam_chirality_from_c3.py` (`SEAM.S3.FROM-P1.01`).** The edge chirality ($c_-{\neq}0$, the S3 phase) is *forced* by the one-sidedness that defines $c_3{=}1/(8\pi)$ (P1) (G6). **[E]** c_3's "8" is the one-sided count $8{=}|\mathbb{Z}_2|\cdot(\int K/\pi){=}2\cdot 4$; **[E]** a reflection sends the Chern integer $C{\to}{-}C$ ($+1{\to}{-}1$ on the $p{+}ip$ model); **[E]** a two-sided boundary forces $C{=}{-}C{=}0$. **[C]** the one-sided seam (no reflection) forces $C{\neq}0$, magnitude $c_-{=}8{=}\,\mathrm{rank}\,E_8$ — S3 is a consequence of P_1; `SEAM.EQUIV.01` existence stays **[O]**. The arithmetic is Wolfram-mirrored.

- **`v457_seam_character_killtest.py` (`SEAM.EQUIV.KILLTEST.01`).** An explicit charac-

ter/sector kill test of $(E_8)_1$ against $SO(16)_1$ (G7). **[E]** the $(E_8)_1$ character $\chi{=}E_4/\eta^8$ with tower degeneracies $\{1, 248, 4124, 34752, \dots\}$; **[E]** the weight-1 count 248=120+128 (not the free 120); **[E]** the sector count $|\det \text{Cartan}\, E_8|{=}1$ (not $|\det \text{Cartan}\, D_8|{=}4$). **[X]** a measured 120, four sectors, or a wrong coefficient would *falsify* $(E_8)_1$; the data give $248/1/\{1, 248, 4124\}$ (passed). `SEAM.EQUIV.01` stays **[O]**. Wolfram-mirrored.

- **`v458_seam_mmst_citation_audit.py` (`SEAM.EQUIV.MMST.AUDIT.01`).** The exact hypothesis-by-hypothesis citation audit of MMST (`arXiv:2107.13834`) against the collar (G3, the "exakter Zitat-Abgleich"). **[C]** H1 CAR class, H2 massless limit, **[E]** H3 per-copy $c{=}\frac{1}{2}$ (Thm 4.11), **[C]** H4 decoupled additivity to $c{=}8$, H5 the WZW range $8{\le}8{\le}16$ with the 120 bilinear currents all match; **[O]** the single residual H6 is the 128-spinor extension $SO(16)_1{\to}(E_8)_1$ ($\det K\, 4{\to}1$), outside MMST's bilinear method. `SEAM.EQUIV.01` stays **[O]**, its open part narrowed to this one holomorphic extension.

- **`v459_seam_lattice_voa_route.py` (`SEAM.EQUIV.LATTICEVOA.01`).** The second, independent route to $(E_8)_1$ supplying exactly the residual of `v458` (G5). **[E]** E_8 even unimodular ($\det{=}1$); **[E]** the weight-1 space 248=8+240; **[E]** the 240 roots split 112 integer + 128 half-integer (the spinor, 248=120+128). **[C]** the lattice net A_{E_8} (Adamo–Giorgetti–Tanimoto, `arXiv:2506.01008`; OS `arXiv:2407.18222`) constructs the holomorphic extension; **[O]** the two routes leave one shared realisation input (collar = A_{E_8}, tied to P_1 by `v456`). `SEAM.EQUIV.01` stays **[O]**. Wolfram-mirrored.

- **Lean (`FORM.SEAM.RESIDUAL.01`, `TfptCarrier/SeamResidualAxiom.lean`).** Kernel-checks the whole G-block arithmetic (the one-sided $8{=}|\mathbb{Z}_2|\int K/\pi$, the reflection $C{\to}{-}C$, $c_{-}{=}8$, the MMST range $8{\le}8{\le}16$, decoupled additivity, 248=8+240=120+128, 240=112+128, $\det K$ 1 vs 4) by `decide`, reducing the `SEAM.EQUIV.01` residual to ONE named TFPT-internal realisation axiom (`CollarRealizedAsLatticePhase`) plus two cited external theorems (MMST `v458`, AGT `v459`); `lake build` OK.

- **Wolfram + integration.** Six exact mirrors added to `tfpt_readouts_extension.wl` (Sugawara $c{=}8$ for 120/15 and 248/31; the one-sided $8{=}2\cdot 4$ and reflection $C{=}{-}C \Rightarrow 0$; the $(E_8)_1$ tower $\{1, 248, 4124, 34752\}$ with 248=120+128 and $|\det \text{Cartan}|$ 1 vs 4; the root split 112+128 and 248=8+240) — extension total now 360/360. Registry, ledger, master index, website mirror and manifests updated in the same change.

2026-06-30 (`v449`–`v453` + Lean + Wolfram — the edge CFT named end-to-end: uniform-in-N convergence, three independent reads of $c_{-}{=}8$, the Ising operator tower, the $(E_8)_1$ modular data, and μ_4-symmetry derived from the four marks)

- **`v449_seam_mmst_uniform.py` (`SEAM.EQUIV.MMST.UNIFORM.01`).** Strengthens the cited MMST hypothesis from convergence at a single cross-ratio (`v444`/`v448`) to the *uniformity* the theorem actually asserts, on the same `v367`/`v368` collar. **[E]** cross-ratio A $Q(1,2,4,7)/12 \to 2{+}\sqrt{3}$, the deviation shrinking monotonically; **[E]** an *independent* cross-ratio B $Q(1,3,5,9)/12 \to$ its own value 2, so convergence is not a tuned point; **[E]** a uniform $1/N$ rate $|Q(N){-}Q_\infty|\cdot N \le 4$ for *both* observables at every $N_x \in \{36, 48, 60\}$ (one N-independent constant); **[E]** the bulk control is off-scale. **[C]** the MMST uniform-convergence hypothesis is empirically satisfied; `SEAM.EQUIV.01` stays **[O]**.

- **`v450_seam_edge_entanglement.py` (`SEAM.EQUIV.EDGE.ENTANGLEMENT.01`).** A *third* independent reading of the edge central charge (after the correlator `v444` and the topology `v447`). **[E]** the Calabrese–Cardy entanglement law gives $c_{\text{Dirac}}{=}1.000$ for the critical free-fermion chain; **[E]** so c per Majorana = 1/2; **[E]** a gapped control saturates ($c{=}0$). **[C]** bulk–edge: $c_{-}{=}N_{\text{Maj}}\cdot\frac{1}{2}{=}16\cdot\frac{1}{2}{=}8{=}g_{\text{car}}{+}N_{\text{fam}}$. `SEAM.EQUIV.01` stays **[O]**.

- **`v451_seam_edge_cardy_tower.py` (`SEAM.EQUIV.EDGE.CARDY.01`).** Names the edge CFT by its *operator content*. **[E]** the Cardy finite-size spectrum gives the spin dimension $h_\sigma{=}1/16$ (the

periodic–antiperiodic ground-energy split) and [**E**] the energy dimension h_ϵ=1/2 (the lowest NS pair excitation), both [**E**] L-independent (conformal $1/L$ scaling). [**C**] $\{c, h_\sigma, h_\epsilon\}$=$\{\frac{1}{2}, \frac{1}{16}, \frac{1}{2}\}$ is the chiral Ising (free-Majorana) minimal model $M(3,4)$; 16 copies give c_-=8. `SEAM.EQUIV.01` stays [**O**].

- **`v452_seam_e8_modular.py` (`SEAM.EQUIV.E8.MODULAR.01`).** The $(E_8)_1$ torus modular signature, extending `v156`'s q-expansion to the transformation data. [**E**] the character χ=E_4/η^8 is S-invariant $\chi(-1/\tau)$=$\chi(\tau)$ (one primary, S=[1]); [**E**] T-phase $\chi(\tau{+}1)/\chi(\tau)$=$e^{-2\pi i/3}$=$e^{-2\pi i c/24}$ (c=8); [**E**] leading power $\chi\, q^{1/3} \to 1$ so $q^{-c/24}$=$q^{-1/3}$ $\Rightarrow$ c=8; [**E**] holomorphic c=8 $\equiv$ 0 mod 8 with T-phase order $24/\gcd(8,24)$=3. [**C**] the $(E_8)_1$ identity is now pinned on the torus *and* the edge. `SEAM.EQUIV.01` stays [**O**].

- **`v453_seam_mu4_from_marks.py` (`SEAM.RIGIDITY.MU4FROMMARKS.01`).** Folds the last structural premise of the rigidity chain (`v446`'s residual `QGEO.SYM.01`) into the existing realisation premise. [**E**] the four Gauss–Bonnet marks *are* μ_4= roots of $z^4{-}1$; [**E**] the order-4 clock $z \to iz$ is a 4-cycle permuting them (the `v445` clock); [**E**] it preserves the cross-ratio λ=2 (in the D_4 stabiliser, `v168`); [**E**] the form basis ω_k=$z^{k-1}dz/(z^4{-}1)$ $\to$ $i^k\omega_k$ is μ_4-graded (weights $(1,2,3)$=A_3 exponents). [**C**] so `QGEO.SYM.01` follows from marks= μ_4 + the existing `QGEO.REALIZE.01` — no new premise; the chain (`v453`$\to$`v446`$\to$`v445`) closes modulo realisation. `SEAM.EQUIV.01` stays [**O**].

- **Lean: `TfptCarrier/SeamEdgeChern.lean` (`FORM.SEAM.EDGE.CHERN.01`).** Kernel-hardens the integer backbone the four edge reads converge on: the FHS Chern integers $(+1/0/-1)$, the bulk–edge channel count $|C|$=1, N_{Maj}=$2^{g_{\mathrm{car}}-1}$=16, the assembly $2c_-$=$N_{\mathrm{Maj}} \cdot |C|$ (so c_-=8), c_-=8=g_{car}+N_{fam}=rank E_8, holomorphic $c \equiv 0$ mod 8 and the order-3 modular T-phase — all by kernel `decide` (`edge_kernel_facts`, axioms `propext` only). `lake build` OK, `#print axioms` clean.

- **Wolfram mirror.** `tfpt_readouts_extension.wl` gains nine exact checks (`v452` modular data $\times$4, `v450` entanglement assembly, `v451` Ising Kac weights, `v453` marks= μ_4 $\times$3): 345 $\to$ 354 checks, all passing (`GATE.WOLFRAM.02`). E_4 is evaluated via its q-series since `EisensteinE` stays symbolic for complex τ.

- **Papers.** The adversarial audit (`tfpt_5_redteam`, Target A) and the research-contracts note record the edge CFT as named end-to-end — central charge c_-=8 from three independent observables (correlator `v444`, topology `v447`, entanglement `v450`), the Ising operator tower (`v451`), the $(E_8)_1$ torus modular data (`v452`), and the μ_4 premise derived from the marks (`v453`) — while the only residual that stays [**O**] remains the cited continuum-existence theorem (`v336`).

2026-06-30 (`v446`/`v447`/`v448` + Lean + Wolfram — deriving clock-invariance from μ_4-symmetry, an independent topological reading of the edge c_-=8, and pinning the one external MMST fact at four-point order)

- **`v446_seam_clock_invariance.py` (`SEAM.RIGIDITY.CLOCKINV.01`).** Closes `v445`'s single residual one notch further: the operator condition $[\rho, \Lambda_\Sigma]$=0 (clock-invariance) is shown to be a *consequence* of a structural symmetry, not an independent assumption. [**E**] the OS canonical transfer Λ=$C(1)C(0)^{-1}$=e^{-h} (fixed by RP+reflection, `v54`) of a μ_4-invariant RP covariance satisfies $[\rho, \Lambda]$=0 (swept N=4..32); [**E**] a μ_4-breaking covariance gives $[\rho, \Lambda] \neq 0$ (iff-sharp neg control); [**E**] Λ is a positive contraction (spectrum in $(0,1]$); [**E**] the free modular Hamiltonian K=$\log((1{-}C)C^{-1})$ (`v258`) of the invariant state has $[\rho, K]$=0 (Tomita–Takesaki), broken control $\neq 0$. [**C**] so with `v445` (commute $\Leftrightarrow$ block-diagonal), RP-definability *plus* the four marks *force* the block-diagonal Λ_Σ; the residual reduces from an operator commutation to "the seam RP data is μ_4-symmetric" (`QGEO.SYM.01`). `SEAM.EQUIV.01` stays [**O**].

- **`v447_seam_edge_chern.py` (`SEAM.EQUIV.EDGE.CHERN.01`).** An *independent* (topological)

reading of the edge central charge, corroborating v444's correlator route from a different observable. [**E**] the $M{=}1$ collar has integer Chern number $C{=}{+}1$ (Fukui–Hatsugai–Suzuki link-variable method, no fit); [**E**] $C{=}0$ in the trivial phase and $C{=}{-}1$ at $M{=}{-}1$ (opposite chirality); [**E**] the open strip carries exactly $|C|{=}1$ chiral edge branch per boundary; [**E**] the Li–Haldane entanglement spectrum has in-gap modes at 1/2 only in the topological phase. [**C**] bulk–edge: $c_{-}{=}N_{\mathrm{Maj}}\cdot\frac{1}{2}\cdot|C|{=}16\cdot\frac{1}{2}{=}8{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}$ — the same $c_{-}{=}8$ as v444. SEAM.EQUIV.01 stays [**O**].

- **v448_seam_edge_fourpoint.py (SEAM.EQUIV.EDGE.FOURPOINT.01).** Pins the one external MMST fact (continuum existence of the chiral scaling limit) empirically at the next order. [**E**] the ground-state correlation matrix is an exact projector $G^2{=}G$ ($\sim 10^{-13}$), so the state is quasi-free (the MMST CAR hypothesis holds); [**E**] the edge four-point cross-ratio $Q{=}[G_{01}G_{23}]/[G_{02}G_{13}]$ at commensurate positions is Cauchy-convergent under refinement; [**E**] Q converges to the free-fermion conformal cross-ratio $\sin(5\pi/12)/\sin(\pi/12){=}2{+}\sqrt{3}\approx 3.732$; [**E**] the bulk control is off-scale (exponential). [**C**] so the one external fact is supported at *both* two-point (v444) and four-point order. SEAM.EQUIV.01 stays [**O**].
- **Lean: TfptCarrier/SeamRigidityForcing.lean (FORM.SEAM.RIGIDITY.FORCING.01).** Kernel-hardens v445 (and v442's uniformity): the entrywise forcing iff clock4 $a{=}$clock4 $b \Leftrightarrow a{=}b$ (i.e. $i^a{=}i^b \Leftrightarrow a \equiv b \bmod 4$, residue-only hence uniform in N), the four distinct eigenphases, the exact commutant dimension $\sum n_s^2{=}64{<}256$ and the order discriminator ($128{>}64$) are all proved by kernel decide (forcing_kernel_facts, no domain axioms); only the operator-level lift to the full L^2 is the named premise. lake build OK, #print axioms clean.
- **Wolfram mirror.** tfpt_readouts_extension.wl gains three exact v445 checks (entrywise forcing for all $a,b\in 0..31$; commutant dimension $64{<}256$; order-2 commutant $128{>}64$ swept $N{=}4..64$): 342 → 345 checks, all passing (GATE.WOLFRAM.02).
- **Papers.** The adversarial audit (tfpt_5_redteam, Target A) records the five-direction narrowing of the SEAM.EQUIV.01 keystone (edge scaling v444, forcing converse v445 + Lean, derived clock-invariance v446, independent topological $c_{-}{=}8$ v447, four-point pinning v448) explicitly as "not promoted": the only residual that stays [**O**] is still the cited continuum-existence theorem (v336). The research-contracts note lists the same as "eleven convergent attacks".

2026-06-30 (v445 — the rigidity FORCING converse: commuting with the order-4 clock is *exactly* μ_4-block-diagonality, upgrading the rigidity step from "permit" to "force")

- **v445_seam_rigidity_forcing.py (SEAM.RIGIDITY.FORCING.01).** The rigidity route (v398/v442) showed a μ_4-block-diagonal K *commutes* with the order-4 clock $\rho{=}\mathrm{diag}(i^n)$. This module proves the *converse* and the exact structure, isolating the single remaining physical premise. [**E**] FORCING (entrywise iff): $[\rho,K]{=}0$ iff $K_{ij}{=}0$ whenever $i{\not\equiv}j \bmod 4$, verified entrywise and swept $N{=}4..128$ — commuting *forces* μ_4-block-diagonality, it does not merely permit it. [**E**] EXACT COMMUTANT DIMENSION: $\dim\{K:[\rho,K]{=}0\}{=}\sum_s n_s^2$ (the ad_ρ nullspace), a proper subspace of the full N^2 algebra ($N{=}16$: $64{<}256$) — a genuine quantitative restriction. [**E**] OFF-BLOCK ⇔ NON-COMMUTING: every off-block matrix unit has $\|[\rho,E_{ij}]\|{\geq}\sqrt{2}{>}0$ (zero slack). [**E**] THE ORDER IS THE DISCRIMINATOR: the order-4 clock gives 4 sectors; the order-2 clock leaves a strictly larger commutant ($N{=}16$: 64 vs 128) — the four Gauss–Bonnet marks (order $4{=}|\mu_4|{=}h(A_3)$) force strictly more structure than two, and only the index-4 extension is $(E_8)_1$ ($\det{=}1$; v281/v154/v351). [**E**] NONDEGENERACY: exactly 4 distinct eigenvalues $\{1,i,-1,-i\}$ for all $N{\geq}4$. [**C**] SYNTHESIS: commuting with the order-4 μ_4 clock is *equivalent* to μ_4-block-diagonality, so RP-definability (v54/v398) + clock-invariance *force* the block-diagonal Λ_Σ — v442's "permit" upgraded to "force".

[**O**] the single residual is now precisely that the physical transfer lies in that commutant (clock-invariance = `QGEO.SYM.01`, `v323`/`v335`); `SEAM.RIGIDITY.01`/`SEAM.EQUIV.01` stays [**O**]. Python-only (numpy; det facts Wolfram-mirrored via `v89`/`v281`/`v422`). Registered in `run_all.py`, `script_registry.csv`, the ledger, and cited in `tfpt_research_contracts` (route (xii) of the convergent attacks, and the sharpened residual in the Seam State Rigidity contract).

2026-06-30 (Lean kernel-hardening of two of the seven `SEAM.EQUIV.01` routes — the Borchers/Wiesbrock standard-pair relations and the applicability-ledger count, machine-verified with clean axioms; rigor without new mathematics)

- **Context.** Two of the seven convergent attacks on the keystone residual are turned from numerical witnesses into kernel-checked facts, in the audit-contract style of `SeamScalingLimit.lean` / `BWKeystone.lean`: the load-bearing algebra/bookkeeping is proved by the Lean kernel, while the cited theorems and the one open continuum input remain named axioms. `lake build` OK; `#print axioms` clean. Tracked as two new `FORM.*` ledger rows; `\veri`-context in `tfpt_research_contracts`; Lean manifest refreshed.
- **`FORM.SEAM.BW.HSMI.01` (`TfptCarrier/SeamStandardPair.lean`) — standard-pair relations, kernel-checked (hardens `v438`).** The four Borchers/Wiesbrock relations on the centred 5-mode ladder over $\mathbb{Z}$ are proved by kernel `decide` (axioms `propext` / `Classical.choice` / `Quot.sound` only, no domain axiom): $[K,P]{=}P$, $J^2{=}1$, $JKJ{=}{-}K$, $JPJ{=}P_+$ (and $\mathrm{tr}\,K{=}0$). The cited Borchers/Wiesbrock theorem and the one continuum input (the continuum standard pair) are named axioms; the composition pins face (ii) of `SEAM.EQUIV.01` to exactly those two — intrinsic Bisognano–Wichmann, no rotation-covariance presupposed.
- **`FORM.SEAM.APPLICABILITY.01` (`TfptCarrier/SeamApplicabilityLedger.lean`) — the audit count, kernel-checked (hardens `v441`).** The headline bookkeeping is itself machine-verified: of the 12 cited-theorem hypotheses (MMST/NPW26/Adamo, mirrored in order) exactly 11 are internal and 1 external (`total_count`=12, `internal_count`=11, `external_count`=1), plus `mmst_range` ($8 \le 8 \le 16$) and `npw_outside_bucket` ($\det K{=}1 \neq 4$) — ALL with NO axioms (pure kernel `decide`). The single external fact (continuum scaling-limit existence) is the named axiom; the keystone reduction pins its external dependence to exactly that one fact.
- **Net.** Both modules are imported by the root `TfptCarrier.lean`, axiom-audited in `AxiomCheck.lean`, and signature-locked in `AuditContract.lean`; full `lake build` succeeds. No new mathematics — the same facts `v438`/`v441` verify numerically are now kernel-proved; `SEAM.EQUIV.01` stays [**O**].

2026-06-30 (the chiral-edge scaling motor `v444` — the edge half of the one `SEAM.EQUIV.01` continuum residual: the lattice edge already carries the conformal data of a chiral $c{=}1/2$ Majorana CFT and converges under refinement, grounding the EDGE the way `v439` grounded the BULK; the keystone stays [O])

- **Context.** `v439` grounded the *bulk* half of the cited MMST existence (uniform gap, exponential clustering, finite light cone) and explicitly narrowed the one open analytic step to the chiral massless *edge* scaling limit. `v444` attacks exactly that residual on the SAME explicit `v367`/`v368` $p{+}ip$ collar. `vN`-registered, runs in `run_all.py` (suite ends `ALL CHECKS PASSED`), `\veri`-cited in the body of `tfpt_research_contracts` (the G_{net} contract). Python-only (numpy; no new exact identity to Wolfram-mirror).
- **`v444_seam_edge_scaling.py` (`SEAM.EQUIV.CONTINUUM.EDGE.01`) — chiral-edge motor.** The lattice edge carries the kinematic and conformal data of a chiral $c{=}1/2$ Majorana CFT: [**E**] a linear, chiral in-gap dispersion (velocity $v{\approx}0.98$, $R^2{=}0.9999$, with *opposite* per-edge

velocities — one chiral Majorana per edge); **[E]** an *algebraic* equal-time edge two-point $\sim r^{-\eta}$ with $\eta\approx1$, versus the *exponential* bulk row and the exponential trivial-phase (M=3, no edge) control; **[E]** a Cauchy-convergent scaling-collapse under refinement (η=1.001±0.007, successive profile spread 0.028→0.013 across N_x=36..60) — the numerical hallmark of an existing scaling limit. **[C]** η=1 ⇔ one chiral Majorana (c=1/2) per edge, so the 16-copy carrier has a chiral c_-=8 $(E_8)_1$ edge CFT ($c_-=g_{\mathrm{car}}+N_{\mathrm{fam}}$). **[O]** the rigorous uniform-convergence theorem (cited MMST, `v336`) is not supplied.

- **Net.** With `v439` (bulk) and `v444` (edge), *both halves* of the one continuum residual are numerically grounded on our explicit model and pinned to a single cited existence theorem; `SEAM.EQUIV.01` stays **[O]**. Ledger row, `script_registry.csv` index and the generated surfaces (`verification.tex`, `ScriptIndex.tsx`) updated in the same change; manifest refreshed last.

2026-06-30 (six convergent attacks on the one `SEAM.EQUIV.01` continuum residual — intrinsic Bisognano–Wichmann, a Lieb–Robinson continuum motor, the β-homotopy, an applicability audit, uniform rigidity, and falsifiable kill tests; none closes the keystone, together they reduce it to a single analytic fact and make the claim falsifiable)

- **Context.** The keystone `SEAM.EQUIV.01` is "closed modulo a cited theorem": its one open input is the continuum existence of the seam's chiral massless scaling limit. Six new modules attack that residual from independent directions; each is honest about *not* closing it. All six are `vN`-registered, run in `run_all.py` (suite ends `ALL CHECKS PASSED`), and are `\veri`-cited in the body of `tfpt_research_contracts` (the G_{net} contract). Python-only (no new exact identity to Wolfram-mirror).

- `v438_seam_hsmi_borchers.py` (`SEAM.EQUIV.BW.HSMI.01`) — **intrinsic BW.** The seam supplies a Longo *standard pair* / +half-sided modular inclusion (positive lightlike translation $P\geq0$, boost K with $[K,P]=P$, reflection J with $JKJ=-K$, $JPJ=P_-$), so by Borchers' theorem the modular flow is geometric *intrinsically — without* presupposing rotation-covariance of the vacuum (complementary to Adamo–Giorgetti–Tanimoto). **[E]** ladder algebra exact, $sl(2,\mathbb{R})$ positivity (min eig >0); **[C]** synthesis; **[O]** the continuum standard pair.

- `v439_seam_lieb_robinson.py` (`SEAM.EQUIV.CONTINUUM.LR.01`) — **continuum motor.** The explicit gapped $p+ip$ collar (`v367`) has a uniform-in-N gap, exponential clustering with N-independent ξ, and a finite Lieb–Robinson light cone, so by Nachtergaele–Sims its *bulk* thermodynamic limit exists — grounding the bulk half of the cited MMST existence in our model and narrowing the open step to the chiral *edge* scaling limit. **[E]**/**[C]**, **[O]** the edge CFT.

- `v440_seam_lto_rp_beta.py` (`SEAM.EQUIV.LTORP.BETA.01`) — **β-homotopy.** The (LTO-RP)/BW data ($\Theta K\Theta=-K$, $[\rho,K]=0$) are β-independent, and along the whole path $C_\beta=1/(1+e^{\beta K})$ from the NPW26-proven *trace* state (β=0) to the seam KMS state (β=1) every covariance is reflection-related, μ_4-invariant and valid — a norm-continuous, obstruction-free homotopy, upgrading the single point of `v426` to a path argument for sub-step (ii) of `v424`.

- `v441_seam_applicability_ledger.py` (`SEAM.EQUIV.APPLICABILITY.01`) — **audit.** A hypothesis-by-hypothesis ledger of the three cited theorems (MMST / NPW26 / Adamo–Giorgetti–Tanimoto) finds 11 of 12 hypotheses are TFPT-internal computed facts and exactly one is external — the continuum existence of the chiral scaling limit; covariance is an Adamo consequence (defuses the `v215` circularity), not an input. The keystone's external dependence is pinned to that one analytic fact.

- `v442_seam_rigidity_uniform.py` (`SEAM.RIGIDITY.UNIFORM.01`) — **uniform rigidity.**

The band-limited μ_4-character rigidity of `v398` is uniform in N: exact 4-sector commutation to N=256 (not merely the $N \leq 64$ sweep), a uniform gap, and an N-independent clock-breaking lower bound $\|[\rho, V]\|$=2 — so it is not a small-N artifact (the full-L^2 forcing stays **[O]**).

- **`v443_seam_e8_discriminator.py` (`SEAM.E8.DISCRIMINATOR.01`) — kill tests.** Mode D of the closure-mode classification (`v347`) as concrete *falsifiable* discriminators against the nearest c=8 rival $SO(16)_1$: torus ground-state degeneracy 1 vs 4, level-1 currents 248 vs 120, lattice theta 240 vs 112 norm-2 roots; c=8 alone cannot decide (`v277`), only $\det K$=1 (the genus-1 statement, `v90`) selects E_8. A seam with GSD>1 would falsify $(E_8)_1$. **[X]**/**[C]**.
- **Status discipline.** `SEAM.EQUIV.01` stays **[O]** throughout; the six modules reduce, audit and falsify the residual but do not supply the one cited continuum-existence theorem. Ledger rows, the `script_registry.csv` index, and the generated surfaces (`verification.tex`, `ScriptIndex.tsx`) are updated in the same change; manifest refreshed last.

2026-06-30 (shadow-export coherence — the `tfpt5-shadow` mirror is now reproducible *as shipped*: the audit is shadow-aware, the content maps are non-destructive on the website-free subset, and an export gate blocks any incoherent mirror)

- **Context (reviewer finding, Alessandro).** On the exported `tfpt5_shadow.zip` the independent surfaces reproduced (`build.sh notes` → `11 ok`; Wolfram base/extension; Red Team; Lean transcript), but the *package* layer did not: `make_manifest.py -check` reported a stale `verification/website_map.csv` and `build.sh audit` crashed because it unconditionally required `website/lib/release.ts`, which the shadow subset deliberately excludes.
- **`make_docs_map.py` is now shadow-aware.** `website_map.csv` mirrors the `website/` sources; on a tree without `website/` the generator now *skips* it instead of regenerating it down to a single `README.md` row. This removes the silent corruption that made `build.sh notes`/`gen` leave `website_map.csv` stale against the manifest on the export (it now matches the established skips in `make_script_index.py` and `make_changelog_web.py`).
- **`audit_sync.py` is now shadow-aware.** When `website/` is absent it skips the website mirror (section D), the website version stamps (E) and the website Wolfram-count card (F), and the website generated-surface freshness in (A), printing exactly which sections were skipped. `build.sh audit` therefore runs to a clean `AUDIT OK` on the shadow subset instead of raising `FileNotFoundError`.
- **Export coherence gate.** `scripts/export-shadow.sh` now re-runs `make_manifest.py -check` and the shadow-aware `audit_sync.py` *on the filtered tree* before the mirror is published. The manifest check catches the earlier `v235-v298` symptom at the source: a module referenced by `run_all.py`/`script_registry.csv`/`manifest.sha256` but not committed (and thus dropped by `git ls-files`) now fails the export loudly instead of shipping a package that says `ALL CHECKS PASSED` while `run_all.py` stops at the first missing script.
- **Manifest refreshed.** `manifest.sha256` was regenerated as the last release step so the shipped digests match the committed `changelog.tex` and `website_map.csv`; the full Python suite reproduces end-to-end on the regenerated export. No theory content, status marker or `vN` module changed — this is an export/infrastructure fix only.

2026-06-29 (discoverability — an SEO/GEO pass on the public website: a generated `/llms.txt`, explicit AI-crawler `robots.txt`, query-targeted metadata, and the first real backlinks from the README and the Zenodo deposit)

- **New `/llms.txt` context file** for AI / answer engines (ChatGPT, Perplexity, Claude, Gemini), generated at build time from the same data layer as the site (`papers.ts`, `predictions.ts`,

`version.ts`) so it never drifts. It carries the one-line summary, the "claims / does not claim" split, the document set with PDF links, the headline predictions, and an explicit *disambiguation* from the unrelated Brouwer–Lefschetz "topological fixed point theory" of mathematics.

- `robots.txt` now lists the major search and AI crawlers (Googlebot, Google-Extended, Bingbot, GPTBot, ChatGPT-User, OAI-SearchBot, PerplexityBot, ClaudeBot, anthropic-ai, ...) explicitly as allowed; the site-wide meta description and keywords are reworded toward natural search queries ("derivation of the fine-structure constant", "Standard Model from first principles", "where does 137 come from") and carry the same physics/mathematics disambiguation.
- **Backlinks (the missing piece).** The root `README.md` and the Zenodo deposit description (`zenodo_description.html`) previously contained *no* outbound links; both now link to `fixpoint-theory.com` and the source repository, and the README gains a citation block with the archived deposit DOI `10.5281/zenodo.20846087`.
- Website / documentation infrastructure only — no theory, suite, ledger or status change.
- **New module `v437` (`E8.DEGREE.JOINT.01`) — a consolidation of `v6`/`v66`/`v355`/`v431` that derives more from the E_8 Casimir degrees *without* crossing the `v354`/`v355` anti-numerology line.** It adds *no* new per-degree coincidence — only one joint lock and a reconstruction.
  - **[E] all of E_8 is fixed by its degree multiset.** From $\deg(E_8) = \{2, 8, 12, 14, 18, 20, 24, 30\}$ alone: $\text{rank} = \#\deg = 8$, $h = \max\deg = 30$, #positive roots $= \sum\exp = 120$, #roots $= 240$, $\dim E_8 = 248$, $|W(E_8)| = \prod\deg = 696{,}729{,}600$, $\sum\deg = 128 = \dim S^+$.
  - **[E] the joint quadratic (the new lock).** The two TFPT structural integers $(g_{\text{car}}, N_{\text{fam}})$ are the *unique* root pair of $x^2 - (\text{rank}\,E_8)\,x + (h/2) = x^2 - 8x + 15 = (x-3)(x-5)$: sum of roots $= \text{rank}\,E_8 = 8$ (the rank-fill, `v6`), product $= h/2 = 15 = g_{\text{car}}N_{\text{fam}}$ ($2g_{\text{car}}N_{\text{fam}} = h$, `v431`); $\{3, 5\}$ is the unique positive-integer factor pair of 15 summing to 8. So $g_{\text{car}} = 5$ and $N_{\text{fam}} = 3$ are forced *together* by two E_8 degree-invariants, not chosen one at a time. The three primary-readout degrees $\{2, 8, 30\}$ are exactly $\{\min, \text{rank}, \max\}$.
  - **[E] discipline upheld:** no new per-degree mining beyond the joint quadratic; the five non-primary degrees keep their `v431` family homes; the `v354`/`v355` line is not crossed.
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (`E8.DEGREE.JOINT.01`); cited in the `tfpt_5` degree-audit winbox beside `v431`; mirrored in `wolfram/tfpt_readouts_extension.wl` (extension 339 → 342). Exact integer/lattice; no new physical number, no status change.

2026-06-29 (`v436` — hardening the unconditional "is it numerology?" floor to an assumption-minimal counting floor; plus a literature note grounding the α^3 Chern level)

- **New module `v436` (`OVERDET.FLOOR.02`) — it *hardens* `OVERDET.FLOOR.01` (`v432`) so the numerology verdict no longer rests on any subjective chance assignment.** `v432`'s nominal $\sim 10^{-10}$ multiplies the counting census by hand-chosen plausibilities (input-"8" independence 0.1^3, seed 0.012, kaon 0.15) a skeptic can contest; `v436` shows the conclusion survives without them.
  - **[E] assumption-minimal floor.** The α cubic-class census (`v100` layer D) is *pure counting*: of $N = 94500$ complexity-matched $F_{U(1)}$ variants only *one* lands in the 4×10^{-8} CODATA window — the TFPT instance, reproducing $\alpha^{-1} = 137.0359992$ — so $p_\alpha \le 1/94500 = 1.06 \times 10^{-5} \sim 4.40\sigma$ with *no* subjective probability, prior or multi-observable grammar.

  - **[E] monotone concession ladder.** $-\log_{10} p$ from generous to adversarial: full scorecard+α $\sim$ 30.7 $\rightarrow$ scorecard-only $\sim$ 25.8 $\rightarrow$ α-census-only $\sim$ 4.98; each rung uses strictly less evidence and the "not chance ($> 4\sigma$)" verdict holds at the *most adversarial* rung, so it never rests on one contestable piece. Stripping *all* of `v432`'s subjective factors ($\sim 1.8 \times 10^{-6}$) leaves the floor at 4.40σ — they only sharpen it.
  - **[C] honest 5σ gap.** The conventional 5σ line ($p = 5.7 \times 10^{-7}$) is crossed unconditionally by the census times exactly *one* further independent factor ≤ 0.054 (the input-"8" fourfold over-determination, or one foreign witness) — stated, not hidden. **[C]** Firewall unchanged: bounds only "is it numerology?", not the physics bridge (`v187`).

- **Literature note (research log, not a status change).** A `next.txt` entry grounds the α^3 Chern-level reading of `v435` in named theorems — Dai–Freed (*η-invariants and determinant lines*, the exponentiated η-invariant as a section of the inverse Quillen determinant line), Bismut–Freed (determinant-line curvature = the α^2 Calderón side) and Redlich/Stora–Zumino (the cubic $\mathrm{Tr}\, F^3$ anomaly descends to a Chern–Simons level, quantised to $\mathbb{Z}$). It records that the `EM1` proof obligation reduces to constructing the seam Dirac family — i.e. it is a *facet* of `SEAM.EQUIV.01` — and remains **[O]** external math (no fabricated closure).

- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (`OVERDET.FLOOR.02`); cited in the `tfpt_safeguards` "unconditional floor" keybox beside `v432`; mirrored on the website. Python-only; no new physical number, no status change.

2026-06-29 (`v435` — a fourth honest step on the α^{-1} Quillen target: a π-power test isolates the cubic α^3 as the unique metric-independent (topological) rung, and types its coefficient as a conditional integer level)

- **New module `v435` (`ALPHA.QUILLEN.PROGRESS.04`) — it attacks the *single* remaining [O] left after `v434` (residual (2), the cubic α^3 Chern/Maxwell moment) and *sharpens* it; it does *not* close it.** The target `ALPHA.QUILLEN.EXACT.01` (`v382`) stays **[O]**; $\alpha^{-1} = 137.0359992$ stays **[E]**.
  - **[E] π-power / metric-independence test.** The three EM-closure coefficients carry π-powers exactly $\{0, 3, 6\}$, equal to their c_3-orders: Maxwell $k_0 = 1 \rightarrow \pi^0$, Calderón $-2c_3^3 = -1/(256\pi^3) \rightarrow \pi^3$, transport $-8b_1c_3^6 = -41/(327680\pi^6) \rightarrow \pi^6$. So the Maxwell a^3 is the *unique* π^0/c_3^0 (metric-*independent*) term. A heat-kernel boundary coefficient always carries the $(4\pi)^{-d/2}$ measure; a metric-independent term carries none — the signature of a *topological* term. The test thus partitions the closure into *one* topological level (a^3) + *two* heat-kernel boundary terms, confirming `v342`/`v434`'s "a^3 is a level, not a Seeley–DeWitt curvature integral" from a *test* rather than a claim.
  - **[C] conditional level quantisation.** *If* the a^3 term is the seam $U(1)$ Chern–Simons/η boundary level (the `EM1` reading, `v48`), large-gauge invariance quantises the level to $\mathbb{Z}$, so $k_0 = 1$ is the *unit* (minimal) level — an integer, not a tunable real. This converts "the coefficient happens to be 1" into "the coefficient is an integer level and 1 is the unit".
  - **[O] not closed:** the `EM1` lemma itself — the from-first-principles Chern–Simons boundary proof that $\partial_\tau \log Z_{\mathrm{Maxwell}} = a^3$ — stays external math (type A, `v384`); the seam F-normalisation stays the one **[C]** (residuals 1 & 3, `v434`); α^{-1} stays **[E]**.

- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (`ALPHA.QUILLEN.PROGRESS.04`); cited in the `tfpt_1` EM-closure subsection beside `v433`/`v434`; mirrored on the website residual-gap lab. Python-only; no new physical number.

2026-06-28 (`v434` — a third honest step on the α^{-1} Quillen target: b_1 is the a_4 heat-kernel coefficient, and the three residuals collapse to one [C] + one [O])

- **New module `v434` (`ALPHA.QUILLEN.PROGRESS.03`) — it settles the status of the three residuals named after `v433` and shows they are *not* three independent open problems.** The target `ALPHA.QUILLEN.EXACT.01` (`v382`) stays **[O]**; $\alpha^{-1} = 137.0359992$ stays **[E]**.
  - **[E] b_1 *is* the $U(1)$ a_4 heat-kernel coefficient (the $\beta = a_4$ theorem).** The one-loop β function of a gauge coupling *is* the a_4 (Seeley–DeWitt a_2 in d=4) heat-kernel coefficient of the charged-field operator (Vassilevich). Computed from the carrier hypercharge content with the standard spin weights ($\frac{2}{3}$ per Weyl fermion, $\frac{1}{3}$ per complex scalar), $\sum \frac{2}{3}Y_f^2 + \frac{1}{3}Y_s^2 = 41/6$ (SM) and $(3/5)\cdot$ that $= 41/10 = b_1$ (GUT). So the b_1 in the counterterm $8b_1c_3^6$ is the a_4 coefficient of the carrier $U(1)_Y$ content (the same number as the β, `v159`/`v246`), the *same kind* of heat-kernel object as the α^2 Calderón a_4 (`v342`) — not a fitted multiplicity.
  - **[E] the geometry factors off:** $8b_1c_3^6 = b_1 \cdot (\operatorname{rank} E_8) \cdot c_3^6$, with $\operatorname{rank} E_8 = 8$ the χ=2 seam boundary count (`v216`) and c_3^6 the six boundary insertions (the π-power test, `v342`).
  - **[C] residual (1) ≡ residual (3).** With b_1 and the c_3-powers heat-kernel-forced, the only freedom left in the exact seam a_4 value is how the gauge curvature couples to the boundary measure — the seam F-normalisation. So "the actual seam $a_4 = 8b_1c_3^6$" and "the seam F-normalisation" are *one* **[C]** obligation, not two.
  - **[E] residual (2) located (not closed):** the determinant-line variation factors $a^3 - 2c_3^3a^2 = a^2(a - 2c_3^3)$; a^2 is the Gilkey gauge-curvature (Quillen/Chern) density (`v342`), so the leading $a^3 = a \cdot a^2$ is the *first moment* of the Chern density — a topological level at boundary order c_3^0, not a Seeley–DeWitt curvature integral. Its from-first-principles (Chern–Simons-type) origin stays **[O]**.
  - **[E] net reduction:** the EM-Ward residual is *one* **[C]** (residuals 1 & 3) + *one* **[O]** (residual 2), not three independent problems — a genuine narrowing of the open surface. **[O]** `ALPHA.QUILLEN.EXACT.01` is *not* closed (the α^3 Chern level and the from-first-principles proof stay external, type A, `v384`).
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (`ALPHA.QUILLEN.PROGRESS.03`); cited in the `tfpt_1` EM-closure subsection beside `v433`; mirrored on the website residual-gap lab. Python-only; no new physical number.

2026-06-27 (`v433` — a second honest step on the α^{-1} Quillen target: a solvable 4D model reaches the a_4 order, connecting `v391` and `v342`)

- **New module `v433` (`ALPHA.QUILLEN.PROGRESS.02`) — a second honest step on the external target `ALPHA.QUILLEN.EXACT.01` (`v382`); it *grounds and connects*, it does *not* close it.** The target stays **[O]** (external math, type A, `v384`); the value $\alpha^{-1} = 137.0359992$ stays **[E]**. The earlier honest attempt `v391` gave a solvable model on a 1D circle whose ζ-determinant variation reached only the low Seeley–DeWitt orders a_0, a_1; separately, `v342` grounded the quadratic Calderón term in the Gilkey gauge-curvature coefficient a_4 ($30/360 = \frac{1}{12}$, $\Omega^2/12$). This module supplies the missing link between the two.
  - **[E] a solvable 4D model reaches the a_4 order:** the flat operator $\Delta = -\partial^2 + m^2$ on a 4-torus of volume V has heat trace $\operatorname{Tr} e^{-t\Delta} = V/(4\pi)^2\, t^{-2} e^{-tm^2}$, with Gilkey coefficients $a_0 = V/(4\pi)^2$, $a_2 = -Vm^2/(4\pi)^2$ and $a_4 = Vm^4/(2(4\pi)^2)$; the t^0 (conformal-anomaly) order a_4 is *non-zero* — exactly the order `v391`'s 1D circle toy could not reach.
  - **[E] it is the order `v342` uses:** so `v391` ("a ζ-determinant variation *is* closed heat-kernel data") and `v342` ("the seam term *is* the Gilkey a_4") are unified — the variation is heat-kernel data *at the a_4 order*, not only the low orders v391 reached.

  - **[E] the measure is c_3-built:** the universal 4D heat-kernel measure $(4\pi)^{-2} = 1/(16\pi^2) = (2c_3)^2 = 4c_3^2$ in the seam's one-sided 2D-boundary normalisation $c_3 = \frac{1}{2}(4\pi)^{-1}$; and the `v382` split is re-verified at $\alpha^{-1} = 137.0359992$.
  - **[O] not closed:** computing the *actual* seam $U(1)$ ζ-determinant a_4 coefficient ($= 8b_1c_3^6$) on the μ_4 seam moduli, and the origin of the cubic α^3 Maxwell/Chern moment, stay open (external math, type A, `v384`); the exact seam F-normalisation fixing the coefficient *value* stays **[C]** (`v342`). A second honest step that grounds and connects, it does not close.

- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (`ALPHA.QUILLEN.PROGRESS.02`); cited in the `tfpt_1` EM-closure subsection beside `v382`/`v391`; mirrored on the website (the residual-gap lab). Python-only (like `v391`); no new physical number.

2026-06-27 (`v431`, `v432` — the "unmapped" E_8 degrees are forced spine+flavor structure; the unconditional over-determination floor)

- **New module `v431` (`E8.DEGREE.LADDER.01`) — the reverse-audit "unmapped" E_8 Casimir degrees are NOT diffuse overhead but a forced two-family decomposition.** The structural complement of the reverse audit (`v354`/`v355`) and its sheet/deck complement (`v430`): $\deg(E_8) = 6 \cdot \text{spine}\{2,3,4,5\} \sqcup (\{2\} \cup \text{det-ladder}\{8,14,20\})$. **[E]** the two-family split is *forced* by $h(E_8) = 30 = 2 \cdot 3 \cdot 5$ — the exponents are the $\varphi(30) = 8$ totatives of 30, hence coprime to 6, hence $\equiv \pm 1 \bmod 6$, so the degrees occupy only the residue classes $\{0,2\} \bmod 6$. **[E]** the $6k$ family $\{12,18,24,30\}/6 = \{2,3,4,5\}$ is the `v91` spine; the $6k{+}2$ family $\{8,14,20\} = (\det R, \det C, \det L)$ is the winding line $\det M(s,0) = 6s + 8$ of the flavor diamond (`v135`). **[E]** "$18 = 6N_{\text{fam}}$" is not a holdout (spine family), and the clean split is special to E_8 among the simply-laced exceptionals ($E_6/E_7/D_5$ fail). **[C] honest `v355` line:** the arithmetic decomposition is exact, but the *functorial* claim "$6k{+}2$ Casimir stratum = flavor sheet operators" needs a representation-theoretic map and stays **[C]** ($12, 24$ admit two canonical readings each) — a structure theorem, not a forced functorial flavor map. It reclassifies the `v354` overhead to zero orphan degrees and closes no gate.

- **New module `v432` (`OVERDET.FLOOR.01`) — the unconditional improbability floor and the L1↔L2 bridge.** The complement of the conditional null model (`v100`) and the multiply-vs-compress framework (`v427`/`v428`): the defensible over-determination number that survives even if a skeptic *rejects* the declared formula grammar. **[E]** compression removed (`v428`), the genuinely multiplicative pieces are the input "8" forced four independent ways plus the foreign witness $\alpha^{-1} \sim 137$. **[C]** multiplying only across disjoint pieces (input over-determination $\times$ the α census $1/94500$ $\times$ the φ_0^{ret}-seed cluster counted once $\times$ one downstream witness) gives a floor $\leq 10^{-6}$ across a grid of generous assignments, nominal $\sim 10^{-10}$. **[E]** that floor sits ~ 20 orders *above* the `v100` conditional $10^{-30.7}$; the gap is exactly the evidential value of proving the forms forced, so the α form-derivation (`ALPHA.QUILLEN.EXACT.01`, `v382`) and this floor are the *same* lever. **[C] honest scope:** bounds only the "is it numerology?" question, not the physics bridge (firewall `v187`); a conservative floor, not a posterior.

- *Surfaces synced in the same change:* `v431`/`v432` added to `run_all.py`, `script_registry.csv` (master index + website `ScriptIndex`), `status_ledger.csv`, `tfpt_5_redteam` (reverse-audit winbox + Casimir-degree figure) and `tfpt_safeguards` (the over-determination layer); `v431` mirrored in `wolfram/tfpt_readouts_extension.wl` (`v432` is a conservative bound, Python-only).

2026-06-27 (v430 — the seam's "other side" is forced-disjoint from E_8's unmapped Casimir degrees)

- **New module v430 (E8.OTHERSIDE.AUDIT.01) — the sheet/deck complement of the reverse audit (v354/v355).** The reverse audit found five E_8 Casimir degrees $\{12, 14, 18, 20, 24\}$ with no primary readout ("hull overhead"); the double cover has an *other side* (the one-sided $\mathbb{Z}_2$ collar / conjugate half-spinor S^-), so the sharp adversarial follow-up is *"is the leftover where the second sheet lives?"* The answer is a disciplined **negative**. [**E**] the deck is the degree-2 invariant: the one-sided cover contributes $|\mathbb{Z}_2| = 2$ (the $\frac{1}{2}$ in $c_3 = 1/(|\mathbb{Z}_2|\,4\pi) = 1/(8\pi)$, v58/v216), and $2 = \min\deg(E_8)$ is the quadratic/metric — one of the *matched* primary readouts $\{2, 8, 30\}$, never one of the unmapped degrees. [**E**] the two sheets are the 128-spinor: $\dim S^+ = \dim S^- = 16$, the spinor block $128 = \operatorname{rank} E_8 \cdot \dim S^+ = 2^{\mathrm{rank}-1} = \sum \deg$, the spinor half of $248 = 120 + 128$; S^- enters 248 as the conjugate $(\overline{16}, \overline{4})$ ($128 = 2 \cdot 64$), not a spare singlet. [**E**] forced-disjoint: the sheet/deck set $\{2, 16, 32, 128\}$ has empty intersection with $\{12, 14, 18, 20, 24\}$, its only degree contact the matched 2. [**O**] each unmapped degree, "explained" from sheet atoms, needs an unforced coefficient and admits ≥ 2 readings (the v355 discriminator), so there is no forced sheet→unmapped-degree identity.
- **A structural negative, not a status change.** It consolidates the S^--dark-matter downgrade (v227) and the no-spare-singlet WIMP no-go of the frontier E_8 scan (tfpt_4_frontier): the other side is matched conjugate matter, the five unmapped degrees are the hull overhead, and the two do not meet. The only live "other side" question — whether the 128 phase/glue channel hosts a dark sector (v227) — concerns *matched* structure, not the unmapped five. Closes no gate, upgrades nothing.
- **Surfaces synced.** run_all.py, script_registry.csv, status_ledger.csv; the reverse-audit Outcome winbox + degree-audit table caption and the Target D winbox of tfpt_5_redteam, the magnitude/phase keybox of tfpt_3_e8_audit_bootstrap, the WIMP-no-go passage of tfpt_4_frontier; the E8 node of the website verification DAG; the Wolfram extension/README/ledger counts and the website verification count; manifests refreshed.

2026-06-26 (v429 — the 'unmapped' golden E_8 structure is the geometry of the one external input θ_i)

- **New module v429 (DM.AXION.PENTAGON.01) — a geometric re-reading answering "why does so much idle E_8 structure exist?" for the one case where the idle structure has a job.** The reverse audit (v354) had argued the golden ratio φ is *unmapped* (it appears only internally — the affine-E_8 attractor angle v312, the icosian lattice v348 — and in no physical readout), and used exactly that to conclude φ cannot be numerology. [**E**] this sharpens that statement: because $N_{\mathrm{fam}} = g_{\mathrm{car}} - 2$, the axion misalignment *spine* angle (v211, DM.AXION.SPINE.01) is $\theta_i = \pi N_{\mathrm{fam}}/g_{\mathrm{car}} = (g_{\mathrm{car}} - 2)\pi/g_{\mathrm{car}}$, which is *exactly* the interior angle of the regular g_{car}-gon — for $g_{\mathrm{car}} = 5$ the pentagon, $\theta_i = 3\pi/5 = 108°$. [**E**] its cosine is golden, $\cos(3\pi/5) = (1-\sqrt{5})/4 = -1/(2\varphi)$ (partner $2\cos(2\pi/5) = 1/\varphi = \varphi - 1$, $\varphi = 2\cos(\pi/5)$). [**E**] the golden character is *unique* to $g_{\mathrm{car}} = 5$: among the small regular n-gons only $n = 5$ has an irrational interior-angle cosine ($n = 3, 4, 6$ give $1/2, 0, -1/2$), so θ_i is golden *because* the carrier is 5 (P2).
- **An honest [C] bridge, not a status change.** It connects two standing facts — φ is the unmapped $g_{\mathrm{car}} = 5$ signature (v354/v313) and $\theta_i = 3\pi/5$ is the one external cosmological input (v211) — and *refines* v354 from "φ in no physical readout" (it is the unmapped $g_{\mathrm{car}} = 5$ signature, v313) to "φ touches *only* the [**C**] spine misalignment angle" (conditional, branch-level, never a frozen prediction). It is a geometric motive for the spine over the hilltop $\theta_i \approx 170°$ (whose cosine is not a clean pentagon value), consistent with — not replacing — the

finite-T solver that decides the branch (v185/v211). It does *not* derive θ_i, does *not* upgrade DM.AXION.SPINE.01 (stays [**C**]) and closes no gate.

- **Surfaces synced.** run_all.py, script_registry.csv, status_ledger.csv, the dark-matter section of tfpt_4_frontier; the Wolfram extension/README/ledger counts (GATE.WOLFRAM.02) and the website verification count; manifests refreshed.

2026-06-25 (External-works references appendix — web-verified)

- **New shared appendix tex-artefacts/tfpt_references.tex** ("External works built upon"), \input into the six math-heavy documents (tfpt_1, tfpt_2, tfpt_4, tfpt_horizon_readouts, origin_theory, tfpt_research_contracts). The active papers named the foundational results they build on by author only, with no reference list; the appendix supplies web-verified locators (journal, volume, year / arXiv) grouped by topic: modular theory & AQFT (Tomita–Takesaki LNM 128 1970; Bisognano–Wichmann J. Math. Phys. 16/17; Osterwalder–Schrader CMP 31/42; Doplicher–Haag–Roberts CMP 23/35; Haag–Ruelle; Longo CMP 126/130; Kawahigashi–Longo–Müger CMP 219), conformal nets/VOAs/lattices (Frenkel–Kac Invent. Math. 62; Segal CMP 80; MMST; Adamo–Moriwaki–Tanimoto; Adamo–Giorgetti–Tanimoto 2506.01008/2508.07109; Naaijkens–Penneys–Wallick), lattices/quadratic forms (Conway–Sloane; Siegel Ann. Math. 36 1935), singularities (Brieskorn Invent. Math. 2 1966; Milnor AMS 61 1968) and spectral geometry/gravity/thermal time (Chamseddine–Connes CMP 186; Connes–Rovelli CQG 11; Wald PRD 48; Casini–Huerta–Myers JHEP 05 2011). Added to the make_manifest.py TEX list. No claim or status change — a provenance/citation completeness pass.

2026-06-25 (External AQFT citations + the four-fold μ_4 identity sharpened)

- **Two recent Adamo–Giorgetti–Tanimoto theorems cited where the keystone builds on them.** The open SEAM.EQUIV.01 continuum/realisation legs (v336) now cite, alongside MMST and Adamo–Moriwaki–Tanimoto (arXiv:2407.18222, already in use): **(a)** the construction and classification of two-dimensional conformal nets from even lattices (arXiv:2506.01008, 2025) — "the lattice is a conformal net" is a theorem of the same operator-algebraic lineage, strengthening the lattice-realisation leg; and **(b)** automatic positive-energy/diffeomorphism covariance of conformal-net representations (arXiv:2508.07109, 2025), which defuses the "Bisognano–Wichmann presupposes the covariance it should produce" circularity (v215) on the intrinsic-BW residual (v424). Added to v336 docstring/registry, the SEAM.EQUIV.CONTINUUM.01 ledger external_data, and tfpt_research_contracts.
- **The four-fold μ_4 identity made explicit (v422).** v422 already proved that the seam deck/divisor μ_4, the carrier Galois group $\mathrm{Gal}(\mathbb{Q}(\zeta_5))$, the discriminant $\mathrm{disc}(D_5)=\mathrm{disc}(A_3)=\mathbb{Z}/4$ and the $(E_8)_1$ simple-current glue are ONE cyclic $\mathbb{Z}/4$ (with the cyclic-vs-Klein negative control). It is now named the *four-fold μ_4 identity* and typed, per v428, as a *compression* (one object read four ways), NOT four independent witnesses – in v422, the registry, the SEAM.GALOIS.NET.01 ledger row and origin_theory. (No new module: a separate v429 would have duplicated v422 and violated the very compression discipline.)

2026-06-25 (v428 — honest self-correction of the over-determination map)

- **v428 (OVERDET.WITNESS.RECLASS.01) — applying v427's own multiply-vs-compress test to v427 itself.** A pattern search and the right objection exposed that calling the seven arithmetic witnesses "disjoint grammars that *multiply*" overstates independence. [**E**] by the Brieskorn classification (v236) the $(2,3,5)$ singularity is the ONE generator behind the order-30 clock (Milnor $\mu=(2-1)(3-1)(5-1)=8=\mathrm{rank}\,E_8$, $30=2\cdot3\cdot5=h(E_8)$, $\varphi(30)=8$), so

each witness is a facet of that same object (primes $2, 3, 5$ of 30, the clock 30, or E_8): all seven collapse to one generator ⇒ *compression*, not seven multiplications. **[E]** what genuinely multiplies is the input forced four independent ways — the "8" in $c_3{=}1/(8\pi)$ from rank E_8, $h(D_5){=}2(5{-}1)$, $\varphi(30)$ and the Milnor number — plus the foreign witness $\alpha^{-1}{\approx}137$. **[C]** refines and partly walks back `v427`, consistent with `v236`/`v305`/`v313`.

- **Surfaces corrected (same change).** `v427`'s prose, the ledger row (`OVERDET.WITNESS.MAP.01`, now noting the refinement), `tfpt_safeguards` Layer 3 (the seven compress one object; a second keybox states what multiplies), the witness-map figure (redrawn), the website (`papers.ts`, `Safeguards.tsx`, `release.ts`, the intro-video description) and the Zenodo description. The over-determination claim is now the narrower, harder-to-attack "one richly over-determined input + a foreign readout".

2026-06-25 (Safeguards visuals + paper expansion + the website safeguards band)

- **Five safeguards figures** (`make_figures.py`, in `manifest.sha256`): the layered defense, the α^{-1} uniqueness scan (one $F_{U(1)}$ variant of 94,500 in the CODATA window), the reverse audit (3/8 vs 5/8), the witness-independence map (seven disjoint grammars), and the look-elsewhere null model (13/13 vs $\leq$ 5/13).
- **`tfpt_safeguards` expanded.** Layer 5 now states the null model explicitly ($\prod p_i{=}10^{-25.8}$, 2×200k pseudo-theories $\leq$ 5/13, $\varphi_0^{\rm ret} \pm 10\% \to$ 6/13, shuffle 1.2) and the α uniqueness ($\leq 10^{-30.7}$, ∼102 bits, `v100`), with the data watchdog (`v307`) and kill board (`v321`); the five figures are embedded.
- **Website: a prominent animated Safeguards band** (`components/Safeguards.tsx`) in the home flow (after the trust beat) — headline statistics, the seven layers, the figures, links to the paper and the suite — to address the coincidence/numerology suspicion proactively.
- **Koide flow-time investigation (honest negative).** Post-`v425`: the flow-time is *not* native and does *not* reduce to one scale — `v99` already rejects a scale reading as look-elsewhere; the only theory-side content is the conditional integer reading $t{=}N_{\rm fam}{=}3 \Rightarrow m_\tau{=}1776.94$ MeV, a Belle II kill test. No new module.

2026-06-25 (Three closure-strategy modules `v425`–`v427` + the Safeguards paper)

- **`v425` (`DYN.TRANSFER.UNIVERSAL.01`) — the single-flow reduction of $F_{\rm transfer}$.** The four frontier transfers do not each need a separate external solver: their *dynamics* is the ONE native flow (the seam modular/recovery semigroup, `v238`/`v240`, gap $6\ln(3/2)$) restricted to each sector. **[E]** the `v99` Koide generator linearised at $q{=}2$ has rate $-\Delta$, time-1 contraction $(2/3)^6$; Boltzmann washout is that contraction (CP phase $4\pi/3$ native, `v320`); QCD rate $b_3{=}-7$. **[C]** only the anchors ($v_{\rm geo}$, C_p, M_R) stay external; $F_{\rm relic}$'s cosmological IC θ_i is the one wall. A reduction (extends `v383` static→dynamic), not internalisation; the firewall holds.
- **`v426` (`SEAM.EQUIV.LTORP.KMS.01`) — the invertible $\beta{=}1$-KMS variant of (LTO-RP).** The corner NPW26 leave open (they prove it for topologically-ordered models; the seam is invertible + KMS, not a trace). **[E]** a gapped reflection-symmetric μ_4-even collar at $\beta{=}1$ KMS satisfies $\Theta K \Theta^{-1}{=}{-}K$ ($u_\Theta{=}J$) and $[\rho, K]{=}0$, so (LTO-RP) holds by direct Tomita–Takesaki; $|\det \mathrm{Cartan}(E_8)|{=}1$ (no anyons), state not a trace; neg controls have teeth. Advances `v424` sub-step (ii); `SEAM.EQUIV.01` stays **[O]** (continuum sub-step (i) remains).
- **`v427` (`OVERDET.WITNESS.MAP.01`) — the honest over-determination accounting.** A reviewer correction made machine-explicit: witnesses from *disjoint* grammars multiply, projections of the *same* generator compress. **[E]** seven disjoint-grammar witnesses each reproduce a carrier-skeleton integer from their own structure — Gauss $N(3{+}2i){=}13$, Eisenstein $N(3{+}2\omega){=}7$, cyclotomy $N(3{+}2\zeta_5){=}55$, Galois $|(\mathbb{Z}/5)^\times|{=}4$, $|\det \mathrm{Cartan}(E_8)|{=}1$, Pascal

$\binom{4}{0}+\binom{4}{1}+\binom{4}{2}=11$ ($2^4=16$), Coxeter $h(E_8)=30=2\cdot3\cdot5$ with $\varphi(30)=8$ — and the anchor $a=(1,1,2)$ is the compression generator. Extends `v305`.

- **New paper `tfpt_safeguards` (companion S).** A dedicated methodology paper collecting, in one language, every mechanism by which TFPT defends a claim against coincidence and numerology: the status calculus + ledger + audit, no-free-pattern + reverse audit, the over-determination map (`v427`), the firewall + No-Unit theorem, frozen predictions + null model, the independent Wolfram and Lean paths, and the red team — with a final kill-test summary. Registered in `build.sh`, `make_docs_map.py`, `make_manifest.py` and the document-set box; the document set is now **eleven** deliverables (five numbered papers + four companions + introduction + changelog).
- **Surfaces synced.** `run_all.py` (421 modules), the registry, the ledger, `tfpt_research_contracts` (the F_{transfer} and keystone sections), `tfpt_safeguards`; manifests refreshed.

2026-06-25 (The (LTO-RP) reduction of the keystone `SEAM.EQUIV.01`: `v424`)

- **New module `v424` (`SEAM.EQUIV.LTORP.01`).** An honest MMST-style reduction of the one open keystone — `SEAM.EQUIV.01`'s intrinsic Bisognano–Wichmann lemma (the `v308`/`v309` residual) — to the recent commuting-projector theorem of Naaijkens–Penneys–Wallick (`arXiv:2605.10693`, [NPW26]), with the central mapping *corrected*. **[E]** the μ_4 clock is a modular *symmetry* ($[\rho,K]=0$, `v309`/`v199`), not the flow — $\sigma^{\psi}=\rho$ is type-wrong (a one-parameter group versus a finite order-4 element). **[E]** NPW26's axiom (LTO-RP) means $u_\Theta=J$ (reflection = modular conjugation) = the intrinsic BW condition $\theta K\theta=-K$, *distinct* from $\omega\circ\rho=\omega$: a positive character-block-diagonal K has $[\rho,K]=0$ yet never $\theta K\theta=-K$, so clock-invariance does *not* imply (LTO-RP) (refuting that identification). **[O]** make-or-break: NPW26 prove (LTO-RP) for *topologically ordered* models (Toric Code $|\det K|=4$ via a trace; Levin–Wen anyonic), while the seam is invertible ($|\det \mathrm{Cartan}(E_8)|=1$, no anyons) and a $\beta=1$ KMS state (not a trace) — in neither proved bucket. **[C]** verdict: the BW residual reduces to NPW26 Theorem A modulo two open sub-steps (realise the seam as a $\mathbb{Z}_2$-reflection commuting-projector LTO; extend (LTO-RP) to the invertible KMS case) — a reduction, *not* a closure; `SEAM.EQUIV.01` stays **[O]**. Cited in `tfpt_research_contracts`; Python-only (numpy; the det discriminators mirrored via `v89`/`v281`).
- **Validation provenance.** The three pivotal references of the closure-strategy note were independently verified real and correctly described: NPW26 (`arXiv:2605.10693`), the RP–Yang–Mills complete-monotonicity reconstruction (Int. J. Geom. Meth. Mod. Phys. 2026), and Marolf (`arXiv:2203.07421`).
- **Surfaces synced.** `run_all.py`, the script registry, the status ledger and the `tfpt_research_contracts` keystone section; manifests refreshed.

2026-06-25 (No-Ambient-Dependency: no frozen readout is computed from `QG.AMB.01`: `v423`)

- **New module `v423` (`QGAMB.NODEP.01`).** Sharpens the `v369`/`v384` prose redundancy ("`QG.AMB.01` is a **[C]** redundancy, not dynamics") into a machine-checked DAG+script fact, told honestly. **[E]** gap-decoupling margin $\Delta_{\text{eff}}=6\ln\frac{3}{2}-31/(4\pi^2)\approx1.648>0$ ($31=2^{g_{\text{car}}}-1$), susceptibility $\chi=729/665$ finite (negative control $\text{gap}\to0\Rightarrow\chi\to\infty$); **[E]** `QG.AMB.01` is a certification *sink* (every claim that lists it is a gravity-gate/QG row, never a frozen prediction) and *not a value source* (no producing script, while α^{-1}, the mixing angles and cosmology are each *computed* by a compiler script, `v3`/`v9`/`v268`/`v7`); **[E]** the one transitive link is architectural — of the frozen readouts only θ_{13} references it, via the 4d-QFT contract spine (`PS.DIRAC`→`CONTRACT.QFT4D`→`QG.G2`→`GATE.METRIC`), a coherence edge, not a numerical input. **[C]** verdict: `QG.AMB.01` is gap-decoupled certification — `v369` as a DAG/script fact;

honest scope — this does *not* discharge `SEAM.EQUIV.01`, on which the readouts still depend. Cited in `tfpt_4`; Python-only (gap algebra mirrored via `v337`).

- **Paper sharpening (seam selector, up front).** The `tfpt_4` quantum-gravity keybox now states explicitly that the open `SEAM.EQUIV.01` premise is the *holomorphic* $|\det K|{=}1$ point that selects $(E_8)_1$ — the central charge $c{=}8$ alone does *not*, being shared by the non-holomorphic $SO(16)_1$ counterexample (`v277`/`v281`).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, `tfpt_4`; manifests refreshed.

2026-06-25 (The Galois↔Net bridge: `v422`)

- **New module `v422` (`SEAM.GALOIS.NET.01`).** The seam $\mu_4 = \mathrm{Gal}(\mathbb{Q}(\zeta_5))$ is the SAME cyclic $\mathbb{Z}/4$ as the $(E_8)_1$ simple-current glue — not a mere order-4 coincidence — bridging the Galois gearbox of `v419` to G_{net}/`SEAM.EQUIV.01`. **[E]** $\mathrm{disc}(A_3) = \mathrm{disc}(D_5) = \mathbb{Z}/4$ (one Smith invariant factor 4 = cyclic, $\det 4$); D_n disc is $\mathbb{Z}/4$ for n odd, $\mathbb{Z}_2{\times}\mathbb{Z}_2$ for n even, so the carrier D_5 (rank $5 = g_{\mathrm{car}}$, odd) is cyclic while D_8 (rank 8, even) is Klein; **[E]** $(\mathbb{Z}/5)^\times = \langle 2\rangle$ is cyclic order 4 (only the carrier prime 5 gives order 4); **[E]** in $\mathbb{Z}_4{\times}\mathbb{Z}_4$ ($16 = \dim S^+$) the glue $(1,1)$ is order 4 with Lagrangian quotient $16/4^2 = 1 = (E_8)_1$ (`v89`/`v154`); **[E]** negative control: the Klein / order-2 $\langle(2,2)\rangle$ gives $16/2^2 = 4 = |\mathrm{disc}(D_8)| = SO(16)$ ($\det 4$), NOT E_8 ($\det 1$) — cyclic $\mathbb{Z}/4 \to E_8$, Klein$\to D_8$. **[C]** the bridge: one cyclic $\mathbb{Z}/4$ threads the seam deck μ_4 (`v419`), $\mathrm{Gal}(\mathbb{Q}(\zeta_5))$ and the $(E_8)_1$ glue — the Galois gearbox $(\mathbb{Z}/30)^\times$ IS the holomorphic net's glue, cyclicity forced by 5, Klein excluded. Cited in `origin_theory`; Wolfram-mirrored ($324 \to 327$).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, `origin_theory`; the Wolfram extension/README/ledger counts (327/327) and the website verification count; manifests refreshed.

2026-06-25 (F_pole external: the Koide source→pole transfer is one-loop EM: `v420`)

- **New module `v420` (`FR.POLE.QED.01`).** The honest external-physics follow-up to `v93` and the clock No-Go: the Koide source→pole transfer is a one-loop *electromagnetic* effect, not an algebraic step. **[E]** Koide Q is rescaling-invariant ($Q(c\,m) = Q(m)$), so flavor-universal running leaves it fixed and the source→pole shift is purely flavor-split; **[C]** the tree deficit $\frac{2}{3} - Q_{\mathrm{src}} = 0.00220$ is one-loop-EM-sized ($= \varphi_0/24$ to 0.6%, $= \alpha/\pi$ to ∼5%), the PDG pole sits at 2/3 (the IR attractor); **[C]** a 1-loop QED pole$\leftrightarrow\overline{\mathrm{MS}}$ per-lepton-log estimate shifts Q by EM-loop order toward 2/3 (the `v82` direction); **[O]** the exact flow time $t \approx 2.84$ (`v93`) is scheme/scale-dependent external QED, only sized. **[C]** verdict: F_{pole} is a one-loop EM effect — the lens forces the rate $(2/3)^6$ (`v327`), external EM supplies the flow; `F_pole` stays **[C]**, firewalled (`v187`), so F_{transfer} is confirmed genuinely external — the open theme sharpened (QED), not closed. Cited in `tfpt_4`; numerical (mpmath), Python-only (no Wolfram check, by the suite convention like `v93`/`v371`).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Koide section of `tfpt_4`; manifests refreshed.

2026-06-25 (The seam μ_4 is the carrier's Galois group: `v419` + figure)

- **New module `v419` (`SEAM.GALOIS.01`).** A positive resolution of the cyclic/Galois asymmetry (`v409`, `RES.COXETER.SYMMETRY.01`): why the seam μ_4 sits on the Galois side, not on the cyclic ζ_{30} dial. **[E]** $h(E_8) = 30$ is squarefree, so $\mathbb{Z}/30$ has element orders $\{1,2,3,5,6,10,15,30\}$ — *no* order 4, so the seam μ_4 cannot be a rotation hand and is forced onto the Galois side; **[E]** by CRT $(\mathbb{Z}/30)^\times = (\mathbb{Z}/2)^\times \times (\mathbb{Z}/3)^\times \times (\mathbb{Z}/5)^\times = \mu_4 \times \mathbb{Z}_2$ (order $\varphi(30) = 8 = \mathrm{rank}\,E_8$),

and the μ_4 ($\mathbb{Z}/4$) factor IS $(\mathbb{Z}/5)^\times$ (the carrier prime 5), the $\mathbb{Z}_2$ IS $(\mathbb{Z}/3)^\times$ (family 3 = CP conjugation, `v316`); **[E]** so the seam $\mu_4 = \mathrm{Gal}(\mathbb{Q}(\zeta_5))$, realised on the carrier clock C_5 (`v418`) by the Frobenius $\zeta_5 \mapsto \zeta_5^2$ — the explicit order-4 operator G with $GC_5G^{-1} = C_5^2$, $G^4 = I$, $G^2C_5G^{-2} = C_5^{-1}$. **[C]** this is the positive content of `v409`: the 2^2=4 is the order of $(\mathbb{Z}/5)^\times$, and "no contraction rate" is because the seam is an *automorphism*, not a hand; so $\mathbb{Q}(\zeta_{30})$ carries both E_8 invariants (cyclic h=30, Galois rank =8). Cited in `origin_theory`; Wolfram-mirrored (321 → 324).

- **New figure `coxeter_galois.pdf`.** The order-30 cyclic dial with the four flavor hands (sheet/family/carrier/CP) beside the Galois gears ($\mu_4 = \mathrm{Aut}(\zeta_5)$ the seam, $\mathbb{Z}_2 = \mathrm{Aut}(\zeta_3)$ the CP conjugation); added to `make_figures.py`, the manifest figure list and `origin_theory`.
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Wolfram extension/README/ledger counts (324/324) and the website verification count; manifests refreshed.

2026-06-25 (The cyclotomic norm triple + the carrier-5 clock: `v418`)

- **New module `v418` (`ARITH.NORMTRIPLE.01`).** The carrier split $(3, 2)$ read in the three atom-rings gives $(7, 13, 55)$ =(scalaron, Δ_Q, quark numerator), and the carrier-5 clock missing from `v417` is found as the 4×4 Φ_5-companion C_5. **[E]** C_5^5=I, $\mathrm{tr}\, C_5$=−1, $\det C_5$=1 (the four primitive 5th roots), with the golden ratio as its real-part data ($2\cos 72^\circ = 1/\varphi$, $2\cos 144^\circ = -\varphi$; output-side, `v313`/`v349`; dim 4, not 3 — this refines `v417`'s gap from "no operator" to "no 3×3 operator"); **[E]** $N_{\mathbb{Z}[\zeta_5]}(3+2\zeta_5) = \det(3I+2C_5) = 55 = 5\cdot 11 = g_{\mathrm{car}} \| \mathrm{Pl}(K) \|_1$, the c_u/c_d numerator (`v410`/`v411`); **[E]** the triple $\det(3I+2\,\mathrm{Comp}(\Phi_n)) = (7, 13, 55)$ for n=3, 4, 5 over $\mathbb{Z}[\omega], \mathbb{Z}[i], \mathbb{Z}[\zeta_5]$ (atoms 3, 2, 5, `v416`); **[E]** NEG control: the anchor $(5, 4) \to (21, 41, 461)$ reaches named values only in the ω- and i-rings (21=$N_\omega(5,4)$, 41=$10b_1$, `v222`) but 461 (prime) in the carrier ring, so the carrier ring separates $(3, 2) \to 55$ from $(5, 4) \to 461$. **[C]** honest: $(3, 2)$ is the *forced* carrier split (`v14`), not the unique clean one — a scan finds $\{(2, 1) \to (3, 5, 11), (3, 2) \to (7, 13, 55)\}$, a small ladder. The sharper discriminator is *cross-sector*: only $(3, 2)$ spans three physics sectors (gravity 7, flavor 13, masses 55), the unique cross-sector seed, while $(2, 1)$ is all compiler-core. Cited in `origin_theory`; Wolfram-mirrored (318 → 321).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, a triple paragraph in `origin_theory`, the Wolfram extension/README/ledger counts (321/321) and the website verification count; manifests refreshed.

2026-06-25 (The Eisenstein/CP operator + the order-30 clock: `v417`)

- **New module `v417` (`DIAMOND.EISEN.01`).** The hexagonal dual of `v415`: the τ=ω flavor side put on operators and unified with the seam into one order-30 Coxeter clock. **[E]** the family rotation $P = (1\,2\,3)$ is the order-3 Eisenstein deck (P^2+P+I = ONES, so ω^2+ω+1=0 on the plane $\perp (1, 1, 1)$), the order-3 dual of J^2=−I; **[E]** it realises the hex CM norm as $(3I+2P)(3I+2P^2) = 7I+6\,\mathrm{ONES}$, Spec$\{7, 7, 25\}$ (7=$N_{\mathbb{Z}[\omega]}(3+2\omega)$= scalaron, 25=$g_{\mathrm{car}}^2$) — the dual of `v415`'s $\{13, 13, 9\}$; **[E]** the CP clock W=−P^2 (ρ of `v233`) has W^2=P, W^3=−I, Spec$\{-1, \zeta_6, \zeta_6^{-1}\}$, giving $\delta_{\mathrm{CKM,lead}} = \arg\zeta_6 = \pi/3$ and $\delta_{\mathrm{PMNS}} = \arg\zeta_6^4 = 4\pi/3$ (`v316`/`v320`); **[E]** all flavor clocks are powers of ζ_{30} (ζ_{30}^{15}=−1, ζ_{30}^{10}=ω, ζ_{30}^{6}=ζ_5, ζ_{30}^{5}=ζ_6) while the seam μ_4=i is the Galois side $(\mathbb{Z}/30)^\times$ of order $\varphi(30)$=8= rank E_8 — so $\mathbb{Q}(\zeta_{30})$ carries both E_8 invariants (h=30 and rank =8). **[C]** honest gap: the carrier μ_5 has no 3×3 operator (the golden φ is an output-side trace, `v313`/`v349`); negatives: no U, V-diamond operator carries ζ_6, $[D, P] \neq 0$ (no μ_{12}), $N(3+2\zeta_6)$=19≠7. Cited in `tfpt_2`; Wolfram-mirrored (315 → 318).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Sheet-Diamond keybox of `tfpt_2`, the Wolfram extension/README/ledger counts (318/318) and the website verification count; manifests refreshed.

2026-06-25 (The Gaussian operator + the atom trichotomy: v415, v416)

- **New module v415 (DIAMOND.GAUSS.01).** The square CM-norm dictionary of v222/v230 is *operator-realised*, and the μ_4 deck becomes an intrinsic diamond object. [**E**] the integer commutator $J := \frac{1}{3}[U,V] = \left(\begin{smallmatrix}-1&1&0\\-2&1&0\\-3&1&0\end{smallmatrix}\right)$ has $\mathrm{Spec}(J) = \{i, -i, 0\}$, $J^2 = -I$ on the rank-2 image and $J^3 + J = 0$ — a μ_4 quarter-turn born from the binary V and ternary U compilers (v410/v95); [**E**] $3I + 2J$ and $5I + 4J$ have spectra $\{3{+}2i, 3{-}2i, 3\}$, $\{5{+}4i, 5{-}4i, 5\}$, so the Gaussian integers $3{+}2i$ (norm $13{=}\Delta_Q$) and $5{+}4i$ (norm $41{=}10b_1$) are *literal eigenvalues*; [**E**] $(aI{+}bJ)(aI{-}bJ)$ reads $(a^2{+}b^2)$ on the μ_4-plane and the real part^2 on the kernel ($\{13, 13, 9\}$, $\{41, 41, 25\}$); [**E**] the intrinsic order-4 deck $D = -I + J - J^2$ ($D^4 = I$, $\chi_D = (x{+}1)(x^2{+}1)$) has χ_2-line $\ker[U,V] = a - \mathbf{1}$. [**C**] the geometric-deck identification sharpens (does not close) the v140 $\mathbb{Z}_3$ residue. Cited in tfpt_2; Wolfram-mirrored.
- **New module v416 (ARITH.TRICHOTOMY.01).** The atoms $\{2, 3, 5\}$ are ramified/inert/split in the two exceptional CM rings. [**E**] in $\mathbb{Z}[i]$ (square/seam) $2{=}|\mathbb{Z}_2|$ ramifies, $3{=}N_{\mathrm{fam}}$ is inert, $5{=}g_{\mathrm{car}}$ *splits* as $(2{+}i)(2{-}i)$; [**E**] in $\mathbb{Z}[\omega]$ (hex/flavor) $3{=}N_{\mathrm{fam}}$ ramifies, $2, 5$ are inert; [**E**] the ramified prime of each ring is its own atom; [**E**] the three quadratic facets $\mathbb{Q}(i), \mathbb{Q}(\sqrt{-3}), \mathbb{Q}(\sqrt{5})$ of $K = \mathbb{Q}(i, \sqrt{-3}, \sqrt{5})$ (v403) ramify over exactly one atom each, product $2{\cdot}3{\cdot}5 = 30 = h(E_8)$, two imaginary (CM/dynamic) + one real (RM/static). [**C**] the seam/flavor/golden reading. Cited in origin_theory; Wolfram-mirrored.
- **Wolfram extension** $308 \to 315$ **+ a pre-existing fix.** Added the seven exact checks for v415/v416 and corrected a one-character sign bug in the v412 CharacteristicPolynomial check (Wolfram returns $-\chi$ for odd rank), so tfpt_readouts_extension.wl now runs genuinely clean at 315/315 (GATE.WOLFRAM.02); README and ledger counts updated.
- **Surfaces synced.** run_all.py, script_registry.csv and status_ledger.csv updated; a new keybox paragraph in the Sheet-Diamond section of tfpt_2 and a trichotomy sentence in origin_theory; manifests refreshed.

2026-06-24 (The sheet generator V as a binary internal compiler: v410–v414)

- **New module v410 (SHEET.GEN.BINARY.01).** The rank-2 sheet axis $V = Q\,\mathrm{diag}(0, 1, 1)$ of the centered cross (v95/v218) is a binary internal compiler: its powers print the carrier spine. [**E**] the closed form $V^n = \left(\begin{smallmatrix}0&2^{n-1}&0\\0&2^n&0\\0&2^{n+1}{-}2&1\end{smallmatrix}\right)$ (proved by the inductive step $F(n)V{=}F(n{+}1)$), so $V^n\mathbf{1} = (2^{n-1}, 2^n, 2^{n+1} - 1) = (1,2,3), (2,4,7), (4,8,15), (8,16,31)$; [**E**] the four bilinear families $\mathbf{1}^T V^n \mathbf{1} = 7 \cdot 2^{n-1}{-}1$, $\mathbf{1}^T V^n a = 7 \cdot 2^{n-1}$, $a^T V^n a = 11 \cdot 2^{n-1}$, $a^T V^n \mathbf{1} = 11 \cdot 2^{n-1}{-}2$ give $(6, 7, 9, 11)$ at $n{=}1$ and $\mathbf{1}^T V^2 \mathbf{1}{=}13{=}\Delta_Q$, $\mathbf{1}^T V^3 \mathbf{1}{=}27{=}\mathbf{1}^T R a$, $\mathbf{1}^T V^4 \mathbf{1}{=}55$, $\mathbf{1}^T V^4 a{=}56{=}\dim \mathbf{56}_{E_7}$; [**C**] the binary spine is FORCED by $\mathrm{Spec}(V){=}\{0, 1, 2\}$, so the carrier/Lie matches ($\mathrm{rank}\, E_8{=}8$, $\dim S^+{=}16$, $248/8{=}31$) and the theta cross-link ($\sigma_3(2){=}9{=}a^T V \mathbf{1}$, $\sigma_3(3){=}28{=}\mathbf{1}^T V^3 a{=}\det(I{+}R)$, $\sigma_3(5){=}126$) are audit-typed (the v95 G_2-center typing).
- **New module v411 (FLAV.UD.VPOWER.01).** The quark ratio is a pure V-power readout, $c_u/c_d = (\mathbf{1}^T V^4 \mathbf{1})/((a^T V \mathbf{1})(\mathbf{1}^T V^2 \mathbf{1})) = 55/(9 \cdot 13) = 55/117$, identical to the established $g_{\mathrm{car}} \|\mathrm{Pl}(K)\|_1/(N_{\mathrm{fam}}^2 \Delta_Q) = 5 \cdot 11/(9 \cdot 13)$ (v94). [**E**] the identity; [**C**] an exact *re-encoding* (no new physical input) — the value and the u/d identification stay conditional, coupled to the (U_{wall}) readout rigidity.
- **New module v412 (DIAMOND.SHEET.SOURCE.J.01).** Names the operator on the $\mathbb{Z}_2$ wall ($t{=}{-}2$, $\det{=}2$, v135), $J{=}M(1,{-}2){=}C{-}2V$, completing the sheet axis $J, K, C, F{=}M(1,{-}2..1)$. [**E**] $\chi_J = \lambda^3 - 6\lambda^2 + 3\lambda - 2 \Rightarrow (\mathrm{tr}, e_2, \det){=}(6, 3, 2)$; $a^T J a{=}30{=}h(E_8)$, $\det(I{+}J){=}12{=}\dim \mathfrak{g}_{\mathrm{SM}}$ and $\det(2I{+}J){=}40{=}|R(D_5)|$; [**C**] the Lie readings audit-typed like the G_2/F_4 shadows.
- **New module v413 (DIAMOND.SHEET.CALCULUS.01).** On $M_t{=}C{+}tV$ the elementary-symmetric coefficients encode the atoms as difference orders: $e_1{=}3t{+}12$, $e_2{=}2t^2{+}15t{+}25$,

$e_3{=}4t^2{+}14t{+}14$ give $\Delta e_1{=}3{=}N_{\rm fam}$, $\Delta^2 e_2{=}4{=}|\mu_4|$, $\Delta^2 e_3{=}8{=}\operatorname{rank} E_8$ (the e_3 curvature is the `v218`/`v224` sheet quadratic; the full triple is new); **[E]** the anchor energy $a^T M_t a{=}52{+}11t$ ($30{\to}41{\to}52{\to}63$, step $11{=}a^T V a$); **[C]** atom/Lie readings audit-typed.

- **New module `v414` (`DIAMOND.CENTER.RESOLVENT.01`).** The center $C{=}M(1,0)$ is a resolvent portal: **[E]** $\det C{=}14{=}\dim G_2$, $\det(I{+}C){=}52{=}\dim F_4$, $\det(2I{+}C){=}120{=}|R^+(E_8)|$, the exceptional shell ladder $G_2{\to}F_4{\to}E_8^+$ as $\det(C{+}kI)$, $k{=}0,1,2$ (with $\operatorname{tr} C{=}12{=}\chi_{E_8}(i)$, $\sum C{=}31{=}2^{g_{\rm car}}{-}1$, `v95`); **[C]** the portal reading.
- **Surfaces synced.** Integrated into the Sheet-Diamond section of `tfpt_2` (new keybox, five `\veri` citations) and the E_8 theta-raster of `tfpt_3`; `run_all.py`, `script_registry.csv` and the ledger updated; Wolfram extension mirror 303 → 308.

2026-06-24 (Two new tracked targets + axion-branch consistency: `v408`, `v409`)

- **New module `v408` (`HYP.PHI0.PUNCTURE.01`).** Names the φ_0 puncture term $48c_3^4$ as the order-$|\mu_4|{=}4$ local boundary heat-kernel (Seeley–DeWitt) contact term of the μ_4 puncture divisor, summed over $\Omega_{\rm adm}{=}N_{\rm fam}\dim S^+{=}48$ admissible states — the analytic complement of the icosahedral tree term (`v396`). **[E]** the exact split $\varphi_0{=}1/(6\pi){+}48c_3^4$, the puncture count, the closed form $3/(256\pi^4)$ and the hypergraph negative (reused from `v396`/`ARCH.QUAD.01`); **[O]** the from-first-principles heat-kernel proof; **[C]** a FACE of the seam boundary analysis (the c_3 Gauss–Bonnet datum), not a new gate. Cited in `origin_theory`; no new number.
- **New module `v409` (`RES.COXETER.SYMMETRY.01` closed as a structural lemma).** prime-2 (the seam sheet involution) is *necessary* but *not autonomous*: $\sigma{=}\operatorname{diag}(1,{-}1,{-}1)$ has spectrum $\{\pm1\}$ and projector $(I{+}\sigma)/2$ spectrum $\{0,1\}$, none in $(0,1)$, so prime-2 carries no contraction rate, whereas the genuine rates $2/3$ and $(1/3)^6$ lie in $(0,1)$ and carry prime-2 only as a factor. **[E]** no prime-2-only attractor exists; **[C]** dynamics begins only after coupling to family-3 or carrier-5 ($|\mu_4|{=}2^2$ the Galois bridge). Resolves the `v394` open question's falsifiable corollary; cited in `tfpt_research_contracts`; the ledger row `RES.COXETER.SYMMETRY.01` retyped **[O]** → **[E]**/**[C]**.
- **Axion branch consistency (`tfpt_4_frontier`, website `predictions.ts`, `v185`).** The dark-matter section's opening + keybox are realigned with the finite-T result already stated later in the same section: the robust dominant-DM branch is the *spine* angle $\theta_i{=}\pi N_{\rm fam}/g_{\rm car}{=}3\pi/5{=}108^\circ$ (untuned, in band), while the seam hilltop $\theta_i{\approx}170^\circ$ over-produces ($\sim 5.4\times$) and is disfavoured; the misalignment angle is retyped from a closed **[E]** input to a **[C]** branch decision. The website prediction card now leads with the spine angle.
- **Lean closure of the prime-2 corollary (`FORM.COXETER.PRIME2.01`).** The v409 result is now also machine-proved in Lean 4 / Mathlib (new module `TfptCarrier/CoxeterPrime2.lean`): an involution eigenvalue ($r^2{=}1$) is ±1, a projector eigenvalue ($r^2{=}r$) is 0 or 1, neither lies in the open interval $(0,1)$, while $2/3$ does — so *no prime-2-only attractor exists.* Built clean (`lake build`); the new theorems are axiom-pure (`#print axioms` = `propext`, `Classical.choice`, `Quot.sound` only) and pass the `AuditCheck`/`AuditContract` signature locks. This lifts `RES.COXETER.SYMMETRY.01` to the machine-proof tier.
- **Verified, no change (Plücker 11).** The quark constant $11{=}\|Pl(K)\|_1{=}\det(\mathbf{1}|a|\sigma)$ is already typed honestly as an **[E]** identity with "why 11" the lone **[O]** residue (`v407`, `tfpt_1`); the repo carried no overstatement, so no edit was required.

2026-06-24 (Cross-surface consistency sweep: papers, website, gates, figures realigned)

- **Ledger hygiene.** Resolved a duplicate active `claim_id`: `GATE.UWALL.06-09` was carried by two distinct chains. It is an ID collision (not a supersede): the monodromy/residue chain

v114–v117 keeps .06-.09 (as asserted by v123_inventory_update.py and run_all.py); the older exterior-leg chain v42–v45 is renumbered to GATE.UWALL.15-.18 (internal depends_on carried along). No claim content changed.

- **Website.** papers.ts document count 8 → 9 (the Red-Team paper was missing from the tally); Red-Team subtitle/highlight *Targets A–E→A–F* with a new Target F section + contribution bullet (the abstract already carried F); StatusMatrix.tsx *Doc 7→Doc 8* for the research contracts and the residual interfaces re-typed v_{geo} **[O]**, G_{net} **[C]**, F_{transfer} **[C]**; objections *Paper 1→Doc 1.*
- **Papers (marker/wording alignment, no canonical status change).** tfpt_1 Strong CP **[E]→[E]/[C]** (given RP) and θ_{12} "genuinely open"→"conditionally derived"; tfpt_2 Starobinsky A_s **[E]→[E]/[C]** (M exact, A_s a band over $N_\star \in [50, 60]$) and the Part-II abstract "opens the one remaining gap"→"closes the last structural gap"; tfpt_research_contracts the QFT4D "nonperturbative measure stays open" line and the "only U_{point} stays open" headline reconciled with the **[C]** redundancy / v_{geo} reduction stated alongside; origin_theory class C re-typed to the ambient-measure **[C]** redundancy; introduction (U_{wall}) → v_{geo}. Forbidden old-paper attributions ("Papers 1–7", "Papers 2/6", "Paper-7", "Paper-1") replaced by current document / neutral wording (the tfpt_3 cascade bridge stays exempt).
- **Suite README.** The residual-gates table (Alessandro-5.0 vintage) brought to the current model (v_{geo} **[O]**; G_{net}/SEAM.EQUIV.01, F_{transfer}, $N_\star$, QG.AMB.01 **[C]**; Wolfram 116/116 + 303/303).
- **Figures.** Regenerated status_heatmap (ledger-driven) and extended the narrative figures script_timeline/residual_chain/qft_skeleton through the v303–v407 arc (typed solvers, parameter-free local Einstein equation v359, QG.AMB.01 **[C]** redundancy, seam closed modulo cited theorems); the qft_skeleton "Stelle ghost" open box is retyped (resolved truncation artefact). Added qft_skeleton/qft_unification to the manifest figure list.

2026-06-24 (The $\tau{=}\omega$ dual keystone + the flavor residuals it homes: v405–v407)

- **[E]/[C]/[O]** v405_seam_equiv_omega.py (SEAM.EQUIV.02): the *dual keystone* at $\tau{=}\omega$ – the family/flavor sector as the order-3 Eisenstein/A_2 face of E_8, dual to the seam = $(E_8)_1$ at $\tau{=}i$ (v404). A named contract (genre of v286/v398), NOT a closure. **[E]** $\chi_{E_8}(\omega) = 0$ (the family CM point degenerates E_8) vs $\chi_{E_8}(i) = 12$; **[E]** A_2 is the $\tau{=}\omega$ Eisenstein lattice (Cartan det $= 3 = N_{\text{fam}}$, $h = 3 = N_{\text{fam}}$); **[E]** R lives in $E_6{\times}A_2$ ($248 = 78 + 8 + 2{\cdot}27{\cdot}3$, $\det R = 8 = \dim A_2 = \operatorname{rank} E_8$); **[E]** $\delta_{\text{PMNS}} = 4\pi/3$ is the $\tau{=}\omega$ CM phase; **[E]** the cusp weights $\{0, 1/3, 2/3\}$ are the order-3 deck = $N_{\text{fam}} = \det Q$. **[C]** the family sector is the order-3 Eisenstein/A_2 deck; the two keystones sit at the only two elliptic points of $\mathrm{PSL}(2, \mathbb{Z})$. **[O]** residual = the canonical $\mathbb{Z}_3$ deck map (v140) + the v_{geo}-class scale. Cited in origin_theory; Python-only.
- **[E]/[C]/[O]** v406_pmns_phase_origin.py (FLAV.PMNS.CLOSE.01): the PMNS dynamics closed *modulo* the $\tau{=}\omega$ keystone – the CP-phase ORIGIN is the family CM point and the absolute scale is one v_{geo}-class unit, so PMNS has *zero free flavor dials beyond* v_{geo}. **[E]** the three angles (v9/v268) → unitary U_{PMNS} (v270); **[C]** $\delta_{\text{PMNS}} = 4\pi/3$ is the $\tau{=}\omega$ hexagonal CM phase (v405), upgrading v270's **[O]** "assembled input" to a keystone consequence; **[C]** $J_{\text{PMNS}} = -0.0297$ derived; **[E]** the absolute scale = one seesaw ratio = v_{geo}-class (v272/v153). **[O]** residual = the seesaw operator realisation (M_R, v184) + the Σm_ν floor kill test. Cited in tfpt_2_standard_model; Python-only.
- **[E]/[C]/[O]** v407_dn_pairings_omega.py (FLAV.SELECTOR.CLOSE.01): the R4′ selector residue folds into the $\tau{=}\omega$ keystone – the three (d, n) pairings ARE the $\tau{=}\omega$ family-slice atoms. **[E]** the frame $(\mathbf{1}, a, \sigma)$ has det $= 11 = \|Pl(K)\|_1$ and $n = (5, -9, 6)$ is the unique covector with $n{\cdot}\mathbf{1} = 2$, $n{\cdot}a = 8$, $n{\cdot}\sigma = 121 = 11^2$ ($\Rightarrow (d, n) \Rightarrow R$, v136/v139); **[E]**

$n{\cdot}a = 8 = \operatorname{rank} E_8 = \dim A_2 = \det R$ (the A_2 block reading R in $E_6{\times}A_2$, `v405`); **[E]** the pairings $\{2, 8, 121\} = \{|Z_2|, \dim A_2{=}\det R, \|Pl(K)\|_1^2\}$. **[C]** deriving (d, n) folds into `SEAM.EQUIV.02` (d anchor-derived, `v139`); **[O]** the lone residue "why $11{=}\|Pl(K)\|_1$". Cited in `tfpt_1_architecture_e8`; Python-only.

2026-06-24 (The arithmetic hull + the E_8-character CM duality: `v403`, `v404`)

- **[E]/[C]** `v403_facet_compositum.py` (`ARITH.HULL.01`): the three single-prime number-field facets of the order-30 clock (`v390`/`v394`, used so far only one at a time) *compose* to one field $K = \mathbb{Q}(i, \sqrt{-3}, \sqrt{5})$. **[E]** the squarefree parts $-1, -3, 5$ are independent in $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$, so $[K : \mathbb{Q}] = 2^3 = 8 = \operatorname{rank} E_8 = g_{\text{car}}{+}N_{\text{fam}}$ with Galois group $(\mathbb{Z}/2)^3$ (order 8); **[E]** exactly 7 quadratic subfields $\{-1, -3, 5, 3, -5, -15, 15\}$, splitting 4 imaginary $= |\mu_4|$ (CM/dynamic) + 3 real $= N_{\text{fam}}$ (RM/static); **[E]** ramified *only* over $\{2, 3, 5\}$ = the atoms = the primes of $h(E_8) = 30$; **[E]** $2^{\omega(h)} =$ rank is E_8-special (the E_7 control $h = 18$ gives $2^2 = 4 \neq 7$). **[C]** K is the abelian $(\mathbb{Z}/2)^3$ *arithmetic* hull dual to the E_8 *lattice* hull; the 7 quadratic subfields are a candidate for the 7 E_8 audit slices. A synthesis composing the established facets, no new number; cited in `origin_theory`; Python-only by the synthesis-module convention (cf. `v390`/`v394`).
- **[E]/[C]** `v404_e8_char_cm_duality.py` (`E8.CMDUAL.01`): the $(E_8)_1$ vacuum character $\chi_{E_8} = E_4/\eta^8 = j^{1/3}$ read at *both* CM points, extending `v282` ($\chi_{E_8}(i) = 12$) to the second elliptic point. **[E]** $\chi_{E_8}^3 = j$ (q-series); **[E]** SEAM $\tau{=}i$ ($\mathbb{Q}(i)$, order 4): cross-ratio $2 \Rightarrow j = 1728 \Rightarrow \chi_{E_8}(i) = 1728^{1/3} = 12$ (E_8 present, `v214`/`v282`); **[E]** FAMILY $\tau{=}\omega$ ($\mathbb{Q}(\sqrt{-3})$, order 6): the equianharmonic λ ($\lambda^2{-}\lambda{+}1 = 0$) $\Rightarrow j = 0 \Rightarrow \chi_{E_8}(\omega) = 0$ (E_4 vanishes, `v220`/`v267`); **[E]** $\tau{=}i, \tau{=}\omega$ are the *only* elliptic points of $\mathrm{PSL}(2, \mathbb{Z})$, so the values are forced. **[C]** the seam/flavor rigidity duality: the seam is rigid where E_8 is present ($\chi = 12$, the keystone `SEAM.EQUIV.01` closes) while the flavor/CP sector is the soft residual where E_8 degenerates ($\chi = 0$: U_{wall}, v_{geo}, the CP texture), a candidate dual keystone at $\tau{=}\omega$. Cited in `origin_theory`; core values $j = 1728/j = 0$ already mirrored (`v214`/`v220`/`v267`/`v282`), Python-only.

2026-06-24 (The five TOE/QFT closure contracts `v398`–`v402` + a ledger rescope)

- **[E]/[O]** `v398_seam_state_rigidity.py` (`SEAM.RIGIDITY.01`, Paper A): the *Seam State Rigidity Theorem* as ONE named target, consolidating the scattered state-invariance reductions (`v177`/`v199`/`v308`/`v309`/`v335`) and *hardening* the finite-block evidence to a multi-mode sweep. **[E]** RP-definability hinge (OS canonical transfer $T{=}e^{-H}$ from RP + reflection alone, `v54`, quasi-free `v155`, no μ_4 normal form); **[E]** band-limited character-block-diagonal core $[\rho, |k|]{=}0$ swept $N{=}4..64$ (beyond the 3-dim H^1); **[E]** holomorphy $\Rightarrow (E_8)_1$ ($c{=}8$, $|\det \text{Cartan}|{=}1$ vs 4, index $4{=}|\mu_4|$). **[O]** the one residual: $[\rho, \Lambda_\Sigma]{=}0$ on the full L^2 (intrinsic Bisognano–Wichmann). NOT a closure of `SEAM.EQUIV.01` (continuum = cited MMST `v336`).
- **[E]/[C]/[O]** `v399_gravity_complete.py` (`GRAVITY.COMPLETE.01`, Paper B): the explicit verdict *Gravity-complete = Boundary-complete + Ambient-redundancy*, combining the parameter-free local equation (`v358`/`v359`) with the ambient redundancy (`v369`). **[E]** the five gravitational readout classes (Newton 8π, Λ, scalaron c_3^7, horizon $1/|\mu_4|$, recovery $(2/3)^6$) are all boundary/seam/gap, so $\mathcal{O}_{\text{phys}} \subset \mathcal{A}_\Sigma$; **[O]** residual = `SEAM.EQUIV.01` + intrinsic BW (no new gate).
- **[E]/[O]** `v400_grav_nonlocal_action.py` (`GRAV.NONLOCAL.01`, Paper C): the nonlocal spectral-gravity action with the local $R{+}R^2$ as its IR projection (consolidating `v304`/`v370`/`v380`/`v386`, adding the IR-matching check). **[E]** entire $a{=}e^u \Rightarrow$ one pole (ghost-free), every truncation carries a Stelle zero (monotone, `v380`), and the IR order reproduces $R{+}R^2$ with the atom-fixed scale $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2{=}c_3^7$ (`v36`/`v253`). **[O]** the entire-a assumption + perturbative-only.

- **[E]/[O]** `v401_metrology_closure.py` (`METROLOGY.CLOSURE.01`, Paper D): the final input set $\{a, \pi, v_{\rm geo}\}$ certified — 0 dimensionless dials + 1 unit (consolidating `v153`/`v274`/`v364`/`v384`). **[E]** dimensional bookkeeping mass/$1G/\Lambda/U_{\rm point}/(m/\mu)$= number $\times v_{\rm geo}^{1,2,4,1,0}$; **[E]** the G-vs-Λ over-determination 0.11% (`v274`); **[O]** $v_{\rm geo}$ theorem-forbidden (TOE$_{\rm strict}$=TOE$_{\rm dimensionless}$+$[v_{\rm geo}]$).
- **[E]/[C]** `v402_ftransfer_suite.py` (`FTRANSFER.SUITE.01`, Paper E): the four frontier sectors as ONE functor $F_{\rm transfer}$, the composition $F_{\rm obs} \circ F_{\rm thr} \circ F_{\rm RG}$ (consolidating the `FR.*.SOLVE` solvers), each with an exact atom source, a committed kill test, and the firewall (physical prediction stays **[C]**/**[O]**, only the algebraic source **[E]**; never re-pressed into the compiler, `v187`).
- **Ledger rescope (repo hygiene):** `GATE.METRIC.01` no longer reads "full covariant metric-sector field equation open" — the *local* covariant equation is closed parameter-free (`GRAV.NONLINEAR.01`/`v359`); the gate is rescoped so the open item is *only* the global nonperturbative ambient measure (`QG.AMB.01`, a **[C]** redundancy `v369`), removing the stale contradiction with the status card.

2026-06-24 (φ_0 leading term from icosahedral combinatorics: `v396`)

- **[E]/[C]** `v396_phi0_icosahedral.py` (`HYP.PHI0.01`): the tree-level seed $\varphi_0^{\rm tree} = 1/(6\pi)$ is exactly $F/(4h\pi) = F/(|\,\mathrm{Aut}\,|\pi) = (F/(g_{\rm car}N_{\rm fam}))c_3$ on the icosahedral hypergraph ($F = 20$ triangular hyperedges, $h = 30$, $|\,\mathrm{Aut}\,| = 120$); $F/h = |Z_2|/N_{\rm fam} = 2/3$ (the same survival ratio as `v327`). **[E]** Gauss–Bonnet consistent with $c_3 = 1/(|Z_2|4\pi)$ (`v216`/`v342`). **[E]** honest negative: the puncture $48c_3^4$ is still analytic, not a hypergraph fraction. **[C]** $\varphi_0^{\rm tree}$ is a combinatorial reinterpretation, not a new independent injection; the puncture stays **[O]**.

2026-06-24 (Coupled hypergraph rewrite: `v395` unifies carrier × family in one local rule)

- **[E]** `v395_hypergraph_coupled.py` (`HYP.COUPLED.01`): one coupled local rewrite on a 9×3 labelled grid (affine-E_8 nodes × family channels) unifies `v299` (carrier growth), `v327` (branching rule M) and `v324` (fiber product). Micro-step = network lazy diffusion on each cusp column + M on each node's family vector ($T_{\rm micro} = T_{\rm net} \otimes M$); joint attractor marks$\otimes e_0$; one clock hand (6 family steps) gives recovery $(2/3)^6$; `v324`'s $T_{\rm net} \otimes T_{\rm cusp}$ emerges as the cusp-readout basis. **[O]** the 3-arm seed (P_2) and analytic φ_0 remain irreducible (`v312`).

2026-06-24 (A falsifiable clock probe of the frontier: `v397` how external is the "external physics"?)

- **`v397` (`FR.CLOCK.PROBE.01`) — do the "external" frontier scheme-numbers land on the $\{2, 3, 5\}$ clock?** A falsifiable, anti-numerology probe (strict No-Free-Pattern, `v100`: "on the clock" means an *exact* match, not a wide-band fit). **[E]** the η_B decuple $A_\Lambda = 10 = |Z_2|g_{\rm car}$ is ON the clock (`v212`); **[E]** the Koide rate $(2/3)^6 = (|Z_2|/N_{\rm fam})^6$ is ON the clock (`v303`); **[E]** the axion spine angle $\theta_i = \pi N_{\rm fam}/g_{\rm car} = 3\pi/5$ is ON the clock (`v211`/`v373`); **[E]** the proton $C_p \sim 2.83 \pm 0.15$ is *not* a discriminating clock value — within its ±5.3% band several simple values fit $(2^{3/2}, 17/6, 14/5)$, so by No-Free-Pattern it is declared EXTERNAL (the firewall `v187`/`v374` is correct there). **[C]** verdict: 3 of the 4 frontier scheme-numbers land *exactly* on the clock, so the irreducible "external physics" residue shrinks to essentially the proton's non-perturbative QCD number C_p. Cited in `tfpt_4_frontier`; a falsifiable probe whose answer could have been "none on the clock"; Python-only.

2026-06-24 (Correction error-bars: v393 the typed budget as actual first-correction magnitudes)

- **v393 (CORRECTIONS.NUMERIC.01) — the typed correction budget (v388) as actual numbers.** The short-term, testable harvest of v388: instead of merely typing each prediction's correction, this gives the magnitude, so a parameter-free central value can be quoted as "value ± computed first correction". **[E]** FIXED-POINT band (the φ_0-derived $\theta_{12}, \theta_{13}, \lambda_C, \beta_{\rm rad}$) = the explicit φ_0 puncture term $36c_3^4/c_3 = 9/(128\pi^3) \approx 0.227\%$ (exact; = θ_{12}'s ε first-correction, $\varepsilon = (3/4)\varphi_0 = c_3 + 36c_3^4$); **[E]** SEAM-GAPPED band (Koide $F_{\rm pole}$, recovery) $= (2/3)^6 \approx 8.78\%$, the QG-decoupling bound capped by $\chi - 1 = 64/665 \approx 9.62\%$ (v337); **[E]** EXACT-IDENTITY band = 0 structurally ($\det R = 8$, $N_\Phi = 1$, $\theta_{\rm eff} = 0$); **[E]** EXTERNAL-RATE band = external physics, quoted (m_p/m_e ±7.5% v374; n_s/r the $N_\star \in [50, 60]$ band; η_B 1.07× v212), fenced by v187. **[C]** the assembled budget gives each prediction a computed theoretical-uncertainty band. Cited in origin_theory, mirrored on the website prediction surface; a harvest of v388, no new number; Python-only.

2026-06-24 (Attacking the last open lemma: v392 the scaling-limit central charge *converges*)

- **v392 (SEAM.S3.SCALINGLIMIT.01) — an honest attack on SEAM.EQUIV.01's one open lemma (continuum existence, v336).** It does *not* close v336 (the abstract existence is the cited MMST theorem); it strengthens the case with the existence-relevant observable v376 did not test: that the finite-size data *converge*. **[E]** the Calabrese–Cardy entanglement slope $c_1(L)$ at $L = 66, 130, 258, 386, 514$ is a monotone Cauchy-like sequence approaching 1 ($|c_1(L) - 1|$ strictly decreasing, $4.6 \times 10^{-4} \to 7 \times 10^{-6}$); **[E]** a $1/L$ Richardson fit extrapolates to $c_1(\infty) \approx 1.000$, so the 16-Majorana collar $c = 8c_1 \approx 8.00 = g_{\rm car} + N_{\rm fam}$ is reached *in* the limit; **[E]** the successive differences are themselves strictly decreasing (a numerical Cauchy sequence). **[O]** convergent data are evidence, not a proof of existence (the cited MMST theorem, v336), and do not pin $(E_8)_1$ over the same-c $SO(16)_1$ (the holomorphy discriminator, v377/v378). Strengthens the residual, does *not* close it. Cited in tfpt_research_contracts; Python-only.

2026-06-24 (Lean 4: FORM.SPECTRALGAP.01 extended to the finite-dimensional Perron–Frobenius case)

- **FORM.SPECTRALGAP.01 — the multi-dimensional case, reduced to the scalar core.** The Lean module TfptCarrier/SpectralGapAttractor.lean now also proves the *finite-dimensional* gapped-operator statement (lake build OK, no sorry): **[E]** component_tendsto (a non-dominant eigencomponent $|r| < 1$ decays, $r^n c \to 0$); **[E]** deviation_tendsto (with the dominant eigenvalue normalised to 1, the deviation vector — all non-dominant components — tends to 0, so the iterate converges to the dominant eigendirection, the unique attractor, for any start); **[E]** deviation_unique (two starts give deviation vectors both $\to 0$, so the selected eigendirection is independent of the start — parameter-freeness in finite dimensions); **[E]** flavor_deviation_tendsto (a concrete instance on the cusp-transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$, v56/v82/v383). Honest scope: this is the finite-dimensional *diagonalisable* case (the genuine multi-dimensional content, reduced to the scalar core); the general non-diagonalisable / infinite-dimensional Perron–Frobenius theorem stays the cited statement. Referenced in origin_theory.

2026-06-24 (Website: an animated "Universal Gap Lab" visualising v383–v391)

- **Website — the UniversalGapLab animation on /verification.** A new client component (Framer Motion, no new dependency) with three linked, module-grounded views: (A) every sector as the gap contraction $\text{iter}_n = x_\star + r^n(x_0 - x_\star)$ racing to its attractor at $r = (2/3)^6$

(seam-gapped, v303/v387/v388) or the golden $(\varphi+2)/4$ (v312), with a gapless $r=1$ track that never forgets its start (the Lean theorem FORM.SPECTRALGAP.01) — the rate *is* the first-correction size (v388); (B) the entire-form-factor graviton ratio e^{-u} falling faster than any power, ghost-free and UV-soft (v386/v389); (C) the residual matrix as certification, not construction — every open item external, zero open internal mechanisms (v384). Mirror-only: every number is grounded in a module; no theory change.

2026-06-24 (Honest attempt at an external target: v391 reduces it, not a closure)

- **v391 (ALPHA.QUILLEN.PROGRESS.01) — an honest attempt at the external target ALPHA.QUILLEN.EXACT.01 (v382, door 5).** It REDUCES the open step to a sharper named sub-target via a solvable model but does *not* close it: ALPHA.QUILLEN.EXACT.01 stays [**O**] (external math, type A, v384). [**E**] a solvable model $\det_\zeta(-\partial_\tau^2+m^2)$ on a circle of circumference β equals $(2\sinh(\beta m/2))^2$, whose m-variation $\beta\coth(\beta m/2) = 2/m+(\beta^2/6)m-(\beta^4/360)m^3+\dots$ is a closed heat-kernel/Laurent series (the m^3 coefficient $-1/360$ verified). [**E**] so a ζ-determinant's coupling-variation is structured heat-kernel data, and v382's counterterm $8b_1c_3^6$ (a Gauss–Bonnet heat-kernel power) is the right *kind* of object; the v382 split is re-verified at $\alpha^{-1}=137.0359992$. [**E**] the reduction: the open step now reduces to the named sub-target "the seam $U(1)$ operator's Seeley–DeWitt coefficient $= 8b_1c_3^6$". [**O**] *not closed*: the actual seam-operator ζ-determinant proof stays open, and SEAM.EQUIV.01 continuum existence (MMST, v336) stays the other external [**O**]; the value α^{-1} stays [**E**]. An attempt that reduces, does not close. Cited in tfpt_1_architecture_e8; Python-only.

2026-06-24 (Lean 4: FORM.SPECTRALGAP.01 formalises the scalar core of the universal spectral-gap principle)

- **FORM.SPECTRALGAP.01 — a Lean 4 proof of the gap ⇒ unique-attractor ⇒ parameter-free core (door 4).** New module TfptCarrier/SpectralGapAttractor.lean (lake build OK, no sorry, only the three standard kernel axioms), formalising the meta-theorem of v303/v383/v387 at the one-dimensional reduction — the affine gap contraction $x\mapsto x_\star + r(x-x_\star)$ with multiplier $r=\lambda_2/\lambda_1$. [**E**] iter_sub: the closed form $\mathrm{iter}_n - x_\star = r^n(x_0-x_\star)$ (general n); [**E**] iter_tendsto: $0\le r<1$ (the gap) ⇒ $\mathrm{iter}_n\to x_\star$ (the unique attractor); [**E**] starts_agree: two starts differ by $r^n(a-b)\to 0$, so the attractor is independent of the initial state (parameter-freeness); [**E**] no_gap_no_forget: $r=1$ (no gap) leaves $\mathrm{iter}_n - x_\star = x_0 - x_\star$ constant — the initial condition is then a free parameter, so the gap is exactly what removes it; [**E**] the seam rate $(2/3)^6$ and the golden $(\varphi+2)/4=(5+\sqrt{5})/8$ both lie in $[0,1)$. Honest scope: the scalar reduction (the contraction core v303/v383 iterate numerically); the full multi-dimensional Perron–Frobenius statement stays the cited theorem. Referenced in origin_theory; ledger row FORM.SPECTRALGAP.01.

2026-06-24 (Completing the Coxeter clock: v390 the prime-2 (Gaussian / $\mathbb{Z}/4$-seam) facet of $30=2{\cdot}3{\cdot}5$)

- **v390 (COXETER.PRIME2.01) — the prime-2 facet of the order-30 Coxeter clock.** v383/v387 named two number-field facets of the clock $30=2{\cdot}3{\cdot}5=h(E_8)$ — the golden $\varphi\in\mathbb{Q}(\sqrt{5})$ (prime-5, carrier) and $(2/3)^6\in\mathbb{Q}$ (prime-3, family). This names the *prime-2* facet and identifies it with structure already in the theory: the Gaussian field $\mathbb{Q}(i)$, i.e. the CM point $\tau=i$ and the order-4 $\mu_4=\mathbb{Z}/4$ *square* seam deck (v282/v288/v333). [**E**] $h(E_8)=30$ factors into exactly three primes; one quadratic facet field per prime, each ramified at its prime: $\mathrm{disc}\,\mathbb{Q}(i)=-4$ (2 ramifies), $\mathrm{disc}\,\mathbb{Q}(\sqrt{-3})=-3$ (3 ramifies), $\mathrm{disc}\,\mathbb{Q}(\sqrt{5})=5$ (5 ramifies); the ramified primes are $\{2,3,5\}$ with product $30=h(E_8)$. [**E**] the prime-2 (Gaussian) facet is the $\mathbb{Z}/4$ seam deck: min. poly x^2+1, $j(i)=1728=12^3$, $\tau=i$ fixed by the modular S of order $4=2^2$, $|\mu_4|=4=2^2$ — the static, even, seam-orientation facet. [**C**] so the three primes of

the clock now each carry a facet (square/hexagon/pentagon = seam/CP/carrier); the prime set $\{2,3,5\}$ is forced by $h(E_8)=30$, not chosen. Cited in `origin_theory`; a synthesis, no new number; Python-only.

2026-06-24 (Graviton loop finiteness: v389 the entire form factor is UV-finite at the loop level by power counting)

- **v389 (GRAV.LOOP.FINITE.01) — the entire-form-factor graviton is UV-finite at the *loop* level.** The honest next step past v386 (which established the tree amplitude), extending v304/v370/v380/v386 from the propagator to the superficial degree of divergence of loop graphs. This is the standard infinite-derivative / nonlocal-gravity finiteness statement (Tomboulis; Biswas–Mazumdar–Siegel; Modesto), *not* a full proof of perturbative quantum gravity. [**E**] super-polynomial falloff: $p^k e^{-p^2/M^2} \to 0$ for every degree k. [**E**] one dressed line turns a quadratic divergence into the finite $(M^2/2)e^{-p_0^2/M^2}$, while the GR $\int p^3/p^2\,dp$ diverges quadratically. [**E**] the superficial degree $\to -\infty$ at every loop order L (the form factor multiplies the integrand by $e^{-(\#\text{internal})p^2/M^2}$; dressed cutoff-ratio $\to 1$ vs GR $\to 4$). [**E**] the regulator is the seam scale $M = M_{\rm scal} = c_3^{7/2}\bar{M}_{\rm Pl}$ (v253/v259), not a dial. [**C**][**O**] loop *unitarity* of the nonlocal form factor (Lorentzian continuation / macro-causality, Pius–Sen / Briscese–Modesto) stays the honest open point — tree unitarity is v386; not claimed solved. Cited in `tfpt_4_frontier`; Python-only.

2026-06-24 (Typed correction budget: v388 the gap correction is keyed to the mechanism class, not a uniform band)

- **v388 (CORRECTIONS.BUDGET.01) — the gap-driven correction (v387) is a *typed* budget.** A machine-checked answer to "does the $(\lambda_2/\lambda_1)^n$ correction apply to *all* predictions?" — the answer is *no*, and for one whole class a band would be *wrong*. Because the size is the distance from a gapped operator's leading eigenvector, it is defined only where a subleading λ_2 exists, so the prediction surface splits into four classes. [**E**] *seam-gapped* (Koide $F_{\rm pole}$ v303, recovery v221, the QG bound capped by $\chi = 729/665$ v337): rate $(2/3)^6$; the discrete compiler: golden $(\varphi+2)/4$ (v312). [**E**] *exact identity* ($\det R = 8$, $N_\Phi = 1$, $\theta_{\rm eff} = 0$, $\sin^2\theta_{13} = \varphi_0 e^{-5/6}$): no λ_2, so the correction is 0 structurally and a band would contradict the [**E**] status. [**E**] *external-rate* (η_B washout, axion freeze, m_p/m_e RG): the gapped *shape* with an external rate — only $F_{\rm pole}$ has the seam rate (v303), firewalled (v187). [**E**] *fixed-point*: the texture is already explicit (e.g. $\sin^2\theta_{12}$, $\varepsilon = (3/4)\varphi_0 = c_3 + 36c_3^4$). [**E**] the Koide first correction (door 2): $F_{\rm pole}$ is the one seam-gapped prediction, rate $(2/3)^6 \approx 0.0878$ (v371), the source quotient 0.33% below 2/3 — the rate is the suppression *scale*, not the residual. [**C**] so a uniform band on all predictions is wrong; the size is keyed to the mechanism class. A typing of v387 (as v384 is of v383); cited in `origin_theory`; Python-only.

2026-06-24 (Harvesting the QFT/gravity machinery: v385 the UV-branch kill-test surface, v386 the graviton-exchange amplitude, v387 the gap-driven correction calculus)

- **v385 (UVBRANCH.KILLTEST.01) — the carrier-Pati–Salam UV branch on the all-order footing.** A consolidation that ties the all-order EG/BRST closure (v381) to the optional UV branch and computes the one genuinely new thing: the *proton-decay safety hierarchy*. [**E**] the carrier content is the SM with no new states, so the SM-non-unification (v246) is now all-order-robust; [**X**] the plain-SM 4D-GUT stays killed ($\sin^2\theta_W = 3/8$ vs 0.231); [**C**] the optional carrier-PS branch sits at $M_{PS} = \kappa M_{\rm scal} \approx 3\times 10^{13}$ GeV ($\kappa \in [1.0, 1.2]$, v253); [**E**] minimal PS gauge bosons are $SU(4)$ leptoquarks that mediate rare LFV ($K_L \to \mu e$), *not* $p \to e^+\pi^0$, so the branch clears the binding bound by $M_{PS}/M_{\rm bound} \sim 10^7$ – proton-safe, with *no* fake τ_p window (v265). Cited in `tfpt_5_redteam`; Python-only.

- **v386 (GRAV.AMPLITUDE.01) — the graviton-exchange amplitude is finite and UV-soft.** Extends v304/v370/v380 from the propagator to the actual tree amplitude. [**E**] the dressed spin-2 propagator $e^{-p^2/M^2}/p^2$ has its only pole at $p^2 = 0$ (residue $1 > 0$, ghost-free), is exponentially softened in the UV (the ratio e^{-u} falls faster than any power) and reduces to GR + a calculable $-p^2/M^2$ correction at low energy ($M = M_{\rm scal}$, v253); [**C**] a single positive-residue pole gives tree unitarity, so perturbative graviton scattering $A \sim (T{\cdot}P_2{\cdot}T')e^{-p^2/M^2}/p^2$ is an explicit finite computation (perturbative-only; QG.AMB.01 stays a [**C**] redundancy, v369). Cited in tfpt_4_frontier; Python-only.
- **v387 (CORRECTIONS.GAP.01) — the gap-driven correction calculus.** The practical harvest of v383: the same spectral gap that makes a sector parameter-free also sets the *size* of its first correction (correction$_n \sim (\lambda_2/\lambda_1)^n$). [**E**] flavor, horizon recovery and the QG bound all share the $(2/3)^6$ rate (the family-3 facet; QG susceptibility $\chi = 729/665$), while the discrete compiler transient decays at the golden $(\varphi + 2)/4$ (the carrier-5 facet) — the two number-field facets of v314/v383. [**C**] the gap does double duty (why no free parameter *and* how big the first correction), so sub-leading magnitudes are organised, not free. Cited in origin_theory; Python-only.

2026-06-23 (Bird's-eye synthesis: two new structural patterns — v383 the universal spectral-gap principle and v384 the residual is certification, not construction)

- **v383 (DYNAMICS.UNIVERSAL.01) — the universal spectral-gap / Perron–Frobenius principle.** A fresh bird's-eye pass found that *every* TFPT sector is the *same* structural object: a gapped operator with a *unique* leading attractor (the physics) and a spectral gap (the reason there is no free parameter). v303 named this only for $F_{\rm transfer}$; this module extends it to the sectors made parameter-free *after* it. [**E**] three gaps are computed — the flavor transfer $\{1, (2/3)^6, (1/3)^6\}$ with gap $6\ln(3/2)$ (v56/v82); the affine-E_8 adjacency with Perron eigenvalue 2 (Kac marks) and subleading $2\cos(\pi/5) = \varphi$, gap $2 - \varphi = 1/\varphi^2$ (v312/v313); the decoupling margin $\Delta_{\rm eff} \approx 1.648$ (v76/v337). [**E**] the two number-field facets (v314): golden $\varphi \in \mathbb{Q}(\sqrt{5})$ (carrier-5, static) and $(2/3)^6 \in \mathbb{Q}$ (family-3, dynamic) are the prime-5 and prime-3 facets of the order-30 Coxeter clock $30 = 2{\cdot}3{\cdot}5 = h(E_8)$. [**C**] the other sectors are the same structure — gravity ($\delta S{=}0$ equilibrium, v358/v359), the QG measure (Gaussian saddle, v365), the boundary QFT (holomorphic $c{=}8$ RG fixed point, v157/v158), the all-order S-matrix (adiabatic limit, v381), recovery (Petz fixed point, v221), $\tau{=}i$ (v333). [**C**] the meta-theorem: gapped ⇒ unique attractor ⇒ parameter-free, theory-wide. Cited in origin_theory; a synthesis (no new number), Python-only.
- **v384 (RESIDUAL.CERTIFICATION.01) — the residual is certification, not construction.** After v369/v381/v382 the *kind* of every open item has shifted: there is *no* open TFPT physics mechanism left. A ledger-CI audit classifies every residual into exactly three external kinds: [**E**] (A) external math proof (SEAM.EQUIV.01 continuum existence = the cited MMST/Adamo theorem v336; ALPHA.QUILLEN.EXACT.01 v382; QG.AMB.01 the constructive measure, a [**C**] redundancy v369), [**E**] (B) theorem-forbidden ($v_{\rm geo}$, the No-Unit theorem v153/v364), [**E**] (C) external physics ($F_{\rm transfer}$, v371–v374, firewalled v187). The two load-bearing facts are verified (No-Unit dimensionlessness; the gap-decoupling margin $1.648 > 0$), and the count of open *internal* TFPT mechanisms is **zero**. [**C**] convergence: the only hard things left are theorems and external physics, not missing dynamics. Cited in tfpt_research_contracts; an audit (no new number), Python-only.

2026-06-23 (External-review closure: the two genuine residual targets named — v381 the all-order Epstein–Glaser/BRST contract for S_{pert}, and v382 the Quillen determinant-line variation as a tracked target)

- **v381 (QFT4D.EG.ALLORDER.01) — the all-order 4D perturbative-QFT closing statement.** A mathematical-physics review of the QFT/TOE layers identified exactly one genuinely new, buildable item: the *all-order* Epstein–Glaser/BRST contract for the perturbative 4D S-matrix S_{pert} (the previous documents had the S_{pert} skeleton and one-loop, v269/v271/v273/v278, but no named all-order claim). This module types the standard causal-perturbation-theory closure for the TFPT finite interaction class — it does *not* build a path integral. [**E**] power-counting: every gauge+matter vertex (Yang–Mills + Dirac–Yukawa + Higgs + ghost) is a mass-dimension-4 marginal operator, so the EG singular order $\omega \leq 4$ and the counterterm space is finite at every order; [**E**] BRST nilpotency s^2=0 verified exactly via the Jacobi identity for $su(2)$ and $su(3)$ (the carrier weak+colour content); [**E**] the seam gap $\Delta = 6\log\frac{3}{2} > 0$ (v302) gives the IR-safe adiabatic limit. [**C**] the two heavy legs are imported rigorous theorems — all-order T_n existence by causal factorisation (Epstein–Glaser 1973) and all-order BRST invariance (Piguet–Sorella / Kugo–Ojima) — applied to the TFPT class. Matter+gauge *only*: the R^2/Weyl^2 gravity sub-sector is the entire-form-factor resummation (v304/v370/v380), and S_{pert} is not the ambient measure QG.AMB.01 (a [**C**] redundancy, v369). NET [**C**], prerequisites [**E**]; not [**E**] and not a path integral. Cited in tfpt_5_redteam and tfpt_2; Python-only.
- **v382 (ALPHA.QUILLEN.EXACT.01) — the Quillen determinant-line variation, named as a target.** Per the same review, the "why *this* $F_{U(1)}$ functional" obligation (previously tracked only as EM.WARD.01's residual, v341/v342) is elevated to its own citable target: $\delta_\tau(\log\det_\zeta \Delta_{U(1)} + 8b_1 c_3^6 \log\varphi_{\text{seam}}) = 0$ on the determinant line over the μ_4 seam moduli. [**E**] the Quillen split holds at the α^{-1}=137.0359992 root with every coefficient a named atom (index $8b_1$, heat-kernel $c_3^{3,6}$, discriminant $q(D_5)$+$q(A_3)$=2); [**E**] the value stays closed (v3); [**O**] the from-first-principles exactness proof is the target. [**C**] it is a *face* of SEAM.EQUIV.01 (v336) — the seam $U(1)$ determinant line lives on the same holomorphic net — so it adds *no* second independent gate; it makes the existing residual trackable. No new number (reuses the Wolfram-mirrored v341 identities). Cited in tfpt_1_architecture_e8; Python-only.

2026-06-23 (Track 2 of the forward plan v2 — the AMBIENT REDUNDANCY statement (v369): the holographic route to Gravity-complete that sidesteps building the general Euclidean QG measure)

- **v369 (QGAMB.REDUNDANCY.01) — Track 2 (forward plan v2).** Instead of constructing the non-perturbative ambient measure (the conformal-factor swamp, v332/v334), assemble the established discriminators that prove the ambient sector carries *no* physical degrees of freedom relative to the holomorphic boundary algebra, so the QG measure (gate C7/QG.AMB.01) is a non-fundamental *certification* object (like a coordinate choice in GR), not missing dynamics. [**E**] holomorphy $\Rightarrow$ DHR$((E_8)_1)$ = Vec: #primaries = $|\det \text{Cartan}| = 1$ for E_8 vs 4 for $SO(16)$ $\Rightarrow$ no nontrivial superselection charge a bulk could carry invisibly; [**E**] no torus ground-state degeneracy ($|\det K| = 1$); [**E**] finite Petz recovery rate $(2/3)^6$ (v221); [**E**] the gap-decoupling margin $\Delta_{\text{eff}} = 6\log\frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337). [**C**] with the Bisognano–Wichmann reconstruction map (v258/v309/v323) the redundancy statement $\mathcal{O}^{\text{bulk}}_{\text{phys}}/\text{Diff} \cong \mathcal{A}^{\mu_4}_{\Sigma}$ is well-posed. [**O**] residual: the two consumed premises — SEAM.EQUIV.01 (the S3 lattice input, v366) and the BW intrinsicality of the raw collar (v329); it does *not* construct QG.AMB.01 but provides the *alternative* (holographic redundancy) route: Gravity-complete = Boundary-complete + Ambient-redundancy. Cited in tfpt_research_contracts; Python-only.

2026-06-23 (Status propagation: the v365–v380 + FORM.SEAM.MMST.01 status carried into the remaining papers, the README and the Zenodo description)

- **Deep-sync of the post-S3 status.** The canonical surfaces (`tfpt_status`, `introduction`) were already current; this pass propagated the same honest wording into `origin_theory`, `tfpt_1`–`tfpt_5`, `tfpt_research_contracts`, the root `README.md`, the Zenodo description and `website/lib/papers.ts`: `SEAM.EQUIV.01` is *closed modulo a cited theorem* (lattice `v367`/`v368` + the S_3 stack `v376`–`v379` + Lean `FORM.SEAM.MMST.01`; residual **[O]** = the cited MMST/OS existence only); `QG.AMB.01` is a **[C]** *redundancy* (`v369`/`v379`), a certification object not missing dynamics; perturbative graviton unitarity is established (the Stelle ghost is a Seeley–DeWitt truncation artefact, `v304`/`v370`/`v380`); and the frontier transfers are typed runnable solvers with kill tests (`v371`–`v374`) + the observatory CI (`v375`). No new claim — a wording propagation so every surface matches the ledger; `AUDIT OK`.

2026-06-23 (v380 — the KMS Entire Hessian: the Stelle ghost is EXACTLY the Seeley–DeWitt truncation, and resummation pushes it to infinity)

- **`v380` (`GRAV.KMS.HESSIAN.01`).** Upgrades `v304`/`v370` from the *assumption* "the resummed graviton form factor is entire" to a derived statement. **[E]** the seam KMS cutoff $f(u)=e^{-u}$ (`v259`) gives the dressed spin-2 propagator $e^{-p^2/M^2}/p^2 = 1/(p^2 a)$ with $a(u)=e^u$ entire and nowhere zero (only the $p^2=0$ pole). **[E]** each finite Seeley–DeWitt truncation is a Taylor partial sum of a and carries a Stelle-ghost zero ($T_1=1+u \to u=-1$; $T_2 \to -1\pm i$); the entire a has none. **[E]** the modulus of the nearest truncation-zero grows monotonically with the order ($1, 1.41, \ldots, 3.07$ for $n=1..8$) $\to \infty$ — the ghost decouples, "entire" = "do not truncate". **[C]** $\Rightarrow$ perturbative graviton unitarity; **[O]** the exact off-shell curved-space Hessian identification is the cited heat-kernel resummation, perturbative-only. Cited in `tfpt_5_redteam` (Target F); Python.

2026-06-23 (Lean: the S3 continuum leg formalised as "closed modulo a cited theorem" — FORM.SEAM.MMST.01)

- **`FORM.SEAM.MMST.01` — Lean 4 formalisation.** The S3 continuum leg is now machine-formalised in a new module (`SeamScalingLimit.lean`), lake build OK and `#print axioms` clean, in the audit-contract style of `SeamEquivChain.lean`. The MMST applicability hypotheses for the seam collar are *provable* by kernel `decide` (no axioms): $D = 16$, rank $= c = 8$, the range $8 \le 8 \le 16$, $c_- = 8 \neq 0$, and $\det K = 1$ vs $SO(16)$'s 4. The MMST scaling-limit theorem (CMP 2022) and the Adamo–Moriwaki–Tanimoto OS reconstruction (2024) are *named cited axioms*; the conclusion $(E_8)_1$ follows by a well-typed composition whose residual (`#print axioms`) is exactly those two published theorems plus the carrier gap — no hidden assumption, no `sorry`. Mirrors the S3 closure stack `v376`–`v379`; so `SEAM.EQUIV.01` is now "closed modulo a cited theorem", the honest formalisation ceiling. Cited in `tfpt_research_contracts`; `lean_manifest` refreshed.

2026-06-23 (S3 closure stack — the target of the seam scaling limit pinned at EVERY computable level: central charge c=8 from the lattice (v376), the $(E_8)_1$ character (v377), the genus-1 torus count = 1 (v378), and reflection positivity (v379); only the abstract continuum existence (cited MMST) stays external)

- **`v376` (`SEAM.S3.CENTRALCHARGE.01`).** The Calabrese–Cardy finite-size entanglement-entropy scaling of the critical free-fermion content gives $c = 1.0000$ per complex mode (Peschel correlation-matrix EE; the $L=4m+2$ sizes avoid the half-filling zero-mode), so the 16-Majorana (= 8 complex) collar has $c = 8 = g_{\mathrm{car}} + N_{\mathrm{fam}}$, chiral ($c_-=8$, `v367`/`v368`). Numerical $c=8$ from the lattice. Python (numpy).

- **v377 (SEAM.S3.E8CHARACTER.01).** The exact affine character $\chi = E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$ gives $c = 8$ and 248 level-1 currents (one primary); the μ_4/GSO promotion $248 = 240 + 8 = 120 + 128$ ($240 = 16{\cdot}5{\cdot}3$) pins $(E_8)_1$ over the same-c rival $SO(16)_1$ (120 currents, 4 primaries). Python (sympy q-series).
- **v378 (SEAM.S3.MODULAR.01).** The genus-1 discriminator: KLM condensation μ(carrier)=16 $\to$ $\mu(E_8)=16/4^2=1$, $|\det \mathrm{Cartan}(E_8)|=1$ vs $D_8=4$, and the single character's modular T-eigenvalue $e^{-2\pi i/3}$ give torus ground-state degeneracy 1 (holomorphic = $(E_8)_1$) — the "hard part" v344 isolated. Python (sympy).
- **v379 (SEAM.S3.RP.01).** Reflection positivity (the OS axiom) of the explicit gapped collar: the positive spectral measure makes the Hankel/reflection Gram matrix PSD (a ghost mode breaks it). With c=8 + $(E_8)_1$ in hand, the OS inputs are complete and C7/QG.AMB.01 is then discharged by the redundancy theorem v369. Python (numpy).
- **Net:** S3 is the master key — on the explicit lattice model the target net $(E_8)_1$ is pinned exactly (central charge, character, genus-1 count, RP); the only remaining residual is the abstract *continuum existence* of the scaling limit (the cited MMST theorem v336), which lives outside the computational suite. Cited in tfpt_research_contracts.

2026-06-23 (Track 4 — the prediction OBSERVATORY: a status-typed CI over the frozen prediction registry that makes the falsifiability surface machine-checkable, with a live data scorecard (v375))

- **v375 (OBSERVATORY.REGISTRY.01).** A CI over freeze_file.csv: [**E**] every prediction carries a kill criterion (falsifiability complete, no orphan claims); [**E**] the headline values re-derive from $\{\varphi_0^{\rm ret}\}$ ($\sin^2\theta_{12}{=}1/3{-}\varphi_0^{\rm ret}/2$, $\sin^2\theta_{13}{=}\varphi_0^{\rm ret}e^{-5/6}$, $\beta_{\rm rad}{=}\varphi_0^{\rm ret}/(4\pi)$); [**C**] a live scorecard records θ_{12} vs the JUNO first run (0.28σ, compatible), θ_{13} vs NuFIT 6.0 NO (2.0σ, the flagged most-tensioned core prediction), $\beta_{\rm rad}$ vs ACT DR6 (0.37σ, a hint), $r < 0.036$ (compatible). Tooling that makes the closed pieces usable and the open ones honestly fenced; no new physics. Cited in tfpt_5_redteam; Python (stdlib).

2026-06-23 (Track 3 — the $F_{\rm transfer}$ functor promoted from contracts to TYPED runnable solvers with kill tests: Koide source→pole (v371), η_B Boltzmann (v372), axion finite-T relic (v373), m_p/m_e uncertainty budget (v374))

- **v371 (FR.POLE.SOLVE.01).** A clean *negative*: standard QED/EW running cannot be the Koide source→pole transfer (it pushes Q above 2/3; the required ϵ is negative while QED's is positive), so the transfer is the non-QED 53/54 operator / $(2/3)^6$ Moebius generator (v183/v82). [**C**], plain running excluded as the kill. Python (numpy+scipy).
- **v372 (FR.BOLTZMANN.SOLVE.01).** The integrated Buchmueller–Di Bari–Pluemacher Boltzmann ODE at the TFPT-frozen $M_1{=}M_{\rm scal}(\varphi_0^{\rm ret})^2/A_\Lambda$ (v212) gives η_B within a factor $\sim$1.1 of $6.1{\times}10^{-10}$ with no M_R dial; replaces the v326 toy washout. [**C**]/[**X**].
- **v373 (FR.RELIC.SOLVE.01).** The finite-T axion misalignment ODE decides the branch: the frozen spine angle $3\pi/5$ gives $\Omega_a h^2{\sim}0.12$ (in $[0.08, 0.16]$, lands on $\Omega_{\rm DM}$ untuned) while the hilltop 170° over-produces ($\sim$5.4×); replaces the v326 toy. [**C**]/[**X**](lattice $\chi(T)$ input).
- **v374 (FR.QCD.BUDGET.01).** The m_p/m_e uncertainty budget: with $b_3 = -7$ [**E**] and the two declared externals ($\alpha_s(M_Z)$, C_p, $\sim$5.3% each), the central $\sim$1824 (v262) carries a propagated $\pm$7.5% band $[\sim 1690, \sim 1960]$ CONTAINING 1836.15; lattice tightening is the kill. The falsifiability companion to v262; [**C**].

2026-06-23 (Track 1 — the S3 input made CONCRETE: an explicit gapped chiral-Majorana (p+ip) lattice model with numerical Chern $|C|$=1 realises the c_-=8 edge (v367), and a strip shows the chiral edge exists by anomaly inflow (v368))

- **v367 (SEAM.S3.LATTICE.01) — Track 1.** The abstract S3 residual of v356/v366 ("the collar realised as a genuine lattice chiral free-fermion invertible phase") is turned into a RUNNABLE model. **[E]** the p+ip Bloch model is gapped at M=1 with a numerically computed Fukui–Hatsugai–Suzuki Chern number $|C|$=1 (C=0 trivially at M=3); **[E]** $N_{\rm Maj}=\dim S^+$=16 copies give c_-=16/2=8=$g_{\rm car}$+$N_{\rm fam}$, with the edge K-matrix $SO(16)_1$ ($\det 4$) $\to$ $(E_8)_1$ ($\det 1$) by the order-4 μ_4 clock (v351/v369). **[C]** the μ_4 condensation (v154/v351); **[O]** the continuum scaling-limit existence (MMST, v336) stays open. Cited in tfpt_research_contracts; Python (numpy + sympy).
- **v368 (SEAM.S3.INFLOW.01) — Track 1.** The bulk-edge / anomaly-inflow leg (S4) made concrete on the v367 model: **[E]** a finite strip has an edge-localised in-gap chiral branch ($\min_{k_x}|E|\sim 0$) in the topological phase and a clean gap in the trivial phase (neg control); **[E]** bulk-edge correspondence gives one chiral Majorana per edge (c_-=8). **[C]** so by anomaly inflow the chiral edge EXISTS as a consequence of the invertible bulk, discharging the v336/v351 "does the edge exist?" residual on the lattice. **[O]** the continuum scaling limit stays open. Cited in tfpt_research_contracts; Python (numpy).

2026-06-23 (Track 2 — perturbative graviton unitarity sector-by-sector: the Barnes–Rivers spin decomposition localises the Stelle ghost in the spin-2 sector, cured by the entire KMS form factor; the spin-0 mode by the GHP contour (v370))

- **v370 (GRAV.SPIN2.UNITARITY.01) — Track 2.** Extends v304 (the scalar $1/(p^2(p^2{+}M^2))$ pole algebra) to the actual tensor spin structure. **[E]** the Barnes–Rivers projectors are complete ($\mathrm{tr}\,P_2=(d{-}2)(d{+}1)/2$, $\mathrm{tr}\,P_1=d{-}1$, $\mathrm{tr}\,P_{0s}=\mathrm{tr}\,P_{0w}=1$ sum to $d(d{+}1)/2$; d=4 split 5+3+1+1=10), the EH propagator is $P_2/p^2-\frac{1}{d-2}P_{0s}/p^2$, and the Stelle ghost is a *spin-2* state. **[E]** the entire KMS form factor e^{p^2/M^2} (v304/v259) makes the spin-2 sector ghost-free; **[C]** the wrong-sign spin-0 conformal mode (v332) is cured separately by the GHP contour (v334). **[C]** so the full *perturbative* graviton is ghost-free sector-by-sector (conditional on v304's entire-analyticity assumption). **[O]** residual: perturbative-only + the open analyticity assumption; not the non-perturbative measure. **[O]** DECLINE: $\mathrm{tr}\,P_2$=5 is dimension counting, not $g_{\rm car}$ (v354/v355). Cited in tfpt_5_redteam (Target F); Python-only.

2026-06-23 (Directions 3 & 7 — the two hard functional-analysis residuals ADVANCED: the QG measure as one-loop fluctuations around the parameter-free saddle (v365), and the seam collar verified IN MMST's free-fermion class so the scaling limit is $(E_8)_1$ (v366))

- **v365 (QGAMB.SADDLE.01) — Direction 3.** With the parameter-free saddle (v358/v359) the QG *measure* (gate C7/QG.AMB.01) organises as a one-loop Gaussian fluctuation determinant. **[E]** the fluctuation stiffness is $1/c_3=8\pi$ (no free dial, so the one-loop problem is now well-posed); **[E]** the fluctuation operator is gapped $M\geq\Delta_{\rm eff}=6\log\frac{3}{2}-\frac{31}{4\pi^2}\approx 1.648>0$ (v76/v337) $\Rightarrow$ mode-by-mode convergence, finite model log-det $\mathrm{Tr}'\log M=6\log\frac{9}{2}$; **[C]** the conformal mode converges on the GHP contour (v334); **[E]** the admissible projective limit exists ($\chi=729/665$, v330) with 0 new dials (v364). **[O]** residual: the full non-perturbative projective limit on the ambient sector. Sharpens C7 from "diffuse full QG" to a one-loop Gaussian problem around the parameter-free saddle; does not close it. Cited in tfpt_research_contracts; Python-only.

- `v366` (`SEAM.MMST.INCLASS.01`) — **Direction 7.** The seam collar is verified IN MMST's free-lattice-fermion scaling-limit class hypothesis-by-hypothesis: [**E**] $D{=}\dim S^{+}{=}16$ Majoranas, $c{=}g_{\rm car}{+}N_{\rm fam}{=}8$, $\operatorname{rank} E_8{=}8$ so rank $\le c \le D$ is $8 \le 8 \le 16$ (in range); [**E**] gapped ($\Delta{=}6\log\frac{3}{2}{>}0$, derived `v302`); [**C**] quasi-free CAR class (`v155`/`v160`); [**E**] chiral ($c_{-}{=}D/2{=}8 \neq 0$). [**C**] the MMST scaling-limit theorem then applies, and [**E**] the order-4 μ_4 clock pins the target to $(E_8)_1$ ($\det K{=}1$ vs $SO(16)$ $\det K{=}4$, `v351`). [**O**] residual: the single S3 input (the collar as a genuine lattice chiral invertible phase, the seam one-sidedness, `v297`/`v356`). Reduces the continuum side of `SEAM.EQUIV.01` to one named input; ADVANCES, does not close. Cited in `tfpt_research_contracts`; Python-only.

2026-06-23 (Consistency pass: the parameter-free *full covariant* Einstein equation (`v358`/`v359`) propagated across ALL stale paper and website surfaces; new `v359` keybox; homepage gravity animation)

- **Deep-sync of the parameter-free gravity status.** A cross-surface audit found several documents and website components still framing gravity as "exact only in the $R + R^2$ regime" or the metric-sector field equation as "open". These were reconciled with the canonical post-`v358`/`v359` status: the *local* covariant field equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ is PARAMETER-FREE (both coefficients fixed); what remains open is the Jacobson equation-of-state status, the global ambient measure (`QG.AMB.01`/C7), and the single unit $v_{\rm geo}$. Touched: `introduction`, `tfpt_2_standard_model`, `tfpt_3_e8_audit_bootstrap`, `tfpt_4_frontier` (summary table), `tfpt_research_contracts`, `origin_theory`.
- **New `v359` keybox** added to `tfpt_research_contracts` (full covariant equation: fixed-volume ⇒ Einstein tensor, Lovelock ⇒ conservation an output); the `v358` residual corrected from "linear→nonlinear" (stale — `v359` closed it) to the equation-of-state + C7 + $v_{\rm geo}$ residual, including the generated script index (`script_registry.csv`).
- **Website.** The `GravityEmergence` animation (three origins of c_3 converging to $1/(8\pi)$; the parameter-free Einstein equation) was updated to the full covariant result and surfaced as a dedicated homepage section (`/#gravity`); gravity-status wording aligned across `ReconstructionChain`, `VerificationDag`, `StatusMatrix`, `WhatTfptClaims`, `ThreeDecoderMap`, `StatusPyramid`, `papers.ts`, `glossary.ts` and the review page.

2026-06-23 (Direction 6, $v_{\rm geo}$ sharpened: after the parameter-free gravity, the *gravity* sector adds NO new dimensionful input either, so $v_{\rm geo}$ is the SINGLE dimensionful input of the complete theory — final tally $\{v_{\rm geo}, \pi\}$, ZERO dimensionless dials: `v364`)

- `v364` (`VGEO.SHARPEN.01`) — **Direction 6.** The No-Unit Theorem (`v153`) already made $v_{\rm geo}$ a forced metrology primitive. [**E**] The SHARPENING after the parameter-free gravity (`v358`/`v359`/`v361`): the Einstein coefficient $1/c_3 = 8\pi$ and the Λ prefactor $(8\pi)^2 \cdot 48 c_3^4 = 3/(4\pi^2)$ are dimensionless, so $1/G \sim v_{\rm geo}^2$ and $\Lambda \sim$ (dimensionless) $v_{\rm geo}^4$ reduce to $v_{\rm geo}$ times atoms — the gravity sector adds no new scale. [**E**] FINAL TALLY: every dimensionful quantity = (a dimensionless ratio)×(a power of $v_{\rm geo}$); TFPT's free content is $\{v_{\rm geo}, \pi\}$ with ZERO dimensionless dials, a $\sim 26 \to 1$ reduction vs the SM. [**O**] $v_{\rm geo}$ stays irreducibly primitive (No-Unit; $1/G$ UV-sensitive, Sakharov `v68`). Direction 6 is clarity — the last dimensionful anchor named and shown to be the only one. Cited in `tfpt_research_contracts`; Python-only.

2026-06-23 (Direction 5, the matter–gravity backreaction: the carrier's gravitational anomaly is forced c_-=8, and the matter vacuum-energy backreaction is finite ($\to$ the forced Λ from α, not $M_{\rm Pl}^4$) — the loop closes; no new SM-quantum-number read-out (declined): `v361`)

- `v361` (`GRAV.BACKREACT.01`) — **Direction 5.** With the parameter-free equation (`v359`) and the explicit J3 flux (`v358`), the backreaction of TFPT's carrier matter is computed. **[E]** the carrier's gravitational anomaly is forced: $c_- = 16/2 = 8 = g_{\rm car}{+}N_{\rm fam}$ (the chiral central charge of the 16-Majorana content). **[E]** the backreaction is FINITE $\to \Lambda$ from α: the matter vacuum energy through $G_{ab} = c_3^{-1}T_{ab}$ is gap-suppressed (Decoupling Thm `v337`) to $\rho_\Lambda/\bar{M}_{\rm Pl}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$ (`v60`), the 123 orders $= 119.03 + 3.92$ — *not* the $M_{\rm Pl}^4$ catastrophe; the loop (`v359`+`v60` + the carrier vacuum) closes. **[E]** J3 explicit: $\delta\langle K_B\rangle = (8\pi^2R^4/15)\delta\langle T_{00}\rangle$, carrier T_{ab} = the SM's (`v159`). **[C]** the SM trace anomaly is reproduced (content = SM). **[O]** DECLINE: no new atom-forced read-out of individual SM quantum numbers beyond c_-=8 and Λ — per `v354`/`v355` declined. The backreaction is finite and consistent, adding no new dial. Cited in `tfpt_4`; Python-only.

2026-06-23 (Direction 2, the disciplined result: the gap-induced GR correction is the known R^2 scalaron — already falsifiable via n_s/r; a *new* forced near-term deviation does NOT survive the discriminator and is declined: `v360`)

- `v360` (`GRAV.GAPCORR.01`) — **Direction 2, run honestly.** Does the seam transfer gap $\{1, (2/3)^6, (1/3)^6\}$ yield a NEW, calculable, FORCED deviation from GR (a near-term kill test) beyond the known R^2 scalaron? With the `v354`/`v355` discriminator the answer is *no*, and this is recorded rather than manufacturing a coefficient. **[E]** the leading higher-curvature term IS the R^2 scalaron (spectral-action $a_4{=}R^2/72$, `v36`; $M_{\rm scal}^2/\bar{M}_{\rm Pl}^2 = c_3^7$, exponent $7{=}g_{\rm car}{+}N_{\rm fam}{-}1$) — the calculable beyond-Einstein correction, already in TFPT. **[C]** it is the entanglement-equilibrium NEXT order (the Wald first law $\to f(R){=}R{+}R^2$, `v28`/`v359`). **[E]** it is ALREADY falsifiable via $n_s{=}1{-}2/N_\star$, $r{=}12/N_\star^2$ (live CMB-S4 kill test). **[E]** the gap $\Delta{=}6\ln\frac{3}{2}$ is a dimensionless rate, so a distinct gap-induced curvature^2 term sits at $\xi\, v_{\rm geo} \sim$ the seam/Planck scale (not near-term observable). **[O]** DECLINE: a new dimensionless coefficient is not atom-forced (the scalaron exponent $7{=}g{+}N{-}1$ and the gap exponent $6{=}2N_{\rm fam}$ are different; c_3^7 and $(2/3)^6$ unrelated) — per `v354`/`v355` declined. Direction 2 yields no new near-term forced prediction; the answer is the existing scalaron. Cited in `tfpt_4`; Python-only.

2026-06-23 (the FULL covariant Einstein equation, parameter-free: the fixed-volume entanglement equilibrium gives $G_{ab}+\Lambda g_{ab} = c_3^{-1}T_{ab}$ with BOTH coefficients TFPT-fixed ($8\pi = 1/c_3$ from `v358`, Λ from α via `v60`); B6 closed at the local level: `v359`)

- `v359` (`GRAV.NONLINEAR.01`) — **Direction 1, the full covariant equation.** Extends `v358` from the linearised (fixed-radius, Ricci-level) result to the FULL covariant equation. **[C]**[**E**] Jacobson 2015: demanding stationarity of the small-ball entanglement entropy at fixed *volume* brings in the EINSTEIN TENSOR G_{ab} (not just R_{ab}) — trace $g^{ab}G_{ab} = (1-d/2)R = -R$ in $d{=}4$, the structure that carries Λ. **[E]** Lovelock: G_{ab} is the unique symmetric divergence-free second-order metric tensor, so $\nabla^a(G_{ab}+\Lambda g_{ab}) = 0$ forces $\nabla^a T_{ab} = 0$ — conservation is an output. **[E]** coefficient 1: $8\pi G \to 8\pi = 1/c_3$ fixed (`v358`). **[E]** coefficient 2: Jacobson leaves Λ free, TFPT fixes it — $\rho_\Lambda/\bar{M}_{\rm Pl}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$, prefactor $(8\pi)^2{\cdot}48c_3^4 = 3/(4\pi^2)$ (`v60`); so BOTH coefficients are TFPT-fixed, not free integration constants. **[C]** all timelike directions $\to$ the covariant tensor equation; the matter side is the assembled CHM ball modular flux (`v358`/`v323`). NET: the original B6 "full covariant Einstein-side field equation" is now PARAMETER-FREE at the local level. **[O]** residual: the Jacobson equation-of-state status (not a from-action quantisation) + the global ambient measure (C7/`QG.AMB.01`, separate,

gap-decoupled); v_{geo} the one unit. Cited in `tfpt_research_contracts`; mirrored in Wolfram (`303/303`). (`next-plan.md` Direction 4 folded in: the Λ here matches the α-readout.)

2026-06-23 (the classical gravity field equation, parameter-free: the entanglement first law $\delta S = \delta\langle K\rangle$ with TFPT's atoms gives the LINEARISED Einstein equation with G from c_3; the thermodynamic and geometric origins of c_3 coincide via $|\mu_4| = |\mathbb{Z}_2|\chi = 4$; the two formerly-open pieces (matter flux J3, entropy-density coefficient) closed/atom-fixed: `v358`)

- **`v358` (`GRAV.ENTROPY.EQUILIBRIUM.01`) — the bridge to the dynamic world, instantiated on the gravity side.** Carrying out the Jacobson-2015 / Faulkner-et-al. "first law of entanglement ⇒ Einstein equation" with TFPT's atoms. **[E]** PARAMETER-FREE COEFFICIENT: $\delta S = \delta\langle K\rangle$ gives $G_{ab} = c_3^{-1} T_{ab}$ with $c_3^{-1} = 8\pi$ fixed; all four c_3-roles consistent (Unruh $1/(2\pi)=4c_3$, EH $1/(16\pi)=c_3/2$, Einstein $8\pi=1/c_3$, Bekenstein $1/4=1/|\mu_4|$) — no free dimensionless Newton dial. **[E]** THERMO = GEO COINCIDENCE (new): the first law's coefficient $2\pi/\eta$ equals the geometric $|\mathbb{Z}_2|2\pi\chi$ iff $|\mu_4| = |\mathbb{Z}_2|\chi = 4$ (`v216`) — the thermodynamic and geometric origins of c_3 are ONE. **[C]** J3 SOLVED: the CHM ball modular Hamiltonian (boost via BW, `v323`) gives the matter flux $\delta\langle K_B\rangle = \frac{8\pi^2 R^4}{15}\delta\langle T_{00}\rangle$. **[E]** ENTROPY DENSITY ATOM-FIXED: $1/4=1/|\mu_4|$ (`v57`) + central charge $c = g_{\text{car}}+N_{\text{fam}} = 8$ (form `v59`). **[O]** residual: linear→full non-linear (the original B6) + the v_{geo} unit. Closes the LINEARISED Einstein equation parameter-free; cited in `tfpt_research_contracts`; mirrored in Wolfram (`302/302`).

2026-06-23 (the two frontier walls attacked directly: Direction A assembles the continuum chain of `SEAM.EQUIV.01` theorem-by-theorem and ties its residual to the seam one-sidedness; Direction B was found ALREADY closed by `v40` (harmonic metric = finite linear algebra, not a PDE): `v356`)

- **`v356` (`SEAM.EQUIV.CONTINUUM.03`) — Direction A, the continuum chain assembled.** After the reverse-audit/bandwidth work showed no hidden *discrete* lever remains, the only fundamental progress is to assemble the *continuum* chain. Link-by-link: **S1** the gapped ($\Delta=6\ln(3/2)>0$, `v302`) quasi-free 16-Majorana collar (`v155`/`v160`) → **S2** invertible (gapped free fermions have no intrinsic topological order, Kitaev, `v301`) → **S3** the *nontrivial* invertible phase ($c_-=8\neq 0$) → **S4** a *non-gappable* anomalous chiral edge (anomaly inflow / bulk–edge), so edge *existence* is a CONSEQUENCE of S2+S3, not a separate assumption — this directly reduces the `v336`/`v351` "does the collar have a chiral edge?" residual → **S5** its chiral-CFT scaling limit (MMST, `v336`; range rank $\leq c \leq D$, $8 \leq 8 \leq 16$) → **S6** the order-4 μ_4 clock + the AQFT stack pin $(E_8)_1$ (`v351`/`v297`). **[E]** the only non-theorem link is S3, and **[C]** S3 is the seam being *one-sided* (chiral, $c_-\neq 0$) — the *same* one-sidedness that makes $c_3=1/(8\pi)$ a one-sided Gauss–Bonnet ($|\mathbb{Z}_2| \oint K$, `v58`). **[O]** the one remaining input is "the abstract collar is realized as a genuine lattice chiral free-fermion invertible phase" (the `v297` Flat-Away). NET: `SEAM.EQUIV.01` reduced to ONE realization input tied to the constant that *defines* the theory; it ADVANCES, it does not close.
- **Direction B — found ALREADY closed by `v40`.** The U-wall "transcendental Hitchin solve" (`v31`) is the worst case for a *generic* Higgs bundle; but at the *polystable* point TFPT selects, Mehta–Seshadri makes the bundle *unitary*, so the harmonic metric is the constant invariant Hermitian form (*finite linear algebra*, Higgs $\Phi=0$) and $\det R=8$ is read off — not a PDE. Only the absolute amplitude U_{point} (an anchor) remains. A redundant new B module was built, found to duplicate/contradict `v40`, and *deleted* (no padding). Honest outcome: B needs no new work.

2026-06-22 (bandwidth search of the unmapped E_8 region, with a strict forced-identity discriminator: the genuine overlooked structure is COLLECTIVE ($\sum\deg = 128 = \dim S^+$, $\prod\deg = |W(E_8)|$, exponents = totatives of 30); the per-degree atom matches are unforced and DECLINED; no new physical hit – reconciling `v66` and `v354`: `v355`)

- **`v355` (`E8.UNMAPPED.BANDWIDTH.01`) — the next step after `v354`, run honestly.** The obvious follow-up ("search those five degrees for hits we overlooked") is itself the numerology temptation `v354` warns about: a rich object like E_8 returns a numerical match for almost anything. So the scan uses a STRICT discriminator – a candidate counts only if it is a FORCED identity (a theorem about E_8), never a coincidence. [**E**] FORCED, SET-LEVEL (the genuinely overlooked structure, collective not per-number): $\sum\deg = 128 = \dim S^+$, the spinor half of $248 = 120 + 128$ (theorem: $\sum\deg$ = #pos roots+rank= $120 + 8$) – the invariant budget of E_8 equals the fermion content the carrier reads; $\sum\exp = 120 =$ #pos roots; $\prod\deg = |W(E_8)| = 696{,}729{,}600$; exponents = the $\varphi(30)$=8 totatives of 30 (`v223`), so the unmapped degrees' exponents are part of the forced set; $\max\deg = 30 = h(E_8)$. [**C**] SUGGESTIVE, not a theorem: 12=$h(E_6)$, 18=$h(E_7)$, 30=$h(E_8)$ are all degrees of E_8 (the $2T/2O/2I$ McKay ladder, `v219`), but "deg(E_8) contains its subalgebras' Coxeter numbers" is not a general theorem – flagged, not claimed. [**O**] DECLINED (the numerology temptation): the per-degree atom matches 14=$\dim G_2$, 20=$\det L$, 24=$|W(A_3)|$, 12=$|R(A_3)|$, 18=$p_4(a)$ (`v66` already flagged $12, 18$ as fittable) each admit ≥ 2 readings, and the lone extra prime in $|W(E_8)| = 2^{14}3^5 5^2\,7$ "=" the scalaron exponent is one of many readings of 7 – all declined. [**E**] RECONCILES `v66` (the 8 degrees coincide with the load-bearing integers) and `v354` (only $2, 8, 30$ are primary readouts): the search finds NO new forced physical hit, and the disciplined decline IS the anti-numerology result. (This also sharpens `v354`: the five are "no PRIMARY readout / hull overhead", not literally "unmapped" – they coincide with structural atoms, but unforced.) Cited in `tfpt_5_redteam` (with a full audited degree table); mirrored in Wolfram (`301/301`).

2026-06-22 (the REVERSE numerology audit, and what the golden ratio means: of E_8's 8 Casimir invariants only 3 feed a primary readout and 5 are hull overhead; the golden ratio is UNMAPPED structure, so it cannot be numerology: `v354`)

- **`v354` (`E8.REVERSE.AUDIT.01`) — the reverse complement of `v305`.** TFPT's forward discipline (`v305`) is one-directional: every physical READOUT must map onto an E_8 structure (anti-fitting). The sharp adversarial question runs the OTHER way — how much E_8 structure maps onto NOTHING, and is the recurring golden ratio numerology? [**E**] THE REVERSE AUDIT: of E_8's eight Casimir invariant degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ (exponents $+1$), exactly THREE feed a readout — 2 (metric), 8 (rank $\to c_3$=$1/(8\pi)$), 30 (Coxeter $\to g_{\rm car}$=5 and the order-30 cycle) — and FIVE carry no primary readout $\{12, 14, 18, 20, 24\}$ (they coincide with structural atoms per `v66`, but unforced – reconciled in `v355`); so 3/8 are primary readouts, 5/8 hull overhead. [**E**] ECONOMY, NOT PROMISCUITY: a small fixed set (rank 8, h=30, $\det K$=1, $D_5\oplus A_3$, the 16 half-spinor) generates the ENTIRE readout set, and the higher Casimirs are never reached into to fit — a numerological use would be PROMISCUOUS (mine the rich structure), TFPT is ECONOMICAL (few invariants, many readouts) and leaves most of E_8 untouched, the anti-numerology signature being precisely the large unused remainder. [**E**] THE GOLDEN RATIO IS UNMAPPED $\Rightarrow$ NOT NUMEROLOGY: $\varphi = 2\cos(\pi/5)$ appears only in internal structure (the affine-E_8 attractor eigenvalue `v312`, the icosian lattice `v348`) and in NO physical readout; numerology means a number FITTED to an observation, φ is fitted to nothing — it is the algebraic signature of the 5-fold (h=30=$2\cdot 3\cdot 5$), the carrier announcing it is icosahedral. [**O**] the 5/8 unmapped is the HULL OVERHEAD (E_8 is the minimal even-unimodular hull, not the world group); where a future higher-invariant readout could live, or genuine over-capacity — stated plainly,

not hidden. Cited in `tfpt_5_redteam`; mirrored in Wolfram (`300/300`).

2026-06-22 (the bird's-eye rethink: TFPT is a unique self-consistent LOOP with ZERO free dials, not a linear axiom system; π is a fixed geometric constant, not a dial; the one residual is the continuum closure: `v353`)

- **`v353` (`TFPT.SELFLOOP.01`) — refining `v352`.** Stepping all the way back: TFPT is not a LINEAR theory (axioms → theorems) but a CLOSED SELF-CONSISTENT LOOP whose outputs fix its own inputs. **[E]** it is a loop: g_{car}=5= max prime of the output $h(E_8)$=30, and the 8 in c_3= rank E_8 – the outputs fix the inputs (`v6`), with a unique fixed point (`v56`); "which is the axiom?" is the wrong question for a loop. **[E]** ZERO free adjustable dimensionless dials: every load-bearing number is forced/over-determined. **[E]** π is a FIXED geometric constant (the boundary 2-sphere, $\oint K{=}4\pi$), not a dial. **[C]** this refines `v352`'s "framework + π": π is not a free input and the framework is a SELECTION PRINCIPLE (the unique self-consistent loop), not a free meta-axiom – the honest characterisation is "a unique self-consistent loop with zero free dials". **[O]** the one residual is the CONTINUUM closure of the loop (the chiral-edge existence, `v336`/`v351`); structural-realism (loop complete) vs constructivism (continuum open) is a philosophical, not numerical, fork – no free numbers either way. A capstone synthesis.

2026-06-22 (two consequences: the c=8 which-net ambiguity resolved by the order-4 clock, and BOTH axioms shown bootstrap-forced so the only irreducibles are the framework + π: `v351`, `v352`)

- **`v351` (`SEAM.EQUIV.CONTINUUM.02`) — the which-net ambiguity falls.** The long-standing "c=8 ambiguity" (E_8 det 1 vs $SO(16){=}D_8$ det 4, both c=8, flagged open in `v277`/`v344`) is RESOLVED, non-circularly, by the seam's ORDER-4 μ_4 clock: the index-4 simple-current extension of $D_5{\oplus}A_3$ is E_8 (`v154`), the index-2 would be $SO(16)$, so the order-4 clock (the four Gauss–Bonnet marks, `v216`) selects E_8 (full μ_4 condensation 16→4→1) not $SO(16)$ (partial $\mathbb{Z}_2$ 16→4). The bootstrap agrees ($h(E_8)$=30 max prime 5=g_{car} vs $h(D_8)$=14 max prime 7). **[E]** so $\det K{=}1$ is a consequence, not an open input. **[O]** the residual of `SEAM.EQUIV.01` is then PURELY the chiral-edge / scaling-limit existence (`v336`), not which-net.
- **`v352` (`TFPT.IRREDUCIBLE.01`) — the inputs are not axioms.** The deepest answer to "the inputs are back-determined": BOTH P1 and P2 are bootstrap-forced fixed points. **[E]** g_{car}=5 is forced three ways (`v6`); the 8 in c_3 four ways (rank $E_8 = h(D_5) = \phi(30) = \det R$, `v54`); the E_8 hull is forced (unique even unimodular rank-8, `v83`/`v154`). **[E]** so the ONLY irreducibles are the FRAMEWORK (a discrete reflection-positive boundary with a finite carrier) and the transcendental π. **[O]** the framework is the META-AXIOM — not math-reducible, the foundational stance (like "spacetime is a manifold"), minimal, tested by its consequences not proved. TFPT's true input is ONE framework + π. Wolfram mirror +2 (297 → 299).

2026-06-22 (correction: the inputs are bootstrap-forced fixed points, not free axioms – g_{car}=5 IS the icosahedral 5 of $h(E_8)$=30, and the golden ratio is emergent from the μ_4-glue, not external: `v350`)

- **`v350` (`SEAM.EQUIV.BOOTSTRAP.01`) — correcting `v349`.** An honest correction prompted by the right objection that the inputs are not axioms but are back-determined by the theory. **[E]** the inputs ARE bootstrap-forced (`v6`): g_{car}=5 is over-determined three ways – rank-fill ($g_{\text{car}}{+}N_{\text{fam}}$=8), Coxeter-match ($g_{\text{car}}$= max prime of $h(E_8)$=30), Pascal ($2^4{=}C(5,0){+}C(5,1){+}C(5,2)$). **[E]** so g_{car}=5 IS the icosahedral 5-fold (the 5 in $h(E_8){=}30{=}2{\cdot}3{\cdot}5$), not the D_5 rank; the atoms $(2,3,5)$ ARE the primes of the icosahedral Coxeter number 30. **[E]** the golden ratio is EMERGENT from the glue: D_5 (h=8)/A_3 (h=4) are non-golden alone, but $D_5{\oplus}A_3 \to E_8$ (μ_4 index-4, `v154`) gives $h(E_8)$=30 (golden) – the seam's own μ_4 produces the 5-fold, so `v349` checked the un-glued pieces, the wrong object,

and its pentagon-vs-count dichotomy was false. [**O**] the residual relocates: the golden is bootstrap-forced, the only open thing is the physical continuum realisation (`v336`), NOT the golden ratio. Honest caveat: self-consistency within the framework, not from-nothing. Wolfram mirror +1 (296 → 297).

2026-06-22 (the honest test – does the raw seam carry φ? – answer NO: the bare carrier gives $\sqrt{2}$, golden is the icosahedral input; the keystone reduces to "is g_{car}=5 a pentagon or a count?": `v349`)

- **`v349` (`SEAM.EQUIV.GOLDEN.01`).** The honest, computed test of the `v348` question, and a clarifying NEGATIVE: the raw seam does *not* carry the golden ratio. [**E**] the golden ratio $2\cos(\pi/5)$ is the signature of a 5-fold rotation ($5 \mid h$); the carrier D_5 has h=8 (eigenvalue $\sqrt{2}$) and A_3 has h=4 – no 5-fold; the dynamical data (cusp $\{0, 1/3, 2/3\}$, recovery $(2/3)^6$, seed φ_0) are rational/transcendental. [**E**] $5 \mid h$ holds only for A_4=$SU(5)$ (h=5), H_3 (h=10), E_8 (h=30) – the output/icosahedral side, so φ is the icosahedral INPUT. [**E**] φ alone would not suffice ($2I \neq C_5 \times \mu_4$, $20 \neq 120$). [**O**] the resolution: the keystone reduces to one question – is g_{car}=5 a PENTAGON (a 5-fold rotation ⇒ golden ⇒ icosahedral $2I = E_8$) or just a COUNT (the rank of $D_5 \Rightarrow \sqrt{2}$)? The "absurdly simple solution" is real but is a *reading* of axiom P2 (the golden ratio is the pentagon content of g_{car}=5, not derivable from g_{car}=5-as-a-count); there is no hidden φ. L2 is foundational: P2-as-pentagon closes it (mode C), else a rigidity theorem must produce the 5-fold (mode B); not closed. Wolfram mirror +1 (295 → 296). A NEGATIVE/clarification; prevents a false closure.

2026-06-22 (Route B carried out: rigidity + the icosian-ring identity reduce the whole keystone to ONE question – does the raw seam carry the golden ratio? – the simplest form of L2: `v348`)

- **`v348` (`SEAM.EQUIV.RIGID.01`).** Executes the rigidity route (mode B of `v347`) and lands on a strikingly simple statement. [**E**] rigidity closes the crystallographic part: the order-4 clock has the unique normal form $z \mapsto iz$ (Nielsen/Kerckhoff, `v180`), Troyanov gives the unique flat pillowcase metric (`v284`). [**E**] the downstream is a LATTICE identity, not just a graph: by Conway–Sloane the ring of icosians (120 unit icosians = $2I$, golden-ratio quaternion norm) IS the rank-8 E_8 lattice. [**E**] the single bridge is the golden ratio: the crystallographic restriction allows only orders $\{1, 2, 3, 4, 6\}$ (the 5-fold absent), so $\varphi = 2\cos(\pi/5)$ is exactly what extends the crystallographic μ_4 to the non-crystallographic icosahedral $2I$ (the quasicrystal). [**O**] so the entire keystone reduces to ONE question: *does the raw seam carry the golden ratio?* – if yes, μ_4+$\varphi \Rightarrow 2I \Rightarrow E_8$ and rigidity fixes the rest. [**O**] honest subtlety: TFPT's φ is currently E_8-side (`v312`/`v313`); the raw-seam data so far are not manifestly golden, so showing the RAW seam carries φ non-circularly is the residual (= L2, sharpest form). Wolfram mirror +1 (294 → 295). An analysis/reduction; closes nothing.

2026-06-22 (can the one open arrow be *fully* solved? the closure-mode classification: the flat→spherical base locus, four modes, no free theorem – only rigidity or axiom P_3: `v347`)

- **`v347` (`SEAM.EQUIV.CLOSURE.01`).** A direct, honest answer to "is there any other way to fully solve the one open arrow L2?". [**E**] the precise locus: the seam gives the *flat* pillowcase base $S^2(2,2,2,2)$ ($\chi_{\mathrm{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$, v216) while the $2I/E_8$ structure is the Seifert base over the *spherical* $S^2(2,3,5)$ ($\chi_{\mathrm{orb}} = 2 - (\frac{1}{2} + \frac{2}{3} + \frac{4}{5}) = \frac{1}{30}$), so L2 bridges two geometric types – why it needs the carrier $(2,3,5)$ input. [**E**] as a "physics-realises-geometry" statement L2 has exactly four closure modes: A construct (the analytic wall), B rigidity (a unique-realisation theorem – the only theorem-route w/o new input, pursued v284/v300/v331), C axiom P_3 (then TFPT is complete relative to 3 axioms), D empirical. [**E**] the "no new state" discipline

(v246) blocks deriving L2 from extra physics, so no route makes it a free theorem. [**E**] verdict: full closure is either a rigidity theorem (B) or axiom status (C), validated either way by (D); only B closes it as a theorem. [**O**] NOT closed – classifies the modes, locates the difficulty. Wolfram mirror +1 (293 → 294). An analysis/roadmap; closes nothing.

2026-06-22 (the geometric bridge made explicit: the keystone is ONE arrow – raw-seam ⇒ $\det K{=}1$ is a six-link chain, five theorems + one open orbifold realisation, the group forced by the $(2,3,5)$ atoms: `v346`)

- **`v346` (`SEAM.EQUIV.GEOM.01`).** The honest capstone of the `SEAM.EQUIV.01` investigation: it lays out the full path raw-seam ⇒ $\det K{=}1$ as six links and shows exactly one is open. [**E**]/[**C**] L1 raw seam → μ_4 + four marks (Gauss–Bonnet, `v216`); [**O**] L2 μ_4 + the $(2,3,5)$ atoms → the $2I$ orbifold $\mathbb{C}^2/\Gamma$ (THE ONE OPEN ARROW = the geometric content of `SEAM.EQUIV.01`); [**E**] L3 $2I$ → affine E_8 McKay (`v219`); [**E**] L4 $2I$ → E_8 du Val (`v232`); [**E**] L5 $\mathbb{C}^2/2I$ → Poincaré link $H_1{=}0$ (`v345`); [**E**] L6 $H_1{=}0$ → $\det K{=}1$. [**E**] five of six links are theorems; [**E**] the group is FORCED by the seam's own atoms (the exceptional $SU(2)$ axis signatures $(2,3,3)/(2,3,4)/(2,3,5)$ → $2T/2O/2I$; $(2,3,5)$ unique to $2I$, and $(|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}}){=}(2,3,5)$ ARE those axes); [**E**] everything downstream of L2 is then forced. [**O**] the irreducible open content is L2 alone ("the raw seam normal bundle is an ADE orbifold point $\mathbb{C}^2/\Gamma$") = `SEAM.EQUIV.01`; not closed, but the bridge is now demonstrably ONE arrow with the group pinned and all downstream proven.

2026-06-22 (executing route R3: the $(2,3,5)$-rewrite attractor link computed to be the Poincaré homology sphere ($\det K{=}1$), the combinatorial core complete, only the geometric bridge open: `v345`)

- **`v345` (`SEAM.DETK.02`).** Carries out the combinatorial core of the hypergraph-homotopy route (R3, the most promising fresh angle from `v344`) as far as it computably goes. By plumbing topology (Brieskorn/Hirzebruch) the link of a tree plumbing with the ADE Cartan intersection form has H_1 = coker(Cartan). [**E**] the Smith normal form of the E_8 Cartan matrix is the identity, so $\text{coker}(E_8) = 0$ – the E_8-graph plumbing boundary is an INTEGRAL HOMOLOGY SPHERE (the genus-1 meaning of $\det E_8{=}1$); among ADE only E_8 does this ($A_n \to \mathbb{Z}/(n{+}1)$, $D_n \to \mathbb{Z}/4$, $E_6 \to \mathbb{Z}/3$, $E_7 \to \mathbb{Z}/2$). [**E**] $\pi_1 = 2I = SL(2,5)$ is PERFECT (commutator subgroup = the whole order-120 group, computed directly), so $H_1 = 0$ and the link is THE Poincaré sphere, $|2I| = 120 = |\mu_4|\, h(E_8)$. So R3's combinatorial half is [**E**]-complete: the E_8 attractor link is forced to be the Poincaré sphere ($\det K{=}1$), uniquely among ADE. [**O**] what stays open is only the geometric realisation bridge (the raw rewrite seam attractor *is* this E_8 du Val plumbing) = the `SEAM.EQUIV.01` content; not closed, but the keystone is now reduced to that single geometric identification. Wolfram mirror +1 (292 → 293).

2026-06-22 (the one keystone bit $\det K{=}1$ mapped: six equivalent faces, the genus-1 obstruction, three access routes, the hypergraph-homotopy route the most promising fresh angle: `v344`)

- **`v344` (`SEAM.DETK.01`).** A bird's-eye, full-spectrum map of `SEAM.EQUIV.01`'s only open residual, the holomorphy bit $\det K{=}1$. [**E**] it is ONE statement with six equivalent faces (holomorphy / homology-sphere link $H_1{=}0$ / one 1-dim irrep / det Cartan = 1 / full μ_4 condensation / $\tau{=}i$ CM), unified by $|\det \text{Cartan}| = |H_1(\text{link})| = \#(\text{1-dim irreps})$ across ADE ($A_n{\to}n{+}1$, $D_n{\to}4$, $E_6{\to}3$, $E_7{\to}2$, $E_8{\to}1$) – only E_8 gives 1. [**E**] the GENUS-1 OBSTRUCTION explains why it is hard: every easy argument (plane vacuum, gap, free-fermion) is genus-0, necessary but not sufficient (`v87`); $\det K$ is the torus degeneracy. [**E**] THREE genus-1 access routes: R1 continuum-modular (`v336`), R2 anyon-condensation (`v281`), R3

hypergraph-homotopy. [**C**] the HYPERGRAPH route is the most promising fresh angle: the $(2,3,5)$ rewrite attractor is the affine-E_8 graph (`v312`) and the E_8 du Val link is the Poincaré sphere (`v232`), so $\det K{=}1$ becomes a finite combinatorial statement (no continuum limit). [**O**] none closes it; the gap is the combinatorial-to-geometric bridge = the `SEAM.EQUIV.01` content. A synthesis/roadmap; closes nothing.

2026-06-22 (the four black-hole-birth completion routes investigated: none closes all or is 100% provable alone; all four converge on `SEAM.EQUIV.01` + the decoupled QG contour: `v343`)

- **`v343` (`FOUR.ROUTES.01`).** A direct, honest answer to "which black-hole-birth route closes ALL problems and is 100% provable": *none individually* — but the four converge on the same two residuals as `v335`. [**E**] **(A)** the finite causal diamond reframes `QG.AMB` to a finite thermal trace (de Sitter $S_{\rm dS}=32\pi^4 e^{2\alpha^{-1}}$ finite, $\chi=729/665$) but inherits the GHP contour (`v334`, [**O**]); does not touch the seam. [**C**] **(B)** Carlip–Cardy gives a second route to existence+c ($c{=}8$ consistent, but the Carlip ambiguity and no holomorphy $\det K{=}1$) — inherits `SEAM.EQUIV.01`. [**E**] **(C)** the modular/thermal flow (`v239`) is well-defined but its geometric content is `QGEO.SYM.01`, a corollary of `SEAM.EQUIV.01` (`v335`) — a reduction. [**E**]/[**O**] **(D)** the unique attractor (`v56`) gives parameter-freeness but the cyclic dynamics is [**C**] interpretation. [**E**] **Verdict:** independent closures = 0; irreducible residuals after all four = 2 (unchanged from `v335`); the horizon premise makes both more natural but eliminates neither — focus = `SEAM.EQUIV.01`. An analysis module; closes nothing, fabricates nothing.

2026-06-22 (the heat-kernel origin of the α terms: the Calderón quadratic is the Gilkey gauge-curvature coefficient; the cubic Maxwell moment stays the `EM.WARD.01` residual: `v342`)

- **`v342` (`EM.WARD.02`) — the heat-kernel sharpening of `EM.WARD.01`.** Derives the ORIGIN and STRUCTURE of the $F_{U(1)}$ determinant-line terms from textbook Seeley–DeWitt / Gilkey coefficients, advancing the EM-Ward origin from [**C**] (conjectured) to [**E**] for the term structure; no new gate. [**E**] $c_3=1/(8\pi)$ *is* the boundary Gauss–Bonnet coefficient ($\oint_{S^2}K=2\pi\chi=4\pi$, $\chi{=}2$, `v216`). [**E**] the α^2 (Calderón) term *is* the Gilkey gauge-curvature coefficient ($30\,\Omega^2/360=\frac{1}{12}\Omega^2$, $\Omega{=}i\alpha F\Rightarrow-(\alpha^2/12)F^2$), so it is forced, not an ansatz. [**E**] the c_3-orders $\{0,3,6\}$ are an arithmetic ladder and $2c_3^3=1/(256\pi^3)$ carries π^3 (three boundary insertions). [**C**] the exact coefficient needs the seam F-normalisation. [**O**] the cubic α^3 Maxwell moment (a topological/Chern level) + the full "this is the seam $U(1)$ ζ-determinant" identification stay the `EM.WARD.01` residual, tied to `SEAM.EQUIV.01`. α^{-1} stays [**E**] (`v3`). Wolfram mirror +1 (290 → 291).

2026-06-22 (α as the stationarity of a $U(1)$ determinant line: every ingredient an index / heat-kernel coefficient / discriminant form, the open EM-Ward obligation named: `v341`)

- **`v341` (`ALPHA.QUILLEN.01`).** Answers the external review's sharpest valid point — "derive the *form* of α, not more decimals". The EM closure $F_{U(1)}=\alpha^3-2c_3^3\alpha^2-\frac{4}{5}c_3^6 M\log(1/\varphi_{\rm seam})=0$ is rewritten as a *stationarity* of the boundary $U(1)$ determinant line: $[\alpha^3-2c_3^3\alpha^2]_{\,\delta\log\det\Delta_{U(1)}}=[8b_1c_3^6]_{\,\text{counterterm}}\cdot[\log(1/\varphi_{\rm seam})]_{\,\text{seam response}}$. [**E**] every ingredient is a NAMED object, not a knob: $M{=}41$ is a $U(1)$ index ($\frac{4}{5}M=8b_1$, $b_1{=}41/10$ the SM $U(1)_Y$ beta, EM Ward `v48`); $c_3{=}1/(8\pi)$ the Gauss–Bonnet boundary coefficient ($8{=}\operatorname{rank}E_8$); $-5/4=-q(D_5)$ a discriminant form and $\varphi_{\rm base}{=}1/(6\pi){=}\frac{4}{3}c_3$ carries the other, $c_3/\varphi_{\rm base}{=}3/4{=}q(A_3)$; $\delta_{\rm top}{=}\Omega_{\rm adm}c_3^4$. The Quillen split is verified at the $\alpha^{-1}{=}137.0359992$ root. **No new gate:** α^{-1} stays closed [**E**]; v341 adds no open item — it *sharpens* the pre-existing [**O**]/physical-origin obligation `EM.WARD.01` ("why *this* function", already conjectural). [**O**] the one still-open part *is* that

residual: the from-first-principles AQFT proof that $F_{U(1)}$ is the EXACT Quillen functional (the variational principle derived, not assembled), unchanged in scope. Wolfram mirror +1 (289 → 290).

2026-06-22 (A_s reconciled with the archived derivation: predicted & dimensionless (so it does not define a mass), the tension is the reheating channel: `v340`)

- **`v340` (`AS.RECONCILE.01`).** Answers "why don't we have A_s / the old versions derived it" and "can a ratio define mass itself". **[E]** A_s *is* predicted and *dimensionless*: $A_s = N_\star^2(M_{\rm scal}/\bar{M}_{\rm Pl})^2/(24\pi^2) = N_\star^2 c_3^7/(24\pi^2)$ with $M_{\rm scal}/\bar{M}_{\rm Pl} = c_3^{7/2} = 1.26\times10^{-5}$ a pure ratio, so A_s pins the e-fold count $N_\star$, NOT $\bar{M}_{\rm Pl}$ — it cannot define an absolute mass (No-Unit, `v153`, stands). **[C]** the -11.4σ at the slow Higgs channel ($N_\star$=51.4) is the slow-vs-fast reheating dichotomy; the A_s-preferred $N_\star$=56.2 ∼ the instantaneous bound. **[C]** the archived TFPT v4.5/v4.6 reported a clean $A_s = 2.124\times10^{-9}$ using the algebraic $N_\star = 3/\varphi_0^{\rm ret} - 1 = 55.42$; at that $N_\star$ the current normalisation gives 2.047×10^{-9} (-1.9σ) — the old clean A_s is near-instantaneous reheating, not the slow channel. **[E]** nothing lost: the current theory replaced the posited algebraic $N_\star$ with one derived from reheating physics and exposed honestly that A_s requires fast (pre)heating. Record band $N_\star \in [50, 60]$ (`v84`) untouched.

2026-06-22 (the first-principles boundary, mapped: can $M_{\rm Planck}$ be computed? can $F_{\rm transfer}$ be first-principles? both NO, with the exact reason: `v339`)

- **`v339` (`FIRSTPRINCIPLES.BOUNDARY.01`).** An honest boundary audit answering two head-on questions and fabricating nothing. **[E]** *The Planck mass cannot be computed* — $v_{\rm geo}/M_{\rm Planck}$ is forbidden by theorem (No-Unit, `v153`): a mass is mass-dimension 1 while every TFPT datum is dimension 0, so no dimensionless map outputs an absolute scale. It is not a gap but over-determined (gravity = dark energy to 0.11%, `v274`); every dimensionless *ratio* is derived, only the one unit is not. **[E]** $F_{\rm transfer}$ *cannot be made first-principles* either: for m_p/m_e the structure is in place (`v262`, carrier b_3=−7 + closed electron) but two inputs stay external. **[X]** $\alpha_s(M_Z)$ is external *because of a tension* — the SM (= TFPT content, no new states) does not unify (1-loop spread ∼8.8 at 2×10^{16} GeV, `v246`). The lattice $C_p \sim 2.83$ is parameter-free but lattice-only (`v35`); η_B/axion ride on cosmological initial conditions; Koide Q=2/3 is structural **[E]**. **[E]** The residuals classify into four honest kinds: forbidden-by-theorem, external-and-tensioned, external-but-parameter-free, cosmological. A classification, not a solution.

2026-06-22 (next-steps round: the one open lemma literature-anchored, the Decoupling Theorem, the holomorphy discriminator hardened in Lean, and the θ_{13} budget: `v336`, `v337`, `v338`, `FORM.CARTAN.DET.01`)

- **`v336` (`SEAM.EQUIV.CONTINUUM.01`) — the one open lemma, literature-anchored.** SEAM.EQUIV.01's only open input (the continuum OS reconstruction of the gapped quasi-free collar) is split into a CITABLE part and a narrow open part. **[C]** continuum leg: Osborne–Stottmeister (*CFT from Lattice Fermions*, doi:10.1007/s00220-022-04521-8, CMP **398** (2023) 219) give, building on the MMST (Morinelli–Morsella–Stottmeister–Tanimoto) operator-algebraic (wavelet) renormalization framework (CMP 387 (2021) 299; PRL 127 (2021) 230601), the chiral-CFT scaling limit of free lattice fermions (Koo–Saleur → Virasoro, correct c); WZW range rank $\le c \le D$, and $(E_8)_1$ (c=8, rank 8, D=16 Majoranas, $SO(16)_1 \to (E_8)_1$) is inside it. **[C]** OS leg: Adamo–Moriwaki–Tanimoto (arXiv:2407.18222, 2024) prove the conformal OS axioms for unitary lattice VOAs. **[E]** target pinned by $c = g_{\rm car} + N_{\rm fam} = 8$, $\det \mathrm{Cartan}(E_8) = 1$. **[O]** narrowed residual = "the raw collar's massless scaling limit is the holomorphic $(E_8)_1$ net" ($\det K = 1$), not a from-scratch construction.

- **v337 (DECOUPLING.THEOREM.01) — the Decoupling Theorem.** [**E**] every dimensionless readout factors through the gapped admissible transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$; finite susceptibility $\chi = 1/(1-(2/3)^6) = 729/665$ gap-suppresses the ambient contribution (margin $6\ln\frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$, with $2\dim(E_8)c_3^2 = 31/(4\pi^2)$ exactly), so no readout depends on the ambient measure ⇒ QG.AMB.01 decoupled; negative control (gap closed ⇒ χ diverges) makes it non-vacuous. This is the theorem that makes the QG.AMB.01 reclassification (v335) honest; consolidates v76/v311/v330/v332.
- **v338 (THETA13.BUDGET.01) — the θ_{13} budget, fenced.** The $+2\sigma$ pressure point (v328) quantified against *both* NuFIT 6.0 NO variants: $+2.0\sigma$ vs the no-SK central (0.02195) but $+1.7\sigma$ vs the with-SK best fit (0.02215 ± 0.00057), a $\sim 0.3\sigma$ systematic spread; [**E**] 0.023108 lies inside the with-SK 3σ band (upper 0.02388), an upper-edge pressure point not a falsification; [**X**] JUNO's sub-1% precision turns it into a $\sim 6\sigma$ kill-or-confirm.
- **Lean (FORM.CARTAN.DET.01).** CartanDeterminants.lean extended with the explicit $D_8 = SO(16)$ Cartan matrix and a fully-proven (kernel-only) holomorphy discriminator: D_8 is even and symmetric just like E_8 (same c=8) but $4 = \det\mathrm{Cartan}(D_8)$ is genuinely not a unit while $\det\mathrm{Cartan}(E_8)$ is — so unimodularity (one anyon) is the load-bearing selector. lake build green.

2026-06-22 (keystone unification + status reframe: ONE open theorem, ONE decoupled foreign problem: v335, FORM.QGEO.BW.01 qgeoSymIsCorollary)

- **v335 (SEAM.EQUIV.UNIFY.01) — the keystone unification.** A bird's-eye consolidation of the v308/v323/v325/v329/v333 arc that collapses the "two open structural items" narrative. [**E**] **QGEO.SYM.01 is a COROLLARY of SEAM.EQUIV.01** (no extra premise): a chiral conformal net has, by axiom, a Möbius-covariant vacuum (the unique invariant positive-energy vector); rotations $U(1)$ are the compact subgroup of the Möbius group, so the vacuum is rotation-invariant, and the μ_4 clock = rotation by $\pi/2$ ($(e^{i\pi/2})^4$=1) fixes it ⇒ $\omega \circ \rho = \omega$. So SEAM.EQUIV.01 ⇒ QGEO.SYM.01 with NO extra premise (the v323 "rotation-invariant vacuum" is a net axiom) — the two bedrock items collapse to ONE. [**E**] **the one keystone theorem**: SEAM.EQUIV.01 = a construction theorem ($c = g_{\mathrm{car}} + N_{\mathrm{fam}} = 8$, $\det\mathrm{Cartan}(E_8) = 1$, gap $6\ln\frac{3}{2} > 0$ derived) with EXACTLY one open continuum lemma (the rigorous continuum OS reconstruction + invertibility of the raw collar). [**E**] **QG.AMB.01 decoupled, not a gate**: margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76/v311/v330/v332) ⇒ no TFPT prediction depends on the ambient measure; it is the *general* Euclidean-QG conformal-factor problem (Gibbons–Hawking–Perry 1978), inherited not TFPT-specific — reclassified from "TFPT structural frontier" to "decoupled general QG problem". [**E**] **collapsed status**: open structural items $2 \to 1$; live residual = $v_{\mathrm{geo}} + G_{\mathrm{net}}$(=SEAM.EQUIV.01) + F_{transfer}. [**O**] the one residual = SEAM.EQUIV.01's continuum OS reconstruction.
- **Lean (FORM.QGEO.BW.01, qgeoSymIsCorollary).** BWKeystone.lean extended with the cited conformal-net vacuum axiom conformal_net_vacuum_rotation_invariant; with it, StateInvariance (=QGEO.SYM.01) follows from SeamIsE8 alone, so QGEO.SYM.01 is a COROLLARY of SEAM.EQUIV.01 with no independent open premise (#print axioms confirms only the chain + cited BW/net axioms). lake build green (3362 jobs).
- **Status deep-sync.** The ledger (QG.AMB.01 reclassified to "decoupled general QG; non-TFPT"; QGEO.SYM.01 marked a corollary; new SEAM.EQUIV.UNIFY.01 row), the canonical status card (tfpt_status.tex), introduction (status core + gate table), origin_theory, tfpt_4_frontier, tfpt_5_redteam, tfpt_research_contracts, README, next.txt and the website (OpenGates, FAQ, papers.ts) all moved to "one solvable theorem (SEAM.EQUIV.01) + one decoupled foreign problem (QG.AMB.01)". No falsifiable prediction changed; this is a status-framing sharpening backed by v335 + the Lean corollary.

2026-06-22 (closing round: the CM-selection mechanism (the last keystone residual), the conformal-mode resolution, and the Conway–Sloane even-unimodular side in Lean: `v333`, `v334`, `FORM.CARTAN.DET.01`)

- **`v333_cm_selection.py` (`QGEO.CMSELECT.01`, [E]/[O]).** The CM-selection mechanism for the last keystone residual: $\tau{=}i$ (the square) is the *unique* order-4 modulus — $S{=}\left(\begin{smallmatrix}0 & -1\\ 1 & 0\end{smallmatrix}\right)$ has $S^4{=}I$ and fixes i, and an order-4 deck selects $j{=}1728$ (vs the order-3 hexagon $j{=}0$) — AND the square is the Fekete attractor of symmetric mark-equilibration (four points minimizing the log-energy converge to equal $\pi/2$ spacing). So given the μ_4 deck, $\tau{=}i$ is forced; the residual sharpens from "why the square" to "the deck acts geometrically" (= `QGEO.SYM.01`). NOT a closure.
- **`v334_conformal_resolution.py` (`QGAMB.CONFORMAL.01`, [E]/[C]/[O]).** The conformal-mode resolution for the QG metric sector: the Gibbons–Hawking–Perry contour rotation $\phi \to i\phi$ flips the wrong-sign kinetic term, making the divergent conformal Gaussian $\int e^{+|c|\phi^2}$ into the convergent $\int e^{-|c|\phi^2}{=}\sqrt{\pi/|c|}$; the entire IDG form factor e^{p^2/M^2} dresses the propagator pole-free (the polynomial $R{+}R^2$ ghost as control). Together they give a *conditional* mode-by-mode construction of the metric-sector measure; the residual is only the contour's nonperturbative validity (gap-decoupled, blocks no test). NOT a closure.
- **`CartanDeterminants.lean` extended (`FORM.CARTAN.DET.01`, [E]).** The Conway–Sloane lattice side: the E_8 Cartan matrix is proved to be an EVEN (diagonal = 2), symmetric, unimodular ($|\det|{=}1$) integer Gram matrix — the defining data of an even unimodular rank-8 lattice — all by kernel `decide` / the integer-inverse argument. The remaining content (Witt's uniqueness: E_8 is the only even unimodular lattice in dim < 16) is the cited classification residual.

2026-06-22 (endgame round: the necessity of H reduced to the order-4 CM selection, the QG metric sector characterized as the conformal-mode problem, $|\det E_8|{=}1$ and the μ_4 commutation grounded in Lean: `v331`, `v332`, `FORM.MU4COMM.01`)

- **`v331_necessity_of_H.py` (`QGEO.NECESS.01`, [E]/[O]).** The necessity companion to `v329`: a concrete Steklov (DtN) computation shows the seam $\Lambda{=}|k|{+}M_f$ commutes with the μ_4 clock *iff* the four marks sit at the square ($\tau{=}i$) — moving a mark off the square breaks $\mathbb{Z}_4$ and $[\rho, \Lambda] \neq 0$. The square has cross-ratio 2 ($j{=}1728$), the unique 4-config with an order-4 automorphism (`v214`/`v267`), so $H \Leftrightarrow$ square $\Leftrightarrow \mu_4$ and "necessity of H" sharpens to: the raw dynamics places the four Gauss–Bonnet marks (`v216`) at the order-4 CM point. Residual = that dynamical selection.
- **`v332_qgamb_metric_sector.py` (`QGAMB.METRIC.01`, [E]/[C]/[O]).** The open metric sector of `QG.AMB.01` is characterized precisely as the Gibbons–Hawking–Perry conformal-factor problem: the conformal mode has a negative kinetic coefficient $-(D{-}1)(D{-}2)/4$ ($= -\frac{3}{2}$ at $D{=}4$), so the bare Euclidean measure diverges over it. The obstruction is in the conformal/gauge direction, gap-insulated from the admissible sector (`v330`/`v76`); the only route is the conditional IDG taming (`v304`), with the polynomial $R{+}R^2$ ghost as the control. Open, but pinned to one named obstruction.
- **`CartanDeterminants.lean` extended (`FORM.CARTAN.DET.01`, [E]).** The rank-8 holomorphy discriminator $|\det \mathrm{Cartan}(E_8)|{=}1$ (E_8 has one anyon, vs D_8's four) is now proved principally and kernel-only: `cartanE8 * cartanE8inv = 1` by `decide` on the 8×8 PRODUCT gives `IsUnit (det E8)` via $\det E_8 \cdot \det E_8^{-1} {=} \det 1{=}1$ — no 8! determinant, no `native_decide` (Mathlib itself leaves `E8_det = 1` a `proof_wanted`).
- **`Mu4Commutation.lean` (`FORM.MU4COMM.01`, Lean 4, [E]).** The v201/v323 mark-locality criterion as exact linear algebra over the Gaussian integers $\mathbb{Z}[i]$: $i^4{=}1$, $i^2{=}-1$, and (on

concrete 6×6 matrices) an offset-4 coupling commutes with the μ_4 clock while an offset-2 coupling does not — all by kernel `decide`. Built `lake build` on Lean v4.29.1 + Mathlib.

2026-06-22 (frontier round: the OS step discharged at the many-body level, the ambient measure built on the gap-decoupled sector, and the carrier Cartan determinants grounded in Lean: `v329`, `v330`, `FORM.CARTAN.DET.01`)

- **`v329_os_gap_reduction.py` (`QGEO.OSGAP.01`, [E]/[O]).** The one analytic input the v308 keystone chain still cited — "the OS-reconstructed bulk gap = the transfer gap" — is discharged at the many-body level: the OS/Euclidean Hamiltonian is the second quantization $d\Gamma(-\log T)$, whose spectrum is the set of sums of single-mode energies, so the bulk gap = the smallest positive single-mode energy $= 6 \ln \frac{3}{2}$, *independent of system size.* Together with $H \Rightarrow \omega \circ \rho{=}\omega$ and $H \Rightarrow \mathrm{SRE} \Rightarrow (E_8)_1$, this exhibits the rotation-invariant flat-pillowcase premise H as the *single sufficient premise* of both bedrock IDs; residual = the necessity of H on the raw collar.
- **`v330_qgamb_admissible_measure.py` (`QGAMB.MEASURE.01`, [E]/[O]).** The constructive step `v275`'s roadmap calls for: on the gap-decoupled *admissible* sector the ambient measure is built as a bona fide OS object — reflection-positive (symmetric PSD transfer), exponentially clustering ($C(n)/C(n{-}1) \to (2/3)^6$), with finite susceptibility $\chi = 729/665$ so $\mathrm{Var}(S_\Lambda)/\Lambda \to \chi$ converges $\Rightarrow$ uniform tightness (Prokhorov) $\Rightarrow$ the projective limit exists; OS reconstruction gives $H_{\mathrm{OS}} \geq 0$ with gap $6 \ln \frac{3}{2}$. The full $\mathrm{QG}_{\mathrm{amb}}$ (the metric sector) stays open, gap-decoupled — a genuine *partial* construction, not a closure.
- **`CartanDeterminants.lean` (`FORM.CARTAN.DET.01`, Lean 4, [E]).** The carrier piece of the holomorphy/anyon discriminator is grounded in real `Matrix.det` computations (kernel `decide`, no extra axiom): $|\det \mathrm{Cartan}(A_3)| {=} 4 {=} |\mu_4|$ ($3{\times}3$, `det_fin_three`) and $|\det \mathrm{Cartan}(D_5)| {=} 4$ ($5{\times}5$, `decide`), giving the carrier $|\det \mathrm{Gram}| {=} 4 \cdot 4 {=} 16$ (the `v281` anyon count). The rank-8 E_8/D_8 case stays the Nat discriminator — Mathlib's own `Data.Matrix.Cartan` leaves `E8_det = 1` as a `proof_wanted`. Built `lake build` on Lean v4.29.1 + Mathlib.

2026-06-22 (next-steps round: the keystone collapsed to one shared premise (Lean + one citable theorem), the F_{transfer} solver suite, the cusp weight derived from a minimal rewrite, and the θ_{13} pressure point: `FORM.QGEO.BW.01`, `v325`, `v326`, `v327`, `v328`)

- **`BWKeystone.lean` (`FORM.QGEO.BW.01`, Lean 4, [E]/[O]).** The `v323` Bisognano–Wichmann linkage formalised: the nodes RotationInvariantVacuum $\to$ GeometricModularFlow $\to$ Mu4ModularSymmetry $\to$ StateInvariance (=`QGEO.SYM.01`) composed over the `v308` chain. The theorem `bedrockReducesToRotationInvariance` shows the WHOLE bedrock `QGEO.SYM.01` reduces to the SINGLE shared open premise `RotationInvariantVacuum` — the two open bedrock items collapse to one — with `#print axioms` confirming the residual = the cited chain steps + recovery gap + the named BW steps + that one premise. Plus a `decide`-proven discriminator $(1 \mid 4) \wedge \neg(4 \mid 1)$ (BW geometric flow strictly stronger than `v309` clock invariance). Built `lake build` on Lean v4.29.1 + Mathlib.
- **`v325_pillowcase_keystone.py` (`QGEO.KEYSTONE.01`, [E]/[O]).** The keystone as ONE citable theorem: IF the raw seam state is the flat $\tau{=}i$ pillowcase (rotation-invariant) state THEN the whole bedrock closes (4 square marks, μ_4 deck, $\omega \circ \rho{=}\omega$, holomorphic $c{=}8 \Rightarrow (E_8)_1$, with the recovery gap as input); all pieces machine-checked (incl. the pillowcase $\chi_{\mathrm{orb}}{=}0$), the single open lemma shared by both bedrock IDs and Lean-pinned. An assembly certificate, not a closure.
- **`v326_ftransfer_suite.py` (`FR.SUITE.01`, [E]/[C]).** The four F_{transfer} readouts productised into one runnable harness, each a gapped relaxation to a unique attractor: F_{pole} (Koide, seam

rate $(2/3)^6$), $F_{\text{Boltzmann}}$ (η_B, H-theorem), F_{relic} (axion adiabatic invariant), F_{QCD} (m_p/m_e, RG to the UV fixed point), plus the F_{RareKaon} bridge. Only F_{pole} carries the seam rate; the four external rates are honestly fenced (`v187`/`v303`).

- **`v327_hypergraph_rewrite.py` (`HYP.REWRITE.01`, [E]/[O]).** The cusp weight 2/3 is DERIVED from a minimal rewrite: a 3-channel/1-absorbing family rule has spectrum $\{1, 2/3, 0\}$, so $2/3=(N_{\text{fam}}-1)/N_{\text{fam}}=|\mathbb{Z}_2|/N_{\text{fam}}$ emerges from the rule arity and $(2/3)^6=64/729$ is the recovery gap; with a PROOF that 2/3 is not a root of the affine-E_8 charpoly ($p(2/3)=-3520/19683\neq 0$, sharpening `v312`). `v324`'s injected datum is reduced to the anchor arity $\{2, 3\}$.
- **`v328_theta13_pressure.py` (`FLAV.TH13.PRESSURE.01`, [C]/[X]) + `v307` hardening.** The honest weak spot named and quantified: $\sin^2\theta_{13}=\varphi_0^{\text{ret}}e^{-5/6}=0.023108$ is $+2.0\sigma$ above NuFIT 6.0 NO, a SEPARATE carrier-trace channel (`v268`), not relieved by any freeze-respecting correction (RG $< 1\%$, exponent 5/6 exact), pre-registered as the kill. The `v307` watchdog gains a data provenance/date check and the rare-kaon F_{transfer} bridge row.
- **Wolfram mirror** 285 → 288. `v325` ($\chi_{\text{orb}}=0$) and `v327` (rewrite spectrum $\{0, 2/3, 1\}$, $(2/3)^6=64/729$, and the non-graph-spectral proof) added to `tfpt_readouts_extension.wl`; `GATE.WOLFRAM.02` and the Wolfram README updated to 288/288.

2026-06-22 (consistency + de-accretion editorial pass: current-state alignment across papers, website and ledger; no new modules)

- **Keystone naming reconciled to `SEAM.EQUIV.01`.** Following `v323`/`QGEO.BW.01` (the deck premise is downstream of the seam being a chiral net), the single named open keystone is now `SEAM.EQUIV.01` *everywhere*, with `QGEO.SYM.01` stated as its *downstream* conformal-deck face. The ledger row `QGEO.SYM.01` was updated (it no longer calls itself "the single structural residual of the whole theory"; it now depends on `SEAM.EQUIV.01`/`QGEO.BW.01`), and the superseded "`GATE.QGEO` open gate" / "two research gates (U_{wall}) and G_{metric}" framing was replaced by the current Rest $= v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$ with the two structural items `SEAM.EQUIV.01` + `QG.AMB.01` (intro status card, `tfpt_4_frontier`, `origin_theory`, `tfpt_5_redteam`, `tfpt_research_contracts`, and the website).
- **De-accretion: the sprint-by-sprint reduction history was moved to this changelog, where it belongs.** Several reader-facing sections had grown by accretion (each round appended a clause). They were rewritten to state the *current* result once, with the round-by-round narrative left here: the introduction closure ledger ("honest bottom line") and hard-reduction subsection; the `origin_theory` "residual inventory" (the five "Update/…/Terminal" passes collapsed to one current inventory); the `tfpt_5_redteam` Target-A Outcome; and on the website the FAQ "What is still open?", the `OpenGates` closing-condition block, the `papers.ts` research-contracts/red-team entries, the verification-DAG `qft` node, the glossary `G_net` entry, the Rosetta lexicon, and the verification/review pages. No scientific content or `\veri` citation was dropped; only the embedded version-stamp logs were removed.
- **Stale facts corrected.** Version label `5.0`→`v5.2` (intro, single-sourced via `\TFPTversion`); Lean build `1886`→`3357` jobs (`FORM.SEAMEQUIV.01`); README Python suite `v213`/ 212→`v324`/ 322 modules; Wolfram extension `269/269` / `249/249`→`285/285` (README, website replication page, `zenodo_description.html`); residual "Paper-3" old-paper attributions in `tfpt_2` neutralised.

(Original 2026-06-22 entry follows.)

2026-06-22 (external-review pass: a registry bug fix, the NA62 2026 rare-kaon update, CP-phase prose hygiene, and three firewall/visualisation enhancements: `v202`, `v203`, `v305`, `v307`, `v321`)

- **Bug fix — $\delta_{\rm CKM}$ display value.** The predictions table (`tex-artefacts/predictions.tex`) showed $\delta = \frac{\pi}{3} + 3\lambda^2 = 66.96°$, but that formula evaluates to 68.65° (the canonical frozen `DELTA_CKM_RAD`= 1.198232 rad, used by `v88/v202/FLAV.CP.01` everywhere else). Corrected to 68.65°. An external review correctly flagged the inconsistency; it was an isolated display typo (the website was already clean).
- **`v202_rare_kaon.py` — NA62 2016–2024 update ([E]/[C]).** The charged-mode comparison is moved from the historical NA62 2016–2022 observation $(13.0^{+3.3}_{-3.0}) \times 10^{-11}$ ($\sim 1.2\sigma$ above) to the full 2016–2024 combination $\mathrm{BR}(K^+) = (9.6^{+1.9}_{-1.8}) \times 10^{-11}$ (La Thuile 2026), on which the prediction 9.45×10^{-11} lands at $+0.08\sigma$. Framed honestly: a strong *consistency* hit for the closed CKM point, but an $F_{\rm transfer}$ bridge — *not* a unique TFPT-vs-SM discriminator (the SM $(8.6 \pm 0.4) \times 10^{-11}$ is also inside the NA62 error). Propagated to the ledger (`FR.RAREKAON.01`), `freeze_file` (new `rare_kaon` kill row), the watchdog (`v307`, a new $F_{\rm transfer}$ bridge row), the forward board (`v321`, rank 9), and `tfpt_2_standard_model` + the website.
- **CP-phase prose hygiene.** The Galois CP lock is tightened from "a relation to the *measured* quark phase" to "a relation to the *leading* ($\pi/3$) component of the quark phase": the lock is to the structural $\pi/3$, not the full measured $\gamma = \delta^{\rm lead}_{\rm CKM} + 3\lambda^2_C \approx 68.7°$, so $\delta_{\rm PMNS} = 240°$ (not 248.7°). Fixed in `tfpt_2_standard_model`, `origin_theory`, `papers.ts` and `predictions.ts`.
- **`v305_witness_independence.py` — the No-Golden-Dynamics rule ([E]).** The number-field split is made an explicit anti-numerology firewall rule: $\phi = 2\cos(\pi/5) \in \mathbb{Q}(\sqrt{5})$ is the *static* carrier ($\mathbb{Z}/5$) field, every *dynamic* rate is rational ($\mathbb{Q}$, the $\mathbb{Z}/6$ hand), so a dynamic rate explained by ϕ is a cross-field false friend (`v314`/`v315`).
- **`v203_eht_achromatic.py` — the μ_4 Fourier fourth null ([X]).** The EHT residual pre-registration gains a fourth null: the azimuthal residual must carry power *only* at $m \equiv 0 \bmod 4$ (the μ_4 fingerprint of the four marks, `v201`); a μ_4-orbit profile has ~ 0 off-mode power while a $\mathbb{Z}_2$ control leaks at $m = 2$.
- **Seed-hyperplane figure (in the introduction).** A new figure visualises `v306`: every scorecard observable inverts *independently* to the one latent seed φ_0, the error-weighted mean matching the axiom to 0.04%, with $\sin^2\theta_{13}$ the $\sim 2\sigma$ pressure point. Added to `make_figures.py`, `tfpt_5_redteam` and the manifest.

2026-06-22 (Wolfram mirror of the arithmetic arc + three sharpening steps: the CP-lock made quantitative, the Bisognano–Wichmann keystone, the minimal hypergraph fibre: `v322`, `v323`, `v324`)

- **Wolfram mirror of the cyclotomic/Galois arithmetic arc (the independent second path).** The exact algebraic results of the arc are now mirrored in `verification/wolfram/tfpt_readouts_extension.wl` (ten new `checkExact` blocks, 275 → 285, *all passing*): `v313` (the affine-E_8 characteristic polynomial with the golden factor x^2-x-1; the $(2,3,5)$ atoms as $2\cos(\pi/k)$; the 31/30 selection), `v314` (the rational kernel vs $\mathbb{Q}(\sqrt{5})$ dynamic-rate split), `v315` (30=5·6, $\varphi(30)$=8, the Galois group $\mu_4 \times \mathbb{Z}_2$, the $\sqrt{5}$ Gauss sum), `v316` (ρ=ζ_6=ζ_{30}^5, $\rho^4 = -\rho$, Gal=$\mathbb{Z}_2$= CP conjugation), `v317` ($N_{\rm fam}$=3 as the μ_3 orbit, the 1+2 Galois split), `v318` ($\varphi_0^{\rm ret}$=$(4/3)c_3$+$48c_3^4$ a pure function of π) and `v320` (the CP lock $\delta_{\rm PMNS}$=240°). Ledger `GATE.WOLFRAM.02` and the Wolfram README updated to 285/285.
- **`v322_cp_lock_sharpened.py` (`GALOIS.CP.SHARPEN.01`, [E]/[C]/[X]).** The `v320` CP-lock sharpened into a quantitative, pre-registered kill test. [**E**] the leading phase sits on

the six hexagonal nodes $\arg\rho^k{=}k\cdot 60°$, and *which* one is selected is the deck order: $\delta^{\rm lead}_{\rm PMNS}{=}|\mu_4|$ $\delta^{\rm lead}_{\rm CKM}{=}4(\pi/3)$ (node 4); **[C]** the sub-leading correction is bounded by the quark analogue $3\lambda^2\approx 8.7°$, so the prediction is the band $240°\pm\sim 9°$; **[C]** against NuFIT 6.0 (NO, $\delta_{CP}{=}212^{+26}_{-41}{}^\circ$) the leading value is $+1.08\sigma$, the band edge $+0.74\sigma$ – compatible today; **[X]** the nearest wrong node is 60° away, far beyond the budget, so DUNE/Hyper-K ($\sim$ 5–15°) discriminate cleanly. Cited in `origin_theory`; ledger `GALOIS.CP.SHARPEN.01`.

- **`v323_bw_geometric_modular.py` (`QGEO.BW.01`, [E]/[C]/[O]).** The keystone pushed one honest step via Bisognano–Wichmann. **[E]** the μ_4 clock is a geometric rotation $\rho{=}\mathrm{diag}(i^n){=}\exp(i\frac{\pi}{2}L)$, $\mu_4\subset U(1)_{\rm rot}$; **[E]** for a rotation-covariant covariance $C{=}f(L)$ the modular Hamiltonian $K{=}\log((1{-}C)/C){=}g(L)$, so $[K,L]{=}0$ (the modular flow is geometric) and the clock is a modular symmetry *for free* (the discriminator: a period-4 curvature preserves the clock but its flow is not geometric, $[K,L]\neq 0$); **[C]** given the seam is the $(E_8)_1$ chiral net (`v308`), BW/Hislop–Longo make the vacuum modular flow geometric, so `QGEO.SYM.01` ($\omega\circ\rho{=}\omega$) is *downstream* of `SEAM.EQUIV.01` + a rotation-invariant vacuum, not an independent axiom; **[O]** residual: the raw collar vacuum is rotation-invariant (cleaner than `v309`, shared with `v308`). Cited in `origin_theory`; ledger `QGEO.BW.01`.
- **`v324_hypergraph_fiber.py` (`HYP.FIBER.01`, [E]/[C]).** The constructive `v312` follow-up: the *minimal* substrate that carries both the E_8 skeleton and the recovery gap is a *product*. **[E]** the carrier network $T_{\rm net}{=}(A{+}2I)/4$ (attractor = Kac marks = skeleton) fibred by the 3-node family cusp $T_{\rm cusp}{=}\mathrm{diag}((1{-}w)^{2N_{\rm fam}})$, $w\in\{0,\frac{1}{3},\frac{2}{3}\}$ (spectrum $\{1,(2/3)^6,(1/3)^6\}$); the fibred $T_{\rm net}\otimes T_{\rm cusp}$ has top eigenvalue 1 (eigenvector marks $\otimes(w{=}0)$) and $(2/3)^6$ as a genuine eigenvalue (marks $\otimes(w{=}\frac{1}{3})$) – the $N_{\rm fam}{=}3$ cusp is the minimal fibre and the substrate = carrier $\times$ family = the $30{=}g_{\rm car}(2N_{\rm fam}){=}5{\times}6$ of `v315`; **[C]** honestly (`v312`) the cusp weight is still *injected*, not derived from the graph – a consistent realisation, not a derivation of 2/3 from a pure rewrite. Cited in `origin_theory`; ledger `HYP.FIBER.01`.

2026-06-22 (the forward kill-test board: the decisive upcoming measurements ranked, each locked to a freeze row — incl. the new Galois-CP relation: `v321`)

- **`v321_killtest_board.py` (`KILL.BOARD.01`, [E]).** The forward companion to `v307` (the current-data board): the decisive *upcoming* measurements, ranked and each locked to a `freeze_file.csv` kill row – #1 JUNO $\sin^2\theta_{12}{=}0.306747$ (fastest), #2 CMB-S4/LiteBIRD r (kill $r{>}0.01$), #3 DESI/Euclid w (kill $w{\neq}{-}1$), #4 DUNE/Hyper-K $\delta_{\rm PMNS}{=}240°$ (the new `v320` Galois-CP relation), #5 DUNE/LEGEND/nEXO ordering+$m_{\beta\beta}$, #6 PSI n2EDM, #7 JUNO $\sin^2\theta_{13}$ ($\sim 2\sigma$ now), #8 LiteBIRD β. A forward decision ledger (publicly-stated experimental reach), complementing `v307`; no new physics. Cited in `origin_theory`; ledger row `KILL.BOARD.01`.

2026-06-22 (Lean: the `SEAM.EQUIV.01` composition chain formalised — the keystone residual machine-pinned to the named cited steps: `FORM.SEAMEQUIV.01`)

- **`SeamEquivChain.lean` (`FORM.SEAMEQUIV.01`, Lean 4, [E]/[O]).** A Lean 4 mirror of the `v308` closing chain: the four nodes GapPositive → InvertibleSRE → HolomorphicC8 → SeamIsE8 assembled as a well-typed composition theorem `seamEquivChain`, with the three cited literature steps (Kitaev free-fermion invertibility; Müger / Kawahigashi–Longo–Müger holomorphy; Conway–Sloane E_8 uniqueness) as named `axioms` and the carrier recovery gap (the OS step) as the single open input. **[E]** `#print axioms seamEquivChain` confirms the residual is exactly those named steps + `recovery_gap` (no hidden assumption); **[E]** a `decide`-proven discriminator primariesE8 $=1\neq$ primariesD8 $=4$ (the $|\det\text{Cartan}|$ that separates the holomorphic E_8 from the same-c rival $D_8{=}SO(16)$). Built end-to-end (`lake build`, 3357 jobs) on Lean v4.29.1 + Mathlib. **[O]** The value is the machine-checked composition + axiom audit (pinning

the keystone residual), *not* a from-scratch proof — it does NOT close `SEAM.EQUIV.01`. Ledger row `FORM.SEAMEQUIV.01`; mirrors `v308`.

2026-06-22 (a NEW falsifiable prediction from the Galois structure: the two CP phases are locked, $\delta_{\rm PMNS} = \delta^{\rm lead}_{\rm CKM} + \pi$: `v320`)

- **`v320_galois_cp_relation.py` (`GALOIS.CP.PREDICT.01`, [E]/[C]/[X]).** The `v316` follow-up turned into a testable cross-prediction. **[E]** both leading CP phases are powers of the *one* hexagonal unit $\rho = \zeta_6$ of the family Galois factor: $\delta^{\rm lead}_{\rm CKM} = \arg\rho = \pi/3$ (60°), $\delta_{\rm PMNS} = \arg\rho^4 = 4\pi/3$ (240°), and $\rho^4 = -\rho$ locks them as $\delta_{\rm PMNS} = \delta^{\rm lead}_{\rm CKM} + \pi$ (the $\mathbb{Z}_2$ sheet flip); **[C]** this upgrades the previously assigned $\delta_{\rm PMNS} = 240°$ to a Galois-forced relation to the measured quark phase (the quark and lepton leading CP phases are not independent); **[X]** kill test: a $\delta_{\rm PMNS}$ robustly incompatible with 240° ($> 3\sigma$ at DUNE/Hyper-K/JUNO) falsifies the Galois-CP organisation; 240° is currently within the broad NuFIT 6.0 range. A genuine new falsifiable consequence of the `v313`–`v319` arithmetic arc (also a `freeze_file.csv` `cp_phase_relation` kill row). Cited in `origin_theory`; ledger row `GALOIS.CP.PREDICT.01`.

2026-06-22 (the translation clock: the discrete↔dynamic bridge *is* the order-30 Coxeter element, a clock with two coprime hands 5×6, read law-inclusive (0..5) or live-only (1..5): `v319`)

- **`v319_translation_clock.py` (`TRANSLATE.CLOCK.01`, [E]/[C]).** A synthesis of the `v313`–`v318` arc with `v124`/`v55`, answering the question "is the static↔dynamic translation a clock running $0,1,2,3,4,5$ or $1,2,3,4,5$?". **[E]** the only object holding both the static carrier ($\mathbb{Q}(\sqrt{5})$) and the dynamic family (rational) data is the order-30 Coxeter element (`v314`); $30 = h(E_8) = g_{\rm car}(2N_{\rm fam}) = 5 \cdot 6$ and $\gcd(5,6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (two coprime hands, `v315`). **[E]** the *dynamic* hand is $\mathbb{Z}/6 = 2N_{\rm fam}$ (ticks $0,\ldots,5$): the recovery exponent is exactly $6 = 2N_{\rm fam}$, rate $(2/3)^6 = (|\mathbb{Z}_2|/N_{\rm fam})^{2N_{\rm fam}}$, order-6 generator c^5 ($\zeta_6 = \zeta_{30}^5$, `v316`). **[E]** position 0 is the *conserved law* — the resummed clock rate$(n) = -p_2 \ln(1 - n/N_{\rm fam})$ (`v124`) has rate$(0) = 0$ (the attractor, eigenvalue 1, `v56`), the live decaying modes from n=1, and the E_8 live phases (totatives of 30) start at 1 ($\{1,7,\ldots,29\}$), the 0 phase not a totative — so "$0,1,2,3,4,5$" is law-inclusive, "$1,2,3,4,5$" live-only. **[E]** the *static* hand is $\mathbb{Z}/5 = g_{\rm car}$ (ticks $1,\ldots,5$): the golden field $\mathbb{Q}(\sqrt{5})$ ($\varphi = 2\cos(\pi/5)$), field-obstructed (no rational dynamic image, `v314`) — static-only; $\zeta_5 = \zeta_{30}^6$ carries no phase. **[E]** one full turn $= \mathrm{lcm}(5,6) = 30 = h(E_8)$, rank $E_8 = 8 = \varphi(30)$ live phases pairing into $|\mu_4| = 4$ invariant planes (`v55`). **[C]** verdict: the discrete→dynamic translation *is* the order-30 clock = (static 5-ring) × (dynamic 6-ring), the dynamic hand carrying the rate (exponent $6 = 2N_{\rm fam}$), the static hand the golden carrier (no rate), and "0 vs 1" = law-inclusive vs live-only; the arithmetic is **[E]**, the "it is one clock" reading **[C]**. Python-only (`sympy`), like the rest of the arc. New figure `figures/translation_clock.pdf` (concentric 5- and 6-hands). Cited in `origin_theory`; ledger row `TRANSLATE.CLOCK.01`.

2026-06-21 (capstone: the SM structural sector is $\mathbb{Q}(\zeta_{30})$+ Galois $\mu_4 \times \mathbb{Z}_2$, and the complete input is $\{a, \pi, v_{\rm geo}\}$ — zero dimensionless free parameters: `v318`)

- **`v318_arithmetic_capstone.py` (`ARITH.CAPSTONE.01`, [E]/[O]).** The synthesis of the `v313`–`v317` arc. **[E]** the SM *structural* sector (the 3 generations, the 2 CP phases, their orbit/hierarchy) is the cyclotomic field $\mathbb{Q}(\zeta_{30})$ with Galois group $\mu_4 \times \mathbb{Z}_2$ (degree $\varphi(30) = 8 =$ rank E_8, $30 = |\mathbb{Z}_2| N_{\rm fam} g_{\rm car} = h(E_8)$), forced by the anchor atoms $\{2,3,5\} = e(a)$; **[E]** the magnitude seed $\varphi_0 = (|\mu_4|/N_{\rm fam})c_3 + \Omega_{\rm adm} c_3^4 = \frac{4}{3}c_3 + 48c_3^4$ ($c_3 = 1/(8\pi)$) is a pure function of π (no free parameters, anchor-derived coefficients), so the magnitudes ($\sin^2\theta_{12} = \frac{1}{3} - \varphi_0/2$) are $\{a,\pi\}$-determined, not free; **[E]** free inventory: *zero* dimensionless free parameters, π the unique transcendental primitive (`ARCH.CORE.01`), $v_{\rm geo}$ the unique dimensionful input (No-Unit

Theorem) — complete input $\{a, \pi, v_{\text{geo}}\}$. **[O]** honest residual: this rigidity holds *given* the axioms and does not realise the structure on the raw seam (`QGEO.SYM.01`/`SEAM.EQUIV.01` stay **[O]**). Cited in `origin_theory`; ledger row `ARITH.CAPSTONE.01`.

2026-06-21 (are the 3 generations a μ_3/Galois orbit? Yes (μ_3), Galois-refined 1+2 with the fixed generation = the recovery attractor: `v317`)

- **`v317_galois_family.py` (`GALOIS.FAMILY.01`, [E]/[C]).** The `v316` follow-up to the generation structure. **[E]** the three cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$ (with $\mathrm{Spec}(Q_+) = \{1,2,3\}$, the A_3 exponents) are the cube roots of unity $\{1, \zeta_3, \zeta_3^2\}$ under $w \mapsto e^{2\pi i w}$, and the cyclic family symmetry $w \mapsto w+\frac{1}{3}$ is a *transitive* μ_3 orbit (the family IS the cube-root orbit); **[E]** the transfer eigenvalues $(1-w)^6 = \{1, (2/3)^6, (1/3)^6\}$ attach to the weights (attractor = eigenvalue 1 at w=0); **[E]** the family Galois $\mathbb{Z}_2$ ($\zeta_3 \mapsto \zeta_3^2 = \bar{\zeta}_3$, = the `v316` CP conjugation) refines the orbit to 1+2 — fixing w=0 (rational) and swapping w=$\frac{1}{3} \leftrightarrow \frac{2}{3}$; **[E]** so the Galois-FIXED generation is the recovery ATTRACTOR (the law) and the Galois-conjugate pair the decaying hierarchy. **[C]** verdict: the family STRUCTURE is the same arithmetic as the CP phase (a μ_3 orbit, Galois-refined 1+2), the mass MAGNITUDES remaining the analytic seed. Cited in `origin_theory`; ledger row `GALOIS.FAMILY.01`.

2026-06-21 (does the Galois $\mu_4 \times \mathbb{Z}_2$ organize the readouts? Yes — the CP phases are the family-factor cyclotomic data, Galois $\mathbb{Z}_2$ = CP conjugation: `v316`)

- **`v316_galois_readout.py` (`GALOIS.READOUT.01`, [E]/[C]).** The `v315` follow-up to phenomenology: does the Galois coupling $\mu_4 \times \mathbb{Z}_2$ reach the physics? **[E]** it does, through the *family* factor — the hexagonal CP unit $\rho = \zeta_6 = \zeta_{30}^5$ is exactly the order-$6 = 2N_{\text{fam}}$ factor c^5 of `v315` (the carrier being $\zeta_5 = \zeta_{30}^6$); **[C]** the leading CP phases are its cyclotomic arguments $\delta_{\text{CKM}}^{\text{lead}} = \pi/3 = \arg\zeta_6$ and $\delta_{\text{PMNS}} = 4\pi/3 = \arg\zeta_6^4$ (power $4 = |\mu_4|$), with $\zeta_6^4 = -\zeta_6$ the $\mathbb{Z}_2$ sheet flip and $\mu_6 = \mu_3 \times \mu_2$ (inherits `v231`/`v233`); **[E]** the family Galois $\mathrm{Gal}(\mathbb{Q}(\zeta_6)/\mathbb{Q}) = \mathbb{Z}_2$ ($\zeta_6 \mapsto \zeta_6^5 = \bar{\zeta}_6$) *is* CP conjugation $\delta \mapsto -\delta$; **[E]** the carrier factor ζ_5 (golden $\sqrt{5}$) carries no phase (μ_4 static), and the magnitudes ($\sin^2\theta_{12} = \frac{1}{3} - \varphi_0/2$, transcendental φ_0) are not cyclotomic. **[C]** verdict: the Galois bridge reaches phenomenology in the phase sector predicted by the `v314` field split. Cited in `origin_theory`; ledger row `GALOIS.READOUT.01`.

2026-06-21 (couple or factorize? the order-30 Coxeter element couples the carrier and family sectors as a cyclotomic compositum, Galois = $\mu_4 \times \mathbb{Z}_2$: `v315`)

- **`v315_coxeter_coupling.py` (`COX.COUPLE.01`, [E]/[C]).** Answers the `v314` question — does $30 = g_{\text{car}}(2N_{\text{fam}}) = 5{\cdot}6$ couple the carrier (5-fold, golden, $\mathbb{Q}(\sqrt{5})$) and family ($6 = 2{\cdot}3$, rational) sectors, or just factorize? **[E]** the cyclic group *factorizes*: $\gcd(5,6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (direct product, group-decoupled, in line with the `v314` field obstruction); **[E]** the E_8 exponents = totatives(30) split as totatives(5)× totatives(6), $\varphi(30) = 4 \cdot 2 = 8 =$ rank E_8; **[E]** but the *field* couples: $\mathbb{Q}(\zeta_{30})$ is the compositum of the carrier $\mathbb{Q}(\zeta_5) \ni \sqrt{5}$ and the family $\mathbb{Q}(\zeta_3)$, degree 8 = rank E_8; **[E]** the Galois group is exactly $(\mathbb{Z}/5)^\times \times (\mathbb{Z}/3)^\times = \mathbb{Z}/4 \times \mathbb{Z}/2 = \mu_4 \times \mathbb{Z}_2$ (order 8, max element order 4), so the μ_4 seam clock *is* the Galois group of the carrier's golden field and $\mathbb{Z}_2$ that of the family field (refining `v223`); **[E]** $\sqrt{5} = \phi + 1/\phi$ is the quadratic Gauss sum in $\mathbb{Q}(\zeta_5)$. **[C]** verdict: $30 = 5 \cdot 6$ does not dynamically entangle the sectors — it couples them as the cyclotomic compositum, the static↔dynamic bridge being *Galois*, not a dynamical map. Cited in `origin_theory`; ledger row `COX.COUPLE.01`.

2026-06-21 (do the discrete-kernel and dynamic rates translate? An exact number-field split $\mathbb{Q}$ vs $\mathbb{Q}(\sqrt{5})$: `v314`)

- **`v314_rate_translation.py` (`RATE.TRANSLATE.01`, [E]/[C]).** Answers, exactly, whether the discrete kernel (the affine-E_8 network, carrying the golden ratio) and the dynamic part (the

seam transfer, carrying the recovery rate $(2/3)^6$) translate. [**E**] FIELD SPLIT: the adjacency spectrum is the rational part $\{0, \pm 1, \pm 2\} \subset \mathbb{Q}$ plus the golden part $\{\pm\phi, \pm 1/\phi\} \subset \mathbb{Q}(\sqrt{5})$, while the transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ is *entirely* rational; [**E**] the family/sheet rates DO translate (the rational angles $2\cos(\pi/2){=}0 = |\mathbb{Z}_2|$ and $2\cos(\pi/3){=}1 = N_{\text{fam}}$ build $(2/3)^6 = (|\mathbb{Z}_2|/N_{\text{fam}})^{2N_{\text{fam}}} \in \mathbb{Q}$, same field, same atoms), but [**E**] the carrier 5-fold (golden, $\mathbb{Q}(\sqrt{5})$) has NO rational dynamic image – a field obstruction (2/3 is neither an adjacency nor a lazy-update eigenvalue; heat-kernel $t = \Delta/(2-\phi) \approx 6.37$ not clean), so the carrier is *static-only*; [**E**] the only object holding both is the order-30 Coxeter element $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$, and the ϕ-quasicrystal $E_8 = H_4{+}\phi H_4$ ($240 = 2 \cdot 120$) carries the carrier but not the (rational, 3-fold) cusp-weight family dynamics. Cited in `origin_theory`; ledger row `RATE.TRANSLATE.01`.

2026-06-21 (the golden ratio made precise: the $(2,3,5)$ atoms are the affine-E_8 network spectral angles, $\phi = 2\cos(\pi/5)$ the $g_{\text{car}}{=}5$ signature: `v313`)

- **`v313_golden_atoms_spectral.py` (`GOLD.ATOMS.01`, [E]/[C]).** A follow-up to `v312`'s observation, made exact. [**E**] the affine-E_8 adjacency characteristic polynomial factors as $x(x^2{-}4)(x^2{-}1)(x^2{-}x{-}1)(x^2{+}x{-}1)$, so the golden minimal polynomial $x^2{-}x{-}1$ divides it and $\phi = 2\cos(\pi/5) = (1{+}\sqrt{5})/2$ is an *exact* eigenvalue; [**E**] the non-trivial eigenvalues are $2\cos(\pi/k)$ for exactly $k \in \{2,3,5\} = \{|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}}\}$ ($2\cos(\pi/2){=}0$, $2\cos(\pi/3){=}1$, $2\cos(\pi/5){=}\phi$, $2\cos(2\pi/5){=}1/\phi$), so the golden ratio is *specifically* the $g_{\text{car}}{=}5$ (5-fold) signature and $g_{\text{car}}{=}5$ gains a new spectral witness distinct from Pascal/Riemann–Roch (raising the `v305` multiplicity); [**E**] the $(2,3,5)$ spherical excess $1/2{+}1/3{+}1/5 = 31/30 > 1$ (the condition selecting the finite $2I \Rightarrow E_8$) ties the `v63`/`v76` gap-margin numerator $31 = 2^{g_{\text{car}}}{-}1 = 248/8 = 1{+}h(E_8)$ to the Coxeter $30 = h(E_8)$ — a unification of already-anchored quantities, NOT new evidence (honest anti-numerology, `v305`); [**E**] honest null: ϕ is a structural lattice/network quantity, not a phenomenological readout. [**C**] direction: ϕ is the E_8-quasicrystal (Elser–Sloane cut-and-project) ratio, a candidate aperiodic substrate for the `v312` question. Cited in `origin_theory`; ledger row `GOLD.ATOMS.01`.

2026-06-21 (TOE-roadmap solver round: the `SEAM.EQUIV.01` keystone certificate, the modular bedrock, the carrier→SM closure, the gap-decoupled measure, and the sharpened hypergraph substrate: `v308`–`v312`)

- **`v308_seam_equiv_chain.py` (`SEAM.EQUIV.CHAIN.01`, [E]/[O]).** TOE attack point 1: the keystone "the raw seam-Calderón net is the holomorphic $c{=}8$ $(E_8)_1$ net" assembled into ONE well-typed logical chain gap>0 $\to$ invertible/SRE (Kitaev) $\to$ holomorphic $c{=}8$ (Müger/KLM) $\to$ $(E_8)_1$ (Conway–Sloane), with the exact discriminators $|\det \text{Cartan}(E_8)|{=}1$ vs $|\det \text{Cartan}(D_8{=}SO(16))|{=}4$, the carrier tower $4{\cdot}4{=}16 \to 4 \to 1$ and $c{=}g_{\text{car}}{+}N_{\text{fam}}{=}8$ all machine-checked; the input gap is the derived recovery gap $6\ln(3/2){>}0$. The only undischarged premise is the cited OS step — an assembly/reduction certificate (a Lean target), not an end-to-end closure. Cited in `tfpt_research_contracts`; ledger row `SEAM.EQUIV.CHAIN.01`.
- **`v309_modular_bedrock.py` (`QGEO.MODHAM.01`, [E]/[O]).** TOE attack point 2: the bedrock $\omega{\circ}\rho{=}\omega$ via an explicit modular Hamiltonian, non-circular — the four marks come from Gauss–Bonnet (`v216`) and the order-4 clock from cross-ratio 2 (`v214`), so μ_4 is derived; the quasi-free modular Hamiltonian $K = \log((1{-}C)/C)$ round-trips to the seam operator and $[\rho, K]{=}0 \Leftrightarrow \omega{\circ}\rho{=}\omega$ (a $\mathbb{Z}_2$ profile breaks it). Reduces the bedrock to the Bisognano–Wichmann intrinsicality of the raw collar; not a closure of `QGEO.SYM.01`. Cited in `tfpt_research_contracts`; ledger row `QGEO.MODHAM.01`.
- **`v310_carrier_sm_anomaly.py` (`CAR.SM.ANOM.01`, [E]).** TOE attack point 3: the carrier half-spinor $16 = \Lambda^{\text{even}}(\mathbb{C}^5) = 1{+}10{+}5$ is exactly one anomaly-free Standard-Model generation

($SU(5) \subset SO(10)$: $10{+}\bar{5}{+}1$), with $\sum Y = \sum Y^3 = [SU(2)]^2U(1) = [SU(3)]^2U(1) = 0$ (exact) and the one-loop $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$; a fourth family shifts b_1, so $N_{\text{fam}}{=}3$ pins the RG. Closes the spectrum/anomaly/β sub-question; the interacting Epstein–Glaser S-matrix and the Yukawa sector stay **[C]**/**[O]**. Cited in `tfpt_research_contracts`; ledger row `CAR.SM.ANOM.01`.

- **`v311_gap_decoupled_measure.py` (`QGAMB.CLUSTER.01`, [E]/[C]/[O]).** TOE attack point 4: the physical QG measure as a gap-decoupled, exponentially-clustering object — the transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ gives geometric clustering at rate $(2/3)^6$ (correlation length $1/(6\ln(3/2))$, finite susceptibility 729/665), and the `v76` margin $6\ln(3/2) - 31/(4\pi^2) \approx 1.648 > 0$ decouples the physical sector from the un-built ambient. `QG.AMB.01` stays **[O]** (gap-decoupled, inert). Cited in `tfpt_research_contracts`; ledger row `QGAMB.CLUSTER.01`.

- **`v312_hypergraph_substrate.py` (`HYP.INJECT.01`, [E]/[O]).** TOE attack point 5: the hypergraph substrate sharpened — the affine-E_8 Kac marks are the Perron eigenvector ($A\,m{=}2m$) and the subleading eigenvalue is exactly $2\cos(\pi/5)$ = the golden ratio, but the recovery rate $(2/3)^6$ is *not* any eigenvalue and φ_0 is not a rational edge fraction; so a full-structure rewrite must inject exactly two non-graph-spectral data — the cusp weight $2/3{=}|\mathbb{Z}_2|/N_{\text{fam}}$ and the analytic seed φ_0. The open question is precisely delimited, not closed. Cited in `introduction`; ledger row `HYP.INJECT.01`.

2026-06-21 (adversarial-review follow-ups: a structural anti-numerology firewall, an out-of-sample seed cross-validation, an operational data watchdog, and the Target-A holomorphy kill test: `v305`–`v307`)

- **`v305_witness_independence.py` (`WITNESS.ECON.01`, [E]).** The *structural* anti-numerology firewall complementing the `v100` phenomenological null test, against the sharpest self-deception worry ("the same integer counted many times as if independent"). Every headline integer ($|\mu_4|{=}4$, $g_{\text{car}}{=}5$, $|\mathbb{Z}_2|{=}2$, $N_{\text{fam}}{=}3$, $\dim S^+{=}16$, $\operatorname{rank} E_8{=}8$, 240, 248, $|2I|{=}120$, $h(E_8){=}30$, $|W(D_5)|{=}1920$, $\Omega_{\text{adm}}{=}48$, the gap exponent 6) is a documented image of the single anchor $a{=}(1,1,2)$ with π the only transcendental primitive (13 atoms from 2 free primitives, 6.5× compression made explicit), so the recurring $2/3{=}|\mathbb{Z}_2|/N_{\text{fam}}$ and $(2/3)^6$ are *one* ratio; the E_8 spine is overdetermined (≥ 3 structurally independent witnesses) while the pure-seed readouts are single-witness (scored jointly by `v100`, never multiplied); the negative control shows α^{-1}'s "137" is foreign to the anchor family (an independent witness via the $F_{U(1)}$ cubic, `v3`). Cited in the `tfpt_5_redteam` body; ledger row `WITNESS.ECON.01`. Python-only.

- **`v306_seed_crossval.py` (`SEED.CROSSVAL.01`, [E]).** A leave-one-out cross-validation turning the "many observables, one seed φ_0" worry into a falsifiable out-of-sample test: back-solving φ_0 from each of the five seed-grammar observables ($\sin^2\theta_{12}$, $\sin^2\theta_{13}$, β, Ω_b, λ_C; NuFIT 6.0 / ACT DR6 / Planck / PDG) and error-weighted-averaging reproduces the axiom seed $\varphi_0 = 1/(6\pi) + 48c_3^4 = 0.0531720$ to 0.04%; leave-one-out predicts every held-out observable within 3σ out of sample (most-tensioned = $\sin^2\theta_{13}$, $\sim 2\sigma$); a data↔formula shuffle explodes the χ^2 by $> 100\times$. Cited in `tfpt_5_redteam`; ledger row `SEED.CROSSVAL.01`. Python-only.

- **`v307_data_watchdog.py` (`DATA.WATCHDOG.01`, [E]).** The operational decision pipeline over the frozen registry (`v84`): it classifies all twenty registry entries PASS/WATCH/TENSION/KILL by live n_σ against the repo-documented current bests (CODATA 2022, NuFIT 6.0, ACT DR6, Planck 2018, PDG 2024, BK18). No decidable prediction is in KILL; the watch items are α^{-1} (1.9σ), s_{23} (1.8σ) and the most-tensioned core prediction $\sin^2\theta_{13}$ (2.0σ, the honest current pressure point); $\sin^2\theta_{12} = 0.306747$ is compatible (0.02σ, the JUNO falsifier), the $r \in [0.0033, 0.0048]$ band is below the BK18 limit and the kill, and $w{=}{-}1$ is flagged as the DESI dark-energy watchdog. Cited in `origin_theory`; ledger row `DATA.WATCHDOG.01`. Python-only.

- **Target-A holomorphy kill test (`freeze_file.csv`).** A new `seam_holomorphy` freeze row records the sharpened *physical* kill criterion for Target A: a raw quasi-free seam bulk showing topological ground-state degeneracy on the torus ($|\det K| > 1$, the same-c rival $SO(16)_1$) would break holomorphy and kill "seam net = $(E_8)_1$" — since holomorphic c=8 $\Leftrightarrow$ E_8 is machine-proven (`v83`/`v143`) and $\det K$=1 $\Leftrightarrow$ invertible/SRE (`v237`/`v301`/`v302`).

2026-06-19 (an infinite-derivative narrowing of the Stelle-ghost attack, the double-pushout figure, and the "constancy of c" framing of `SEAM.EQUIV.01`: `v304`)

- **`v304_idg_nonlocal_ghost.py` (`QGAMB.IDG.01`, [C]/[O]).** A conditional, parameter-free narrowing of the red-team `rt_F`/`v278` Stelle-ghost attack on the perturbative gravity sector. [**E**] the local R+R^2/Weyl^2 four-derivative propagator $1/(p^2(p^2{+}M^2))$ carries a negative-residue massive spin-2 mode (the Stelle ghost) only as a *truncation artefact*; [**E**] for an *entire* graviton form factor $a(p^2) = e^{p^2/M^2}$ (nowhere zero) the dressed propagator $1/(p^2 a)$ has no extra pole (Tomboulis 1997; Biswas–Mazumdar–Siegel 2006), whereas any *polynomial* truncation has a real zero = the ghost; [**C**] the spectral-action cutoff is the seam KMS weight $f(u) = e^{-u}$ (`v259`), so the un-truncated trace points to a nonlocal form factor with *no new parameter*. [**O**] It does *not* close `QG.AMB.01`: the trace regulator is not yet shown to resum to an entire $a(\Box)$, and a ghost-free *perturbative* graviton is not the nonperturbative ambient measure. Cited in the red-team Target F body; ledger row `QGAMB.IDG.01`.
- **Double-pushout figure (`tfpt_1_architecture_e8`).** A new shared TikZ macro (`TFPTdoublepushout` in `tex-artefacts/tfpt_figures.tex`) draws the commuting square showing that the order-$|\mu_4|$=4 glue is *one* operation in two categories: the lattice pushout $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ (`v1`) and the conformal-net / anyon-condensation extension $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1$ (`v92`/`v281`/`v234`/`v277`/`v260`). Expository only; no new claim.
- **Framing (`introduction`, `tfpt_research_contracts`).** `SEAM.EQUIV.01` is now stated explicitly as the *one* irreducible structural postulate of the theory — the role the constancy of c plays in relativity (cf. `v267`). Prose sharpening; no status change.

2026-06-19 (the discrete→dynamic lens on `F_transfer`: one shape, one seam-rate, `v303`)

- **`v303_ftransfer_dynamics.py`** asks whether the main-branch mechanism — a gapped, positivity-preserving relaxation to a *unique* attractor (the local-averaging update → the E_8 marks, gap $6\ln(3/2)$) — is also the shape of the four `F_transfer` instances, or an unrelated bolt-on. It is the *same* shape: [**E**] F_pole (Koide) *is* the seam dynamics (Möbius multiplier F'=$(2/3)^6$=λ_2, the seam-clock eigenvalue ⇒ a unique attractor at the seam rate, `v82`/`v54`); [**C**] F_Boltzmann (η_B) is a gapped positive contraction (washout $\kappa_f \in (0,1)$, H-theorem); [**C**] F_relic (axion) freezes at an adiabatic fixed point; [**E**] F_QCD (m_p/m_e) is the 1-loop RG flow with the carrier $b_3 = -7$ (asymptotic freedom, Gaussian UV attractor). [**O**] Verdict: one shape underlies all four; F_pole has the seam rate $(2/3)^6$ *exactly*, the other three share only the shape with external rates (thermal/cosmological/RG). So `F_transfer` is the readout end of the one discrete→dynamic principle, *not* a bolt-on; the theory's simplicity is preserved because the external rates are honestly fenced (`v187`), not reduced to the seam. [**E**] suite `v303` 5/5, `AUDIT OK`.

2026-06-19 (the last lever: the seam mass gap *is* the derived Recovery gap $6\ln(3/2)$, `v302`)

- **`v302_seam_gap.py`** closes the chain on the single statement `v301` had left: "the quasi-free seam bulk is gapped". That gap is *not* a free input — it is the established *Recovery gap* of the seam transfer operator, derived from the carrier. [**E**] the frozen transfer spectrum

is spec$(T) = \{1, (2/3)^6, (1/3)^6\}$ (`v160/v162`), a simple Perron eigenvalue 1 separated from the rest; **[E]** the gap is $\Delta = -\ln((2/3)^6) = 6\ln(3/2) = 2N_{\text{fam}}\ln(3/2) \approx 2.4328 > 0$ (multiplicative gap $1-(2/3)^6 = 665/729$; $(2/3)^6$ the μ_4-deck transfer eigenvalue forced by the carrier); **[E]** with the decoupling margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (`v76`) it is robust, and Perron–Frobenius primitivity makes the leading mode simple; **[C]** by Osterwalder–Schrader / quasi-free clustering a transfer-operator gap $\Delta > 0$ *is* a mass gap of the reconstructed Hamiltonian. **[O] Bottom line:** with `v300/v301`, `SEAM.EQUIV.01`'s entire residual is now a composition of *standard cited theorems* (OS/quasi-free clustering + Kitaev free-fermion invertibility + the `v297` AQFT stack) over *established* TFPT facts (16 quasi-free Majoranas; transfer gap $6\ln(3/2)$) — *no undischarged TFPT-internal assumption remains.* It stays **[O]** (not machine-proved end-to-end), but the open front is reduced to citable machinery over a spectral gap that is provably positive. **[E]** suite `v302` 5/5, `AUDIT OK`.

2026-06-19 (Route A's last hypothesis discharged: the gapped quasi-free bulk is invertible by the free-fermion classification, `v301`)

- **`v301_route_a_invertible.py`** attacks the single remaining *unconditional* gap of `SEAM.EQUIV.01`. Since `v300` collapsed Route B into Route A's rationality, that gap is Route A's open hypothesis (`v297` LIT-A): "the raw quasi-free seam bulk is invertible (SRE)". A *gapped* free-fermion (Gaussian) 2+1D bulk cannot carry intrinsic topological order — free fermions give the integer (IQHE/Chern) phases, all *invertible*; intrinsic topological order needs interactions (Kitaev's periodic table, class D). **[E]** the carrier is 16 Majoranas, $c = g_{\text{car}} + N_{\text{fam}} = 8 = 16/2$ (`v148`); **[E]** $SO(16)_1 \supset (E_8)_1$ ($\dim \mathfrak{so}(16) = 120$ currents $+128$ spinor $= 248$, `v154`); **[C]** so a gapped quasi-free bulk is invertible automatically; **[E]** the invariant $|\det K| = \#\text{anyons}$ is 1 for E_8 (invertible) vs 4 for the gauged $D_8{=}SO(16)$ (anyons), which free fermions cannot produce. **[O]** LIT-A's 'invertible bulk' now *follows* from 'gapped 16-Majorana quasi-free bulk' (`v148/v160`+the mass gap), so the residual changes from a topological-order statement to a spectral one. **[O] Bottom line:** with `v300`, the whole open content of `SEAM.EQUIV.01` reduces to the *single spectral statement* "the quasi-free seam bulk is gapped" — the topological-order obstruction (hidden anyons) is removed; the gap is not manufactured. **[E]** suite `v301` 6/6, `AUDIT OK`.

2026-06-19 (the Flat-Away attack: a hard discrete obstruction + the pin derived from $(E_8)_1$, `v300`)

- **`v300_flataway_rigid.py`** attacks the single external fact the three Flat-Away routes (`v291/v293/v295`) left open, on two fronts. *Harden:* the flat seam was only a *soft* minimum ("the spectral mismatch grows with ε"); here the obstruction becomes a *discrete* on/off invariant. **[E]** the flat fingerprint — spec$(|D_\theta|)$ is the integer ladder with every level $n \geq 1$ of multiplicity exactly 2; **[E]** the resonant mode $4k$ splits level $2k$ by gap $= g_k$ (sympy 2×2 $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$, + an $O(g^2)$ common shift), validated against the full Toeplitz operator; **[E]** any $g_k \neq 0$ lifts the resonant degeneracy ($2 \to 1{+}1$) and changes the spectral *multiset*, so preserving the flat ladder *with* its mult-2 degeneracies forces every $g_k = 0 \Rightarrow f = 0$ — strictly stronger than the "deviation grows" scan; **[E]** with the `v295` convexity, flat is isolated analytically *and* discretely. *Derive the pin:* **[C]** the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$ has integer q-coefficients (248 at level 1; E_8 even unimodular $\Rightarrow$ integer weights), so by 2d Steklov conformal rigidity the integer ladder $\Leftrightarrow$ flat seam, i.e. the external pin *is* the $(E_8)_1$ rationality (Route A). **[O]** verdict: Flat-Away carries *zero* new open content beyond Route A — the two `SEAM.EQUIV.01` routes converge on one rationality fact. **[E]** suite `v300` 6/6, `AUDIT OK`.

2026-06-19 (the hypergraph/Wolfram bridge hardened: the $(2,3,r)$ seed reaches E_8 as the last finite point, v299)

- **v299_hypergraph_0d_autonomous.py** hardens the Wolfram/hypergraph reading into a precisely typed bridge (builds on v219/v298): [**E**] the icosahedron is a 3-uniform hypergraph ($|\mathrm{Aut}(\text{graph})| = 120 = |I_h| = |A_5 \times \mathbb{Z}_2|$; the McKay group is the $SU(2)$ binary lift $2I$, a *different* order-120 group whose McKay graph is affine E_8; 30 edges $= h(E_8)$); [**E**] the $1 \to 4$ subdivision rewrite is equivariant (all 120 symmetries at every scale); [**E**] the $2I$ fusion tensor is a weighted 3-uniform hypergraph whose 2-dim slice is the affine E_8 McKay graph; [**E**] Smith ($\rho \le 2 \Leftrightarrow$ affine ADE) + Collatz–Wielandt ($\rho \le 2 \Leftrightarrow$ a local witness $Am \le 2m$); [**E**]* the autonomous rule (diffusion-generated witness + *local* gate $Am \le 2m$, Perron only as audit) runs $E_6 \to E_7 \to E_8 \to \hat{E}_8$; [**E**] the negative controls $\rho(A_n) = 2\cos\frac{\pi}{n+1}$ and $\rho(D_n) = 2\cos\frac{\pi}{2n-2}$ are < 2 for *all* n (closed form), so neither family ever reaches E_8. Honest type: [**O**] the $(2,3,\cdot)$ seed is the single irreducible input = P2 — *the rule reaches E_8 as the last finite point on this seed; it is not a derivation of P2.* [**E**] build green, AUDIT OK.

2026-06-18 (E8 as a network attractor — the Wolfram-program reading, v298)

- **v298_e8_network_attractor.py** is an honest audit of the "TFPT as a hypergraph/rewrite system" idea, on the v219 McKay bedrock. [**E**] the affine-E_8 Kac marks $(1,2,3,4,5,6,4,2,3)$ are the *unique dynamical attractor* of a minimal local-averaging update $m \leftarrow (A+2I)/4\, m$ on the $(2,3,5)$ McKay graph (converges from any positive start; $A\,m = 2m$, top eigenvalue 2); the spherical tower terminates at E_8 and the cascade is nested-subgraph coarse-graining. [**C**] the Wolfram reading "E_8 is the universal attractor of a minimal $(2,3,5)$ network" holds at the Coxeter-skeleton level. [**O**] a minimal hypergraph *rewrite* reproducing the *full* structure stays open; honest negatives: φ_0^{ret} is the analytic seed $\frac{4}{3}c_3 = 1/(6\pi)$, not an edge fraction, and recovery is a 1-D Möbius fixed point, not a rewrite. [**E**] build green, AUDIT OK.

2026-06-18 (a_2 closed form + Route A literature stack, v296–v297)

- **v296_flataway_a2_closed.py (FLATAWAY.A2.CLOSED.01)** evaluates the exact a_2 coefficient in *closed form*: the operator tails telescope geometrically ($C_k + D_k e^{-4kt} = 4kt(1-e^{-4kt})$) to $W_k^{\mathrm{tail}} = t(1-e^{-4kt})/(8k(1-e^{-t}))$, leaving a finite $(4k-1)$-term middle; e.g. $W_1(t) = t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8$. Manifestly positive, small-t law $W_k = t/2 + O(t^3)$ (the t^2 term cancels). Reproduces v295 to machine precision.

- **v297_route_a_literature_stack.py (SEAM.EQUIV.A.LIT.01)** writes Route A's one import as a citable stack: LIT-A (Kitaev; Freed–Hopkins) → LIT-B (Müger; Kawahigashi–Longo–Müger) → LIT-C (Conway–Sloane; Dong–Xu) compose into "invertible bulk $\Rightarrow (E_8)_1$", all legs established; the single open hypothesis (seam-bulk invertibility) coincides with Route B's Flat-Away, and the stack is non-circular with Route B (the v286 firewall holds). [**E**] build green, AUDIT OK.

2026-06-18 (Flat-Away heat route made exact + Lean-formalised, v295 / FORM.FLATAWAY.01)

- **v295_flataway_a2_exact.py (FLATAWAY.A2.01)** upgrades the v292 positive-definiteness from a numerical scan to a *proof*: the first variation of $\mathrm{Tr}\, e^{-t\Lambda}$ vanishes (zero-mean off-mark curvature, Gauss-Bonnet), and since $\Lambda \mapsto \mathrm{Tr}\, e^{-t\Lambda}$ is convex (Klein/Peierls), ε=0 is the *global* minimum — so $\Delta\mathrm{Tr} \ge 0$ for all deformations, $= 0$ iff $f = 0$. The exact second-order coefficient is the divided-difference kernel of e^{-tx} (Duhamel), validated to rel. err $< 10^{-6}$.

- **Lean FORM.FLATAWAY.01** (SeamDeckClosure.lean Part 5): the formal Δ=0 $\Leftrightarrow$ flat step — a positive-weighted sum of squares is positive-definite (heatDeviation_eq_zero_iff); built and axiom-checked (only propext, Classical.choice, Quot.sound). So the heat route's

positive-definiteness is now both analytically proved and machine-formalised; only the single external fact (the seam fixes a_2) remains. [**E**] build green, `AUDIT OK`.

2026-06-18 (Flat-Away attacked on all three routes, v292–v294)

- **`v292_flataway_heat_reduction.py` (`FLATAWAY.HEAT.01`)**: the heat route, made precise. The heat-trace deviation $\Delta\mathrm{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$ is a *positive-definite* quadratic form in the off-mark curvature ($c_k(t) > 0$ for every $\mathbb{Z}_4$ mode, positive on all combinations), leading invariant $\|f\|_{L^2}$ — so Flat-Away reduces to the single fact "the seam fixes the a_2 heat coefficient".
- **`v293_flataway_spectral_hessian.py` (`FLATAWAY.SPEC.01`)**: the spectral route, upgraded from one direction to the *full* smooth-$\mathbb{Z}_4$ space. Mode $4k$ splits the degenerate Steklov level $|2k|$ at first order; the Hessian of the spectral mismatch over modes $\{4, 8, 12\}$ is positive-definite (eigenvalues $\{0.497, 0.500, 0.503\}$), so the flat seam is a strict isolated minimum — the "only one direction" objection is answered.
- **`v294_flataway_rp_energy.py` (`FLATAWAY.ENERGY.01`)**: the variational route. With the rigid mark divisor (4 cone angles π, Gauss-Bonnet 4π), Troyanov uniqueness gives a unique flat cone metric, and the curvature energy $\int f^2$ is strictly convex (Hessian $2\pi I$) with a unique minimiser — Flat-Away reduces to "RP+gap realises the energy-minimiser". The three routes converge on one selection principle for the flat metric; closing any one closes Route B. [**E**] build green, `AUDIT OK`.

2026-06-18 (Route-B red-team: the smooth $\mathbb{Z}_4$ adversary + Flat-Away as a mini-theorem, v290–v291)

- **`v290_z4_smooth_curvature_adversary.py` (`SEAM.ADVERSARY.01`)** is the honest red-team that $\mathbb{Z}_4$ block-diagonality is *not* mark-locality: a smooth $\mathbb{Z}_4$ off-mark curvature $f = \varepsilon\cos(4\theta)$ [**E**] passes the `v288` commutator ($\|[\rho, \Lambda]\| = 0$, same as mark-local) yet [**E**] shifts the DtN spectrum ($\max|\Delta\lambda| = 0.155$) and the heat trace ($\Delta = 0.019$). So the commutator is necessary, not sufficient; the modulus is excluded by the spectral data, not $[\rho, \Lambda]{=}0$ — which is exactly why Flat-Away cannot be read off the commutator.
- **`v291_flataway_contract.py` (`FLATAWAY.RP.01`)** states Flat-Away as its own named mini-theorem and types three independent proof routes: [**E**] spectral-rigidity and [**E**] heat-kernel (the flat seam is the *unique* spectral/heat match of the smooth $\mathbb{Z}_4$ family — deviation zero only at $\varepsilon{=}0$) and [**C**] RP-energy/Troyanov (constant-curvature minimiser). Closing any one closes Route B. [**E**] build green, `AUDIT OK`.

2026-06-18 (Route B sharpened: raw mark-locality decomposed + reviewer firewalls, v289 / v287–v288)

- **`v289_marklocal_raw.py` (`MARKLOCAL.RAW.01`)** decomposes the one remaining Route-B residual ('why is the raw seam sub-principal term mark-local?') into a 5-lemma chain and isolates the *single* open analytic lemma: [**C**] L1 conic parametrix (b-calculus/Melrose, imported); [**O**] L2 **Flat-Away** — RP+gap+4 marks ⇒ curvature vanishes away from the marks; [**E**] L3 orbit-source (`v195`/`v216`), L4 Fourier-support on $4\mathbb{Z}$ (`v264`/`v288`), L5 full-operator $[\rho, \Lambda]{=}0$ (`v288`). 4/5 discharged — Route B reduces to exactly L2.
- **`v288` reviewer firewall**: a SCOPE FIREWALL check makes the exact-label split explicit (Fourier lemma Formal/Exact; numerical test Numerical Exact; transfer to the raw conic DtN Conditional, `v264`; mark-locality from the RP seam Open Selection, `v289`) — so it can never be read as 'full-L^2 QGEO proved'.

- **v287 literature matrix**: L3 ('invertible bulk ⇒ single-sector') is typed as a composition of citable results (LIT-A Kitaev/Freed–Hopkins, LIT-B Müger/Kawahigashi–Longo–Müger, LIT-C Kawahigashi–Longo) conditional on one open hypothesis — seam-bulk invertibility, which coincides with Route B. [**E**] build green, `AUDIT OK`.

2026-06-18 (closing sprint: the Seam Equivalence Theorem named + both routes attacked, v286–v288)

- **`v286_seam_equivalence_contract.py` (`SEAM.EQUIV.01`)** names the one open theorem — *the raw RP seam state is the holomorphic* $(E_8)_1$ *net at* $\tau{=}i$ — and concentrates the whole open structure on it: four objects, six arrows (2 closed, 1 conditional `v282`, 1 open Gral), an **import firewall** proving the two routes (`v276`/`v280`/`v284` vs `v277`/`v281`/`v285`) never import each other (no 'E_8 into the geometry and back' circularity), and an exact-label taxonomy. Writes `seam_equivalence.json`.
- **`v287_free_rp_bulk_to_holomorphic_boundary.py` (`SEAM.EQUIV.A01`)**: Route A (AQFT) dependency checker — the 5-lemma chain RP+gap+chirality+Gaussian+invertible ⇒ $\mu{=}1$ ⇒ holomorphic ⇒ $(E_8)_1$ is logically complete *modulo one* standard theorem: *invertible (SRE) Gaussian bulk ⇒ single-sector boundary* (the AQFT form of $\det K{=}1$). 4/5 discharged.
- **`v288_full_l2_subprincipal_z4.py` (`SEAM.EQUIV.B01`)**: Route B (DtN) — *proves* the full-L^2 lift of the sub-principal $\mathbb{Z}_4$ block-diagonality (a mark-sourced curvature has Fourier support only on $m \equiv 0 \bmod 4$, so $\Lambda = |n| + M_f$ commutes with $\rho = \mathrm{diag}(i^n)$ on the full operator, $\|[\rho, \Lambda]\| \sim 2{\times}10^{-15}$; generic curvature $\to 4.44$), lifting `v201`/`v284` to L^2. The residual shrinks to one sharper question: why is the raw seam sub-principal term mark-local? [**E**] build green, `AUDIT OK`.

2026-06-18 (the two proof routes decomposed: Route (i) and Route (ii) into lemma chains, v284/v285)

- **`v284_route_i_rp_uniqueness.py` (`QGEO.ROUTEI.01`)** turns Route (i) (RP-state uniqueness) into a 6-lemma chain: [**E**] 5/6 lemmas are already discharged — L1 RP→DtN (`v194`), L2 marks→pillowcase (`v195`/`v216`/`v214`), L3 flat Steklov $= \sqrt{-\Delta_{\mathrm{flat}}}$ (verified here), L4 covariance from DtN (`v258`), L5 μ_4-invariance (`v276`/Lean/`v280`); [**C**] Troyanov ($\chi_{\mathrm{orb}}{=}0 \Rightarrow$ unique flat $\tau{=}i$) reduces the residual to one lemma. [**O**] the one open lemma: *the raw seam is the constant-curvature geometric state.*
- **`v285_route_ii_seam_condensation.py` (`QGAMB.ROUTEII.01`)** turns Route (ii) (seam-net condensation) into a 4-lemma chain: [**E**] 3/4 discharged (no abelian sector ⇔ $\det K{=}1$ ⇔ holomorphic ⇔ E_8; the tower 16→4→1), and **its open lemma coincides with Route (i)'s** — both are "the raw seam is the $(E_8)_1$-at-$\tau{=}i$ object" (`v282`), so proving *either* route closes *both*. [**O**] the unified open lemma: a canonical equivalence between the raw seam KMS/DtN state and the holomorphic $(E_8)_1$ net at $\tau{=}i$. Build green, `AUDIT OK`.

2026-06-18 (self-investigation round: pillowcase DtN, modular data, the E8-at-$\tau{=}i$ unification, live scorecard, v280–v283)

- **`v280_pillowcase_steklov.py` (`QGEO.STEKLOV.01`)** runs the QGEO chain on the actual flat $\tau{=}i$ pillowcase orbifold: the order-4 deck permutes the 4 cone points (2 fixed + 1 swap) and the geometric chain $[\rho, \Delta]{=}0 \Rightarrow [\rho, H]{=}0 \Rightarrow [\rho, C]{=}0 \Rightarrow \omega \circ \rho{=}\omega$ is numerically exact — route-(i) evidence for `QGEO.OBLIG.01`; the premise stays [**O**].
- **`v281_holomorphy_modular_data.py` (`QGAMB.MODULAR.01`)**: #anyons $= |\det \mathrm{Gram}|$ (E8→1, SO(16)→4), the anyon-condensation tower $16 \to 4 \to 1$, Gauss–Milgram $\to c{=}8$; holomorphy ($\det K{=}1$) is the single Tier-B discriminator.

- **`v282_e8_tau_i_unification.py` (`QGAMB.UNIFY.01`)**: the candidate *simplification* — $\chi_{E_8}=j^{1/3}$ gives $\chi_{E_8}(i)=1728^{1/3}=12$, so $\tau=i$ is both the order-4 CM point (QGEO) and the $(E_8)_1$ modulus (QG.AMB): the two obligations are two faces of one object (the seam is $(E_8)_1$ at $\tau=i$), **obligation count** $2 \to 1$.
- **`v283_live_killtest_scorecard.py` (`PRED.LIVE.01`)**: the recent round's computable predictions vs current data — all consistent $< 2\sigma$, with two sharp near-term kill tests (Σm_ν floor 0.0586 eV; proton decay 1.55×10^{35} yr).
- Wolfram 275/275 (v281/v282 exact-mirrored); build green, `AUDIT OK`.

2026-06-18 (the Grál: QGEO.SYM.01 as one precise lemma + the all-orders step Lean-formalised, v279/`FORM.QGEO.04`)

- **`v279_qgeo_obligation_lemma.py` (`QGEO.OBLIG.01`) writes the bedrock as ONE precise constructive-QFT lemma.** Given the premise *the raw seam carries the flat $\tau=i$ pillowcase metric* (Troyanov-unique, $\Lambda_\Sigma=\sqrt{-\Delta_{\text{flat}}}$), then $\omega_\Sigma \circ \rho = \omega_\Sigma \to$ mark-locality $\to E_8$. **[E]** the reduction DAG from 'free RP seam' has exactly *one* open leaf — 'the raw seam state is the flat state'; **[E]** the geometric side is fixed (flat DtN diagonal, ρ commutes to all orders), so the obligation is a state-identification, with **[C]** two proof routes (RP-state uniqueness, or Troyanov + OS).
- **`FORM.QGEO.04`: the all-orders step is now machine-proved in Lean.** `SeamDeckClosure.lean` gains `diagonal_commutesClock` and `flat_all_orders_clock`: any spectral function $g(\Delta_{\text{flat}})$ of the Fourier-diagonal flat Laplacian commutes with the carrier clock $\rho=\text{diag}(i^n)$ — so the `v276` all-orders closure is a Lean theorem (only `propext`, `Classical.choice`, `Quot.sound`; no `sorry`; full `lake build` green, 3356 jobs). The premise (the seam *is* flat) stays the **[O]** `QGEO.SYM.01` input. Build green, `AUDIT OK`.

2026-06-18 (4D QFT: the $S_{\text{pert}} \to S_{\text{phys}}$ LSZ bridge + one-loop unitarity, v278)

- **`v278_lsz_bridge_unitarity.py` (`QFT4D.SPERT.04`) connects the perturbative S-matrix to the physical asymptotic one and verifies one-loop unitarity.** **[E]** LSZ bridge: the EG T-products of S_{pert} (`v269`) *are* the OS Wick-rotated correlators (`v240`), so amputate+on-shell-residue gives $S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$; **[C]** the gap $\Delta=6\log(3/2)>0$ gives Haag–Ruelle asymptotic states. **[E]** *one-loop optical theorem*: the bubble discontinuity $\text{Im}\,B(s)/\pi = x_+ - x_- = \sqrt{1-4m^2/s}$ *exactly* = the two-body phase space, so $2\,\text{Im}\,M = \sum\!\int d\Pi |M|^2$ (cutting rule) holds — the matter+gauge sector is unitary, with the cut turning on at the physical two-particle threshold. **[O]** the gravity sector's Stelle ghost (`v269/rt_F`) is non-unitary → `QG.AMB.01`. Exact core mirrored in Wolfram (273/273). Build green, `AUDIT OK`.

2026-06-18 (QG.AMB.01 Tier-B made concrete: the seam-Calderón $\to (E_8)_1$ match, v277)

- **`v277_seam_calderon_e8_match.py` (`QGAMB.TIERB.01`) reduces Tier-B to ONE bit.** **[E]** the full $(E_8)_1$ matching target is computed and matches: $c=8$, $\det\text{Cartan}(E_8)=1$, the holomorphic character $E_4/\eta^8=j^{1/3}=q^{-1/3}(1+248q+\dots)$ with a single primary and exactly $248=\dim E_8$ weight-1 currents (240 roots). **[E]** the discriminator is holomorphy — the $c=8$ counterexample $(D_8)=SO(16)_1$ has $\det\text{Cartan}=4$ (non-holomorphic), so the one condition is $|\det K|=1$. **[C]** the seam side matches via `v276` (flat $\tau=i$) + `v260` (Kummer/K3). **[O]** residual = the single holomorphy bit; reduced, not closed. Exact core mirrored in Wolfram (272/272). Build green, `AUDIT OK`.

2026-06-18 (QGEO.SYM.01 narrowed: the flat pillowcase closes the commutator to all orders, v276)

- **v276_qgeo_flat_closes_commutator.py (QGEO.SYM.03) collapses the bedrock to ONE geometric postulate.** The state-level residual the whole bedrock rests on is the operator equation $[\rho, H]$=0. **[E]** if $H = f(\Delta_{\text{flat}})$ and the order-4 deck ρ ($z \mapsto iz$) is a flat-τ=i isometry, then $[\rho, \Delta]$=0 $\Rightarrow$ $[\rho, H]$=0 *exactly*, to all orders (verified on the discretised square torus: ρ^4=I, $[\rho, \Delta]$=0, $[\rho, \sqrt{-\Delta}]$=0; generic non-isometry fails) — upgrading the leading-order v198/v201 to the full H. **[C]** τ=i is forced by order-4 (j=1728; hexagonal j=0 is order-6, excluded). **[O]** QGEO.SYM.01 therefore collapses to the single statement "the raw seam IS the flat τ=i pillowcase quasi-free state" — one human constructive-QFT proof, or an honest axiom. Narrows, does not close. Build green, AUDIT OK.

2026-06-18 (Red Team Target F — adversarial audit of the v269–v275 round, rt_F)

- **redteam/rt_F_qft4d.py (REDTEAM.F.01) stress-tests the just-published round** (the perturbative S_{pert} layer, the PMNS Jarlskog, the ν-scale, the over-determination, the QG.AMB.01 roadmap). SURVIVES (NARROWED): two attacks land and are folded in. **(1)** the gravity R^2/Weyl^2 sub-sector is power-counting renormalizable but *non-unitary* (the Stelle ghost: the four-derivative propagator $1/(p^2(p^2 + M^2))$ has a negative-residue massive spin-2 mode), so S_{pert} is unitary for the matter+gauge sector only — which is exactly why the non-perturbative QG.AMB.01 is the real frontier (caveat added to v269). **(2)** the 0.11% anchor over-determination is conditional on the Λ-branch readout (LAMBDA.BRANCH.01, [C]), the gravity route being unconditional (caveat added to v274). Added as Target F in tfpt_5_redteam + the red-team table; run_redteam.py green.

2026-06-18 (Zenodo release 5.2 (rev 191) — the perturbative 4D-QFT layer + scale closure, v269–v275)

- **Published to Zenodo.** Version 5.2 (rev 191) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20743347, DOI 10.5281/zenodo.20743347. This release carries the perturbative 4D-QFT layer (the Epstein–Glaser S_{pert} skeleton v269, the concrete one-loop quartic v271 and gauge β-coefficients v273), the complete complex PMNS matrix and leptonic CP strength v270, the honest absolute ν-mass-scale typing v272, the over-determination of the single mass anchor v274, and the QG.AMB.01 obligation roadmap v275. Suite 273 modules (highest ID v275; v186/v226 skipped), Wolfram 271/271, run_all.py green, AUDIT OK, npm build clean. The Zenodo description (zenodo_description.html) and the public website were synced in the same release.

2026-06-18 (scale + frontier round: ν-mass scale, EG gauge running, over-determination, QG roadmap, v272–v275)

- **v272_nu_mass_scale.py (FLAV.NUSCALE.01) types the absolute neutrino-mass scale honestly** (no fabricated Σm_ν — would contradict the v184 firewall). **[E]** the absolute scale reduces to *one* seesaw ratio $m_3 = (y_\nu v_{\text{EW}})^2/M_R$ (one UV input, like v_{geo}, v153); **[C]** the observed m_3=0.050 eV with the TFPT PS$\to$GUT window $M_R \in [4.2{\times}10^{13}, 2.4{\times}10^{15}]$ GeV (v249) needs $y_\nu \in [0.26, 2.0]$ — $O(1)$, no tuning; **[O]** M_R is not a clean $\bar{M}_{\text{Pl}}$ power, so the absolute value stays open; **[X]** the NO floor Σm_ν=0.0586 eV is the cosmological kill test. Closes the PMNS bookkeeping (the last **[O]** is one seesaw ratio).

- **v273_eg_gauge_running.py (QFT4D.SPERT.03) derives the SM one-loop b_i via the EG gauge self-energy.** **[E]** the gauge 2-point is the same scaling-degree-4 extension as the quartic (v271), giving $(b_1, b_2, b_3) = (41/10, -19/6, -7)$ from the carrier/SM content (the same

41 as the carrier algebra, `v159`); **[O]** two-loop + all-order stay open. Exact b_i mirrored in Wolfram.

- **`v274_scale_overdetermination.py` (`SCALE.OVERDET.01`) shows the single mass anchor is over-determined.** **[E]**/**[C]** two independent routes to $\bar{M}_{\rm Pl}$ agree — gravity $(8\pi G)^{-1/2} = 2.4353{\times}10^{18}$ GeV and cosmology (the compiler prediction $\rho_\Lambda/\bar{M}_{\rm Pl}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$ with $\rho_\Lambda^{1/4} = 2.24$ meV) $= 2.438{\times}10^{18}$ GeV, to 0.11%; the seam being a conformal c=8 CFT makes "no derivable absolute scale" a theorem, not a gap. Sharpens `v153`/`v78`.
- **`v275_qgamb_roadmap.py` (`QGAMB.ROADMAP.01`) gives the open gate `QG.AMB.01` a module.** **[E]** Tier A gap decoupling ($\Delta_{\rm eff} = 1.648 > 0$, `v36`/`v76`); **[E]**/**[C]** Tier B reduction to the holomorphic $(E_8)_1$ boundary net (`v77`); **[C]** the perturbative layer in hand (`v269`/`v271`/`v273`); **[O]** the nonperturbative ambient measure stays the one structural frontier.
- Suite `271/271` Wolfram, `run_all` green, `AUDIT OK`, build green.

2026-06-18 (4D QFT: a concrete Epstein–Glaser one-loop quartic renormalization, `v271`)

- **`v271_eg_oneloop_quartic.py` (`QFT4D.SPERT.02`) takes the `v269` $S_{\rm pert}$ skeleton from "exists" to an actual EG number** (no path integral). **[E]** the first renormalization is the order-2 product T_2; the massless propagator has scaling degree sd=2 in d=4, the one-loop bubble has sd=4=d, so ω=sd−d=0 ⇒ exactly one logarithmic local counterterm ($C(\omega{+}d, d){=}1$, finite, no infinite tower); the loop factor $\Omega_3/(2(2\pi)^4) = 1/(16\pi^2)$ exactly — the *same* factor that normalises the spectral-action a_4 geometric quartic. **[C]** the ϕ^4 one-loop $\beta_\lambda = 3\lambda^2/(16\pi^2)$ (symmetry 3 = s,t,u channels), EG initial condition $\lambda_\Phi({\rm UV}) = 1/(16\pi^2)$. **[O]** order-2/one-loop only; the all-order EG construction and the nonperturbative measure `QG.AMB.01` stay open. Exact core mirrored in Wolfram (270/270). **[E]** build green, `AUDIT OK`.

2026-06-18 (PMNS: the complete complex matrix + the leptonic Jarlskog CP strength, `v270`)

- **`v270_pmns_jarlskog_assembly.py` (`FLAV.PMNS.03`) assembles the four TFPT channels** ($\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\rm ret}}{2}$, $\sin^2\theta_{23}{=}\frac{1}{2}$, $\sin^2\theta_{13}{=}\varphi_0^{\rm ret}e^{-5/6}$, $\delta{=}4\pi/3$) **into one exactly-unitary complex PMNS matrix and derives the leptonic CP strength.** **[E]** $U = R_{23}U_{13}(\delta)R_{12}$ is unitary; **[C]** the Jarlskog $J_{\rm PMNS} = \Im(U_{e1}U_{\mu2}U_{e2}^*U_{\mu1}^*) = -0.02965$ ($J_{\max} = 0.03424$) is a *derived* number, the leptonic analogue of the CKM J (`v88`); **[C]** data: $J_{\max}$ vs NuFIT 5.2 0.0332 ($\sim 3\%$), J vs best fit -0.026, the negative CP-violating sign reproduced. **[O]** δ is the μ_6/triality phase (assembled, not from M_ν/D_F) and the absolute ν-mass scale stays open — closes the CP-*strength* readout, not the phase origin nor the scale. **[E]** build green, `AUDIT OK`.

2026-06-18 (4D QFT: the perturbative S-matrix as an Epstein–Glaser / pAQFT skeleton, `v269`)

- **`v269_spert_paqft_skeleton.py` (`QFT4D.SPERT.01`) builds the honest middle layer of the three-layer S-matrix** (the first concrete 4D-QFT step beyond the spectral-action contract), *not* a nonperturbative closure. **[E]** three distinct layers: $S_{\rm top}$ (2d DHR braiding $|M|$=1, statistics, `v243`), $S_{\rm pert}$ (4d Epstein–Glaser/BV S-matrix of the spectral action), $S_{\rm phys}$ (LSZ on the OS Wightman functions, `v240`). **[E]** the spectral-action a_4 interaction is power-counting renormalizable (all operators dim ≤ 4; 7 marginal + 3 relevant, a finite counterterm basis), so **[C]** by the Epstein–Glaser theorem the perturbative S-matrix exists order by order (causal factorisation + finite local extension; no path integral, no UV divergences). **[C]** the gap $\Delta = 6\log(3/2)$ gives an IR-safe adiabatic limit ($S_{\rm pert} \to S_{\rm phys}$ via LSZ). **[O]** perturbative only — the ambient measure `QG.AMB.01` stays open, the canonical 4d reading remains boundary-only (`v265`). **[E]** build green, `AUDIT OK`.

2026-06-18 (PMNS: the reactor-angle exponent closed as the carrier hypercharge trace, v268)

- **`v268_theta13_carrier_trace.py` (`FLAV.TH13.01`) closes the θ_{13} exponent origin.** The value $\sin^2\theta_{13} = \varphi_0^{\rm ret} e^{-5/6} = 0.0231$ was already checked (`v16`); now the exponent is exact: **[E]** $\gamma = 5/6 = \mathrm{tr}_E Y^2$, the carrier hypercharge-squared trace ($Y = \mathrm{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$, roots of $6Y^2 - Y - 1 = 0$; complement $\frac{1}{6}$ the neutrino ratio). **[E]** own channel: the $\mu\tau$-breaking $\varepsilon = \frac{3}{4}\varphi_0^{\rm ret}$ induces only $\sim 1.6 \times 10^{-3} \ll 0.023$, so θ_{13} is a separate carrier-trace channel — correcting the tempting (and wrong) "θ_{13} from M_ν" route flagged by the new script atlas. **[O]** the CP magnitude and the absolute ν-mass scale remain the open PMNS items. Exact core mirrored in Wolfram. **[E]** build green, `AUDIT OK`.

2026-06-18 (Infrastructure: a generated causal "script atlas" so the suite is navigable at a glance)

- **`verification/make_script_atlas.py` generates `verification/script_atlas.md`** — a single grouped, dependency-aware view fusing the existing single sources (`status_ledger.csv` for claim→status→dependencies→supersedes, `script_registry.csv`/`script_clusters.csv` for the theme grouping, `docs_map.csv` for where each script is cited). It lists every registered script by cluster with its claim(s), four-class marker, one-liner, resolved dependencies and citing papers, plus a *supersede map* (which old claim each new one replaces) and a dependency/most-depended-on overview — the antidote to duplicating work or forgetting superseded results. No new data, a re-projection only; wired into `bash build.sh gen`. **[E]** build green, `AUDIT OK`.

2026-06-18 (The bedrock in its minimal-axiom form: a rigidity theorem for `QGEO.SYM.01`, v267)

- **`v267_qgeo_rigidity_minimal_axiom.py` (`QGEO.SYM.02`) sharpens the one bedrock postulate to its irreducible kernel** (it does *not* close it). **[E]** the *single* bare statement "the seam admits the carrier order-4 clock as a conformal symmetry permuting its four marks" ($= \omega \circ \rho = \omega$) forces *everything*: an order-4 Möbius orbit $\{a, ia, -a, -ia\}$ has cross-ratio 2 independent of a; cross-ratio 2 $\Leftrightarrow$ j=1728 (only $\lambda \in \{-1, \frac{1}{2}, 2\}$) $\Leftrightarrow$ the order-4 CM automorphism on $y^2{=}x^3{-}x$; and the unique flat pillowcase metric, mark-locality, $[\rho, \Lambda]{=}0$ and $\omega \circ \rho{=}\omega$ all follow (`v214`/`v201`/`v264`/`v198`). **[E]** neg controls: generic config $j \neq 1728$ ($\mathbb{Z}_2$ only), hexagonal j=0 ($\mathbb{Z}_6$, order 6) — only the square (order-4) realises the μ_4 clock. **[C]** second, independent reason: $\{1, i, -1, -i\}$ over $\mathbb{Q}$ with j=1728 (CM by $\mathbb{Z}[i]$), Belyi/dessins Galois-rigidity. **[O]** the bare order-4 symmetry stays the one foundational postulate (like c=const) — the *reduction* is closed (axiom $\Rightarrow$ everything), the axiom is not. **[E]** build green, `AUDIT OK`.

2026-06-18 (Branch B decided: the explicit PS threshold model + proton-decay window — minimal content killed, $+(15,1,1)$ survives, Hyper-K decisive, v266)

- **`v266_ps_threshold_proton.py` (`PS.PROTON.02`/`PS.THRESHOLD.02`) decides branch B of the `v265` fork.** The explicit two-step Pati–Salam thresholds give d=6 $\tau_p(p{\to}e^+\pi^0)$ (writes `ps_threshold_proton.json`): **[X]** the minimal 16-Higgs content ($M_{GUT}{\sim}2.4{\times}10^{15}$) gives $\tau_p{\sim}4.4{\times}10^{33}$ yr < Super-K and is *killed* even at the high hadronic band; **[C]** the E_8-allowed $(15,1,1)$ [the single 45] raises $M_{GUT}{\sim}5.9{\times}10^{15}$ $\Rightarrow$ $\tau_p{\sim}1.6{\times}10^{35}$ yr, proton-safe *now*; **[X]** but that sits at the Hyper-K reach ($\sim 1.4{\times}10^{35}$ yr) — a *dated* kill test decisive within the decade. So branch B is promoted from "well-mannered candidate" to *content-selected (must be $+(15,1,1)$), Hyper-K-decisive UV branch*; **[O]** structurally marginal (only one 45 in the hull, `v247`) with an $O(3)$ τ_p band. Branch A (boundary-only) is unchanged and canonical. **[E]** build green, `AUDIT OK`.

2026-06-18 (The 4D-QFT fork frozen as a branch policy: boundary-only canonical, Pati–Salam a falsifiable UV branch, SM-only GUT killed, v265)

- **v265_qft4d_fork_freeze.py (QFT4D.FORK.01) turns the 4D-QFT fork from an open ambiguity into a frozen decision tree** (qft4d_fork_freeze.json), no new physics. **A (canonical):** boundary-only is the default 4D reading — a 4D-GUT is *not claimed by default* (E_8 is the audit hull). [**X**] SM-only 4D-GUT unification is killed (1-loop: at the $\alpha_1{=}\alpha_2$ scale α_3^{-1} off by $\sim$5.6, spread $\sim$8.8 at $2{\times}10^{16}$ GeV, re-deriving v246) — the canonical expectation, not a failure. **B (conditional):** carrier-native Pati–Salam is a falsifiable UV branch only with the E_8-allowed content $\{1, 10, 16, 45\}$ (no **126**, v247), $\sin^2\theta_W{=}3/8$ (v248), the $O(1)$ $\kappa{=}M_{PS}/M_s$ band (v249/v253), a threshold model and a proton-decay kill test ([**X**], no fake τ_p window; carrier gauging stays the contract). [**E**] a prose guard (QFT4D.NO_OVERCLAIM.01) forbids bare 4D over-claims and requires the scoped policy wording. New ledger rows QFT4D.BOUNDARY.DEFAULT.01, QFT4D.SMONLYGUT.01, PS.UVBRANCH.01, PS.THRESHOLD.01, PS.PROTON.KILL.01. A fork-policy box added to tfpt_research_contracts; [**E**] build green, AUDIT OK.

2026-06-18 (Three open-residual advances: the $F_{\rm QCD}$ solver, M_ν from D_F, and the raw-seam⇒mark-locality reduction, v262–v264)

- **v262_fqcd_mp_me.py (FR.MPME.02) implements the one contract-only $F_{\rm transfer}$ pipeline.** A real two-loop α_s solver runs $\alpha_s(M_Z){=}0.1179$ with n_f thresholds (carrier $b_3{=}-7$, [**E**]) to $\alpha_s(2\,\mathrm{GeV}){\sim}0.30$ and $\Lambda^{(3)}_{\overline{\rm MS}}{\sim}0.33\,\mathrm{GeV}$; with the lattice bridge C_p (external $O(1)$) and the closed m_e it reproduces $m_p/m_e{\sim}1824$ vs the measured 1836.15 within the $\sim 7\%$ external budget. A [**C**] transfer reproduction (firewall: no compiler integer for 1836), advancing FR.MPME.01 ([**O**]) to FR.MPME.02 ([**C**]).
- **v263_mnu_from_df.py (FLAV.PMNS.02) makes M_ν a seesaw operator of D_F.** [**E**] $M_\nu = -m_D M_R^{-1} m_D^\top$ is symmetric, seesaw-suppressed and (for $\mu\tau$-symmetric inputs) $\mu\tau$-symmetric; [**C**] it reproduces $\theta_{23}{=}45^\circ$ and $\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\rm ret}}{2}$ (v9) under the seam misalignment; [**O**] $\theta_{13}{=}0$ in the leading texture, the CP *magnitude* (240°) is the μ_6/triality phase (v231/v233), and M_R is the free seesaw scale. An operator advance, not a closure of full PMNS dynamics.
- **v264_raw_seam_marklocal.py (QGEO.ENERGY.03) sharpens the bedrock.** Composing v216+v201+v198 on the canonical pillowcase: the flat-pillowcase curvature lives only at the 4 marks, so the DtN sub-principal symbol is $\mathbb{Z}_4$-invariant (Fourier on $4\mathbb{Z}$, FFT-verified), block-diagonal, and $[\rho, \Lambda]{=}0 \Rightarrow \omega \circ \rho{=}\omega$ on this metric. So the whole bedrock collapses to the single metric statement "the raw RP seam carries the uniformising flat pillowcase metric" (=QGEO.SYM.01). A reduction/sharpening, [**O**] — it does *not* prove the seam must carry that metric (negative control: curvature off the μ_4 orbit breaks the chain). Paper citations added in tfpt_4_frontier, tfpt_2_standard_model and tfpt_research_contracts. [**E**] AUDIT OK, build green.

2026-06-18 (Paper consistency pass: the closure ledger and the status/QFT sections updated for the Modular Spectral Closure, with an explicit "what is *not* closed" line)

- **Five paper sections that predated the v238–v261 round are brought up to date** (additive, no status marker moved). The introduction closure-ledger table and the "five questions" item now carry a *boundary QFT (Modular Spectral Closure)* row — one relative object $\mathsf{TFPT}_{\rm QFT} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\rm rel})$ closed modulo QGEO.SYM.01 (v258–v261) — plus an explicit list of what the closure does *not* reach: the full measured pole masses, the absolute QCD scales (m_p/m_e), the complete PMNS dynamics, and the real 4D interacting QFT / ambient measure. The tfpt_2 "ambient closure" section now states

that the cutoff of its $S_{\rm rel}$ *is* the seam KMS weight (f_2/f_0=1, `v259`) and $D_{\rm rel}$ the covariance induction (`v258`); the `tfpt_2` "(4) Constructive QFT closure" section adds the relative-object pointer; `tfpt_4_frontier`'s full-quantum-gravity section records that the cutoff moment f_0 is pinned by the seam state (`v255`/`v259`); and `origin_theory`'s honest-status keybox notes that the whole boundary QFT collapses onto the same `QGEO.SYM.01` postulate (no sixth structural class). An editorial consistency pass so every status surface agrees with the ledger. [**E**] build green (`10 ok, 0 failed`), `AUDIT OK`.

2026-06-18 (Consistency pass: the Modular Spectral Closure surfaced structurally across the website and the intro status card)

- **The `v258`–`v261` closure is now reflected on every "what is open" surface, not just the research-contracts paper.** The homepage open-gates section and compiler-pipeline diagram, the hostile-referee FAQ, the glossary $G_{\rm net}$ entry, the reviewer page and the verification residual-chain caption now state that the *entire* emergent-QFT layer (Dirac, cutoff, gauging, glue, orientability) collapses onto the *same* premise `QGEO.SYM.01` via the Modular Spectral Closure — the boundary QFT is one relative object that adds *no new open item*, ambient QG separate. The introduction status card ("the structural residual is one condition") carries the same sentence. No status marker moved; this is an editorial consistency pass so the public surfaces agree with the ledger and the paper. [**E**] build green, `AUDIT OK`.

2026-06-17 (The Modular Spectral Closure: Dirac as covariance induction, the modular cutoff, the Kummer/K3 unification, and the assembly certificate, `v258`–`v261`)

- **`v258_dirac_covariance_induction.py` (`PS.DIRAC.03`) closes the "posited Yukawa vs derived operator" gap of `v250`/`v252`.** The finite Dirac operator D_F is the modular/covariance *induction* of the boundary seam quasi-free (KMS, β=1) state: a Gaussian state is fixed by its covariance C (a positive contraction) with one-particle modular Hamiltonian $H=\log((1-C)C^{-1})$ (Araki; Casini–Huerta), inverse to the KMS occupation $C=(1+e^{H})^{-1}$. [**E**] the inversion identity holds exactly; [**E**] C_F is a positive contraction; [**E**] the induced operator is chirality-odd; [**E**] it is unitarily equivalent to the `v252` D_F (same mass spectrum); [**E**] the Majorana/seesaw m_D^2/M_R is an off-diagonal covariance entry. [**C**] $C_F=P_F C_\Sigma P_F$, so D_F is the Kasparov shadow $[D_\Sigma]\widehat{\otimes}_{(E_8)_1}[\mathcal{K}_{\rm car}]$ of the boundary geometry and the `v18` Yukawas are the readout of C_Σ.
- **`v259_modular_cutoff_kappa.py` (`PS.SPECACT.02`) fixes the last open spectral-action input.** The cutoff f is the seam's own modular/KMS weight: by Tomita–Takesaki/Bisognano–Wichmann the seam is KMS at β=1 (`v239`) and $2\pi=1/(4c_3)$ (`v58`) fixes β, so the heat-kernel cutoff $f(u)=e^{-u}$ is *derived*. [**E**] its moments give $f_2/f_0=f_4/f_2=1$ exactly; [**E**] a generic Gaussian cutoff gives $\sqrt{\pi}/2\neq 1$ (the choice is non-vacuous); [**E**] hence $\kappa=\sqrt{(f_2/f_0)\,c_{PS}/c_{\rm grav}}=\sqrt{c_{PS}/c_{\rm grav}}$ loses its scheme factor. [**C**] the open [**O**] cutoff freedom of `v253`/`v255` becomes a finite trace ratio over the 96-dim H_F (the RG $\kappa\approx1.3$ pins $c_{PS}/c_{\rm grav}\approx1.7$).
- **`v260_k3_kummer_unification.py` (`ARCH.K3.01`) unifies seam, carrier and lattice on one surface.** The pillowcase is the elliptic-involution quotient $T^2_{\tau=i}/(z\mapsto -z)$ (`v214`); squaring it gives the Kummer/K3 surface, whose three facets are three TFPT objects. [**E**] the E_8 Cartan is even/det 1/positive-definite; [**E**] $H^2(\mathrm{K3},\mathbb{Z})=U^3\oplus E_8(-1)^2$ (rank 22, det -1, even, signature $(3,19)$); [**E**] the seam is $T^2/(z\mapsto -z)$ with 4 marks (orbifold Euler char 0); [**E**] the carrier-16 are the $|A[2]|=2^4=16$ Kummer nodes $=\dim S^+$ (`v197`), $16=4\times4$; [**E**] honest note — the rank-16 Kummer lattice is *not* the $E_8(-1)^2$ summand. [**C**] so the `v1` "glue $D_5\oplus A_3+\mu_4\Rightarrow E_8$" is the lattice shadow of a single object; E_8 appears both locally (du Val $\mathbb{C}^2/2I$, `v232`/`v236`) and globally (H^2 summand). A unification, not a closure.

- **`v261_modular_spectral_closure.py` (`QFT.MSC.01`) is the assembly capstone.** The whole QFT round (`v238`–`v260`) is certified as *one* relative object $\mathsf{TFPT}_{\mathrm{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\mathrm{rel}})$ reduced to *one* open premise. [**E**] the cross-consistency invariants agree: $4 = [B{:}A] = |\mu_4| = 2\chi = |(\mathbb{Z}/2)^2|$ (index = marks = fixed points = glue order), one carrier-16 (spinor = lattice = triple), one gap $6\log(3/2)$ (OS = recovery = Lindblad); [**E**] the three closures re-derived; [**E**] the ten-component manifest present. [**O**] the QFT layer reduces to `QGEO.SYM.01` (seam realisation), the ambient QG measure `QG.AMB.01` kept separate. [**C**] the *Modular Spectral Universality* master statement: the unique RP holomorphic index-4 KMS seam state induces the 96-dim KO-6 triple whose inner fluctuations give the gauged Pati–Salam spectral action — the honest sense in which the boundary QFT is complete.

2026-06-17 (Infrastructure: the changelog is now a public website page, generated from `changelog.tex` and audit-gated)

- **The canonical changelog now also drives a public `/changelog` website page.** A new generator `verification/make_changelog_web.py` parses this file (dated `\subsection*` entries, `itemize` items, and the inline markup `\textbf`/`\emph`/`\texttt`/`\vref`, the status markers [**E**]/[**C**]/[**O**]/[**X**], inline math and accents) into the typed data file `website/lib/changelog.ts`, rendered server-side (math via KaTeX) by `website/app/changelog/page.tsx` + `website/components/Changelog.tsx`. The generator is wired into `bash build.sh gen`; its freshness is enforced by `audit_sync.py` (section A.generated), so the page can never drift from this file – editing the changelog and re-running `gen` is the only supported path (the `.ts` mirror is never hand-edited). `/changelog` is linked from the navbar, footer and sitemap. The `tfpt-workflow`, `website-sync` and `sync-maps` cursor rules were updated to record the new single-source generation step. [**E**] build (`npm run build` green, `/changelog` prerendered static) and `bash build.sh audit` (`AUDIT OK`).

2026-06-17 (Orientability + Poincaré duality: the honest, literature-confirmed status — strict fail, weakened hold, `v257`)

- **`v257_quasiorient_modpd.py` (`PS.NCG.ORIENT.02`) corrects the slightly too-generous `v256` typing.** The published result (Cacic–Stephan arXiv:0902.2068; CCM07; SM-vacuum hep-th/0601192) is that the Chamseddine–Connes–Marcolli Standard-Model finite triple (KO-dimension 6, with three right-handed neutrinos — exactly the TFPT seesaw case) *genuinely fails* strict orientability and strict Poincaré duality, and instead satisfies the physically correct weakened versions. Verified on the explicit `v252` triple: [**E**] strict orientability fails ($\gamma \notin \mathrm{span}\{\lambda(a)\rho(b)\}$); [**E**] *quasi-orientability* holds (the left/right A-irrep boxes are disjoint by chirality: $L = H$ weak-doublet, $R = \mathbb{C}$ singlet, so $L^{\mathrm{LR}}(\mathcal{H}^{\mathrm{even}}, \mathcal{H}^{\mathrm{odd}}) = \{0\}$); [**E**] strict Poincaré duality fails (KO-6 skew intersection form, corank 1, the equal-L/R-neutrino case); [**C**] modified Poincaré duality holds (CCM), Stephan's alternative triple (St06) restoring strict PD with identical physics. A known property of the NCG-SM model class, not a TFPT defect — no faked closure.

2026-06-17 (Remaining NCG obligations discharged or honestly typed: inner fluctuations, spectral action, orientability/Poincaré, `v254`–`v256`)

- **`v254_inner_fluctuations.py` (`PS.NCG.FLUCT.01`) closes `v250` #5 + the no-junk obligation.** The inner fluctuations $\Omega^1_D(A_F)$ of the finite Dirac operator are computed explicitly: [**E**] they are purely chirality-odd (scalars), and for the SM algebra sit *exactly* in the one Higgs doublet $(1,2)_{1/2}$ — no junk, no triplet, no $(10,1,3)/\mathbf{126}$. [**E**] the σ (ν_R Majorana) direction is absent for the SM algebra but present for Pati–Salam (the CCvS beyond-first-order scalar), and [**E**] σ is an SM singlet $(1,1,0)$ with $B-L = 2$. [**C**] σ = scalaron.

- **`v255_spectral_action_expansion.py` (`PS.SPECACT.01`) closes the spectral-action expansion (`v252` #11).** The Seeley–DeWitt expansion $S = 2f_4\Lambda^4 a_0 + 2f_2\Lambda^2 a_2 + f_0 a_4$: **[E]** structure ($a_0$ cosmological, a_2 Einstein–Hilbert+Higgs-mass, a_4 Yang–Mills+Weyl^2 + R^2+Higgs-quartic, $N = 96$); **[E]** $b/a^2 = 1/3$ (top-dominated); **[E]** $\sin^2\theta_W = 3/8$; **[E]** the R^2 spin-0 mode = the Starobinsky scalaron; **[C]** Higgs quartic $\to \sim$125 GeV with σ; **[C]** $\kappa = \sqrt{(f_2/f_0)c_{PS}/c_{\rm grav}}$ explicit; **[O]** exact decimal needs the cutoff moments.
- **`v256_orientability_poincare.py` (`PS.NCG.ORIENT.01`) — honest status, no faked proof.** KO-dimension 6 forces the intersection form $\mathrm{Tr}(\gamma\,\pi(p)J\pi(q)J^{-1})$ to be *antisymmetric*: **[E]** skew form (verified), **[E]** rank 2, corank 1, null $\sim (1, 1, -1)$ on the SM K-theory $\mathbb{Z}^3$; **[E]** the orientability local condition $[\gamma_F, \pi(a)] = 0$. **[C]** the canonical hypercharge-faithful SM/PS triple satisfies orientability and Poincaré duality (van Suijlekom; CCM); **[O]** a self-contained machine proof needs the complete canonical bimodule.

2026-06-17 (Full 96-dim finite spectral triple built + NCG axioms verified, and κ pinned to $O(1)$, `v252`–`v253`)

- **`v252_full_finite_triple.py` (`PS.DIRAC.02`) advances the Dirac contract `v250` from a 7-obligation contract to a concrete object discharging 5 to [E].** The full 96-dim finite spectral triple $(A_F, H_F, D_F, J, \gamma)$ — H_F the particle⊕antiparticle doubling of three $SO(10)$ **16**-plets — is built explicitly and the NCG axioms are checked by direct computation: **[E]** reality (J^2=+1, JD=DJ), grading (γ^2=1, $[\gamma, a]$=0, $J\gamma$=$-\gamma J$), KO-dimension 6, order-zero, the first-order condition (HOLDS for the SM algebra including the Majorana; VIOLATED for the Pati–Salam algebra exactly by the Majorana, localized to $\{\nu_R, e_R\}$ — the CCvS beyond-first-order mechanism on the full triple), and the spectrum of D_F reproducing the fermion masses and the type-I seesaw $m_{\rm light}$=m_D^2/M_R. **[C]** Poincaré duality; **[O]** orientability, no-junk, spectral-action expansion.
- **`v253_kappa_scalaron_ps.py` (`PS.KAPPA.01`) discharges residual 6(b) of `v249`.** κ in $M_{PS} = \kappa\, M_s$ ($M_s = c_3^{7/2}\bar{M} \approx 3.06{\times}10^{13}$ GeV, exact) is pinned by an RG scan (PDG errors × the two E_8-allowed contents) to a tight $O(1)$ interval (≈ 1.3). **[C]** heat-kernel origin: $\kappa = \sqrt{(f_2/f_0)\, c_{PS}/c_{\rm grav}}$ is a scale-free ratio of heat-kernel traces over the 96-dim H_F (Λ cancels). **[X]** the **126**-content drives κ out of the $O(1)$ window (content-selective). **[O]** the exact decimal needs f_2/f_0 fixed.

2026-06-17 (Spectral triple, first step: the first-order condition is violated exactly by the Majorana/σ — the CCvS Pati–Salam mechanism, `v251`)

- **`v251_first_order_sigma.py` (`PS.DIRAC.01`) advances the Dirac contract `v250` from a contract to an exhibited mechanism.** On a faithful lepton-block model ($H = H_F \oplus H_F^c$, $H_F = \mathbb{C}^3$), the real even structure is checked (J^2=+1, γ^2=1, $D\gamma$=$-\gamma D$, KO-dim-6 signs), order-zero holds, the first-order condition HOLDS for the Yukawa block, and is VIOLATED *exactly* by the Majorana ($\nu_R\nu_R^c = \sigma$) block — the Chamseddine–Connes–van Suijlekom "beyond first order" mechanism by which the inner fluctuations promote the SM to Pati–Salam and generate σ. Honest correction to the review: σ does *not* rescue the first-order condition; its controlled violation *is* the mechanism. **[C]** σ = scalaron (`v249`); **[O]** the full 96-dim triple + spectral action.

2026-06-17 (Carrier-native Pati–Salam program: E_8 forbids the 126, PS algebra, unification kill-test, Dirac contract, `v247`–`v250`)

- **Four scripts turning the gauge-unification finding into typed claims with negative controls.** `v247_e8_branching_no126.py` (`PS.E8BRANCH.01`): the E_8 hull branches under $SO(10){\times}SU(4)$ as $(45,1)+(1,15)+(10,6)+(16,4)+(\overline{16},\bar{4})$, so the $SO(10)$ content is exactly

$\{1, 10, 16, 45\}$ and the **126** is *absent* — the renormalisable 126_H is algebraically forbidden and 10+16+45 is E_8-supplied (only one 45 $\Rightarrow$ proton-marginal). `v248_ps_algebra.py` (`PS.ALGEBRA.01`): $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C}) \to SU(2)_L \times SU(2)_R \times SU(4)_c$, exact hypercharges $Y = T_{3R} + (B-L)/2$, matching $\alpha_1^{-1} = \frac{3}{5}\alpha_{2R}^{-1} + \frac{2}{5}\alpha_4^{-1}$, $\sin^2\theta_W = 3/8$ (colour $SU(4)_c \subset D_5$, distinct from the family A_3). `v249_ps_unification.py` (`PS.RGTEST.01`): the two-step PS$\to$ $SO(10)$ unification kill-test with negative controls (SM-only fails; 126 breaks the scalaron match; proton decay selects content), $M_{PS}/M_s = 1.0$–1.7. `v250_dirac_triple_contract.py` (`CONTRACT.QFT4D.DIRAC.01`): the spectral-triple contract ($H_F = 3{\times}16$, the seven NCG axioms, σ =scalaron) that would turn "carrier is gauged" from a fork into a theorem. The root-level `qft.txt` carries the full analysis.

2026-06-17 (Honest data cross-check + emergent-QFT figures: the unification tension, v246)

- **`v246_unification_data_crosscheck.py` (`QFT4D.RGTEST.01`) — the first hard data confrontation of the spectral-action prediction.** TFPT's gauge-charged content *is* the Standard Model (`v159`) and the theory adds no new states; the E_8 cascade is an arithmetic spine (`v5`), not a threshold tower. So the run *is* the SM run: at one loop the couplings spread $\sim 10^{13}$–10^{17} GeV (residual $\Delta\alpha^{-1} \approx 9$ at $2{\times}10^{16}$ GeV), and at two loops (RK4) the minimal spread is $\Delta\alpha^{-1} \approx 3.2$ at $1.7{\times}10^{14}$ GeV — the three never meet. Typed **[X]**/**[O]**: with "no new state" there is *no* admissible threshold source, so the spectral-action 4d-GUT route is in genuine tension (same status as NCG-SM), likely falsified unless extended — never a confirmation, and *not* rescued by a cascade. Two new figures (`figures/qft_skeleton.pdf`, `figures/qft_unification.pdf`) added via `make_figures.py` (PDF + website PNG) and embedded in `tfpt_research_contracts.tex`; a root-level `qft.txt` summarises the emergent-QFT skeleton, what each script proves, the data status and the open questions.

2026-06-17 (Spectral-action matter content: the $SO(10)$ **16** = one anomaly-free SM generation, v245)

- **`v245_spectral_matter_content.py` (`QFT4D.MATTER.01`) discharges the computable core of the 4d contract `v244`.** The carrier half-spinor (the $SO(10)$ **16**) is verified, in exact rational arithmetic, to be one Standard-Model generation: $\mathbf{16} = SU(5)(\mathbf{10}{+}\bar{\mathbf{5}}{+}\mathbf{1})$, the SM multiplet dimensions sum to $16 = \dim S^+$ (15 Weyl fermions $+\nu_R$), the electric charges $Q = T_3 + Y$ are exactly the SM values, all four gauge anomalies cancel ($\sum Y = \sum Y^3 = [SU(2)]^2U(1) = [SU(3)]^2U(1) = 0$), the spectral-action normalisation gives $\sin^2\theta_W = 3/8$ ($g_3 = g_2 = \sqrt{5/3}\, g_1$), and the Higgs is the $(1,2)_{1/2}$ inner fluctuation. So the matter-content + tree-relation half of `CONTRACT.QFT4D.01` is now **[E]**; only the seam$\to$Dirac realisation and the run-down couplings stay **[O]**. Wired into `run_all.py`, the ledger, the registry, cited in `tfpt_research_contracts.tex`, mirrored to the website.

2026-06-17 (The static$\to$dynamic flip: modular flow + a recovery Lindblad generator, v238)

- **New `v238_modular_lindblad_dynamics.py` (`DYN.SEMIGROUP.01`, cluster registry).** A companion to the modular round (`v196`/`v198`) and the recovery code (`v221`/`v64`) that reads the *same* seam objects *dynamically* rather than statically. **[E]** the modular flow $\sigma_t = e^{it\Lambda_\Sigma}$ is a unitary one-parameter group (Tomita–Takesaki) whose conservation of the μ_4 clock, $\rho\sigma_t\rho^{-1} = \sigma_t \Leftrightarrow [\rho, \Lambda_\Sigma] = 0$, IS the `v198` state-invariance — so the static commutator is the conservation law of modular time. **[E]** the recovery channel's generator $L = \log T$ is a doubly-stochastic/CPTP-classical Markov generator ($e^L = T$, $e^{L\tau} \to$ the democratic projector = the arrow of time). **[E]** keystone: the dissipative gap $= -\log((2/3)^6) = 6\log(3/2) =$ the OS mass gap Δ (`v64`) — the static recovery bound (`v221`) and the dynamical mass gap

(v64) are one number, one exponentiated. **[E]** the GKSL split $G = i\Lambda_\Sigma + L$ (imaginary + real≤ 0 spectrum). **[O]** the residual is the *geometric* modular flow (Connes–Rovelli "thermal time = physical time") on the full seam algebra = QGEO.SYM.01 (v198) — a target, not a closure. Wired into run_all.py, status_ledger.csv, script_registry.csv and cited in tfpt_research_contracts.tex; mirrored to website/public/verification.

2026-06-17 (Red-team layer re-synced to the paper; shadow export mirrors figures/)

- **Red Team Target A re-synced to tfpt_5_redteam (rt_A_e8net.py, redteam_table.txt, redteam/README.md, infrastructure).** The Python red-team surface still emitted the historical *three-residual* "A reduced, not closed" form, lagging the reader-facing note. It now encodes the sharper narrowing already in the paper: the residual collapses to the *single* statement "the seam–Calderón boundary net is holomorphic with $c = 8$" ($\Leftrightarrow$ the index-4 simple-current extension), with the index-4 $(D5)_1 \times (A3)_1 \to (E8)_1$ glue realised at Lie level (v143) and its odd sectors twisted/Ramond ($E_8 = 120_{\mathrm{NS}} + 128_{\mathrm{R}}$, v148), while the $q(A_3)$ normalisation collapses into the one declared anchor (v152). Two exact adversarial checks added (120+128 = 248; glue index = $|\mu_4| = N_{\mathrm{fam}}+1 = 4$); status stays **[C]**/**[O]** (reduced, not closed).
- **Shadow export now mirrors figures/ but not website/ (scripts/export-shadow.sh, verification/make_script_index.py, infrastructure).** export-shadow.sh ships figures/*.pdf (so make_manifest.py --check passes literally on the exported tree) while still excluding website/ and the compiled paper PDFs; the SHADOW_MIRROR.md table is corrected accordingly. make_script_index.py now detects a missing website/ and skips its ScriptIndex.tsx mirror, so build.sh notes runs on the subset without crashing.

2026-06-17 (CP phases as μ_6 powers — a reduction of red-team Target D)

- **Both CP phases are μ_6 powers of one hexagonal CM unit, split by the sheet (v231_cp_mu6_phases, tfpt_5_redteam, [E]/[C]).** Sharper than v220/v225 (which only *located* the CP residual). With $\rho = e^{i\pi/3}$ the primitive 6th root (the j=0 Eisenstein CM unit): δ_{CKM} leading = $\arg(\rho^1) = \pi/3$ (quark) and $\delta_{\mathrm{PMNS}} = \arg(\rho^4) = 4\pi/3$ (lepton, the phase lattice), with $\rho^4 = -\rho$, so $\delta_{\mathrm{PMNS}} = \delta_{\mathrm{CKM,lead}} + \pi$ = (CM unit) $\times$ ($\mathbb{Z}_2$ sheet, ρ^3=-1). The two CP phases are *one* hexagonal unit on the two sheets, not two independent inputs; the dual-frame Jarlskog orientations are sheet-flipped ($\pm 21 \sin(\pi/3)$). A structural reduction of Target D (removes one of its two phase inputs); the quark seam correction $3\lambda_C^2$ stays the v88 misalignment, the deck stays $\mathbb{Z}/4$, and the physical CP derivation stays **[C]** (CP.MU6.01).
- **The seam as the E_8 Kleinian singularity (v232_e8_kleinian_seam, origin_theory, [E]/[C]).** The du Val side of the McKay correspondence (v219) gives the open seam-realisation premise QGEO.REALIZE.01 a canonical algebraic-geometric model: dropping the trivial-rep node of the affine-E_8 McKay graph of $2I$ leaves the finite E_8 Dynkin = the dual intersection graph of the eight exceptional $\mathbb{P}^1$'s of the minimal resolution of $\mathbb{C}^2/2I$; the eight curves = rank $E_8 = g_{\mathrm{car}}+N_{\mathrm{fam}}$ (a fourth reading of the 8), their negated intersection form is the E_8 Cartan (det 1, even, (-2)-curves), and the link is the Poincaré homology 3-sphere $S^3/2I$ ($2I$ perfect $\Rightarrow H_1$=0, π_1=$2I$ order 120). The raw seam ($\mathbb{P}^1$ with marks) is canonically a du Val exceptional curve — a model for the bedrock, not a P2 proof (it presupposes E_8; TOPO.E8.01, bedrock stays **[O]**).
- **CP is the universal family/triality phase, only sheet-split (v233_cp_triality_phase, tfpt_5_redteam, [E]/[C]).** One step past v231 in the $\mu_6 = \mu_3$(family/triality)$\times\mu_2$(sheet) factorisation: $\rho = \omega^2(-1)$ with $\omega = e^{2\pi i/3}$ the $\mathbb{Z}_3$ triality centre (the 2/3 cusp weight). Since $\rho^4 = \rho^1\rho^3$ with $\rho^3 \in \mu_2$, the quark (ρ^1) and lepton (ρ^4) CP phases share the *same*

family/triality class and differ only by the sheet — so CP *is* the universal triality phase, sheet-split, not a free power choice (`CP.TRIALITY.01`).

- **The Seam-Holomorphy selection certificate: the structural residual is one gate (`v234_seam_holomorphy_selection`, `tfpt_research_contracts`, [E]/[O]).** The whole structural residual is *one* condition — "the seam carries no nontrivial abelian sector" — with three provably-equivalent faces, all forcing E_8: (i) holomorphy (μ-index 1; Target A), (ii) the seam link is a homology 3-sphere (Γ perfect $\Leftrightarrow 2I$; `v232`), (iii) exactly one 1-dim irrep (`v219`). All equal $\#(\text{mark-1}) = |\Gamma^{\text{ab}}| = |H_1|$, which is 1 only for E_8; and holomorphic $c{=}8{=}g_{\text{car}}{+}N_{\text{fam}}$ gives the unique even unimodular rank-8 lattice E_8. So P2, G_{net}, Target A and the Kleinian seam are one gate. The stated closing theorem ([O]): RP+gap+chirality ⇒ quasi-free bulk (`v160`) ⇒ holomorphic boundary ⇒ E_8; the single residual analytic step is "free bulk ⇒ holomorphic boundary" (`GATE.HOLO.01`).
- **The closing step in Chern–Simons language: holomorphic ⇔ $\det K{=}1$ (`v235_seam_chern_simons`, `tfpt_research_contracts`, [E]/[O]).** A genuine reduction (not a closure) of `GATE.HOLO.01` into standard anyon-condensation physics: a free gapped bosonic 2+1d bulk is an even integer K-matrix with #anyons = $|\det K|$, edge $c = \text{signature}(K)$, and the edge net holomorphic ⇔ $|\det K|{=}1$. The TFPT extension tower (`v92`) *is* the condensation tower read by determinant: carrier $D_5{\oplus}A_3$ ($\det 16$, 16 anyons) $\to SO(16)_1{=}D_8$ ($\det 4$) $\to (E_8)_1$ ($\det 1$, holomorphic) — all rank 8, $c{=}8{=}g_{\text{car}}{+}N_{\text{fam}}$ — so holomorphic = E_8 = the Kitaev E_8 quantum-Hall state. Holomorphy ⇔ condensing the order-$|\mu_4|$ Lagrangian glue ($\det 16{\to}1$, vs. a partial order-2 isotropic $\to 4{=}D_8$). The residual narrows to one integer condition "the free RP seam condenses the Lagrangian glue ($\det{=}1$)" = the sheet/QGEO selection, now with holomorphy⇔ $\det 1$ a theorem (`GATE.HOLO.02`).
- **The $(2,3,5)$ Brieskorn singularity is the one generator of the skeleton (`v236_brieskorn_capstone`, `origin_theory`, [E]).** Capstone of the icosahedral round: the singularity $x^2{+}y^3{+}z^5$, whose exponents are the atoms $(|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}}){=}(2,3,5)$, has Milnor number $\mu{=}(2{-}1)(3{-}1)(5{-}1){=}8{=}\operatorname{rank} E_8$ (a *fifth* origin of the "8"), Milnor monodromy = the order-30 E_8 Coxeter element (eigenvalues the primitive 30th roots = the E_8 exponents, charpoly Φ_{30}), Milnor lattice E_8, link the Poincaré sphere; the two clocks are facets of this one monodromy — the μ_3 triality (CP, `v233`) is h^{10} in $\langle h\rangle{=}\mathbb{Z}/30$, the μ_4 clock (`v223`) is the order-4 Galois automorphism of $\mathbb{Q}(\zeta_{30})$ (`TOPO.BRIESKORN.01`).
- **The closing step as a physical condition: no topological degeneracy ⇔ $\det K{=}1$ ⇔ the seam is SRE (`v237_seam_sre_closure`, `tfpt_research_contracts`, [E]/[O]).** Sharpens `GATE.HOLO.02`'s residual from definitional to falsifiable: the K-matrix ground-state degeneracy on a genus-g surface is $|\det K|^g$ (torus: carrier 16, D_8 4, E_8 1), so "no topological degeneracy on any closed surface" ⇔ $\det K{=}1$; among rank-8 even positive-definite K ($c{=}8{=}g_{\text{car}}{+}N_{\text{fam}}$) only E_8 is short-range-entangled (the Kitaev E_8 phase). A unique vacuum on the plane (RP+gap+cluster) is necessary but not sufficient (the plane cannot see $\det K$); so the one open step is the physical statement "the free RP seam is SRE (no topological order)" (`GATE.HOLO.03`).

2026-06-16 (the icosahedral bedrock, CM-norm duality, and a structural-finds round)

- **McKay bedrock: why the atoms are $\{2,3,5\}$ (`v219_icosahedral_mckay`, `origin_theory`, [E]).** E_8 is the exceptional *top* of the McKay tower of finite $SU(2)$ subgroups ($2T{\to}\hat{E}_6$, $2O{\to}\hat{E}_7$, $2I{\to}\hat{E}_8$). Built *from the group*: the 120 unit-quaternion icosians close to the binary icosahedral group $2I$ (element orders $\{1,2,3,4,5,6,10\}$ — it contains the $2,3,5$ rotation-axis orders), 9 conjugacy classes; the 9 irreducible character degrees (Dixon, from the class algebra) are $\{1,2,2,3,3,4,4,5,6\}$, sum $30{=}h(E_8)$, sum of squares $120{=}|R^+(E_8)|$; the McKay

adjacency from the natural 2-dim rep *is* the affine E_8 graph with Kac marks = the degrees. A backward certificate of the closed E_8, not a P2 proof (`MCKAY.E8.01`).

- **CM-norm duality: 41 (square) and 7 (hex) (`v222_cm_norm_duality`, `origin_theory`, [E]).** The square seam (Gaussian $\mathbb{Z}[i]$, j=1728) gives $N(g_{\text{car}}+i|\mu_4|)=5^2+4^2=41=10b_1$ (the EM index *is* the Gaussian norm of the carrier-glue vector) and $N(3+2i)=13=\Delta_Q$; the hexagonal partner (Eisenstein $\mathbb{Z}[\omega]$, j=0) gives $N(3+2\omega)=7=$ scalaron. The $(3,2)$ split generates $(5,6,7,13)$ by sum/product/Eisenstein/Gauss norm; rings rigid (`CMNORM.DUAL.01`).
- **The μ_4 clock is the order-4 character of the E_8 Coxeter cycle (`v223_coxeter_totative_clock`, `origin_theory`, [E]).** $(\mathbb{Z}/30)^\times$ are the E_8 exponents; the order-4 generator 7 ($\langle 7\rangle=\{1,7,13,19\}$) is a literal μ_4 clock permuting the four conjugate planes; only E_8 has $\varphi(h)$=rank and $h=2{\cdot}3{\cdot}5$ (`COX.CLOCK.01`).
- **248=120+128 as a magnitude/phase channel typing (`v227_degree_exponent_channel_split`, `tfpt_5_redteam`, [E]).** Exponents sum $120=|R^+(E_8)|$ (magnitude channel), degrees sum $128=\text{rank}\cdot\dim S^+$ (phase/glue channel); the red-team Target D magnitude bijection lives in 120, the un-covered CP phases in 128; the dark-sector reading of 128 stays Frontier (replaces the over-strong S^- reading; `E8.CHAN.01`).
- **P2 as the mode space of a degree-4 seam divisor (`v228_rr_index_gate`, `origin_theory`, [E]/[C]/[O]).** A Riemann–Roch index gate: $\deg D=4=|\mu_4|$ on $\mathbb{P}^1$ forces $h^0=5=g_{\text{car}}$, $\text{rank}\,H_1=3=N_{\text{fam}}$, $h^0+H_1=8=\text{rank}\,E_8$, $\Lambda^{\text{even}}(\mathbb{C}^5)=16=\dim S^+$ — bedrock language for P2, conditional on the seam producing the degree-4 divisor (`QGEO.RR.01`).
- **The seam as a finite recoverability code (`v221_seam_qecc`, `origin_theory`, [E]/[C]).** Code dimension $\dim S^+=16$; the gapped transport is a CPTP doubly-stochastic contraction with spectrum $\{1,(2/3)^6,(1/3)^6\}$, recovery bound $\|T^n\delta\|\le(2/3)^{6n}\|\delta\|$ (Knill–Laflamme type); free ratio breaks it. Holographic/Petz reading [C], gated on the Seam–Horizon theorem (`QEC.SEAM.01`).
- **CP lives in the hexagonal phase fiber (`v220_cp_hexagonal_modulus`, `tfpt_5_redteam`, [E]/[C]).** The seam deck is the square modulus j=1728 ($\mathbb{Z}/4$); its partner is the hexagonal j=0 ($\mathbb{Z}/6$, $\arg\rho=\pi/3$). The frozen $\delta=\pi/3+3\lambda_C^2=68.654^\circ$ has leading term $\pi/3=\arg\rho$; the deck stays $\mathbb{Z}/4$, CP is the $\mathbb{Z}/6$ fiber (`CP.MOD.01`).
- **The dual normal frame (d,n) with CP as orientation (`v225_dual_normal_frame`, `tfpt_5_redteam`, [E]/[C]).** $d=a^\top R^{-1}=(-\frac12,-\frac12,1)=-\frac12$ Nariai; $n=(5,-9,6)$ reads det on the first-generation column; $\det(\mathbf{1},d,n)=21=3{\cdot}7$; $\text{Im}\det(\mathbf{1},d,\rho n)=21\sin(\pi/3)$ — CP as orientation (`DUAL.FRAME.01`).
- **F_{transfer} as one path on the Sheet Diamond (`v224_diamond_ftransfer_path`, `tfpt_2_standard_model`, [E]/[C]).** $M(s,t)=R+Q\,\text{diag}(s,t,t)$: K,C,F on the sheet axis (curved, 2nd diff 8), the winding axis flat (slope 6), Plücker steps $(1,8,10),(1,8,16)$; the Koide source→pole is the 1-D projection (`FTR.PATH.01`).
- **The charged leptons as an étale Frobenius algebra (`v229_lepton_frobenius_algebra`, `tfpt_2_standard_model`, [E]/[C]).** $(c_e,c_\mu,c_\tau)=(\frac{16}{7},\frac43,\frac76)$, ring closure $c_ec_\tau=\frac83=|\mathbb{Z}_2|c_\mu$, product $\frac{32}{9}$; $A=\mathbb{Q}[t]/(m)$ is étale (trace-form Gram nondegenerate), a commutative Frobenius algebra over the μ_6 resolvent (`LEP.FROB.01`).
- **The center budget $(7,11,13)$ as three local norms (`v230_center_budget_norms`, `tfpt_2_standard_model`, [E]).** $C=R+Q\,\text{diag}(1,0,0)$ row sums $(7,11,13)$: $7=N_{\mathbb{Z}[\omega]}(3+2\omega)$ (hex), $13=N_{\mathbb{Z}[i]}(3+2i)$ (square), $11=\binom40+\binom41+\binom42$ (QBL Fock count); the two CM rings and the boundary count meet in one operator center (`CENTER.NORM.01`).

2026-06-15 (the diamond axis geometry — two axes + the transfer corner)

- **The Sheet Diamond is a discrete geometry with two axes (`v218_diamond_axis_geometry`, `tfpt_2_standard_model`, [E]).** Sharpens `v94`/`v95` with three exact [**E**] lemmas (no new numbers). **(1) Axis curvature:** along the centered cross the determinant is linear on the winding axis $\det(C{+}xU) = 14{+}6x$ (slope $6{=}|R^+(A_3)|$) and quadratic on the sheet axis $\det(C{+}yV) = 14{+}14y{+}4y^2$ (2nd difference $8{=}\operatorname{rank} E_8$); the anchor block has $\det B(C{+}xU) = 2\det(C{+}xU)$ and sheet curvature $6{=}|R^+(A_3)|$ — two curvatures $(8,6) = (\operatorname{rank} E_8, |R^+(A_3)|)$. **(2) Plücker transfer ladder:** $\mathrm{Pl}(K) \to \mathrm{Pl}(C) \to \mathrm{Pl}(F)$ lifts in steps $(1,8,10) = (N_\Phi, \operatorname{rank} E_8, A_\Lambda)$ then $(1,8,16) = (N_\Phi, \operatorname{rank} E_8, \dim S^+)$ — the decuple then the full generation (identity [**E**], source→pole reading [**C**]). **(3) Spectral ramification:** for Q, K, C, F the cubic discriminant factors as $q(r)^2 \operatorname{Disc}(q)$ with squares $(1,3,4,6)$ and kernels $(13, 48, 65, 105)$, F carrying $105 = 3{\cdot}5{\cdot}7$. The anchor-defect spectrum and pair-sum/staircase blocks are honest *audit* fingerprints ($28{=}2\dim G_2$, $52{=}\dim F_4$ stay audit-only, not spine, as in `v94`/`v95`); F as the transfer-completion corner is a *heuristic*, not a hard filter. Ledger `DIAMOND.AXIS.01`, `DIAMOND.PLUCKER.01`, `DIAMOND.SPECTRAL.01`; Wolfram-mirrored (249/249).

2026-06-15 (QGEO: the four marks emerge from Gauss–Bonnet — K1 executed)

- **The four marks emerge from Gauss–Bonnet — count derived, not assumed (`v216_marks_gauss_bonnet`, `tfpt_research_contracts`, [E]; ledger `QGEO.MARKS.03`).** The K1 lever of the `v215` kill-test ("exactly four marks") is *executed*: if the marks are $\mathbb{Z}_2$ branch points (deficit π) on the flat seam sphere ($\chi = 2$, the same χ as the 8 in c_3), Gauss–Bonnet forces $n\pi = 2\pi\chi = 4\pi$, so $n = 2\chi = 4 = |\mu_4| = N_{\mathrm{fam}} + 1$. The closed Euclidean sphere 2-orbifolds are exactly $(2,3,6)$, $(2,4,4)$, $(3,3,3)$, $(2,2,2,2)$, and three independent criteria — all cone orders $= 2$ (the $|\mathbb{Z}_2|$ sheet branch), $\operatorname{rank} H^1 = \#\mathrm{marks} - 1 = 3 = N_{\mathrm{fam}}$ (the three generations pick the 4-mark square over the 3-mark hexagonal), and a free order-4 deck — converge on the pillowcase $(2,2,2,2)$. The numerical sibling `v217_marks_emergence_scan` ([**C**], ledger `QGEO.EMERGE.01`) confirms it on the actual DtN/state (number n and positions free, the 4-square config is the unique joint minimiser of clock-invariance + 3-family cohomology + gap $(2/3)^6$). The mark *count* and *equality* move from input to derived; only the square *modulus* (cross-ratio 2, the order-4 clock) stays the one carrier input, and the from-zero raw-seam emergence `QGEO.REALIZE.01` stays [**O**]. `v216` exact (Wolfram extension → 248/248); `v217` numerical (Python-only). Suite → 217 modules.

2026-06-15 (QGEO: the bedrock recast as a falsifiable kill-test)

- **`QGEO.SYM.01` as a kill-test, not a proof-promise (`v215_seam_deck_killtest`, `tfpt_research_contracts`, [E]/[O]; ledger `QGEO.KILL.01`).** A foundational symmetry cannot be derived from nothing (Bisognano–Wichmann would presuppose the covariance it should produce), so the bedrock $\omega \circ \rho = \omega$ is recast as four independently-falsifiable predictions about the raw seam DtN. The non-circular reduction (`v198`/`v199`/`v201`) gives $\omega \circ \rho = \omega \Leftrightarrow [\rho, C] = 0$ with $|k|$ commuting exactly, so the residual is the bounded sub-principal M_f (equivalently the raw Gaussian bulk form A is μ_4-deck-invariant). Levers: **K1** exactly four marks ($b_1 = \#\mathrm{marks} - 1 = 3 = N_{\mathrm{fam}} \Rightarrow \#\mathrm{marks} = 4 = |\mu_4|$); **K2** the square config (cross-ratio $2 \Rightarrow j = 1728$, $\mathrm{Aut} = \mathbb{Z}/4$; order 4 selects $\mathbb{Z}/4$ not the $j = 0$ $\mathbb{Z}/6$; generic $\Rightarrow \mathbb{Z}/2$); **K3** mod-4 sub-principal support ($[\rho, M_f] = 0 \Leftrightarrow f$ $\mathbb{Z}_4$-invariant); **K4** transfer gap $(2/3)^6 = 64/729$. Freeze-bound via `freeze_file.csv` (`seam_deck_symmetry`), carrier-side [**E**] (regression guards), with the *decisive* kill deferred to `QGEO.REALIZE.01` [**O**] (the raw collar DtN must *produce* four square marks without inputting them — an emergence test). A kill-test, not a closure. Python-only (falsification layer; the exact sub-parts $j = 1728$ and $(2/3)^6$ are Wolfram-mirrored via `v214`/`v54`/`v56`). Suite → 215 modules.

2026-06-15 (QGEO: the pillowcase reduction — two residuals become one canonical metric)

- **Pillowcase reduction of `QGEO.SYM.01` (`v214_seam_pillowcase`, `tfpt_research_contracts`, [E]/[O]; ledger `QGEO.PILLOW.01`).** The two separately-tracked open QGEO residuals — `QGEO.ISO.01` (`v180`: the carrier clock is an order-4 *isometry* of the seam metric) and `QGEO.SUBPRIN.01` (`v201`: the DtN sub-principal symbol is *mark-local*, the seam flat away from the μ_4 marks) — are shown to be *one* statement: the seam carries the uniformising flat orbifold (*pillowcase*) metric. **(1)** The four marks make the Euclidean orbifold $S^2(2,2,2,2)$ ($\chi_{\rm orb} = 2 - 4(1-\frac{1}{2}) = 0$), so by Troyanov uniformisation the metric is flat (unique up to scale) with curvature only at the cone points (Gauss–Bonnet $4\pi = 2\pi\chi$) — this *is* the `v201` "flat away from the marks". **(2)** New link: cross-ratio$(\mu_4) = 2$ (`v168`) $\Rightarrow j = 1728$ ($j(\lambda) = 256(\lambda^2-\lambda+1)^3/(\lambda^2(\lambda-1)^2)$; all six harmonic cross-ratios give 1728) $\Rightarrow$ the *square* modulus $\tau = i$, whose CM by $\mathbb{Z}[i]$ *derives* the order-4 clock $z \mapsto iz$ (explicit $(x,y) \mapsto (-x, iy)$ on $y^2 = x^3 - x$); generic cross-ratio gives $j \notin \{0, 1728\}$ and only $\mathbb{Z}/2$. **(3)** Unification: "seam = uniformising flat pillowcase metric" implies *both* `v201` and `v180`, so two residuals collapse to one canonical-metric premise. **[O]** It does *not* close `QGEO.SYM.01`: the choice "RP seam metric = the constant-curvature representative" stays open, milder and more canonical than the bare isometry premise. Wolfram-mirrored (the exact orbifold/j-invariant/CM arithmetic); Klein-J values mpmath-numerical. Suite → 214 modules.

2026-06-15 (the F_transfer functor contract CONTRACT.F.01)

- **$F_{\rm transfer}$ formalised as a typed functor with four axioms (`v213_ftransfer_functor`, `tfpt_research_contracts`, [C]).** The four frontier transfers are consolidated into ONE functor $F_{\rm transfer} = F_{\rm observable} \circ F_{\rm threshold} \circ F_{\rm RG}$, each instance a discrete compiler kernel **[E]** composed with an external continuous solver **[C]**: **(1)** μ_4-deck equivariance (the Koide multiplier $\lambda_2 = (2/3)^6 = 64/729$ is the deck transfer eigenvalue); **(2)** Plücker preservation ($53 = a^T(R+Q)\mathbf{1}$, ${\rm Pl}(F) = (1, 22, 30)$, $\|{\rm Pl}(K)\|_1 = 11$); **(3)** positivity/stochasticity (spec T positive, $\kappa_f \in (0,1)$); **(4)** external modules explicit ($b_3 = -7$ a carrier output). The four instances are $F_{\rm pole}$ (`v183`), $F_{\rm Boltzmann}$ (`v212`), $F_{\rm relic}$ (`v211`), $F_{\rm QCD}$ (`v164`). `CONTRACT.F.01` is the third research contract alongside ($U_{\rm wall}$) and ($G_{\rm metric}$), the structural complement of the `v187` typing guard — an architectural consolidation, *not* a closure of any frontier number. Ledger `CONTRACT.F.01`.

2026-06-15 (frontier: two sharper [C] scenario branches — axion angle + leptogenesis)

- **Axion DM spine-angle branch $\theta_i = \pi N_{\rm fam}/g_{\rm car} = 3\pi/5$ (`v211_axion_spine_angle`, `tfpt_4_frontier`, [C]).** A more *robust* misalignment angle than the determinant-line hilltop $\theta_i \approx 170^\circ$ (`v185`, which over-produces $\Omega_a h^2 \approx 0.66$): the central spine quotient gives $\theta_i = 3\pi/5 = 108^\circ$ exactly (no fit), the finite-T solver reaches $\Omega_{\rm DM}$ at $\approx 106^\circ$, and 108° sits in the *mild*-anharmonic regime (62° below the hill-top) — not exponentially sensitive. An alternative ansatz to $\theta_i = \pi(1 - \varphi_\Sigma)$ (mutually exclusive, the full solver decides, `DM.AXION.SPINE.01`); **[X]** $\Omega_a h^2$ outside $\sim[0.08, 0.16]$ demotes it. A sharper scenario, not a derivation.
- **Leptogenesis scalaron-decuple branch (`v212_leptogenesis_decuple`, `tfpt_4_frontier`, [C]).** Both Boltzmann inputs share the decuple $A_\Lambda = 10 = |E(K_5)|$: $M_1 = M_{\rm scal}(\varphi_0^{\rm ret})^2/A_\Lambda \approx 8.65 \times 10^9$ GeV and $\tilde{m}_1 = m_3/A_\Lambda \approx 5$ meV. M_1 uses *only* $\{M_{\rm scal}, \varphi_0^{\rm ret}, A_\Lambda\}$ — no hidden seesaw scale, cleaner than `v184`'s $M_R\varphi_0^4$ — and lands in the thermal window where $\eta_B \approx 1.2 \times 10^{-9}$ brackets the observed 6.1×10^{-10}. The coupling mechanism is posited not derived, so η_B stays **[C]**; **[X]** a precise Boltzmann/RGE solve outside the band falls the route, not the theory (`FR.ETAB.04`; Route A independent).

2026-06-15 (Lean: QGEO.COHOM.01 character grading + MODULE parity machine-proved)

- **The cohomology character grading is now Lean-formalised** (`CohomologyGrading.lean`, ledger `FORM.QGEO.03`, [**O**]). A new module machine-proves the COHOM node of the QGEO proof tree (`v177`; AUDIT: PASS, no `sorry`/`admit`, axioms `propext, Classical.choice, Quot.sound` only): the three $H^1(\mathbb{P} \setminus \mu_4)$ eigenforms $\omega_k = z^{k-1}/(z^4-1)$ (k=1, 2, 3) have pullback character $\rho^*\omega_k = i^k\omega_k$ under the clock $\rho : z \mapsto iz$ (each an exact rational-function identity, the denominator clock-invariant since i^4=1), so the grading is $(i, -1, -i)$ with weights $(1, 2, 3)$ = the A_3 exponents = $\mathrm{Spec}(Q_+)$, rank 3 = N_{fam}. The MODULE *parity* is also proved: the reflection $\sigma : z \mapsto 1/z$ satisfies $\sigma^*\omega_1 = \omega_3$, $\sigma^*\omega_2 = \omega_2$, $\sigma^*\omega_3 = \omega_1$ (swap $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed). Signature-locked in `AuditContract.lean`. The geometric COHOM/MODULE-parity nodes are now [**E**]; the MARKS/KERNEL obligations and the MODULE *uniqueness* stay [**O**].

2026-06-15 (Lean: QGEO.UNIFORM.01 geometric normal form machine-proved)

- **The Möbius uniformisation normal form is now Lean-formalised** (`Mobius Uniformisation.lean`, ledger `FORM.QGEO.02`, [**O**]). A new module machine-proves the constructive UNIFORM node of the QGEO proof tree (`v177`; AUDIT: PASS, no `sorry`/`admit`, axioms `propext, Classical.choice, Quot.sound` only): the clock $\rho : z \mapsto iz$ has ρ^4=id and $\rho^2 \neq$id (order exactly 4); the reflection $\sigma : z \mapsto 1/z$ is an involution with $\sigma\rho\sigma = \rho^{-1}$ (so $\langle \rho, \sigma \rangle$ is the dihedral D_4 of order 8); a non-fixed orbit $\{a, ia, -a, -ia\}$ scales to $\mu_4 = \{1, i, -1, -i\}$; σ permutes μ_4 (fixes $1, -1$, swaps $i, -i$); and the multiplier classification "$z \mapsto \zeta z$ has order exactly 4 $\Leftrightarrow$ $\zeta = \pm i$". Signature-locked in `AuditContract.lean`. This formalises the geometric half of the seam realisation *given* the four marks and the clock; the raw-seam marking obligation `QGEO.MARKS.01` stays [**O**], *not* closed.

2026-06-15 (Lean: the QGEO.SYM.01 conditional theorem machine-proved)

- **The mark-local $\Rightarrow \omega \circ \rho = \omega$ implication is now Lean-formalised** (`SeamDeckClosure.lean`, ledger `FORM.QGEO.01`, [**O**]). A new module in the carrier-rigidity Lean project machine-proves the algebraic core of `v201`/`v210` (AUDIT: PASS, no `sorry`/`admit`, no domain axioms — `#print axioms` = `propext, Classical.choice, Quot.sound` only): (i) the 4th-root character orthogonality $\sum_{j<4} \zeta^j$ = (if ζ=1 then 4 else 0) for ζ^4=1 (so a μ_4-mark sum is supported only on modes $\equiv 0 \bmod 4$); (ii) the clock generator $-i$ has order 4; (iii) `markLocal_blockDiagonal`: a mark-local Toeplitz symbol connects only equal clock-characters, $[\rho, M_f] = 0$; (iv) the premise is a typed `structure SeamDeckPremise` (the physical seam DtN *is* mark-local) — consumed, *not* proved, exactly as `CalderonProjector` encodes the Paper-1 analytic input; (v) `SeamDeckPremise.clock_invariant`: *given* the premise, the clock commutes with the DtN, whence $\omega \circ \rho = \omega$. So the *implication* is now [**E**] (formally proved); the *premise* (the physical seam being mark-local) stays [**O**] — the one fundamental seam-identification postulate, *not* closed.

2026-06-15 (QGEO sharpening: mark-local DtN certified on realistic profiles)

- **Mark-local DtN $\Rightarrow \omega \circ \rho = \omega$, certified on realistic Steklov profiles and at the state level (`v210_mark_local_dtn`, `tfpt_research_contracts`, [**O**]).** A numerical sharpening of `QGEO.SUBPRIN.01`/`v201`: a real von-Mises curvature bump summed over the four μ_4 marks, $f(\theta) = \sum_{j=0}^{3} g(\theta - j\pi/2)$, builds the Steklov DtN $\Lambda = |D_\theta| + M_f$ whose off-(mod 4) Fourier support vanishes to $< 10^{-16}$, $\|[\rho, \Lambda]\| < 10^{-15}$, and — the genuinely new step — the quasi-free covariance $C = \frac{1}{2}(1 + \mathrm{sgn}\, H_1)$ is clock-invariant ($\rho C \rho^\dagger = C$ to $< 10^{-13}$), i.e. the actual $\omega \circ \rho = \omega$. Negative controls (single off-centre bump, $\mathbb{Z}_3$ source, four generic marks) break both by $O(1)$; convergence stable for $N \in \{16, 32, 64\}$. This certifies the *implication*; the premise that the physical seam *is* mark-local stays [**O**] (it does *not* close `QGEO.SYM.01`; net existence

and full-cone RP remain the [**E**] of `v175`). Ledger `QGEO.MARKLOCAL.01`. Python-only (the exact mark-sum identity is Wolfram-mirrored via `v201`).

2026-06-15 (archive integration #5: six dormant results revived from the old papers)

- **Muon $g-2$ as a seam vertex (`v204_muon_seam_g2`, `tfpt_4_frontier`, [C]).** The carrier's second-order topological defect $\delta_2 = \frac{5}{4}\delta_{\text{top}}^2$ projected through the seam-loop phase gives $a_\mu^{\text{seam}} = \delta_2/(2\pi) = 45/(524288\,\pi^9) \approx 2.879 \times 10^{-9}$ (0.81σ vs the dispersive Δa_μ; $\sim 1.5\sigma$ vs lattice/CMD-3). The value is exact [**E**]; the vertex identification is the [**C**] bridge. Ledger `FR.MUONG2.01`; Wolfram-mirrored.
- **An independent gravitational 3/4 (`v205_xi_threequarter`, `tfpt_4_frontier`, [E]/[C]).** $\xi = c_3/\varphi_{\text{tree}} = 3/4$ exactly (Einstein limit of $\kappa^2 = \xi\varphi_0^{\text{ret}}/c_3^2$) — the same $3/4 = q(A_3) = \ln(m/\mu)$ as the seam-determinant replica (`v152`), an over-determination. Ledger `GRAV.XI.01`; Wolfram-mirrored.
- **Quantitative H_0 from the Λ branch (`v206_h0_lambda_branch`, `tfpt_4_frontier`, [C]).** $\delta_\Sigma = 48\,e^{-2\pi\alpha^{-1}/3} \approx 1.085 \times 10^{-123}$, the $(8\pi)^2$ Planck-convention split, and $H_0 = 66.5$–67.1 km/s/Mpc (below Planck, far below SH0ES — the Hubble tension is *not* relieved). Ledger `GRAV.H0.01`.
- **Asymptotic Safety as a second QG route (`v207_asymptotic_safety`, `tfpt_4_frontier`, [O]).** A complementary continuum-FRG argument (a non-Gaussian UV fixed point from the same axioms; $y^2 = 16c_3^2 = 1/(4\pi^2)$) — a plausibility cross-check, *not* load-bearing, does *not* close G_{metric}. Ledger `GRAV.ASYMP.01`.
- **Black-hole thermodynamics from the seam** (`v208_bh_thermodynamics`, `tfpt_horizon_readouts`, [E]/[C]). Modular $T_H = \kappa/(2\pi)$ (the $2\pi = 1/(4c_3)$ seam unit, ties to `v198`/`v199`) and the scalaron-corrected Wald entropy $S_W = (A/4G)(1 + R_h/3M_s^2)$ (exact from `v28`). Ledger `HOR.BHTHERMO.01`; Wolfram-mirrored.
- **The black hole as a seam fixed point** (`v209_bh_regular_core`, `tfpt_horizon_readouts`, [C]). No singularity ($\varphi \to \varphi_\star$ at the gapped attractor $(2/3)^6$); the maximal BH = the anchor (Nariai roots $(1, 1, -2)$); information via Page recovery; a Planck-scale holographic floor. The superseded RN/torsion metric is *not* resurrected. Ledger `HOR.BHCORE.01`.

2026-06-15 (archive integration #4: Möbius ≅ Lorentz foundation citation)

- **The Möbius seam is the boundary Lorentz structure (`tfpt_3`, [O]).** Added a paragraph to the "Möbius origin of π" section: $\text{Möb}(\hat{\mathbb{C}}) \cong \text{PSL}(2,\mathbb{C}) \cong \text{SO}^+(3,1)$ (Penrose & Rindler, *Spinors and Space-time*, Vol. 1, §1.2–1.4), the celestial sphere of null directions via stereographic projection. This gives the self-consistency loop a spacetime meaning — the carrier clock $z \mapsto iz$ is an elliptic $\text{PSL}(2,\mathbb{C})$ rotation (`v180`) and the RP→OS causal cone is the same Lorentz structure — so "Möbius origin of π" and "one Lorentzian cone" are two faces of one boundary $\text{PSL}(2,\mathbb{C})$. A foundation citation, not a new claim (π stays primitive).

2026-06-15 (archive integration #3: axion coupling $g_{a\gamma\gamma} = -4c_3$ + retired small-angle reading)

- **Haloscope coupling tied to c_3 (`tfpt_4_frontier`, [C]).** Made explicit: in the determinant-line normalization the axion–photon anomaly coefficient is $g_{a\gamma\gamma} = -4c_3 = -1/(2\pi)$ ($y^2 = 16c_3^2 = 1/(4\pi^2) \approx 0.0253$), so the haloscope $F\tilde{F}$ coupling is set by the *same* c_3 that fixes α and $\beta_{\text{rad}} = \varphi_0/4\pi$ — a [**C**] structural relation (the coefficient, not a parameter-free physical $g_{a\gamma\gamma}$). The earlier small-angle $\theta_i = \varphi_0$ reading (old determinant-line note, a much higher axion mass) is *superseded* by the closed near-hilltop $\theta_i = \pi(1 - \varphi_{\text{seam}}) = 170.4°$, recorded as

a retired reading and guarded by `v188` (a new sentinel forbids that stale axion mass from re-entering the active docs).

2026-06-15 (archive integration #2: the EHT achromatic polarization intercept — v203)

- **`HOR.EHT.01` (`v203`, [E]/[C]/[X]).** The surviving core of the old UFE/black-hole notes (*not* the RN-type metric — superseded by the Nariai/seam=horizon readouts `v101`–`v104`/`v190`; only the polarization signature survives): a local achromatic polarization-rotation intercept around a horizon-scale black hole, $\beta_{\rm BH}(r) = 16c_3^4\, Q_e^{\rm eff} Q_m^{\rm eff}/r^2$. The coupling $16c_3^4 = 1/(256\pi^4) = \delta_{\rm top}/3$ is *exact* ([**E**], Wolfram-mirrored) — the *same* top-form coefficient $\delta_{\rm top} = 48c_3^4$ that fixes the α-kernel precision-zone correction, one row from the cosmic-birefringence seed $\beta_{\rm rad} = \varphi_0/4\pi$. TFPT fixes the $1/r^2$ shape, achromaticity and sign-flip ([**C**]); the amplitude Q_eQ_m is an MHD/GR weight, never a pixel prediction. **Dated kill test** [**X**]: the residual intercept must pass three nulls (frequency, spatial $1/r^2$, sign-flip) after honest GRMHD subtraction. Integrated into `tfpt_horizon_readouts` (new EHT section), the ledger, registry, Wolfram (243/243), and the website (`predictions.ts`). Suite 202 modules.

2026-06-15 (archive integration #1: the rare kaon sector $K \to \pi\nu\bar{\nu}$ — v202)

- **`FR.RAREKAON.01` (`v202`, [C]).** The closed TFPT CKM point (s_{12}=λ_C, s_{23}=$\varphi_0/(1+\lambda_C)$=$|V_{cb}|$, s_{13}=$\lambda_C^3/3$, δ=$\pi/3+3\lambda_C^2$=1.19823 rad; `v18`/`v88`, no flavor fit) feeds the cleanest FCNC probe. With standard external short-distance input (Brod–Gorbahn–Stamou $X_t, P_c, \kappa_\pm$) it gives $\mathrm{BR}(K^+ \to \pi^+\nu\bar{\nu}) = 9.45 \times 10^{-11}$ and $\mathrm{BR}(K_L \to \pi^0\nu\bar{\nu}) = 3.33 \times 10^{-11}$ — a [**C**] downstream flavor readout, *not* a compiler power (the SD functions are external). Recomputed from the *current* CKM point (an earlier archive scaffold quoted $\mathrm{BR}(K_L) = 3.47 \times 10^{-11}$ from an $\Im(\lambda_t)/\lambda^5$ slip, corrected here). Grossman–Nir ratio $0.352 \ll 4.3$; NA62 $(13.0^{+3.3}_{-3.0}) \times 10^{-11}$ is $\sim 1.2\sigma$ above. **Dated kill test** [**X**]: a stable NA62 $\mathrm{BR}(K^+)$ outside $[7, 12] \times 10^{-11}$, or a KOTO-II $\mathrm{BR}(K_L)$ off the GN point, breaks the flavor bridge (core untouched). Integrated into `tfpt_2_standard_model` (§rare kaon), the ledger, `v187` guard (new `FR.RAREKAON.` family), registry, and the website (`predictions.ts`). Python-only. Suite 201 modules.

2026-06-15 (bedrock: the $\omega \circ \rho = \omega$ residual reduced once more — v201)

- **`QGEO.SUBPRIN.01` (`v201`, [E]/[O]).** The `v199` bedrock residual — "the bounded sub-principal symbol of the raw seam DtN has no off-character (mod 4) matrix elements" — is reduced one genuine step further: block-diagonality is *not* an independent postulate. Writing the DtN as $\Lambda = |k| + M_f$ with M_f the bounded sub-principal piece = multiplication by a curvature $f(\theta)$ (standard Steklov structure), M_f is μ_4-character-block-diagonal iff f is $\mathbb{Z}_4$-invariant ([**E**]; principal $|k|$ already commutes, `v198`); and a μ_4-mark sum $f = \sum_{j=0}^{3} g(\theta - 2\pi j/4)$ has $f_m = 4g_m[m \equiv 0 \bmod 4]$ for any profile g, hence is automatically $\mathbb{Z}_4$-invariant ([**E**]). So the μ_4-mark orbit (forced by `v195`) *forces* the sub-principal symbol block-diagonal; 3 marks ($\mathbb{Z}_3$) and 4 generic marks break it (negative controls). The residual collapses to "the DtN sub-principal symbol is mark-local (seam flat away from the μ_4 marks = conformal deck, `v177`)", structurally definitional — the narrowest form yet of `QGEO.SYM.01`, [**O**]. Registered in `run_all.py`, the ledger, `tfpt_research_contracts`, `origin_theory`; Wolfram-mirrored (the mark-sum Fourier identity is exact; extension now 242/242). Suite 200 modules.

2026-06-15 (F_transfer workshop: the axion relic solver — an honest overproduction tension)

- **Axion relic solver (`experiments/ftransfer/axion_relic/relic_solver.py`, [C]).** The first real $F_{\rm transfer}$ pipeline computation (work package C / review §6.2): a standard misalignment relic with a numerically-computed anharmonic factor $F(\theta_i) \approx 4$ at the hilltop. Honest

finding: at the *predicted* $\theta_i \approx 170°$ the estimate *over*-produces dark matter, $\Omega_a h^2 \sim 0.6$–2.3 (robustly > 0.12), because the large $\theta_i^2 \approx 8.8$ times F overshoots. So the determinant-line axion as the *dominant* DM is in *mild tension* (it would prefer $\theta_i \sim 120$–$130°$ or extra dilution) — this *refines* `v185`'s optimistic "can be all-DM" toward an over-production tension. The near-hilltop regime is O(1)-uncertain, so a full finite-T solve decides; over-production kills only the axion-as-dominant-DM *branch*, not the compiler. Reflected in `tfpt_4_frontier` (dark-matter section) and the `v185` docstring; experiments-only, not a suite/ledger claim. No refitted exponents; $\theta_i = \pi(1 - \varphi_{\text{seam}})$ only.

- **Full finite-T solve confirms it (`axion_relic/full_finiteT_solve.py`).** A converged nonlinear misalignment solve ($\theta'' + (3 + \mathrm{d}\ln H/\mathrm{d}N)\theta' + (m_a(T)/H)^2 \sin\theta = 0$, lattice $\chi(T) \propto T^{-8.16}$, realistic $g_*(T)$; normalised so $\theta_i{=}1 \to \Omega_a h^2 \approx 0.03$, the standard value) gives $\Omega_a h^2 \approx 0.66$ at the predicted $\theta_i \approx 170°$ — $\sim$5.5$\times$ above $\Omega_{\text{DM}} = 0.12$, reached only at $\theta_i \approx 106°$. The over-production is thus *confirmed* (not the optimistic "all-DM"); `tfpt_4_frontier`, `v185` docstring and the website updated accordingly. Stays **[C]**; kills only the axion-as-dominant-DM branch.

2026-06-14 (work packages A+B: attacking $\omega \circ \rho = \omega$ + the variational scan — v199, v200)

- **`v199_seam_state_invariance.py` [E]/[O] (claim `QGEO.STATE.01`, work package A).** Attacking $\omega \circ \rho = \omega$ directly on the *raw* seam. (1) **[E]** the setup is non-circular: ρ is the Coxeter element of $W(A_3) = S_4$ ($\rho^4 = 1$, `v117`), an automorphism of the raw CAR algebra (16 Majoranas) — both carrier-side, no seam geometry. (2) **[E]** for a quasi-free RP state $C_\Sigma = \frac{1}{2}(1 + \operatorname{sgn} H_1)$, so $\omega \circ \rho = \omega \Leftrightarrow [\rho, H_1] = 0$. (3) **[E]** $[\rho, H_1] = 0 \Leftrightarrow H_1$ is block-diagonal in the four μ_4-character classes $\{n{=}r \bmod 4\}$ (verified). (4) **[O]** the residual is sharpened to "the bounded sub-principal symbol has no off-character matrix elements" — a bounded, finite-character condition, not a diffuse geometry; not a closure. Cited in `tfpt_research_contracts`; Wolfram-mirrored.

- **`v200_seam_variational_scan.py` [C]/[O] (claim `QGEO.VARI.02`, work package B).** The discretised variational test: with the clock angle free, $E_{\text{fail}}(\theta)$ has its unique minimum at $\theta = \pi/2$ (the μ_4 clock $z \mapsto iz$) — order-4 *and* the three distinct characters $i, -1, -i =$ the $(1, 2, 3)$ grading; decoys $(0, \pi, \pi/3, 2\pi/3)$ all lose. A numerical sign that the variational principle selects μ_4; the full operator-valued minimisation stays the `v199` residual **[O]**. Cited in `tfpt_research_contracts`.

- *Also (this pass, `experiments/` only, not the ledger): the F_{transfer} four-pipeline scaffold (`experiments/ftransfer/`) with one CONTRACT per interface, and confirmation that the FRB preregistration (`frb_tfpt_v1.yaml`) and its status semantics already exist.* Suite now 199 modules, highest ID `v200`; Wolfram extension 241/241.

2026-06-14 (cracking the bedrock to a state-invariance: principal symbol + Tomita–Takesaki — v198)

- **`v198_modular_commutator_reduction.py` [O] (claim `QGEO.MODULAR.01`).** The decisive crack on the last residual. (1) **[E]** the DtN principal symbol $|k| = \sqrt{-\partial_\theta^2} = \operatorname{diag}(|n|)$ and the clock $\rho{:}z \mapsto iz = \operatorname{diag}(i^n)$ are both diagonal in the Fourier basis, so $[\rho, |k|] = 0$ *exactly* on all of L^2 (verified on 33 modes) — the leading-order commutation is free on the whole boundary, never the open part. (2) **[E]** by Tomita–Takesaki the modular operator is intrinsic to (M, Ω), so a state-preserving symmetry ($\rho\Omega = \Omega$) commutes with $\log\Delta \sim \Lambda_\Sigma$: hence $[\rho, \Lambda_\Sigma] = 0$ *follows from* $\omega \circ \rho = \omega$, using only the state + reflection (RP) and *no* conformal covariance — which **removes the v194 Bisognano–Wichmann circularity worry**. (3) **[E]** the finite H^1 block is already μ_4-invariant (`v177`). So the entire residual collapses to

the single *state-invariance* "the raw quasi-free seam state is μ_4-invariant" ($\omega \circ \rho = \omega \Leftarrow$ the Gaussian bulk measure is deck-invariant) — the maximally-operational, non-circular form of `QGEO.SYM.01`, **[O]**. *Not* a full closure (a foundational symmetry cannot be derived from nothing, like $c = \text{const}$), but the sharpest falsifiable form, with the circularity gone. Cited in `tfpt_research_contracts`; `OpenGates` updated. Suite now 197 modules, highest ID `v198`; Wolfram extension 240/240.

2026-06-14 (three review levers: Marks-via-Lefschetz, E_fail functional, RR→D5 — v195–v197)

- `v195_marks_lefschetz_character.py` **[E]/[O] (claim `QGEO.MARKS.02`).** The four seam marks are *forced* (not posited): $\rho{:}z \mapsto iz$ fixes only $\{0, \infty\}$, $H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$ (the A_3 exponents, $\mathrm{Tr}(\rho|H^1) = i + i^2 + i^3 = -1$) has no trivial component ⇒ no puncture at a fixed point, and $\mathrm{rank}\, H^1 = n - 1 = 3 \Rightarrow n = 4 \Rightarrow$ a single free μ_4 orbit (cross-ratio 2). So MARKS reduces from the geometric posit "D_4 marks" to the milder premise "H^1 carries the A_3-exponent rep" — less circular, still **[O]**. Cited in `tfpt_research_contracts`.
- `v196_seam_energy_functional.py` **[E]/[O] (claim `QGEO.VARI.01`).** The failure functional $E_{\text{fail}} = \|[\rho, \Lambda_\Sigma]\|^2_{HS} + \|\rho^4 - 1\|^2 + \|\Theta\rho\Theta - \rho^{-1}\|^2$, zero locus = the `QGEO.SYM.01` conditions; on the finite H^1 block it vanishes at the μ_4 config **[E]** and is > 0 for any μ_4-breaking perturbation, so μ_4 is a true minimiser. The full-operator minimisation is the `v194` Bisognano–Wichmann residual — a concrete discretisable *target* on the open bedrock, **[O]**. Cited in `tfpt_research_contracts`.
- `v197_rr_carrier_clifford_d5.py` **[E]/[C] (claim `ARCH.RRCAR.02`).** Hardens the Riemann–Roch carrier from g_{car} to D_5 itself: $\Lambda^{\text{even}}(C_{\text{car}}) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 16 = 2^{g_{\text{car}} - 1} = \dim S^+$, the $\mathfrak{so}(10) = D_5$ half-spinor — so D_5 is the Clifford spinor of the 5-dim mode space, closing $\mu_4 \to g_{\text{car}} \to D_5$. **[E]** arithmetic, **[C]** identification (as `v189`). Cited in `origin_theory`.
- All three are validated external-review levers (the axion-hilltop "$\pi - 3\phi_0$" proposal was *rejected* as a $\pi \approx 3$ coincidence, not built). Suite now 196 modules, highest ID `v197`; Wolfram extension 239/239.

2026-06-14 (subclaim 2 of ENERGY.02 tackled: raw-seam-DtN RP-definability — v194)

- `v194_raw_seam_dtn_rp.py` **[O] (claim `QGEO.DTN.01`).** The next hard lever — "is the raw RP-seam DtN definable from RP data alone?" (sub-claim 2 of `QGEO.ENERGY.02`) — was tackled, and the honest result is a *reduction*, not a closure. (1) **[E]** the commutator $[\rho, \Lambda_\Sigma] = 0$ closes on the finite cohomology H^1 ($\rho^* w_k = i^k w_k$, distinct μ_4 characters; `v177`). (2) **[E]** the *hinge* is met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, `v54`) on a quasi-free seam state (`v155`), so Λ_Σ is RP-definable without naming $\mathbb{P}^1 \setminus \mu_4$. (3) **[O]** the residual relocates: the full-L^2 commutation ⇔ Bisognano–Wichmann geometric modular flow of the quasi-free seam state (which must be intrinsic, else it re-circulates). So `QGEO.SYM.01` reduces from a definitional postulate to "intrinsic BW geometric modular covariance" — one notch sharper, still **[O]**; the last structural fog point sharpens, it does *not* close to **[E]**. Cited in `tfpt_research_contracts`; `OpenGates` updated. Suite now 193 modules, highest ID `v194`.

2026-06-14 (QGEO.ENERGY.02 energy-commutator + QFT-wording hardening — v193)

- `v193_qgeo_energy_commutator.py` **[O] (claim `QGEO.ENERGY.02`).** The non-circular sharpening of `QGEO.ENERGY.01`: the energy commutator $[\rho, \Lambda_\Sigma] = 0$ on the rank-8 Calderón polarisation, with three sub-claims — (1) **[E]** ρ is carrier-defined (Coxeter of $W(A_3) = S_4$, `v117`), not imported from the seam; (2) **[O]** the non-circularity hinge: Λ_Σ must be the *raw*

RP-seam DtN, not the μ_4 normal form; (3) **[E]** the commutator forces the $(1,2,3)$ grading on H^1 (`QGEO.COHOM.01`). Exact **[E]** rider (Wolfram-mirrored): the induced EH coefficient $k = (\ln m)/(12\pi)$ (`v150`) reproduces $c_3/2$ iff $\ln m = q(A_3) = \frac{3}{4}$ (*family*), not $q(D_5) = \frac{5}{4}$ (carrier, off by 5/3) — gravity is family-geometry-induced. A *proof target*; if sub-claim (2) holds, `QGEO.SYM.01` becomes an energy-rigidity theorem. Cited in `tfpt_research_contracts`. Suite now 192 modules, highest ID `v193`; Wolfram extension 236/236.

- **QFT-wording hardening (review point 7).** `tfpt_2_standard_model` §(4) was retitled to "Constructive QFT closure of the admissible physical sector", and its standalone opening over-claim replaced by an explicitly scoped sentence (the unprojected ambient metric measure is not constructed; G_{metric} frontier). The two superseded standalone phrasings were added to the `v188` prose guard so a frozen release cannot re-introduce them.

2026-06-14 (geometry round: Riemann-Roch carrier, Nariai bound, branch line, energy clock — v189–v192)

- `v189_riemann_roch_carrier.py` **[E]/[C] (claim `ARCH.RRCAR.01`).** The pair $(g_{\text{car}}, N_{\text{fam}}) = (5,3)$ are the two canonical invariants of *one* object, the four-point seam divisor $D = \mu_4$ on $\mathbb{P}^1$: the cycle side $\operatorname{rank} H_1(\mathbb{P}^1 \setminus \mu_4) = \deg -1 = 3 = N_{\text{fam}}$ (already `QGEO.COHOM.01`) and the function side $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg +1 = 5 = g_{\text{car}}$, matched by $\operatorname{rank}(D_5) = 5$. The carrier-as-mode-space reading is **[C]** (geometric, does not displace the over-determined $g_{\text{car}} = 5$). Cited in `origin_theory`; Wolfram-mirrored.

- `v190_nariai_entropy_bound.py` **[E] (claim `HOR.NARIAI.02`).** A variation-free one-line proof of the SdS entropy floor: $3(x^2+1) - 2\Phi_3(x) = (x-1)^2 \geq 0$, so $S_{\text{tot}} \geq \frac{2}{3} S_{dS}$ with equality iff $x = 1$ (the Nariai merge). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.

- `v191_universal_branch_line.py` **[C] (claim `HOR.BRANCHLINE.01`).** The affine map $q = \frac{7}{2} + \frac{N_{\text{fam}}^2}{2} m$ carries the SdS branch points $\pm 1/N_{\text{fam}}$ onto the flavor labels $\{2,5\}$. *Honestly typed **[C]*** — an exact *relabeling*, not a theorem (a decoy pair $\{3,4\}$ admits an equally exact map; the shared content is the known 2/3 ramification). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.

- `v192_qgeo_energy_clock.py` **[O] (claim `QGEO.ENERGY.01`).** An operator-checkable restatement of the bedrock: $\rho^* \Lambda_\Sigma \rho = \Lambda_\Sigma$ (the clock preserves the DtN energy form). *Honestly typed:* it *relocates* `QGEO.SYM.01` to a more falsifiable surface but does *not* eliminate it (plausibly equivalent unless ρ is independently defined and the invariance proven). Cited in `tfpt_research_contracts`. Suite now 191 modules, highest ID `v192` (v186 skipped).

- *All four are validated external-review proposals (`problem_b`); each is built with the honest typing determined on review — the branch line is recorded as an alignment, not a theorem, and the energy clock as a relocation, not a closure.*

2026-06-14 (Zenodo release 5.1 (rev 120) — F_transfer firewall, v184–v188)

- **Published to Zenodo.** Version `5.1 (rev 120)` of the ten-PDF document set was published as a new version of the TFPT deposit — record `20687564`, DOI `10.5281/zenodo.20687564`. This version carries the F_transfer-firewall round (`v184`–`v188`): the honest η_B anchored-Boltzmann test (sharper scenario, not a closure), the axion hilltop-relic typing, the ledger `v187` and prose `v188` guards, and the Route B/Koide/dark-matter wording fixes plus the P1/P2-vs-`QGEO.SYM.01` two-layer clarification. The uploader's `DEFAULT_RECORD_ID` was advanced to the new record for the next release. Suite 187 modules (highest ID `v188`; v186 skipped). `run_all.py` green, audit clean, website built.

2026-06-14 (F_transfer round: eta_B honest test v184, axion relic v185, discipline guard v187)

- `v184_etaB_anchored_boltzmann.py` **[C] (claim `FR.ETAB.03`).** An honest test of the external-review η_B proposal ($M_1 = M_R\varphi_0^4$, $\tilde{m}_1 = m_3/A_\Lambda$). Firewall verdict: *sharper scenario, not a closure.* The washout anchors plausibly ($\tilde{m}_1 = m_3/10 \approx 5$ meV, $A_\Lambda = 10$ atom, in the strong-washout window) **[C]**; but M_1 does *not* — M_R is the seesaw scale $\sim v^2/m_3$ and the nearest clean TFPT powers of $\bar{M}_{\rm Pl}$ both miss it (c_3-/φ_0-powers off $\sim$75%/$\sim$40%), so $M_1 = M_R\varphi_0^4$ merely relocates the free input. η_B stays **[C]**, the route viable but with one (not zero) free input. Cited in `tfpt_4_frontier`.
- `v185_axion_relic_solver.py` **[C] (claim `FR.DM.03`).** Honest misalignment estimate: the determinant-line axion $f_a = M_{\rm scal}/128 \approx 2.39\times10^{11}$ GeV ($m_a \approx 23.8\,\mu$eV) CAN be all dark matter, but the harmonic misalignment under-produces ($\sim$21% at $\theta \sim O(1)$), so the $\theta_i \approx 170^\circ$ hilltop anharmonic enhancement is *required* — where the relic is exponentially sensitive. Hence a **[C]** scenario, not a sharp prediction; the relic contract is a kill test for the BRANCH, not the theory. Cited in `tfpt_4_frontier`.
- `v187_ftransfer_laws.py` **[E] (claim `FR.TRANSFER.01`).** A CI-style firewall guard: the four continuous-transfer frontier observables (Koide, η_B, axion, m_p/m_e) are read from the ledger and required to carry a **[C]**/**[O]** marker — never a primitive **[E]** compiler prediction (exact sub-parts like the Koide 53/54 factor or $b_3 = -7$ may be **[E]**, the physical prediction never is). Records the four interfaces $F_{\rm pole}/F_{\rm Boltzmann}/F_{\rm relic}/F_{\rm QCD}$; prevents any future edit from quietly promoting a frontier number to a closed output. Cited in `tfpt_4_frontier`.
- `v188_frontier_wording_guard.py` **[E] (claim `AUDIT.WORDING.01`).** A prose sentinel (no new physics): `v187` protects the ledger *typing*, this guard protects the *prose*. It scans the 10 active documents and forbids five stale phrasings whose semantics earlier rounds superseded (the old **[E]**-typed Route B leptogenesis claim and its heavy-scale wording → v184 **[C]**; the old "open relic scale" dark-matter line → v185; the old "near, not exact" Koide line → v183; and the module-count/ID conflation), so a frozen Zenodo release can never re-introduce wording that was true yesterday and is half-true today. The same pass updated the `tfpt_4_frontier` Route B (to **[C]**), the Koide and dark-matter summary lines, and added the P1/P2-vs-`QGEO.SYM.01` two-layer clarification to the introduction. Suite now has **187 modules**, highest verification ID `v188` (v186 was intentionally skipped as redundant — m_p/m_e is already typed **[O]** as a QCD matching contract, `QCD.LAMBDA.01`).
- *Review framing points (P1/P2 as projections of the seam-deck postulate, $v_{\rm geo}$ as unit gauge, $G_{\rm net}$ a corollary of `QGEO.SYM.01`, grammar tuning = data) were validated as already implemented in the prior rounds; m_p/m_e is already typed **[O]** as a QCD matching contract (`QCD.LAMBDA.01`), so no redundant module was added.*

2026-06-14 (operator origin of the Koide 53/54 factor: the missing Sheet-Diamond corner, v183)

- `v183_koide_f_corner_transfer.py` **[E]/[C] (claim `FR.KOIDE.07`).** An external-review proposal, validated exactly: the Koide source→pole factor $\frac{53}{54}$ is not a bare arithmetic guess but an *operator* identity on the carrier, $u_{\rm pole}/u_{\rm source} = a^T(R+Q)\mathbf{1}/(2\,\mathbf{1}^T Ra) = 53/(2{\cdot}27) = \frac{53}{54}$. Here $\mathbf{1}^T Ra = 27$ is the $E_6{\times}A_2$ cubic family block ($248 = \dim E_6 + \det R + 2(\mathbf{1}^T Ra)N_{\rm fam} = 78{+}8{+}2{\cdot}27{\cdot}3$) and $a^T(R{+}Q)\mathbf{1} = 53 = 52{+}1$, where $F = R{+}Q$ is the *missing* Sheet-Diamond corner with $\det B_F = 52 = \dim F_4$ (`v80`). Hard negative control: $a^T M\mathbf{1}$ for $M \in \{R, Q, K, L\}$ is $\{32, 21, 35, 56\}$ — only the absent corner F gives 53. This upgrades `FR.KOIDE.03` from "near miss + conjecture" to "operator-sourced source→pole transfer": the *factor* is **[E]** (exact, mirrored in Wolfram, 232/232); the *mechanism* (the leptonic source→pole transfer reads the F corner) stays **[C]**, an $F_{\rm transfer}$ special case. The prediction itself stays **[C]**. Cited in `tfpt_4_frontier` (Koide section). Suite now 183 modules.

- **Review point 2 noted as already-present:** the claim that flavor and Nariai share one double-cover ClockFunctor (flavor rate $(2/3)^{2N_{\text{fam}}}$ vs gravity curvature $(2/3)^{N_{\text{fam}}}$, exponent ratio $|\mathbb{Z}_2|$, "one clock, two geometries") is exactly `HOR.TRISECT.01/v103` — a rediscovery of an existing result, no new module.

2026-06-14 (review recommendations 1/3/4 + the four concerns A–D mapped to the named residuals)

- `v182_reviewer_residual_map.py` **[E]/[C] (claim `REVIEW.MAP.01`).** A careful external review raised four concerns (A mapping $2D \to 4D$, B UV-gravity abstractness, C grammar-tuning/meta-look-elsewhere, D Koide Möbius flow). This module shows three of them *reduce*, by the same discipline as `v175`–`v181`, onto the *two already-named* residuals $\{$`QGEO.SYM.01`, $F_{\text{transfer}}\}$: A (Plücker-11 = QBL boundary count `v174` + holographic reduction), B ($\equiv$ `QGEO.SYM.01`), D (Möbius forced by $\text{Aut}(\mathbb{P}) = \text{PSL}(2,\mathbb{C})$ on the deck; the missing generator = F_{transfer}). Only C is genuinely *experiment-only* (frozen + minimal + over-determined grammar; residual falls to JUNO/CMB-S4). A unification of the concern *structure*, not a closure. Cited in `tfpt_5_redteam`. Suite now 182 modules.
- **Review recommendation #3 — a "Rosetta" translation lexicon** (website `RosettaLexicon`): TFPT's compiler-speak (seam, deck, readout, microcode, audit hull, gap, anchor, F_{transfer}, v_{geo}, `QGEO.SYM.01`) mapped onto standard QFT / mathematical-physics language (Wilsonian boundary CFT, orbifold deck, index/intersection number, RG irrelevance rate, renormalization condition, fundamental postulate) — a reading aid for mainstream physicists, not a new claim.
- **Review recommendation #1 — kill-tests up front** (website `TrustContract`): the frozen pre-data predictions ($\sin^2\theta_{12} = 0.3067$, JUNO; $r \approx 0.004$, CMB-S4) and the null model ($P \le 10^{-30.7}$) are now a prominent strip directly under the hero.
- **Review recommendation #4 — the bedrock as a postulate.** `QGEO.SYM.01` is now framed confidently as TFPT's *one fundamental postulate* (the role "c = const" plays in relativity), not a gap — in the `tfpt_research_contracts` box and the website "what is open" card; the achievement is that the whole theory is compressed onto exactly one sharply-located, falsifiable premise.

2026-06-14 (external-review response: CSV fix, G_net 3-level, QGEO.SYM.01 page, claim detox)

- **Ledger CSV bug fixed + guarded.** Three rows (`AX.P1.01`, `AX.P2.01`, `QGEO.ISO.01`) had *unquoted commas* in a status/external field (e.g. "a=(1,1,2)", "PSL(2,C)"), so a strict CSV parser read 11–12 columns instead of 10 — a real machine-readability defect in the single source of truth. All offending fields are now quoted (every row parses to exactly 10 columns), and `audit_sync.py` gained a `check_csv_wellformed` guard (ledger, registry, clusters, freeze must all parse to constant width) so it cannot regress.
- G_{net} **typed in three explicit levels** (website "What is open" card): (1) *algebra* **[E]** $(D_5)_1 \otimes (A_3)_1 \rtimes \langle(1,1)\rangle \cong (E_8)_1$; (2) *AQFT machinery* **[E]** (net existence + full-cone RP, CAR functor, `v175`); (3) *seam instantiation* **[O]** (`QGEO.SYM.01`). The target object's mathematics is closed; only the physical coupling of the raw seam is open.
- `QGEO.SYM.01` **now has its own one-page box** in `tfpt_research_contracts`: what is identified, why it is weaker than `CONF/ISO`, what follows automatically (the conditional theorem), and *what would falsify it* (genus > 0 double; non-deck transport; $N_{\text{fam}} \neq 3$); with the explicit conditional framing "*given* `QGEO.SYM.01`, the closure follows" rather than "the theory derives the seam geometry".

- **Claim-language detox.** "no free fundamental number" softened to "no free *dimensionless compiler dial* beyond the anchor and π (dimensionful scale + transfer physics stay typed)" (`origin_theory`, `introduction`, website); the site title softened from "the Standard Model Derived" to "the Standard-Model *Skeleton* Derived".
- **External-reviewer map** added to the website verification page: four boxes (exact kernel [**E**], conditional physics [**C**], open premises [**O**], kill tests [**X**]) so a reviewer sees the attack surface without reading 181 scripts. Suite unchanged at 181.

2026-06-14 (Zenodo release 5.1 (rev 114) — v175–v181 bedrock + figures + P1/P2)

- **Published to Zenodo.** Version `5.1 (rev 114)` of the ten-PDF document set was published as a new version of the TFPT deposit — record `20686284`, DOI `10.5281/zenodo.20686284`. This version carries the AQFT-closure-to-bedrock chain (`v175`–`v181`, the structural residual reduced to the single definitional premise `QGEO.SYM.01`), the two overview figures (`residual_chain`, `script_timeline`), and the P1/P2 anchor-reduction provenance on the axiom surfaces. The `README` and the Zenodo description were refreshed (181 modules), and the uploader's `DEFAULT_RECORD_ID` was advanced to the new record for the next release.

2026-06-14 (overview figures + P1/P2 reduction reflected on the axiom surfaces)

- **Two overview figures** (`make_figures.py`; PDF + website PNG). (i) `residual_chain` — the structural-residual reduction chain `v175`–`v181` as a descending staircase from "build a quantum-gravity measure" to the single definitional bedrock `QGEO.SYM.01`; embedded in `tfpt_research_contracts`. (ii) `script_timeline` — the eight-phase journey of the ∼181 scripts (foundations → SM readouts → seam= horizon → horizon/flavor geometry → R1–R5/premise-(A) → PyR@TE cross-checks → AQFT bridges → AQFT closure), what each did mathematically and physically; embedded in `introduction` ahead of the master script index, and both shown on the website verification page.
- **P1/P2 reduction now reflected on the axiom surfaces.** The reduction of the two axioms to the single parabolic anchor $a = (1, 1, 2)$ plus π (with $g_{\mathrm{car}} = 5$ an over-determined bootstrap fixed point — forced three ways, `v6`, Pascal-unique and Lean-formalised, `FORM.LADDER.01`) was already established (`ANCHOR.GEN.01/v23`, `ARCH.CORE.01/v53`) and stated in the papers/glossary, but the *axiom surfaces* still read as bare "declared inputs". Fixed: the ledger rows `AX.P1.01/AX.P2.01` now carry the anchor/bootstrap reduction and point to it, and the website dependency-graph nodes P1/P2 now show the anchor input ($c_3 = 1/(2e_1(a)\pi)$; $g_{\mathrm{car}} = e_2(a)$, bootstrap-forced) rather than "declared axiom". No marker change (still declared inputs), only honest provenance.
- Manifests refreshed (two new figures added to the `FIG` list). Suite unchanged at 181.

2026-06-14 (the final reduction to bedrock v181 + full non-changelog status sweep)

- **`v181_clock_is_conformal_symmetry.py` [E]/[O] (claim `QGEO.SYM.01`).** The isometry premise `QGEO.ISO.01` (`v180`) is weakened one more genuine step — to a conformal-*symmetry* premise — via *equivariant uniformisation* (Nielsen realisation; for cyclic groups Kerékjártó + uniformisation): a *conformal* automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form $z \mapsto iz$. Since the clock is the Coxeter element of $W(A_3) = S_4$ (`v117`), the μ_4 deck, and a deck transformation preserves the pulled-back conformal structure by definition, the residual collapses to the *definitional* identification `QGEO.SYM.01`: "the seam's conformal structure *is* the μ_4-deck structure of the carrier clock". This is bedrock — a

physical/definitional statement tying P2 to the seam's conformal geometry, *not* a finite computation and not reducible further without relabeling. The chain `REALIZE(v175)`→ ⋯ →`SYM.01(v181)` ends here; we stop honestly. Python-only. Suite now 181 modules.

- **Full non-changelog status sweep.** The current bedrock framing was propagated to every live status surface (not just the changelog): the `introduction` "latest reductions" prose, `tfpt_5_redteam` Target A, the `tfpt_research_contracts` central-theorem keybox, `origin_theory`; and on the website `OpenGates`, `VerificationDag`, `papers.ts` (Target A ×2, G_{net}), the `glossary` G_{net} entry and the `faq` live-residual answer — all now end at `QGEO.SYM.01`.

2026-06-14 (the conformal premise reduces to a milder isometry premise v180)

- `v180_clock_is_mobius.py` **[E]/[O] (claim `QGEO.ISO.01`).** Cracks `QGEO.CONF.01` one level further: the conformal-realisation premise is *replaced*, via three classical theorems, by a strictly *milder, non-circular* premise. (1) *Uniformisation* (Poincaré–Koebe): a genus-0 Riemann surface is conformally $\mathbb{P}$, so the genus-0 seam double (P1, `v179`) with its RP-metric conformal structure *is* $\mathbb{P}$ — the target is *produced*, not assumed; (2) *Kerékjártó* (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of S^2 is conjugate to a rotation, so the order-4 clock ($h(A_3)$=4) is topologically $z \mapsto iz$; (3) *isometry⇒ conformal⇒Möbius*: an orientation-preserving isometry of a Riemannian surface is holomorphic, $\mathrm{Aut}(\mathbb{P}) = \mathrm{PSL}(2,\mathbb{C})$; (4) an order-4 Möbius map is $z \mapsto iz$ (verified: projective order 4, fixed $\{0,\infty\}$, multiplier i, free orbit μ_4). Hence `QGEO.CONF.01` is *implied* by the milder premise `QGEO.ISO.01`: "the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric" — which does not name $\mathbb{P}/\mu_4$ (uniformisation produces them) and is the natural statement that the seam transport is metric-compatible (a holonomy preserving the RP inner product). That milder premise is the single, now milder, structural residual of the whole theory; its surface realisation from the raw seam stays honestly **[O]**. Cited in `tfpt_research_contracts`; Python-only. Suite now 180 modules.

2026-06-14 (the two obligations collapse to ONE geometric premise v179)

- `v179_conformal_realisation.py` **[E]/[O] (claim `QGEO.CONF.01`).** Investigating the two residual premises left by `v178` shows they are *not* independent — they collapse onto a single geometric premise (honest unification; the premise itself stays **[O]**). Marks's residual is not three independent assumptions: (1) genus-0 *is* P1 — the one-sided Gauss–Bonnet $c_3 = 1/(|\mathbb{Z}_2|\,2\pi\,\chi(S^2)) = 1/(8\pi)$ uses $\chi(S^2) = 2$ and $\chi = 2 - 2g \Rightarrow g = 0$, so "genus-0" is the same $\chi = 2$ that gives the "8" in c_3 (`v58`/`v73`); (2) the order-4 clock *is* the carrier — the Coxeter element of $W(A_3) = S_4$ (`v117`), order $h(A_3) = 4 = |\mu_4|$; (3) marks = one orbit is *forced* (the parabolic marks are not the two elliptic fixed points, a free order-4 orbit has exactly $4 = e_1(a)$ points). Kernel then *follows* from the same geometry (DtN symbol $|k|$, Lee–Uhlmann `v156`; $c = 8$ rigidity `v158`; the cusped-surface residual = $H^1 = \mathbb{C}^3 = N_{\text{fam}}$). So the *entire* structural residual of the theory is the *single* geometric premise `QGEO.CONF.01`: "the raw RP seam normal double is conformally $(\mathbb{P}, \mu_4)$ with the carrier clock as the order-4 Möbius automorphism". The two doors of `v177`/`v178` are one geometric realisation, left honestly **[O]**. Cited in `tfpt_research_contracts`; Python-only. Suite now 179 modules.

2026-06-14 (an honest attempt at the two obligations v178 — deeper reduction, not closure)

- `v178_marks_kernel_attempt.py` **[E]/[O] (claim `QGEO.REDUCE.01`).** An honest attempt at `QGEO.MARKS.01` and `QGEO.KERNEL.01`. These are genuine constructive-geometry/AQFT statements, *not* finite computations — neither is closed. What the module does is push each as far as computation allows: it closes the finite **[E]** core of each and reduces each to *one*

sharper premise. *Marks core* [**E**]: given a genus-0 double with an order-4 clock ρ (fixed points $\{0,\infty\}$) and reflection τ, the marks are forced to be a generic ρ-orbit $\{1,i,-1,-i\}=\mu_4$ (*not* the fixed points), distinct iff $b_1=4-1=3=N_{\text{fam}}$ (any collision drops b_1), with $\tau\rho\tau=\rho^{-1}$ giving a faithful D_4 of order 8 — so MARKS reduces to the single topological/clock premise. *Kernel core* [**E**]: on the 3-dim multiplicity-free H^1 the μ_4 characters $(1,2,3)$ are distinct, so by Schur a μ_4-equivariant operator is forced *diagonal* and completely determined by its eigenvalue per character; two such with the forced transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$ (`v162`) and the unique residue-normalised basis (`v177`) are *equal as operators*, not merely isospectral — the spectrum-vs-operator worry *vanishes* on the finite block. So KERNEL reduces to the single full-operator-reduction premise (principal symbol $|k|$, Lee–Uhlmann/`v156`; no drift `v158`). Net: neither obligation closed; both sharpened to milder premises with their finite cores [**E**]. Cited in `tfpt_research_contracts`; Python-only (Möbius/Schur symbolic + numeric). Suite now 178 modules.

2026-06-14 (QGEO proof tree v177 — the residual sharpened into two named obligations)

- **`v177_seam_marking_kernel.py` [E]/[O] (claims `QGEO.TREE.01`, `QGEO.MARKS.01`, `QGEO.KERNEL.01`).** A second external residual-class audit pointed out that the `v176` statement takes the marks/D_4 as a *hypothesis* — a near-circular trap. This module factors the central theorem honestly as REALIZE = MARKS·UNIFORM·COHOM·MODULE·KERNEL·NET and closes *four* nodes *constructively* (all validated symbolically before adoption): UNIFORM [**E**] — an order-4 Möbius map conjugates to $z\mapsto iz$, a non-fixed orbit scales to μ_4, and $\sigma\rho\sigma=\rho^{-1}$ for $\sigma: z\mapsto 1/z$ gives a faithful D_4 (order 8), so a genus-0 four-marked faithful-D_4 boundary *is* $(\mathbb{P},\mu_4)$ (strengthening `v168`'s cross-ratio to the full uniformisation); COHOM [**E**] — $\rho^*\omega_k=i^k\omega_k$, grading $(1,2,3)=A_3$ exponents; MODULE [**E**] — a *unique* μ_4-equivariant iso (multiplicity-free + residue normalisation; reflection $\omega_1\leftrightarrow\omega_3$, ω_2 fixed); NET [**E**] (`v154`/`v175`). The structural residual is thereby sharpened from one diffuse premise into *exactly two* named open obligations, neither a finite computation: `QGEO.MARKS.01` (the raw RP seam collar canonically produces the genus-0 four-marked D_4 boundary) and `QGEO.KERNEL.01` (the raw seam Calderón operator *is* the μ_4-equivariant free gapped contraction, as an operator). Cited in `tfpt_research_contracts` (the central-theorem keybox); Python-only (Möbius/cohomology symbolic; numbers mirrored via `v168`/`v137`/`v162`/`v154`/`v175`). Suite now 177 modules.

2026-06-13 (Seam Collar Realisation Theorem v176 — the one central proof obligation)

- **`v176_seam_collar_realisation.py` [E]/[O] (claim `QGEO.REALIZE.02`).** Following an external residual-class audit, the single remaining structural item is formulated as one central theorem and certified as a *reduction* (not a fabricated proof). *Theorem:* given an oriented one-sided RP seam collar with a sheet-odd Calderón involution, faithful D_4 marks, four parabolic marks with μ_4 decomposition and admissible c=8 data, its conformal boundary is Möbius-equivalent to $\mathbb{P}\setminus\mu_4$ with H^1 grading $(1,2,3)$, the Calderón contraction is the rank-8 K_Σ polarisation and the index-4 extension is $(E_8)_1$. The proof decomposes into six steps, *five already* [**E**]: geometry/uniformisation (`v168`), Calderón DtN symbol $|k|$ (`v156`), H^1 character = generation grading $(1,2,3)=A_3$ exponents (`v137`/`v168`), μ_4-equivariant contraction with gap $(2/3)^6$ (`v162`), AQFT closure $(D_5)_1\otimes(A_3)_1\rtimes\mu_4=(E_8)_1$ + full-cone RP all m (`v154`/`v175`). The whole structural residual thus reduces to the *one* open premise `QGEO.REALIZE.01`: that the physical seam collar's boundary *is* this $\mathbb{P}\setminus\mu_4$ — a constructive-geometry statement, left honestly [**O**]. Cited in `tfpt_research_contracts` (the central theorem keybox); assembly/reduction certificate, Python-only (its exact identities are mirrored via `v168`/`v156`/`v162`/`v154`/`v175`). Suite now 176 modules. *Audit note:* the metrology residual

the same review flags is already discharged — `v153` (No-Unit Theorem) is the metrology-line typing ($v_{\mathrm{geo}} \mapsto \lambda^{-1} v_{\mathrm{geo}}$, $1/G \mapsto v_{\mathrm{geo}}^2$, dimensionless data invariant), and the Koide $u \to \frac{53}{54} u$ source→pole transfer is already typed **[C]** (`FR.KOIDE.03`); neither is re-derived.

2026-06-13 (Zenodo release 5.1 (rev 105) — v174 + v175 + doc refresh)

- **Published to Zenodo.** Version `5.1 (rev 105)` of the ten-PDF document set (introduction, origin_theory, tfpt_1–5, horizon, research_contracts, changelog) was published as a new version of the TFPT deposit — record `20681451`, DOI `10.5281/zenodo.20681451`. This version carries the AQFT closure round (v170–v175): the seam Fock-space readings and the heavy-artillery discharge of net existence + full-cone reflection positivity to **[E]**. The `README` and the Zenodo description were refreshed to the current state (175 modules, Wolfram 116/116 + 231/231), and the uploader's `DEFAULT_RECORD_ID` was advanced to the new record for the next release.

2026-06-13 (the heavy artillery v175: net existence + full-cone RP discharged to [E])

- `v175_net_existence_full_cone.py` **[E]/[O] (claim `QFT.NETRP.01`).** The two structural residuals that the premise-(A) chain had relocated to *already-open* items — A_2 net existence and full-cone reflection positivity — are now *discharged to a theorem*, leaving the seam realisation as the single irreducible open premise (nothing fabricated). (1) *Full-cone RP for all m*: the CAR second-quantisation functor $\Gamma(t) = \bigoplus_m \Lambda^m(t)$ lifts the one-particle seam contraction t to the complete Fock space $\Lambda^\bullet(\mathbb{R}^{16})$ ($\dim 2^{16} = 65536$); by the exterior-power spectrum theorem every eigenvalue is a subset product of $\mathrm{spec}(t)$, so all 2^{16} products lie in $[0,1]$ (a contraction = full-cone RP), with eigenvalue-1 multiplicity $2^8 = 256$ (wedges of the rank-8 K_Σ polarisation) and sub-leading = the one-particle gap $(2/3)^6$ — so full-cone RP reduces *for every m* (not only $m \le 6$, `v161`) to the one-particle contraction (Shale–Stinespring), discharging the `v160` scope premise. (2) A_2 *assembled + verified*: the $(E_8)_1$ character $E_4/\eta^8 = 1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + 1057504q^5$ (level-1 = 248, extends the order-4 `v156` check by one order), the embedding $248 = 120 + 128$, and the E_8 Cartan matrix even with $\det 1$ (the unique rank-8 even unimodular lattice, `v83`); the net *exists* by cited theorems (free-fermion net, lattice VOA, index-4 Q-system, `v125`). (3) Both then need the *same* input — the seam realisation `QGEO.REALIZE.01` (finite half proven, `v168`) — which is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly **[O]**. **Net:** net existence and full-cone RP become **[E]**; the one irreducible structural premise is the seam realisation. Cited in `tfpt_research_contracts` (premise (A) closure) and `tfpt_5_redteam` (Target A); the exact parts (Cartan unimodular, character order 5, functoriality integers) mirrored in the Wolfram extension (231/231). Suite now 175 modules.

2026-06-13 (seam Fock-space readings v174: CAR cone, Witten index, g-theorem)

- `v174_seam_fock_readings.py` **[E] (claim `QFT.FOCK.01`).** Three more exact audit bridges from the same source note. (1) *CAR cone compression*: the even CAR algebra of the 16 seam Majoranas has $\dim \Lambda^{\mathrm{even}}(\mathbb{R}^{16}) = 2^{15} = 32768$ directions, but a quasi-free Bogoliubov transport reduces the whole cone to a 16-dim one-particle problem ($2^{15}/16 = 2^{11}$) — the explicit count behind `v161`. (2) *Witten index = the Plücker norm* 11: the four μ_4-corner fermions span a $2^4 = 16 = \dim S^+$ Fock space; QBL (deg ≤ 2) keeps $\sum_{k\le 2} \binom{4}{k} = 11 = \dim S^+ - g_{\mathrm{car}}$, so $c_u/c_d = 55/117$ reads a topological state-space dimension. (3) *Affleck–Ludwig g-theorem*: $c_{UV} = 5 + 3 = 8 = c_{IR}((E_8)_1)$, so $\Delta c = 0$ forces an exact isometry — the boundary-entropy form of the `v157`/`v158` rigid fixed point. Cited in `tfpt_2` (Exterior Leg Lemma); mirrored in the Wolfram extension (230/230). Suite now 174 modules. (The categorical reframings in the source note remain framing, not asserted claims.)

2026-06-13 (CFT/QFT bridges: slice compression v170, OS moment v171, trace-anomaly seed v172, Pfaffian CP v173)

- `v170_diamond_slice_compression.py` **[E] (claim `E8.SLICE.01`).** E8 Slice Compression Lemma: the seven E_8 audit slices are *one* projection of two alphabets — anchor power sums $P = (3, 4, 6, 10)$ ($p_n = 2 + 2^n$) and Sheet-Diamond determinants $(3, 4, 8, 14, 20, 32)$ summing to $81 = N_{\text{fam}}^4$. All seven 248-slices, the K_4 edge graph, and the row-budget cross $(7, 11, 13) + s(3, 3, 3) + t(1, 2, 3)$ follow; $78 = \dim E_6 = p_2(p_0 + p_3)$. Cited in `tfpt_3` (slice atlas). Exact; reading is audit (anti-numerology).
- `v171_os_moment_cluster.py` **[E]/[C] (claim `QFT.OSMOMENT.01`).** Atomic OS Moment Lemma: $\text{spec}(T) = \{1, 64/729, 1/729\}$ makes every correlator a positive moment sequence, so the OS Hankel matrices factorise as Vandermonde Grams (a clean RP certificate); cluster 729/665, $\xi = 0.4111$. Sugawara Gap Safety: $31 = 1 + h^\vee(E_8)$, $c((E_8)_1) = 248/31 = 8$, $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - 31/(4\pi^2) > 0$. Cited in `tfpt_research_contracts` (G5).
- `v172_trace_anomaly_seed.py` **[E] (claim `SEED.ANOMALY.01`).** The seed prefactor $\frac{4}{3}$ is the halved $(E_8)_1$ Weyl anomaly: over the seam S^2 ($\chi = 2$) the $c = 8$ trace anomaly is $c\chi/6 = \frac{8}{3}$, halved by the one-sided $|\mathbb{Z}_2|$ to $\frac{4}{3} = 16/12$. Plus QCD $b_3 = 7 =$ scalaron exponent $48 - 41$ (confinement $\leftrightarrow$ inflation), and the Volkov–Wipf $-98/45 = -2 \cdot 7^2/\dim(D_5)$ factorisation. Cited in `tfpt_2` (seed).
- `v173_pfaffian_cp_car.py` **[E] (claim `FLAV.STRONGCP.02`).** Strong CP $\theta_{\text{eff}} = 0$ as Pfaffian reality in the self-dual 16-Majorana CAR model: $\{D, \Sigma\} = 0$ pairs the spectrum, γ_5-Hermiticity makes the Pfaffian real, and RP picks the positive branch. Cited in `tfpt_2` (Strong CP section).
- All four validated computationally before adoption; the exact identities (v170/v171/v172) are mirrored in the Wolfram extension (229/229). Suite now 173 modules. (The categorical reframings in the source note — compiler-as-single-functor, $E_8 = K_0$, GIT/monadic/Langlands — are recorded as framing, not asserted as machine-checked claims.)

2026-06-13 (Zenodo release 5.1 (rev 97))

- **Published to Zenodo.** Version `5.1 (rev 97)` of the ten-PDF document set (introduction, origin_theory, tfpt_1–5, horizon, research_contracts, changelog) was published as a new version of the TFPT deposit — record `20678568`, DOI `10.5281/zenodo.20678568`. The uploader's `DEFAULT_RECORD_ID` was advanced to the new record for the next release. Also fixed `build.sh zenodo` on macOS bash 3.2 (empty-array expansion under `set -u`).

2026-06-13 (QGEO rigidity theorem v168 + η_B leptogenesis interface v169)

- `v168_qgeo_rigidity.py` **[E]/[C] (claims `QGEO.RIGID.01`, `QGEO.REALIZE.01`).** Hardens the *finite* half of the one remaining Tier-1 hinge `GATE.QGEO` (the seam = flavour = horizon identification). Exact, machine-checked: $\mu_4 = \{1, i, -1, -i\}$ has cross-ratio 2 and a faithful Möbius stabiliser of order $8 = |D_4|$; $H^1(\mathbb{P}^1 \setminus \mu_4)$ has rank $4 - 1 = N_{\text{fam}} = 3$; and the forms $\omega_k = z^{k-1} dz/(z^4 - 1)$ ($k = 1, 2, 3$) are μ_4-eigenforms with characters χ_1, χ_2, χ_3 of weights $(1, 2, 3) = A_3$ exponents $= \text{Spec}(Q_+)$. So a genus-0 boundary with faithful D_4, four parabolic marks and this μ_4-decomposition is Möbius-equivalent to $\mathbb{P}^1 \backslash \mu_4$ — the QGEO Rigidity Theorem (`QGEO.RIGID.01`, **[E]**). The seam-collar realisation stays the one premise (`QGEO.REALIZE.01`, **[C]**). Mirrored in the Wolfram extension (226/226). Cited in `tfpt_research_contracts` (U0). *Not* a new structural class — it hardens the existing hinge (consistent with v167's completeness claim).
- `v169_etaB_boltzmann_interface.py` **[C] (claim `FR.ETAB.02`).** Operationalises the Tier-3 F_{transfer} leptogenesis path. Type-I thermal leptogenesis $\eta_B \sim 0.96 \times 10^{-2} \varepsilon_1 \kappa_f$ (sphaleron

28/79), fed by TFPT's normal-ordered neutrino spectrum and the predicted $\delta_{CP} = 240°$. Over natural $M_1 \in [3\times10^9, 3\times10^{10}]$ GeV and washout, η_B brackets the observed 6.1×10^{-10} (a canonical point gives 6.0×10^{-10}, untuned) — the route is *viable*. But M_1/washout are scenario inputs, so η_B stays **[C]**; if a precise solve excluded the window the leptogenesis *route* falls, not the theory (the downstream Ω_b readout is independent). Cited in `tfpt_4_frontier` (Route B). Python-only.

2026-06-13 (Higgs quartic from the free seam: SM near-criticality explained, v166)

- `v166_higgs_free_seam.py` **[E]/[C] (claim `HIGGS.FREESEAM.01`).** A new prediction tying the Higgs mass to premise (A). The seam UV is the free chiral c=8 fixed point (v158, Gaussian, no relevant/marginal interactions), so the one marginal SM scalar coupling vanishes there: $\lambda(M_{\rm seam}) = 0$ and $\beta_\lambda(M_{\rm seam}) = 0$ — the Shaposhnikov–Wetterich double-criticality, here *derived* from the free seam, not assumed. Integrating the PyR@TE-confirmed *two-loop* SM RGEs from M_Z up with the measured (m_H, m_t) gives $\lambda(\bar{M}_{\rm Pl}) \approx 0.002$ and $\beta_\lambda(\bar{M}_{\rm Pl}) \approx 0$: the famous SM near-criticality is a *consequence* of the free seam (2-loop sharpens $\lambda(\bar{M}_{\rm Pl})$ from $\sim$0.022 at one loop to $\sim$0.002). The double condition predicts $m_H \sim 129$–134 GeV; the same condition at the scalaron scale gives $\sim$107 GeV (too low), so the boundary condition lives at the reduced Planck scale (seam = horizon = Planck). Fills the Higgs-sector gap (was only $N_\Phi = 1$); cited in `tfpt_4_frontier`.
- PyR@TE-sourced: the reproducer's SM run supplies the two-loop β_λ, reduced to top-only (`verification/pyrate/sm_higgs_betas.txt`). Suite now 166 modules. Python-only (numerical 2-loop ODE).

2026-06-13 (PyR@TE $F_{\rm transfer}$ follow-ups: flavor scale-tagging v163, QCD scale v164)

- `v163_rg_stability_flavor.py` **[E]/[C] (claim `FLAV.RGSTAB.01`).** Scale-tagging of the flavor sector. The lepton-mixing seeds ($\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0}{2} = 0.3067$, $\sin^2\theta_{13} = \varphi_0 e^{-5/6} = 0.0231$) are *RG-stable*: the PMNS running is y_τ^2-suppressed (the only mixing-rotating Yukawa term is $\frac{3}{2}Y_eY_e^\dagger Y_e$, PyR@TE SM RGEs), so $\kappa_\tau = y_\tau^2/(16\pi^2)\ln(M_{\rm scal}/M_Z) \sim 1.8\times10^{-5}$ and $|\Delta\sin^2\theta_{ij}| \lesssim 10^{-4}$ — two to three orders below precision. Seam value = measured value; the y_t^2-driven quark mass ratios ($m_t/m_b \sim 41$) by contrast carry a scale tag. Cited in `tfpt_2` (solar/reactor angle box).
- `v164_qcd_scale.py` **[E]/[O] (claim `QCD.LAMBDA.01`).** The QCD confinement scale is parameter-free from the carrier $b_3 = -7$ (v159): running $\alpha_s(M_Z) = 0.1179$ down (two-loop, n_f thresholds) reproduces $\alpha_s(m_b) \simeq 0.22$, $\alpha_s(m_c) \simeq 0.38$ and confinement at ~ 0.5 GeV by dimensional transmutation — so the QCD-scale *numerator* of m_p/m_e is independently sourced. The $O(1)$ proton/Λ factor is lattice, so $m_p/m_e = 1836$ stays the honest "**[O]** — not forced" frontier ratio. Cited in `tfpt_4_frontier`.
- Both are PyR@TE-backed (the reproducer now also extracts the SM Yukawa β's, `verification/pyrate/sm_yukawa_betas.txt`). Suite now 164 modules. Python-only (no Wolfram mirror: magnitude bound / numerical running).

2026-06-13 (PyR@TE external cross-check of the $F_{\rm transfer}$ gauge inputs — v159)

- **New module** `v159_pyrate_gauge_crosscheck.py` **[E]/[C] (claim `EM.BUDGET.02`).** The one-loop gauge coefficients used by the transfer functor are confirmed by an *independent* third-party RGE generator. Feeding the carrier / Standard-Model field content into the textbook one-loop formula (GUT normalisation) gives $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$, and **PyR@TE 3** (`arXiv:2007.12700`, `github.com/LSartore/pyrate`) writes $\beta_{g_1} = (\frac{41}{10})g_1^3$, $\beta_{g_2} = -\frac{19}{6}g_2^3$,

$\beta_{g_3} = -7g_3^3$ verbatim (plus the full standard 2-loop matrix). So $b_1 = 41/10$ is pinned *three* independent ways — carrier algebra $10b_1 = g_{\rm car}2^{g_{\rm car}-2}+1 = 41$, the $U(1)$ hypercharge index $\sum \frac{3}{5}Y^2$, and PyR@TE — and the split $10b_1 = 40_{\rm (ferm.)} + 1_{\rm (Higgs)} = 41$ reproduces the EM-budget Gate 1 of `tfpt_2`. Cited there with `\veri`; the gauge sector of $F_{\rm transfer}$ is thus not a free knob.

- **PyR@TE reproducer (not vendored).** `verification/pyrate/reproduce.sh` clones PyR@TE on demand into `experiments/pyrate` (git-ignored), neutralises its interactive `pdflatex` end-command, runs the SM at one and two loops, and re-extracts the gauge β functions; the committed evidence is `verification/pyrate/sm_gauge_betas.txt`. PyR@TE is the right external tool for the transfer/RGE gates (gauge running, and — with a seesaw model — the neutrino-Yukawa running behind η_B); it is *not* a tool for premise (A) (a 2d boundary-CFT question). Mirrored in the Wolfram extension; suite now 159 modules.

2026-06-13 (terminal residual inventory: (A) no longer a structural class; one hinge + F_transfer + irreducible core, v167)

- **New module `v167_inventory_post_closure.py` [E]/[C].** The terminal residual inventory (line `v105`→`v123`→`v167`), re-pinned after the (A)-chain. With (A) discharged (`GATE.METRIC.16-19`) the `v123` *five* structural classes collapse: the seam clock (R1), the index-4 net (R2) and the Q-realisation (R5) all merge into *one* Tier-1 hinge — the seam=flavour=horizon identification = A_2 net existence (assembly of established net constructions) + `GATE.QGEO` (seam boundary = $\mathbb{P} \setminus \mu_4$, cohomology b_1=$N_{\rm fam}$=3); the EH step (R3) folds into $v_{\rm geo}$ (`v152`) and the ambient measure into $G_{\rm net}$ (`v76`). **Terminal structure:** Tier 0 = the irreducible core $\{\pi, v_{\rm geo}\}$ (a theorem, No-Unit `v153`); Tier 1 = one hinge; Tier 2 = folded; Tier 3 = $F_{\rm transfer}$, one functor with four typed conditional interfaces (Koide, η_B, axion, m_p/m_e), η_B carrying the most open computational room. Completeness: a sixth independent structural class — or a new irreducible — fails the script (ledger `SYNTH.INVENTORY.03`). A unification + re-typing, NOT an unconditional proof. Python-only (reads the ledger).

- **Bookkeeping.** Suite at 167 modules, all green; Wolfram extension unchanged at 224/224; `origin_theory` (residual-inventory keybox) and `next.txt` moved together.

2026-06-13 (premise A maximal honest closure: A2 by assembly, R1 = GATE.QGEO, zero independent open gates, v165)

- **New module `v165_premise_a_closure.py` [C]/[O].** Pins the two residual gates of the `v160`–`v162` (A)-chain to their floor. A_2 **(net existence) closes by assembly** of established mathematics [**C**]: the 16 free Majoranas give a free-fermion conformal net (Buchholz–Mack–Todorov), the holomorphic c=8 lattice net is $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the μ_4 extension is a Longo Q-system of index 4 (`v125`), $\mu(B)$=16/16=1, 248=120+128, character E_4/η^8=$j^{1/3}$ (`v156`); the one TFPT-specific input (seam Calderón kernel = free-fermion two-point function) is done via DtN = $|k|$ (`v156`/`v157`). R_1 **(seam clock) reduces to `GATE.QGEO` [C]**: a μ_4-equivariant flat gapped generator is *unique* (= $A_0^\star$, `v115`/`v116`), so the gap $(2/3)^6$ is forced and the residual is the one geometric identification "seam boundary = $\mathbb{P} \setminus \mu_4$" = `GATE.QGEO` (cohomology established, `v137`, b_1=3=$N_{\rm fam}$). **Final accounting:** by the No-Unit Theorem (`v153`) the irreducibles stay $\{\pi, v_{\rm geo}\}$; premise (A) = (A_2 assembly) + (R_1=`GATE.QGEO`) + the $\{\pi, v_{\rm geo}\}$ core, so it carries *zero independent open gates* and ceases to be its own gate — a unification and re-typing, *not* an unconditional proof (ledger `GATE.METRIC.19`). Python-only (numpy/sympy + stdlib).

- **Bookkeeping.** Suite at 165 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (the final identification: premise A factors into the already-open A2 + R1, no new open content, `v162`)

- **New module `v162_seam_transport_identification.py` [E]/[O].** Pushes the `v161` residual as far as it rigorously goes and shows it carries *no new open content*. The one-particle symbol splits into a gap value (β) and a fixed space (α). (β) **is forced:** a μ_4-equivariant generator is diagonal in the cusp basis (the deck average is the diagonal projection, $\sum_k U^k X U^{-k} = |\mu_4| \operatorname{diag} X$ for $U = \operatorname{diag}(1, i, -i)$; the commutant of U is exactly the diagonal algebra), so its spectrum is the cusp weights $\operatorname{spec}(A_0^\star) = \{0, \frac{1}{3}, \frac{2}{3}\}$ (`v115`) and the transfer eigenvalue $(1-\alpha)^{p_2}$ with $p_2 = 6 = |R_+(A_3)|$ gives $\{1, (2/3)^6, (1/3)^6\}$, gap $6 \log \frac{3}{2}$ — all carrier data; the lift to the 16 Majoranas is the A_3 subnet grading (`v137`). (α) is the rank-8 Calderón polarisation of $\omega_k = |k|$ (`v113`/`v156`). **Closure:** with `v160`+`v161`, (A) closes given only μ_4-equivariance + admissibility, whose two non-derived inputs are the already-open A2 (net existence, Kawahigashi–Longo) and R1 (seam clock, `HOR.CLOCK`). So premise (A) factors exactly into $A_2 + R_1$ — it adds *no* independent open item and is closed *conditionally* on them (a unification, not an unconditional proof; ledger `GATE.METRIC.18`). Python-only (numpy/sympy + stdlib).
- **Bookkeeping.** Suite at 162 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (the cone reduces to one particle: premise A's scope premise discharged to a 16-dim statement, `v161`)

- **New module `v161_one_particle_reduction.py` [E]/[O].** *Discharges* the scope premise of `v160` ("the transport is gapped on the *full* Schwinger cone") via Bogoliubov second quantisation. A quasi-free transport is $T_\Sigma = \Gamma(t)$, the second quantisation of a one-particle contraction t on the 16-dim Majorana space; $\Gamma(t)$ acts on the degree-m correlator sector as the exterior power $\Lambda^m(t)$. Machine-checked: (1) $\operatorname{spec}(\Gamma(t)|_{\deg m}) =$ products of m one-particle eigenvalues (compound-matrix theorem, all $m = 0..6$); (2) **the cone gap equals the one-particle gap** — the sub-leading eigenvalue over the whole cone is $(2/3)^6$ exactly, so premise 5 on the cumulant space *is* premise 5 on the 16-dim space; (3) the eigenvalue-1 sector is Λ^* of the fixed subspace (disconnected Wick powers), with uniqueness closed by `v113` (one polarisation $\Leftrightarrow$ one vacuum); (4) quasi-freeness is *forced* — the $120 = \dim \mathfrak{so}(16)$ chiral currents are the Bogoliubov generators and `v158` makes every non-Gaussian deformation irrelevant. **Consequence:** `v160`'s three full-cone premises collapse to ONE finite (16-dim) one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8 K_Σ, gap $(2/3)^6$); all numbers supplied, only the physical identification (Kawahigashi–Longo class = the A2 net residual) stays open — the infinite-dimensional cone is eliminated (ledger `GATE.METRIC.17`). Python-only (numpy + stdlib).
- **Bookkeeping.** Suite at 161 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (premise A as a conditional fixed-point theorem; the load relocates to one scope premise, `v160`)

- **New module `v160_seam_gaussianity_from_pf.py` [E]/[O].** Formalises the proposed closure of premise (A) ("the seam bulk is a reflection-positive free field") as a fixed-point theorem and types it honestly. The *exact* parts: (1) a quasi-free state has *all* higher connected cumulants $\kappa_{2n} = 0$ — verified by the *signed* set-partition/Pfaffian moment$\leftrightarrow$cumulant inversion on a concrete pure Majorana covariance (the `v113` "one kernel = net" property at the cumulant level); (2) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (3) the Perron–Frobenius dichotomy (a *fixed* κ_{2n} is a second fixed ray $\Rightarrow$ breaks uniqueness; a *non-fixed* one decays by the gap) forces Gaussianity; (4) the qualitative claim

needs only gap>0, the value $(2/3)^6$ only sets the rate. **The honest residual:** `v56`'s gapped PF uniqueness lives on the *three-dimensional* scalar transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$, while the 16-Majorana seam already carries $\binom{16}{4}$=1820 and $\binom{16}{6}$=8008 higher-cumulant directions — so (A) reduces to the single internal, exact-testable statement "the admissible TFPT seam transport is primitive + spectrally gapped on the *full* Schwinger cone, not only the readout spectrum of `v56`", which the suite does *not* yet establish (ledger `GATE.METRIC.16`). Python-only (numpy + stdlib); no new exact-identity content for the Wolfram mirror.

- **Bookkeeping.** Suite at 160 modules, all green; Wolfram extension unchanged at 224/224 (`v160` is numerical, Python-only); `tfpt_research_contracts` (the premise-A keybox) and `next.txt` moved together.

2026-06-13 (changelog date-order audit + reorder pass)

- **Changelog sort enforced.** `audit_sync.py` now checks that every dated `\subsection{YYYY-MM-DD...}` block in this file is non-increasing (newest first; range titles such as `2026-06-07 / 2026-06-08` sort by the later date). A manual prepend had left five `2026-06-12` entries above seven `2026-06-13` blocks — reordered; future mis-sorts fail `build.sh audit` as `[B.changelog]`.

2026-06-13 (premise A: the destination fixed point is provably stable, `v158`)

- **The free chiral c=8 fixed point is provably stable (`v158`, exact dimension count).** The finite core of (A)'s "reaches-the-fixed-point" residual: the chiral Majorana has h=$\frac{1}{2}$, the bilinear current h=1, the quartic h=2. The relevant window $0 < h < 1$ for bosonic deformations is *empty*, the h=1 currents are chiral (`v157`, not $(1,1)$-marginal), and the lowest interaction (quartic, h=2) is irrelevant — so the fixed point is isolated and stable against all interactions. Premise (A) reduces to a basin-of-attraction statement with a *proven* stable target (ledger `GATE.METRIC.15`). *Tooling note:* PyR@TE (4d gauge-Yukawa RGEs) is the wrong tool for this 2d boundary-CFT residual, but the right tool for the F_{transfer} interfaces (η_B leptogenesis RGEs, gauge-coupling running, the b_1=41/10 cross-check).
- **Bookkeeping.** Suite at 158 modules, all green; Wolfram extension 224/224; contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (premise A reframed: freeness is a rigid boundary fixed point, `v157`)

- **The free-bulk premise becomes a fixed-point property (`v157`, simplicity-first).** Instead of postulating Gaussianity, three facts click: (i) the DtN/Calderón symbol is *universally* $|k|$ (Lee–Uhlmann — order-1 ΨDO, principal symbol $|\xi|_g$; interactions live in the irrelevant lower-order symbol), so the free dispersion is bulk-detail-independent; (ii) a holomorphic c=8 theory is *rigid* — every field has $\bar{h}$=0, so no marginal $(1,1)$ field exists and the fixed point is isolated, with no knob for an interaction; (iii) the same $|\mathbb{Z}_2|$ that halves 8π in $c_3 = \frac{1}{8\pi}$ is the Ramond/GSO projection giving μ=1 (248=120_{NS}+128_{R}). With $(E_8)_1$ unique (`v83`), premise (A) shrinks from "prove the seam is Gaussian" to "the gapped (Δ=$6\log\frac{3}{2}$) boundary flow reaches the holomorphic fixed point" — freeness then forced by rigidity, not fine-tuned (ledger `GATE.METRIC.14`).
- **Bookkeeping.** Suite at 157 modules, all green; Wolfram extension 223/223; contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (Route 3: the seam net constructed and $(E_8)_1$ verified, `v156`)

- **The constructive route carried to its exact endpoint (`v156`).** "$B \cong (E_8)_1$" is no longer an assertion: the chiral seam net is built explicitly and the identification *checked*. (a) The

Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian is the free chiral dispersion $\omega_k = |k|$ (harmonic extension $e^{ikx-|k|y}$); (b) 16 free Majoranas give $c = 8 = 5{+}3$; (c) the weight-1 currents are $248 = 120_{\mathrm{NS}} + 128_{\mathrm{R}}$ ($SO(16)$ bilinears + Ramond spinor, Frenkel–Kac); (d) the holomorphic character $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + 34752q^3 + 213126q^4) = j^{1/3}$ (verified by q-series) — the 248 at level 1 is $\dim E_8$, so the net *is* $(E_8)_1$. *Residual:* exact given (i) the free-bulk premise (DtN = $|k|$, `v155`) and (ii) the operator-algebraic existence of the net from the seam kernel (a Kawahigashi–Longo-type theorem, not a finite computation) — where TFPT meets constructive QFT, now sharply located (ledger `GATE.METRIC.13`).

- **Bookkeeping.** Suite at 156 modules, all green; Wolfram extension 222/222 (the DtN = $|k|$ identity, the $(E_8)_1$ character $E_4/\eta^8 = j^{1/3}$ and the 248=120+128 split mirrored); contracts / redteam (Target A) / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (the seam-side identification reduced to one shared premise, `v155`)

- **G_{net}'s last premise reduces to a single shared physical premise (`v155`, Python-only).** The seam-side identification of the Simple-Current Extension Theorem (`v154`) splits (`v110`) into (a) "the Calderón involution is sheet-odd" (the double-cover deck, structural) and (b) "the seam 2-point kernel is quasi-free". Part (b) is *not* independent: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression $P\Gamma P$ of the covariance, and the compression of a contraction is a contraction — so the seam state is automatically quasi-free (Wick inherited). Hence (b) *follows* from "the seam bulk is a reflection-positive free field", the *same* premise already load-bearing in the area law (`v59`) and the EH mechanism (`v150`/`v151`). **The entire seam-side residual is therefore one physical premise**, shared by G_{net}, the area law and R3 — not reducible by a finite computation, but no longer three separate items (ledger `GATE.METRIC.12`).

- **Bookkeeping.** Suite at 155 modules, all green; Wolfram unchanged at 221/221 (`v155` is numerical/numpy, Python-only); contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (external review formalised: No-Unit Theorem + Simple-Current Extension Theorem, `v153`–`v154`)

- **v_{geo} re-typed: the No-Unit Theorem (`v153`).** Every load-bearing compiler datum is mass-dimension 0 and invariant under a unit rescaling $L \mapsto \lambda L$, whereas a mass / $1/G$ is not — so a dimensionless boundary compiler *provably cannot* select an absolute scale. v_{geo} is therefore an *irreducible metrology primitive*, not an open gap, and the three apparent residuals collapse onto it ($U_{\mathrm{point}} \sim v_{\mathrm{geo}}$, $1/G \sim v_{\mathrm{geo}}^2$, $m/\mu = e^{3/4}$ a pure ratio); the irreducibles stay {one unit, π} (ledger `ANCHOR.VGEO.02`).

- **G_{net} stated as the central closing theorem (`v154`).** The Simple-Current Extension Theorem: $A{=}(D_5)_1{\otimes}(A_3)_1$ extended by the isotropic glue $L{=}\langle(1,1)\rangle$ has $[B{:}A]{=}|L|{=}4$, $c{=}5{+}3{=}8$, $\mu(B){=}\mu(A)/|L|^2{=}16/16{=}1 \Rightarrow$ holomorphic $\Rightarrow B \cong (E_8)_1$ ($SO(16)_1$ excluded). Every algebraic ingredient is machine-checked (`v89`/`v92`/`v125`/`v143`/`v148`); the only residue is the seam-side identification. Pulled to the front of `tfpt_research_contracts`, `tfpt_5_redteam` (Target A final form) and `origin_theory` (ledger `GATE.METRIC.11`).

- **Status surfaces re-typed** (review): the canonical status card now reads v_{geo} as a metrology *primitive*, G_{net} as the simple-current extension $\cong (E_8)_1$ ([**E**]/[**C**]), and F_{transfer} as one functor with four typed interfaces — "not structural gaps". The final one-line framing (dimensionless boundary compiler; irreducibles $\{\pi, v_{\mathrm{geo}}\}$; one seam-net theorem; frontier = F_{transfer}) is in `origin_theory`, the introduction, the README and `next.txt`.

- **Bookkeeping.** Suite at 154 modules, all green; Wolfram extension 221/221; intro/README/`origin_theory`/contracts/redteam/status-card/website moved together.

2026-06-13 (currency sweep: remaining old-series "Paper N" cross-references removed)

- **No old-version framing left in the active docs.** A consistency sweep found a residual layer of old-series "Paper N" attributions and removed them: four literal "Paper 6" references (`tfpt_4_frontier` leptogenesis/axion, `tfpt_2` seam determinant), the old "Paper 4" (OS/admissibility) and "Paper 5" (APS/Hodge) references in `tfpt_2`, the old "Paper 3" transport-kernel attributions in `tfpt_2`, and the internal "Paper 1/2/4" cross-references in `tfpt_3`, `tfpt_4` and `tfpt_horizon_readouts` — all rewritten as content-based phrasing (the flavor sector, the architecture document, the E_8 decoder, the anchor characteristic polynomial, etc.). The active documents now attribute results to the current documents / axioms / scripts only, as the workflow rule requires; the machine-checked sync (scripts ↔ papers ↔ ledger ↔ website) was already current (152 scripts, AUDIT OK).

2026-06-13 (fifth round: the R3 normalisation is the dimensionful anchor, v152)

- **R3: the $q(A_3)$ normalisation merges into the one anchor (v152).** The replica EH coefficient is a *log-ratio* $k = \ln(m/\mu)/(12\pi)$ ($\ln m$ alone is scale-ambiguous, only m/μ is physical); pinning $k = c_3/2$ fixes the dimensionless ratio $m/\mu = e^{3/4}$, and in d=2 that ratio *is* the induced Newton constant $1/G$ (`v68`). So v_{geo} (`v78`), $1/G$ (`v68`) and m/μ are *one* dimensionful anchor in three readings — the irreducibles stay {one scale, π}, and `SEAM.THEOREM.01`'s residual reduces to the kernel-identification premise alone. *Audit (not promoted):* the log-ratio is numerically $\frac{3}{4} = q(A_3)$, the target value the anchor takes in seam units — not a derivation (ledger `SEAM.EHMODEL.03`).
- **Bookkeeping.** Suite at 152 modules, all green; Wolfram extension 219/219; intro/README/`origin_theory/next.txt`/website moved together.

2026-06-12 (Zenodo version label: semantic + git rev)

- **Release version string.** `build.sh` now writes `tex-artefacts/release-version.txt` as `{TFPTversion} (rev {git count})` (matches `\TFPTdatestamp` without the date). `scripts/zenodo_upload.py` and `build.sh zenodo`/`zenodo-publish` use this label on Zenodo; cursor rules extended for when `zenodo_description.html` must move with deposit-facing changes.

2026-06-12 (Zenodo API upload workflow)

- **Automated Zenodo versioning.** Added `scripts/zenodo_upload.py` and `.github/workflows/zenodo-release.yml`: creates a new draft version of record `20579275` (DOI `10.5281/zenodo.20579275`), uploads the ten active paper PDFs, refreshes `zenodo_description.html`, bumps the version from `tex-artefacts/version.tex`, and optionally publishes via GitHub Actions (`ZENODO_TOKEN` secret).

2026-06-12 (Overleaf shadow mirror: tfpt5-shadow)

- **One-way GitHub mirror for Overleaf sync.** Added `scripts/export-shadow.sh` and `.github/workflows/shadow-sync.yml`: on every push to `main`, a filtered copy is pushed to `sthamann/tfpt5-shadow` (papers, `tex-artefacts/`, `figures/`, `verification/`, `experiments/`; excludes `_archive/`, `website/`, all `*.pdf`, `.cursor/`). ∼270 files / ∼3 MB — within Overleaf GitHub-sync limits.

2026-06-12 (origin_theory style alignment + box reduction)

- **Fewer boxes, embedded in text.** `origin_theory` from 22 to 16 boxes: the narrative/interpretation callouts ("Why the interpretation is internally consistent", "Dependency chain of the birefringence test", "The precise reformulation", "The three theses honestly graded", "The whole story", "One-line summary") become run-in bold prose; the boxed reformulation becomes a `keyeq`. The exact result keyboxes (`v53`–`v56`, `v101`, `v105`, `v113`, ...) and the master-principle / Seam–Horizon target boxes are kept.
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline `vN` references; stale literal `[A]`/`[I]`/`[L]` markers in running text replaced by the 4-class **[O]**/**[E]**.

2026-06-12 (tfpt_5 / horizon / research-contracts pass + build hardening)

- **Forked document-set box removed (horizon).** `tfpt_horizon_readouts` carried a hand-written, *stale* copy of the document-set card ("five short documents", missing `tfpt_5`) *and* the canonical `\input{tex-artefacts/tfpt_docset}`; the fork is deleted so the single-source card (all five papers + three companions) appears once.
- **Fewer boxes, embedded in text.** Narrative/result callouts converted to run-in bold prose (with the `\veri{}` citation moved out of the bold title into the body): `tfpt_horizon_readouts` from 22 to 7 boxes (the ten-box "clock" wall becomes prose); `tfpt_research_contracts` from 19 to 10 (the nine progress/result winboxes become prose, the named-theorem keyboxes kept); `tfpt_5_redteam` from 10 to 9 (the outcome summary; the per-target verdict boxes are kept as the structured red-team output).
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline `vN` references in all three documents; the horizon "Scope/typing" marker line and a stale literal `[A]` in the research contracts updated to the 4-class markers.
- **Build hardening.** `build.sh` now removes a document's `.aux`/`.toc`/`.out` *on failure*, so a half-written cross-reference file from a crashed compile cannot poison the next build; artefacts are still kept on success.

2026-06-12 (fourth round: the Calderón transfer answered — the BFK split, `v151`)

- **R3: the Calderón/DtN kernel is conically clean (`v151`).** Reflection doubling derives that the Kac corner constant is boundary-condition independent ($C_N = C_D$), so by the Burghelea–Friedlander–Kappeler gluing ledger the conical deficit of the two-sided Calderón jump determinant *vanishes identically* ($C_{\mathrm{cone}}(\gamma) - 2C_D(\gamma/2) = 0$): the seam-reduced effective action inherits the `v150` EH form *verbatim*, with all of it localised in the *local* half-space determinants — the "Calderón version" demanded by `v150` needs no separate derivation. `SEAM.THEOREM.01` narrows to the $q(A_3)$ normalisation plus the standing kernel premise; the gate stays **[O]** (ledger `SEAM.EHMODEL.02`).
- **Bookkeeping.** Suite at 151 modules, all green; Wolfram extension 218/218; intro/README/`origin_theory/next.txt` moved together.

2026-06-12 (third round: both normals anchor data; the R3 mechanism exists, `v149`–`v150`)

- **R4′: both dual normals are anchor data (`v149`).** In the cusp frame the torsion normal pairs to $(6, 3, 5) = (p_2, p_0, e_2)(a)$ — one anchor invariant per Q_+ line (the cusp matrix is unimodular, so this determines n uniquely); with $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$ the dual normal pair is anchor data through and through. Equivalences to the $(2, 8, 121)$ frame data and the w_0-lift exact. R4′ residue = *one* discrete assignment (ledger `GATE.UWALL.14`).

- **R3: the mechanism exists at the gapped-model level (`v150`) — the first movement on `SEAM.THEOREM.01`.** The conical heat-trace deficit is exact and t-independent (Cheeger image sum $(N^2-1)/(12N)$); for a *gapped* operator the ζ-regularised replica variation is *finite and cutoff-independent*, $\Delta \log \det' = 2C(\gamma) \ln m$, and its linearisation is the Einstein–Hilbert form $(\ln m/12\pi) \int \sqrt{g}R$. The `v73` normalisation becomes one exact target equation: $k = c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$ (audit, not asserted). Remaining: the Calderón (boundary) version and that normalisation — the gate narrows but stays **[O]** (ledger `SEAM.EHMODEL.01`).
- **Bookkeeping.** Suite at 150 modules, all green; Wolfram extension 217/217; status surfaces (intro reductions + card, `origin_theory`, contracts, `next.txt`, website) moved together.

2026-06-12 (R1–R5 second round: every class at its floor, `v145`–`v148`)

- **R4′: the pairing values are atom identities (`v145`).** Reading the `v139` lift structurally ($n = w_0(\sigma) + |\mu_4| e_3$, $w_0 =$ the A_3 exponent duality), all three pairings become atom arithmetic — including the *new* closure $|\mu_4| + g_{\mathrm{car}} = N_{\mathrm{fam}}^2$ (4+5=9, $\sigma \perp w_0(a)$) and $\|\sigma\|^2 = 110 = |\mathbb{Z}_2| g_{\mathrm{car}} \|\mathrm{Pl}(K)\|_1$. R4′ = derive *one* map (ledger `GATE.UWALL.13`).
- **R5: the Möbius D_4 realisation (`v146`).** The *full* dihedral symmetry $\langle z \mapsto iz,\ z \mapsto 1/z \rangle$ of the seam curve acts on H^1 exactly as the integer model: ι^* is the exponent duality with parity $+1$ on the self-conjugate line ($= T_A$ in the cusp basis, glue-swap parity), and Σ is the $\delta\iota$ class — the premise reduces to the module identification itself (ledger `FLAV.QGEO.07`).
- **R1: the clock is a Gaussian zero-mode integral, and the bend is det′-clean (`v147`).** Variance ratio $(1-\alpha)$ (forced by norm = area) gives the Born-squared ratio $(1-\alpha)^{p_2}$, the ln series *is* the `v127` ring sum; the two quantum weights share *one* Nariai geometry, so the bend $\log_{3/2} 3$ carries no determinant correction — the `v144` obstruction is localised to the $\alpha = 0$ reference (ledger `HOR.CLOCK.13`).
- **R2: the NS/R sector census (`v148`).** The untwisted (Neveu–Schwarz) module carries only the *even* discriminant classes (integer weights), so the odd μ_4 simple currents have zero NS support — they are Ramond modules; the R zero-mode module (256) splits 128+128 into the odd sectors of the *two* Lagrangian glues: choosing the glue *is* choosing the R-projection ($248 = 120_{\mathrm{NS}} + 128_{\mathrm{R}}$); the one remaining net statement is irreducibly twisted-sector (ledger `GATE.METRIC.10`). *Correction note (same day):* the first version of `v148` graded the Ramond labels by the sum $q_D + q_A$ (not the glue label) and called that module untwisted; the census was rewritten as above — conclusion unchanged, support corrected, the R-sector sheet split is new.
- **Bookkeeping.** Suite at 148 modules, all green; Wolfram extension 215/215 (README + `GATE.WOLFRAM.02` + website counter in lock-step); status surfaces (intro card, `tfpt_1`, `origin_theory` third update, `next.txt`) moved together.

2026-06-12 (external review integrated: Λ typing, marker migration, website parity)

- **Λ normalisation made unambiguous.** The external review flagged the adjacent reduced/unreduced displays in `origin_theory` as a type trap (the formulas were individually correct — $\bar{M}_{\mathrm{Pl}}$ vs M_{Pl} — but easy to misread). Both normalisations are now shown explicitly side by side with their order counts ($3/(4\pi^2) \Rightarrow 120.147$ reduced; $3/(256\pi^4) \Rightarrow 122.948 = 119.028 + 3.920$ unreduced) plus a typing note.
- **Stale references cleaned.** "How each of the seven papers simplifies" → "How the legacy layers simplify (bridge map)" with the lead-in "(Paper 6)" reference replaced by the scalaron-reheating chain (`v86`); "(Paper~6/7)" in the `tfpt_3` bridge table reworded; the `tfpt_1` "single remaining geometric input (H2)" passage and the `tfpt_2` "single remaining input (U)" passage

carry dated status updates pointing at the current decomposition (v118–v122, v139, v141, v142, v75).

- **Marker migration completed in the shared figures.** The claim stack, anchor cockpit, E_8 glue, 2/3 motif and K_5 figures (tex-artefacts/tfpt_figures.tex) now show the four public classes **[E]**/**[C]**/**[O]** instead of the legacy $[A]$/$[I]$/$[L]$/$[B]$/$[P]$/$[R]$ badges.
- **Red-team front table pulled to the final state.** The Target-A residual column now states the one remaining statement (boundary-net holomorphy + c=8 $\Leftrightarrow$ index-4 inclusion; v83/v87/v89, Lie-level realisation v143); the historical three-residual form is kept below, dated.
- **Website parity.** The Red Team document is now a first-class entry (papers.ts number 5, slug redteam) and the changelog PDF a download row; Appendix H / Origin Theory / Research Contracts are labelled *companions* (never "Paper 5–7"); the marker system on every website surface migrated to $[E]$/$[C]$/$[O]$/$[X]$ (fine types named in prose; the script index keeps quoting the scripts' internal fine-grained typing); the verification page counters are live (144 checks generated from the registry via lib/suite.ts, Wolfram 116/116 + 211/211 — now guarded by the sync audit) and the reproduce block matches the current pipeline. The papers.ts section bodies of the affected papers carry the v141–v144 outcomes (sheet question: deck derived; G_{net}: Lie-level realisation; resummed clock: in-family det-ratio cancellation; selector triangle: two line pairings).
- **Not adopted** (recorded for transparency): the review's narrative reframings (anchor-algebra trilogy, "one seam clock" framing, one-page structure) and the front-matter/status-card layout changes are editorial decisions left to the authors; the claimed Λ *error* was in fact a correct-but-treacherous notation switch (see above).

2026-06-12 (R1–R5 attack round: four residual classes sharpened, v141–v144)

- **R5 closed at the discrete level (v141, deck selection theorem).** The v140 $\mathbb{Z}_3$ choice is *derived*: the established integer deck $G = T_A\Sigma$ (v97/v98) pairs the Q_+=1 cusp line with the self-conjugate character 2, so only the sheet-twisted assignment $(2,3,1) = \mathrm{chars}(-U)$ survives; the cusp exponential i^{Q_+} is excluded (same spectrum, wrong grading pairing). Explicit conjugacy constructed; under $k \to -k$ only the established sheet $\mathbb{Z}_2$ flips. Consequence: Euler grading $= Q_+ \circ \rho$. GATE.QGEO keeps only its realisation premise — no discrete freedom left (ledger FLAV.QGEO.06).
- **R4′ narrowed: three pairings → two (v142, frame integrality).** Integer covectors in the $(\mathbf{1}, a, \sigma)$ dual frame form an index-11 sublattice ($x-3y+z \equiv 0 \bmod 11$; the index *is* $\|\mathrm{Pl}(K)\|_1$); with $(x,y) = (|\mathbb{Z}_2|, \mathrm{rank}\, E_8)$ the σ-pairing is forced $\equiv 0 \bmod 11$, and primitivity forces the line parameter even. Cramer reductions for both line pairings recorded as restructuring (ledger GATE.UWALL.12).
- **R2 realised at the finite Lie level (v143, graded Frobenius).** The $\mathbb{Z}_4$-graded hull carries exact coset duality ($-C_1 = C_3$; Killing pairing nondegenerate per $g_k \times g_{-k}$), the glue average is exactly the carrier projector (index 4), and each glue sector is one Weyl orbit (simple currents, fusion $= \mathbb{C}[\mathbb{Z}_4]$) — the v125 Q-system realised by the hull; residue = one conformal-net statement (ledger GATE.METRIC.09).
- **R1's det-ratio step derived in-family (v144).** e_2-rigidity of the traceless SdS cubic gives $r_b r_c = 1 - \Delta^2/3$ exactly; with the v132 anomaly the non-zero-mode determinant ratio is $(1-\Delta^2/3)^{4/3}$ — no first-order term in the horizon split (the v131 cancellation derived, not assumed); deficit coefficient $\frac{32}{27} = |\mu_4|(\frac{2}{3})^3$ (audit). Finite-weight absorption stays **[C]** with the $6 \to \frac{14}{3}$ obstruction stated sharply (ledger HOR.CLOCK.12).
- **R3 attempted, honestly unchanged.** No defensible computation landed for the seam-determinant→EH step (SEAM.THEOREM.01); the gate keeps its **[O]** typing untouched.

- **Wolfram mirror extended** to v141–v144: extension file now 211/211 (GATE.WOLFRAM.02 and the wolfram README updated in lock-step); Python suite at 144 modules, all green.

2026-06-12 (sync infrastructure: single-source script index, content maps, one audit)

- **Single-source script index.** The master script→check table (tex-artefacts/verification.tex) and the website ScriptIndex.tsx are now *generated* from one registry (script_registry.csv + script_clusters.csv in verification/) by make_script_index.py; both targets carry a generated-file header and are never edited by hand.
- **Content maps without splitting the papers.** verification/make_docs_map.py generates docs_map.csv (every paper section with line range, the vN scripts cited there, a content hash and a last-changed date) and website_map.csv (which website file mentions which scripts/documents) — the machine-readable enumeration of all sync surfaces.
- **One sync audit.** verification/audit_sync.py bundles and extends the previous loose shell checks, now in *both* directions: suite ↔ run_all.py ↔ registry ↔ ledger; every script cited in a paper *body* (the master index alone no longer counts); no stale \veri/\vref targets; every new module mentioned in this changelog; website mirror byte-identical (PDFs, scripts, release.ts hashes, version stamps); wolfram counts; overfull boxes. Frozen exceptions live in audit_baseline.json (remove-only).
- **Build pipeline.** build.sh gains gen / website / audit / release steps; notes now regenerates the single-source surfaces first; website copies all active PDFs (now incl. tfpt_5_redteam.pdf and changelog.pdf, with new release.ts entries) and the vN scripts, and stamps version, date and git revision into website/lib/version.ts (shown in the site header) and release.ts.
- **Citation gaps closed.** The audit immediately found three scripts cited only in the master index: v17 and v85 are now cited in their tfpt_2 sections (hexagonal resolvent; master cover), and two short \veri{v19} citations were expanded to the full script name. v91 (spine tetrahedron) had no paper section at all — it is now integrated: a keybox in tfpt_3 ("Three views of one spine": anchor closure, edge/face products, volume 120, K_6 negative control, the tautology/sub-grammar caveats) and a closing note in the tfpt_1 Spine Quotient Lemma; the ARCH.SPINE.01 ledger location is corrected to tfpt_1;tfpt_3 and the baseline exception list is empty again.
- **Rules.** .cursor/rules updated (tfpt-workflow, website-sync) and a new sync-maps rule added describing the agentic procedure: *regenerate* the maps first, enumerate the affected surfaces from them, audit as the exit gate. The mirror map (website_map.csv) also covers the root README.md and next.txt (bare vN ids resolved to full script names), and both files are named explicitly as status surfaces that move with every finding.

2026-06-12 (tfpt_3 / tfpt_4 style alignment + box reduction)

- **Shared header/footer on tfpt_3.** tfpt_3_e8_audit_bootstrap dropped its custom \footmarkers footer (the post-collapse four-[**E**] artefact) and inherits the shared tfpt_style header/footer with the version stamp; the box-orphan needspace guards are kept.
- **Fewer boxes, embedded in text.** Narrative/commentary callouts converted to run-in bold paragraphs (boxed one-line formulas to keyeq): tfpt_3 from 39 to 27 boxes (e.g. "Purpose and discipline", the five "(i)–(v)" audit-lemma boxes, "Net", "The connection in one sentence", "The unification", "The question and the honest answer", "Net: two axioms"); tfpt_4 from 18 to 7 (the seven-box Koide stack and the two dark-matter conjecture boxes become prose). The per-item "Honest status of . . . " keyboxes are kept — tfpt_4 is the status authority for the frontier.

- **Colour-marked module references.** `\vref` applied to the inline `vN` references in tfpt_3 and tfpt_4 (TikZ node names excluded).
- **Marker hygiene.** A stale literal `[P]` in tfpt_4 (Koide flow-time) replaced by the 4-class [**C**]; status tables checked against the ledger.

2026-06-12 (tfpt_1 / tfpt_2 style alignment + box reduction)

- **Shared header/footer everywhere.** `tfpt_1_architecture_e8` dropped its custom `\footmarkers` footer (which after the marker collapse showed four redundant [**E**]) and now inherits the shared `tfpt_style` header/footer with the version stamp, identical to every other document.
- **Fewer boxes, embedded in text.** The narrative/commentary callouts of both documents are converted to run-in bold paragraphs (and the boxed one-line formulas to the colour-marked `keyeq`): tfpt_1 from 21 to 14 loud boxes (+3 `keyeq`); tfpt_2 from 52 to 43 (e.g. "In one sentence", "The one-paragraph version", the four "Sharpening" notes, "What the simplification means net", "The disciplined main statement", "What improves…", "Scope and honesty"). Only genuine results / theorems / gates keep a box.
- **Colour-marked module references.** `\vref` now also colours the inline `vN` references throughout tfpt_1 and tfpt_2 (TikZ node names accidentally matching the pattern were excluded).
- **Status currency.** tfpt_2's inflation-pivot table gains the `v86` scalaron-reheating point ($N_\star$=51.4 $\Rightarrow$ n_s=0.9611, r=0.0045, A_s-disfavoured) and states the band-of-record explicitly, matching `COSMO.NSTAR.01` and the introduction; the box is re-typed [**C**]. The other status tables were checked against the ledger and are current under the 4-class markers.

2026-06-12 (marker simplification to 4 classes + status reconciliation)

- **Four-class display markers.** The reader-facing marker system is collapsed from eleven symbols to four: [**E**] exact / machine-proven (identity, Lie/lattice, formalised, numerical FP), [**C**] conditional (physical / bridge / readout under named hypotheses), [**O**] open / axiom, and [**X**] kill test. `tex-artefacts/tfpt_style.tex` defines `\mE/\mC/\mO/\mX` and keeps the legacy `\tI ...\tX` names as aliases that render their class. All marker call-sites across the nine documents and the shared fragments were rewritten, collapsing same-class combinations (e.g. `[I]/[L]` $\rightarrow$ [**E**], `[N]/[P]` $\rightarrow$ [**E**]/[**C**]). The fine-grained per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) is unchanged in `status_ledger.csv` — the single source of truth — so no fidelity is lost. The marker key, badge bar, README marker table and the workflow rule were updated to match.
- **Status tables reconciled against the ledger.** The introduction's inflation/$N_\star$ rows are made consistent with `v86/COSMO.NSTAR.01`: r and n_s are shown as the frozen *bands* over $N_\star \in [50, 60]$ with the scalaron-reheating conditional point $N_\star$=51.4 (n_s=0.9611, r=0.0045), replacing the stale point values $N_\star$=55 (predictions table) and $N_\star \simeq 57$ (residual card); the closure-ledger inflation row is re-typed [**E**] $\rightarrow$ [**C**](conditional on $M_{\rm Star}$=$M_{\rm scal}$ + reheating), matching the other two tables and the ledger `[P]` typing.

2026-06-12 (version system, predictions/K5, readability pass B3)

- **Authors, auto-date and auto-version on every paper.** New single source `tex-artefacts/version.tex` defines `\TFPTauthors` (Stefan Hamann, Alessandro Rizzo), a curated `\TFPTversion`, and a title/footer date stamp (`\today` + version + revision). `build.sh` writes the auto build revision (git commit count) into the gitignored `tex-artefacts/version-auto.tex`, so every document's title page and page footer carry

author, date and v5.2 (rev N) automatically. All nine active documents and changelog.tex now set \author{\TFPTauthors}.

- **Builder keeps build artefacts.** build.sh no longer deletes the .aux/.log/.out/.toc after a successful build (the .log is needed for the overfull-box audit and the .aux/.toc for correct cross-references on the next incremental build).
- **Predictions table outsourced.** The running/upcoming-experiment predictions table is moved from introduction.tex into the shared fragment tex-artefacts/predictions.tex (\input in the predictions section).
- **K5 figure fixed.** The \TFPTactionkfive mnemonic (K_5 on the five carrier slots) is redrawn so the complete graph and the Pascal-row reading list sit side by side instead of overlapping.
- **Colour-marked script references.** New \vref macro colours inline module references (e.g. v108–v113, v49/v71) in the introduction, predictions and verification fragments, matching the blue of the \veri citation, so machine-checked references are visible at a glance.
- **Lighter, non-breaking, embedded boxes.** The tcolorbox families are restyled to a light tint + thin frame + light title band with dark coloured title (no heavy white-on-saturated bars); a neutral navbox carries the document map and a colour-marked keyeq sets off load-bearing display formulas. The introduction's callouts use the non-breaking variants (keyboxnb/winboxnb/gapboxnb) so no box splits across a page, and the shared status card is non-breaking.
- **Readability / sectioning.** The dense run-in prose of the introduction is broken into \subsection* headings (In plain words; one anchor and E_8 as a compiler hull; the companion documents; the compact status formula; the hard reduction; the Pascal action grammar; the closure ledger) with added whitespace.
- **Orphan results linked to scripts.** The Glue theorem, the "Reduction in one line" and the fermion-spectrum callouts now carry inline \veri{} citations (v1, v6/v14/v23, v18/v20/v46).

2026-06-12 (tex-artefacts refactor + introduction visual pass B2)

- **Shared imports moved to tex-artefacts/.** The shared, imported-only TeX fragments (tfpt_style.tex, tfpt_figures.tex, tfpt_docset.tex, tfpt_status.tex) now live in a dedicated tex-artefacts/ folder; all nine active documents \input them via tex-artefacts/<name> (and tfpt_style loads tex-artefacts/tfpt_figures). changelog.tex stays at the repo root because it is also a standalone build.sh notes deliverable. The make_manifest.py TEX list and the workflow rule were updated to the new paths.
- **Master script index outsourced (tex-artefacts/verification.tex).** The long vN→check table is moved out of introduction.tex into the shared fragment tex-artefacts/verification.tex (\input in the verification appendix); the paper-consistency audit now also reads that file, so every registered script stays cited exactly once.
- **Introduction box reduction (B2).** The introduction's eleven keyboxes are reduced to the single headline "In one sentence" card; the explanatory keyboxes (*In plain words*, *Sharper still*, *compact status formula*, *hard reduction*, *Cosmology Pascal*, $E_6 \times A_2$, *closure ledger*, *Plausibility balance*) are converted to inline run-in paragraphs, the *Glue theorem* is re-typed as a winbox (a result, not a claim), and the explanatory *plausible/follows next* winboxes become prose. The loud boxes now carry only genuinely load-bearing callouts (one keybox, five winboxes, one gapbox) plus the two shared orientation cards.
- **Document-count consistency.** The introduction's document map gained the missing tfpt_5_redteam row; "seven/eight companion documents" and the ledger "six documents"

wording are corrected to the true count (nine active documents); `README.md` ("10 ok", 140 scripts) and the workflow rule ("9 docs") are reconciled.

2026-06-12 (visual pass: box reduction B1)

- **Box reduction (B1).** The 26 non-load-bearing aside boxes (audit / recorded / interpretation: `auditbox`/`audbox`/`intbox`/`resbox`/`roadbox`) across tfpt_1, tfpt_2, tfpt_3 and `origin_theory` are converted to inline bold run-in paragraphs — the reviewer's prescription that an audit fingerprint must not carry the same visual volume as a theorem. The loud box families are now only `keybox` (claim), `winbox` (result) and `gapbox` (open/caveat).

2026-06-12 (visual pass: shared figures, badges, marker typology)

- **Shared figure library `tfpt_figures.tex`** (`\input` by `tfpt_style`): reusable TikZ macros for the architecture *claim stack* (`\TFPTclaimstack[zone]`, B3), the *anchor cockpit* $a = (1, 1, 2)$ (B4), the E_8 *glue* diagram (B5), the 2/3 *motif map* (B6) and the K_5 *action lattice* (B7). The claim stack and a marker *badge bar* are now shown near the front of *every* document, each highlighting that document's layer; the topical figures are placed in their natural homes (cockpit + K_5 in the introduction, cockpit + glue in tfpt_1, motif in the horizon note).
- **Marker typology (A2).** `tfpt_style` adds the sharper sub-types **[E]** (exact algebra), **[E]** (exact number), **[C]** (readout / standard physics in TFPT units) to the existing **[C]** (bridge) and **[X]** (kill test); the marker key and the new `\TFPTbadges` bar show the full typology. (Inline body markers retain **[E]** where they are genuine exact identities.)

2026-06-12 (review pass on list.txt)

- **Claim hygiene (review section A).** The introduction's one-sentence claim is softened from "derives all constants" to "constructs a discrete compiler; physical readouts run through named, status-typed bridges"; the ledger zombie in `FLAV.QRATIO.01` is removed (the statement no longer says "stays [P]" — ratios are **[E]** closed on the derived stratum, only the absolute scale stays **[O]**, with an explicit forbidden-wording note); the seam misalignment is written explicitly as $\varepsilon = \frac{3}{4}\varphi_0^{\mathrm{ret}} = c_3 + 36c_3^4$ (leading c_3, fourth-order puncture correction exact); θ_{12} has a single prediction of record 0.306747 with the seam/non-linear values typed as scheme diagnostics; JUNO's first 59.1-day result (0.3092 ± 0.0087, compatible, precision pending) is recorded and NuFIT 6.0 is marked the pre-JUNO baseline; the ACT DR6 birefringence hint carries a systematics caveat (not a validation target); CODATA-2022 is dated; the null-model figure is moved out of the main text and typed as a grammar-internal bound. "Complete solutions of the five open questions" becomes "Closure status of the five former open questions".
- **Interface language unified (review sections A5/C2/C3/C6/C7/C8).** The live residual is stated everywhere as $v_{\mathrm{geo}} \oplus G_{\mathrm{net}} \oplus F_{\mathrm{transfer}}$ (historical $U_{\mathrm{wall}}/G_{\mathrm{metric}}/F_{\mathrm{frontier}}$ kept only for ledger continuity); `tfpt_research_contracts` is retitled and its recommended order rewritten (selector-triangle pairings $\to v_{\mathrm{geo}} \to G_{\mathrm{net}}$); F_{transfer} is named as one missing functor with Koide/η_B/axion/m_p/m_e as its four instances; full QG closure is framed as a certification layer, not a prerequisite. Document numbering aligned with file names (the Red Team paper is now in the canonical set; document $N \equiv$ `tfpt_N`). Website mirrored throughout (`papers.ts`, `predictions.ts`, `faq.ts`, `glossary.ts`, `OpenGates`, `Hero`, orientation components).

2026-06-12 (later)

- **Shared style and central status artifact.** Two new single-source fragments: `tfpt_style.tex` (the canonical palette, status markers, marker key, `\veri` reference, the four tcolorbox families with back-compat aliases, and one running header/footer) is

now `\input` by all eight documents — the per-paper duplicated preambles are gone; and `tfpt_status.tex` ("TFPT status at a glance") gives one canonical overall-status card imported near the front of every document, replacing the divergent per-paper status statements. Both registered in `build.sh` and the manifest `TEX` list.

2026-06-12

- **Margin theorem (v122).** The established frozen selectors (D_4 annihilator $n = (5, -9, 6)$, $\det R = 8$, $\det K = 4$) pin R uniquely among hexagon matrices ($64 \to 12 \to 1$); the atom margins become theorems — H2 introduces no new residual class.
- **Inventory update (v123).** Residual table re-pinned post v110–v122: exactly five structural classes (R1 clock, R2 index-4 net, R3 EH step, R4′ selector establishment — replacing the discharged H2 dictionary — and R5 Q realisation); R2 carries three loads (metric + carrier + QBL: one theorem, three doors).
- **Resummed clock + glue Q-system (v124/v125).** R1's bend in closed form: $\mathrm{rate}(n) = -p_2 \ln(1 - n/N_{\mathrm{fam}})$ (pole at N_{fam} forces the three-level spectrum); the index-4 μ_4 extension exists explicitly as a Longo Q-system on $\mathbb{C}[\mathbb{Z}_4]$ (all Frobenius axioms exact) — R2 sharpens from "find" to "identify".
- **Clock–wall bridge (v126) and ring resummation (v127).** The resummed-clock weights are the Mehta–Seshadri parabolic weights ($\mathrm{spec}\, A_0^\star$); $\det(\mathbf{1} - A_0^\star) = \frac{2}{9}$ = the entropy curvature; the clock is a log-determinant (RPA rings, one tower per hexagon site, coupling = parabolic weight).
- **Graded hull (v128).** E_8 is a $\mathbb{Z}_4$-graded Lie algebra over the carrier (grading = glue Q-system; exact on all 6720 root-pair sums); ring multiplicity 6 = sheet-doubled Ginsparg–Perry zero modes. *Notation fix in the same round:* the clock factor 6 is $p_2(a)$, not $2p_2$ — labels corrected across modules, ledger, papers, website (numbers unchanged).
- **Entropy power law → zeta budget (v129–v133).** The clock is $\Gamma_n \propto (S_n/S_{dS})^{p_2}$ (Gibbons–Hawking form); $p_2 = 2h$ with $h = N_{\mathrm{fam}}$ from mode counting + the Born rule (v130); the zero-mode norm is the area, $\|Y_{1m}\|^2 = A/(4\pi)$ exactly (v131); the det-ratio anomaly is the Koide constant, $\zeta(0)|_{\det'} = -\frac{2}{3}$ (v132); both $\zeta(0)$ budget routes computed exactly — the reduced seam route carries pure anchor ratios ($-\frac{4}{3}$ = minus the seed gain), the naive 4d route ($-\frac{109}{45}$) carries no atom (v133).
- **Dual anchor + determinant surface (v134/v135).** $d := a^\top R^{-1} = a^\top L^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1)$ with $(1, 1, -2) = -2d$ — the inverse flavor response is the Nariai root, invariance exact via Sherman–Morrison (third algebraic flavor–horizon bridge leg); $\det M(s,t) = 3s(t{+}1)(t{+}2) + t^2 + 5t + 8$ with iso-volume walls at the $\mathbb{Z}_2/\mu_4$ atoms, winding line $(8, 14, 20)$, $\det F = 32$, row-budget axes with $\mathrm{rowsum}(V) = \mathrm{Spec}(Q_+)$; micro-identities $41 = 16 + 25$, $52 = 16 + 36$.
- **Dual-normal selector, Q_+ cohomology, Volkov–Wipf firewall (v136–v138).** (d, n) pins R *columnwise* (each column the unique lattice point of the address box; kernel $(6, 8, 7)$); the $A_0^\star$ zero mode carries the spectral normal $\sigma = (2, -9, 5)$ with a $W(A_3)$ orbit control excluding the naive lift; $H^1(\mathbb{P}^1 \setminus \mu_4) = \chi_1 \oplus \chi_2 \oplus \chi_3$ with degrees $(1, 2, 3) = \mathrm{Spec}(Q_+)$ = the A_3 exponents and cusp weights = $\mathrm{spec}\, A_0^\star$; the external full-4d graviton/ghost value $-\frac{98}{45}$ is pinned (arXiv:2506.02142 = Volkov–Wipf hep-th/0003081) and its *edge* part is exactly two copies of the reduced seam budget — R1 typed closed as an edge-sector object.
- **Selector triangle + canonical map (v139/v140).** $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$ is pure anchor data (first selector *derived*); the frame $(\mathbf{1}, a, \sigma)$ has volume 11 and n is the unique covector with atom pairings $(2, 8, 121)$ — the R4′ residue is three pairings; the $H^1 \to$ generation equivariant map exists and is Schur-rigid (only sign-twisted μ_4 actions match), so the `GATE.QGEO` residue collapses to one $\mathbb{Z}_3$ choice.

- **Editorial and consistency passes.** Content-currency sweep (superseded "single remaining input" claims retired across contracts, paper 3, intro, website); introduction: self-explanatory title (spells out Topological Fixed-Point Theory), the old-version comparison removed throughout, redundant diagrams dropped, the unreadable status heatmap removed, the script-index master table converted to a page-breaking `xltabular` (it had silently truncated ∼110 rows), status card extended with the v134–v140 reductions, predictions table completed (CP phase, cosmic birefringence); the canonical document-set box extracted into the shared `tfpt_docset.tex`; paper 1 cleaned of all 35 old-series references, every section now opens with its machine-check script references; smaller box fonts across all eight documents; this `changelog.tex` introduced as a dual-mode (standalone + imported) document.

2026-06-11

- **External-review integration (v83) and release hygiene.** Manifest `-check` mode and the release rule; holomorphy pins $(E_8)_1$ (the unique even unimodular rank-8 lattice theory).
- **Sheet diamond and centered form (v94/v95).** All four flavor operators are points of one two-parameter surface $M(s,t)$; centered cross $R, L = C \mp U$, $K, F = C \mp V$ with the cofactor seam normal $n = (5, -9, 6)$; documentation pass ("The Sheet Diamond and the Winding Line" in paper 3).
- **Horizon/anchor round (v101–v103).** The maximal (Nariai) black hole is the anchor: roots $(1, 1, -2)$, entropy bound $\frac{2}{3} = |\mathbb{Z}_2|/N_{\text{fam}}$; one repeller/attractor orientation in both sectors; trisection normal form (the canonical cosine coordinate).
- **Clock/ladder round (v104–v108) + Lean hardening.** The classical Nariai clock speaks anchor ($\chi_{\text{clock}} = (\lambda - 1)(\lambda + 2)$); machine-pinned residual inventory; first review validation; quantum-clock target made quantitative; Pascal Ladder Theorem ($g = 2K + 1 \Leftrightarrow$ the closure).
- **QBL theorem chain (v109–v113).** Sheet-Pairing Lemma; Calderón-sheet selection (a scalar seam datum exists iff the involution is sheet-odd); Quadratic-Transport Theorem (degree exactly 2); Self-Counting Channel (the Pascal closure as an identity); quasi-free kernel (one 2-point kernel determines the whole net, rank $= c$) — the QBL input merges with the R2/holomorphy premise; communicated across all status surfaces.
- U_{wall} **made exact (v114–v118).** Torsion normal form and $\delta = \frac{1}{2}$ theorem; exact anchor residue $A_0^\star$ (diag $= a/|\mu_4|$, off-weights $(8, 0, 5)/144$); resonance theorem ($M_\infty = \mathbf{1} \iff a_{13} = 0$, one gauge orbit); the holonomy is the exact 24-element $W(A_3) = S_4$; the hypercharge hexagon is the sign-twisted family spectrum with the lepton coefficients as resolvent determinants.
- **Second review validation + address pinning (v119–v121).** Anchor ratio triad $(\frac{2}{3}, \frac{4}{3}, \frac{5}{3})$; the 121 audit lemma; the lepton words are the compiler atoms $(8, 5, 3)$ with divmod-hexagon addresses; R is the unique hexagon matrix with atom margins and $\det 8$ — the address table carries zero residual information.

2026-06-10

- **B/C/D round closed.** Koide attractor toy model (v93), glue uniqueness (v92), the Wolfram extension as the independent second verification path, Lean hardening; papers and website synced with the round.
- **Content rounds v96–v100.** Sheet conjugation bridge, discriminant dictionary, Koide flow time, and the look-elsewhere-corrected numerology null test (v100: joint null probability $\leq 10^{-30.7}$, conditional on the declared grammar).

2026-06-09

- **Blind registry frozen (v84).** Machine-enforced prediction registry (`predictions_frozen.json`): exactly one θ_{12} number of record before JUNO; conditional entries carry their hypotheses by name. Research-contracts document added to the set.

2026-06-07 / 2026-06-08

- **Compiler-closure consolidation.** The TFPT development is consolidated into the current document set with the verification suite (modules v1–v82: E_8 glue, carrier Pascal, EM fixed point, flavor matrix, cascade, bootstrap, gravity/cosmology, transport, masses, registry and gate audits) as the single proof layer; manifests (`manifest.sha256`, `lean_manifest.sha256`) as content identity; website mirrors the suite (interactive DAG, in-browser script reproducer, script index).

2026-04-27

- **Initial commit.** TFPT papers and website baseline: electromagnetic closure and flavor transport (UFE bridge, birefringence seed, dyonic intercept), PDF-download tracking, accessibility and metadata passes.